

Nuclear Structure and Decay Data for A=35 Isobars*

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Abstract: This work presents a comprehensive and critical evaluation of experimental nuclear spectroscopic data from reactions and decays for all 11 known nuclides with mass number 35 (Ne, Na, Mg, Al, Si, P, S, Cl, Ar, K, Ca). Recommended values are produced for level energies, spins and parities, half-lives, and radiation properties including energies, branching ratios, and multiplicities of γ rays, as well as characteristics of β radiation decays, based on a rigorous assessment of all available experimental data. Discrepancies among existing results are carefully addressed. This work supersedes earlier full evaluations of A=35 published by [2011Ch48](#), [1990En08](#) (also [1998En04](#) update) and [1978En02](#).

^{35}Cl remains the most extensively studied nuclide, investigated through 28 different reactions and decays, followed by ^{35}S , ^{35}P , and ^{35}Ar . For ^{35}Na , only two excited states have been observed, and its β^- -decay scheme has not yet been established. For ^{35}Mg , no γ rays have been placed with certainty, and no decay dataset exists on $^{35}\text{Mg} \rightarrow ^{35}\text{Al}$. ^{35}Ar resonances of astrophysical interest have been studied via transfer reactions and ^{35}K $\varepsilon + \beta^+$ decays, but their properties are yet to be determined. For ^{35}K , many excited states discovered via ^{35}Ca $\varepsilon + \beta^+$ decay are unresolved. For ^{35}Ca , no γ rays have been observed and only one excited state has been identified. Notably, ^{35}Ca is the only nuclide in this mass chain that decays by $\varepsilon + \beta^+$ -delayed two protons, although the emission mechanism remains unclear.

Cutoff Date: Literature available up to September 30, 2025 has been consulted, with the help of the primary bibliographic source, the NSR database ([2014Pr09](#)) available at Brookhaven National Laboratory web page: www.nndc.bnl.gov/nsr/.

General Policies and Organization of Material: See the April 2026 issue of the *Nuclear Data Sheets* or <https://www.nndc.bnl.gov/nds/docs/NDSPolicies.pdf>.

General Comments: The statistical analysis of γ -ray data and deduced level schemes is carried out through a suite of current versions of computer codes available at the IAEA-NDS web page of ENSDF codes (https://www-nds.iaea.org/public/ensdf_pgm) such as GLSC, Java-RULER, BrIccMixing, and ConsistencyCheck. The direct feedings to excited states in β^- and ε decays have generally been computed from $I(\gamma + ce)$ intensity balances at each level; the associated $\log ft$ values are calculated using the BetaShape code. Theoretical total conversion coefficients are calculated using version v2.3e of BrIcc code ([2008Ki07](#)) for frozen-orbit option with an implicit uncertainty of 1.4% when not stated. Q values and particle separation energies have been adopted from [2021Wa16](#) (AME2020). Static magnetic dipole moments are taken from [2019StZV](#) (long-lived states) and [2020StZV](#) (short-lived states) evaluations, and static electric quadrupole moments from [2021StZZ](#) evaluation, when available in these publications. Nuclear rms charge radii have been adopted from [2013An02](#) evaluation. In cases where weighted averaging procedures have been used, the assigned uncertainty in the result is generally not lower than the lowest uncertainty in the data points used in the procedure.

Acknowledgements: Earlier evaluations ([2011Ch48](#), [1998En04](#) update, [1990En08](#), [1978En02](#), [1973EnVA](#), [1967En05](#), [1962En05](#), [1957En01](#), [1954En19](#)), and compilations available in the XUNDL database provided valuable resources for the present work. The evaluators are grateful to the reviewers at different stages of this work, and to the personnel at the National Nuclear Data Center (NNDC) at Brookhaven National Laboratory for facilitating this work.

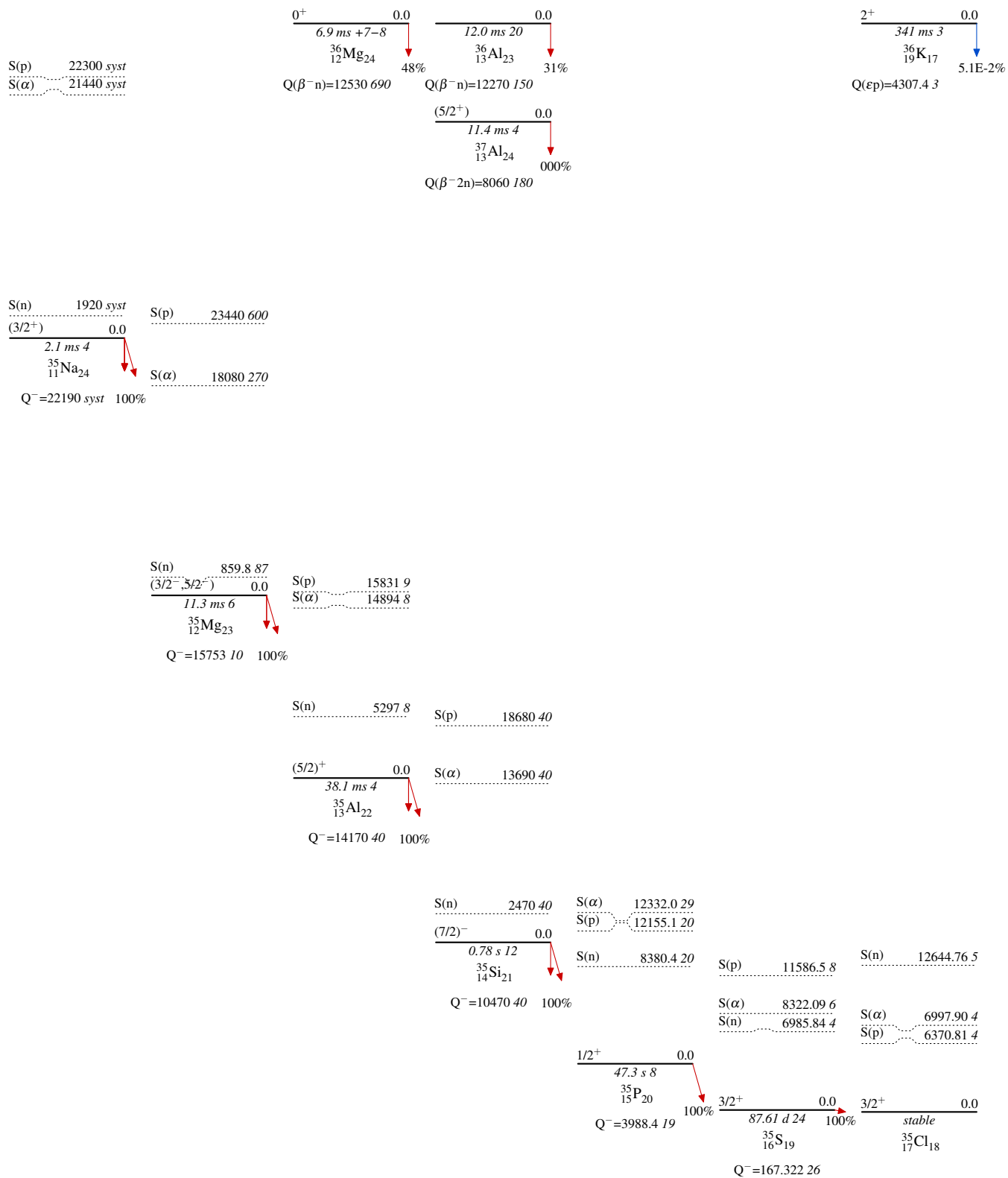
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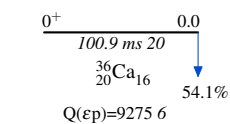
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Skeleton Scheme for A=35

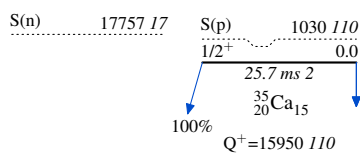


Skeleton Scheme for A=35 (continued)

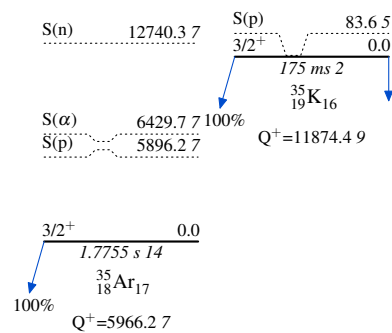


S(n) 17770 syst

S(α) 8940 110



S(α) 6563.3



Ground-State and Isomeric-Level Properties				
Nuclide	Level	J π	T _{1/2}	Decay Mode
³⁵ Na	0.0	(3/2 ⁺)	2.1 ms 4	%β ⁻ =100; %β ⁻ n>0; %β ⁻ 2n=?...
³⁵ Mg	0.0	(3/2 ⁻ , 5/2 ⁻)	11.3 ms 6	%β ⁻ =100; %β ⁻ n=52.46; %β ⁻ 2n=?
³⁵ Al	0.0	(5/2 ⁺)	38.1 ms 4	%β ⁻ =100; %β ⁻ n=37.4; %β ⁻ 2n=?
³⁵ Si	0.0	(7/2 ⁻)	0.78 s 12	%β ⁻ =100; %β ⁻ n<5
³⁵ P	0.0	1/2 ⁺	47.3 s 8	%β ⁻ =100
³⁵ S	0.0	3/2 ⁺	87.61 d 24	%β ⁻ =100
³⁵ Cl	0.0	3/2 ⁺	stable	
³⁵ Ar	0.0	3/2 ⁺	1.7755 s 14	%ε+%β ⁺ =100
³⁵ K	0.0	3/2 ⁺	175 ms 2	%ε+%β ⁺ =100; %εp=0.37 15
³⁵ Ca	0.0	1/2 ⁺	25.7 ms 2	%ε+%β ⁺ =100; %εp=95.83; %ε2p=4.23
³⁶ Mg	0.0	0 ⁺	6.9 ms +7-8	%β ⁻ n=48 12
³⁶ Al	0.0	0 ⁺	12.0 ms 20	%β ⁻ n<31
³⁷ Al	0.0	(5/2 ⁺)	11.4 ms 4	%β ⁻ 2n≥000
³⁶ K	0.0	2 ⁺	341 ms 3	%εp=5.1×10 ⁻²
³⁶ Ca	0.0	0 ⁺	100.9 ms 20	%εp=54.1 12

Adopted Levels: not observed

$Q(\beta^-)=24430$ *calc*; $S(n)=-2370$ *calc*; $S(p)=29480$ *calc* [2019Mo01](#)

$S(2n)=-470$ ([2019Mo01](#), FRDM).

[2019Ah07](#): $^9\text{Be}(^{48}\text{Ca}, X)$ using a 345-MeV/nucleon, 450-pnA ^{48}Ca beam provided by the cascade operation of the RIBF accelerator complex at RIKEN and impinging on a 20-mm-thick beryllium target. Projectile fragments were separated and identified using ΔE -tof- $B\rho$ by the large-acceptance two-stage separator BigRIPS. Tof was measured using two thin plastic scintillators placed at the intermediate and final foci of the second stage of BigRIPS. $B\rho$ was measured from position measurement at the intermediate focus using the plastic scintillator. ΔE was measured using a stack of six silicon detectors installed at the final focus. Optimum settings of $B\rho$ were tuned to transmit ^{33}F for 14 hours and $^{36}\text{Ne}+^{39}\text{Na}$ for 7.8 hours. The Be target was irradiated with 1.4×10^{17} and 7.8×10^{16} ^{48}Ca ions, respectively. Measured Z vs A/Z particle-identification plot. No ^{35}Ne events were observed in either setting. Under ^{33}F setting, the expected ^{35}Ne yields obtained from LISE++ are 177 53 using the production $\sigma=37.8$ fb from EPAX 2.15 systematics and 69 17 using the production $\sigma=14.8$ fb 36 from Q_g systematics.

[2022Ah02](#): Same experimental setup as [2019Ah07](#) with 540-pnA ^{48}Ca beam. Optimum settings of $B\rho$ were tuned to transmit ^{39}Na for 46.1 hours and ^{36}Ne for 25.3 hours. Measured Z vs A/Z particle-identification plot. No ^{35}Ne events were observed in either setting. Also see [2025Ku15](#).

[2023Ra22](#): Relativistic Hartree-Bogoliubov model with nuclear energy density functional (EDF) calculations of binding energies.

[2020Mi15](#): VS-IMSRG ab initio calculations of ground-state energies and $S(2n)$.

 ^{35}Ne Levels

<u>E(level)</u>	<u>Comments</u>
0?	$\%0n=?$; $\%2n=?$ Evaluators estimate the probability of not observing ^{35}Ne events by chance is 2.6×10^{-23} using the lowest expected yield of 52 events (2019Ah07) and Poisson probability distributions. ^{35}Ne is determined to be unbound at a confidence level of $1-2.6\times 10^{-23}$. The heaviest bound neon isotope is ^{34}Ne. J^π: $5/2^-$ calculated projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). $T_{1/2}$: 2.7 ms calculated with respect to Gamow-Teller QRPA transitions and phenomenological first-forbidden contributions (2019Mo01).

Adopted Levels, Gammas

$Q(\beta^-)=22190$ syst; $S(n)=1920$ syst; $S(p)=22300$ syst; $Q(\alpha)=-21440$ syst [2021Wa16](#)

$\Delta Q(\beta^-)=720$, $\Delta S(n)=300$, $\Delta S(p)=840$, $\Delta Q(\alpha)=860$ (syst,[2021Wa16](#)).

$S(2n)=2090$ 810, $Q(\beta^-n)=21440$ 670 (syst,[2021Wa16](#)).

Isotope discovery ([2012Th10](#)): Ir(p,X) ^{35}Na at CERN ([1983La12](#)).

^{35}Na production:

[2025Ku15](#),[2022Ah02](#),[2019Ah07](#): $^9\text{Be}(^{48}\text{Ca},X)$ at RIKEN.

[2002LuZT](#): Ta($^{48}\text{Ca},X$) at GANIL.

^{35}Na decay measurements:

[2022Cr03](#): $^9\text{Be}(^{48}\text{Ca},X)$ at FRIB. Measured $T_{1/2}$.

[2013StZY](#): $^9\text{Be}(^{48}\text{Ca},X)$ at RIKEN. Measured $T_{1/2}$ and β^- -delayed γ rays.

[1983La12](#),[1984La03](#): Ir(p,X) at CERN. Measured $T_{1/2}$ and β^- -delayed neutrons.

^{35}Na mass measurements: None.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π , mass, $T_{1/2}$, etc.): [2024Lu06](#),
[2023Zh15](#), [2023Ba27](#), [2022Ot01](#), [2020Ch21](#), [2020Ts03](#), [2013Li39](#), [2013Sh05](#), [2009Ly01](#), [2004Ge02](#), [2004Lu10](#), [2002Sa08](#),
[1997Mo25](#), [1991Pa19](#), [1991Pa21](#), [1989Ly01](#), [1987SaZQ](#), [1985Ly02](#), [1975Ca27](#).

 ^{35}Na LevelsCross Reference (XREF) Flags

A $^9\text{Be}(^{48}\text{Ca},^{35}\text{Na})$
B $\text{C}(^{36}\text{Mg},^{35}\text{Na}\gamma)$

E(level)	$J^\pi^\dagger$	$T_{1/2}$	XREF	Comments
0.0 ‡	(3/2 $^+$)	2.1 ms 4	AB	$\% \beta^- = 100$; $\% \beta^- n > 0$; $\% \beta^- 2n = ?$; $\% \beta^- 3n = ?$; $\% \beta^- 4n = ?$ ^{35}Na β^- -delayed neutrons have been observed by 1983La12 . Experimental $\% \beta^- n$ values are unknown. Theoretical $\% \beta^- 0n = 1.4$, $\% \beta^- 1n = 73.5$, $\% \beta^- 2n = 20.1$, $\% \beta^- 3n = 4.8$ (2021Mi17). Theoretical $\% \beta^- 0n = 14.0$, $\% \beta^- 1n = 73.0$, $\% \beta^- 2n = 10.0$, $\% \beta^- 3n = 3.0$ (2019Mo01). $T_{1/2}$: weighted average of 2.4 ms 3 (stat) 2 (syst) (2022Cr03 , implant- β correlation), 2.4 ms 3 (stat) 6 (syst) (2013StZY , implant- β correlation), and 1.5 ms 5 (1983La12 , 1984La03 , decay curve of βn -coin).
373 ‡ 5	(5/2 $^+$)		B	
1014 ‡ 17	(7/2 $^+$)		B	

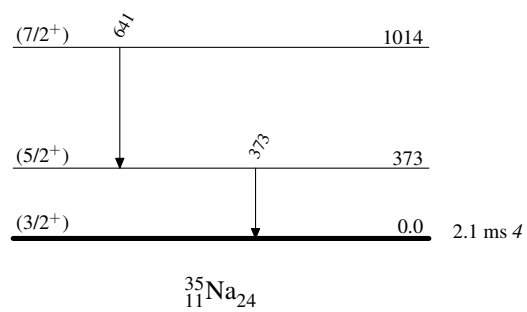
† From Monte-Carlo shell-model calculations using the SPDF-M effective interaction ([2014Do05](#)).

‡ Band(A): $K^\pi=(3/2^+)$ rotational band predicted by the shell model ([2014Do05](#)).

 $\gamma(^{35}\text{Na})$

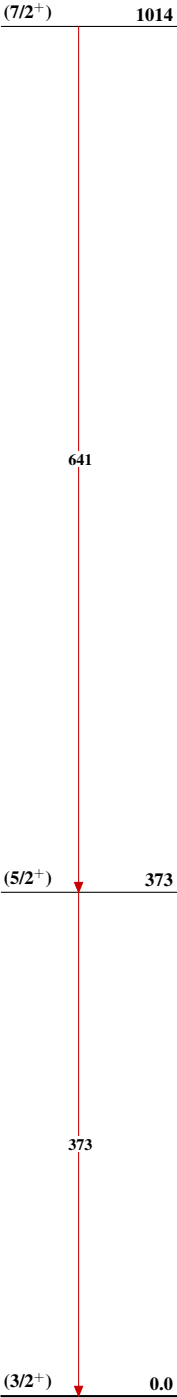
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	E_f	J_f^π	
373	(5/2 $^+$)	373 5	0.0	(3/2 $^+$)	
1014	(7/2 $^+$)	641 16	373	(5/2 $^+$)	

† From $\text{C}(^{36}\text{Mg},^{35}\text{Na}\gamma)$.

Adopted Levels, GammasLevel Scheme

Adopted Levels, Gammas

Band(A): $K^\pi=(3/2^+)$
rotational band
predicted by the shell
model (2014Do05)



³⁵Na₂₄

$^9\text{Be}(^{48}\text{Ca}, ^{35}\text{Na})$ 2013StZY,2022Cr03

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $\text{B}\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured $\text{E}\gamma$, $\beta\gamma$ -coin, and implant- β correlation, and deduced the $T_{1/2}$ of ^{35}Na . Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}\text{Ca}^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr_3 detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event-by-event by energy loss in the upstream PIN detector (Z determination) and by time-of-flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- β correlation and deduced $T_{1/2}$ of ^{35}Na . Comparisons with QRPA and shell-model calculations.

 ^{35}Na Levels

<u>E(level)</u>	<u>$T_{1/2}$</u>	<u>Comments</u>
0.0	2.1 ms 4	$T_{1/2}$: From the Adopted Levels of ^{35}Na . Values from this dataset: 2.4 ms 3 (stat) 2 (syst) (2022Cr03, implant- β correlation); 2.4 ms 3 (stat) 6 (syst) (2013StZY, implant- β correlation).

$\text{C}(^{36}\text{Mg}, ^{35}\text{Na}\gamma)$ 2014Do05

2014Do05: A 345-MeV/nucleon ^{48}Ca primary beam was accelerated by the superconducting ring cyclotron (SRC) at RIKEN. A 236-MeV/nucleon ^{36}Mg secondary beam was produced by the fragmentation of ^{48}Ca on a Be target and separated by the BigRIPS separator using the $\text{B}\rho$ - ΔE - $\text{B}\rho$ method. The secondary reaction targets were 2.54 g/cm² thick carbon and 2.13 g/cm² thick CH₂ polyethylene. The secondary reaction products were identified by the magnetic spectrometer ZeroDegree using the $\text{B}\rho$ - ΔE - $\text{B}\rho$ method. γ rays in coincidence with ^{35}Na were detected using the DALI2 array of 186 large NaI(Tl) detectors. Measured E_γ with Doppler correction, I_γ , and $\gamma\gamma$ -coin. Deduced levels, J , π , and bands. Compared with shell-model calculations using the SPDF-M effective interaction.

 ^{35}Na Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0 [#]	(3/2 ⁺)
373 [#] 5	(5/2 ⁺)
1014 [#] 17	(7/2 ⁺)

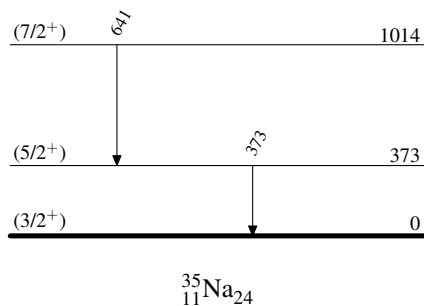
[†] From a least-squares fit to γ -ray energies.

[‡] From shell-model calculations.

[#] Band(A): $K^\pi=(3/2^+)$ rotational band predicted by the shell model.

 $\gamma(^{35}\text{Na})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
373 5	373	(5/2 ⁺)	0	(3/2 ⁺)
641 16	1014	(7/2 ⁺)	373	(5/2 ⁺)

 $\text{C}(^{36}\text{Mg}, ^{35}\text{Na}\gamma)$ 2014Do05Level Scheme

C($^{36}\text{Mg}, ^{35}\text{Na}\gamma$) 2014Do05

Band(A): $K^\pi=(3/2^+)$
rotational band
predicted by the shell
model

($7/2^+$) 1014

641

($5/2^+$) 373

373

($3/2^+$) 0

$^{35}_{11}\text{Na}_{24}$

Adopted Levels, Gammas

$Q(\beta^-)=15753$ 10; $S(n)=859.8$ 87; $S(p)=2.344\times 10^4$ 60; $Q(\alpha)=-1.808\times 10^4$ 27

$Q(\beta^-), S(n), S(p), Q(\alpha)$: Deduced by the evaluators using mass excesses of 15529.5 71 for ^{35}Mg measured by 2025Ly01, and 8318 5 for ^{34}Mg ; a weighted average of 8323 7 (2019As04) and 8315 5 (2025Ly01); -224 7 for ^{35}Al , 31680 600 for ^{34}Na , and 31180 270 for ^{31}Ne from 2021Wa16. Values from 2021Wa16: $Q(\beta^-)=15860$ 270, $S(n)=750$ 270, $S(p)=23330$ 660, $Q(\alpha)=-17970$ 380.

$S(2n)=5576.0$ 76, $Q(\beta^-n)=10455.8$ 74, from mass excesses of 15529.5 71 for ^{35}Mg measured by 2025Ly01; 4962.9 27 for ^{33}Mg and -2997.6 21 for ^{34}Al from 2021Wa16. Values from 2021Wa16: $S(2n)=5470$ 270, $Q(\beta^-n)=10570$ 270.

$S(2p)=45070$ 660 (syst) (2021Wa16).

Isotope discovery (2012Th10): $\text{Ta}(^{48}\text{Ca}, X)$ projectile fragmentation at GANIL (1989Gu03, 1991Or01).

^{35}Mg production:

2012Kw02: ^{35}Mg produced by $\text{natNi}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar})=140$ MeV/nucleon at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2011FuZZ: ^{35}Mg produced by $^9\text{Be}(^{48}\text{Ca}, X)$ fragmentation at $E(^{48}\text{Ca})=345$ MeV/nucleon at RIKEN. Measured thick target fragmentation, deduced production cross sections, and compared with EPAX-2.15 systematics.

2007Ts09: Analyzed fragmentation σ of n-rich Na, Mg isotopes (including ^{35}Mg) from $^9\text{Be}, ^{181}\text{Ta}(^{48}\text{Ca}, X)$. Used systematics based on average binding energy to extrapolate towards drip line, predicting σ for ^{40}Mg and discussing ^{39}Na . Compared different extrapolation models.

^{35}Mg decay measurements:

2013StZY: $^9\text{Be}(^{48}\text{Ca}, X)$ at RIKEN. Measured $T_{1/2}$.

1999YoZW: $^9\text{Be}(^{48}\text{Ca}, X)$ and $^{181}\text{Ta}(^{48}\text{Ca}, X)$ at RIKEN. Measured $T_{1/2}$ and $\% \beta^- n$.

^{35}Mg radius measurements:

2011Ka01: ^{35}Mg produced by $^9\text{Be}(^{48}\text{Ca}, X)$ fragmentation at GSI. Measured interaction cross sections with C and CH_2 targets at 900 MeV/nucleon. Deduced rms matter radii.

2006Kh08: ^{35}Mg produced by $^{181}\text{Ta}(^{48}\text{Ca}, X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Mg mass measurements: 2025Ly01, 2007Ju03, 2001Sa72, 2000Sa21, 1991Or01.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J, π , mass, $T_{1/2}$, etc): 2023Ra22,

2021Ka07, 2020Mi15, 2016Ba59, 2016Sa46, 2016Sh05, 2015Sh21, 2014Ga13, 2014Wa14, 2013Ch31, 2013Li39, 2013Sh05,

2012Fo27, 2012Ho19, 2007Ha53, 2006Zh19, 2005Ch71, 2004Kh16, 1996Re10, 1991Pa19, 1991Pa21.

The level scheme is based on that of 2011Ga15 and 2017Mo26.

 ^{35}Mg LevelsCross Reference (XREF) Flags

- A ^{35}Na β^- decay (2.1 ms)
 B $^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$
 C $\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(3/2^-, 5/2^-)$	11.3 ms 6	BC	$\% \beta^- = 100$; $\% \beta^- n = 52$ 46; $\% \beta^- 2n = ?$ $\% \beta^- n$: from 1995ReZZ, 2008ReZZ. Other: 52 11 (1999YoZW, preliminary). Theoretical $\% \beta^- 0n = 29$, $\% \beta^- 1n = 66$, $\% \beta^- 2n = 5$ (2021Mi17). Theoretical $\% \beta^- 0n = 65$, $\% \beta^- 1n = 32$, $\% \beta^- 2n = 3$ (2019Mo01). J^π : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+2p $_{1/2}$ interactions (2017Mo26). Nearly degenerate 30-keV $3/2^-$ and $5/2^-$ g.s. from Monte Carlo shell-model calculations with the SDPF-M interaction (2011Ga15), and $3/2^-$ g.s. from shell-model calculations with the SDPF-U interaction (2011Ga15). $3/2^-$ from projection of the odd-neutron angular momentum along the symmetry axis and parity of the wave function (2019Mo01). Others: $3/2^+$ from antisymmetrized molecular dynamics (AMD) calculations with the Gogny DIS force (2017Mo26).

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Mg Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>XREF</u>	<u>Comments</u>
			T _{1/2} : 11.3 ms 5 (stat) 4 (syst) (2013StZY, implant-β correlation). Other: 72 ms 43 (2008ReZZ, 1995ReZZ) and ≈9 ms (1999YoZW, implant-β correlation, preliminary). Reduced strong absorption radius r ₀ ² =1.64 fm ² 15 from the energy-integrated σ of Si(³⁵ Mg,X) (2006Kh08). The rms matter radius R _{rms} ^m =3.40 fm 24 from the interaction σ of C,CH ₂ (³⁵ Mg,X) (2011Ka01).
0+x		BC	E(level): x≤80 keV (2011Ga15 detection threshold); x≤200 keV (2017Mo26 detection threshold). J ^π : Shell-model calculations with the SDPF-M+2p _{1/2} interaction predict a 1/2 ⁻ level at 141 keV (2017Mo26). Shell-model calculations with the SDPF-M interaction predict a 5/2 ⁻ level at 84 keV (2017Mo26). Monte Carlo shell-model calculations with the SDPF-M interaction predict a 3/2 ⁻ level at 30 keV (2011Ga15).
0+y?		 C	E(level): y≤200 keV (2017Mo26 detection threshold). 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the l=3 orbit in ³⁶ Mg in the inclusive parallel momentum distribution.
206+x 8		C	J ^π : 2017Mo26 states that based on the observed weak γ-ray intensity, this level is not the 1/2 ⁻ level at 141 keV predicted by shell-model calculations with the SDPF-M+2p _{1/2} interaction.
445+x 5	(3/2 ⁺ , 5/2 ⁺)	BC	J ^π : L(³⁶ Mg, ³⁵ Mg)=(2) from 0 ⁺ . 3/2 ⁺ from shell-model calculations with the SDPF-M+2p _{1/2} interaction (2017Mo26).
619+x 7	(1/2 ⁻ , 3/2 ⁻)	BC	J ^π : L(³⁶ Mg, ³⁵ Mg)=(1) from 0 ⁺ . 3/2 ⁻ from shell-model calculations with the SDPF-M+2p _{1/2} interaction (2017Mo26).
670+x 8		BC	

[†] From E_γ data in (³⁸Si, ³⁵Mgγ) and (³⁶Mg, ³⁵Mgγ), (³⁷Al, ³⁵Mgγ).

γ(³⁵Mg)

<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_γ</u>	<u>I_γ</u>	<u>E_f</u>	<u>Comments</u>
206+x		206 8	100	0+x	E _γ : From 2017Mo26.
445+x	(3/2 ⁺ , 5/2 ⁺)	445 5	100	0+x	E _γ : weighted average of 443 7 (2017Mo26) and 446 5 (2011Ga15).
619+x	(1/2 ⁻ , 3/2 ⁻)	619 7	100	0+x	E _γ : weighted average of 616 8 (2017Mo26) and 621 7 (2011Ga15).
670+x		670 [†] 8	100	0+x	E _γ : From 2011Ga15, as this γ is not resolved from the 616γ in 2017Mo26, but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670γ remains vague.

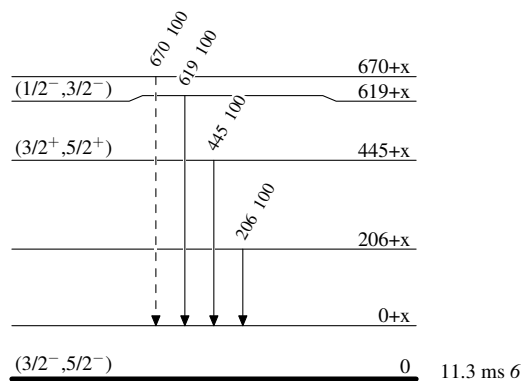
[†] Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{12}\text{Mg}_{23}$

^{35}Na β^- decay (2.1 ms) 2013StZY,2022Cr03,1983La12

Parent: ^{35}Na : $E=0$; $J^\pi=(3/2^+)$; $T_{1/2}=2.1$ ms 4; $Q(\beta^-)=22190$ syst; $\% \beta^-$ decay=100

^{35}Na - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Na .

^{35}Na - $Q(\beta^-)$: 22190 720 (syst, 2021Wa16).

^{35}Na - $\% \beta^-$ decay: $\% \beta^- = 100$, $\% \beta^- n > 0$.

2013StZY: ^{35}Na was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on a 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , $\beta\gamma$ -coin, and implant- β correlation, and deduced the $T_{1/2}$ of ^{35}Na . Comparisons with QRPA and shell-model calculations.

2022Cr03: ^{35}Na was produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA $^{48}\text{Ca}^{20+}$ primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr_3 detectors, and the VANDLE array of 88 neutron detectors. Two Si PIN detectors and a plastic scintillator were placed 1.5 m upstream in the last diagnostic detector box of the beamline. Ions were identified event by event by energy loss in the upstream PIN detector (Z determination) and time of flight between a plastic timing scintillator at the start of stage 3 of ARIS and the timing scintillator in the diagnostic detector box (A/Q determination) over a flight path of 33.5 m. Measured implant- β correlation and deduced $T_{1/2}$ of ^{35}Na .

Comparisons with QRPA and shell-model calculations.

1983La12,1984La03: ^{35}Na was produced via the fragmentation of a 30 g/cm² iridium target by 10-GeV protons from the CERN synchrotron. The recoiling nuclear fragments are thermalized in heated graphite. The alkali elements selectively diffuse out of the target, are surface ionized, and analyzed by the online mass spectrometer, then focused at the bottom of a well-shaped NE213 liquid scintillator neutron detector. β particles were detected using a well-shaped thin plastic scintillator. Measured β and neutron counts. Deduced the $T_{1/2}$ of ^{35}Na . P_n values were deduced for other isotopes but not for ^{35}Na .

^{35}Na β^- -delayed neutrons have been observed by 1983La12. Experimental $\% \beta^- n$ values are unknown.

 $\gamma(^{35}\text{Mg})$

E_γ	Comments
^x 661	2013StZY observed this γ ray in the background-subtracted γ -ray spectrum in a time window of 0-10 ms after the implantation of ^{35}Na but did not observe it in later time windows. 2013StZY proposed that it could either be the deexcitation of the first 2^+ to 0^+ g.s. in ^{34}Mg or the deexcitation of an excited state in ^{35}Mg . Based on theoretical calculations (2019Mo01,2021Mi17), the $\beta^- 1n$ should be the dominant branch, implying that this γ ray is more likely from ^{34}Mg .

^x γ ray not placed in level scheme.

$^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$ **2011Ga15** $^{38}\text{Si} \rightarrow 2\text{p} + 1\text{n} + ^{35}\text{Mg}$.

2011Ga15: A ^{38}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target and selected by the A1900 separator at NSCL, MSU. $E(^{38}\text{Si})=83$ MeV/nucleon in the middle of the **376(4)**-mg/cm² ^9Be secondary target. The reaction leading to ^{35}Mg from ^{38}Si is likely dominated by the two-proton knockout into the continuum of ^{36}Mg and subsequent neutron emission. The reaction residues were selected and identified by the S800 spectrograph using the ΔE -ToF method. The γ rays in coincidence with reaction residues were detected using a 32-fold segmented high-purity germanium detector array (SeGA) at 90° and 37°. Measured Doppler-corrected $E_\gamma(>80 \text{ keV})$ and deduced levels. Results were compared with Monte Carlo shell-model calculations using the SDPF-M effective interaction that allows for unrestricted mixing of neutron particle-hole configurations across the $N = 20$ gap, and with conventional shell-model calculations with the SDPF-U effective interaction that does not include neutron intruder configurations in the model space.

 ^{35}Mg Levels

E(level) [†]	J^π	Comments
0	(5/2 ⁻)	J^π : From shell-model calculations with the SDPF-M interaction.
0+x		
0+y		
446+x 5		
621+y 7		
670+y 8		

[†] From measured E_γ . $x \leq 80$ keV and $y \leq 80$ keV (detection threshold). Shell-model calculations with the SDPF-M interaction predicts a 3/2⁻ level at 30 keV.

 $\gamma(^{35}\text{Mg})$

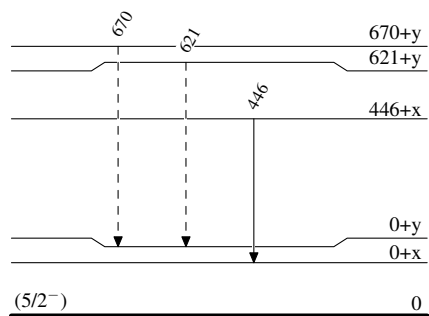
2011Ga15 states that the γ -ray transitions they observed were not in coincidence and depopulate excited states at ≈ 500 and ≈ 700 keV to the ground state or the alleged near-degenerate excited state below 100-keV excitation energy. **2017Mo26** states that the difference of the γ -ray emission between 1n-knockout and 1p1n-removal reaction channels suggests that the 621 and 670-keV γ rays were emitted from two different excited states. Another possible placement of the 621 and 670-keV γ rays: **2011Ga15** states that it might be possible that the two transitions originate from the same level and that their energy difference of **49(11)** keV corresponds to the energy spacing of the alleged (5/2⁻, 3/2⁻) doublet near the ground state. In Adopted Gammas, all γ rays in this dataset are placed to feed the same final levels based on suggestions by **2017Mo26**.

E_γ	$E_i(\text{level})$	E_f	Comments
446 5	446+x	0+x	E_γ : strong γ transition observed in 2011Ga15 ; could correspond to a transition to the g.s. or the shell-model predicted low-lying excited state from any of the shell-model predicted levels at 350 keV with $J^\pi=7/2^-$; 360 keV with $J^\pi=3/2^+$, 400 keV with $J^\pi=1/2^-$, or 560 keV with $J^\pi=7/2^-$.
621 [†] 7	621+y	0+y	
670 [†] 8	670+y	0+y	

[†] Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{38}\text{Si}, ^{35}\text{Mg}\gamma)$ 2011Ga15

Legend

Level Scheme-----► γ Decay (Uncertain) $^{35}_{12}\text{Mg}_{23}$

$\text{C}(^{36}\text{Mg}, ^{35}\text{Mg}\gamma), (^{37}\text{Al}, ^{35}\text{Mg}\gamma)$ **2017Mo26**

$^{36}\text{Mg} \rightarrow 1\text{n} + ^{35}\text{Mg}$ from $J^\pi=0^+$ ^{36}Mg ground state.

$^{37}\text{Al} \rightarrow 1\text{p}1\text{n} + ^{35}\text{Mg}$ from $J^\pi=(5/2^+)$ ^{37}Al ground state.

2017Mo26: A secondary beam composed of ^{36}Mg and ^{37}Al was produced via the projectile fragmentation of a 345-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target and selected by the BigRIPS separator at RIKEN. $E(^{36}\text{Mg})=235$ MeV/nucleon and $E(^{37}\text{Al})=246$ MeV/nucleon in front of the 2.54 g/cm² carbon secondary target. The reactions leading to ^{35}Mg from ^{36}Mg and ^{37}Al are likely 1n-knockout and 1p1n-removal reactions, respectively. The reaction residues were selected and identified by the Zero Degree spectrometer using the B ρ - ΔE -ToF method. The γ rays in coincidence with ^{35}Mg residues were detected using the DALI2 array of 186 NaI(Tl) crystals at 20°–150°. Measured $E_\gamma(>200\text{ keV})$, I_γ , $(^{35}\text{Mg})\gamma$ -coin, the inclusive one-neutron knockout cross section and exclusive γ -ray emission cross sections, and parallel momentum distributions of ^{35}Mg in coincidence with γ rays. Deduced levels, L-transfers, J, and π . Compared with shell-model calculations using the SDPF-M interaction in the sd shell with $\nu 1f_{7/2}$ and $\nu 2p_{3/2}$ orbits, and the SDPF-M interaction in a model space up to $\nu 2p_{1/2}$, and antisymmetrized molecular dynamics (AMD) model calculations using the Gogny D1S force.

 ^{35}Mg Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$L^\ddagger$	Comments
0			J^π : $3/2^-$ from shell-model calculations with the SDPF-M and SDPF-M+ $2p_{1/2}$ interactions; $3/2^+$ from AMD with the Gogny D1S force (2017Mo26).
0+x			$E(\text{level})$: <200 keV; the detection threshold in 2017Mo26 . The original placement in 2017Mo26 of all γ rays is to the ground state, which cannot be confirmed with existing data.
0+y?	($5/2^-$, $7/2^-$)		$E(\text{level})$: <200 keV; the detection threshold. 2017Mo26 suggested a low-lying level from the observed 42(1)% neutron knockout from the $l=3$ orbit in the inclusive parallel momentum distribution.
206+x 8			J^π : γ -ray intensity is too low to be assigned to the $1/2^-$ level at 141 keV from shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
443+x 7	($3/2^+$, $5/2^+$)	(2)	
616+x 8	($1/2^-$, $3/2^-$)	(1)	
670+x 8			

[†] From E_γ data. **2017Mo26** suggested that all the four observed γ rays were emitted independently and fed either the ground state or a low-lying excited state below 200 keV. But the suggested experimental level scheme in FIG.4 of **2017Mo26** (with no γ rays shown) indicates all four γ rays feeding the ground state, which is not adopted by the evaluators since it cannot be confirmed with existing data.

[‡] **2017Mo26** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of ^{35}Mg residues. J^π options are deduced accordingly.

 $\gamma(^{35}\text{Mg})$

2017Mo26 states that no clear $\gamma\gamma$ coincidence was observed.

It is possible that the final levels of 206 γ , 443 γ , 616 γ , and 670 γ are not the same level from authors' comparisons with shell-model calculations with different interactions.

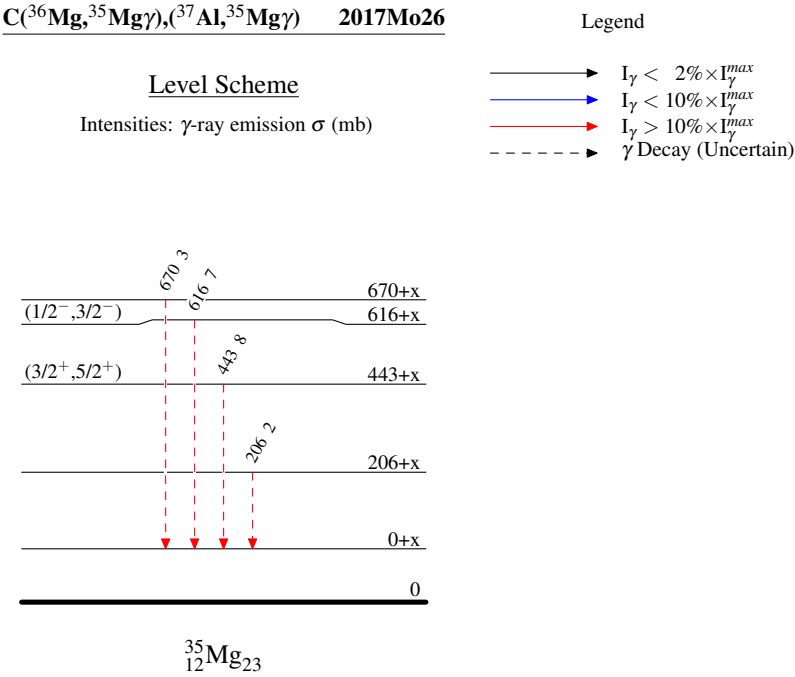
E_γ	$I_\gamma^\dagger$	$E_i(\text{level})$	J^π_i	E_f	Comments
206 ‡ 8	2 1	206+x		0+x	
443 ‡ 7	8 1	443+x	($3/2^+$, $5/2^+$)	0+x	2017Mo26 assigned this γ as the $3/2^+$ at 788 keV to $3/2^-$ g.s. transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
616 ‡ 8	7 1	616+x	($1/2^-$, $3/2^-$)	0+x	2017Mo26 assigned this γ as the $3/2^-$ at 664 keV to $1/2^-$ at 141 keV transition in shell-model calculations with the SDPF-M+ $2p_{1/2}$ interaction (2017Mo26).
670 ‡ 8	3 1	670+x		0+x	E_γ : From 2011Ga15 , as this γ is not resolved from the 616 γ in 2017Mo26 , but its presence is indicated in the fit of the spectrum. 2017Mo26 states that the origin of the 670 γ remains vague.

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C(³⁶Mg,³⁵Mgγ),(³⁷Al,³⁵Mgγ) 2017Mo26 (continued)

γ(³⁵Mg) (continued)

[†] γ-ray emission σ (mb).
[‡] Placement of transition in the level scheme is uncertain.



Adopted Levels, Gammas

$Q(\beta^-)=14170$ 40; $S(n)=5297$ 8; $S(p)=15831$ 9; $Q(\alpha)=-14894$ 8 [2021Wa16](#)

$S(p), Q(\alpha)$: Deduced by evaluators using mass excesses of 8318 5 for ^{34}Mg : a weighted average of 8323 7 ([2019As04](#)) and 8315 5 ([2025Ly01](#)), and 12245.3 26 for ^{31}Na measured by [2025Ly01](#); -224 7 for ^{35}Al from [2021Wa16](#). Values from [2021Wa16](#): $S(p)=15836$ 10, $Q(\alpha)=-14895$ 16.

$S(2p)=38400.5$ 85, $Q(\epsilon)=-15753$ 10, from mass excesses of 23598.8 43 for ^{33}Na and 15529.5 71 for ^{35}Mg measured by [2025Ly01](#); -224 7 for ^{35}Al from [2021Wa16](#). Values from [2021Wa16](#): $S(2p)=38580$ 450, $Q(\epsilon)=-15860$ 270.

$S(2n)=7869$ 10, $Q(\beta^-n)=11697$ 7 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{C}(^{40}\text{Ar}, \text{X})$ projectile fragmentation at Berkeley ([1979Sy01](#)).

^{35}Al production:

[2015Mo17](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

[2012Kw02](#): $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

[2012Zh06](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

[2007No13](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

^{35}Al decay measurements:

[2017Ha23](#): $^9\text{Be}(^{40}\text{Ar}, \text{X})$ at HIRFL. Measured $T_{1/2}$.

[2005Ti11](#), [2006AnZW](#): $(^{36}\text{S}, \text{X})$ at GANIL. Measured $T_{1/2}$, β^- -delayed γ and neutron spectroscopy.

[2001Nu01](#), [2002Nu02](#): $\text{U}(p, \text{X})$ at CERN. Measured $T_{1/2}$, β^- -delayed γ and neutron spectroscopy.

[1999YoZW](#): $^9\text{Be}(^{48}\text{Ca}, \text{X})$ and $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ at RIKEN. Measured $T_{1/2}$ and $\% \beta^-n$.

[1995ReZZ](#), [2008ReZZ](#): $^{232}\text{Th}(p, \text{X})$ at LAMPF. Measured $T_{1/2}$ and $\% \beta^-n$, and average E_n .

[1989Le16](#), [1989MuZU](#): $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ at GANIL. Measured $T_{1/2}$ and $\% \beta^-n$.

[1988Mu08](#), [1988MuZY](#), [1988BaYZ](#): $^{181}\text{Ta}(^{86}\text{Kr}, \text{X})$ at GANIL. $T_{1/2}$ and $\% \beta^-n$.

^{35}Al radius measurement:

[2006Kh08](#): ^{35}Al produced by $^{181}\text{Ta}(^{48}\text{Ca}, \text{X})$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

^{35}Al knockout-reaction measurements:

[2012No05](#): $^{33,34,35,36}\text{Al}$ produced by $\text{Be}(^{48}\text{Ca}, \text{X})$ fragmentation at $E(^{48}\text{Ca})=900$ MeV/nucleon at GSI. Measured $1n$ removal cross sections and longitudinal momentum distributions of the residues. Deduced single-particle occupancies in the ground states of $^{33,34,35}\text{Al}$. $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=75$ mb 4 and $\sigma(^{36}\text{Al} \rightarrow ^{35}\text{Al})=95$ mb 5.

[2010Ro23](#): ^{35}Al produced by $^9\text{Be}(^{40}\text{Ar}, \text{X})$ fragmentation at $E(^{40}\text{Ar})=700$ MeV/nucleon at GSI. Measured $1n$ knockout cross sections and longitudinal momentum distributions of the residues. $\sigma(^{35}\text{Al} \rightarrow ^{34}\text{Al})=65$ mb 18.

^{35}Al in-beam γ spectroscopy:

[2006FuZX](#): ^{35}Al produced by $\text{Be}, \text{C}(^{40}\text{Ar}, \text{X})$ and observed one 760.1-keV 21 γ ray from $\text{He}(^{35}\text{Al}, ^{35}\text{Al}\gamma)$ without placing it in the level scheme.

^{35}Al mass measurements: [2017Ga20](#), [2007Ju03](#), [1991Or01](#), [1991Zh24](#), [1987Gi05](#).

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J , π , mass, $T_{1/2}$, etc): [2016Sa46](#), [2014Ca21](#), [2013Li39](#), [2013Sh05](#), [2011Ki12](#), [2009Yo05](#), [2004Kh16](#), [1994Po05](#).

 ^{35}Al LevelsCross Reference (XREF) Flags

A	$^{36}\text{Mg} \beta^-n$ decay (6.9 ms)	D	$\text{Pb}(^{35}\text{Al}, ^{34}\text{Al}n\gamma)$
B	$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$	E	Coulomb excitation
C	$\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$		

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Al Levels (continued)

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(5/2)^+$	38.1 ms 4	AB DE	$\% \beta^- = 100$; $\% \beta^- n = 37$ 4; $\% \beta^- 2n = ?$ J^π : From shell-model calculations and $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$ from 0^+ . $T_{1/2}$: weighted average of 38.4 ms 3 (2017Ha23, implant- β correlation), 36.8 ms 5 (2005Ti11, 2006AnZW, implant- β correlation), 38.6 ms 4 (2001Nu01, 2002Nu02, three β -counting rates and one γ -counting rate), 30 ms 4 (1995ReZZ, 2008ReZZ, implant- β/n correlation), 170 ms +90-50 (1989Le16, 1989MuZU, β -decay curve), 30 ms 10 (1988DuZT, beam pulsations), and 130 ms +100-50 (1988Mu08, 1988BaYZ, β counting rate). $\% \beta^- n$: Unweighted average of 41 2 (2005Ti11, 2006AnZW), 41 13 (2001Nu01, 2002Nu02), 26 4 (1995ReZZ, 2008ReZZ), and 40 10 (1989Le16, 1989MuZU). Others: 43 9 (1999YoZW, preliminary), 87 +37-25 (1988Mu08, 1988BaYZ) with a non-physical upper bound. Note that the original value reported in 2005Ti11 is 38 2 (2006AnZW report 38 3) and the evaluators have added intensities of γ rays deexciting the 4379 (0.9% 2) and 4255 (2.0% 8) levels in ^{34}Si not included in their reported $\% \beta^- n$ value, as the authors state that these levels are fed by either neutrons with energy below their detection threshold or by weak transitions that cannot be identified in the neutron spectrum. $\% \beta^- 2n$: No evidence was found for any $A=33$ activity and $2n$ emission was thus found negligible (2001Nu01). Reduced strong absorption radius $r_0^2 = 1.188 \text{ fm}^2$ 14 from the energy-integrated σ of $\text{Si}(^{35}\text{Al}, X)$ (2006Kh08). Major configurations for $J^\pi = 5/2^+$ of ^{35}Al g.s. from $\text{Pb}(^{35}\text{Al}, ^{34}\text{Aln}\gamma)$: (g.s., 4^- in ^{34}Al) $\otimes$ $\nu p_{3/2}$, $S=0.36$ 9 (2017Ch36); (46 keV, 1^+ in ^{34}Al) $\otimes$ $\nu d_{3/2}$, $S=1.47$ 22 (2017Ch36); (1.4 MeV, 2^+ in ^{34}Al) $\otimes$ $\nu s_{1/2}$, $S=0.16$ 1 (2021Bh12); (2.5 MeV, 3^- in ^{34}Al) $\otimes$ $\nu p_{3/2}$, $S=1.48$ 18 (2021Bh12).
802 3			AB	
1007 4			B E	XREF: E(1020) B(E2) $\uparrow = 0.0142$ 52 from Coulomb excitation assuming E2 excitation mode (1999Ib01). Others: B(E1) $\uparrow \leq 0.00020$ 9, B(E2) $\uparrow \leq 0.0125$ 56, B(M1) $\uparrow \leq 0.0024$ 11.
1866 4			B	
1975 4	$3/2^+, 5/2^{+\ddagger}$		B	
2734 7			B	
3245 5	$3/2^+, 5/2^{+\ddagger}$		B	
4275? 9	$3/2^+, 5/2^{+\ddagger}$		B	XREF: B(?) Level exists assuming that the 4275 γ feeds the ground state of ^{35}Al (2014St18).

 † From a least-squares fit to γ -ray energies. ‡ From $L(^{36}\text{Si}, ^{35}\text{Al}) = 2$ of parallel momentum distributions of ^{35}Al residues (2014St18). $\gamma(^{35}\text{Al})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
802		802 4	100	0	$(5/2)^+$	
1007		1006 6	100	0	$(5/2)^+$	E_γ : weighted average of 1003 4 from $(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ and 1020 9 from Coulomb excitation.
1866		859 4	100 8	1007		
		1064 4	22 6	802		
1975	$3/2^+, 5/2^+$	968 4	59 4	1007		
		1174 5	37 4	802		
		1972 6	100 7	0	$(5/2)^+$	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Al})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
2734		1932 6	100	802		
3245	$3/2^+, 5/2^+$	2237 6	100 8	1007		
		2440 7	18.0 26	802		
		3250 8	42 5	0	$(5/2)^+$	
4275?	$3/2^+, 5/2^+$	4275 ‡ 9	100	0	$(5/2)^+$	2014St18 states that the 4275γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

† From $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$, unless otherwise noted.

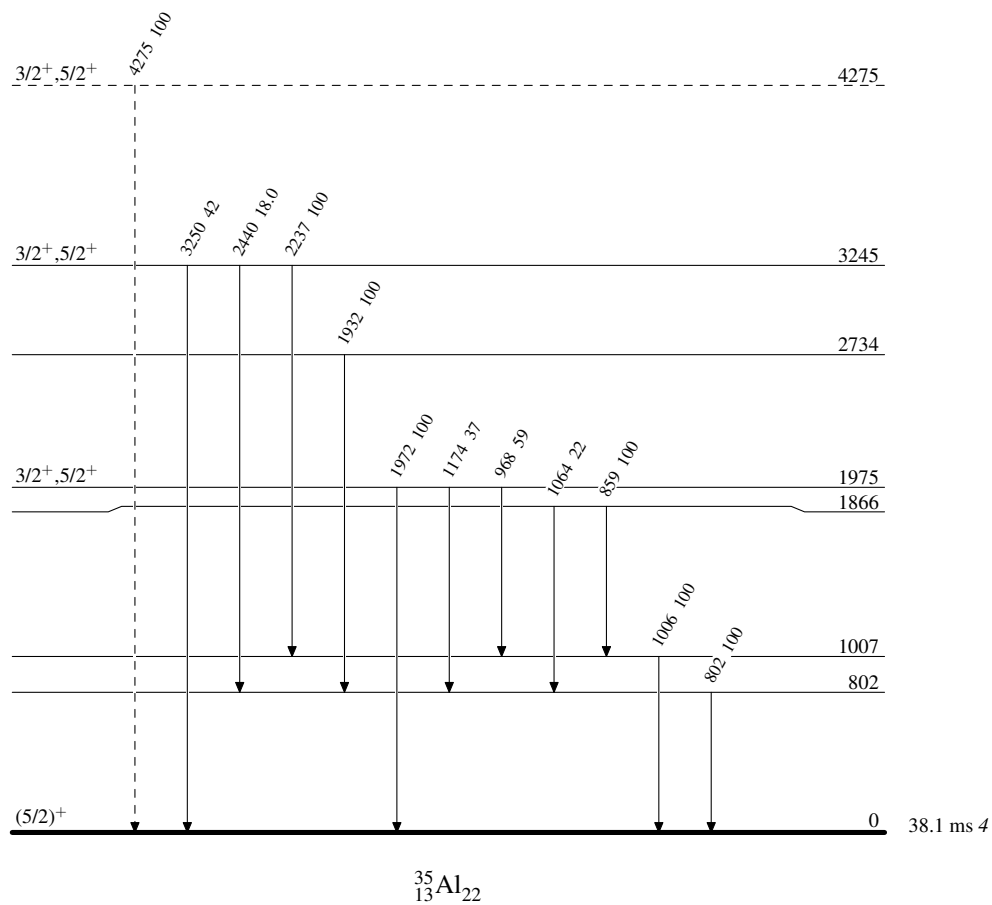
‡ Placement of transition in the level scheme is uncertain.

Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)

^{36}Mg β^- -n decay (6.9 ms) 2023Lu07

Parent: ^{36}Mg : $E=0$; $J^\pi=0^+$; $T_{1/2}=6.9$ ms $+7-8$; $Q(\beta^-n)=1.253\times 10^4$ 69; $\%\beta^-n$ decay=48 12

^{36}Mg - $T_{1/2}$: Weighted average of 3.9 ms 13 (2004Gr20,2003Gr22, implant- β correlation), 7.6 ms $+5-8$ (2013StZY, implant- β correlation, original $T_{1/2}=7.6$ ms 1 (stat) $+5-8$ (syst)), 7.2 ms 12 (2022Cr03, implant- β correlation, original $T_{1/2}=7.2$ ms 1 (stat) 12 (syst)), and 6.8 ms 10 (2023Lu07, implant- $\beta\gamma$ correlation). Other: ≈ 5 ms (1999YoZW, implant- β correlation, preliminary).

^{36}Mg - $Q(\beta^-n)$: From 2021Wa16.

^{36}Mg - $\%\beta^-n$ decay: From 1999YoZW (preliminary; not adopted in Adopted Levels of ^{36}Mg).

2023Lu07: Exp 1: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA ^{48}Ca primary beam from the NSCL cyclotrons impinging on a 642-mg/cm²-thick ^9Be target. The secondary cocktail beam centered around ^{33}Na was selected by the A1900 separator and implanted into a CeBr₃ scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 LaBr₃ detectors. Exp 2: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA ^{48}Ca primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr₃ detectors, and the VANDLE array of 88 neutron detectors. Measured E_γ , I_γ , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, implant- $\beta\gamma$ correlation. Deduced $T_{1/2}$ of ^{36}Mg g.s., ^{36}Al g.s. and a ^{36}Al isomer. Comparisons with FSU shell-model calculations.

2013StZY: ^{36}Mg was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , $\beta\gamma$ -coin, and implant- β correlation. Deduced parent $T_{1/2}$. Comparisons with QRPA and shell-model calculations.

 ^{35}Al Levels

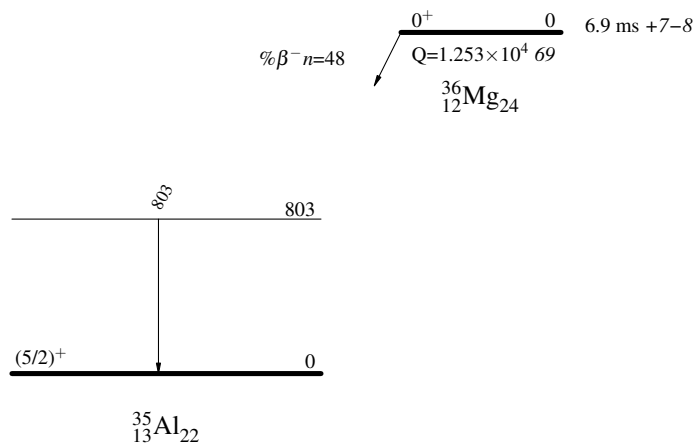
<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0 803	(5/2) ⁺	J^π : From the Adopted Levels.

 $\gamma(^{35}\text{Al})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
803	803	0	(5/2) ⁺	E_γ: From 2023Lu07. 2023Lu07 observed the 803γ within the 100-ms window following the arrival of a ^{36}Mg ion in both NSCL and FRIB experiments. 2013StZY observed a 804-keV γ ray that was not in coincidence with other γ rays. 2013StZY states that the 804γ is a prompt line that predominantly appears in the time window 0-10 ms after an implantation of ^{36}Mg, so it must originate from a deexcitation in one of the daughter nuclei $^{34,35,36}\text{Al}$.

^{36}Mg β^-n decay (6.9 ms) 2023Lu07

Decay Scheme



$\text{He}(^{35}\text{Al}, ^{35}\text{Al}'\gamma)$ **2006FuZX**

2006FuZX: ^{35}Al was produced via the projectile fragmentation of a 63-MeV/nucleon ^{40}Ar primary beam impinging on 1-mm-thick C and Be targets and selected by the RIKEN Projectile-Fragment Separator (RIPS). The secondary cocktail beam was directed onto a 150-mg/cm² liquid He target and the outgoing particles were identified by energy loss, kinetic energy, and time-of-flight measured using a Si detector, a NaI calorimeter, and a couple of plastic scintillators. The positions and directions of incident and outgoing particles were measured using PPACs. Only outgoing particles were identified so that the excited states could be populated by any allowed reactions. Deexcited γ rays emitted from outgoing particle moving at $v/c \approx 0.3$ were detected by the Gamma-Ray detector Array with Position and Energy sensitivity (GRAPE) consisting of segmented Ge detectors surrounding the liquid He target.

 $\gamma(^{35}\text{Al})$

$E_{\gamma}^{\dagger}$
^x760.1 21

[†] From **2006FuZX**.

^x γ ray not placed in level scheme.

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al})$ **2014St18**

$^{36}\text{Si} \rightarrow \text{p} + ^{35}\text{Al}$ from $J^\pi=0^+$ ^{36}Si ground state.

2014St18: A ^{36}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target at NSCL, MSU and was selected by the A1900 separator. The states of ^{35}Al and ^{35}Si were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Si beam at a midtarget energy of 97.7 MeV/nucleon 5 on a 287-mg/cm² ^9Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt γ rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected E_γ , I_γ , $(^{35}\text{Al})\gamma$ -coin, $\gamma\gamma$ -coin, the parallel momentum distributions of populated states in ^{35}Al residues. Deduced levels, J , π , L-transfers, inclusive and exclusive knockout cross section for producing ^{35}Al from ^{36}Si . Calculations using eikonal model and shell model calculations with SDPF-U and SDPF-MU interactions.

 ^{35}Al Levels

Total knockout $\sigma=22$ mb *I* for producing ^{35}Al from ^{36}Si .

E(level) [†]	J^π [‡]	L [‡]	Comments
0	$(5/2)^+$	2	J^π : interpreted as the $1d_{5/2}$ proton removal from ^{36}Si (2014St18). Partial knockout $\sigma=13$ mb 2.
801 3			Partial knockout $\sigma=1.0$ mb 7.
1005 3			Partial knockout $\sigma=0.8$ mb 9.
1865 4			Partial knockout $\sigma=1.0$ mb 2.
1973 4	$3/2^+, 5/2^+$	2	Partial knockout $\sigma=3.2$ mb 5.
2733 7			Partial knockout $\sigma=0.5$ mb <i>I</i> .
3244 5	$3/2^+, 5/2^+$	2	Partial knockout $\sigma=2.6$ mb 3.
4275? 9	$3/2^+, 5/2^+$	2	Partial knockout $\sigma=0.5$ mb <i>I</i> .

[†] From a least-squares fit to γ -ray energies.

[‡] **2014St18** deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. J^π options are deduced accordingly.

 $\gamma(^{35}\text{Al})$

E_γ	I_γ	$E_i(\text{level})$	J^π_i	E_f	J^π_f	Comments
802 4	10 1	801		0	$(5/2)^+$	
859 4	3.6 3	1865		1005		
968 4	4.4 3	1973	$3/2^+, 5/2^+$	1005		
1003 4	19 1	1005		0	$(5/2)^+$	
1064 4	0.8 2	1865		801		
1174 5	2.8 3	1973	$3/2^+, 5/2^+$	801		
^x 1473 5	1.1 2					
1932 6	2.5 3	2733		801		
1972 6	7.5 5	1973	$3/2^+, 5/2^+$	0	$(5/2)^+$	
2237 6	7.8 6	3244	$3/2^+, 5/2^+$	1005		
2440 7	1.4 2	3244	$3/2^+, 5/2^+$	801		
^x 3060 8	1.6 4					
3250 8	3.3 4	3244	$3/2^+, 5/2^+$	0	$(5/2)^+$	
4275 [†] 9	3 1	4275?	$3/2^+, 5/2^+$	0	$(5/2)^+$	2014St18 states that the 4275 γ transition could populate the ground state or one of the lowest two excited states. In the absence of any observed $\gamma\gamma$ coincidences, they tentatively place the 4275-keV γ transition feeding the ground state.

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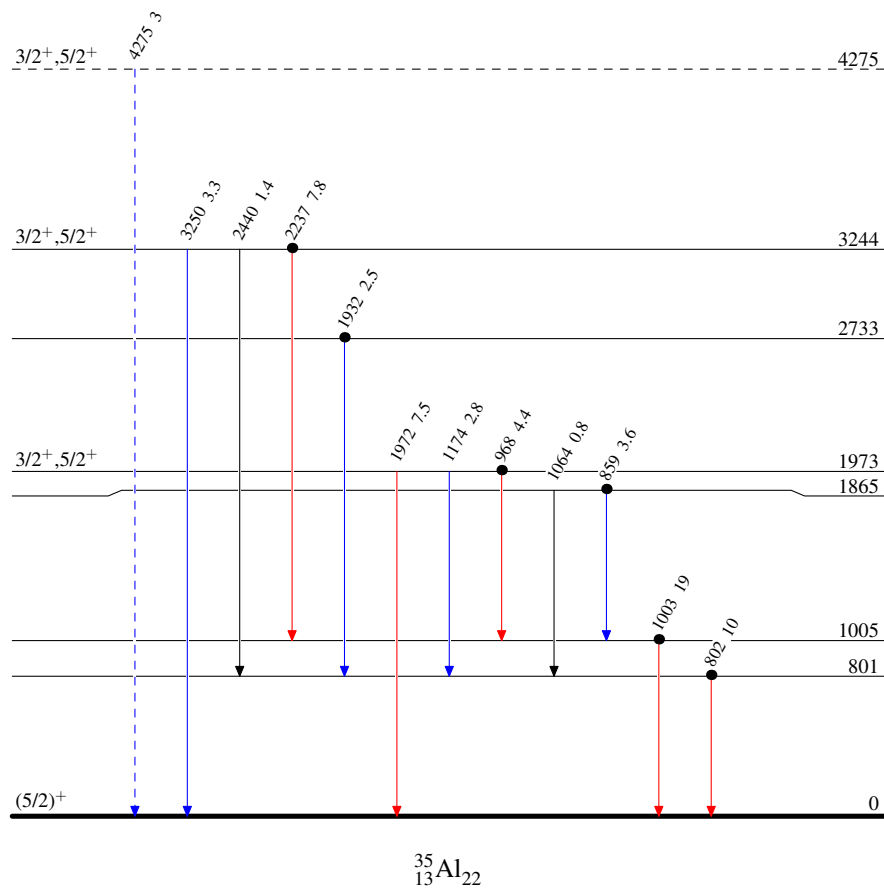
$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ 2014St18 (continued) $\gamma(^{35}\text{Al})$ (continued)[†] Placement of transition in the level scheme is uncertain.^x γ ray not placed in level scheme. $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Al}\gamma)$ 2014St18

Level Scheme

Intensities: Yield/100 ions

Legend

- $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
- $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
- $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
- - - - - γ Decay (Uncertain)
- Coincidence



Pb($^{35}\text{Al}, ^{34}\text{Al}n\gamma$) 2017Ch36,2021Bh12

Coulomb dissociation of ^{35}Al on Pb target.

2017Ch36,2014ChZZ: ^{35}Al was produced via the projectile fragmentation of a 531-MeV/nucleon ^{40}Ar primary beam from the Heavy Ion Synchrotron (SIS18) at GSI. The secondary cocktail beam was separated by the FRS separator and impinged on a 2 g/cm² Pb target and a 0.93 g/cm² C target. Projectiles and reaction fragments were detected using 8 DSSDs, separated by a large-area dipole magnet (ALADIN) and tracked using two large scintillator fiber detectors (GFIs). Neutrons from the excited projectiles were detected using the high-efficiency Large Area Neutron Detector (LAND). γ rays from the deexcited projectile and projectile-like fragments were detected using a spherical 4 π Crystal Ball detector array of 162 NaI(Tl) crystals. Measured $E(\text{fragment})$, E_n , E_γ , Coulomb dissociation cross sections. Deduced relative populations of ^{34}Al , ^{35}Al g.s. configuration. Comparisons with shell-model calculations with the SDPF-M interaction. The measured inclusive differential CD cross section (integrated up to 5.0 MeV relative energy) for $^{35}\text{Al} \rightarrow ^{34}\text{Al}+n$ using a Pb target is 78 mb 13.

2021Bh12: A further analysis of the data from **2017Ch36**. The $^{35}\text{Al}(\gamma,n)^{34}\text{Al}$ photoabsorption cross section was obtained from fitting the direct breakup model to the measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al core excited states. The $^{34}\text{Al}(n,\gamma)^{35}\text{Al}$ neutron capture cross sections were obtained from the photoabsorption cross sections using the detailed balance theorem.

 ^{35}Al Levels

<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0	(5/2 ⁺ , 3/2 ⁺ , 1/2 ⁺)	J^π: From comparisons of measured differential Coulomb dissociation cross section of ^{35}Al breaking up into ^{34}Al in its g.s. and/or 46-keV isomer with theoretical calculations from the direct breakup model using the plane-wave approximation assuming the valence neutron at different orbitals. 2017Ch36 states that the differential CD cross section of $^{35}\text{Al} \rightarrow ^{34}\text{Al}+n$ has been interpreted in the light of a direct breakup model, and it suggests that the possible ground-state spin and parity of ^{35}Al could be, tentatively, 1/2⁺ or 3/2⁺ or 5/2⁺. Major configurations and spectroscopic factor for $J^\pi=5/2^+$ of ^{35}Al g.s.: (g.s., 4⁻ in ^{34}Al)$\otimes$vp_{3/2}, S=0.36 9 (2017Ch36); (46 keV, 1⁺ in ^{34}Al)$\otimes$vd_{3/2}, S=1.47 22 (2017Ch36); (1.4 MeV, 2⁺ in ^{34}Al)$\otimes$vs_{1/2}, S=0.16 1 (2021Bh12); (2.5 MeV, 3⁻ in ^{34}Al)$\otimes$vp_{3/2}, S=1.48 18 (2021Bh12). Other configurations for $J^\pi=1/2^+, 3/2^+$ of ^{35}Al g.s.: (g.s., 4⁻ in ^{34}Al)$\otimes$vf_{7/2}, S=1.03 43 and (46 keV, 1⁺ in ^{34}Al)$\otimes$vs_{1/2}, S=0.62 7; (46 keV, 1⁺ in ^{34}Al)$\otimes$vs_{1/2}, S=0.72 8; (46 keV, 1⁺ in ^{34}Al)$\otimes$vs_{1/2}, S=0.45 7 and (46 keV, 1⁺ in ^{34}Al)$\otimes$vd_{5/2}, S=0.94 22.

Coulomb excitation [1999Ib01,2000PrZX](#)

[1999Ib01](#): $^{197}\text{Au}(^{35}\text{Al}, ^{35}\text{Al}')$ Nuclei of interest were produced via the projectile fragmentation of a 70-MeV/nucleon, ^{48}Ca primary beam from the NSCL K1200 cyclotron impinging on a 285-mg/cm²-thick ^9Be target. The secondary cocktail beam was selected by the A1200 separator and impinged on a 532 mg/cm² ^{197}Au target. The position and direction of each incident beam particle were measured using two upstream parallel-plate avalanche counters (PPAC). Scattered beam particles $\theta_{\text{lab}} < 3.8^\circ$ were detected using a downstream position-sensitive plastic phoswich detector in coincidence with γ rays detected using an array of 38 cylindrical NaI(Tl) detectors centered around the ^{197}Au target. Measured Doppler-corrected E_γ , I_γ , and excitation cross sections. Deduced levels and E2 transition probabilities for ^{35}Al , ^{37}Si , ^{39}P , ^{41}S , ^{43}S , and ^{45}Cl .

[2000PrZX](#): $^{197}\text{Au}(^{35}\text{Al}, ^{35}\text{Al}')$ The same experimental setup as [1999Ib01](#) with an 80-MeV/nucleon ^{48}Ca primary beam. Scattered beam particles $\theta_{\text{c.m.}} < 3.3^\circ$ were detected.

 ^{35}Al Levels

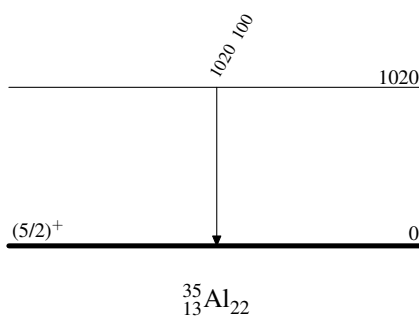
<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0	$(5/2)^+$	J^π : From the Adopted Levels.
1020 9		E(level): From measured E_γ . B(E2) $\uparrow$ =0.0142 52 from 1999Ib01 , assuming E2 excitation mode. Others: B(E1) $\uparrow$ $\leq$ 0.00020 9, B(E2) $\uparrow$ $\leq$ 0.0125 56, B(M1) $\uparrow$ $\leq$ 0.0024 11 (2000PrZX). Measured σ =30 mb 14 (2000PrZX).

 $\gamma(^{35}\text{Al})$

<u>$E_i(\text{level})$</u>	<u>E_γ</u>	<u>I_γ</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
1020	1020 9	100	0	$(5/2)^+$	E_γ : Weighted average of 1006 19 (1999Ib01) and 1023 9 (2000PrZX). 2000PrZX reports E_γ =1023 8 in Table 4.9 and E_γ =1023 9 in Table 4.10.

Coulomb excitation [1999Ib01,2000PrZX](#)Level Scheme

Intensities: % photon branching from each level



Adopted Levels, Gammas

$Q(\beta^-)=10470$ 40; $S(n)=2470$ 40; $S(p)=18680$ 40; $Q(\alpha)=-13690$ 40 2021Wa16

$S(2n)=10020$ 40, $S(2p)=33930$ 40, $Q(\beta^-n)=2090$ 40 (2021Wa16).

Isotope discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ at Dubna (1971Ar32).

^{35}Si production:

2015Mo17: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=95$ MeV/nucleon at RIKEN. Measured angular distributions and transverse momentum distributions of fragments. Deduced formulation for the width of transverse momentum distribution as a function of fragment velocity.

2012Kw02: $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be , natural nickel, and ^{181}Ta targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: $^9\text{Be}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

2006Ro34: $^2\text{H}(^{42}\text{S},X)$ at $E(^{42}\text{S})=99.8$ MeV/nucleon at NSCL. Measured production cross sections.

1997Fo01: $^{208}\text{Pb}(^{37}\text{Cl},X)$ at $E(^{37}\text{Cl})=230$ MeV at Legnaro. Measured yields.

^{35}Si decay measurements:

1986Du07, 1986HuZW, 1987DuZU, 1988DuZS, 1988DuZT: $^9\text{Be}(^{40}\text{Ar},X)$ at GANIL. Measured $T_{1/2}$ and β^- -delayed γ rays.

2007Ne14: Polarized ^{35}Si from $^9\text{Be}(^{36}\text{S},X)$ 1n pickup at GANIL. ^{35}Si g.s. magnetic moment and g -factor using β -NMR.

^{35}Si radius measurements:

2006Kh08: ^{35}Si produced by $^{181}\text{Ta}(^{48}\text{Ca},X)$ fragmentation at $E(^{48}\text{Ca})=60.3$ MeV/nucleon at GANIL. Measured energy-integrated reaction cross sections at 30-65 MeV/nucleon using a silicon telescope as both active target and detector. Deduced reduced strong absorption radii, isospin dependence, and possible halo structure or large deformation.

1999Ai02: $\text{Si}(^{35}\text{Si},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

^{35}Si mass measurements: 1986Fi06, 1986Sm05, 1984Ma49.

Theoretical calculations (binding energies, deformation, quadrupole moments, radii, levels, J^π , etc.): 2011Ka03, 2009No01, 2008Wi11, 2007Ch82, 2004Kh16, 1999Du05, 1994Mo37, 1994Po05, 1987Wa10, 1986Wo02.

 ^{35}Si LevelsCross Reference (XREF) Flags

A	^{35}Al β^- decay (38.1 ms)	D	$^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$
B	^{36}Al β^-n decay (12.0 ms)	E	$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$
C	^{37}Al β^-2n decay (11.4 ms)	F	$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0	$(7/2)^-$	0.78 s 12	ABCDEF	$\% \beta^- = 100$; $\% \beta^- n < 5$ (1995ReZZ, 2008ReZZ) $\mu = (-)1.639$ 4 (2007Ne14, 2019StZV) μ : From original $(-)1.638$ 4 using β -NMR (2007Ne14) with corrections by 2019StZV. J^π : $L=3$ from 0^+ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu f_{7/2}$ configuration from shell model. $T_{1/2}$: From $\beta\gamma(t)$ (1988DuZS, 1988DuZT). Other: 0.87 s 17 by the same group (1986Du07). Reduced strong absorption radius $r_0^2 = 1.261$ fm ² 35 from the energy-integrated σ of $\text{Si}(^{35}\text{Si}, X)$ (2006Kh08) and $r_0^2 = 1.258$ fm ² 92 from the energy-integrated σ of $\text{Si}(^{35}\text{Si}, X)$ (1999Ai02).
909.95 23	$(3/2)^-$	55 ps 14	ABCDEF	J^π : $L=1$ from 0^+ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$ and $\nu p_{3/2}$ configuration from shell model. $T_{1/2}$: From analysis of broadened line shapes in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}Si Levels (continued)

E(level) [†]	J ^π	T _{1/2}	XREF	Comments
973.88 18	(3/2 ⁺)	5.9 ns 6	A EF	J ^π : νd _{3/2} configuration from shell model and 715γ from 1689, 1/2 ⁺ . T _{1/2} : From βγ(t) in ^{35}Al β ⁻ decay.
1689.4 28	1/2 ⁺		E	J ^π : L=0 from 0 ⁺ in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$.
1970 6			E	
2044 5	(1/2) ⁻		DE	J ^π : L=1 from 0 ⁺ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and νp _{1/2} configuration from shell model.
2168.2 4	(5/2 ⁺)		A E	J ^π : From shell model and its possible isobaric analog state in ^{35}P with L=2 from 0 ⁺ in $^1\text{H}(^{34}\text{Si}, \text{p})$ from R-matrix analysis in 2012Im01. L=2,3 from 0 ⁺ in $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$.
2275 6			E	
2377 7			E	
3140	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
3450	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
3611? 8			E	XREF: E(?) Level exists assuming that the 3611γ feeds the ground state of ^{35}Si (2014St18).
3770	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
5190	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
≈5500	(5/2) ⁻		D	E(level): A broad level at ≈5500 deduced from E _p in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$. J ^π : L=3 from 0 ⁺ in $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$ and νf _{5/2} configuration from shell model.
5760	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
6330	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
7360	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	
7690	(3/2 ⁺ , 5/2 ⁺ , 7/2 ⁺) [‡]		A	

[†] From a least-squares fit to γ-ray energies for levels connected with γ transitions; from ^{35}Al β⁻-delayed neutron decays for other levels, unless otherwise noted.

[‡] Possible allowed β⁻ feeding from (5/2)⁺ parent in ^{35}Al .

γ(^{35}Si)

E _i (level)	J ^π _i	E _γ [†]	I _γ [†]	E _f	J ^π _f	Mult.	α [#]	Comments
909.95	(3/2) ⁻	910.11 30	100	0	(7/2) ⁻	[E2]	4.13×10 ⁻⁵ 6	B(E2)(W.u.)=2.4 +8-5 E _γ : others: 910 3 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$, 908 4 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si})$, and 910 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si})$.
973.88	(3/2 ⁺)	64.1 3 973.78 20	100 11.8 24	909.95 0	(3/2) ⁻ (7/2) ⁻	[E1] [M2]	0.0368 8 5.05×10 ⁻⁵ 7	B(E1)(W.u.)=3.52×10 ⁻⁴ +41-34 B(M2)(W.u.)=0.057 +13-12 E _γ : other: 974 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si})$.
1689.4	1/2 ⁺	715 [‡] 4 780 [‡] 4	14.6 [‡] 16 100 [‡] 8	973.88 909.95	(3/2 ⁺) (3/2) ⁻			
1970		1970 [‡] 6	100	0	(7/2) ⁻			
2044	(1/2) ⁻	1134 [‡] 5	100	909.95	(3/2) ⁻			E _γ : Other: 1134 6 from $^2\text{H}(^{34}\text{Si}, ^{35}\text{Si})$.
2168.2	(5/2 ⁺)	1194.2 4	35 8	973.88	(3/2 ⁺)			

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Si})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Comments
2168.2	2168.2 $6^\dagger$	100 20	0	(7/2) ⁻	E_γ : Other: 2164 6 from $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.
2275	2275 $6^\ddagger$	100	0	(7/2) ⁻	
2377	2377 $7^\ddagger$	100	0	(7/2) ⁻	
3611?	3611 $8^\ddagger@$	100	0	(7/2) ⁻	2014St18 observed this surprising γ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering $S(n)=2470$ 40 (2021Wa16). 2014St18 proposes two possible explanations involving hindered neutron emission or multistep process.

† From ^{35}Al β^- decay, unless otherwise noted.

‡ From $^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$.

Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Placement of transition in the level scheme is uncertain.

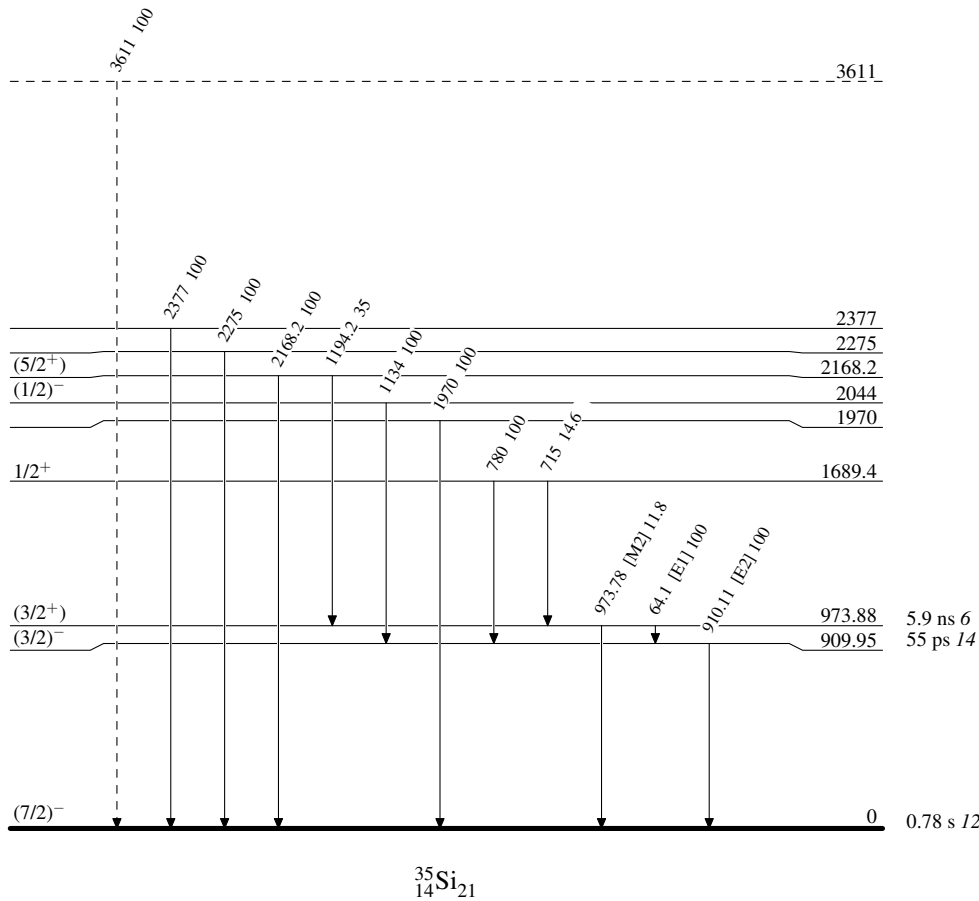
Adopted Levels, Gammas

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



^{35}Al β^- decay (38.1 ms) 2005Ti11,2001Nu01

Parent: ^{35}Al : $E=0$; $J^\pi=(5/2)^+$; $T_{1/2}=38.1$ ms 4; $Q(\beta^-)=14170$ 40; $\% \beta^-$ decay=100

^{35}Al - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Al .

^{35}Al - $Q(\beta^-)$: From 2021Wa16.

2005Ti11,2006AnZW: A ^{35}Al secondary beam at ≈ 2 pps was produced via the fragmentation of a 78-MeV/nucleon ^{36}S primary beam and selected by the LISE3 spectrometer at GANIL. A total of 3.46×10^5 ^{35}Al ions were continuously implanted into an NE102A plastic scintillator also for detecting β . The implantation detector was sandwiched between two silicon detectors for monitoring beam and for veto, respectively. Neutrons were detected using the TONNERRE array consisting of 19 plastic scintillator modules. γ rays were detected using two EXOGAM clover modules and a LEPS detector. Measured E_γ , I_γ , E_n , I_n , $\beta\gamma$ -coin, βn -coin, and $\beta n\gamma$ -coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, $\log ft$, $B(\text{GT})$, and β -delayed neutron emission probability. Comparisons with shell-model calculations.

2001Nu01,2002Nu02: Exp 1: A ^{35}Al secondary beam at 8 pps was produced via the fragmentation of a UC target with 1.4 GeV protons at ISOLDE, CERN with subsequent surface-ionization and mass separation. ^{35}Al ions were collected onto a moving tape. β particles were detected using a thin cylindrical plastic scintillator, γ rays were detected using two Ge detectors, and neutrons were detected using eight low-threshold plastic scintillators. Measured E_γ , I_γ , E_n , I_n , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, βn -coin. Deduced the decay scheme consisting of ^{35}Si and ^{34}Si levels, ^{35}Al $T_{1/2}$, decay branching ratios, $\log ft$, and β -delayed neutron emission probability. Comparisons with shell-model calculations. Exp 2: A lifetime measurement for the 974-keV level in ^{35}Si used a thin plastic scintillator for detecting β and a BaF_2 detector for detecting γ .

Other experimental studies on ^{35}Al $T_{1/2}$ and β -delayed neutron emission probability: 2017Ha23, 1999YoZW,

1995ReZZ/2008ReZZ, 1989MuZU, 1989Le16, 1988MuZY, 1988Mu08, 1988DuZT, 1988BaYZ, 1987DuZU, 1987BaZI.

Theoretical studies involving ^{35}Al decay: 2018Yo06, 2013Li39.

 ^{35}Si Levels

$E(n)$ under comments are deduced from $E(n)_{\text{c.m.}} = E(\text{level})(^{35}\text{Si}) - S(n)(^{35}\text{Si}) - E(\text{level})(^{34}\text{Si})$, with $E(\text{level})(^{35}\text{Si})$ reported in 2005Ti11 based on the neutron time-of-flight spectrum. 2005Ti11 used $E(\text{level})(^{34}\text{Si})=3326$ for $E(n1)$. 2005Ti11 used $S(n)(^{35}\text{Si})=2474$ 43, consistent with $S(n)(^{35}\text{Si})=2470$ 40 from 2021Wa16.

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$	Comments
0	$(7/2)^-$	0.78 s 12	
909.95 23	$(3/2)^-$	55 ps 14	2001Nu01 reports a $\% \beta^- < 0.9$ from 2001Nu01 without considering the conversion electron of the transition from 974 level to 910 level. The net feeding is deduced to be -1.9 9 from the γ +ce intensity balance.
973.88 18	$(3/2^+)$	5.9 ns 6	$T_{1/2}$: lifetime=8.5 ns 9 from $\beta\gamma(t)$ in 2001Nu01, also adopted in the Adopted Levels.
2168.16 36	$(5/2^+)$		
3140	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=666$.
3450	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=976$.
3770	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=1296$.
5190	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=2716$.
5760	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=3286$.
6330	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n0)_{\text{c.m.}}=3856$; $E(n1)_{\text{c.m.}}=530$.
7360	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n1)_{\text{c.m.}}=1560$.
7690	$(3/2^+, 5/2^+, 7/2^+)$		Deduced $E(n1)_{\text{c.m.}}=1890$.

† From a least-squares fit to γ -ray energies for levels connected with γ transitions. For unbound levels, from 2005Ti11 based on the neutron time-of-flight spectrum.

‡ From the Adopted Levels of ^{35}Si .

^{35}Al β^- decay (38.1 ms) [2005Ti11](#), [2001Nu01](#) (continued) β^- radiations

The decay scheme is incomplete due to possible unobserved levels in a large gap of about 6.5 MeV between Q-value=14.17 MeV and the highest populated level at E=7.69 MeV.

E(decay)	E(level)	$I\beta^-$ ^{†‡}	Log ft [‡]	Comments
(6.48×10 ³ 4)	7690	2.7 2	4.5	
(6.81×10 ³ 4)	7360	2.6 2	4.6	
(7.84×10 ³ 4)	6330	6.8 3	4.5	
(8.41×10 ³ 4)	5760	4.5 2	4.8	
(8.98×10 ³ 4)	5190	8.9 3	4.6	
(1.040×10 ⁴ 4)	3770	3.2 2	5.4	
(1.072×10 ⁴ 4)	3450	6.0 3	5.2	
(1.103×10 ⁴ 4)	3140	3.3 2	5.5	
(1.200×10 ⁴ 4)	2168.16	9.5	5.2	$I\beta^-$: 9.2 19 from 2001Nu01 ; 6.7 9 from 2005Ti11 with only one branch 2168 γ observed.
(1.320×10 ⁴ 4)	973.88	52	4.7	$I\beta^-$: 48 9 from 2001Nu01 ; 50 3 from 2005Ti11 .
(1.417×10 ⁴ 4)	0	3.0 10	6.1 2	$I\beta^-$: from 2001Nu01 , which states that the β branch to the ^{35}Si ground state was evaluated by comparing the total γ intensity due to the deexcitation of excited states of ^{35}Si with the decay of ^{35}Si activity and assuming no direct ^{35}Si production.

[†] For the two bound excited states, $I\beta^-$ is from the γ +ce intensity balance at each state. Quoted $I\beta^-$ values without uncertainties are considered upper limits due to the incomplete decay scheme, and the associated log ft values are considered lower limits. For unbound excited states (>2470), $I\beta^-$ is from the absolute $I\beta^-$ in [2005Ti11](#) based on neutron intensities.

[‡] For unbound levels, [2005Ti11](#) did not report uncertainties for neutron energies or level energies. The evaluators estimate the uncertainties of log ft are <0.1, assuming a 200-keV uncertainty in level energies.

[#] Absolute intensity per 100 decays.

 $\gamma(^{35}\text{Si})$

$I\gamma$ normalization: 0.47 4 from $\Sigma I(\gamma\text{+ce to g.s.})=60$ 4, deduced from $100-\% \beta^- \text{ n}-\% I\beta^- (\text{g.s.})$, where $\% \beta^- \text{ n}=37$ 4 from the Adopted Levels of ^{35}Al and $\% I\beta^- (\text{g.s.})=3.0$ 10 from [2001Nu01](#). The deduced normalization factor should be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state. Other: 0.45 from [2001Nu01](#), based on $\% \beta^- \text{ n}=41$ 13.

E_γ [†]	I_γ ^{†‡}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult.	α [#]	Comments
64.1 3	100	973.88	(3/2 ⁺)	909.95	(3/2) ⁻	[E1]	0.0368 8	$\% I\gamma=47$
910.11 30	99.7 19	909.95	(3/2) ⁻	0	(7/2) ⁻	[E2]	4.13×10 ⁻⁵ 6	$\% I\gamma=47$
973.78 20	11.8 24	973.88	(3/2 ⁺)	0	(7/2) ⁻	[M2]	5.05×10 ⁻⁵ 7	$\% I\gamma=5.6$
^x 1130.4 4	3.2 9							$\% I\gamma=1.5$ 2014St18 suggests this γ is likely the 1134 γ from 2042 to 908 transition observed in the $^{36}\text{Si} \rightarrow \text{n} + ^{35}\text{Si}$ knockout reaction.
1194.2 4	5.3 12	2168.16	(5/2 ⁺)	973.88	(3/2 ⁺)			$\% I\gamma=2.5$
2168.2 6	15 3	2168.16	(5/2 ⁺)	0	(7/2) ⁻			$\% I\gamma=7.1$
^x 5629 3	2.4 12							$\% I\gamma=1.1$

[†] From [2001Nu01](#).

[‡] For absolute intensity per 100 decays, multiply by 0.47.

Continued on next page (footnotes at end of table)

$^{35}\text{Al} \beta^-$ decay (38.1 ms) [2005Ti11](#), [2001Nu01](#) (continued)

$\gamma(^{35}\text{Si})$ (continued)

Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multiplicities, and mixing ratios, unless otherwise specified.

^x γ ray not placed in level scheme.

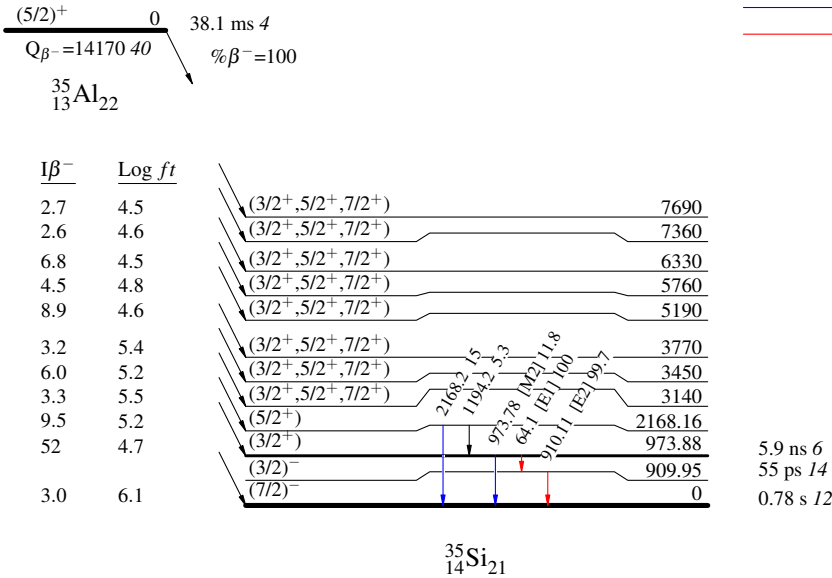
³⁵Al β⁻ decay (38.1 ms) 2005Ti11,2001Nu01

Decay Scheme

Intensities: Relative I_γ

Legend

- I_γ < 2% × I_γ^{max}
- I_γ < 10% × I_γ^{max}
- I_γ > 10% × I_γ^{max}



$^{36}\text{Al} \beta^- \text{n} \text{ decay (12.0 ms) } \quad \text{2023Lu07}$

Parent: ^{36}Al : $E=0$; $T_{1/2}=12.0 \text{ ms } 20$; $Q(\beta^- \text{n})=1.227 \times 10^4 \text{ keV } 15$; $\% \beta^- \text{n} \text{ decay} < 31$

^{36}Al - $T_{1/2}$: From implant- $\beta\gamma$ correlation (2023Lu07). Other: 14.7 ms 10 (2023Lu07, implant- β correlation), 94 ms 37 (1995ReZZ/2008ReZZ, implant- β correlation), $\approx 15 \text{ ms}$ (1999YoZW, implant- β correlation, preliminary). 2023Lu07 adopted 12.0 ms 20 instead of 14.7 ms 10 based on the cleanest determination with the γ -ray gating.

^{36}Al - $Q(\beta^- \text{n})$: From 2021Wa16.

^{36}Al - $\% \beta^- \text{n} \text{ decay}$: From 1995ReZZ/2008ReZZ.

2023Lu07: Exp 1: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 140-MeV/nucleon, 80-pnA ^{48}Ca primary beam from the NSCL cyclotrons impinging on a 642-mg/cm²-thick ^9Be target. The secondary cocktail beam centered around ^{33}Na was selected by the A1900 separator and implanted into a CeBr₃ scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were the SeGA array of 16 segmented Ge detectors and 15 LaBr₃ detectors. Exp 2: ^{36}Mg and ^{36}Al were produced via the projectile fragmentation of a 172.3-MeV/nucleon, 120-pnA ^{48}Ca primary beam from the FRIB linac impinging on an 8.89-mm-thick ^9Be target. The secondary cocktail beam centered around ^{42}Si was selected by the ARIS separator and implanted into a 5-mm-thick YSO segmented scintillator sandwiched between two plastic scintillator veto detectors. Surrounding the implantation array were 11 HPGe clover detectors and 15 fast-timing LaBr₃ detectors, and the VANDLE array of 88 neutron detectors. Both the NSCL and FRIB experiments in 2023Lu07 measured E_γ , I_γ , $\beta\gamma$ -coin, $\gamma\gamma$ -coin, implant- $\beta\gamma$ correlation and deduced $T_{1/2}$ of ^{36}Mg g.s., ^{36}Al g.s. and a ^{36}Al isomer. Comparisons with FSU shell-model calculations.

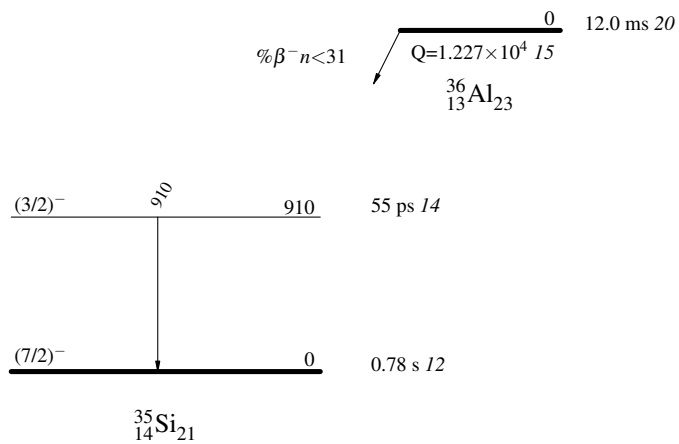
 ^{35}Si Levels

<u>$E(\text{level})$</u>	<u>J^π</u>	<u>$T_{1/2}$</u>
0	$(7/2)^-$	0.78 s 12
910	$(3/2)^-$	55 ps 14

[†] From the Adopted Levels of ^{35}Si .

 $\gamma(^{35}\text{Si})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
910	910	$(3/2)^-$	0	$(7/2)^-$	E_γ: From 2023Lu07. 2023Lu07 observed the 910γ within the 100-ms window following the arrival of a ^{36}Al ion in both NSCL and FRIB experiments. 2023Lu07 also observed the 910γ within the 100-ms window following the arrival of a ^{36}Mg ion in both NSCL and FRIB experiments. 2013StZY observed the 910γ in ^{36}Mg decay with its maximum intensity in the 20-30 ms time window after the implantation of ^{36}Mg, indicating its origin from the granddaughter generation.

^{36}Al β^-n decay (12.0 ms) 2023Lu07Decay Scheme

^{37}Al β^-2n decay (11.4 ms) 2013StZY

Parent: ^{37}Al : $E=0$; $J^\pi=(5/2^+)$; $T_{1/2}=11.4$ ms 4; $Q(\beta^-2n)=8.06\times 10^3$ 18; $\% \beta^-2n$ decay ≥ 000

^{37}Al - J^π : From shell-model predictions (2013StZY).

^{37}Al - $T_{1/2}$: Weighted average of 10.7 ms 13 (2004Gr20,2003Gr22, implant- β correlation), 11.5 ms 4 (2013StZY, implant- $\beta\gamma$ correlation). Other: 11.8 ms 1 (stat) +22-34 (syst) (2013StZY, implant- β correlation).

^{37}Al - $Q(\beta^-2n)$: Deduced from mass excesses in 2021Wa16.

^{37}Al - $\% \beta^-2n$ decay: $\% \beta^-2n \geq 1$ 1 (2013StZY).

2013StZY: ^{37}Al was produced via the projectile fragmentation of a 345-MeV/nucleon, 70-pnA $^{48}\text{Ca}^{20+}$ primary beam from the linear accelerator RILAC and the four cyclotrons RRC, fRC, IRC, and SRC at RIKEN impinging on an 15-mm-thick ^9Be target. The secondary cocktail beam was selected by the BigRIPS separator and the zero-degree spectrometer (ZDS) using the $B\rho$ - ΔE -ToF method, and implanted into the Cylindrical Active Implantation Target for Exotic Nuclei (CAITEN) consisting of a segmented movable hollow-cylindrical-shaped plastic scintillator and a stationary ring of 24 position-sensitive photomultiplier tubes (PSPMTs) arranged on a ring inside the scintillator at the height of the beam line. To reduce background buildup, the scintillator barrel was rotated quickly and axially-moved slowly in vertical direction, resulting in a helix-shaped motion. β particles were detected by the CAITEN and γ rays were detected using three HPGe clover detectors. Measured E_γ , $\beta\gamma$ -coin, and implant- β correlation, and deduced $T_{1/2}$. Comparisons with QRPA and shell-model calculations.

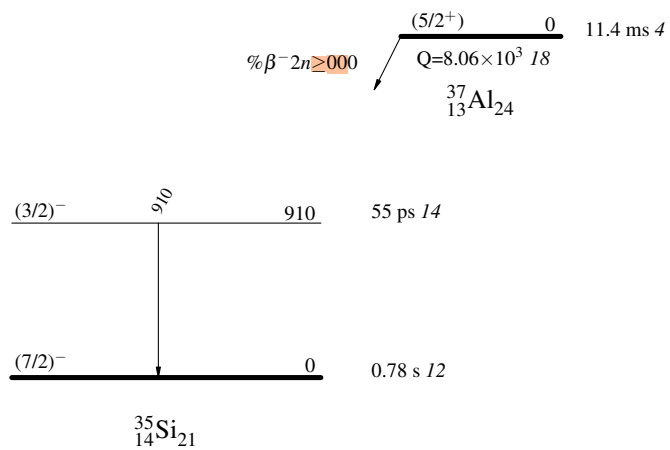
 ^{35}Si Levels

<u>$E(\text{level})$</u>	<u>$J^\pi^\dagger$</u>	<u>$T_{1/2}^\dagger$</u>
0	$(7/2)^-$	0.78 s 12
910	$(3/2)^-$	55 ps 14

† From the Adopted Levels of ^{35}Si .

 $\gamma(^{35}\text{Si})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
910	910	$(3/2)^-$	0	$(7/2)^-$	I_γ : 2013StZY reports 2 3 relative to $I_\gamma=100$ for the 156 γ in ^{37}Si from ^{37}Al decay.

 ^{37}Al $\beta^- 2n$ decay (11.4 ms) 2013StZYDecay Scheme

$2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$ **2014Bu01**

$^{34}\text{Si}(\text{d},\text{p})^{35}\text{Si}$ from $J^\pi=0^+$ ^{34}Si g.s. in inverse kinematics.

2014Bu01: A 20.5-MeV/nucleon, 1.1×10^5 pps, and 95% pure ^{34}Si secondary beam was produced via the fragmentation of a 55-MeV/nucleon $^{36}\text{S}^{16+}$ primary beam impinging on a 1075- μm -thick Be target, separated by the LISE3 spectrometer at GANIL, and incident on a 2.6- mg/cm² 1 CD_2 secondary target. Incoming beam ions were tracked using two position-sensitive multiwire proportional chambers upstream of the target. Outgoing ions were identified using an ionization chamber (IC) downstream of target for energy loss and a plastic scintillator behind the IC for TOF measurements. Protons from the (d,p) reaction were detected using four modules of the MUST2 array placed 10 cm from the target covering polar angles ranging from 105° to 150° with respect to the beam direction and a 16-strip annular Si detector at a distance of 11.3 cm covering polar angles from 156° to 168°. γ rays were detected using four segmented Ge detectors from the EXOGAM array perpendicular to the beam axis at a mean distance of 5 cm, and 9 cm downstream from the target with efficiency $\varepsilon=3.8\%$ at 1 MeV. Measured $\sigma(E_p, \theta)$, Doppler-corrected E_γ , I_γ , (^{34}Si)p-coin. Deduced levels, J , π , L-transfer and spectroscopic factors. Comparisons with shell-model calculations.

2007GeZX: 30-AMeV ^{34}Si beam on 30-mg/cm² CD_2 secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected E_γ , I_γ , $\gamma\gamma$ -coin, and (^{35}Si) γ -coin. Deduced levels, J , π . Compared with shell-model calculations.

 ^{35}Si Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$L^\#$	$S^\#$	Comments
0	$7/2^-$	3	0.56 6	Interpreted as the $1f_{7/2}$ neutron on top of the ^{34}Si core.
910 3	$3/2^-$	1	0.69 10	Interpreted as the $2p_{3/2}$ neutron on top of the ^{34}Si core. E(level): Other: 906 32 from measured E_p . 2014Bu01 deduced that a contamination of the proton spectrum at $E(\text{level})=906$ 32 due to transfer to the $3/2^+$ level at 970 keV is less than 30% of the $3/2^-$ component with a confidence limit of 3σ .
2044 7	$1/2^-$	1	0.73 10	Interpreted as the $2p_{1/2}$ neutron on top of the ^{34}Si core. E(level): Other: 2060 50 from measured E_p . J^π : 2014Bu01 states that J^π is likely to be $1/2^-$ as its large spectroscopic factor value excludes another large $L=1$, $3/2^-$ component.
≈ 5500	$5/2^-$	3	0.32 3	Interpreted as the $1f_{5/2}$ neutron on top of the ^{34}Si core. S: 2014Bu01 reports $S=0.32$ 4 in Fig. 2, $S=0.32$ 2 in text on page 3, $S=0.32$ 3 in text on page 4, and a full error bar ≈ 0.05 in Fig. 3.

[†] From a least-squares fit to γ -ray energies, except for a broad level observed at ≈ 5500 keV from E_p . Another broad structure is observed at 3330 keV 120 from E_p and likely corresponds to the elastic deuteron break-up process, the cross section of which was estimated to be 0.1 mb/MeV (**2014Bu01**).

[‡] As given in **2014Bu01** based on L-transfers and shell-model predictions.

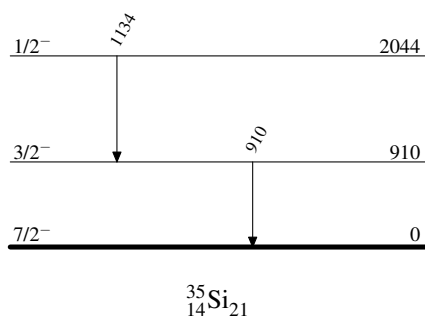
[#] From TWOFN-ADWA analysis of measured proton angular distributions (**2014Bu01**). Additional uncertainties of $\approx 15\%$ in spectroscopic factors due to global potentials in the ADWA calculation are not included.

 $\gamma(^{35}\text{Si})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
910 3	910	$3/2^-$	0	$7/2^-$	E_γ : Other: 905 1 in 2007GeZX , with uncertainty likely statistical only considering it is from a γ -ray spectrum with a bin size of 4 keV.
1134 6	2044	$1/2^-$	910	$3/2^-$	E_γ : Other: 1133 3 in 2007GeZX , with uncertainty likely statistical only considering it is a very weak γ ray seen in a γ -ray spectrum with a bin size of 4 keV.

$^2\text{H}(^{34}\text{Si}, ^{35}\text{Si}\gamma)$ 2014Bu01

Level Scheme



$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ 2014St18, 2015St06

$^{36}\text{Si} \rightarrow n + ^{35}\text{Si}$ from $J^\pi=0^+$ ^{36}Si ground state.

2014St18: A ^{36}Si secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a ^9Be target at NSCL, MSU and was selected by the A1900 separator. The states of ^{35}Al and ^{35}Si were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Si beam at a midtarget energy of 97.7 MeV/nucleon 5 on a 287-mg/cm² ^9Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The position and angle of the residues were measured using two cathode-readout drift chambers. Prompt γ rays from the deexcitation of the residues were detected by the GRETINA Ge array. Measured Doppler-corrected E_γ , I_γ , $(^{35}\text{Si})\gamma$ -coin, $\gamma\gamma$ -coin, the parallel momentum distributions of populated states in ^{35}Si residues. Deduced levels, J , π , L -transfers, inclusive and exclusive knockout cross section for producing ^{35}Si from ^{36}Si . Calculations using eikonal reaction model and shell model spectroscopic strengths with SDPF-U and SDPF-MU interactions.

2015St06: Further theoretical discussions on the data from **2014St18**.

 ^{35}Si Levels

Total knockout $\sigma=81$ mb 2 for producing ^{35}Si from ^{36}Si .

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	$L^\#$	Comments
0	$(7/2^-)$		3	Partial knockout $\sigma=52$ mb 4, including an unresolved population of the $3/2^+$ isomer at 973 keV (2014St18). Partial knockout $\sigma=23$ mb 6, resolved by the fitting parallel momentum distribution subtracting all events that decay by prompt γ emission with a linear combination of the theoretical distributions for neutron removal from the $f_{7/2}$ and $d_{3/2}$ orbits together via the χ^2 minimization method (2015St06).
908 4	$(3/2^-)$	55 ps 14	1	$T_{1/2}$: lifetime=80 ps 20 extracted from fitting Geant4 simulation to broadened line shapes via the maximum likelihood method (2014St18). Partial knockout $\sigma=8$ mb 3.
973 7	$(3/2^+)$			Partial knockout $\sigma=29$ mb 6, resolved by 2015St06 .
1688 6	$1/2^+$		0	Partial knockout $\sigma=13$ mb 1.
1970 6				Partial knockout $\sigma=1.1$ mb 2.
2042 7	$(1/2^-)$		0,1	Partial knockout $\sigma=1.3$ mb 2.
2164 6	$(5/2^+)$		2,3	Partial knockout $\sigma=1.1$ mb 2.
2275 6				Partial knockout $\sigma=1.6$ mb 2.
2377 7				Partial knockout $\sigma=2.1$ mb 2.
				L: 2014St18 showed $L=0,1,2,3$ fits to the measured parallel momentum distribution but did not specify L . $L=0$ seems unlikely.
3611? 8				2014St18 observed a surprising γ transition at 3611 keV, which must depopulate a state that is unbound to neutron emission by at least 1.1 MeV, considering $S(n)=2470$ 40 (2021Wa16). One possible explanation is this ^{35}Si level is populated by $1n$ knockout from the deeply-bound $1d_{5/2}$ orbit in ^{36}Si and followed by the emission of the other $1d_{5/2}$ neutron, resulting in a $2p-2h$ configuration dominated by two holes in the deeply bound neutron $1d_{5/2}$ orbit. Such a configuration should have a small overlap with the ground state in ^{34}Si that has a large contribution of $0p-0h$ configurations, leading to a significantly hindered neutron decay. The γ decay, on the other hand, would be a fast $5/2^+ \rightarrow 7/2^-$ g.s. $E1$ decay and could potentially compete with the neutron emission. A second possible explanation could be that this state is weakly populated in a more complicated multistep reaction pathway. The 3611-keV γ results predominantly from events with residue momenta in the tail of the momentum distribution. The low-momentum tail is associated with more inelastic reactions, which would be consistent with a multistep process. Partial knockout $\sigma=0.8$ mb 2.

[†] From a least-squares fit to γ -ray energies.

$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ 2014St18, 2015St06 (continued) ^{35}Si Levels (continued)

‡ As given in 2014St18 based on L-transfers and shell-model predictions.

$^{\#}$ 2014St18 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

 $\gamma(^{35}\text{Si})$

$E_{\gamma}^{\dagger}$	$I_{\gamma}^{\dagger}$	$E_i(\text{level})$	J_i^{π}	E_f	J_f^{π}	Comments
715 4	1.9 2	1688	$1/2^{+}$	973	$(3/2^{+})$	
780 4	13 1	1688	$1/2^{+}$	908	$(3/2^{-})$	
908 4	25 2	908	$(3/2^{-})$	0	$(7/2^{-})$	$B(E2)_{\downarrow}=0.0017 \text{ }^{+4-5}$ (2014St18)
1134 5	1.5 2	2042	$(1/2^{-})$	908	$(3/2^{-})$	
1970 6	1.4 2	1970		0	$(7/2^{-})$	
2164 6	1.4 2	2164	$(5/2^{+})$	0	$(7/2^{-})$	
2275 6	2.0 3	2275		0	$(7/2^{-})$	
2377 7	2.5 3	2377		0	$(7/2^{-})$	
3611 ‡ 8	1.0 2	3611?		0	$(7/2^{-})$	

† From 2014St18, unless otherwise noted.

‡ Placement of transition in the level scheme is uncertain.

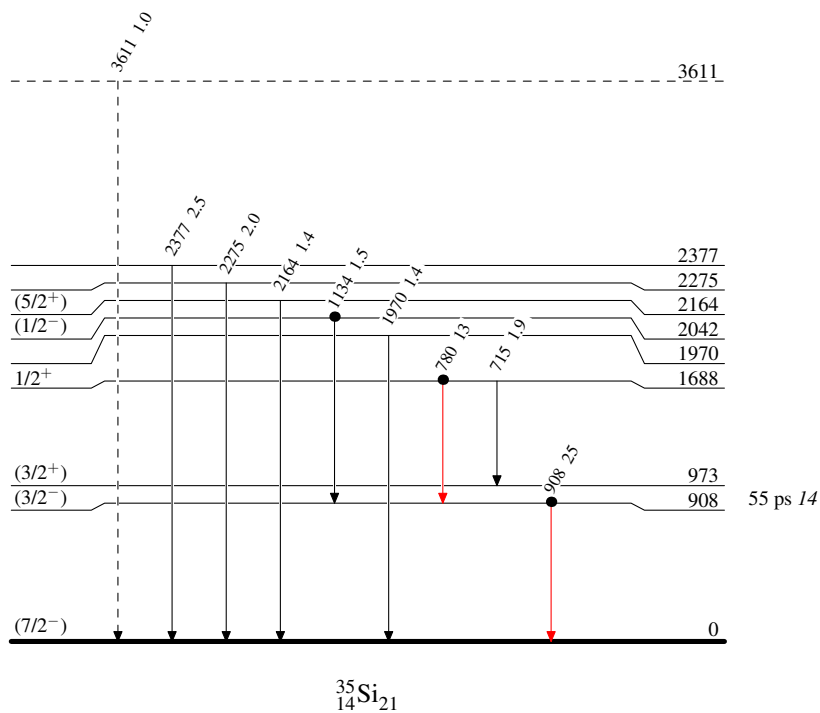
$^9\text{Be}(^{36}\text{Si}, ^{35}\text{Si}\gamma)$ 2014St18,2015St06

Level Scheme

Intensities: Yield/100 ions

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - -→ γ Decay (Uncertain)
- Coincidence



$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$ 2010WaZT

2010WaZT,2010Wa20: A 215-MeV $^{36}\text{S}^{9+}$ beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was $300\text{ }\mu\text{g}/\text{cm}^2$ 99.7% enriched ^{208}Pb with $20\text{ }\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180° . Measured E_γ with Doppler corrections, I_γ , and $(^{35}\text{Si})\gamma$ -coin. Deduced levels. Compared with shell-model calculations.

 ^{35}Si Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0	$(7/2^-)$
910	$(3/2^-)$
974	$(3/2^+)$

† From E_γ data.

‡ As given in 2010WaZT, based on shell-model calculations.

 $\gamma(^{35}\text{Si})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
910	100 25	910	$(3/2^-)$	0	$(7/2^-)$
974 ‡	21 10	974	$(3/2^+)$	0	$(7/2^-)$

† From 2010WaZT. The author report $\Delta E_\gamma=1\text{ keV}$, but it is very likely statistical only since 910γ and 974γ are weak with poor statistics in the γ -ray spectrum, and thus is not adopted by the evaluators.

‡ Placement of transition in the level scheme is uncertain.

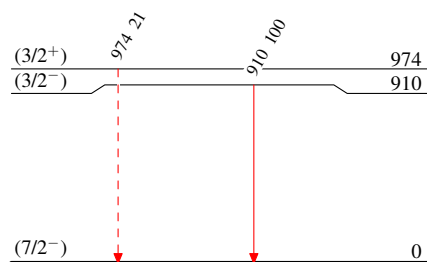
$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{Si}\gamma)$ 2010WaZT

Legend

Level Scheme

 Intensities: Relative I_γ

- ▶ $I_\gamma < 2\% \times I_\gamma^{\max}$
- ▶ $I_\gamma < 10\% \times I_\gamma^{\max}$
- ▶ $I_\gamma > 10\% \times I_\gamma^{\max}$
- - - - -▶ γ Decay (Uncertain)


 $^{35}_{14}\text{Si}_{21}$

Adopted Levels, Gammas

$Q(\beta^-)=3988.4$ 19; $S(n)=8380.4$ 20; $S(p)=12155.1$ 20; $Q(\alpha)=-12332.0$ 29 2021Wa16

$S(2n)=14663.1$ 22, $S(2p)=30938$ 7 (2021Wa16).

Isotope discovery (2012Th10): $^{232}\text{Th}(^{40}\text{Ar},X)$ at Dubna (1971Ar32) and $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz (1971Gr53).

^{35}P production:

2012Kw02: $E(^{40}\text{Ar})=140$ MeV/nucleon impinged on ^9Be and natural nickel targets at NSCL. Measured fragmentation cross sections, parallel momentum transfers, and widths. Compared with empirical formula EPAX, and predictions from internuclear cascade and deep inelastic models using Monte Carlo ISABEL-GEMINI and DIT-GEMINI codes.

2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.

2007No13: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar},X)$ and $(^{40}\text{Ar},X)$ at $E(^{40}\text{Ar})=100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.

1997Vo03: $^{56}\text{Fe}(p,X)$ at $E_p=800$ MeV at LANL. Measured γ radiation. Deduced production cross sections.

^{35}P decay measurements:

1986Wa22: ^{35}P produced via $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured β -delayed γ rays.

1972Ap01: ^{35}P produced via $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$ at LANL. Measured $T_{1/2}$ and β -delayed γ rays.

1972Go31: ^{35}P produced via $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$ and $^{36}\text{S}(t,\alpha)^{35}\text{P}$ at BNL. Measured $T_{1/2}$ and β -delayed γ rays.

1971Gr53: ^{35}P produced via $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ at Mainz. Measured $T_{1/2}$ and γ rays.

^{35}P radius measurement:

1999Ai02: $\text{Si}(^{35}\text{P},X)$ at NSCL. Measured energy-integrated reaction cross sections at $E=38$ -80 MeV/ nucleon. Deduced strong absorption radii.

^{35}P mass measurements:

$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$ and $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$ (1988Or01), $^{36}\text{S}(^6\text{Li},^7\text{Be})$ (1985Dr06), $^{36}\text{S}(d,^3\text{He})$ (1985Kh04), $^{36}\text{S}(^{14}\text{C},^{15}\text{N})$ (1984Ma49).

Theoretical calculations (binding energies, dipole moments, quadrupole moments, radii, levels, J , π , etc.): 2012BoZT, 2009No01, 2004Kh16, 2003Sm02, 1999Du05, 1988Wa04, 1987Wa10, 1986Wo02, 1983Wi08, 1975JeZX.

 ^{35}P LevelsCross Reference (XREF) Flags

A	$^{35}\text{Si} \beta^-$ decay (0.78 s)	E	$^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$	I	$^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$
B	$^1\text{H}(^{34}\text{Si},p):\text{resonances}$	F	$^{34}\text{S}(^{18}\text{O},^{17}\text{F})$	J	$^{160}\text{Gd}(^{37}\text{Cl},X\gamma)$
C	$^2\text{H}(^{34}\text{Si},^{35}\text{P}\gamma)$	G	$^{36}\text{S}(\text{pol } d,^3\text{He})$	K	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$
D	$^2\text{H}(^{36}\text{S},^3\text{He})$	H	$^{36}\text{S}(d,^3\text{He})$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$ or $\Gamma^\#$	XREF	Comments
0	$1/2^+$	47.3 s 8	A CDEFGHIJK	$\% \beta^- = 100$ J^π : L(pol d, ^3He)=0 from 0^+ and analyzing power. $T_{1/2}$: weighted average of 45 s 2 (1971Gr53, $\gamma(t)$), 47.7 s 29 (1972Go31, $\gamma(t)$), 48.2 s 17 (1972Go31, $\gamma(t)$), and 47.4 s 8 (1972Ap01, $\gamma(t)$).
2386.9 11	$3/2^+$	<0.69 ps	A CDEFG I K	XREF: F(2420) J^π : L(pol d, ^3He)=2 from 0^+ and L-1/2 transfer from analyzing power.
3860.4 11	$5/2^+$	<0.69 ps	A CDE GHIJK	J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.
4101.7 11	$(7/2^-)^\ddagger$	>69 ps	A C E JK	
4250 20			I	
4382.0 12	$(5/2^-)$		A E K	XREF: A(?) J^π : possibly allowed β^- feeding from $7/2^-$ parent; 1994.9 γ to $3/2^+$.
4494.1 12	$(7/2^-)^\ddagger$	2.29 ps 49	A C E G JK	XREF: G(4474) J^π : L=(3) in $^9\text{Be}(^{36}\text{S},^{35}\text{P})$ from 0^+ .
4666.2 16	$5/2^+$		DE GHI	XREF: I(4640) J^π : L(pol d, ^3He)=2 from 0^+ and L+1/2 transfer from analyzing power.
4767.0 13	$(9/2^-)^\ddagger$		E K	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}P Levels (continued)				
E(level) [†]	J ^π	T _{1/2} or Γ [#]	XREF	Comments
4869.6 12	(5/2 ⁻ , 7/2 ⁻)		A C K	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 1009.7γ to 5/2 ⁺ .
4962.8 12	(9/2 ⁻) [‡]		A E K	XREF: A(?)
5010 20			I	1988Or01 observed a 5010 20 level in ($^{11}\text{B}, ^{13}\text{N}$) and a 5070 40 level in ($^{18}\text{O}, ^{17}\text{F}$) but did not assign them as the same level.
5090.2 13	(11/2 ⁻) [‡]		EF K	XREF: F(5070)
5199.3 16	5/2 ⁺		DE GHI	XREF: I(5220) J ^π : L(pol d, ^3He)=2 from 0 ⁺ and L+1/2 transfer from analyzing power.
5487.9 13			K	
5561.0 13	(5/2 ⁻)		A K	J ^π : possibly allowed β ⁻ feeding from 7/2 ⁻ parent; 3173.5γ to 3/2 ⁺ .
5709.5 23	(1/2 ⁻)		DE	J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 5709γ to 1/2 ⁺ .
5.86×10 ³ 5			F I	XREF: F(5890)I(5840) E(level): weighted average of 5890 70 from ($^{18}\text{O}, ^{17}\text{F}$) and 5840 50 from ($^{11}\text{B}, ^{13}\text{N}$).
6222.7 13	(7/2 ⁻ , 9/2, 11/2 ⁻)		K	J ^π : 1132γ to (11/2 ⁻) and 1729γ to (7/2 ⁻).
6440 60			F	
7050 60			F	
7440 60			Fg	XREF: g(7520)
7526.9 23	(1/2 ⁻)		E g	XREF: g(7520) J ^π : L=(1) in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$ from 0 ⁺ ; interpreted as the deeply bound 1p _{1/2} proton removal from 0 ⁺ in $^9\text{Be}(^{36}\text{S}, ^{35}\text{P})$; 7526γ to 1/2 ⁺ .
7590 20			I	
7920 60			F	
8390 40			I	
8.60×10 ³ 10			F	
9290 50			F	
14938 24	1/2 ⁺	<12.7 keV	B	J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15161 3	5/2 ⁻ , 7/2 ⁻	<4.4 keV	B	J ^π : L=3 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15306 24	3/2 ⁺ , 5/2 ⁺	<30.4 keV	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
15964 18	3/2 ⁺ , 5/2 ⁺	84 keV 25	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16145 36	1/2 ⁻ , 3/2 ⁻	0.35 MeV 9	B	J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
16605 44	1/2 ⁺	0.22 MeV 15	B	J ^π : L=0 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17254 12	3/2 ⁺ , 5/2 ⁺	<11.6 keV	B	J ^π : L=2 in $^1\text{H}(^{34}\text{Si}, \text{p})$.
17355 15	1/2 ⁻ , 3/2 ⁻	32 keV 22	B	J ^π : L=1 in $^1\text{H}(^{34}\text{Si}, \text{p})$.

[†] From a least-squares fit to γ-ray energies for levels connected with γ transitions, from particle-transfer reactions for other levels, or from proton elastic scattering for resonances.

[‡] Comparisons with shell-model calculations (2019Gr08).

[#] T_{1/2} from the differential recoil-distance method (2019Gr08) in $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ and widths from the R-matrix analysis of ($^{34}\text{Si}, \text{p}$) for resonances, unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$									
$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	δ	$\alpha^\#$	Comments
2386.9	3/2 ⁺	2386.3 6	100	0	1/2 ⁺	[M1,E2]		0.00046 5	E_γ : weighted average of 2386.4 6 from $^{35}\text{Si } \beta^-$ decay, 2386 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 2386 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0023 if M1, B(E2)(W.u.)>1.6 if E2.
3860.4	5/2 ⁺	1473.5 5	15.6 14	2386.9	3/2 ⁺	[M1,E2]		8.3×10 ⁻⁵ 13	E_γ : weighted average of 1473.4 5 from $^{35}\text{Si } \beta^-$ decay, 1473 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1474 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : weighted average of 14.1 33 from $^{35}\text{Si } \beta^-$ decay and 15.9 14 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. B(M1)(W.u.)>0.0012 if M1, B(E2)(W.u.)>2.1 if E2.
		3860.2 10	100.0 32	0	1/2 ⁺	[E2]		1.12×10 ⁻³ 2	B(E2)(W.u.)>0.12 E_γ : weighted average of 3859.5 10 from $^{35}\text{Si } \beta^-$ decay, 3860 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 3861 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : From $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Other: 100 7 from $^{35}\text{Si } \beta^-$ decay.
4101.7	(7/2 ⁻)	241.3 5	100 [†] 7	3860.4	5/2 ⁺	[E1]		0.000665 10	B(E1)(W.u.)<4.4×10 ⁻⁴ E_γ : weighted average of 241.4 3 from $^{35}\text{Si } \beta^-$ decay, 237 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 241 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 100 4 from $^{35}\text{Si } \beta^-$ decay.
		1714.8 6	6.6 [†] 17	2386.9	3/2 ⁺	[M2]		7.93×10 ⁻⁵ 11	B(M2)(W.u.)<0.16 E_γ : weighted average of 1714.7 6 from $^{35}\text{Si } \beta^-$ decay and 1715 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 22 5 from $^{35}\text{Si } \beta^-$ decay.
		4101.4 10	54 [†] 8	0	1/2 ⁺	[E3]		0.000924 13	B(E3)(W.u.)<4.8 E_γ : weighted average of 4100.8 10 from $^{35}\text{Si } \beta^-$ decay and 4102 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. I_γ : other: 135 8 from $^{35}\text{Si } \beta^-$ decay.
4382.0	(5/2 ⁻)	1994.9 6	100	2386.9	3/2 ⁺				E_γ : weighted average of 1994.8 6 from $^{35}\text{Si } \beta^-$ decay, 1995 2 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$, and 1995 1 from $^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$. Placement by 1988DuZS, 2008Wi09, and 2016Mu03. 1988DuZT and 1987Wa10 placed this γ as the 6096 to 4101 transition. 1988Or01 placed this γ as the 6488 to 4493 transition.
4494.1	(7/2 ⁻)	392.3 3	100 5	4101.7	(7/2 ⁻)	[M1+E2]	<0.22	0.000199 12	B(M1)(W.u.)=0.117 +42-29 E_γ : weighted average of 392.3 3 from $^{35}\text{Si } \beta^-$ decay, 391

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)								
$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Mult.	$\alpha^\#$	Comments
4494.1	(7/2 ⁻)	633.6 5	34 5	3860.4	5/2 ⁺	[E1]	4.64×10 ⁻⁵ 7	2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$, and 392 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$. I_γ : From ^{35}Si β^- decay. Other: 100 17 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$. δ : deduced by evaluators from RUL=100 for B(E2)(W.u.). B(E1)(W.u.)=2.8×10 ⁻⁴ +8-6 E_γ : weighted average of 633.7 5 from ^{35}Si β^- decay, 634 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$, and 633 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$. I_γ : weighted average of 38 5 from ^{35}Si β^- decay and 27 7 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
4666.2	5/2 ⁺	804 $\dagger$ 2 2279 $\dagger$ 2 4668 $\dagger$ 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
4767.0	(9/2 ⁻)	273 1	40.0 $\dagger$ 25	4494.1	(7/2 ⁻)			E_γ : weighted average of 274 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 273 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
		664 1	100 $\dagger$ 47	4101.7	(7/2 ⁻)			E_γ : weighted average of 666 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 664 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
4869.6	(5/2 ⁻ , 7/2 ⁻)	374 $\dagger$ 1 487 $\dagger$ 1 767.9 4	60 $\dagger$ 20 <40 $\dagger$ 100 $\dagger$ 20	4494.1	(7/2 ⁻) 4382.0 (5/2 ⁻) 4101.7 (7/2 ⁻)			E_γ : weighted average of 768.0 4 from ^{35}Si β^- decay and 767 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$. I_γ : other: 100 18 from ^{35}Si β^- decay.
		1009.7 5	<20 $\dagger$	3860.4	5/2 ⁺			E_γ : weighted average of 1009.9 5 from ^{35}Si β^- decay and 1009 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$. I_γ : other: 152 32 from ^{35}Si β^- decay.
4962.8	(9/2 ⁻)	468.9 4	100 $\dagger$ 8	4494.1	(7/2 ⁻)			E_γ : weighted average of 468.9 4 from ^{35}Si β^- decay, 469 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$, and 468 2 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
		859 $\dagger$ 3	66 $\dagger$ 9	4101.7	(7/2 ⁻)			
5090.2	(11/2 ⁻)	128 1	50 $\dagger$ 25	4962.8	(9/2 ⁻)			E_γ : weighted average of 127 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 128 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
		322 1	100 $\dagger$ 35	4767.0	(9/2 ⁻)			E_γ : weighted average of 321 2 from $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ and 322 1 from $^{208}\text{Pb}(^{36}\text{S},^{35}\text{P}\gamma)$.
5199.3	5/2 ⁺	1337 $\dagger$ 2 2811 $\dagger$ 2 5202 $\dagger$ 2		3860.4	5/2 ⁺ 2386.9 3/2 ⁺ 0 1/2 ⁺			
5487.9		993 $\dagger$ 1 1387 $\dagger$ 1	100 $\dagger$ 20 60 $\dagger$ 20	4494.1	(7/2 ⁻) 4101.7 (7/2 ⁻)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{P})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ	I_γ	E_f	J_f^π	Comments
5561.0	(5/2 ⁻)	1459.4 7	34 12	4101.7	(7/2 ⁻)	E_γ : weighted average of 1459.7 5 from ³⁵ Si β^- decay and 1458 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). I_γ : From ³⁵ Si β^- decay.
		3173.5 10	100 17	2386.9	3/2 ⁺	3173.5 γ is not observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ), but the weaker 1459 γ deexciting the same level is observed in ²⁰⁸ Pb(³⁶ S, ³⁵ P γ). Further experiments are needed to resolve the discrepancy. E_γ, I_γ : From ³⁵ Si β^- decay.
5709.5	(1/2 ⁻)	5709 [‡] 2		0	1/2 ⁺	
6222.7	(7/2 ⁻ , 9/2, 11/2 ⁻)	1132 [†] 1	<25 [†]	5090.2	(11/2 ⁻)	
		1260 [†] 1	100 [†] 25	4962.8	(9/2 ⁻)	
		1729 [†] 1	100 [†] 25	4494.1	(7/2 ⁻)	
7526.9	(1/2 ⁻)	7526 2		0	1/2 ⁺	

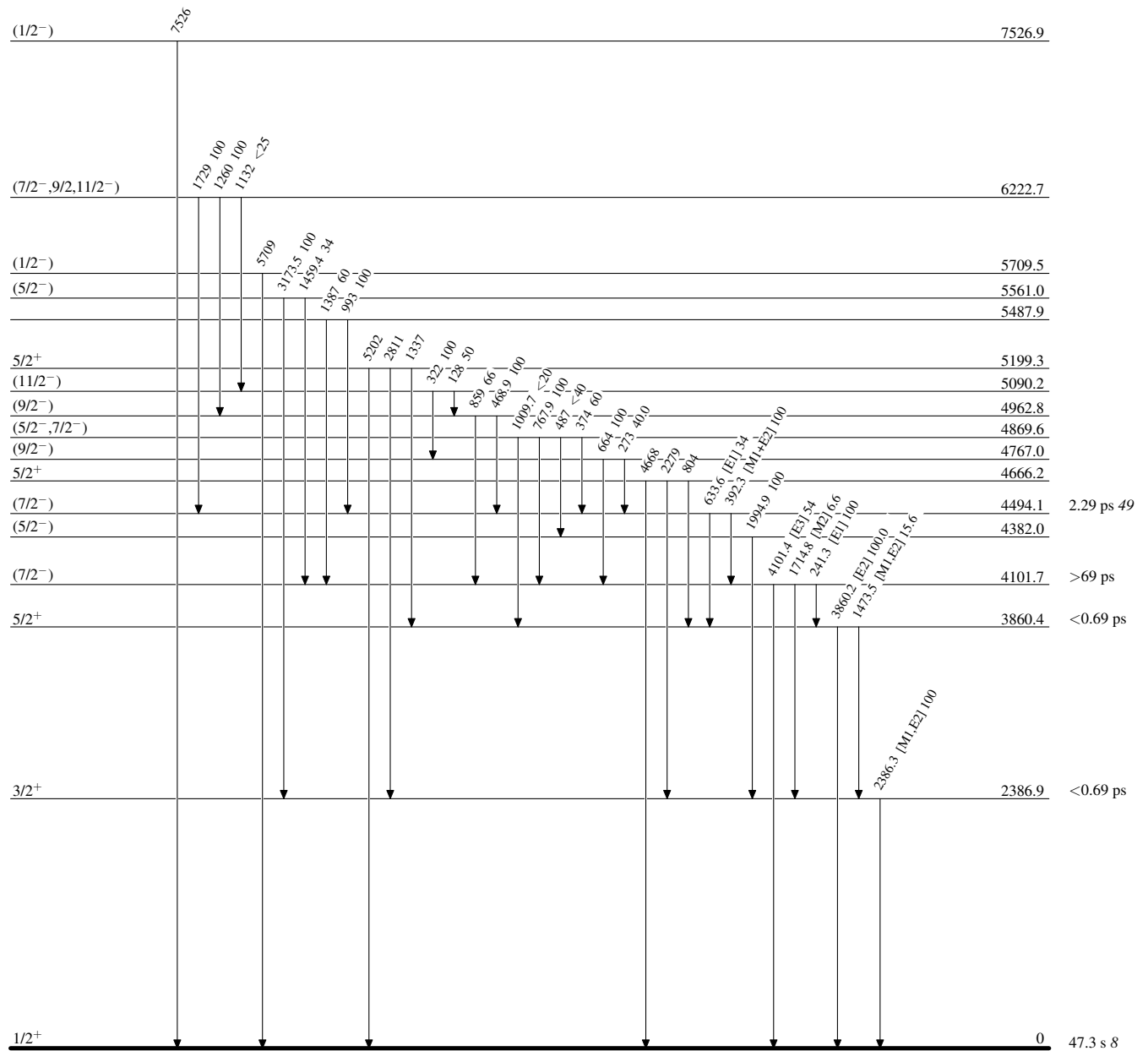
[†] From ²⁰⁸Pb(³⁶S,³⁵P γ).

[‡] From ⁹Be(³⁶S,³⁵P γ).

Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level



^{35}Si β^- decay (0.78 s) 1988DuZS,1986Du07,1988DuZT

Parent: ^{35}Si : $E=0$; $J^\pi=(7/2)^-$; $T_{1/2}=0.78$ s 12; $Q(\beta^-)=10470$ 40; $\% \beta^-$ decay=100

^{35}Si - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Si .

^{35}Si - $Q(\beta^-)$: From 2021Wa16.

1988DuZS,1986Du07,1988DuZT,1987DuZU: ^{35}Si was produced by the fragmentation of an ^{40}Ar beam of 2×10^{11} particles/s at 60 MeV/nucleon on a 190 mg/cm² Be target at GANIL. Decay observed with a 1 mm thick plastic scintillator and a 174 cm³ intrinsic Ge detector with 1.2% absolute efficiency at 1.33 MeV. Measured $\beta\gamma(t)$, $E\gamma$, $I\gamma$. Deduced levels, J , π , parent $T_{1/2}$.

2007Ne14: measured ^{35}Si ground state magnetic moment and g -factor using the β -NMR method.

1987Wa10: shell-model calculations for ^{35}Si β^- decay scheme, ^{35}P levels, decay branching ratios, $\log ft$, and Gamow-Teller transition strengths.

The decay scheme is considered incomplete due to a large gap of about 4.9 MeV between the highest observed level at $E=5561$ and $Q(\beta^-)$ value=10470 40 (2021Wa16). There may be missing transitions from unobserved levels in the gap.

 ^{35}P Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$1/2^+$	47.3 s 8
2386.5 5	$3/2^+$	<0.69 ps
3859.7 5	$5/2^+$	<0.69 ps
4101.2 5	$(7/2^-)$	>69 ps
4381.3? 8	$(5/2^-)$	
4493.5 6	$(7/2^-)$	2.29 ps 49
4869.4 6	$(5/2^-, 7/2^-)$	
4962.4? 7	$(9/2^-)$	
5560.7 7	$(5/2^-)$	

† From a least-squares fit to γ -ray energies.

‡ From the Adopted Levels.

 β^- radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^-^{\ddagger\ddagger}$	$\log ft^\dagger$	Comments
(4.91×10^3) 4)	5560.7	12.7	4.6	$I\beta=12.4$ 18 and $\log ft=4.6$ from 1988DuZS.
(5.51×10^3) 4)	4962.4?	5.1	5.2	Placed based on the 4959 to 4493 transition observed in $^{208}\text{Pb}(^{36}\text{S}, X\gamma)$ (2008Wi09) and adopted 468.9I γ from 1988DuZS to deduce its $I\beta$.
(5.60×10^3) 4)	4869.4	10.8	4.9	$I\beta=10.8$ 16 and $\log ft=4.9$ from 1988DuZS.
(5.98×10^3) 4)	4493.5	16.6	4.9	$I\beta=21.4$ 11 and $\log ft=4.8$ from 1988DuZS. 468.9I γ feeding this level is deducted from its $I\beta$.
(6.09×10^3) 4)	4381.3?	9.7	5.1	$I\beta=9.4$ 13 and $\log ft=5.1$ from 1988DuZS.
(6.37×10^3) 4)	4101.2	46.1	4.6	$I\beta=45.9$ 31 and $\log ft=4.5$ from 1988DuZS.
(8.08×10^3) 4)	2386.5	1.9	8.6 ^{1u}	

† β -feeding from γ -ray intensity balance at each level. Quoted $I\beta^-$ values are considered upper limits due to the incomplete decay scheme, and the associated $\log ft$ values are considered lower limits.

‡ Absolute intensity per 100 decays.

$\gamma(^{35}\text{P})$

I γ normalization: From $\Sigma\%I(\gamma \text{ to g.s.})=100$. The deduced normalization factor of 0.27 should be considered an upper limit due to potential missing γ transitions from unobserved levels in the gap to the ground state.

E_γ [‡]	I_γ ^{‡@}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	α ^{&}	Comments
241.4 3	100 4	4101.2	(7/2 ⁻)	3859.7	5/2 ⁺	[E1]	0.000664 10	%I γ =27
392.3 3	58.2 28	4493.5	(7/2 ⁻)	4101.2	(7/2 ⁻)	[M1+E2]	4.4×10 ⁻⁴ 25	%I γ =16
468.9 ^a 4	18.7 25	4962.4?	(9/2 ⁻)	4493.5	(7/2 ⁻)			%I γ =5.0
								Unplaced γ ray in 1988DuZS . The placement is based on a 466 γ observed in $^{208}\text{Pb}(^{36}\text{S},\text{X}\gamma)$ (2008Wi09).
633.7 5	21.9 28	4493.5	(7/2 ⁻)	3859.7	5/2 ⁺	[E1]	4.64×10 ⁻⁵ 7	%I γ =5.9
768.0 4	15.9 29	4869.4	(5/2 ⁻ , 7/2 ⁻)	4101.2	(7/2 ⁻)			%I γ =4.3
1009.9 5	24 5	4869.4	(5/2 ⁻ , 7/2 ⁻)	3859.7	5/2 ⁺			%I γ =6.5
1459.7 5	12 4	5560.7	(5/2 ⁻)	4101.2	(7/2 ⁻)			%I γ =3.2
1473.4 5	17 4	3859.7	5/2 ⁺	2386.5	3/2 ⁺	[M1,E2]	8.3×10 ⁻⁵ 13	%I γ =4.6
1714.7 6	22 5	4101.2	(7/2 ⁻)	2386.5	3/2 ⁺	[M2]	7.93×10 ⁻⁵ 11	%I γ =5.9
1994.8 ^a 6	36 6	4381.3?	(5/2 ⁻)	2386.5	3/2 ⁺			%I γ =9.7
								Placement from 1988DuZS , consistent with the 1995 γ 4381 to 2386 transition observed in $^{208}\text{Pb}(^{36}\text{S},\text{X}\gamma)$ (2008Wi09) and the 1995 γ 4382 to 2386 transition observed in $^9\text{Be}(^{36}\text{S},^{35}\text{P}\gamma)$ (2016Mu03). 1988DuZT and 1987Wa10 placed this γ as the 6096 to 4101 transition. 1988Or01 placed this γ as the 6488 to 4493 transition.
2386.4 6	117 7	2386.5	3/2 ⁺	0	1/2 ⁺	[M1,E2]	0.00046 5	%I γ =32
3173.5 10	35 6	5560.7	(5/2 ⁻)	2386.5	3/2 ⁺			%I γ =9.5
								1988Or01 suggested that this γ ray cannot be placed into the decay scheme of ^{35}Si , and it could be a transition to the 4101 or 4493 levels based on intensity balances.
^x 3349.1 [#] 10	46 [#] 6							%I γ =12
^x 3590.0 [#] 11	60 [#] 7							%I γ =16
3859.5 10	121 8	3859.7	5/2 ⁺	0	1/2 ⁺	[E2]	1.12×10 ⁻³ 2	%I γ =33
4100.8 10	135 8	4101.2	(7/2 ⁻)	0	1/2 ⁺	[E3]	0.000924 13	%I γ =36

[†] From the Adopted Levels.

[‡] From [1988DuZS](#), unless otherwise noted.

[#] From [1986Du07](#). [1988Or01](#) tentatively suggested that these γ rays deexcite a level at 7450, but this suggestion has not been experimentally confirmed.

[@] For absolute intensity per 100 decays, multiply by 0.27.

[&] Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

$^{35}\text{Si} \beta^-$ decay (0.78 s) [1988DuZS](#),[1986Du07](#),[1988DuZT](#) (continued)

$\gamma(^{35}\text{P})$ (continued)

^a Placement of transition in the level scheme is uncertain.

^x γ ray not placed in level scheme.

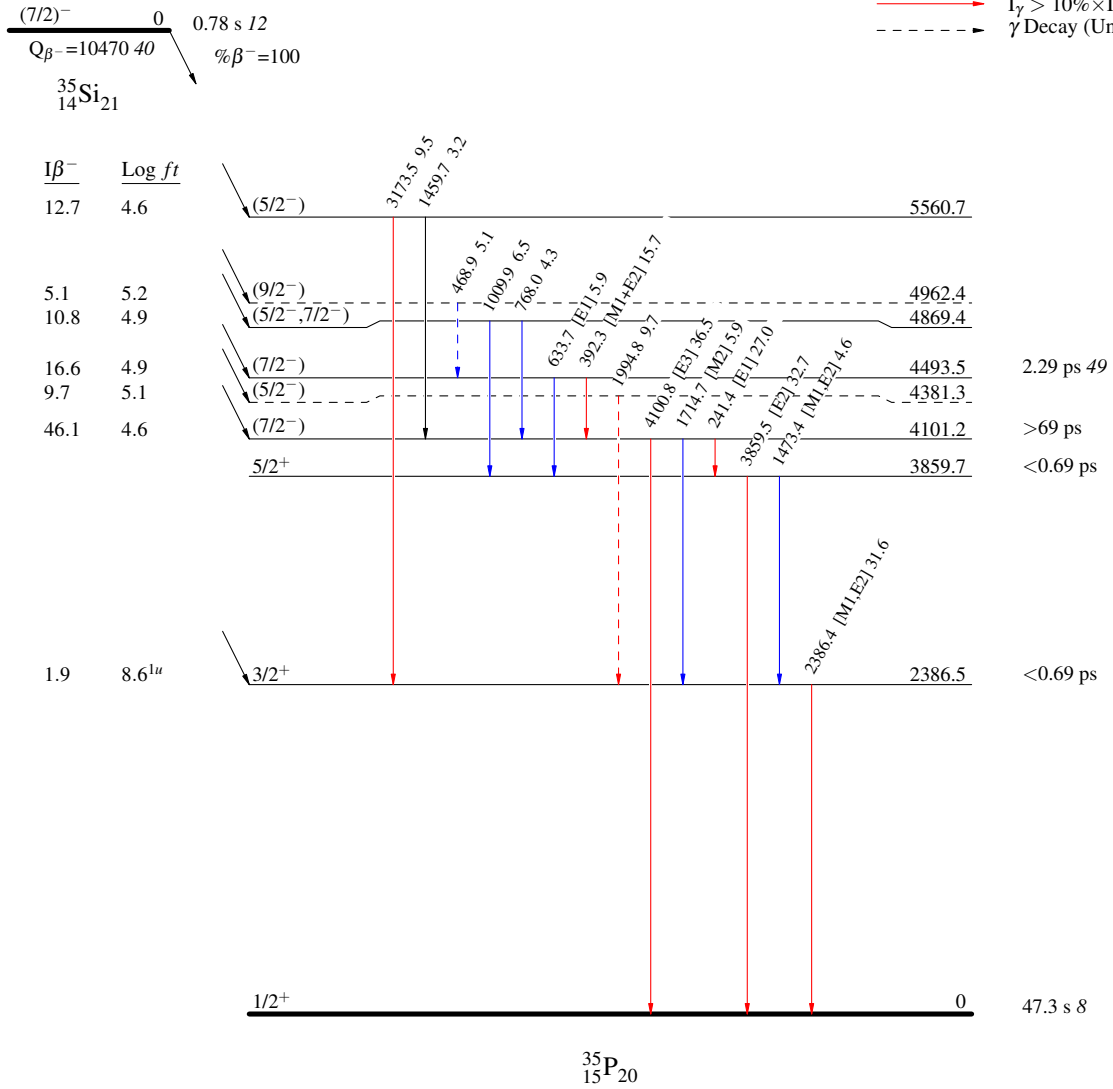
^{35}Si β^- decay (0.78 s) 1988DuZS,1986Du07,1988DuZT

Decay Scheme

Intensities: I_γ per 100 parent decays

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$
 $\cdots$ γ Decay (Uncertain)



$^1\text{H}(^{34}\text{Si},\text{p})$:resonances **2012Im01**

$J^\pi=0^+$ for ^{34}Si ground state.

2012Im01: A ^{34}Si beam at 7×10^4 pps and a purity of 97% was produced by the projectile fragmentation of a 63-MeV/nucleon ^{40}Ar primary beam and separated by the RIPS separator at RIKEN. The secondary target was a 10.9-mg/cm² 5 polyethylene film. An incident energy of 4.4 MeV/nucleon 12 for the ^{34}Si beam was determined by the timing difference between a plastic scintillator and two PPACs placed upstream of the target. The PPACs also record the positions and angles of the projectiles incident upon the target. Outgoing particles were detected and identified by a three-layer ΔE -E telescope consisting of 0.5-mm DSSD, 1.5-mm silicon, and 1.5-mm silicon detectors mounted at 0° with an E_{lab} resolution $\sigma=130$ keV. Measured excitation functions of proton elastic scattering on ^{34}Si for $\theta_{\text{lab}} < 10^\circ$ using thick target inverse kinematics. Deduced E_R , L-transfer, Γ_p , and Γ from R-matrix analysis for 8 resonances in the highly excited states in ^{35}P , which are isobaric analog states of ^{35}Si states.

 ^{35}P Levels

All data are from **2012Im01**.

<u>E(level)[†]</u>	<u>Γ</u>	<u>L</u>	<u>$S^\ddagger$</u>	<u>Comments</u>
14938 24	<12.7 keV	0		Γ : from original value of 4.6 keV 81 in 2012Im01 . $E_R=2783$ 24, $\Gamma_p=4.6$ keV 28.
15161 3	<4.4 keV	3	0.63 16	Γ : from original value of 1.6 keV 28 in 2012Im01 . $E_R=3006$ 2, $\Gamma_p=1.6$ keV 4. IAR of the $7/2^-$ g.s. of ^{35}Si .
15306 24	<30.4 keV	2	0.19 15	Γ : from original value of 10.4 keV 200 in 2012Im01 . $E_R=3151$ 24, $\Gamma_p=3.3$ keV 27.
15964 18	84 keV 25	2	0.79 20	$E_R=3809$ 18, $\Gamma_p=26.7$ keV 69 in 2012Im01 .
16145 36	354 keV 87	1	1.37 32	$E_R=3990$ 36, $\Gamma_p=185$ keV 43.
16605 44	0.22 MeV 15	0	0.45 28	$E_R=4450$ 44, $\Gamma_p=58.4$ keV 370.
17254 12	<11.6 keV	2	0.04 1	Γ : from original value of 3.8 keV 78 in 2012Im01 . $E_R=5099$ 12, $\Gamma_p=3.8$ keV 9.
17355 15	32 keV 22	1	0.12 7	$E_R=5200$ 15, $\Gamma_p=20.9$ keV 120.

[†] Excitation energies are deduced by evaluators from $E_R+S_p(^{35}\text{P})$, where $S_p(^{35}\text{P})=12155.1$ 20 (**2021Wa16**). E_R given in **2012Im01** are in the center-of-mass system.

[‡] Spectroscopic factors are derived from Γ_p using the formula from **1968Th07** as described in **2012Im01**.

$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$ [2007GeZX](#)

$^{34}\text{Si}(\text{d}, \text{n})^{35}\text{P}$ from $J^\pi=0^+$ ^{34}Si g.s. in inverse kinematics.

[2007GeZX](#): 30-AMeV ^{34}Si beam on 30-mg/cm² CD₂ secondary target at GANIL. Heavy ions produced in reactions were identified by the VAMOS spectrometer. γ rays were detected using the EXOGAM germanium clover array. Measured Doppler-corrected E_γ , I_γ , $\gamma\gamma$ -coin, and $(^{35}\text{P})\gamma$ -coin. Deduced levels, J , π . Compared with shell-model calculations.

 ^{35}P Levels

$E(\text{level})^\dagger$

0
2386.5 8
3859.9 8
4100.9 13
4492.9 16
4868.9 13

[†] From a least-squares fit to γ -ray energies, assuming $\Delta E_\gamma=1$ keV since uncertainties were not given in [2007GeZX](#).

 $\gamma(^{35}\text{P})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
241	4100.9	3859.9
392	4492.9	4100.9
1009	4868.9	3859.9
1473 [‡]	3859.9	2386.5
2386	2386.5	0
3860	3859.9	0

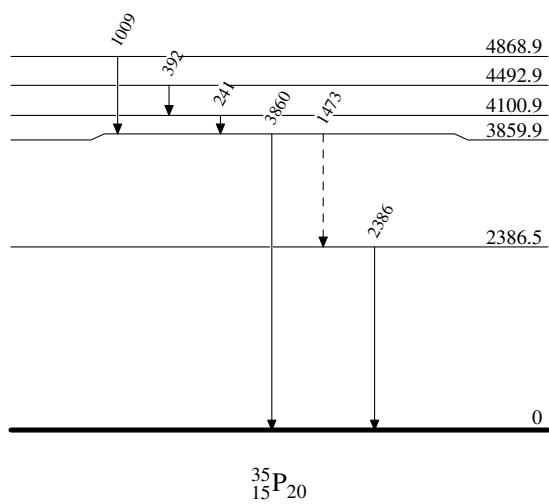
[†] From [2007GeZX](#).

[‡] Placement of transition in the level scheme is uncertain.

$^2\text{H}(^{34}\text{Si}, ^{35}\text{P}\gamma)$ 2007GeZX

Legend

Level Scheme

 -----► γ Decay (Uncertain)


$^2\text{H}(^{36}\text{S}, ^3\text{He})$ 2020Sa44

2020Sa44: $E(^{36}\text{S})=15.3$ MeV/nucleon was provided by the Argonne Tandem Linear Accelerator System. Targets were 81, 127, 529- $\mu\text{g}/\text{cm}^2$ deuterated plastic. The reaction was studied in inverse kinematics with the HELical Orbit Spectrometer (HELIOS). The ^3He ions spiralled in a 2.85-T magnetic field and were detected by a position-sensitive silicon array. The heavy-ion products were measured with a 65- μm thick silicon detector installed between the target and the silicon array. Measured $\sigma(E(^3\text{He}), \theta)$. Deduced relative spectroscopic factors for 6 levels from the PTOLEMY-DWBA analysis of the angular distributions. A potential $1/2^+$ neutron 2p2h excitation bandhead in 2.5-3.6 MeV was investigated, and an upper limit for the transfer reaction cross section to populate such an intruder configuration was deduced.

 ^{35}P Levels

2020Sa44 states that the measurement of the beam current was not made with sufficient accuracy to calculate absolute spectroscopic factors. The C^2S values are relative spectroscopic factors normalized such that the ground-state value is 2.

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$C^2S^\#$</u>
0 1	$1/2^+$	2.0
2388 13	$3/2^+$	0.33 9
3860 2	$5/2^+$	2.9 10
4666 9	$5/2^+$	0.71 34
5202 8	$5/2^+$	1.1 6
5706 38	$(1/2^-)$	0.23 5

[†] From 2020Sa44.

[‡] Assumed by 2020Sa44 for deducing C^2S , consistent with the Adopted Levels.

[#] Uncertainties are dominated by systematic components.

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2016Mu03

$J^\pi=0^+$ for ^{36}S ground state.

2016Mu03: A secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on an 846-mg/cm 2 ^9Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}P were populated by the one-proton knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm 2 ^9Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}P residues were detected by seven modules of the GRETINA Ge array. Measured E_γ , I_γ , $(^{35}\text{P})\gamma$ -coin, $\gamma\gamma$ -coin, the inclusive knockout cross section for producing ^{35}P from ^{36}S , the fractional populations and parallel momentum distributions of 8 populated states in ^{35}P residues. Deduced levels, J , π , L-transfers, partial knockout cross sections and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for proton removal and parallel momentum distributions using the eikonal model. This work is also part of the thesis [2015MuZY](#).

 ^{35}P Levels

b_f values under comments are population fraction in the knock-out reaction, σ_f^{sp} values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in [2014To14](#). The partial cross sections to each level can be deduced from $b_f \times \sigma_{\text{inc}}^{\text{exp}}$.

The inclusive cross section $\sigma_{\text{inc}}^{\text{exp}}$ for producing ^{35}P from ^{36}S is measured to be 51 mb. 95.3% is attributed to a direct knockout process from the p-sd shell, i.e., $\sigma_{\text{inc,KO}}^{\text{exp}} = 0.953 \times \sigma_{\text{inc}}^{\text{exp}}$. The remaining 4.7% is possibly attributed to a charge-exchange or a two-step process: ^{36}S core excitation by inelastic scattering, followed by a proton removal. This is proposed based on their shifted parallel momentum distributions, their decay modes to negative parity states, their high excitation energy (around 4.7 MeV), and the fact that they were not observed in the $^{36}\text{S}(d, ^3\text{He})$ reaction study ([1985Kh04](#)).

E(level) [†]	J^π @	L [#]	C ² S&	Comments
0.0	1/2 ⁺	0	2.2 7	$b_f=31.4\%$ 72, $\sigma_f^{\text{sp}}=13.5$ mb, $R_s=0.55$ 11.
2388 1	3/2 ⁺	2	0.7 3	$b_f=7.8\%$ 36, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.52$ 10.
3862 1	5/2 ⁺	2	2.7 8	$b_f=27.3\%$ 58, $\sigma_f^{\text{sp}}=10.6$ mb, $R_s=0.49$ 10.
4101 2	(7/2 ⁻) [‡]			$b_f=-0.7\%$ 7.
4383 3	(5/2 ⁻) [‡]			$b_f=2.0\%$ 4.
4494 2	(7/2 ⁻)	(3)	0.30 7	L: 3 reported in 2016Mu03 . The parentheses are added by evaluators based on the statement in 2016Mu03 that the parallel momentum distribution has limited statistics and seems more consistent with L=3. $b_f=2.9\%$ 5, $\sigma_f^{\text{sp}}=10.7$ mb, $R_s=0.48$ 10. $b_f=9.5\%$ 10, $\sigma_f^{\text{sp}}=10.3$ mb, $R_s=0.47$ 9.
4667 2	5/2 ⁺	2	1.0 2	$b_f=0.9\%$ 3.
4768 2	(9/2 ⁻) [‡]			$b_f=0.3\%$ 2.
4962 3	(9/2 ⁻) [‡]			$b_f=1.5\%$ 2.
5089 3	(11/2 ⁻) [‡]			$b_f=14.5\%$ 14, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.47$ 9.
5200 2	5/2 ⁺	2	1.5 3	$b_f=1.4\%$ 10, $\sigma_f^{\text{sp}}=10.8$ mb, $R_s=0.47$ 9.
5710 2	(1/2 ⁻)	(1)	0.21 16	J^π : Interpreted as the deeply bound $1p_{1/2}$ proton removal. L: 1 reported in 2016Mu03 . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03 . $b_f=1.2\%$ 3, $\sigma_f^{\text{sp}}=10.2$ mb, $R_s=0.44$ 9. J^π : Interpreted as the deeply bound $1p_{1/2}$ proton removal. L: 1 reported in 2016Mu03 . The parentheses are added by evaluators based on the parallel momentum distribution (Fig. 3) in 2016Mu03 .
7527 2	(1/2 ⁻)	(1)	0.20 6	

[†] From a least-squares fit to γ -ray energies.

[‡] Populated by a two-step reaction mechanism. [2016Mu03](#) states that they are likely to have $J \geq 5/2$ and negative parities.

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2016Mu03 (continued) ^{35}P Levels (continued)

2016Mu03 deduced L by comparing the measured and eikonal-calculated parallel momentum distributions of residuals.

@ From the Adopted Levels, also assumed by 2016Mu03 for deducing C^2S .

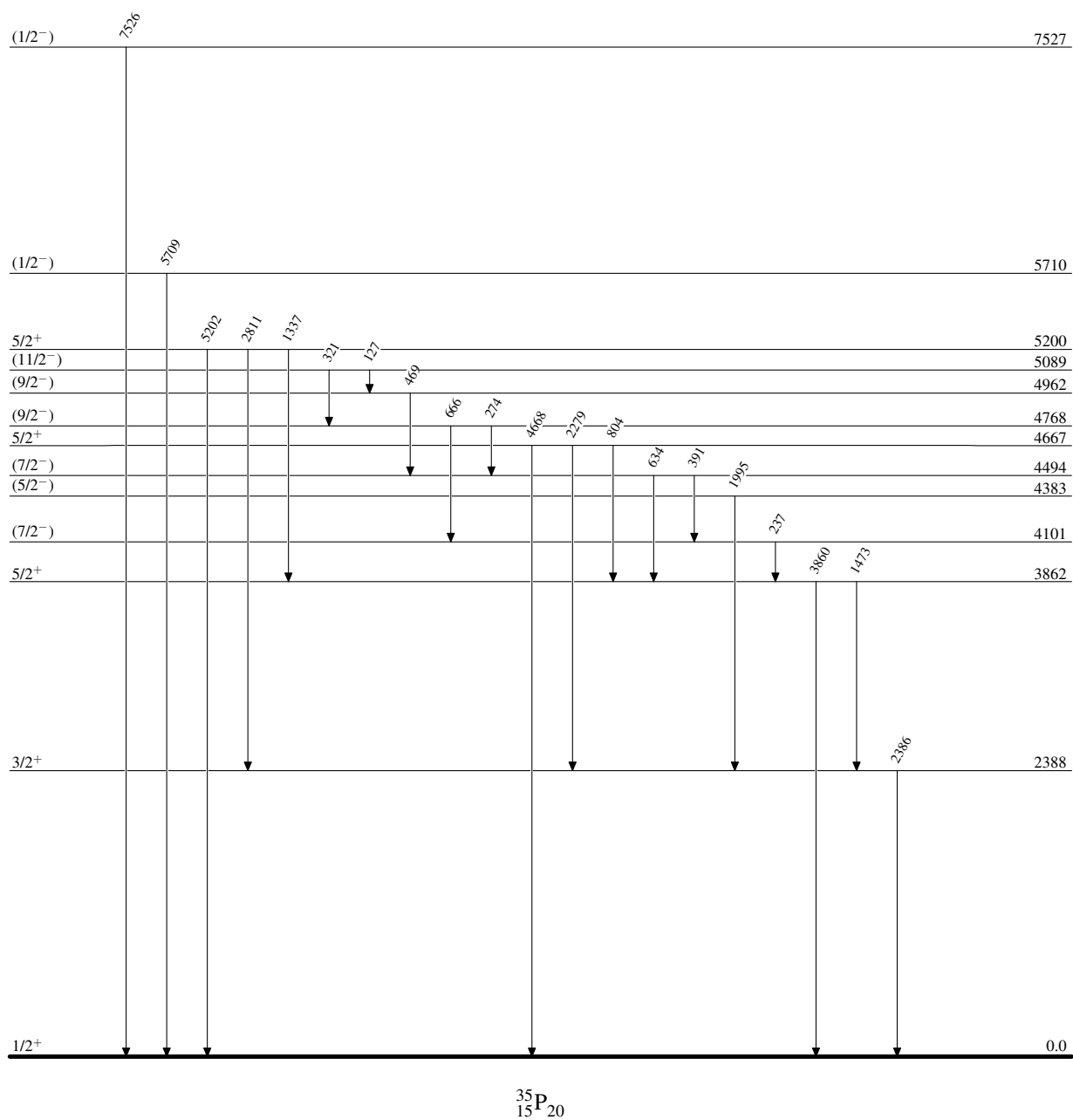
& The spectroscopic factor $\text{C}^2\text{S} = b_f \sigma_{\text{inc}}^{\text{exp}} / R_s \sigma_f^{\text{sp}}$, where b_f is the population fraction, $\sigma_{\text{inc}}^{\text{exp}}$ is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ_f^{sp} is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{P})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π
127 2	5089	(11/2 ⁻)	4962	(9/2 ⁻)	1473 2	3862	5/2 ⁺	2388	3/2 ⁺
237 2	4101	(7/2 ⁻)	3862	5/2 ⁺	1995 2	4383	(5/2 ⁻)	2388	3/2 ⁺
274 2	4768	(9/2 ⁻)	4494	(7/2 ⁻)	2279 2	4667	5/2 ⁺	2388	3/2 ⁺
321 2	5089	(11/2 ⁻)	4768	(9/2 ⁻)	2386 2	2388	3/2 ⁺	0.0	1/2 ⁺
391 2	4494	(7/2 ⁻)	4101	(7/2 ⁻)	2811 2	5200	5/2 ⁺	2388	3/2 ⁺
469 2	4962	(9/2 ⁻)	4494	(7/2 ⁻)	3860 2	3862	5/2 ⁺	0.0	1/2 ⁺
634 2	4494	(7/2 ⁻)	3862	5/2 ⁺	4668 2	4667	5/2 ⁺	0.0	1/2 ⁺
666 2	4768	(9/2 ⁻)	4101	(7/2 ⁻)	5202 2	5200	5/2 ⁺	0.0	1/2 ⁺
804 2	4667	5/2 ⁺	3862	5/2 ⁺	5709 2	5710	(1/2 ⁻)	0.0	1/2 ⁺
1337 2	5200	5/2 ⁺	3862	5/2 ⁺	7526 2	7527	(1/2 ⁻)	0.0	1/2 ⁺

$^9\text{Be}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2016Mu03

Level Scheme



 $^{34}\text{S}(^{18}\text{O}, ^{17}\text{F})$ **1988Or01**

1988Or01: A 108-MeV $^{18}\text{O}^{7+}$ beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched Ag_2S .

Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=250 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured $\sigma(E(^{17}\text{F}))$. Deduced levels and ground state mass excess (−24.87 MeV *4*). Comparisons with shell-model calculations. Proposed a decay scheme of ^{35}Si for the γ -ray transitions observed but not placed in a decay scheme by **1986Du07**.

 ^{35}P Levels

<u>E(level)[†]</u>
0
2420 <i>40</i>
5070 <i>40</i>
5890 <i>70</i>
6440 <i>60</i>
7050 <i>60</i>
7440 <i>60</i>
7920 <i>60</i>
8.60×10^3 <i>10</i>
9290 <i>50</i>

[†] From **1988Or01**.

$^{36}\text{S}(\text{d}, ^3\text{He})$ 1984Th08

$J^\pi=0^+$ for ^{36}S ground state.

1984Th08: A 30-MeV deuteron beam was produced from the BNL Double MP tandem facility. Targets were 20.1 and 23.0 $\mu\text{g}/\text{cm}^2$ sulphur with ^{36}S enriched to 81.1%. Reaction products were momentum-analyzed with the BNL Q3D magnetic spectrometer (FWHM $\approx$ 32 keV) and detected with a multi-wire proportional counter backed by a topping plastic scintillator. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced levels, J, π , L-transfers, spectroscopic factors from the finite-range DWUCK4-DWBA analysis of the angular distributions.

 ^{35}P Levels

Spectroscopic factor $\text{C}^2\text{S}=(2j+1)\times\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/\text{N}$, where j denotes the total angular momentum of the transferred nucleon.

N=29.5 is a normalization factor adopted by 1984Th08. The discrepancy from the original N=2.95 in 1966Ba54 likely arises from different units of cross sections.

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0	0	2.3 12
3864 10	2	1.45, 1.10 [#]
4664 10	2	0.53, 0.41 [#]
5202 10	2	0.40, 0.30 [#]

[†] From 1984Th08.

[‡] From DWBA analysis of measured $\sigma(\theta)$. The uncertainty of C^2S is estimated to be 50% by 1984Th08.

[#] Quoted values are for j=L-1/2 (1d_{3/2}) and L+1/2 (1d_{5/2}), respectively.

$^{36}\text{S}(\text{pol d}, ^3\text{He})$ 1985Kh04

$J^\pi=0^+$ for ^{36}S ground state.

1985Kh04: 52-MeV unpolarized and vector-polarized deuteron beams of 100 nA were produced from the Karlsruhe cyclotron. The target was a 1 mg/cm² ^{208}Pb sulfide with 81.1% enriched ^{36}S on ^{12}C backing. Reaction products were detected using two 300- μm and 1500- μm thick $\Delta\text{E-E}$ surface-barrier detector telescopes (FWHM $\approx$ 90 keV). Measured angular distributions of differential cross sections $\sigma(\text{E}(^3\text{He}), \theta)$ and angular distributions of analyzing powers $i\text{T}_{11}(\text{E}(^3\text{He}), \theta)$. Deduced mass excess, levels, J, π , L-transfers, and spectroscopic factors from standard local, zero-range JULIE-DWBA analysis of $\sigma(\text{E}(^3\text{He}), \theta)$ and $i\text{T}_{11}(\text{E}(^3\text{He}), \theta)$.

 ^{35}P Levels

Spectroscopic factor $\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/\text{N}$, where $\text{N}=2.95$ is a normalization factor adopted by 1985Kh04 from 1974Ma34, originally from 1966Ba54.

$\text{E}(\text{level})^\dagger$	J^π	$\text{L}^\ddagger$	$\text{C}^2\text{S}^\ddagger$	Comments
0	$1/2^+$	0	1.63	
2386 6	$3/2^+^\#$	2	$0.31^\#$	
3857 2	$5/2^+@$	2	$2.91@$	
4474 21			<0.2	$1\text{d}_{3/2}$ proton transfer assumed in DWBA calculations, but 2016Mu03 proposed $1\text{f}_{7/2}$ ($\text{L}=3$) based on the measured parallel momentum distribution.
4665 3	$5/2^+@$	2	$1.06@$	
5189 13	$5/2^+@$	2	$1.38@$	
7520 30			<0.4	

† From 1985Kh04.

‡ From DWBA analysis of measured $\sigma(\theta)$.

$^\#$ L-1/2 transfer from analyzing power measurements; $1\text{d}_{3/2}$ proton transfer assumed in DWBA calculations.

$@$ L+1/2 transfer from analyzing power measurements; $1\text{d}_{5/2}$ proton transfer assumed in DWBA calculations.

 $^{37}\text{Cl}(^{11}\text{B},^{13}\text{N})$ **1988Or01**

1988Or01: An 81-MeV $^{11}\text{B}^{5+}$ beam was produced from the ANU 14UD Pelletron accelerator. Targets were enriched BaCl_2 .

Reaction products were momentum-analyzed with an Enge split-pole spectrometer (FWHM=200 keV at 5.75-10.25°) and detected with a multi-element gas-filled detector at the focal plane. Measured $\sigma(E(^{13}\text{N}))$. Deduced levels. Comparisons with shell-model and DWBA calculations. Proposed a decay scheme of ^{35}Si based on the γ transitions observed by **1986Du07**.

 ^{35}P Levels

E(level) [†]	Comments
0	$d\sigma/d\Omega=120 \mu\text{b/sr}$ 30.
2389 4	$d\sigma/d\Omega=200 \mu\text{b/sr}$ 50.
3860 10	$d\sigma/d\Omega=35 \mu\text{b/sr}$ 9.
4250 20	
4640 20	$d\sigma/d\Omega=30 \mu\text{b/sr}$ 8.
5010 20	
5220 40	$d\sigma/d\Omega\approx 10 \mu\text{b/sr}$ 3.
5840 50	
7590 20	$d\sigma/d\Omega=45 \mu\text{b/sr}$ 11.
8390 40	

[†] From **1988Or01**.

$^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04

Possibly $^{160}\text{Gd} + ^{37}\text{Cl} \rightarrow ^{160}\text{Dy} + ^{35}\text{P} + 2\text{n}$.

1994Fo04: A 167-MeV ^{37}Cl beam was produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm² 98.1% enriched ^{160}Gd backed by 15 mg/cm² gold. γ rays were detected using the Argonne-Notre Dame BGO γ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured E_γ , I_γ , $\gamma\gamma$ -coin. Deduced levels.

 ^{35}P Levels

$E(\text{level})^\dagger$

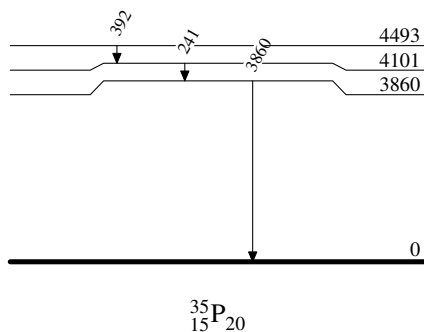
0
3860
4101
4493

[†] From E_γ data.

 $\gamma(^{35}\text{P})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
241	4101	3860
392	4493	4101
3860	3860	0

[†] From 1994Fo04.

 $^{160}\text{Gd}(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04Level Scheme

average intensities relative to a single transitions.

²⁰⁸Pb(³⁶S, ³⁵Pγ) 2008Wi09, 2015Ch56, 2019Gr08

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different

2008Wi09, 2010WiZZ: A 230-MeV ³⁶S beam was produced by the Argonne Tandem Linear Accelerator System (ATLAS) with an intensity of 1.5 pA on a 0.5 mg/cm² ²⁰⁸Pb target and an intensity of 0.3 pA on a 44 mg/cm² ²⁰⁸Pb target. In the thin-target run, binary transfer products were detected using a heavy-ion parallel-plate avalanche counter (PPAC) array (CHICO) (Time resolution ≈0.7ns). The polar angle covered was 12°–85° with respect to the beam. γ rays were detected by Gammasphere consisting of 101 HPGe detectors with FWHM=2-10 keV at E_γ=1 MeV. Event-by-event Doppler shift correction was applied. In the thick-target run, binary transfer products were stopped in the target. γ rays were detected by Gammasphere consisting of 95 HPGe detectors with FWHM=2-3 keV at E_γ=1 MeV. Measured E_γ, I_γ, γγ-coin. Deduced levels. Comparisons with shell-model calculations. Branching-ratio limits were reported for predicted transitions to the 2ħω bandheads in ³⁵P and ³⁴Si. An e-mail reply from Mathis Wiedeking in April, 2010 (**2010WiZZ**) provides relative γ-ray intensities, supplementing **2008Wi09**.

2015Ch56: A 215-MeV ³⁶S beam was produced using the combination of XTU tandem Van de Graaff accelerator and ALPI superconducting linear accelerator at the INFN Legnaro National Laboratory. The target was 300-μg/cm² 99.7% enriched ²⁰⁸Pb on a 20 μg/cm² carbon backing. Projectile-like fragments produced in multinucleon binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 EUROBALL escape-suppressed HPGe clover detectors. Doppler corrections of γ-ray energies were performed event by event. Measured E_γ, I_γ, (³⁵P)γ-coin, and γγ-coin. **2015Ch56** also revisited the γγγ-coin of ³⁶S+¹⁷⁶Yb deep-inelastic data by J. Ollier Ph.D. thesis, University of Paisley (2004) to strengthen the evidence for γ-ray placements (see Ref. [39] in **2015Ch56**). Deduced levels, J, π. Comparisons with shell-model calculations.

2019Gr08: A 225-MeV ³⁶S beam was provided by Tandem-ALPI accelerator complex at the INFN Legnaro National Laboratory. The target was 1 mg/cm² 99.7% enriched ²⁰⁸Pb with 1 mg/cm² Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Doppler corrections of γ-ray energies were performed event by event. Measured E_γ, (³⁵P)γ-coin, and level lifetimes using the differential recoil-distance method (DRDM). Comparison with shell-model calculations.

³⁵P Levels

E(level) [†]	J ^{π‡}	T _{1/2} [#]	Comments
0	1/2 ⁺		
2386.7 7	3/2 ⁺	<0.69 ps	T _{1/2} : estimated mean lifetime τ<1 ps (2019Gr08).
3860.8 7	5/2 ⁺	<0.69 ps	T _{1/2} : estimated mean lifetime τ<1 ps (2019Gr08).
4102.1 7	(7/2 ⁻)	>69 ps	J ^π : 7/2 ⁻ proposed by 2019Gr08 based on comparisons with shell-model calculations.
			T _{1/2} : estimated mean lifetime τ>100 ps (2019Gr08).
4381.9 10	(5/2 ⁻)		
4494.2 8	(7/2 ⁻)	2.29 ps 49	J ^π : 7/2 ⁻ proposed by 2019Gr08 based on comparisons with shell-model calculations.
			T _{1/2} : measured mean lifetime τ=3.3 ps 7 (2019Gr08).
4767.1 10	(9/2 ⁻)		J ^π : 9/2 ⁻ proposed by 2019Gr08 based on comparisons with shell-model calculations.
4869.0 8	(5/2 ⁻ , 7/2 ⁻)		
4962.1 12	(9/2 ⁻)		J ^π : 9/2 ⁻ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5089.8 11	(11/2 ⁻)		J ^π : 11/2 ⁻ proposed by 2019Gr08 based on comparisons with shell-model calculations.
5488.2 10			
5560.1 12	(5/2 ⁻)		
6222.4 11	(7/2 ⁻ , 9/2, 11/2 ⁻)		

[†] From a least-squares fit to γ-ray energies.[‡] From the Adopted Levels.[#] From differential recoil-distance method (DRDM) (**2019Gr08**).

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different

³⁵P₁₅ beam energies, which would result in different

populations of excited states. You can average branching ratios,

but I wouldn't average intensities relative to a single transitions.

³⁵P₁₅²⁰⁻²⁴

²⁰⁸Pb(³⁶S, ³⁵P_γ) **2008Wi09, 2015Ch56, 2019Gr08 (continued)**

$\gamma(^{35}\text{P})$						
E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
128 <i>I</i>	10 5	5089.8	(11/2 ⁻)	4962.1	(9/2 ⁻)	E_γ : weighted average of 128 <i>I</i> (2008Wi09) and 127 <i>I</i> (2015Ch56). I_γ : unweighted average of 14 2 (2008Wi09) and 5.2 6 (2015Ch56).
241 <i>I</i>	61 4	4102.1	(7/2 ⁻)	3860.8	5/2 ⁺	E_γ : From 2008Wi09 and 2015Ch56. I_γ : From 2008Wi09. Other: 32.6 9 (2015Ch56).
273 <i>I</i>	12.8 8	4767.1	(9/2 ⁻)	4494.2	(7/2 ⁻)	E_γ : From 2008Wi09 and 2015Ch56. I_γ : weighted average of 12 2 (2008Wi09) and 12.9 8 (2015Ch56).
322 <i>I</i>	20 7	5089.8	(11/2 ⁻)	4767.1	(9/2 ⁻)	E_γ : weighted average of 321 <i>I</i> (2008Wi09) and 323 <i>I</i> (2015Ch56). I_γ : unweighted average of 27 3 (2008Wi09) and 12.9 8 (2015Ch56).
374 [†] <i>I</i>	3 [†] <i>I</i>	4869.0	(5/2 ⁻ , 7/2 ⁻)	4494.2	(7/2 ⁻)	E_γ : weighted average of 391 <i>I</i> (2008Wi09) and 392 <i>I</i> (2015Ch56). I_γ : unweighted average of 35 3 (2008Wi09) and 24.9 11 (2015Ch56).
392 <i>I</i>	30 5	4494.2	(7/2 ⁻)	4102.1	(7/2 ⁻)	
468 2	16.2 12	4962.1	(9/2 ⁻)	4494.2	(7/2 ⁻)	E_γ : unweighted average of 466 <i>I</i> (2008Wi09) and 469 <i>I</i> (2015Ch56). I_γ : weighted average of 14 2 (2008Wi09) and 16.8 11 (2015Ch56).
487 [†] <i>I</i>	<2 [†]	4869.0	(5/2 ⁻ , 7/2 ⁻)	4381.9	(5/2 ⁻)	I_γ : 1 <i>I</i> from 2008Wi09. E_γ : weighted average of 632 <i>I</i> (2008Wi09) and 633 <i>I</i> (2015Ch56). I_γ : unweighted average of 6 <i>I</i> (2008Wi09) and 10.4 9 (2015Ch56).
633 <i>I</i>	8.2 22	4494.2	(7/2 ⁻)	3860.8	5/2 ⁺	
664 <i>I</i>	32 15	4767.1	(9/2 ⁻)	4102.1	(7/2 ⁻)	E_γ : weighted average of 663 <i>I</i> (2008Wi09) and 665 <i>I</i> (2015Ch56). I_γ : unweighted average of 47 4 (2008Wi09) and 17.8 10 (2015Ch56).
767 [†] <i>I</i>	5 [†] <i>I</i>	4869.0	(5/2 ⁻ , 7/2 ⁻)	4102.1	(7/2 ⁻)	E_γ : unweighted average of 856 <i>I</i> (2008Wi09) and 861 <i>I</i> (2015Ch56). I_γ : weighted average of 13 2 (2008Wi09) and 9.8 12 (2015Ch56).
859 3	10.7 14	4962.1	(9/2 ⁻)	4102.1	(7/2 ⁻)	
993 [†] <i>I</i>	5 [†] <i>I</i>	5488.2		4494.2	(7/2 ⁻)	E_γ, I_γ : From 2015Ch56.
1009 [†] <i>I</i>	<1 [†]	4869.0	(5/2 ⁻ , 7/2 ⁻)	3860.8	5/2 ⁺	
1132 [†] <i>I</i>	<1 [†]	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	5089.8	(11/2 ⁻)	
1260 [†] <i>I</i>	4 [†] <i>I</i>	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	4962.1	(9/2 ⁻)	
^x 1353 <i>I</i>	9.2 11					E_γ, I_γ : From 2015Ch56.
1387 [†] <i>I</i>	3 [†] <i>I</i>	5488.2		4102.1	(7/2 ⁻)	
1458 [†] <i>I</i>	7 [†] 2	5560.1	(5/2 ⁻)	4102.1	(7/2 ⁻)	
1474 <i>I</i>	15.9 14	3860.8	5/2 ⁺	2386.7	3/2 ⁺	
^x 1592 <i>I</i>	7.7 10					E_γ : weighted average of 1473 <i>I</i> (2008Wi09) and 1474 <i>I</i> (2015Ch56). I_γ : weighted average of 15 2 (2008Wi09) and 16.4 14 (2015Ch56). E_γ, I_γ : From 2015Ch56.
1715 [†] <i>I</i>	4 [†] <i>I</i>	4102.1	(7/2 ⁻)	2386.7	3/2 ⁺	
1729 [†] <i>I</i>	4 [†] <i>I</i>	6222.4	(7/2 ⁻ , 9/2, 11/2 ⁻)	4494.2	(7/2 ⁻)	
1995 <i>I</i>	8 6	4381.9	(5/2 ⁻)	2386.7	3/2 ⁺	

E_γ, I_γ : From 2015Ch56.

E_γ : weighted average of 1473 *I* (2008Wi09) and 1474 *I* (2015Ch56).

I_γ : weighted average of 15 2 (2008Wi09) and 16.4 14 (2015Ch56).

E_γ, I_γ : From 2015Ch56.

E_γ : weighted average of 1995 *I* (2008Wi09) and 1994 *I* (2015Ch56, unplaced).

Continued on next page (footnotes at end of table)

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ **2008Wi09,2015Ch56,2019Gr08 (continued)** $\gamma(^{35}\text{P})$ (continued)

E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
2386 1	30 4	2386.7	$3/2^+$	0	$1/2^+$	I_γ : unweighted average of 2 1 (2008Wi09) and 14.2 11 (2015Ch56). E_γ : From 2008Wi09 and 2015Ch56. I_γ : From 2008Wi09. Other: 99.2 28 (2015Ch56). Shell-model calculations indicate a small occupancy of the proton $1d_{3/2}$ orbit in the ground state of ^{36}S .
3861 1	100.0 32	3860.8	$5/2^+$	0	$1/2^+$	E_γ : weighted average of 3861 1 (2008Wi09) and 3860 1 (2015Ch56). I_γ : From 2015Ch56. Other: 100 (2008Wi09).
4102 [†] 1	33 [†] 5	4102.1	$(7/2^-)$	0	$1/2^+$	E_γ : other: 4101 (2015Ch56).

[†] From 2008Wi09.^x γ ray not placed in level scheme.

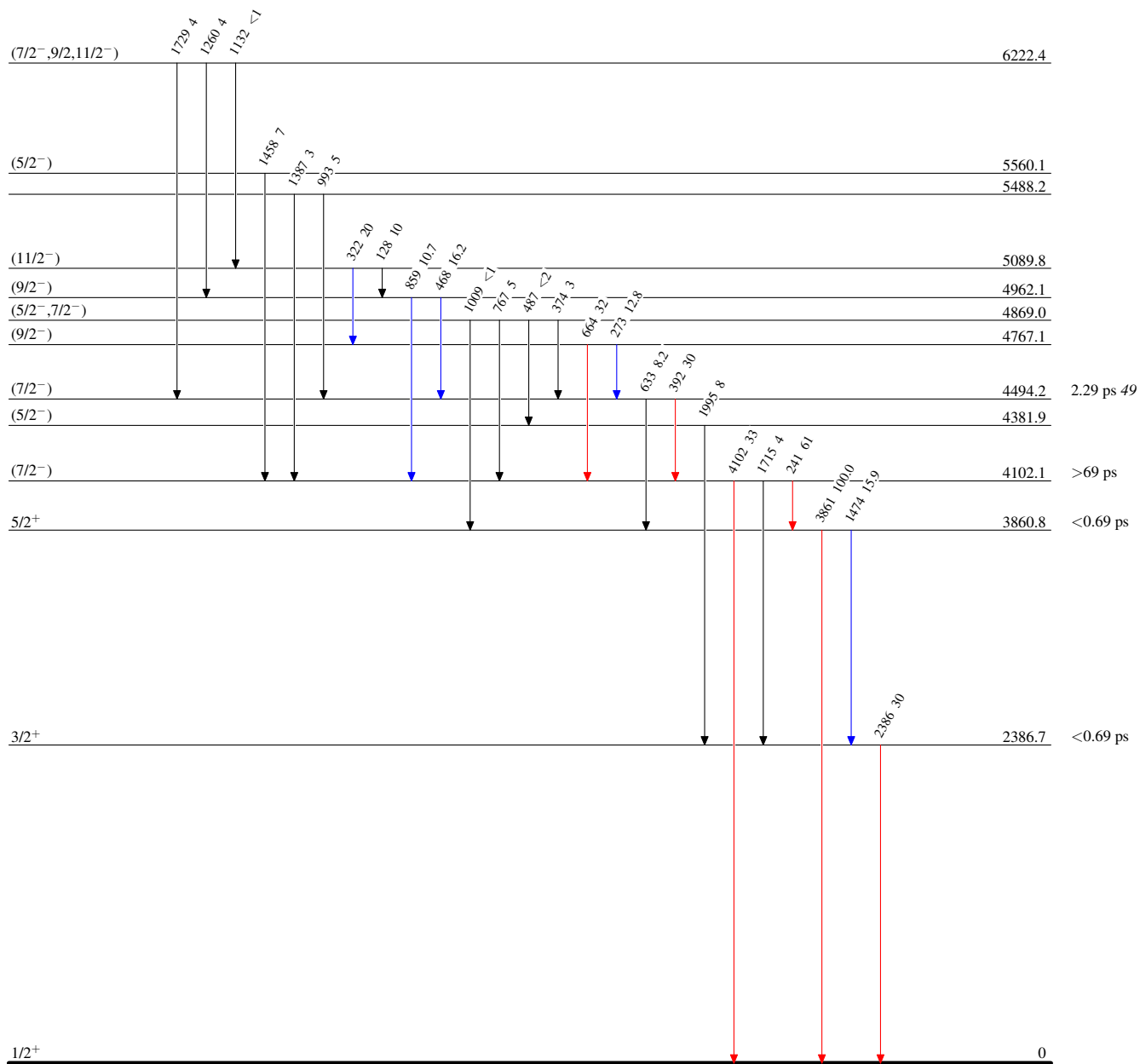
$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{P}\gamma)$ 2008Wi09,2015Ch56,2019Gr08

Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



Adopted Levels, Gammas

$Q(\beta^-)=167.322\ 26$; $S(n)=6985.84\ 4$; $S(p)=11586.5\ 8$; $Q(\alpha)=-8322.09\ 6$ [2021Wa16](#)

$S(2n)=18402.99\ 4$, $S(2p)=22909.8\ 7$ ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): $\text{Cl}(n,X)$ reaction at Physical Institute of Aarhus University ([1936An01](#)).

^{35}S decay measurements:

[2014Si21](#), [2000Ho13](#), [1999Pa18](#), [1999Oh09](#).

^{35}S cross-section measurements:

$^{35}\text{Cl}(n,p)^{35}\text{S}$: [2024Ha18](#), [2023De34](#), [2021WaZY](#), [2020Ku17](#), [2019Ba16](#), [2000Ka30](#), [1999Gl09](#), [1999Gl02](#);

Projectile fragmentation $^9\text{Be}(^{40}\text{Ar},X)$ and $^{181}\text{Ta}(^{40}\text{Ar},X)$: [2012Zh06](#), [2007No13](#);

Spallation $^{136}\text{Xe}(p,X)$: [2007Na31](#);

$^{34}\text{S}(n,\gamma)$ $E=\text{thermal}$: [2023KoZS](#), [2008FiZZ](#).

 ^{35}S LevelsCross Reference (XREF) Flags

A	$^{35}\text{P}\ \beta^-$ decay (47.3 s)	F	$^{34}\text{S}(n,\gamma)$ $E=\text{thermal}$	K	$^{36}\text{S}(p,d)$
B	$^2\text{H}(^{34}\text{S},p\gamma)$	G	$^{34}\text{S}(n,n),(n,\gamma):\text{resonances}$	L	$^{37}\text{Cl}(p,^3\text{He})$
C	$^9\text{Be}(^{36}\text{S},^{35}\text{S}\gamma)$	H	$^{34}\text{S}(\text{pol } d,p)$	M	$^{37}\text{Cl}(d,\alpha\gamma)$
D	$^{24}\text{Mg}(^{14}\text{N},3p\gamma)$	I	$^{34}\text{S}(d,p)$	N	$^{160}\text{Gd}(^{34}\text{S},X\gamma),(^{37}\text{Cl},X\gamma)$
E	$^{26}\text{Mg}(^{18}\text{O},2\alpha n\gamma)$	J	$^{34}\text{S}(d,p\gamma)$	O	$^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$T_{1/2}$</u>	<u>XREF</u>	<u>Comments</u>
0.0	$3/2^+$	87.61 d 24	ABCDEF HIJKLMNO	$\% \beta^- = 100$ $\mu = (+)1.00\ 4$ (1954Bu05 , 2019StZV) $Q = 0.0483\ 3$ (2018Py01 , 2021StZZ) J^π : $L=2$ from 0^+ in $^{34}\text{S}(\text{pol } d,p)$, $^{34}\text{S}(d,p)$, $^{36}\text{S}(p,d)$, and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L-1/2$ transfer from analyzing power. $J=3/2$ also from microwave spectroscopy (1949Co12 , 1951We11). $T_{1/2}$: unweighted average of 88 d 3 (1941Ka01), 87.1 d 12 (1943He01), 86.35 d 17 (1959Co56), 87.16 d 10 (1958Se49), 88.8 d 10 (1959Ca12), 89.0 d 5 (1961Wy01), 87.1 d 9 (1961Oz01), 87.9 d 3 (1965Fl02), 87.39 d 10 (1968Wo06), 87.5 d 4 (1969La34), and 87.38 d 3 (1999Pa18). μ : $+1.00\ 4$ or $-1.07\ 4$ from 1954Bu05 using the microwave absorption method, and the negative value is less likely from systematics of odd-A S isotopes. Q : from 2018Py01 using the microwave absorption method. $-0.483\ 3$ is given in 2021StZZ . Other: 0.045 from 1954Bi40 using the microwave absorption method.
1572.369 9	$1/2^+$	2.29 ps 14	ABCD F HIJKLM O	J^π : $L=0$ from 0^+ in $^{34}\text{S}(\text{pol } d,p)$, $^{34}\text{S}(d,p)$, $^{36}\text{S}(p,d)$, and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$. $T_{1/2}$: lifetime $\tau=3.3$ ps 2 in $^{208}\text{Pb}(^{36}\text{S},^{35}\text{S}\gamma)$ from 2022Gr07 with DRDM. Others: 3.3 ps 5 in $^{34}\text{S},p\gamma$ from 1973Wa10 with DSAM, >1.8 ps in $^{34}\text{S}(d,p\gamma)$ from 1970Bu18 with DSAM, and >1.6 ps in $^{34}\text{S}(d,p\gamma)$ from 1972Fr11 with DSAM.
1991.27 4	$7/2^-$	1.03 ns 5	CDEF HIJKLMNO	J^π : $L=3$ from 0^+ in $^{36}\text{S}(p,d)$ and $^{34}\text{S}(\text{pol } d,p)$, and $L+1/2$ transfer from analyzing power. $T_{1/2}$: lifetime $\tau=1.48$ ns 7: weighted average of 1.7 ns 3 in $^{34}\text{S}(d,p\gamma)$ from 2024Co04 with $p\gamma$ -delayed coin and 1.47 ns 7 in $^{34}\text{S}(d,p\gamma)$ from 1971Pr11 with $p\gamma$ -delayed coin. Other: >4.5 ps in $^{34}\text{S}(d,p\gamma)$ from 1970Bu18 with DSAM.
2347.779 10	$3/2^-$	0.78 ps 12	BCD F HIJK M O	J^π : $L=1$ from 0^+ in $^{34}\text{S}(\text{pol } d,p)$, $^{34}\text{S}(d,p)$, $^{36}\text{S}(p,d)$, and $^9\text{Be}(^{36}\text{S},^{35}\text{S})$ and $L+1/2$ transfer from analyzing power.

Continued on next page (footnotes at end of table)

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use different beam energies, which would result in different populations of excited states. You can

³⁵S
16

NUCLEAR DATA SHEETS

³⁵S
16¹⁹-2

Adopted Levels, Gammas (continued)

A general comment (not necessary to apply to dataset. 2008Wi09 and 2015Ch56 use di

³⁵S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
2717.06 8	5/2 ⁺	69 fs 24	CD F HIJKLM O	T _{1/2} : lifetime $\tau=1.12$ ps 18; weighted average of 1.3 ps 2 in (³⁴ S,py) from 1973Wa10 with DSAM, 0.8 ps 2 in ³⁴ S(d,py) from 1970Bu18 with DSAM, and 1.22 ps 18 in ³⁴ S(d,py) from 1972Fr11 with DSAM. J ^π : L=2 from 0 ⁺ in ³⁴ S(pol d,p), ³⁴ S(d,p), ³⁶ S(p,d), and ⁹ Be(³⁶ S, ³⁵ S); L+1/2 transfer from J-dependence in ³⁶ S(p,d) and L+1/2 transfer from analyzing power. T _{1/2} : lifetime $\tau=100$ fs 35 in ³⁴ S(d,py) from 1972Fr11 with DSAM.
2938.63 5	3/2 ⁺		A C F HIJKLM	J ^π : L=2 from 0 ⁺ in ³⁴ S(d,p), ³⁶ S(p,d), ³⁴ S(pol d,p), and L=(2) in ⁹ Be(³⁶ S, ³⁵ S); L-1/2 transfer from J-dependence in ³⁶ S(p,d); possibly allowed β^- feeding from 1/2 ⁺ parent with log $f_t=5.0$.
3421.2 20	5/2 ⁺	<69 fs	C HIJKLM O	J ^π : L=2 from 0 ⁺ in ³⁴ S(pol d,p), ³⁴ S(d,p), ³⁶ S(p,d), and ⁹ Be(³⁶ S, ³⁵ S); L+1/2 transfer from J-dependence in ³⁶ S(p,d) and L+1/2 transfer from analyzing power. T _{1/2} : lifetime $\tau<100$ fs in ³⁴ S(d,py) from 1972Fr11 with DSAM.
3558.080 28	(5/2 ⁺)		F HIJK M O	J ^π : L=(2) from 0 ⁺ in ³⁶ S(p,d); 1986 γ to 1/2 ⁺ , 1572.370 and 1566.7 γ to 7/2 ⁻ , 1991.27.
3594.62 24	(7/2 ⁺)		CDE IJKLM O	XREF: C(3595)I(3596)K(3592)L(3598) J ^π : 3594.5 γ Q, $\Delta J=2$ to 3/2 ⁺ , g.s.; L=2+4 from 3/2 ⁺ in (p, ³ He): L=2 suggests (1/2 ⁺ to 9/2 ⁺) and L=4 suggests (3/2 ⁺ to 13/2 ⁺); L=(2) from 0 ⁺ in ³⁶ S(p,d) and ⁹ Be(³⁶ S, ³⁵ S). 7/2 ⁺ is suggested by shell model (2014Ay01,2021Go09). If 3594.5 γ is consistent with $\Delta J=0$ to 3/2 ⁺ , the discrepancy between the in-beam γ -ray spectroscopy (¹⁴ N,3py) and transfer reactions may be resolved.
3675? 10 3801.952 16	(1/2 ⁻ ,3/2 ⁻) 3/2 ⁻	25 fs 18	I F HIJKLM	J ^π : L=(1) from 0 ⁺ in ³⁴ S(d,p). XREF: I(3811) J ^π : L=1 from 0 ⁺ in ³⁴ S(d,p), ³⁶ S(p,d), and ³⁴ S(pol d,p) and L+1/2 transfer from analyzing power. T _{1/2} : lifetime $\tau=36$ fs 26 in ³⁴ S(d,py) from 1972Fr11 with DSAM.
3816.01 21	(9/2 ⁻)	0.28 ps 3	CDE H J LM O	XREF: I(3811) J ^π : 1824.7 γ M1+E2 to 7/2 ⁻ , 1991.27; 2061.6 γ D, $\Delta J=1$ from (11/2 ⁺), 5877.70. L=(3) from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ S γ) is inconsistent but the fit on the parallel momentum distribution is poor.
3886.28	(5/2)		CD IJK M	T _{1/2} : in (¹⁴ N,3py) from 2014Ay01 with DSAM. XREF: C(3890)M(3889) J ^π : 1894.9 γ D, $\Delta J=1$ to 7/2 ⁻ , 1991.27; 3886 γ to 3/2 ⁺ , g.s.; L=3 from 0 ⁺ in ³⁶ S(p,d). L=(0) from 0 ⁺ in ⁹ Be(³⁶ S, ³⁵ S γ) is inconsistent.
4023.24 21	(11/2 ⁻)	0.32 ps 3	DE MNO	J ^π : 2032.0 γ E2, $\Delta J=2$ to 7/2 ⁻ , 1991.27; 207.0 γ to (9/2 ⁻), 3816.01.
4027.2 14	(3/2 ⁺ ,5/2 ⁺)		C IJKLM	T _{1/2} : in (¹⁴ N,3py) from 2014Ay01 with DSAM. XREF: I(4025)K(4023) J ^π : L=(2) from 0 ⁺ in ³⁶ S(p,d) and ⁹ Be(³⁶ S, ³⁵ S γ).
4105.6 8	(5/2 ⁺)	<55 fs	F IJKLM	J ^π : L=(2) from 0 ⁺ in ³⁶ S(p,d); 2533.1 γ to 1/2 ⁺ , 1572.370 and 2114.3 γ to 7/2 ⁻ , 1991.27. T _{1/2} : lifetime $\tau<80$ fs in ³⁴ S(d,py) from 1972Fr11 with DSAM.

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2}</u>	<u>XREF</u>	<u>Comments</u>
4180 3	(1/2,3/2,5/2 ⁺)		LM	XREF: l(4186) J ^π : 2608γ to 1/2 ⁺ , 1572.370.
4189.240 33	1/2 ⁻	<35 fs	F HIJKLM	XREF: K(4182)l(4186)M(4187) J ^π : L=1 from 0 ⁺ in $^{34}\text{S}(\text{d,p})$, $^{36}\text{S}(\text{p,d})$, and $^{34}\text{S}(\text{pol d,p})$ and L-1/2 transfer from analyzing power. T _{1/2} : lifetime τ<50 fs in $^{34}\text{S}(\text{d,p}γ)$ from 1972Fr11 with DSAM.
4302.6 7	(1/2 ⁻ ,3/2 ⁻)		I KLM	XREF: L(4290) J ^π : L=(1) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$. L=(2) from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
4477.63	(5/2 ⁺)	<62 fs	F IJ LM	XREF: i(4481)L(4489) J ^π : 2905.1γ to 1/2 ⁺ , 1572.370 and 2486.27γ to 7/2 ⁻ , 1991.27. L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺). L=2 component of L=2,3 doublet from 0 ⁺ in $^{34}\text{S}(\text{d,p})$. T _{1/2} : lifetime τ<90 fs in $^{34}\text{S}(\text{d,p}γ)$ from 1972Fr11 with DSAM.
4486 7	7/2 ⁻		Hi K	XREF: H(4482)i(4481) E(level): from $^{36}\text{S}(\text{p,d})$. J ^π : L=3 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$ and $^{34}\text{S}(\text{pol d,p})$, and L+1/2 transfer from analyzing power. L=3 component of L=2,3 doublet from 0 ⁺ in $^{34}\text{S}(\text{d,p})$.
4575 5	3/2 ⁺		HI KL	E(level): weighted average of 4575 8 from $^{34}\text{S}(\text{d,p})$, 4577 10 from (p, ^3He), and 4574 5 from $^{36}\text{S}(\text{p,d})$. J ^π : L=2 from 0 ⁺ and L-1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$. L=0+2 from 3/2 ⁺ in (p, ^3He): L=0 suggests (1/2 ⁺ ,3/2 ⁺ ,5/2 ⁺) and L=2 suggests (1/2 ⁺ to 9/2 ⁺).
4615 5	3/2 ⁺ ,5/2 ⁺		KL	E(level): weighted average of 4614 5 from $^{36}\text{S}(\text{p,d})$ and 4617 10 from (p, ^3He). J ^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
4822.79 24	(9/2 ⁺)		DE	J ^π : 1228.1γ D, ΔJ=1 to (7/2 ⁺), 3594.62; 1055.1γ D, ΔJ=1 from (11/2 ⁺), 5877.70; 9/2 ⁺ suggested by shell-model calculations (2021Go09).
4839 5	(1/2 ⁺)		HI KL	E(level): weighted average of 4837 8 from $^{34}\text{S}(\text{d,p})$, 4838 5 from $^{36}\text{S}(\text{p,d})$, and 4843 10 from (p, ^3He). Other: 4837 from (pol d,p). J ^π : L=(0) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$; L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
4899.82 27	(9/2 ⁺)		DE	J ^π : 2908.3γ D, ΔJ=1 to 7/2 ⁻ , 1991.27; 977.9γ D+Q, ΔJ=1 from (11/2 ⁺), 5877.70; 9/2 ⁺ suggested by shell-model calculations (2021Go09).
4903.371 16	1/2 ⁺		F HI K	J ^π : L=0 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$. Other: L=1 from 0 ⁺ in $^{34}\text{S}(\text{d,p})$ and $^{34}\text{S}(\text{pol d,p})$ and L-1/2 transfer from analyzing power. 2026Jo01 in $^{36}\text{S}(\text{p,d})$ acknowledges this discrepancy and states that the J ^π is confronted with strong evidence.
4963.082 25	5/2 ⁺		C F HI KL	XREF: C(4948)K(4955) J ^π : L=2 from 0 ⁺ and L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p,d})$. Other: L=1 from 0 ⁺ in $^{34}\text{S}(\text{d,p})$ and $^{34}\text{S}(\text{pol d,p})$ and L+1/2 transfer from analyzing power. 2026Jo01 in $^{36}\text{S}(\text{p,d})$ acknowledges this discrepancy and states that the J ^π is confronted with strong evidence.
4990 10	(⁺)		L	J ^π : L=0+2 from 3/2 ⁺ in (p, ^3He): L=0 suggests (1/2 ⁺ ,3/2 ⁺ ,5/2 ⁺) and L=2 suggests (1/2 ⁺ to 9/2 ⁺).
5010.05 29	(11/2 ⁻)	0.45 ps 8	DE	J ^π : 3018.5γ E2, ΔJ=2 to 7/2 ⁻ , 1991.27; 986.8γ to (11/2 ⁻ ,

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
				4023.24.
5058 8	7/2 ⁻		HI	T _{1/2} : in ($^{14}\text{N}, 3\text{p}\gamma$) from 2014Ay01 with DSAM. E(level): from $^{34}\text{S}(\text{d}, \text{p})$. J ^π : L=3 from 0 ⁺ in $^{34}\text{S}(\text{d}, \text{p})$ and $^{34}\text{S}(\text{pol d}, \text{p})$ and L+1/2 transfer from analyzing power.
5123 5	1/2 ⁺		HI KL	E(level): weighted average of 5126 11 from $^{34}\text{S}(\text{d}, \text{p})$, 5121 5 from $^{36}\text{S}(\text{p}, \text{d})$, and 5127 10 from (p, ^3He). J ^π : L=0 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$; L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
5286 5	(1/2 ⁻ , 3/2 ⁻)		K	
5343 8	(⁻)		C I L	E(level): weighted average of 5342 8 from $^{34}\text{S}(\text{d}, \text{p})$ and 5345 10 from (p, ^3He). Other: 5349 8 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. J ^π : L=3 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁻ to 11/2 ⁻). J ^π : 1389.0γ D, ΔJ=1 to (11/2) ⁻ , 4023.24; 1525.7γ to (5/2), 3886.28; 9/2 ⁺ suggested by shell model (2014Ay01, 2021Go09). XREF: I(5475?)
5412.28 25	(9/2 ⁺)		DE	J ^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$. XREF: I(5542?)
5476 5	3/2 ⁺ , 5/2 ⁺		I K	J ^π : L=3 from 3/2 ⁺ in (p, ^3He): L=3 suggests (1/2 ⁻ to 11/2 ⁻). J ^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$. XREF: I(5740)
5550 10			I L	J ^π : L=3 from 0 ⁺ in $^{34}\text{S}(\text{d}, \text{p})$. E(level): weighted average of 5766 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 5771 10 from (p, ^3He). Other: 5775 4 from $^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$. J ^π : L=2 from 0 ⁺ and L+1/2 transfer from J-dependence in $^{36}\text{S}(\text{p}, \text{d})$; L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺). J ^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$. J ^π : 465.4γ D, ΔJ=1 to (9/2 ⁺), 5412.28; 867.6γ D, ΔJ=0 to (11/2) ⁻ , 5010.05; strong 2283.0γ to (7/2 ⁺), 3594.62. XREF: L(5915?) J ^π : L=2 from 0 ⁺ in $^{34}\text{S}(\text{d}, \text{p})$.
5767 5	5/2 ⁺		C KL	
5841.482 31	(3/2 ⁺ , 5/2 ⁺)		F K	
5877.70 22	(11/2 ⁺)		DE	J ^π : L=1 from 0 ⁺ in $^{34}\text{S}(\text{d}, \text{p})$. E(level): weighted average of 6121 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 6129 10 from (p, ^3He). J ^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$; L=0+2 from 3/2 ⁺ in (p, ^3He): L=0 suggests (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺) and L=2 suggests (1/2 ⁺ to 9/2 ⁺). J ^π : L=(3) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$. XREF: I(6292)
5890 20	3/2 ⁺ , 5/2 ⁺		I L	
5980 10			I	
6018.8 6			F	
6078.475 33	1/2 ⁻ , 3/2 ⁻		F I	
6123 5	(3/2 ⁺ , 5/2 ⁺)		KL	
6218 5	(5/2 ⁻ , 7/2 ⁻)		K	
6293.93 6			F I	
6299.4 4	(11/2)		D	
6337 5	1/2 ⁺		I K	
6345 8	(⁺)		I L	
6352.2 5	(13/2) ⁻	50 fs 10	D	
6354.89 23			F	

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)				
E(level) [†]	J^π [‡]	$T_{1/2}$	XREF	Comments
6419.9 11			F	
6441 5	(1/2 ⁺)		I K	E(level): weighted average of 6446 8 from $^{34}\text{S}(\text{d,p})$ and 6439 5 from $^{36}\text{S}(\text{p,d})$. J^π : L=(0) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
6496 8			I	
6537.7 14			I	
6545.1 13	3/2 ⁺ , 5/2 ⁺		I K	XREF: K(6550) J^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
6584 10			I	
6629.43 9			F	
6635.2 13	3/2 ⁺ , 5/2 ⁺		I K	XREF: K(6639) J^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
6654 10	(⁻)		L	J^π : L=(3) from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁻ to 11/2 ⁻).
6683 5	1/2 ⁺		I KL	E(level): weighted average of 6677 8 from $^{34}\text{S}(\text{d,p})$, 6682 5 from $^{36}\text{S}(\text{p,d})$, and 6696 10 from (p, ^3He). J^π : L=0 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$; L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
6761.0 12			F	
6790 5	(3/2 ⁺ , 5/2 ⁺)		K	J^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
6891.3 14	(⁻)		I	J^π : L=(1,3) from 0 ⁺ in $^{34}\text{S}(\text{d,p})$.
6962 5	(1/2 ⁻ , 3/2 ⁻)		K	J^π : L=(1) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
(6986.097 25)	1/2 ⁺		F	J^π : s-wave resonance capture state.
7018.89 4	3/2 ⁺ , 5/2 ⁺		G I K	XREF: I(7021)K(7019) E(level): from (n,n),(n, γ):resonances. J^π : L=2 from 0 ⁺ in (n,n),(n, γ):resonances. Others: L=(2,3) from 0 ⁺ in $^{34}\text{S}(\text{d,p})$ for 7021 10; L=(1) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$ for 7019 5.
7072.66 5		0.14 eV 4	G	
7074.33 6		0.084 eV 3	G	
7097.72 30	3/2 ⁺ , 5/2 ⁺	0.009 eV 3	G	
7099.85 20	3/2 ⁺ , 5/2 ⁺	0.033 eV 5	G	
7100.73 4	3/2 ⁻		G K	J^π : other: L=(1) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
7143.27 11	(5/2 ⁻ , 7/2 ⁻)	0.42 eV 1	G I KL	XREF: I(7150)K(7151)L(7151?) J^π : L=(3) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$. Others: L=(0,3) from 0 ⁺ in $^{34}\text{S}(\text{d,p})$; L=(4) from 3/2 ⁺ in (p, ^3He) suggests (3/2 ⁺ to 13/2 ⁺).
7179.74 32	(15/2 ⁺)	3.1 ps 12	DE	J^π : 1302.0 γ E2, $\Delta J=2$ to (11/2 ⁺), 5877.70; 827.8 γ to (13/2 ⁻), 6352.2. $T_{1/2}$: in ($^{14}\text{N}, 3\text{p}\gamma$) from 2014Ay01 with DSAM.
7210.42 5	1/2 ⁻ , 3/2, 5/2 ⁺		G i k	XREF: i(7205)k(7215) J^π : others: L=(1,2) from 0 ⁺ in $^{34}\text{S}(\text{d,p})$ for 7205 35; L=2 from 0 ⁺ in $^{36}\text{S}(\text{p,d})$ for 7215 5.
7218.0 4		0.33 eV 2	G i k	XREF: i(7205)k(7215)
7234.0 5		0.19 eV 1	G i	XREF: i(7205)
7239.8 6		0.06 eV 2	G i	XREF: i(7205)
7253.2 6	(3/2 ⁺ , 5/2 ⁺)	0.35 eV 2	G K	XREF: K(7249) J^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$.
7275.93 5	1/2 ⁺		G k	XREF: k(7279)
7279.13?		0.10 eV 3	G k	J^π : other: L=(0) from 0 ⁺ in $^{36}\text{S}(\text{p,d})$ for 7279 5.
7289.8 8		0.31 eV 3	G	XREF: G(?)k(7279)
7294.19 6	3/2 ⁻		G	
7331.00 5	1/2 ⁺		G K	XREF: K(7326)

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Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
				J ^π : other: L=0 from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$.
7338.18 20	1/2 ⁻ , 3/2, 5/2 ⁺		G	
7343.91 20		0.72 eV 3	G k	XREF: k(7349)
7347.2 5		0.44 eV 6	G k	XREF: k(7349)
7347.9 5		0.30 eV 5	G k	XREF: k(7349)
7353.9 5		0.44 eV 4	G k	XREF: k(7349)
7357.6 5		0.16 eV 4	G	
7370.56 6	1/2 ⁻		G 1	XREF: l(7375)
7371.97?		0.12 eV	G 1	XREF: G(?)l(7375)
7396.0 6	3/2 ⁺ , 5/2 ⁺	0.44 eV 33	G	
7404.6 6		0.21 eV 5	G	
7408.68 11	3/2 ⁻		G	
7411.55 6			G	
7416.50 4			G	
7429.1 6		0.12 eV 3	G	
7433.56 5			G	
7436.4 4	1/2 ⁻ , 3/2, 5/2 ⁺		G	
7442.14 5	1/2 ⁺		G K	XREF: K(7441)
				J ^π : other: L=0 from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$.
7462.2 6		0.33 eV 4	G	
7481.15 7	1/2 ⁻		G I	E(level): weighted average of 7481.15 6 from (n,n),(n,γ):resonances and 7482.7 13 from $^{34}\text{S}(\text{d},\text{p})$.
				J ^π : other: L=(1,2) from 0 ⁺ in $^{34}\text{S}(\text{d},\text{p})$.
7494.5 6	1/2 ⁻ , 3/2 ⁻	0.17 eV 4	G K	XREF: K(7490)
				J ^π : L=1 from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$.
7542.3 6		0.21 eV 3	G i	XREF: i(7570)
7604.0 6		0.90 eV 8	G i	XREF: i(7570)
7609.26 5	3/2 ⁻		G	
7648.9 6		0.21 eV 3	G i	XREF: i(7640)
7655.7 6		0.80 eV 5	G i	XREF: i(7640)
7663.91 11	1/2 ⁻ , 3/2, 5/2 ⁺		G i	XREF: i(7640)
7678.3 7		0.45 eV 7	G i	XREF: i(7640)
7731.0 7		0.31 eV 13	G L	XREF: L(7712?)
7749.7 4	3/2 ⁺ , 5/2 ⁺		G K	XREF: K(7753)
				J ^π : L=2 from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$.
7761.46 7	3/2 ⁻		G 1	XREF: l(7770)
7776.17 11	1/2 ⁻		G 1	XREF: l(7770)
7797.99 9	1/2 ⁺		G	
7812.24 7	3/2 ⁺		G	
7853.25 7	3/2 ⁻		G	
7861.8 12		1.13 eV 19	G	
7880.8 12		0.67 eV 17	G k	XREF: k(7885)
				J ^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$ for 7885 5.
7889.0 12		0.57 eV 25	G k	XREF: k(7885)
				J ^π : L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$ for 7885 5.
7894.63 7	3/2 ⁻		G	
7899.70 30			G	
7934.1 15		0.5 eV 3	G	
7939.5 15		0.89 eV 4	G	
7954.96 11	3/2 ⁻		G	
7974.19 11	3/2 ⁻		G K	XREF: K(7980)
				J ^π : other: L=1 from 0 ⁺ in $^{36}\text{S}(\text{p},\text{d})$.
8019.5 2	3/2 ⁺ , 5/2 ⁺		G	
8024.0 7	(17/2 ⁺)	0.15 ps 4	DE	J ^π : 844.2γ (M1), ΔJ=1 to (15/2 ⁺), 7179.74. 17/2 ⁺ suggested by shell-model calculations (2021Go09).

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Adopted Levels, Gammas (continued)

^{35}S Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2}	XREF	Comments
8041.20 20	3/2 ⁺ , 5/2 ⁺		G	T _{1/2} : in ($^{14}\text{N}, 3\text{p}\gamma$) from 2014Ay01 with DSAM.
8076.46 20	3/2 ⁺ , 5/2 ⁺		G k	XREF: k(8073)
8077.91 30	3/2 ⁺ , 5/2 ⁺		G k	J ^π : other: L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$ for 8073 5. XREF: k(8073)
8093.0 5	1/2 ⁻		G KL	J ^π : other: L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$ for 8073 5. XREF: K(8103)L(8103)
8143.66 20	3/2 ⁺ , 5/2 ⁺		G	E(level): from (n,n),(n,γ):resonances. Others: 8103 5 from $^{36}\text{S}(\text{p}, \text{d})$ and 8103 10 from (p, ^3He). J ^π : other: L=1 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$; L=(1+3) from 3/2 ⁺ in (p, ^3He).
8160 10			L	J ^π : L=(1) from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁻ to 7/2 ⁻).
8187.17 30	1/2 ⁻		G	
8211.25 30			G	
8224.2 4	1/2 ⁻		G K	XREF: K(8221) J ^π : other: L=1 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$.
8228.3 4	3/2 ⁺ , 5/2 ⁺		G	
8243.98 30	3/2 ⁺ , 5/2 ⁺		G	
8256.31 20	3/2 ⁻		G	
8262.4 4	5/2 ⁺		G	
8273.3 4	(1/2 ⁻)		G K	XREF: K(8270) J ^π : other: L=1 from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$.
8298.27 30	3/2 ⁺ , 5/2 ⁺		G	
8333.8 4	3/2 ⁻		G	
8335.9 4	3/2 ⁺ , 5/2 ⁺		G	
8391.3 4	3/2 ⁻		G	
8393.64 30	3/2 ⁺ , 5/2 ⁺		G	
8406.55 30	3/2 ⁺ , 5/2 ⁺		G k	XREF: k(8410)
8418.8 4	3/2 ⁺ , 5/2 ⁺		G kL	XREF: k(8410)L(8430) J ^π : others: L=(2) from 0 ⁺ in $^{36}\text{S}(\text{p}, \text{d})$; L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
8465 7	(3/2 ⁺ , 5/2 ⁺)		K	
8509 5	3/2 ⁺ , 5/2 ⁺		K	
8557 7	(5/2 ⁻ , 7/2 ⁻)		K	
8602 5	(5/2 ⁻ , 7/2 ⁻)		K	
8649 7	(5/2 ⁻ , 7/2 ⁻)		K	
8707 5	(3/2 ⁺ , 5/2 ⁺)		K	
8755.9 11	(17/2)	<1 ps	E	J ^π : 1576.3γ D, ΔJ=1 to (15/2 ⁺), 7179.74. T _{1/2} : in ($^{18}\text{O}, 2\alpha\text{p}\gamma$) from 2021Go09 with residual Doppler energy shifts observed for the 732γ and 1576γ peaks.
8761 7	(3/2 ⁺ , 5/2 ⁺)		K	
8822 5	(1/2 ⁻ , 3/2 ⁻)		K	
8876 5	(1/2 ⁻ , 3/2 ⁻)		K	
8947 7	(1/2 ⁻ , 3/2 ⁻)		K	
9032 15	(1/2 ⁻ , 3/2 ⁻)		K	
9065 15	(1/2 ⁻ , 3/2 ⁻)		K	
9126 7	1/2 ⁺		K1	T=5/2 XREF: l(9155) J ^π , T: isobaric analog state of 1/2 ⁺ , T=5/2, g.s. of ^{35}P (2026Jo01). XREF: K(9180)l(9155)
9180 5			K1	J ^π : L=2 from 3/2 ⁺ in (p, ^3He) suggests (1/2 ⁺ to 9/2 ⁺).
9219 15	(1/2 ⁻ , 3/2 ⁻)		K	
9294 8	(1/2 ⁻ , 3/2 ⁻)		K	
9339 8	(1/2 ⁻ , 3/2 ⁻)		K	

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued) ^{35}S Levels (continued)

E(level) [†]	J ^π [‡]	XREF	Comments
9395 5	(1/2 ⁻ ,3/2 ⁻)	K	
9446 7	(1/2 ⁻ ,3/2 ⁻)	K	
9510 10	(1/2 ⁻ ,3/2 ⁻)	K	
9565 20	(1/2 ⁻ ,3/2 ⁻)	K	
9615 10	(1/2 ⁻ ,3/2 ⁻)	K	
9666 10	(1/2 ⁻ ,3/2 ⁻)	K	
9716 10	(1/2 ⁻ ,3/2 ⁻)	K	
9774 10	(1/2 ⁻ ,3/2 ⁻)	K	
9849 10	(1/2 ⁻ ,3/2 ⁻)	K	
9911 10	(1/2 ⁻ ,3/2 ⁻)	K	
9989 10	(1/2 ⁻ ,3/2 ⁻)	K	
10048.6 9	(19/2 ⁺)	E	J ^π : 2868.7γ Q, ΔJ=2 to (15/2 ⁺), 7179.74.
1.0049×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0166×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0225×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0277×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0350×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0421×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0493×10 ⁴ 10	(5/2 ⁻ ,7/2 ⁻)	K	
1.0563×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0663×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0746×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.0810×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	K	
1.147×10 ⁴ 2	(3/2 ⁺)	K	T=5/2
12469.3 14	(21/2)	E	J ^π ,T: IAS of 3/2 ⁺ , T=5/2, 2387 level in ^{35}P (2026Jo01).
1.288×10 ⁴ 10	(5/2 ⁺)	K	J ^π : 2420.6γ (D), ΔJ=(1) to (19/2 ⁺), 10048.6.
			T=5/2
1.368×10 ⁴ 10	(5/2 ⁺)	K	J ^π ,T: IAS of 5/2 ⁺ , T=5/2, 3860 level in ^{35}P (2026Jo01).
			T=5/2
1.422×10 ⁴ 10	(5/2 ⁺)	K	J ^π ,T: IAS of 5/2 ⁺ , T=5/2, 4666 level in ^{35}P (2026Jo01).
			T=5/2
1.470×10 ⁴ 10		K	J ^π ,T: IAS of 5/2 ⁺ , T=5/2, 5199 level in ^{35}P (2026Jo01).

[†] From a least-squares fit to γ-ray energies with uncertainties for levels connected with γ transitions; from transfer reactions or $^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances for other levels, unless otherwise noted.

[‡] For high-spin states, it is assumed that spin increases as the excitation energy increases. For resonances (7018-8418 keV), J^π is from $^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances, unless otherwise noted; for resonances above 8418 keV, J^π is from $^{36}\text{S}(\text{p,d})$, unless otherwise noted.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	Comments
1572.369	1/2 ⁺	1572.327 11	100	0.0	3/2 ⁺	(M1)		0.0001007 14	B(M1)(W.u.)=0.00248 16 E _γ : weighted average of 1572.252 29 from ³⁵ P β ⁻ decay, 1572.4 3 from (¹⁴ N,3pγ), 1572.333 8 from (n,γ) E=thermal, 1572.24 17 from ³⁴ S(d,pγ), and 1572 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
1991.27	7/2 ⁻	1991.26 5	100	0.0	3/2 ⁺	M2+E3	+0.17 6	0.0001476 22	Mult.: D from (¹⁴ N,3pγ); Δπ=no from level scheme. B(M2)(W.u.)=0.0874 +46-48; B(E3)(W.u.)=3.7 +29-21 E _γ : weighted average of 1991.3 2 from (¹⁴ N,3pγ), 1990.5 4 from (¹⁸ O,2αnγ), 1991.27 5 from (n,γ) E=thermal, and 1991 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
2347.779	3/2 ⁻	356.66 @ 9 775.398 6	<0.084 35.3 27	1991.27 1572.369	7/2 ⁻ 1/2 ⁺	[E2] (E1)		1.15×10 ⁻³ 2 3.54×10 ⁻⁵ 5	Mult.,δ: from ³⁴ S(d,pγ). D+Q or Q+O from γ(θ) in ³⁴ S(d,pγ); E2+M3 is ruled out by RUL; D+Q is inconsistent with level scheme from 7/2 ⁻ to 3/2 ⁺ . E _γ ,I _γ : from (n,γ) E=thermal. B(E1)(W.u.)=4.5×10 ⁻⁴ +9-7 E _γ : from (n,γ) E=thermal. Other: 775.4 4 from (¹⁴ N,3pγ). I _γ : weighted average of 37 5 from (¹⁴ N,3pγ), 35 4 from (n,γ) E=thermal, 37.0 28 from ³⁴ S(d,pγ), and 33.3 27 from (d,αγ).
		2347.700 19	100.0 27	0.0	3/2 ⁺	(E1)		0.000880 12	Mult.: D(+Q) from γ(θ) in ³⁴ S(d,pγ); Δπ=yes from level scheme; M2 ruled out by RUL. B(E1)(W.u.)=4.6×10 ⁻⁵ +8-6 E _γ : weighted average of 2347.8 3 from (¹⁴ N,3pγ), 2347.700 15 from (n,γ) E=thermal, and 2347 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
2717.06	5/2 ⁺	368.7 5	14 8	2347.779	3/2 ⁻	[E1]		0.0002294 33	I _γ : from (d,αγ). Others: 100 8 from (¹⁴ N,3pγ), 100 8 from (n,γ) E=thermal, and 100.0 28 from ³⁴ S(d,pγ). Mult.: D(+Q) from γ(θ) in ³⁴ S(d,pγ); Δπ=yes from level scheme; M2 component is less likely based on RUL. B(E1)(W.u.)=0.022 +20-10 E _γ : weighted average of 368.5 4 from (n,γ) E=thermal and 370 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).
		725.78 1144.67	<5 <5	1991.27 1572.369	7/2 ⁻ 1/2 ⁺	[E1] [E2]		4.10×10 ⁻⁵ 6 3.79×10 ⁻⁵ 5	I _γ : unweighted average of 6.7 23 from (n,γ) E=thermal and 22 4 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	Comments
2717.06	5/2 ⁺	2716.99 15	100 10	0.0	3/2 ⁺	M1+E2	-0.39 5	0.000568 9	B(M1)(W.u.)=0.012 +7-3; B(E2)(W.u.)=0.9 +6-3 E _γ : weighted average of 2717.0 4 from (¹⁴ N,3pγ), 2716.99 16 from (n,γ) E=thermal, and 2717 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). Other: 100 17 from (n,γ) E=thermal. Mult.,δ: D+Q from γ(θ) in ³⁴ S(d,pγ) with δ=-0.39 5 or -15.0 34 for J _i =5/2 (1975VaYG); -15.0 34 is unlikely based on RUL; E1+M2 is ruled out by RUL.
2938.63	3/2 ⁺	590.85 947.35 1366.23 2938.56 11	<10 <20 <15 100	2347.779 1991.27 1572.369 0.0	3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺				E _γ : weighted average of 2938.29 40 from ³⁵ P β ⁻ decay and 2938.58 11 from (n,γ) E=thermal.
3421.2	5/2 ⁺	482.6 704.1 1073.4 1429.9 1848.8 3421 2	<4 <6 <7 <30 <7 100	2938.63 2717.06 2347.779 1991.27 1572.369 0.0	3/2 ⁺ 5/2 ⁺ 3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺	[M1,E2] [M1,E2] [E1] [E1] [E2] [M1,E2]		2.7×10 ⁻⁴ 13 9.5×10 ⁻⁵ 28 1.836×10 ⁻⁵ 26 0.0002243 31 0.000254 4 0.00090 7	E _γ : from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from (n,γ) E=thermal. E _γ : from (n,γ) E=thermal. Other: 1211 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : weighted average of 54 6 from (n,γ) E=thermal and 54 6 from (d,αγ). Other: 18 7 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ : from (n,γ) E=thermal. Other: 1567 2 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). I _γ : from (d,αγ). Others: 100 19 from (n,γ) E=thermal and 100 10 from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from ²⁰⁸ Pb(³⁶ S, ³⁵ Sγ). E _γ ,I _γ : from (n,γ) E=thermal.
3558.080	(5/2 ⁺)	619.23 19 1210.28 4	8.9 17 54 6	2938.63 2347.779	3/2 ⁺ 3/2 ⁻				E _γ ,I _γ : from (¹⁴ N,3pγ). Other: I _γ <11 from ³⁴ S(d,pγ).
		1566.7 3	100 6	1991.27	7/2 ⁻				E _γ : weighted average of 3594.7 7 from (¹⁴ N,3pγ), 3594.4 11 from (¹⁸ O,2αγ), and 3593 2 from
3594.62	(7/2 ⁺)	1986 & 2 3558.1 5 655.98 877.6 5	39 6 18 4 <10 9.3 23	1572.369 0.0 2938.63 2717.06	1/2 ⁺ 3/2 ⁺ 3/2 ⁺ 5/2 ⁺				
		1246.82 1603.31 3594.5 7	<15 <17 100 9	2347.779 1991.27 0.0	3/2 ⁻ 7/2 ⁻ 3/2 ⁺	Q			

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{S})$ (continued)</u>									
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>$\delta^\ddagger$</u>	<u>$\alpha^\#$</u>	<u>Comments</u>
3801.952	3/2 ⁻	863.28 28	1.3 4	2938.63	3/2 ⁺	[E1]		2.82×10 ⁻⁵ 4	²⁰⁸ Pb(³⁶ S, ³⁵ S γ). I γ : from (¹⁴ N,3p γ). Mult.: from $\gamma\gamma(\theta)$ (ADO) in (¹⁴ N,3p γ). B(E1)(W.u.)=2.4×10 ⁻⁴ +30-12 E γ , I γ : from (n, γ) E=thermal. Other: I γ <8.9 from ³⁴ S(d,p γ).
		1084.79 15	2.2 7	2717.06	5/2 ⁺	[E1]		1.801×10 ⁻⁵ 25	B(E1)(W.u.)=2.0×10 ⁻⁴ +28-11 E γ : from (n, γ) E=thermal. I γ : weighted average of 2.1 4 from (n, γ) E=thermal and 11 5 from ³⁴ S(d,p γ).
		1454.09 4	18 5	2347.779	3/2 ⁻	[M1,E2]		8.0×10 ⁻⁵ 12	E γ : from (n, γ) E=thermal. I γ : weighted average of 18 4 from (n, γ) E=thermal and 18 7 from ³⁴ S(d,p γ).
		1810.63	<8.9	1991.27	7/2 ⁻	[E2]		0.0002364 33	B(M1)(W.u.)=0.024 +32-12 if M1, B(E2)(W.u.)=4×10 ¹ +6-2 if E2.
		2229.510 16	88 15	1572.369	1/2 ⁺	[E1]		0.000804 11	B(E1)(W.u.)=9×10 ⁻⁴ +11-4 E γ : from (n, γ) E=thermal. I γ : unweighted average of 110 11 from (n, γ) E=thermal, 93 9 from ³⁴ S(d,p γ), and 61 5 from (d, $\alpha\gamma$).
3816.01	(9/2) ⁻	3801.73 4	100 5	0.0	3/2 ⁺	[E1]		1.60×10 ⁻³ 2	B(E1)(W.u.)=2.2×10 ⁻⁴ +27-10 E γ : from (n, γ) E=thermal. I γ : from (d, $\alpha\gamma$). Other: 100 9 from ³⁴ S(d,p γ), 100 9 from (n, γ) E=thermal.
		1824.7 3	100	1991.27	7/2 ⁻	M1+E2	+0.55 9	0.000200 4	B(M1)(W.u.)=0.0099 +14-12; B(E2)(W.u.)=3.4 +10-9 E γ : weighted average of 1824.6 2 from (¹⁴ N,3p γ), 1824.4 6 from (¹⁸ O,2 $\alpha\gamma$), and 1827 1 from ²⁰⁸ Pb(³⁶ S, ³⁵ S γ).
3886.28	(5/2)								Mult., δ : D+Q and δ from $\gamma(\theta)$ and $\gamma\gamma(\theta)$ (ADO) in (¹⁴ N,3p γ); E1+M2 ruled out by RUL.
		465	<30	3421.2	5/2 ⁺				
		947.64	<12	2938.63	3/2 ⁺				
		1169.20	<15	2717.06	5/2 ⁺				
		1538.4 8	57 33	2347.779	3/2 ⁻				E γ : from (¹⁴ N,3p γ). I γ : unweighted average of 24 10 from (¹⁴ N,3p γ) and 89 9 from (d, $\alpha\gamma$). Other: <18 from ³⁴ S(d,p γ).
		1894.9 4	100 11	1991.27	7/2 ⁻	D			E γ , Mult.: from (¹⁴ N,3p γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\delta^\ddagger$	$\alpha^\#$	Comments
									I_γ : from (d, $\alpha\gamma$). Other: 100 14 from (^{14}N ,3p γ) and 100 from ^{34}S (d,p γ).
3886.28	(5/2)	2313.83	<16	1572.369	1/2 ⁺				
		3886	33 7	0.0	3/2 ⁺				
4023.24	(11/2) ⁻	207.0 5	0.59 30	3816.01	(9/2) ⁻	[M1+E2]	<0.25	0.00124 24	I_γ : from (d, $\alpha\gamma$). Other: <16 from ^{34}S (d,p γ). B(M1)(W.u.)=0.05 +4-3 if M1. δ : calculated by evaluators from the measured half-life of the level and RUL=100 for B(E2)(W.u.).
		2032.0 3	100 6	1991.27	7/2 ⁻	E2		0.000341 5	B(E2)(W.u.)=7.5 +8-7 E_γ : weighted average of 2031.8 3 from (^{14}N ,3p γ) and 2032.4 4 from (^{18}O ,2 α n γ). Other: 2032 2 from ^{208}Pb (^{36}S , ^{35}S γ). Mult.: Q from $\gamma\gamma(\theta)$ (ADO) in (^{14}N ,3p γ) and M2 ruled out by RUL.
4027.2	(3/2 ⁺ ,5/2 ⁺)	469.1	<14	3558.080	(5/2 ⁺)				
		606.0	<28	3421.2	5/2 ⁺				
		1088.6	95 12	2938.63	3/2 ⁺				I_γ : weighted average of 86 22 from ^{34}S (d,p γ) and 97 12 from (d, $\alpha\gamma$).
		1310.1	<22	2717.06	5/2 ⁺				
		1679.4	97 12	2347.779	3/2 ⁻				I_γ : from (d, $\alpha\gamma$). Other: 92 33 from ^{34}S (d,p γ).
		2035.9	<28	1991.27	7/2 ⁻				
		2454.7	100 18	1572.369	1/2 ⁺				I_γ : from (d, $\alpha\gamma$). Other: 100 33 from ^{34}S (d,p γ).
		4027.0	<36	0.0	3/2 ⁺				
4105.6	(5/2 ⁺)	547.5	<4.6	3558.080	(5/2 ⁺)				
		684.4	<4.6	3421.2	5/2 ⁺				
		1167.0	15 7	2938.63	3/2 ⁺				
		1388.5	<12	2717.06	5/2 ⁺				
		1757.8	<5.8	2347.779	3/2 ⁻				
		2114.3	<8.1	1991.27	7/2 ⁻				
		2533.1	<5.8	1572.369	1/2 ⁺				
		4105.3 8	100 7	0.0	3/2 ⁺				E_γ : from (n, γ) E=thermal.
4180	(1/2,3/2,5/2 ⁺)	2608	22 6	1572.369	1/2 ⁺				I_γ : from (d, $\alpha\gamma$).
		4180	100 6	0.0	3/2 ⁺				I_γ : from (d, $\alpha\gamma$).
4189.240	1/2 ⁻	631.32 @ 24	<2.8	3558.080	(5/2 ⁺)				E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <7.7 from ^{34}S (d,p γ).
		1250.61 5	8.4 10	2938.63	3/2 ⁺	[E1]		0.0001047 15	E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <9.6 from ^{34}S (d,p γ).
		1840.2 12	91 10	2347.779	3/2 ⁻	[M1,E2]		0.000221 29	E_γ : from (n, γ) E=thermal. I_γ : weighted average of 87 16 from (n, γ) E=thermal and 92 10 from ^{34}S (d,p γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
4189.240	1/2 ⁻	2616.8 13	9.5 28	1572.369	1/2 ⁺	[E1]	1.04×10 ⁻³ 2	E_γ, I_γ : from (n, γ) E=thermal. Other: I_γ <12 from $^{34}\text{S}(\text{d},\text{p}\gamma)$.
		4188.95 5	100 10	0.0	3/2 ⁺	[E1]	1.75×10 ⁻³ 3	E_γ : from (n, γ) E=thermal. I_γ : other: 100 11 from (n, γ) E=thermal.
4302.6	(1/2 ⁻ , 3/2 ⁻)	1954.8	100 9	2347.779	3/2 ⁻			I_γ : from (d, $\alpha\gamma$).
		4302.3	70 9	0.0	3/2 ⁺			I_γ : from (d, $\alpha\gamma$).
4477.63	(5/2 ⁺)	591.35	<8.2	3886.28	(5/2)		1.5×10 ⁻⁴ 6	
		675.67	<8.2	3801.952	3/2 ⁻	[E1]	4.82×10 ⁻⁵ 7	
		883.00	<8.2	3594.62	(7/2 ⁺)	[M1,E2]	5.4×10 ⁻⁵ 12	
		919.54	<8.2	3558.080	(5/2 ⁺)	[M1,E2]	4.9×10 ⁻⁵ 10	
		1056.4	<16	3421.2	5/2 ⁺	[M1,E2]	3.6×10 ⁻⁵ 6	
		1538.96	<12	2938.63	3/2 ⁺	[M1,E2]	0.000106 16	
		1760.55 11	100 12	2717.06	5/2 ⁺	[M1,E2]	0.000189 25	E_γ : from (n, γ) E=thermal. I_γ : others: 100 15 from (n, γ) E=thermal and 100 11 from (d, $\alpha\gamma$).
		2129.78	<13	2347.779	3/2 ⁻	[E1]	0.000738 10	
		2486.27	<16	1991.27	7/2 ⁻	[E1]	0.000962 13	I_γ : other: 42 11 from (d, $\alpha\gamma$).
		2905.1 4	90 14	1572.369	1/2 ⁺	[E2]	0.000749 10	E_γ : from (n, γ) E=thermal. I_γ : unweighted average of 96 28 from (n, γ) E=thermal, 64 12 from $^{34}\text{S}(\text{d},\text{p}\gamma)$, and 109 11 from (d, $\alpha\gamma$).
4822.79	(9/2 ⁺)	4477	<20	0.0	3/2 ⁺	[M1,E2]	0.00126 8	I_γ : other: 28 6 from (d, $\alpha\gamma$).
		1006.8 2	66 15	3816.01	(9/2) ⁻	D		E_γ : from ($^{14}\text{N}, 3\text{p}\gamma$). Other: 1007.2 11 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$). I_γ : weighted average of 73 9 from ($^{14}\text{N}, 3\text{p}\gamma$) and 36 18 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$).
								Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ($^{14}\text{N}, 3\text{p}\gamma$) and ($^{18}\text{O}, 2\alpha\text{n}\gamma$), consistent with $\Delta J=0$.
		1228.1 3	100 9	3594.62	(7/2 ⁺)	D		E_γ, I_γ : other: 1228.3 6 with $I_\gamma=100$ 18 from ($^{18}\text{O}, 2\alpha\text{n}\gamma$). Mult.: from ($^{14}\text{N}, 3\text{p}\gamma$).
4899.82	(9/2 ⁺)	2831.4 9	36 9	1991.27	7/2 ⁻			
		1305.3 4	55 18	3594.62	(7/2 ⁺)			E_γ : from ($^{18}\text{O}, 2\alpha\text{n}\gamma$). Other: 1304.8 12 from ($^{14}\text{N}, 3\text{p}\gamma$).
		2908.3 5	100 18	1991.27	7/2 ⁻	D		Mult.: from $\gamma\gamma(\theta)(\text{ADO})$ in ($^{14}\text{N}, 3\text{p}\gamma$).
4903.371	1/2 ⁺	1101.92 31	0.43 11	3801.952	3/2 ⁻			
		1964.8 2	4.9 13	2938.63	3/2 ⁺			
		2555.492 14	41 4	2347.779	3/2 ⁻			
		3330.84 13	100 9	1572.369	1/2 ⁺			
		4902.99 5	53 5	0.0	3/2 ⁺			
4963.082	5/2 ⁺	1161.05 20	0.98 14	3801.952	3/2 ⁻			
		1404.967 24	10.7 11	3558.080	(5/2 ⁺)			
		2615.2 2	19.5 18	2347.779	3/2 ⁻			
		2972.0 4	3.2 11	1991.27	7/2 ⁻			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. ‡	$\alpha^\#$	Comments
4963.082	$5/2^+$	3390.62 7 4963.06 22	100 9 47 5	1572.369 0.0	$1/2^+$ $3/2^+$			
5010.05	$(11/2)^-$	986.8 3	100 5	4023.24	$(11/2)^-$	(M1)	3.45×10^{-5} 5	B(M1)(W.u.)=0.036 +8-6 E_γ : weighted average of 986.9 3 from ($^{14}\text{N}, 3\text{p}\gamma$) and 986.6 3 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: D from ($^{18}\text{O}, 2\alpha\text{p}\gamma$) and ($^{14}\text{N}, 3\text{p}\gamma$); $\Delta\pi$ =no from level scheme.
		3018.5 7	43 5	1991.27	$7/2^-$	E2	0.000798 11	B(E2)(W.u.)=0.22 +5-4 Mult.: Q from $\gamma\gamma(\theta)$ (ADO) and M2 ruled out by RUL in ($^{14}\text{N}, 3\text{p}\gamma$).
5343	$(-)$	5343		0.0	$3/2^+$			
5412.28	$(9/2^+)$	1389.0 4	100 10	4023.24	$(11/2)^-$	D		E_γ : other: E_γ =1389.0 8 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: from $\gamma\gamma(\theta)$ (ADO) in ($^{14}\text{N}, 3\text{p}\gamma$).
		1525.7 8 3420.6 9	15.4 26 28.2 26	3886.28 1991.27	$(5/2)$ $7/2^-$			
5752.5	$5/2^-, 7/2^-$	5752.0 8	100	0.0	$3/2^+$			
5767	$5/2^+$	5767		0.0	$3/2^+$			
5841.482	$(3/2^+, 5/2^+)$	4268.65 14	100	1572.369	$1/2^+$			E_γ, I_γ : from (n, γ) E=thermal.
5877.70	$(11/2^+)$	465.4 2	38 6	5412.28	$(9/2^+)$	D		E_γ : weighted average of 465.3 3 from ($^{14}\text{N}, 3\text{p}\gamma$) and 465.5 2 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). I_γ : unweighted average of 31.6 26 from ($^{14}\text{N}, 3\text{p}\gamma$) and 43.8 32 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\text{p}\gamma$).
		867.6 4	50.0 26	5010.05	$(11/2)^-$	D		E_γ : weighted average of 867.3 3 from ($^{14}\text{N}, 3\text{p}\gamma$) and 868.2 4 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). I_γ : other: 62 7 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\text{p}\gamma$).
		977.9 3	43 5	4899.82	$(9/2^+)$	D+Q		E_γ : weighted average of 977.8 3 from ($^{14}\text{N}, 3\text{p}\gamma$) and 978.4 10 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). I_γ : weighted average of 42.1 26 from ($^{14}\text{N}, 3\text{p}\gamma$) and 82 22 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: from ($^{14}\text{N}, 3\text{p}\gamma$).
		1055.1 6	80 7	4822.79	$(9/2^+)$	D		E_γ : unweighted average of 1054.5 3 from ($^{14}\text{N}, 3\text{p}\gamma$) and 1055.6 4 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). I_γ : weighted average of 79 5 from ($^{14}\text{N}, 3\text{p}\gamma$) and 122 32 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$). Mult.: from ($^{18}\text{O}, 2\alpha\text{p}\gamma$).
		1854.4 4	44.7 26	4023.24	$(11/2)^-$			E_γ, I_γ : other: 1854.5 42 with I_γ =88 44 from ($^{18}\text{O}, 2\alpha\text{p}\gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
5877.70	(11/2 ⁺)	2061.6 4	100 5	3816.01	(9/2) ⁻	D		E_γ : weighted average of 2061.5 3 from (¹⁴ N,3p γ) and 2063.3 14 from (¹⁸ O,2 α n γ). I_γ : other: 100 20 from (¹⁸ O,2 α n γ). Mult.: from (¹⁸ O,2 α n γ). E_γ : weighted average of 2282.9 4 from (¹⁴ N,3p γ) and 2283.5 10 from (¹⁸ O,2 α n γ). I_γ : unweighted average of 68.4 26 from (¹⁴ N,3p γ) and 44 10 from (¹⁸ O,2 α n γ).
		2283.0 4	56 12	3594.62	(7/2 ⁺)			
6018.8		3886.0 13	7.9 26	1991.27	7/2 ⁻			
6078.475	1/2 ⁻ ,3/2 ⁻	6018.2 6	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
		3139.9 5	22 7	2938.63	3/2 ⁺			
		6077.87 11	100 12	0.0	3/2 ⁺			
6293.93		6293.2 4	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
6299.4	(11/2)	887.0 7	40 10	5412.28	(9/2 ⁺)	D		
		2275.9 6	100 20	4023.24	(11/2) ⁻			
		2483.6 8	90 20	3816.01	(9/2) ⁻			
6352.2	(13/2) ⁻	2329.5 9	100 20	4023.24	(11/2) ⁻	(M1)	0.000392 5	B(M1)(W.u.)=0.023 +6-5 Mult.: D from $\gamma\gamma(\theta)$ (ADO) in (¹⁴ N,3p γ); $\Delta\pi$ =no from level scheme.
		2535.8 11	50 6	3816.01	(9/2) ⁻	E2	0.000581 8	B(E2)(W.u.)=5.3 +17-11 Mult.: Q from $\gamma\gamma(\theta)$ (ADO) in (¹⁴ N,3p γ) and M2 ruled out by RUL.
6354.89		6355.0 6	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
6419.9		6419.3 11	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
6629.43		6628.5 6	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
6761.0		6760.3 12	100	0.0	3/2 ⁺			E_γ, I_γ : from (n, γ) E=thermal.
(6986.097)	1/2 ⁺	356.66 @ 9	<0.073	6629.43				
		631.32 @ 24	<0.123	6354.89				
		692.16 5	0.238 31	6293.93				
		907.608 23	1.00 18	6078.475	1/2 ⁻ ,3/2 ⁻			
		1144.591 20	0.98 11	5841.482	(3/2 ⁺ ,5/2 ⁺)			
		2022.954 9	19.6 18	4963.082	5/2 ⁺			
		2082.83 17	28.4 27	4903.371	1/2 ⁺			
		2508.39 8	0.68 7	4477.63	(5/2 ⁺)			
		2796.74 4	8.8 7	4189.240	1/2 ⁻			
		3183.95 4	10.9 11	3801.952	3/2 ⁻			
		4638.14 14	100 9	2347.779	3/2 ⁻			
		6985.7 10	0.064 14	0.0	3/2 ⁺			
7179.74	(15/2 ⁺)	827.8 6	2.2 8	6352.2	(13/2) ⁻	[E1]	3.08×10 ⁻⁵ 4	B(E1)(W.u.)=7×10 ⁻⁶ +6-3
		880.2 4	5.9 8	6299.4	(11/2)			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]	$\alpha^\#$	Comments
7179.74	(15/2 ⁺)	1302.0 3	100 6	5877.70	(11/2 ⁺)	E2	5.56×10^{-5} 8	B(E2)(W.u.)=6.6 +41-19 E _γ : weighted average of 1302.2 2 from (¹⁴ N,3pγ) and 1301.6 3 from (¹⁸ O,2αnγ). Mult.: Q from γγ(θ)(ADO) in (¹⁴ N,3pγ) and M2 ruled out by RUL.
8024.0	(17/2 ⁺)	844.2 6	100	7179.74	(15/2 ⁺)	(M1)	4.66×10^{-5} 7	B(M1)(W.u.)=0.24 +9-5 E _γ : weighted average of 843.9 8 from (¹⁴ N,3pγ) and 844.3 6 from (¹⁸ O,2αnγ). Mult.: D from γγ(ADO) in (¹⁸ O,2αnγ); Δπ=(no) from level scheme.
8755.9	(17/2)	731.6 15	57 22	8024.0	(17/2 ⁺)			E _γ ,I _γ : from (¹⁸ O,2αnγ).
		1576.3 14	100 22	7179.74	(15/2 ⁺)	D		E _γ ,I _γ : from (¹⁸ O,2αnγ).
10048.6	(19/2 ⁺)	2868.7 8	100	7179.74	(15/2 ⁺)	Q		E _γ ,I _γ : from (¹⁸ O,2αnγ).
12469.3	(21/2)	2420.6 11	100	10048.6	(19/2 ⁺)	(D)		E _γ ,I _γ : from (¹⁸ O,2αnγ).

[†] For γ rays from low-spin levels (J<9/2), values are from ³⁴S(d,pγ) up to 4478 level and from (n,γ) E=thermal above that, for γ rays from high-spin level (J≥9/2), values are from ²⁴Mg(¹⁴N,3pγ), unless otherwise noted. E_γ values without uncertainties are deduced from level-energy differences.

[‡] Mult and δ are primarily from γ(θ) and γγ(ADO) in ³⁴S(d,pγ), ²⁴Mg(¹⁴N,3pγ), and ²⁶Mg(¹⁸O,2αnγ). The magnetic or electric nature is determined based on RUL and measured T_{1/2} where available, unless otherwise noted.

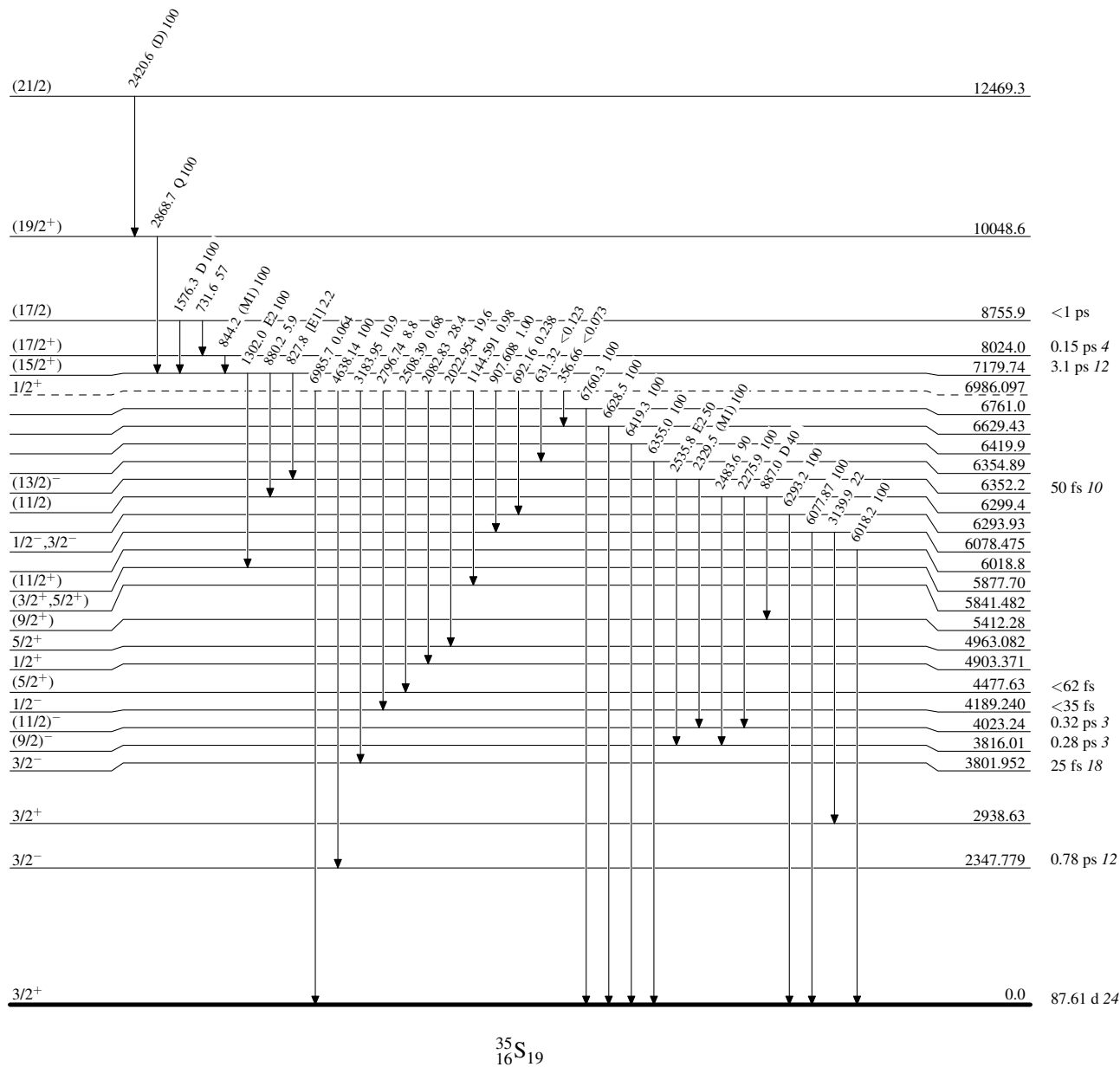
Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

@ Multiply placed.

& Placement of transition in the level scheme is uncertain.

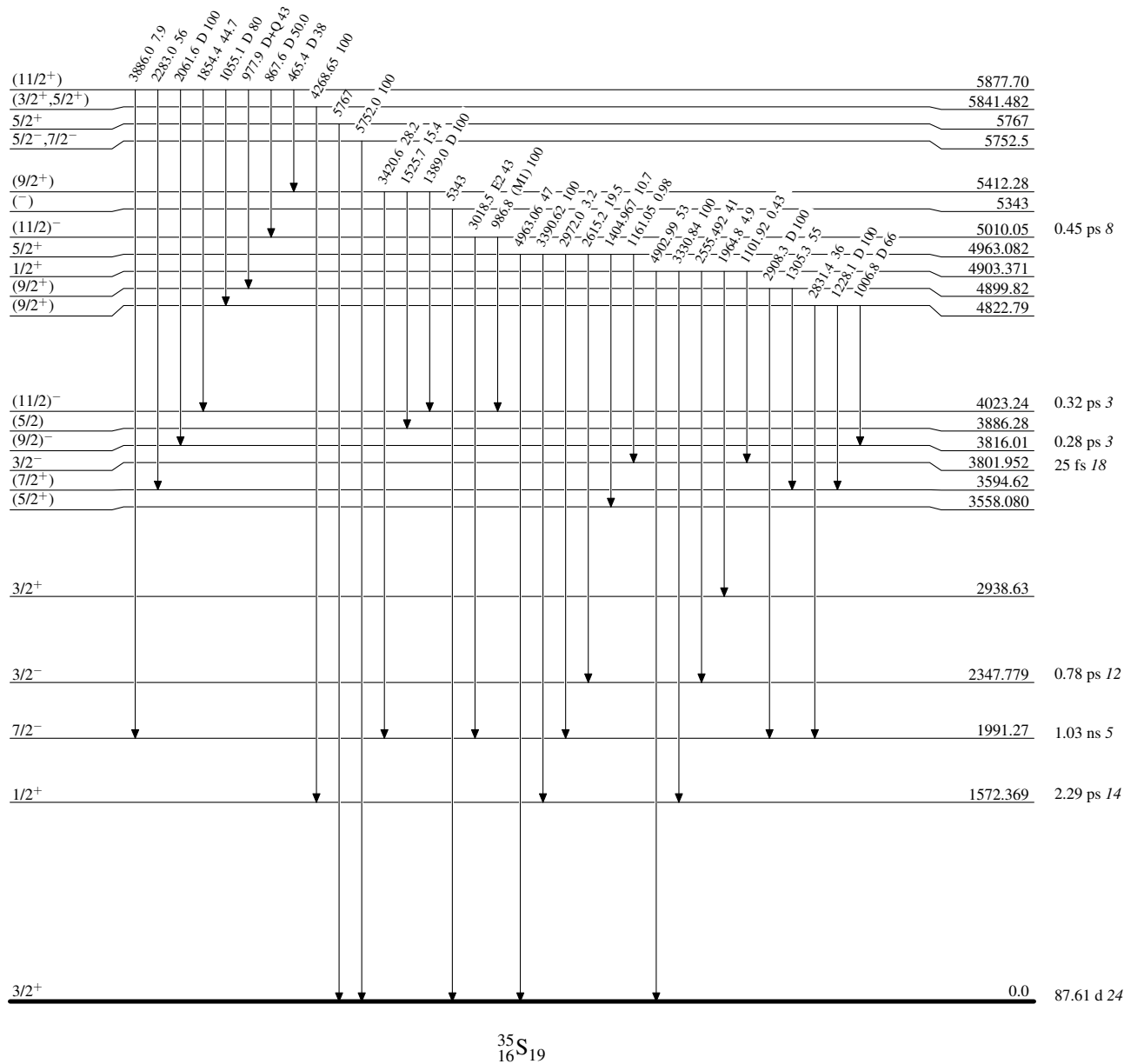
Adopted Levels, GammasLevel Scheme

Intensities: Relative photon branching from each level

 $^{35}_{16}\text{S}_{19}$

Adopted Levels, GammasLevel Scheme (continued)

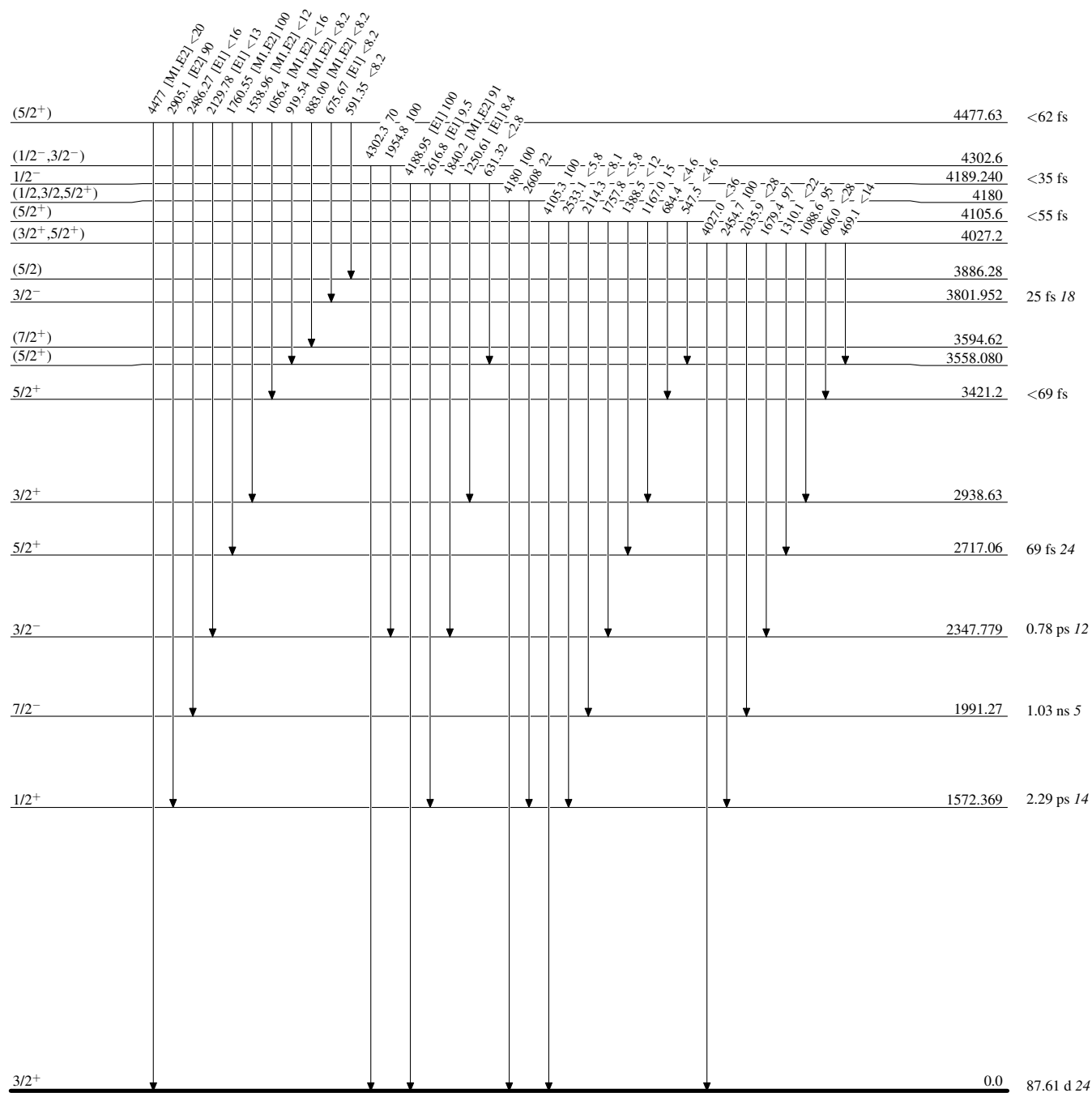
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



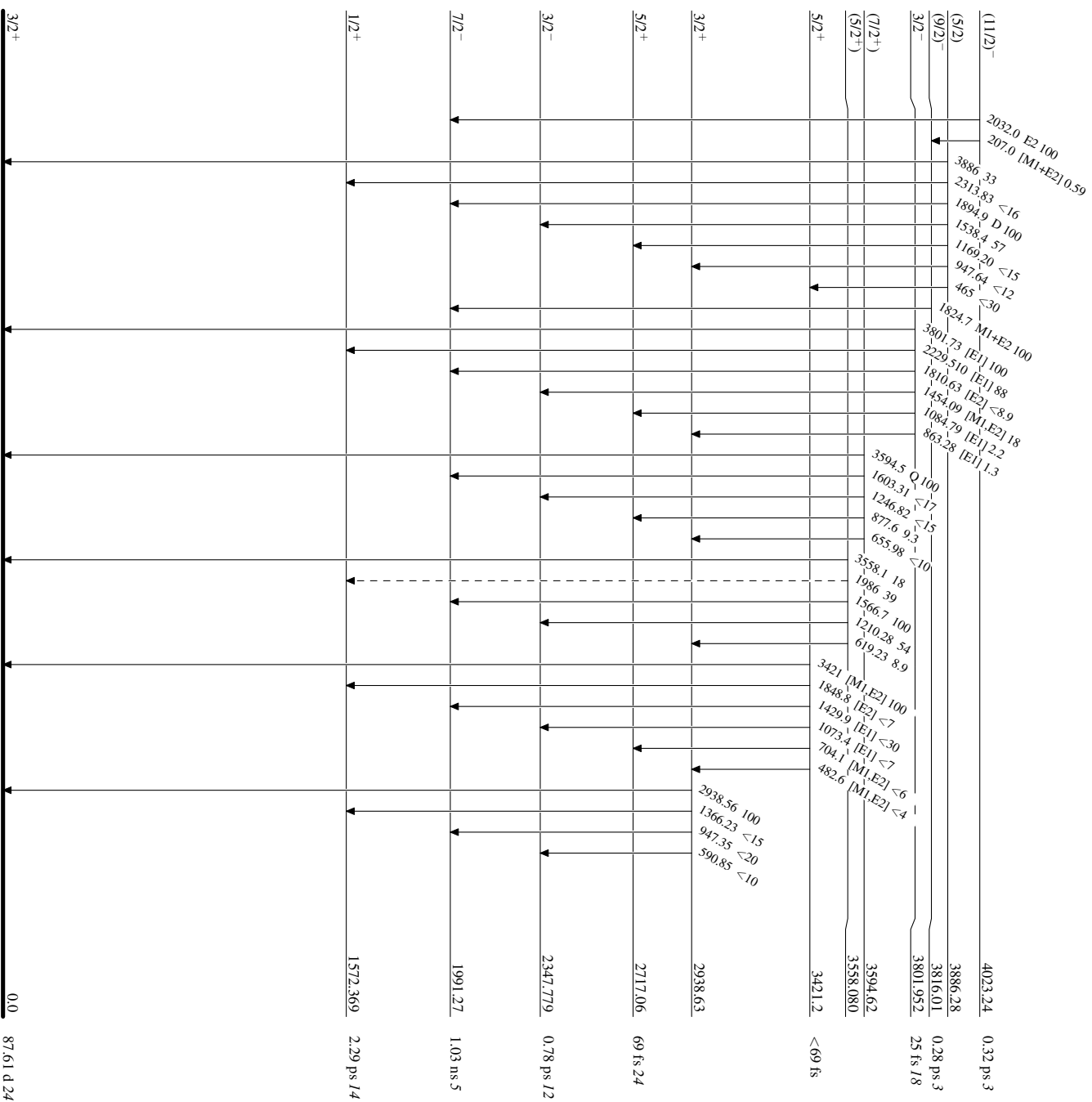
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

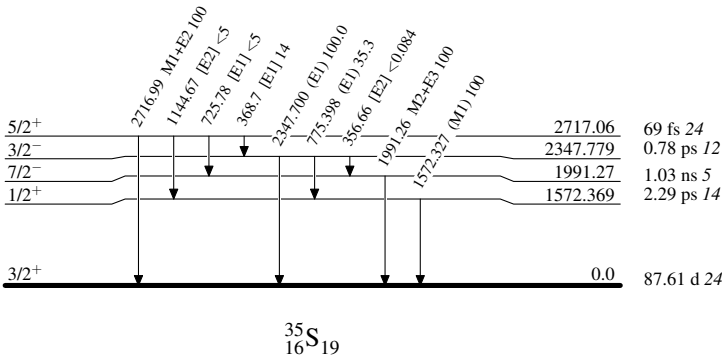
-----▶ γ Decay (Uncertain)



Adopted Levels, Gammas

Level Scheme (continued)

Intensities: Relative photon branching from each level



^{35}P β^- decay (47.3 s) 1986Wa22

Parent: ^{35}P : $E=0$; $J^\pi=1/2^+$; $T_{1/2}=47.3$ s 8; $Q(\beta^-)=3988.4$ 19; $\% \beta^-$ decay=100

^{35}P - J^π , $T_{1/2}$: From Adopted Levels of ^{35}P .

^{35}P - $Q(\beta^-)$: From 2021Wa16.

1986Wa22: ^{35}P was produced via the $^{36}\text{S}(t,\alpha)^{35}\text{P}$ reaction. 81% enriched ^{36}S targets were bombarded with 3.4-MeV tritons from the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. γ rays were detected using an intrinsic coaxial Ge detector. Measured E_γ , I_γ . Deduced levels, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1972Ap01: ^{35}P was produced via the $^{37}\text{Cl}(t,\alpha p)^{35}\text{P}$ reaction. LiCl and NaCl targets were bombarded with 16-MeV tritons from the Los Alamos tandem Van de Graaff facility. γ rays were detected using an 85 cm³ Ge(Li) detector. β particles were detected using an end-window gas-flow proportional counter. Measured E_γ , I_γ , and I_β . Deduced ^{35}P $T_{1/2}$, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1972Go31: ^{35}P was produced via the $^{18}\text{O}(^{19}\text{F},2p)^{35}\text{P}$ and $^{36}\text{S}(t,\alpha)^{35}\text{P}$ reactions at the second tandem of the Brookhaven National Laboratory (BNL) tandem van de Graaff facility. ^{35}P β -delayed γ rays were detected using a 60 cm³ Ge(Li) detector. β particles were detected in coincidence with the 1572 γ using a NE102 scintillator. Measured E_γ , I_γ , and E_β . Deduced ^{35}P $T_{1/2}$, β end-point energy, mass, decay branching ratios, and $\log ft$. Compared with shell model calculations.

1971Gr53: ^{35}P was produced via the $^{37}\text{Cl}(\gamma,2p)^{35}\text{P}$ reaction. 6-g $\text{NH}_4\text{Cl}(\text{nat})$ samples were irradiated with γ rays from a bremsstrahlung spectrum at the Mainz electron linear accelerator. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured E_γ . Deduced $T_{1/2}$ of ^{35}P and levels in ^{35}S .

There is a 1-MeV gap between the highest observed level at $E=2938.4$ and $Q(\beta^-)$ value=3988.4 19 (2021Wa16). There may be missing weak transitions from unobserved levels in the gap. Upper limits for the unobserved β feedings:

1991: <0.08 (1986Wa22), <2.5 (1972Go31), <0.09 (1972Ap01);

2348: <0.08 (1986Wa22), <5 (1972Go31), <0.24 (1972Ap01);

2717: <0.03 (1986Wa22), <7 (1972Go31), <0.38 (1972Ap01);

3421: <0.02 (1986Wa22);

3558: <0.11 (1986Wa22);

3597: <0.02 (1986Wa22);

3675: <0.02 (1986Wa22);

3802: <0.03 (1986Wa22).

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$3/2^+$	87.61 d 24
1572.290 29	$1/2^+$	2.29 ps 14
2938.4 4	$3/2^+$	



[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

 β^- radiations

$E(\text{decay})$	$E(\text{level})$	$I\beta^-^\ddagger$	$\log ft$	Comments
(1050.0 22)	2938.4	0.47	5.0	$I\beta^-$: others: <8 (1972Go31), <0.45 (1972Ap01).
(2416.1 22)	1572.290	99.2	4.1	$I\beta^-$: other: 99 +1-7 from measured γ/β ratio (1972Ap01), 100 (1972Go31).
(3988.4 [#] 24)	0	<0.73	>7.2	$I\beta^-$: converted by evaluators from the shell-model calculated 0.73 in 1986Wa22. Others: <9 in 1972Ap01; not given in 1972Go31.

[†] From I_γ intensity balance for excited levels.

[‡] Absolute intensity per 100 decays.

[#] Existence of this branch is questionable.

$\gamma(^{35}\text{S})$

E_γ [†]	I_γ ^{†#}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	α [@]	Comments
1572.252 29	100	1572.290	1/2 ⁺	0	3/2 ⁺	(M1)	0.0001007 14	%I γ =99.2 E γ : weighted average of 1571.8 2 (1971Gr53), 1572.2 4 (1972Ap01), 1572.36 24 (1972Go31), 1572.47 25 (1972Go31), and 1572.256 24 (1986Wa22).
2938.29 40	0.476 30	2938.4	3/2 ⁺	0	3/2 ⁺			%I γ =0.47

* Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

Energy level diagram for the $^{35}\text{S}_{19}$ - $^{35}\text{P}_{20}$ system. The diagram shows four energy levels for $^{35}\text{P}_{20}$ (top) and $^{35}\text{S}_{19}$ (bottom). Transitions are labeled with energies and half-lives. A legend on the right indicates gamma-ray branching ratios.

Transition	Energy (keV)	Half-life (s)	β^- (%)
$1/2^+ \rightarrow 0$	47.3	8	100
$3/2^+ \rightarrow 1/2^+$	2938.29	0.472	-
$1/2^+ \rightarrow 3/2^+$	1572.252	99.2	-
$3/2^+ \rightarrow 0$	2938.4	2.29	-
$3/2^+ \rightarrow 0$	1572.290	87.61	-

Legend:

- $I_\gamma < 2\% \times I_\gamma^{\text{max}}$ (Black arrow)
- $I_\gamma < 10\% \times I_\gamma^{\text{max}}$ (Blue arrow)
- $I_\gamma > 10\% \times I_\gamma^{\text{max}}$ (Red arrow)

$^2\text{H}(^{34}\text{S},\text{p}\gamma)$ 1973Wa10

$^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200- $\mu\text{g}/\text{cm}^2$ TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm³ Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured E_γ . Deduced $T_{1/2}$ for two ^{35}S levels using Doppler Shift Attenuation lineshape analysis.

 ^{35}S Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$T_{1/2}^\#$</u>	<u>Comments</u>
0	$3/2^+$		
1574	$1/2^+$	2.29 ps 35	$T_{1/2}$: Lifetime=3.3 ps 5 from 1973Wa10.
2350	$3/2^-$	0.90 ps 14	$T_{1/2}$: Lifetime=1.3 ps 2 from 1973Wa10.

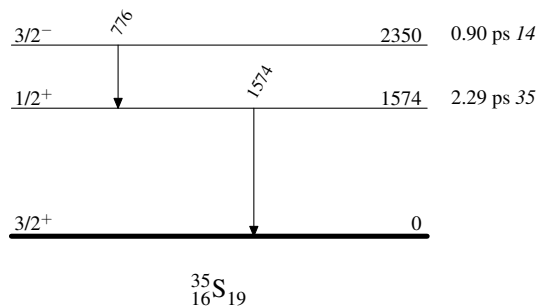
[†] From 1973Wa10.

[‡] From the Adopted Levels.

[#] From Doppler Shift Attenuation Method (DSAM).

 $\gamma(^{35}\text{S})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
776	2350	$3/2^-$	1574	$1/2^+$
1574	1574	$1/2^+$	0	$3/2^+$

 $^2\text{H}(^{34}\text{S},\text{p}\gamma)$ 1973Wa10Level Scheme

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2015MuZY, 2021JoZZ

$J^\pi=0^+$ for ^{36}S ground state.

2015MuZY: a secondary beam of ^{36}S was produced via the projectile fragmentation of a 140-MeV/nucleon ^{48}Ca primary beam impinging on a 846-mg/cm² ^9Be target at the coupled cyclotron facility at NSCL, MSU. The ^{36}S nuclei were selected using the A1900 separator with an intensity of 8.1×10^5 pps and a purity of 89.7%. The excited states of ^{35}S were populated by the one-neutron knockout reaction from the ^{36}S beam at 88 MeV/nucleon (midtarget energy) on a 100-mg/cm² ^9Be secondary target located at the reaction target position of the S800 spectrometer. The projectile-like residues were identified from their energy loss measured by an ionization chamber located at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators situated at the object position and at the focal plane of the S800 spectrometer. Prompt γ rays from the deexcitation of the ^{35}S residues were detected by seven modules of the GRETTINA Ge array. Measured E_γ , I_γ , (^{35}S) γ -coin, $\gamma\gamma$ -coin, the inclusive knockout cross section for producing ^{35}S from ^{36}S , the fractional populations and parallel momentum distributions of 11 populated states in ^{35}S residues. Deduced levels, J , π , L-transfers, partial knockout cross sections and spectroscopic factors.

Calculated single-particle cross sections (σ_{sp}) for neutron removal and parallel momentum distributions using the eikonal model.

2021JoZZ: a more careful reanalysis of **2015MuZY** data.

 ^{35}S Levels

b_f values under comments are population fraction in the knock-out reaction, $\sigma_{\text{f}}^{\text{sp}}$ values are for calculated single-particle cross sections, and R_s values are quenching factors from a systematics study in **2014To14**. The partial cross sections to each level can be deduced from $b_f \times \sigma_{\text{inc}}^{\text{exp}}$.

The inclusive cross section $\sigma_{\text{inc}}^{\text{exp}}$ for producing ^{35}S from ^{36}S is measured to be 63.7 mb **6** (**2015MuZY**) and 67 mb **4** (**2021JoZZ**).

$E(\text{level})^\dagger$	$L^\ddagger$	$C^2S^\#$	Comments
0	2	3.33 76	C^2S : 3.2 7 (2015MuZY). $b_f=44.5\%$ 37 (2021JoZZ), $b_f=43.3\%$ 47, $\sigma_{\text{f}}^{\text{sp}}=13.1$ mb, $R_s=0.66$ 13 (2015MuZY).
1572 1	0	1.26 25	C^2S : 1.1 3 (2015MuZY). $b_f=19.9\%$ 8 (2021JoZZ), $b_f=18.6\%$ 31, $\sigma_{\text{f}}^{\text{sp}}=16.1$ mb, $R_s=0.64$ 13 (2015MuZY).
1990	3	 20	C^2S : <0.2 (2015MuZY). $b_f=1.41\%$ 10 (2021JoZZ), $b_f<2.9\%$, $\sigma_{\text{f}}^{\text{sp}}=12.7$ mb, $R_s=0.69$ 14 (2015MuZY).
2348 1	1	0.34 1	C^2S : 0.4 1 (2015MuZY). $b_f=4.33\%$ 8 (2021JoZZ), $b_f=5.3\%$ 13, $\sigma_{\text{f}}^{\text{sp}}=13.1$ mb, $R_s=0.70$ 14 (2015MuZY).
2717 2	2	0.56 12	C^2S : 0.6 2 (2015MuZY). $b_f=6.14\%$ 7 (2021JoZZ), $b_f=6.4\%$ 16, $\sigma_{\text{f}}^{\text{sp}}=11.4$ mb, $R_s=0.62$ 12 (2015MuZY).
2939 2	(2)	0.31 6	L: 2015MuZY reports $L=0$. C^2S : 0.4 2 (2015MuZY). $b_f=3.99\%$ 3 (2021JoZZ), $b_f=5.7\%$ 16, $\sigma_{\text{f}}^{\text{sp}}=13.5$ mb, $R_s=0.61$ 12 (2015MuZY).
3421 2	2	0.49 11	C^2S : 0.5 1 (2015MuZY). $b_f=5.08\%$ 5 (2021JoZZ), $b_f=5.1\%$ 10, $\sigma_{\text{f}}^{\text{sp}}=11.0$ mb, $R_s=0.61$ 12 (2015MuZY).
3595 5	(2)	0.21 6	C^2S : 0.3 1 (2015MuZY). $b_f=2.13\%$ 5 (2021JoZZ), $b_f=3.0\%$ 7, $\sigma_{\text{f}}^{\text{sp}}=11.0$ mb, $R_s=0.60$ 12 (2015MuZY).
3818 5	(3)	0.25 7	C^2S : 0.3 1 (2015MuZY). $b_f=0.49\%$ 4 (2021JoZZ), $b_f=4.0\%$ 9, $\sigma_{\text{f}}^{\text{sp}}=11.8$ mb, $R_s=0.72$ 14 (2015MuZY).
3890 4	(0)	0.29 9	C^2S : 0.15 4 (2015MuZY). $b_f=3.09\%$ 6 (2021JoZZ), $b_f=2.3\%$ 5, $\sigma_{\text{f}}^{\text{sp}}=13.5$ mb, $R_s=0.72$ 14 (2015MuZY). L: inconsistent with $J^\pi=(5/2^+)$ in the Adopted Levels.
4025 4	(2)	0.34 9	C^2S : 0.6 1 (2015MuZY). $b_f=3.98\%$ 5 (2021JoZZ), $b_f=6.4\%$ 10, $\sigma_{\text{f}}^{\text{sp}}=12.1$ mb, $R_s=0.60$ 12 (2015MuZY).
4948 5			
5349 8			
5775 4			

[†] From **2021JoZZ** based on measured E_γ .

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2015MuZY,2021JoZZ (continued) ^{35}S Levels (continued)

‡ From 2021JoZZ. L-transfers are deduced by comparing the measured and eikonal-calculated parallel momentum distributions of residuals. Parentheses around L values are added by evaluators due to poor fits.

$^{\#}$ From 2021JoZZ. The spectroscopic factor $C^2S = b_f \sigma_{\text{inc}}^{\text{exp}} / R_s \sigma_f^{\text{sp}}$, where b_f is the population fraction, $\sigma_{\text{inc}}^{\text{exp}}$ is the measured total inclusive cross section, R_s is the quenching factor from a systematics study 2014To14, and σ_f^{sp} is the eikonal model calculated single-particle cross section.

 $\gamma(^{35}\text{S})$

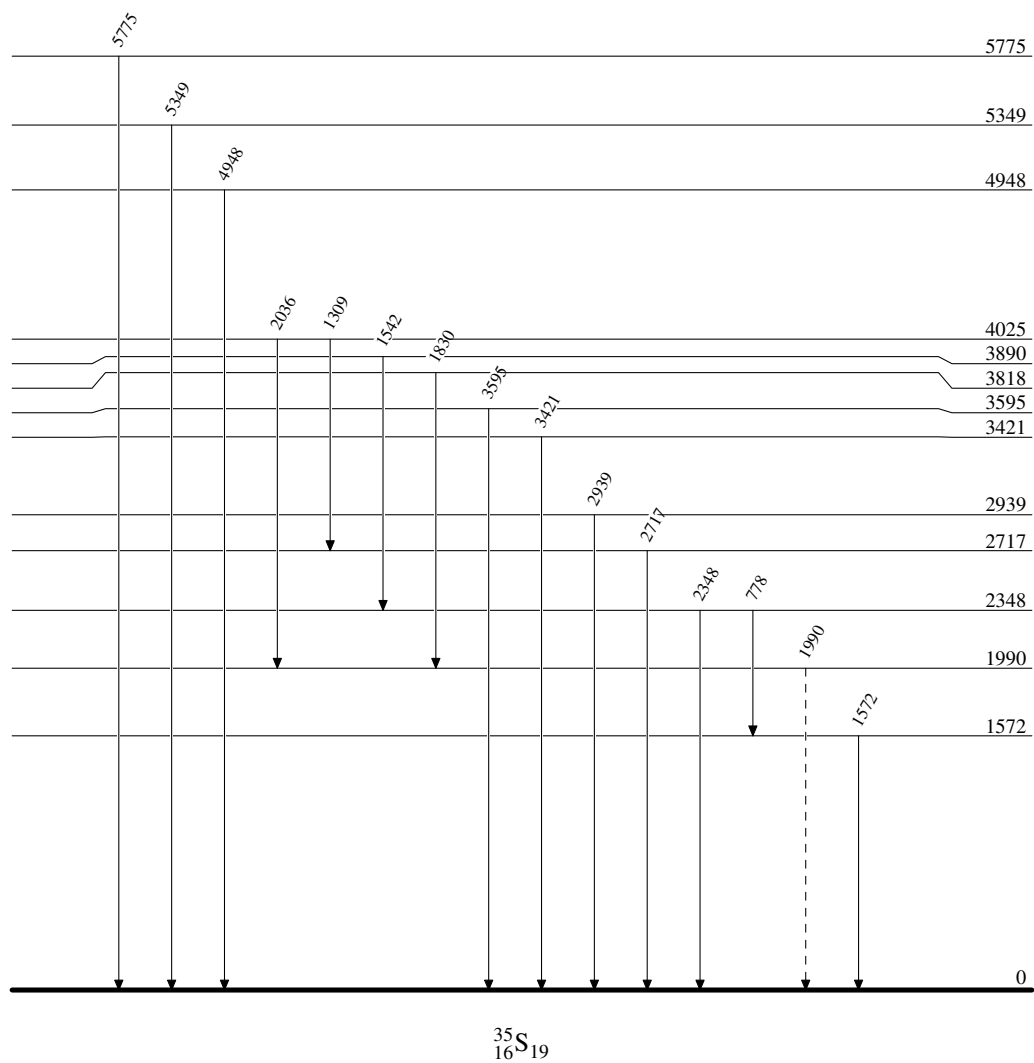
$E_{\gamma}^{\dagger}$	$E_i(\text{level})$	E_f	$E_{\gamma}^{\dagger}$	$E_i(\text{level})$	E_f	$E_{\gamma}^{\dagger}$	$E_i(\text{level})$	E_f	$E_{\gamma}^{\dagger}$	$E_i(\text{level})$	E_f
778	2348	1572	1830	3818	1990	2717	2717	0	4948	4948	0
1309	4025	2717	1990 ‡	1990	0	2939	2939	0	5349	5349	0
1542	3890	2348	2036	4025	1990	3421	3421	0	5775	5775	0
1572	1572	0	2348	2348	0	3595	3595	0			

† From 2021JoZZ level scheme. γ -ray energy uncertainties are not given.

‡ Placement of transition in the level scheme is uncertain.

$^9\text{Be}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2015MuZY,2021JoZZ

Legend

Level Scheme-----► γ Decay (Uncertain)

$^{24}\text{Mg}(^{14}\text{N},3\text{p}\gamma)$ 2014Ay01

2014Ay01: $^{24}\text{Mg}(^{14}\text{N},3\text{p})^{35}\text{S}$ fusion-evaporation reaction. A 40-MeV, 5-pnA ^{14}N beam was delivered from the LNL

XTU-Tandem accelerator and impinged on 1-mg/cm² thick, 99.7% isotopically enriched ^{24}Mg evaporated on an 8-mg/cm² gold layer. γ rays were detected using 4 π -GASP array composed of 40 Compton-suppressed large volume HPGe detectors arranged in seven rings at different angles with respect to beam axis. Events were collected when at least two germanium detectors fired in coincidence. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma\gamma(\theta)$ (ADO). Deduced levels, J , π , level lifetimes via the Doppler shift attenuation method (DSAM). Compared with large-scale shell model calculations.

 ^{35}S Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\ddagger$</u>	<u>$E(\text{level})^\dagger$</u>	<u>$J^\ddagger$</u>	<u>$T_{1/2}^\#$</u>	<u>$E(\text{level})^\dagger$</u>	<u>$J^\ddagger$</u>	<u>$T_{1/2}^\#$</u>
0.0 ^{&}	3/2 ⁺	3816.0 [@] 3	9/2 ⁻	0.28 ps 3	5412.2 3	9/2 ⁺	
1572.5 3	1/2 ⁺	3886.4 4	5/2 ⁺		5877.6 ^{&} 3	11/2 ⁺	
1991.4 [@] 2	7/2 ⁻	4023.2 [@] 3	11/2 ⁻	0.32 ps 3	6299.4 4	11/2 ⁺	
2347.9 3	3/2 ⁻	4822.9 3	9/2 ⁺		6352.2 [@] 5	13/2 ⁻	50 fs 10
2717.1 3	5/2 ⁺	4899.7 4	9/2 ⁺		7179.8 ^{&} 3	15/2 ⁺	3.1 ps 12
3594.7 ^{&} 3	7/2 ⁺	5010.2 3	11/2 ⁻	0.45 ps 8	8023.7 ^{&} 9	17/2 ⁺	0.15 ps 4

[†] From a least-squares fit to γ -ray energies.

[‡] As given in 2014Ay01, which quotes the known J^π assignments for four low-lying levels at 1572, 1991, 2347, and 2717 from 2011Ch48. 2014Ay01 assigns J^π for levels above 3 MeV based on the measured $\gamma\gamma(\theta)$ (ADO) ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

[#] From DSAM line-shape analyses (2014Ay01).

[@] Seq.(A): $\Delta J=1$ sequence based on 7/2⁻.

[&] Seq.(B): Sequence based on 3/2⁺.

 $\gamma(^{35}\text{S})$

$R_{\text{ADO}}=[I_\gamma(34^\circ)+I_\gamma(146^\circ)]/2I_\gamma(90^\circ)$. Expected values are 0.8 for stretched dipole ($\Delta J=1$) and 1.4 for stretched quadrupole ($\Delta J=2$) or unstretched dipole ($\Delta J=0$) transitions.

<u>E_γ</u>	<u>I_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[†]</u>	<u>Comments</u>
207.0 5	0.4 2	4023.2	11/2 ⁻	3816.0	9/2 ⁻		
465.3 3	1.2 1	5877.6	11/2 ⁺	5412.2	9/2 ⁺	D	$R_{\text{ADO}}=0.71$ 9.
775.4 4	1.4 2	2347.9	3/2 ⁻	1572.5	1/2 ⁺	D	$R_{\text{ADO}}=0.68$ 14.
827.8 6	0.3 1	7179.8	15/2 ⁺	6352.2	13/2 ⁻		
843.9 8	1.6 2	8023.7	17/2 ⁺	7179.8	15/2 ⁺	D	$R_{\text{ADO}}=0.8$ 3.
867.3 3	1.9 1	5877.6	11/2 ⁺	5010.2	11/2 ⁻	D	$R_{\text{ADO}}=1.36$ 20, consistent with $\Delta J=0$.
877.6 5	0.4 1	3594.7	7/2 ⁺	2717.1	5/2 ⁺		
880.2 4	0.8 1	7179.8	15/2 ⁺	6299.4	11/2 ⁺		
887.0 7	0.4 1	6299.4	11/2 ⁺	5412.2	9/2 ⁺	D	$R_{\text{ADO}}=0.6$ 3.
977.8 3	1.6 1	5877.6	11/2 ⁺	4899.7	9/2 ⁺	D+Q	$R_{\text{ADO}}=1.43$ 16.
986.9 3	4.4 2	5010.2	11/2 ⁻	4023.2	11/2 ⁻	D	$B(E1)(\text{W.u.})=0.00102 +22-16$; $B(M1)(\text{W.u.})=0.036 +8-6$ $R_{\text{ADO}}=1.30$ 13, consistent with $\Delta J=0$.
1006.8 2	0.8 1	4822.9	9/2 ⁺	3816.0	9/2 ⁻	D	$R_{\text{ADO}}=1.37$ 11, consistent with $\Delta J=0$.
1054.5 3	3.0 2	5877.6	11/2 ⁺	4822.9	9/2 ⁺	D	$R_{\text{ADO}}=0.72$ 11.
1228.1 3	1.1 1	4822.9	9/2 ⁺	3594.7	7/2 ⁺	D	$R_{\text{ADO}}=0.78$ 9.
1302.2 2	13.5 8	7179.8	15/2 ⁺	5877.6	11/2 ⁺	E2	$B(E2)(\text{W.u.})=6.6 +42-19$ $R_{\text{ADO}}=1.40$ 5, $A_2=+0.27$ 5, $A_4=-0.07$ 5.
1304.8 12	0.6 2	4899.7	9/2 ⁺	3594.7	7/2 ⁺		
1389.0 4	3.9 4	5412.2	9/2 ⁺	4023.2	11/2 ⁻	D	$R_{\text{ADO}}=0.79$ 11.

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{14}\text{N}, 3\text{p}\gamma)$ **2014Ay01** (continued) $\gamma(^{35}\text{S})$ (continued)

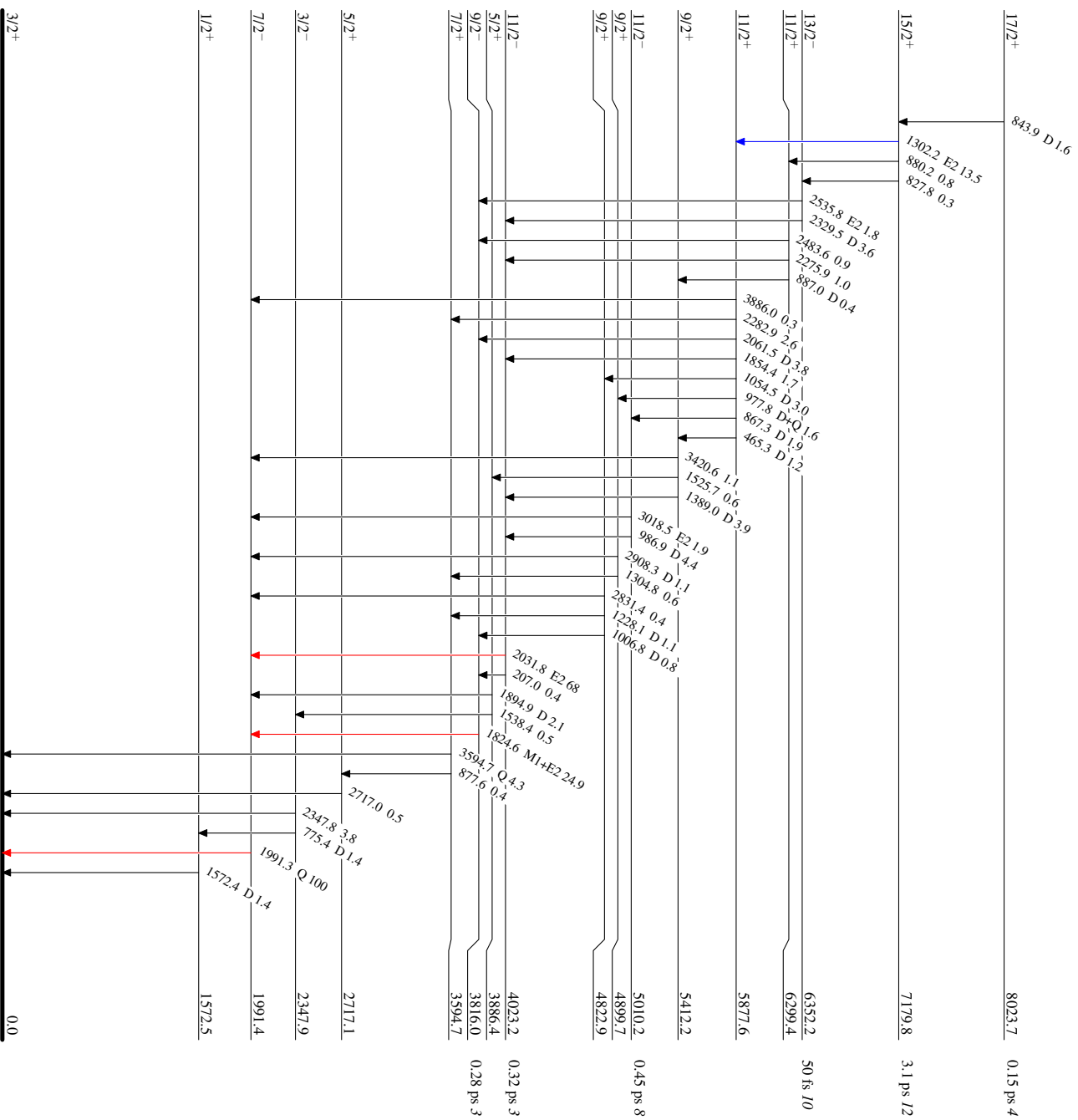
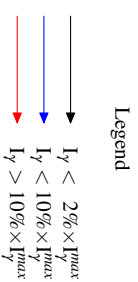
E_γ	I_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [†]	$\delta^\dagger$	Comments
1525.7 8	0.6 1	5412.2	9/2 ⁺	3886.4	5/2 ⁺			
1538.4 8	0.5 2	3886.4	5/2 ⁺	2347.9	3/2 ⁻			
1572.4 3	1.4 4	1572.5	1/2 ⁺	0.0	3/2 ⁺	D		$R_{\text{ADO}}=0.88$ 10.
1824.6 2	24.9 15	3816.0	9/2 ⁻	1991.4	7/2 ⁻	M1+E2	+0.55 9	$B(\text{M1})(\text{W.u.})=0.0099$ +14-12; $B(\text{E2})(\text{W.u.})=3.4$ +10-9 Mult., δ : D+Q and δ from $\gamma(\theta)$ in 2014Ay01 ; E1+M2 ruled out by RUL. $R_{\text{ADO}}=1.58$ 5 indicating $\Delta J=2$ or 0 seems questionable considering $\Delta J=1$ from level scheme. $A_2=+0.48$ 4, $A_4=+0.08$ 6.
1854.4 4	1.7 1	5877.6	11/2 ⁺	4023.2	11/2 ⁻			
1894.9 4	2.1 3	3886.4	5/2 ⁺	1991.4	7/2 ⁻	D		$R_{\text{ADO}}=0.69$ 10.
1991.3 2	100 6	1991.4	7/2 ⁻	0.0	3/2 ⁺	Q		$R_{\text{ADO}}=1.45$ 15.
2031.8 3	68 4	4023.2	11/2 ⁻	1991.4	7/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=7.5$ +8-7 $R_{\text{ADO}}=1.41$ 10, $A_2=+0.29$ 4, $A_4=-0.10$ 6. $R_{\text{ADO}}=0.65$ 9.
2061.5 3	3.8 2	5877.6	11/2 ⁺	3816.0	9/2 ⁻	D		
2275.9 6	1.0 2	6299.4	11/2 ⁺	4023.2	11/2 ⁻			
2282.9 4	2.6 1	5877.6	11/2 ⁺	3594.7	7/2 ⁺			
2329.5 9	3.6 7	6352.2	13/2 ⁻	4023.2	11/2 ⁻	D		$B(\text{E1})(\text{W.u.})=6.7 \times 10^{-4}$ +17-13; $B(\text{M1})(\text{W.u.})=0.023$ +6-5 $R_{\text{ADO}}=0.8$ 3.
2347.8 3	3.8 3	2347.9	3/2 ⁻	0.0	3/2 ⁺			
2483.6 8	0.9 2	6299.4	11/2 ⁺	3816.0	9/2 ⁻			
2535.8 11	1.8 2	6352.2	13/2 ⁻	3816.0	9/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=5.3$ +17-11 $R_{\text{ADO}}=1.5$ 5.
2717.0 4	0.5 3	2717.1	5/2 ⁺	0.0	3/2 ⁺			
2831.4 9	0.4 1	4822.9	9/2 ⁺	1991.4	7/2 ⁻			
2908.3 5	1.1 2	4899.7	9/2 ⁺	1991.4	7/2 ⁻	D		$R_{\text{ADO}}=0.74$ 14.
3018.5 7	1.9 2	5010.2	11/2 ⁻	1991.4	7/2 ⁻	E2		$B(\text{E2})(\text{W.u.})=0.223$ +49-39 $R_{\text{ADO}}=1.35$ 34.
3420.6 9	1.1 1	5412.2	9/2 ⁺	1991.4	7/2 ⁻			
3594.7 7	4.3 4	3594.7	7/2 ⁺	0.0	3/2 ⁺	Q		$R_{\text{ADO}}=1.36$ 16.
3886.0 13	0.3 1	5877.6	11/2 ⁺	1991.4	7/2 ⁻			

[†] From $\gamma(\theta)$, $\gamma\gamma(\theta)$ (ADO) data, and RUL when $T_{1/2}$ is available in [2014Ay01](#).

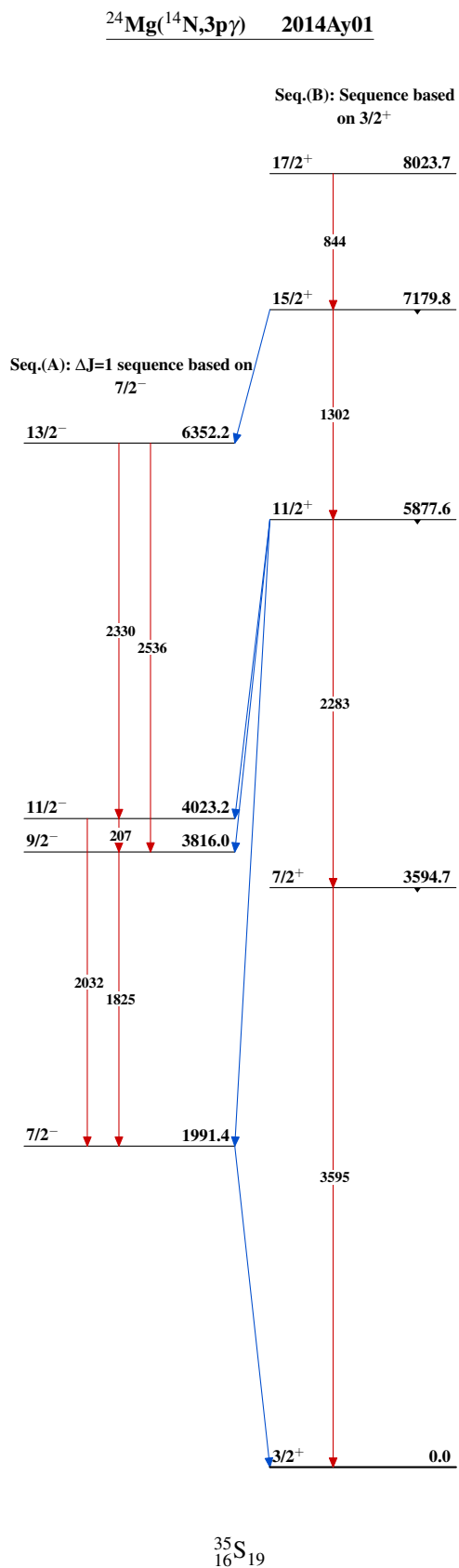
²⁴Mg(¹⁴N,3pγ) 2014Ay01

Level Scheme

Intensities: Relative I_γ



³⁵S₁₉



$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ **2021Go09**

2021Go09: $^{26}\text{Mg}(^{18}\text{O}, 2\alpha n)^{35}\text{S}$ fusion-evaporation reaction. Exp 1 determined the beam energy for producing high-spin states of ^{35}S : an 80-MeV ^{18}O beam was delivered from the JAEA tandem accelerator facility and impinged on a 0.5-mg/cm² thick, self-supporting metallic enriched foil of ^{26}Mg . γ rays were detected using the GEMINI-II array of 14 Ge detectors with BGO Compton-suppressor shields. The reaction channel was selected by detecting the evaporated charged particles with a 4 π silicon detector. Measured E_γ , I_γ , $\alpha\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (ADO). Exp 2 used the same ^{18}O beam energy, the same target, and the same charged-particle detectors as Exp 1. The beam was delivered from the ALTO-Tandem accelerator facility at IPN Orsay. γ rays were detected using the ORGAM array of 13 Ge detectors based on the EUROGAM array with the BGO scintillator for the anti-Compton suppressor. Compared with large-scale shell-model calculations with the SDPF-MSD4 interaction. Also see [2015GoZY](#), [2014GoZZ](#), [2013GoZW](#), and [2012IdZZ](#).

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0.0 [#]	3/2 ⁺		
1990.6 [#] 4	7/2 ⁻		
3594.2 [#] 7	(7/2 ⁺)		
3815.0 6	(9/2 ⁻)		
4023.1 5	(11/2 ⁻)		
4822.4 6	(9/2 ⁺)		
4899.6 7	(9/2 ⁺)		
5009.8 6	(11/2 ⁻)		
5412.4 6	(9/2 ⁺)		
5878.0 [#] 6	(11/2 ⁺)		
7179.6 [#] 7	(15/2 ⁺)		
8023.9 [#] 9	(17/2 ⁺)		
8755.7 [#] 13	(17/2)	<1 ps	$T_{1/2}$: estimated by 2021Go09 from residual Doppler energy shifts observed for the 732 γ and 1576 γ peaks.
10048.4 [#] 10	(19/2)		
12469.1 [#] 15	(21/2)		

[†] From a least-squares fit to γ -ray energies.

[‡] As given in [2021Go09](#), which quotes the J^π assignments for levels below 8.5 MeV from [2014Ay01](#) but adds parentheses. [2021Go09](#) assigns J^π for levels above 8.5 MeV based on the measured $\gamma\gamma(\theta)$ (ADO) ratios.

[#] Seq.(A): γ cascade based on the g.s.

 $\gamma(^{35}\text{S})$

$R_{\text{ADO}}^{\gamma 1} = I_{\gamma 1}(47^\circ \text{ gated by } \gamma_2 \text{ at all angles}) / I_{\gamma 1}(86^\circ \text{ gated by } \gamma_2 \text{ at all angles})$. Expected values are ≈ 0.8 for dipole ($\Delta J=1$) and ≈ 1.0 for stretched quadrupole ($\Delta J=2$) transitions.

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	Comments
465.5 2	14 1	5878.0	(11/2 ⁺)	5412.4	(9/2 ⁺)	D	$R_{\text{ADO}}=0.67$ 11.
731.6 15	8 3	8755.7	(17/2)	8023.9	(17/2 ⁺)		
844.3 6	20 3	8023.9	(17/2 ⁺)	7179.6	(15/2 ⁺)	D	E_γ : 8443(6) in 2021Go09 is likely a misprint. $R_{\text{ADO}}=0.51$ 8. $R_{\text{ADO}}=0.54$ 21.
868.2 4	20 2	5878.0	(11/2 ⁺)	5009.8	(11/2 ⁻)	D	
978.4 10	26 7	5878.0	(11/2 ⁺)	4899.6	(9/2 ⁺)		
986.6 3	26 1	5009.8	(11/2 ⁻)	4023.1	(11/2 ⁻)		$R_{\text{ADO}}=0.91$ 13 consistent with $\Delta J=(0)$.
1007.2 11	4 2	4822.4	(9/2 ⁺)	3815.0	(9/2 ⁻)	(D)	$R_{\text{ADO}}=0.87$ 15.
1055.6 4	39 10	5878.0	(11/2 ⁺)	4822.4	(9/2 ⁺)	D	$R_{\text{ADO}}=0.67$ 18.

Continued on next page (footnotes at end of table)

$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ **2021Go09** (continued) $\gamma(^{35}\text{S})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	Comments
1228.3 6	11 2	4822.4	(9/2 ⁺)	3594.2	(7/2 ⁺)		
1301.6 3	56 11	7179.6	(15/2 ⁺)	5878.0	(11/2 ⁺)	Q	$R_{\text{ADO}}=1.40$ 17.
1305.3 4	5 3	4899.6	(9/2 ⁺)	3594.2	(7/2 ⁺)		
1389.0 8	9 2	5412.4	(9/2 ⁺)	4023.1	(11/2 ⁻)		
1576.3 14	14 3	8755.7	(17/2)	7179.6	(15/2 ⁺)	D	$R_{\text{ADO}}=0.43$ 9. 2021Go09 Table I lists this $\Delta J=1$ ADO ratio for the 731.6 γ , which however is a (17/2) to (17/2 ⁺) transition with $\Delta J=0$. 2021Go09 repeatedly states the 8756 level has $J=(17/2)$ and therefore Evaluators reassign this $\Delta J=1$ ADO ratio to the 1576.3 γ , corresponding to a (17/2) to (15/2 ⁺) transition.
1824.4 6	48 2	3815.0	(9/2 ⁻)	1990.6	7/2 ⁻		
1854.5 42	28 14	5878.0	(11/2 ⁺)	4023.1	(11/2 ⁻)		
1990.5 4	100	1990.6	7/2 ⁻	0.0	3/2 ⁺	Q	$R_{\text{ADO}}=1.30$ 8.
2032.4 4	70 16	4023.1	(11/2 ⁻)	1990.6	7/2 ⁻		
2063.3 14	32 6	5878.0	(11/2 ⁺)	3815.0	(9/2 ⁻)	D	$R_{\text{ADO}}=0.58$ 41.
2283.5 10	14 3	5878.0	(11/2 ⁺)	3594.2	(7/2 ⁺)		
2420.6 11	21 6	12469.1	(21/2)	10048.4	(19/2)	(D)	$R_{\text{ADO}}=0.50$ 42.
2868.7 8	26 2	10048.4	(19/2)	7179.6	(15/2 ⁺)	Q	$R_{\text{ADO}}=0.98$ 13.
3594.4 11	35 2	3594.2	(7/2 ⁺)	0.0	3/2 ⁺	(Q)	$R_{\text{ADO}}=0.93$ 20.

 † From **2021Go09**. ‡ From $\gamma\gamma(\theta)(\text{ADO})$ data in **2021Go09**.

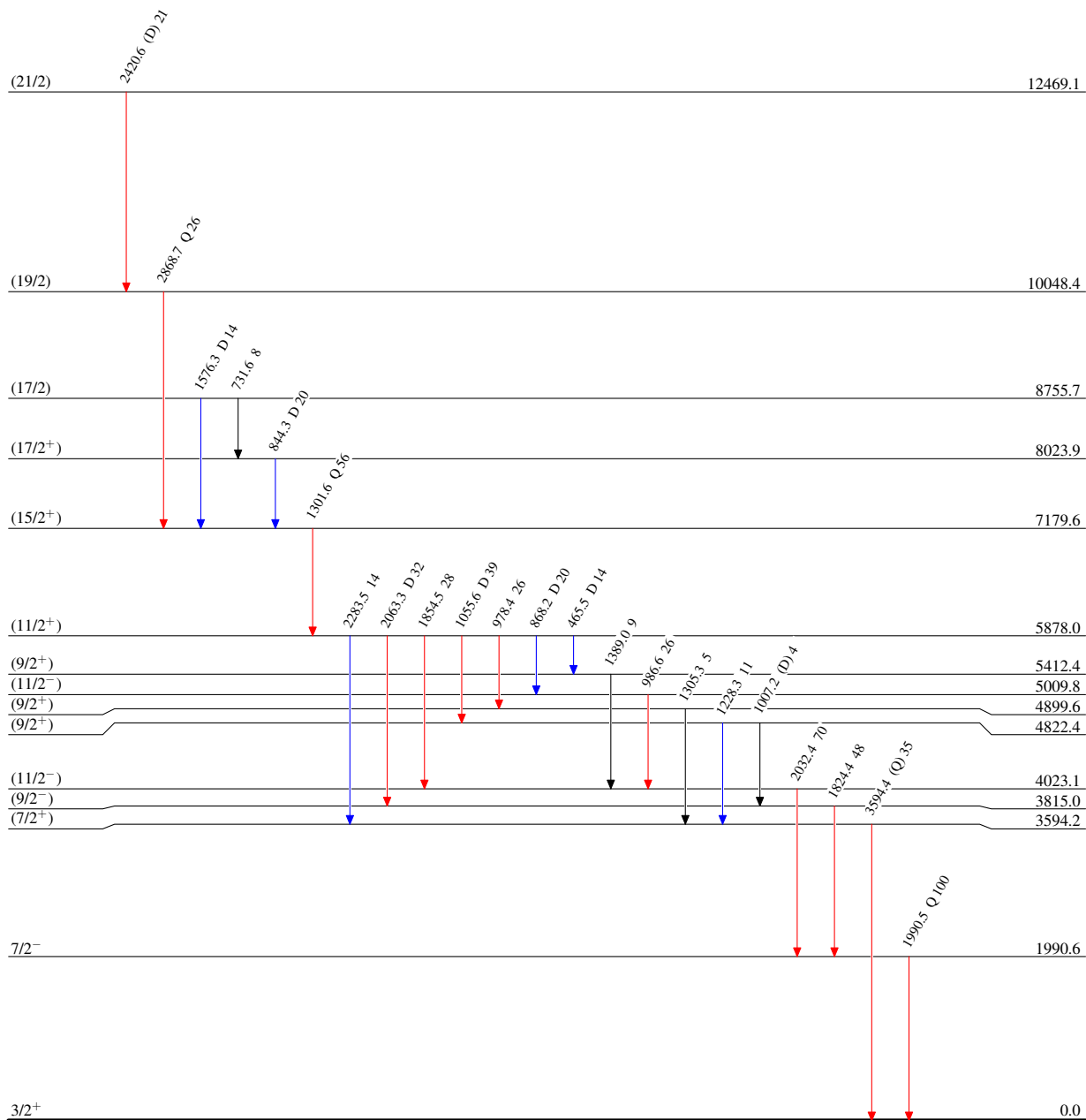
$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ 2021Go09

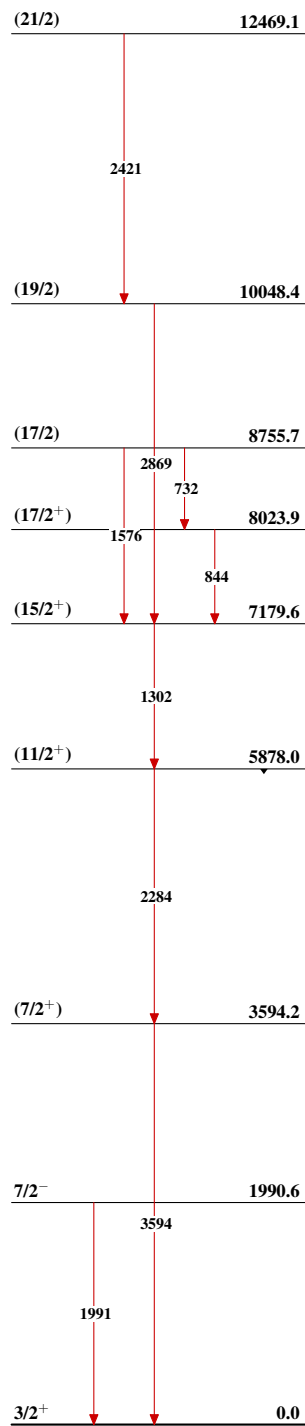
Level Scheme

Intensities: Relative I_γ

Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{16}\text{S}_{19}$

$^{26}\text{Mg}(^{18}\text{O}, 2\alpha n\gamma)$ 2021Go09Seq.(A): γ cascade based on the
g.s $^{35}_{16}\text{S}_{19}$

$^{34}\text{S}(\text{n},\gamma)$ E=thermal 1985Ra15

- 1985Ra15:** 6×10^{11} n/cm²/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.09-g, 77.7% enriched ^{34}S target were detected using a 26 cm³ coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ , E_γ , and I_γ for 61 γ -ray transitions. Deduced 22 levels and γ -ray branching ratios.
- 1972Dz13:** Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute. γ rays following thermal neutron capture by a 1.5-g, 97%-enriched ^{34}S target were detected using a 10-cm³ Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured E_γ and I_γ for 22 γ -ray transitions. Deduced 11 levels.
- 1985Ke08:** 5×10^{12} n/cm²/s thermal neutrons were produced from the McMaster University Reactor. γ rays following thermal neutron capture by a natural sulfur target were detected using a 10% efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured E_γ and I_γ for 13 γ -ray transitions. Deduced 9 levels.
- 1997Be42:** Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium. γ rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched ^{34}S and 5.933%-enriched ^{36}S target were detected using a Ge detector. Measured σ_γ , E_γ , and I_γ for 11 γ -ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.
- All 55 E_γ values from **2007ChZX**: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below. Others: **2000Re01**, **2001ZhZZ**.

 ^{35}S Levels

E(level) [†]	Comments
0	
1572.373 7	
1991.28 4	
2347.782 8	C ² S=0.48 for 3/2 ⁻ from 1997Be42 .
2717.05 9	
2938.63 5	
3558.082 27	
3801.955 15	C ² S=0.08 for 3/2 ⁻ from 1997Be42 .
4105.6 8	
4189.243 33	C ² S=0.15 for 1/2 ⁻ from 1997Be42 .
4477.62 7	
4903.374 15	C ² S=0.49 for 1/2 ⁻ from 1997Be42 .
4963.084 25	C ² S=0.19 for 3/2 ⁻ from 1997Be42 .
5752.5 8	
5841.485 31	This level is introduced in S. Raman to P.M. Endt private communication (mentioned in the Adopted Levels of 2011Ch48). Populated by 1144.6 γ from 6986 level and deexcites by 4268.6 γ to 1572 level.
6018.8 6	
6078.477 32	
6293.93 6	
6354.89 23	
6419.9 11	
6629.43 9	
6761.0 12	
(6986.099 24)	

[†] From a least-squares fit to γ -ray energies.

$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15 (continued)** $\gamma(^{35}\text{S})$

Photons per 100 thermal neutron captures are reported in [1972Dz13](#) and [1985Ke08](#). γ -ray cross sections in mb are reported in [1985Ra15](#), multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures. $\sigma(4638)=163$ mb *15*.
 $I_\gamma(4638)=\sigma(4638)\times 0.340=55$ *5*.

γ -ray intensities per incident neutron for all transitions and γ -ray cross sections in mb for primary transitions are reported in [1997Be42](#). The strongest 4638 γ is used for normalization. $\sigma(4638)=148.6$ mb *55* and
 $I_\gamma(4638)=\sigma(1997\text{Be42})/\sigma(1985\text{Ra15})\times I_\gamma(1985\text{Ra15})=50$ *7*.

E_γ †	I_γ ‡#	$E_i(\text{level})$	E_f	Comments
356.66 @ 9	0.037 @ 4	2347.782	1991.28	
356.66 @ 9	0.037 @ 4	(6986.099)	6629.43	
368.5 4	0.020 7	2717.05	2347.782	
619.23 19	0.042 8	3558.082	2938.63	
631.32 @ 24	0.060 @ 9	4189.243	3558.082	
631.32 @ 24	0.060 @ 9	(6986.099)	6354.89	
x 646.9 ‡ 5	0.7 1			E_γ, I_γ : from 1972Dz13 . 1972Dz13 assigned this γ ray as the 6986 to 6339 transition. This γ ray and its final level 6339 have not been observed in any other $^{34}\text{S}(\text{n},\gamma)$ measurements. Evaluators consider this γ ray questionable.
x 663.41 ‡ 7	0.09 1			
692.16 5	0.133 17	(6986.099)	6293.93	
775.398 6	17.2 19	2347.782	1572.373	E_γ : other: 775.50 15 (1972Dz13). I_γ : weighted average of 19.5 20 (1972Dz13) and 15.6 17 (1985Ra15).
x 803.81 ‡ 9	0.092 14			
863.28 28	0.034 10	3801.955	2938.63	
907.608 ‡ 23	0.56 10	(6986.099)	6078.477	E_γ : weighted average of 908.1 3 (1972Dz13) and 907.607 14 (1985Ra15). I_γ : weighted average of 0.5 1 (1972Dz13) and 0.61 10 (1985Ra15).
1084.79 15	0.054 10	3801.955	2717.05	
1101.92 31	0.033 8	4903.374	3801.955	
1144.591 20	0.55 6	(6986.099)	5841.485	The placement of this γ ray is changed from 2717 to 1572 (1985Ra15) to 6986 to 5841 (S. Raman to P.M. Endt private communication).
1161.05 20	0.055 8	4963.084	3801.955	
1210.28 4	0.252 27	3558.082	2347.782	
1250.61 5	0.214 24	4189.243	2938.63	
x 1381.67 ‡ 24	0.024 8			
1404.967 24	0.60 6	4963.084	3558.082	
1454.09 4	0.45 11	3801.955	2347.782	
1566.7 3	0.47 9	3558.082	1991.28	
1572.333 8	35.0 34	1572.373	0	E_γ : weighted average of 1572.15 16 (1972Dz13) and 1572.333 7 (1985Ra15). I_γ : others: 39 4 (1972Dz13) and 32 4 (1997Be42).
1760.55 11	0.146 21	4477.62	2717.05	
1840.2 12	2.2 4	4189.243	2347.782	E_γ : unweighted average of 1839 1 (1972Dz13) and 1841.426 15 (1985Ra15). I_γ : weighted average of 3.6 8 (1972Dz13) and 2.14 21 (1985Ra15).
1964.8 2	0.37 10	4903.374	2938.63	
1991.27 5	0.54 6	1991.28	0	
2022.954 9	11 1	(6986.099)	4963.084	E_γ : others: 2022.80 20 (1972Dz13) and 2023.0 10 (1985Ke08). I_γ : weighted average of 12.0 15 (1972Dz13), 11 1 (1985Ke08), 11.4 10 (1985Ra15), and 9.1 12 (1997Be42).
2082.83 17	15.9 15	(6986.099)	4903.374	E_γ : unweighted average of 2082.65 16 (1972Dz13), 2083.17 7 (1985Ke08), and 2082.681 12 (1985Ra15).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$ (continued)

E_γ †	I_γ ‡#	$E_i(\text{level})$	E_f	Comments
$^{*}2184.16^{\pm}_{19}$	4.7 5			I_γ: weighted average of 18.0 15 (1972Dz13), 17 2 (1985Ke08), 15.6 17 (1985Ra15), and 12.6 17 (1997Be42). E_γ, I_γ: from 1985Ke08. 1985Ke08 assigned this γ ray as a 4903 to 2719 transition. The final level is closest to the known 2717 level. This γ ray has not been observed in any other $^{34}\text{S}(\text{n},\gamma)$ measurements. Including this γ ray in the level scheme would result in a significant intensity imbalance at its initial level 4903 and its final level 2717 and also lead to poor fitting in the level scheme. Evaluators therefore consider the placement of this γ ray unlikely.
2229.510 16 2347.700 15	2.86 27 49 4	3801.955 2347.782	1572.373 0	E_γ: weighted average of 2347.50 10 (1972Dz13), 2347.71 3 (1985Ke08), and 2347.701 11 (1985Ra15). I_γ: weighted average of 54 5 (1972Dz13), 49 4 (1985Ke08), 50 5 (1985Ra15), and 40 6 (1997Be42).
2508.39 8 2555.492 14	0.38 4 3.14 31	(6986.099) 4903.374	4477.62 2347.782	E_γ: other: 2555.9 4 (1972Dz13). I_γ: weighted average of 3.4 7 (1972Dz13) and 3.09 31 (1985Ra15).
2615.2 2	1.09 10	4963.084	2347.782	E_γ: other: 2614.3 3 doublet (1972Dz13).
2616.8 13	0.24 7	4189.243	1572.373	I_γ: other: <1.3 4 (1972Dz13). E_γ: other: 2614.3 3 doublet (1972Dz13). I_γ: other: <1.3 4 (1972Dz13).
2716.99 16 2796.74 4	0.30 5 4.9 4	2717.05 (6986.099)	0 4189.243	E_γ: weighted average of 2796.83 13 (1985Ke08) and 2796.73 4 (1985Ra15). Other: 2796.6 5 (1972Dz13). I_γ: weighted average of 6.1 8 (1972Dz13), 4.3 4 (1985Ke08), 5.4 5 (1985Ra15), and 4.9 7 (1997Be42).
2905.1 4 2938.58 11 2972.0 4 3139.9 5 3183.95 4	0.14 4 0.89 9 0.18 6 0.078 24 6.1 6	4477.62 2938.63 4963.084 6078.477 (6986.099)	1572.373 0 1991.28 2938.63 3801.955	E_γ: weighted average of 3184.30 20 (1972Dz13), 3183.94 7 (1985Ke08), and 3183.94 4 (1985Ra15). I_γ: weighted average of 7.0 9 (1972Dz13), 5.7 6 (1985Ke08), 6.2 6 (1985Ra15), and 5.7 8 (1997Be42).
3330.84 13	7.6 7	4903.374	1572.373	E_γ: unweighted average of 3331.08 15 (1972Dz13), 3330.64 6 (1985Ke08), and 3330.80 4 (1985Ra15). I_γ: weighted average of 7.8 10 (1972Dz13), 7.7 8 (1985Ke08), and 7.4 7 (1985Ra15).
3390.62 7	5.6 5	4963.084	1572.373	E_γ: weighted average of 3390.86 20 (1972Dz13), 3390.73 7 (1985Ke08), and 3390.55 5 (1985Ra15). I_γ: weighted average of 5.5 8 (1972Dz13), 5.7 6 (1985Ke08), and 5.5 5 (1985Ra15).
3558.1 5 3801.73 4	0.085 17 2.59 23	3558.082 3801.955	0 0	E_γ: weighted average of 3802.0 3 (1972Dz13), 3801.49 14 (1985Ke08), and 3801.74 3 (1985Ra15). I_γ: weighted average of 3.6 7 (1972Dz13), 2.3 2 (1985Ke08), 3.09 31 (1985Ra15), and 2.6 4 (1997Be42).
4105.3 8 4188.95 5	0.048 17 2.54 27	4105.6 4189.243	0 0	E_γ: weighted average of 4188.80 13 (1985Ke08) and 4188.96 4 (1985Ra15). Other: 4189.6 3 (1972Dz13). I_γ: weighted average of 1.9 4 (1972Dz13), 2.72 27 (1985Ra15), and 2.8 4 (1997Be42). Other: 24 2 (1985Ke08).
4268.65 14	0.34 4	5841.485	1572.373	E_γ: other: 4268.4 7 (assigned as 6986 to 2717 in 1972Dz13). The placement of 4268γ is changed from 6986 to 2717 (1985Ra15) to 5841 to 1572 (S. Raman to P.M. Endt, private communication, mentioned in the

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n},\gamma)$ E=thermal **1985Ra15** (continued) $\gamma(^{35}\text{S})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^{\ddagger\#}$	$E_i(\text{level})$	E_f	Comments
Adopted Levels of 2011Ch48 .				
4638.14 14	56 5	(6986.099)	2347.782	I_γ : other: 0.4 2 (1972Dz13). E_γ : unweighted average of 4638.4 2 (1972Dz13), 4638.10 3 (1985Ke08), and 4637.91 4 (1985Ra15). I_γ : weighted average of 56 6 (1972Dz13), 59 5 (1985Ke08), 55 5 (1985Ra15), and 50 7 (1997Be42).
4902.99 5	4.0 4	4903.374	0	E_γ : weighted average of 4903.07 7 (1985Ke08) and 4902.96 4 (1985Ra15). Other: 4903.4 3 (1972Dz13). I_γ : weighted average of 3.2 8 (1972Dz13), 4.3 4 (1985Ke08), 3.7 4 (1985Ra15), and 4.2 6 (1997Be42).
4963.06 22	2.62 27	4963.084	0	E_γ : unweighted average of 4963.2 5 (1972Dz13), 4963.34 10 (1985Ke08), and 4962.63 5 (1985Ra15). I_γ : weighted average of 2.6 8 (1972Dz13), 2.92 27 (1985Ra15), and 2.18 33 (1997Be42). Other: 29 3 (1985Ke08).
5752.0 8	0.018 5	5752.5	0	E_γ : other: 6078.2 10 (1972Dz13). I_γ : weighted average of 0.3 1 (1972Dz13) and 0.36 4 (1985Ra15).
6018.2 6	0.020 7	6018.8	0	
6077.87 11	0.35 4	6078.477	0	
6293.2 4	0.065 14	6293.93	0	
6355.0 6	0.046 7	6354.89	0	
6419.3 11	0.016 5	6419.9	0	
6628.5 6	0.030 6	6629.43	0	
6760.3 12	0.019 8	6761.0	0	
6985.7 10	0.036 8	(6986.099)	0	

 † From **1985Ra15**, unless otherwise noted. ‡ Not observed in **2007ChZX** (PGAA).

Intensity per 100 neutron captures.

@ Multiply placed with undivided intensity.

 x γ ray not placed in level scheme.

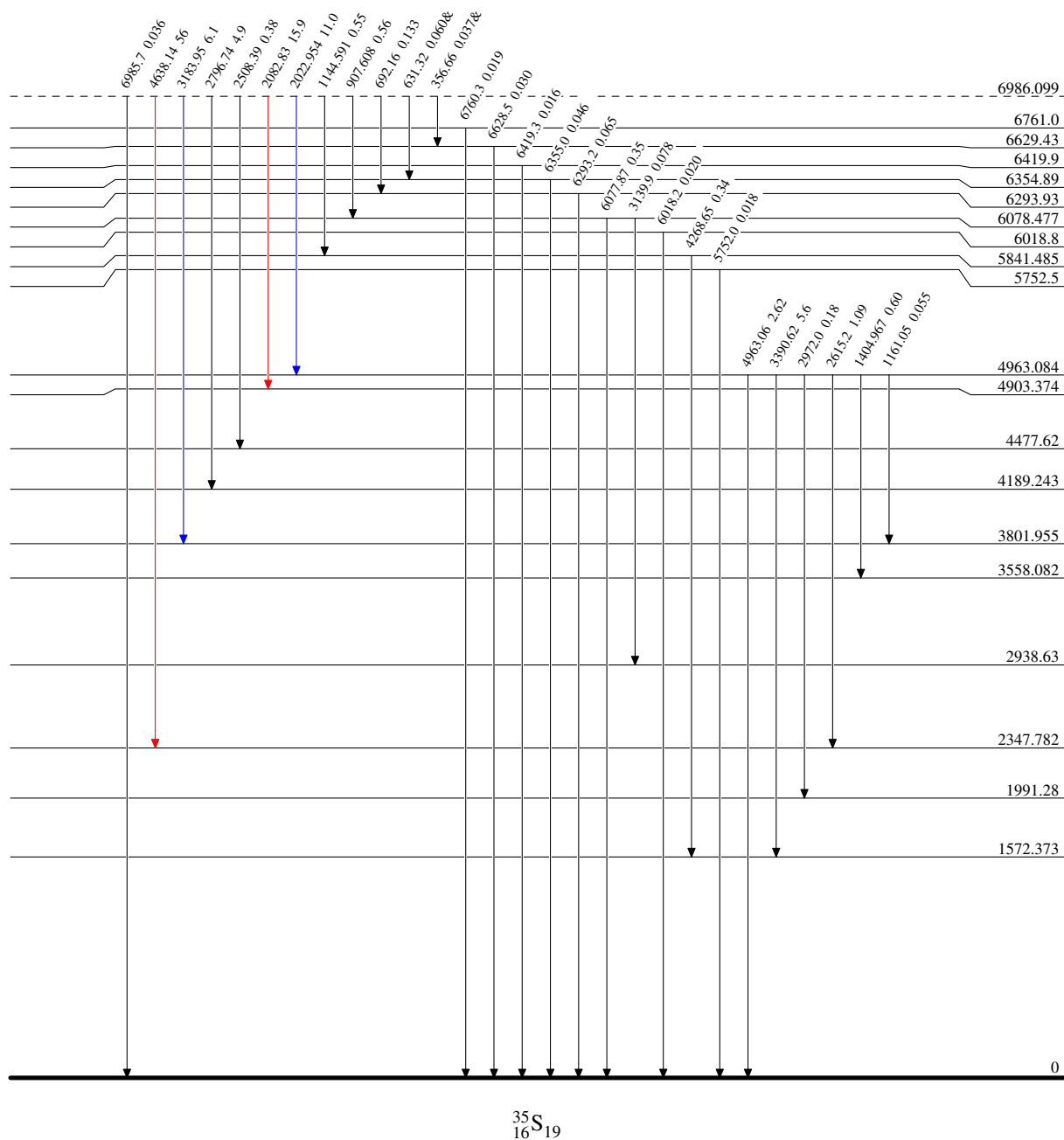
$^{34}\text{S}(\text{n},\gamma) \text{E=thermal}$ 1985Ra15

Level Scheme

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

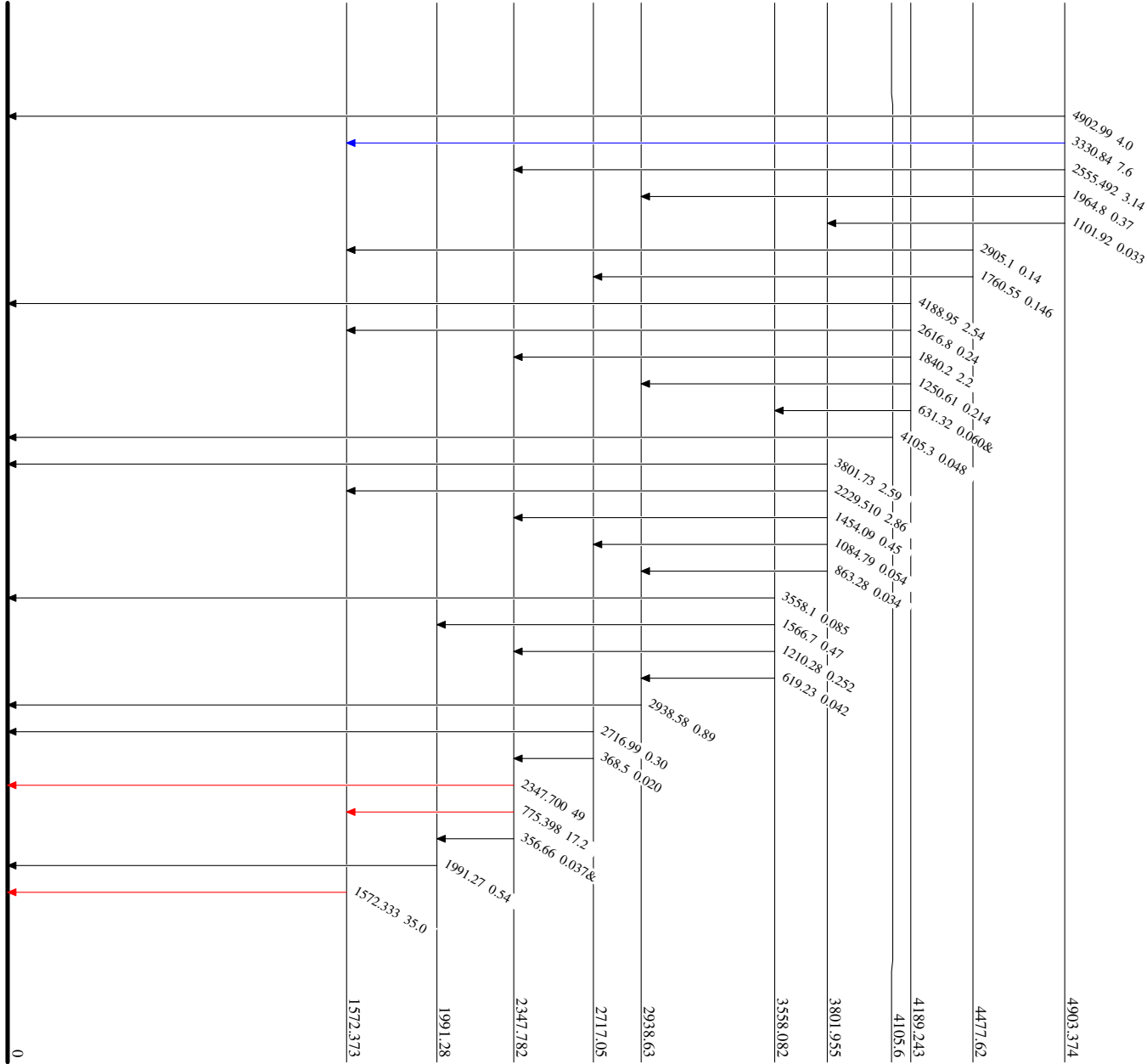
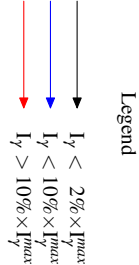
—→ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 —→ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 —→ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$



³⁴S(n,γ) E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given



$^{34}\text{S}(\text{n},\text{n}),(\text{n},\gamma):\text{resonances}$ **1984Ca14**

1984Ca14: 30-1500-keV neutrons were produced from the Oak Ridge Electron Linear Accelerator (ORELA). The target was 94.33%-enriched ^{34}S with a 5.54% impurity of ^{32}S . Neutrons were detected using a 2-cm-thick and 7.5-cm in diameter NE110 proton-recoil detector. Prompt γ rays from neutron capture were detected using two nonhydrogenous fluorocarbon scintillators (NE-226). Measured $\sigma(E_n)$. Deduced 81 resonances, J , π , L , widths and spectroscopic factors by comparing the R-matrix predictions of the cross section with the data.

2018MuZY: Atlas of Neutron Resonances. Evaluation of neutron resonance energies, spins, width parameters, and resonance strengths for nuclei of $Z=1-60$.

 ^{35}S Levels

J^π , L , and resonance parameters E_n , $g\Gamma_n$, Γ_γ , and $g\Gamma_n\Gamma_\gamma/\Gamma$ are from **1984Ca14** due to significant discrepancies, including orders of magnitude inconsistencies, between the **1984Ca14** original data and **2018MuZY** evaluation. $g\Gamma_n=(2J+1)\Gamma_n/2$.

$E(\text{level})^\dagger$	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
6985.54?			0	-0.305	$E(\text{level})$: fictitious level from analysis of resonance spectrum below the $S(n)$ threshold. $\Gamma_\gamma=0.38$ eV.
7018.89 4		0.025 eV 1	2	34.03 1	
7072.66 5		0.14 eV 4		89.40 3	
7074.33 6		0.084 eV 3		91.12 4	
7097.72 30		0.009 eV 3	2	115.2 3	
7099.85 20		0.033 eV 5	2	117.4 2	$g\Gamma_n<1$ eV.
7100.73 4	$3/2^-$			118.30 1	$g\Gamma_n=372$ eV 3; $\Gamma_\gamma=0.70$ eV 4.
7143.27 11		0.42 eV 1		162.1 1	$g\Gamma_n<1$ eV.
7210.42 5			1,2	231.25 2	$g\Gamma_n=62$ eV 4; $\Gamma_\gamma=0.30$ eV 2.
7218.0 4		0.33 eV 2		239.0 4	$g\Gamma_n<4$ eV.
7234.0 5		0.19 eV 1		255.5 5	$g\Gamma_n<4$ eV.
7239.8 6		0.06 eV 2		261.5 6	$g\Gamma_n<2$ eV.
7253.2 6		0.35 eV 2		275.3 6	$g\Gamma_n<1$ eV.
7275.93 5	$1/2^+$			298.70 3	$g\Gamma_n=9600$ eV 150; $\Gamma_\gamma=2.82$ eV 67.
7279.13?		0.10 eV 3		302	
7289.8 8		0.31 eV 3		313.0 8	$g\Gamma_n<6$ eV.
7294.19 6	$3/2^-$			317.51 5	$g\Gamma_n=2500$ eV 25; $\Gamma_\gamma=0.23$ eV 8.
7331.00 5	$1/2^+$			355.41 2	$g\Gamma_n=4810$ eV 80; $\Gamma_\gamma=0.30$ eV 30.
7338.18 20			1,2	362.8 2	$g\Gamma_n=230$ eV 10; $\Gamma_\gamma=1.06$ eV 7.
7343.91 20		0.72 eV 3		368.7 2	$g\Gamma_n<2$ eV.
7347.2 5		0.44 eV 6		372.1 5	$g\Gamma_n<2$ eV.
7347.9 5		0.30 eV 5		372.8 5	$g\Gamma_n<2$ eV.
7353.9 5		0.44 eV 4		379.0 5	$g\Gamma_n<10$ eV.
7357.6 5		0.16 eV 4		382.8 5	$g\Gamma_n<6$ eV.
7370.56 6	$1/2^-$			396.14 4	$g\Gamma_n=6430$ eV 150; $\Gamma_\gamma=3.08$ eV 20.
7371.97?		0.12 eV		397.6	
7396.0 6		0.44 eV 33	2	422.3 6	
7404.6 6		0.21 eV 5		431.2 6	$g\Gamma_n<1$ eV.
7408.68 11	$3/2^-$			435.4 1	$g\Gamma_n=1570$ eV 40; $\Gamma_\gamma=0.91$ eV 8.
7411.55 6				438.35 5	$g\Gamma_n=49$ eV 5.
7416.50 4				443.45 1	$g\Gamma_n=80$ eV 8; $\Gamma_\gamma=0.51$ eV 5.
7429.1 6		0.12 eV 3		456.4 6	$g\Gamma_n<6$ eV; $\Gamma_\gamma=0.12$ eV 3.
7433.56 5				461.01 2	$g\Gamma_n=50$ eV 5; $\Gamma_\gamma=0.035$ eV 20.
7436.4 4			1,2	463.9 4	$g\Gamma_n=296$ eV 30; $\Gamma_\gamma=0.28$ eV 3.
7442.14 5	$1/2^+$			469.85 2	$g\Gamma_n=1050$ eV 35; $\Gamma_\gamma=0.37$ eV 8.
7462.2 6		0.33 eV 4		490.5 6	$g\Gamma_n=50$ eV 10.
7481.15 6	$1/2^-$			510.02 4	$g\Gamma_n=375$ eV 25; $\Gamma_\gamma=2.1$ eV 1.
7494.5 6		0.17 eV 4		523.8 6	$g\Gamma_n<4$ eV.
7542.3 6		0.21 eV 3		573.0 6	$g\Gamma_n<1$ eV.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{n,n}),(\text{n},\gamma)$:resonances **1984Ca14** (continued) ^{35}S Levels (continued)

$E(\text{level})^\dagger$	J^π	$g\Gamma_n\Gamma_\gamma/\Gamma$	L	$E_n(\text{lab})$ (keV)	Comments
7604.0 6		0.90 eV 8		636.5 6	$g\Gamma_n < 10$ eV.
7609.26 5	$3/2^-$			641.93 3	$g\Gamma_n = 2520$ eV 60; $\Gamma_\gamma = 0.51$ eV 6.
7648.9 6		0.21 eV 3		682.7 6	$g\Gamma_n < 30$ eV.
7655.7 6		0.80 eV 5		689.7 6	$g\Gamma_n < 10$ eV.
7663.91 11			1,2	698.2 1	$g\Gamma_n = 305$ eV 50; $\Gamma_\gamma = 0.80$ eV 8.
7678.3 7		0.45 eV 7		713.0 7	$g\Gamma_n < 4$ eV.
7731.0 7		0.31 eV 13		767.3 7	$g\Gamma_n < 4$ eV.
7749.7 4				786.5 4	$g\Gamma_n = 18$ eV 6; $\Gamma_\gamma = 2.65$ eV 18.
7761.46 7	$3/2^-$			798.65 6	$g\Gamma_n = 25350$ eV 300; $\Gamma_\gamma = 1.52$ eV 50.
7776.17 11	$1/2^-$			813.8 1	$g\Gamma_n = 860$ eV 60; $\Gamma_\gamma = 0.19$ eV 13.
7797.99 9	$1/2^+$			836.27 8	$g\Gamma_n = 3600$ eV 160; $\Gamma_\gamma = 0.39$ eV 18.
7812.24 7	$3/2^+$			850.94 6	$g\Gamma_n = 2650$ eV 110; $\Gamma_\gamma = 0.24$ eV 19.
7853.25 7	$3/2^-$			893.17 6	$g\Gamma_n = 4950$ eV 160; $\Gamma_\gamma = 0.38$ eV 16.
7861.8 12		1.13 eV 19		902.0 12	$g\Gamma_n < 3$ eV.
7880.8 12		0.67 eV 17		921.5 12	$g\Gamma_n < 20$ eV.
7889.0 12		0.57 eV 25		930.0 12	$g\Gamma_n < 20$ eV.
7894.63 7	$3/2^-$			935.78 6	$g\Gamma_n = 4100$ eV 160; $\Gamma_\gamma = 3.48$ eV 23.
7899.70 30				941.0 3	$g\Gamma_n = 176$ eV 16; $\Gamma_\gamma = 0.5$ eV 3.
7934.1 15		0.5 eV 3		976.4 15	$g\Gamma_n < 2$ eV.
7939.5 15		0.89 eV 4		982.0 15	$g\Gamma_n < 8$ eV.
7954.96 11	$3/2^-$			997.9 1	$g\Gamma_n = 930$ eV 80; $\Gamma_\gamma = 3.51$ eV 19.
7974.19 11	$3/2^-$			1017.7 1	$g\Gamma_n = 4020$ eV 180; $\Gamma_\gamma = 1.22$ eV 17.
8019.5 2			2	1064.4 2	$g\Gamma_n = 1032$ eV 90.
8041.20 20			2	1086.7 2	$g\Gamma_n = 2650$ eV 160.
8076.46 20			2	1123.0 2	$g\Gamma_n = 5480$ eV 388.
8077.91 30			2	1124.5 3	$g\Gamma_n = 4730$ eV 320.
8093.0 5	$1/2^-$			1140.0 5	$g\Gamma_n = 34880$ eV 830.
8143.66 20			2	1192.2 2	$g\Gamma_n = 5510$ eV 290.
8187.17 30	$1/2^-$			1237.0 3	$g\Gamma_n = 2945$ eV 230.
8211.25 30				1261.8 3	$g\Gamma_n = 975$ eV 86.
8224.2 4	$1/2^-$			1275.1 4	$g\Gamma_n = 2170$ eV 180.
8228.3 4			2	1279.4 4	$g\Gamma_n = 1140$ eV 100.
8243.98 30			2	1295.5 3	$g\Gamma_n = 1980$ eV 150.
8256.31 20	$3/2^-$			1308.2 2	$g\Gamma_n = 12475$ eV 770.
8262.4 4	$5/2^+$			1314.5 4	$g\Gamma_n = 6240$ eV 500.
8273.3 4	$(1/2^-)$			1325.7 4	$g\Gamma_n = 1635$ eV 145.
8298.27 30			2	1351.4 3	$g\Gamma_n = 14470$ eV 650.
8333.8 4	$3/2^-$			1388.0 4	$g\Gamma_n = 28845$ eV 1057.
8335.9 4			2	1390.1 4	$g\Gamma_n = 1955$ eV 170.
8391.3 4	$3/2^-$			1447.2 4	$g\Gamma_n = 27100$ eV 1390.
8393.64 30			2	1449.6 3	$g\Gamma_n = 3025$ eV 260.
8406.55 30			2	1462.9 3	$g\Gamma_n = 3885$ eV 345.
8418.8 4			2	1475.5 4	$g\Gamma_n = 3795$ eV 340.

 † From $E_{\text{c.m.}} + S(\text{n})(^{35}\text{S})$ where $S(\text{n})(^{35}\text{S}) = 6985.84$ 4 (2021Wa16) and $E_{\text{c.m.}} = E_n(\text{lab}) \times m(^{34}\text{S}) / [m_n + m(^{34}\text{S})]$.

$^{34}\text{S}(\text{pol d,p})$ 1977Ab07 $J^\pi=0^+$ for ^{34}S g.s.

1977Ab07: A vector-polarized 11.8-MeV deuteron beam was produced from the University of Wisconsin Lamb-shift polarized ion source and tandem accelerator. The target was a 90% enriched ^{34}S , 610- $\mu\text{g}/\text{cm}^2$ thick, evaporated onto a 100- $\mu\text{g}/\text{cm}^2$ gold foil and covered by a 50- $\mu\text{g}/\text{cm}^2$ gold layer. The reaction products were detected using four ΔE -E counter telescopes of freon-cooled surface-barrier and Si(Li) detectors with FWHM $\approx$ 60 keV. Measured angular distributions of cross sections $\sigma(E_p, \theta)$ and angular distributions of analyzing powers $iT_{11}(E_p, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from finite-range, non-locality corrected DWUCK-DWBA analysis of $\sigma(E_p, \theta)$ and $iT_{11}(E_p, \theta)$. Also see 1976AbZX.

 ^{35}S Levels

Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=1.53$ is a normalization factor (1977Ab07).

E(level) [†]	J^π	L [@]	C^2S [@]	Comments
0	$3/2^+ \ddagger$	2	0.56, 0.48 [‡]	
1572	$1/2^+ \#$	0	0.27, 0.18 [#]	
1992	$7/2^- \#$	3	1.16, 0.73 [#]	
2348	$3/2^- \#$	1	0.56, 0.46 [#]	
2718	$5/2^+ \#$	2	0.03, 0.02 [#]	
2939	$3/2^+, 5/2^+$	2	0.10, 0.06	J^π : 1977Ab07 states that the analyzing power seems to favor $5/2^+$ but they do not think that the evidence is strong enough to support an unambiguous spin-parity assignment. C^2S : extracted from an incoherent superposition of direct and compound cross sections in 1977Ab07.
3421	$5/2^+ \#$	2	0.04, 0.03 [#]	
3563				
3802	$3/2^- \#$	1	0.10, 0.08 [#]	
3818				
4190	$1/2^- \ddagger$	1	0.15, 0.13 [‡]	
4482	$7/2^- \#$	3	0.06, 0.05 [#]	
4575				
4837				
4904	$1/2^- \ddagger$	1	0.48, 0.41 [‡]	
4963	$3/2^- \#$	1	0.19, 0.17 [#]	
5058	$7/2^- \#$	3	0.03, 0.03 [#]	
5126				

[†] As given in 1977Ab07, originally from 1973EnVA rounded to the nearest integer.

[‡] L-1/2 transfer from analyzing power measurements.

[#] L+1/2 transfer from analyzing power measurements.

[@] From DWUCK-DWBA analysis of the measured $\sigma(\theta)$ in 1977Ab07. C^2S calculations use two sets of optical model potentials. Set 1: the deuteron elastic potential and the Becchetti-Greenlees proton parameters; Set 2: Adiabatic model parameters and the Becchetti-Greenlees proton parameters.

$^{34}\text{S}(\text{d},\text{p})$ 2024Ku24,1971Va18,1969Mo12

$J^\pi=0^+$ for ^{34}S g.s.

2024Ku24: An 8-MeV/nucleon deuteron beam was produced from the 9-MV Super-FN tandem Van de Graaff accelerator at FSU.

The target was carbon-backed sulfur enriched in ^{34}S . Protons were spatially dispersed by the Super-Enge Split-Pole Spectrograph and detected using a position-sensitive ionization chamber (ΔE) and a plastic scintillator (E) at the focal plane with FWHM $\approx$ 45 keV. An additional $^{34}\text{S}(\text{d},\text{p}\gamma)^{35}\text{S}$ measurement using the CeBr₃ Array (CeBrA) demonstrator provides information on some overlapping peaks produced from the ^{32}S contaminants in the target. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 23 levels, single-neutron spectroscopic factors and strengths from finite-range PTOLEMY-DWBA analysis.

1971Va18: A 10-MeV deuteron beam was produced from the Utrecht 6-MV tandem accelerator. Targets were 100 $\mu\text{g}/\text{cm}^2$ PbS on carbon plus formvar foils and 5 $\mu\text{g}/\text{cm}^2$ pure ^{34}S embedded in aluminum foil. Protons were detected using one 15-mm long and seven 30-mm long, 0.6-mm thick position-sensitive detectors (PSD) in the focal plane of the Utrecht split-pole magnetic spectrograph. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 17 levels, L-transfers, J, π , spectroscopic factors from DWUCK-DWBA analysis for the ground state and seven excited states of ^{35}S .

1969Mo12: A 6.495-MeV deuteron beam was produced from the ONR-CIT tandem accelerator. The target was an enriched target of 450- $\mu\text{g}/\text{cm}^2$ 25 CdS (85% ^{34}S) evaporated onto a 301- $\mu\text{g}/\text{cm}^2$ 15 gold foil and a natural target of 289- $\mu\text{g}/\text{cm}^2$ 30 Sb²S³ evaporated onto a 289- $\mu\text{g}/\text{cm}^2$ 30 gold foil. Protons were detected using an array of 16 Au-Si surface barrier detectors with FWHM=30 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 37 levels.

1984Pi03: A 12.3-MeV deuteron beam was produced from a cyclotron. The target was CdS (61.9% in ^{36}S , 37.7% in ^{34}S), 100 $\mu\text{g}/\text{cm}^2$ on a 20 $\mu\text{g}/\text{cm}^2$ carbon backing. Protons were analyzed by a multi-angle magnetic spectrograph and detected using 700-mm long nuclear emulsion plates Ilford L4. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 13 levels, L-transfers, J, π , and spectroscopic factors from DWUCK-DWBA analysis for six excited states of ^{35}S .

1979So01: A 3.55-MeV deuteron beam was produced from a Van de Graaff electrostatic generator. The target was 18- $\mu\text{g}/\text{cm}^2$ GeS (80% ^{36}S , 20% ^{34}S) on a 25- $\mu\text{g}/\text{cm}^2$ carbon backing. Protons were analyzed by a magnetic spectrograph and recorded by a spark chamber with FWHM=10-15 keV. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels.

1971Ko33: A 6.6-MeV deuteron beam was produced from a cyclotron. Targets were GeS films of 70-80 $\mu\text{g}/\text{cm}^2$ (98% enrichment in ^{34}S) evaporated onto a 20 $\mu\text{g}/\text{cm}^2$ carbon substrate. Protons were detected using an 800- μm -thick Si(Li). Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from distorted-wave analysis for the ground state and five excited states of ^{35}S .

1971Me12: An 18-MeV deuteron beam was produced from the Yale MP tandem Van de Graaff accelerator. The target was H²S gas of natural isotopic composition (95.06% ^{32}S , 4.18% ^{34}S). Protons were detected using a ΔE -E telescope of silicon surface-barrier detectors. Measured $\sigma(E_{\text{p}},\theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from local, zero-range JULIE-DWBA analysis of $\sigma(E_{\text{p}},\theta)$ for the ground state and two excited states of ^{35}S .

1966Sc09: 9- and 12-MeV deuteron beams were produced from the Argonne tandem accelerator. The target was PbS. Protons were detected using silicon surface barrier detectors. Measured $\sigma(E_{\text{p}},\theta)$ for the ground state and one excited state of ^{35}S .

1958En51: 6.006- and 6.542-MeV deuteron beams were produced from the MIT-ONR electrostatic generator. Targets were Sb²S³ of natural isotopic constitution (4.2% ^{34}S). Energies of charged reaction products emitted from the target at angles of 50, 90, and 130° were measured by a broad-range magnetic spectrograph and nuclear emulsions. Measured $\sigma(E_{\text{p}},\theta)$. Deduced 5 levels.

Other measurements: **1976We29:** $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ angular distributions for the ground state and the first excited state of ^{35}S .

1973Co25: $^{34}\text{S}(\text{d},\text{p})^{35}\text{S}$ excitation functions for the two lowest excited states of ^{35}S .

Theoretical calculations: **1977Os07**, **1974Os02**.

 ^{35}S Levels

C²S from **2024Ku24**. The normalization was determined from the L=3 strengths, $G_{L=3}$, summed up through $E_x=5.1$ MeV and divided by the expected 1f_{7/2} vacancy, i.e., $\Sigma G_{L=3}/(2j+1)$, where $j=7/2$. This is equivalent to $\Sigma C^2S_{L=3}=1$ up to 5.1 MeV. For C²S given under comments, $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N/(2J_f+1)\times(2J_i+1)$, where the normalization factor $N=1.53$. The C²S from **1971Me12** based on $N=1.58$ has been renormalized by evaluators using $N=1.53$.

E(level)	J^π	L^a	C ² S ^a	Comments
0	3/2 ⁺	2	0.54 12	L: 2 also from 1971Va18 , 1971Ko33 , and 1971Me12 . C ² S: others: 0.43 11 (1971Va18), 0.33 (1971Ko33), and 0.48 (1971Me12).
1571.92 [#] 19	1/2 ⁺	0	0.14 7	L: 0 also from 1971Va18 , 1971Ko33 , and 1984Pi03 .

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d,p})$ [2024Ku24,1971Va18,1969Mo12](#) (continued) ^{35}S Levels (continued)

E(level)	J^π	L^a	C^2S^a	Comments
1991.08 [#] 16	7/2 ⁻	3	0.87 9	C ² S: others: 0.17 4 (1971Va18), 0.25 (1971Ko33), and 0.154 (1984Pi03). C ² S: others: 0.68 17 (1971Va18), 0.38 8 (1971Ko33), 0.65 10 (1971Me12), and 0.841 12 (1984Pi03).
2347.59 [#] 15	3/2 ⁻	1	0.62 7	E(level): others: 2347 8 (1969Mo12) and 2336 10 (1971Va18). L: 1 also from 1971Va18 and 1984Pi03 . C ² S: others: 0.52 13 (1971Va18), 0.52 (1971Me12), 0.33 (1971Ko33), and 0.505 (1984Pi03).
2724 8	5/2 ⁺	2	0.02 1	E(level): weighted average of 2722 8 (1969Mo12) and 2726 8 (1971Va18). L: other: (2,3) from 1971Va18 .
2941 10	3/2 ⁺	2	<0.02	E(level): weighted average of 2943 10 (1969Mo12) and 2939 10 (1971Va18).
3420 8	5/2 ⁺	2	0.02 1	E(level): weighted average of 3422 8 (1969Mo12) and 3415 12 (1971Va18).
3560 8				E(level): weighted average of 3563 8 (1969Mo12) and 3555 9 (1971Va18).
3596 [†] 8				E(level): other: 3595 9 (1971Va18).
3675 [‡] 10	(1/2 ⁻ , 3/2 ⁻)	(1)		J^π : assigned from L=1 and also adopted in the Adopted Levels. L: from 1971Va18 . C ² S: <2.5×10 ⁻³ for $J^\pi=1/2^-$ and <1.25×10 ⁻³ for $J^\pi=3/2^-$ (1971Va18).
3801.90 [#] 30	3/2 ⁻	1	0.07 4	E(level): others: 3804 8 (1969Mo12) and 3795 10 (1971Va18). L: 1 also from 1971Va18 . C ² S: 0.09 3 (1971Va18).
3885 [†] 10				E(level): other: tentative 3866 10 (1971Va18) and 3907 10 (1971Va18).
4025 10				E(level): weighted average of 4025 10 (1969Mo12) and 4025 10 (1971Va18).
4109 10				E(level): weighted average of 4112 10 (1969Mo12) and 4105 10 (1971Va18).
4189.87 [#] 26	1/2 ⁻	1	0.17 2	E(level): others: 4190 8 (1969Mo12) and 4196 12 (1971Va18). C ² S: other: 0.12 3 (1971Va18).
4305 8				E(level): weighted average of 4302 8 (1969Mo12) and 4312 12 (1971Va18).
4481 [†] 8	3/2 ⁺ , 7/2 ⁻	2+3	0.07, 0.05	E(level): other: 4477 doublet (2024Ku24). J^π : for two components, as given in 2024Ku24 . L: 2,3 doublet (2024Ku24). C ² S: others: 0.07 2 for 3/2 ⁺ component and 0.05 2 for 7/2 ⁻ component (2024Ku24).
4575 [†] 8				
4837 [†] 8				
4903.28 [#] 16	1/2 ⁻	1	0.78 8	E(level): other: 4903 8 (1969Mo12). L: 1 also from 1984Pi03 . C ² S: other: 0.776 (1984Pi03).
4963.12 [#] 16	3/2 ⁻	1	0.32 3	E(level): other: 4965 8 (1969Mo12). L: 1 also from 1984Pi03 . C ² S: other: 0.218 (1984Pi03).
5058 [†] 8	7/2 ⁻	3	0.08 1	
5126 [†] 11				
5342 [†] 8				
5475 [†] 10				Only observed in 1969Mo12 and not included in the Adopted Levels.
5542 [†] 8				
5740 [@] 20	5/2 ⁻ , 7/2 ⁻	3	0.03 1	J^π : assigned from L=3 and also adopted in the Adopted Levels. 2024Ku24 assigns (5/2 ⁻) and states that the J^π assignment is purely speculative.
5890 [@] 20	3/2 ⁺ , 5/2 ⁺	2	0.11 2	J^π : assigned from L=2 and also adopted in the Adopted Levels. 2024Ku24 assigns (3/2 ⁺) and states that the J^π assignment is purely speculative.
5980 [†] 10				
6078.6 [#] 13	1/2 ⁻ , 3/2 ⁻	1		E(level): other: 6080 8 (1969Mo12).

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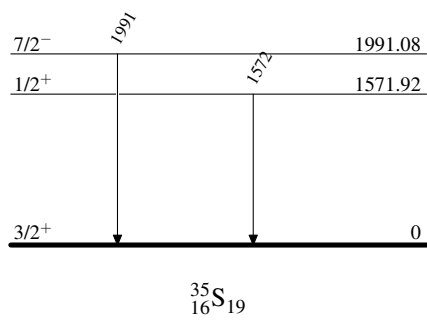
$^{34}\text{S}(\text{d,p})$ [2024Ku24](#), [1971Va18](#), [1969Mo12](#) (continued) ^{35}S Levels (continued)

E(level)	J^π &	L^a	Comments
J^π : assigned from $L=1$ and also adopted in the Adopted Levels. L: from 1984Pi03 . $C^2S=0.086$ for $J^\pi=1/2^-$ and 0.042 for $J^\pi=3/2^-$.			
6292 [†] 8			
6334 [†] 8			
6344 [†] 8			
6446 [†] 8			
6496 [†] 8			
6537.7 [#] 14			
6545.1 [#] 13			E(level): other: 6543 8 (1969Mo12).
6584 [†] 10			
6635.2 [#] 13		(2,3)	E(level): others: 6634 8 (1969Mo12) and 6625 30 (2024Ku24).
6677 [†] 8			
6891.3 [#] 14	(1/2 to 7/2) ⁽⁻⁾	(1,3)	E(level): others: 6892 10 (1969Mo12) and 6915 30 (2024Ku24).
7021 10		(2,3)	E(level): weighted average of 7022 10 (1969Mo12) and 7005 35 (2024Ku24).
7150 [@] 35		(0,3)	
7205 [@] 35		(1,2)	
7482.7 [#] 13		(1,2)	E(level): other: 7495 40 (2024Ku24).
7570 [@] 40		(2,3)	
7640 [@] 40		(2,3)	

[†] From [1969Mo12](#).[‡] From [1971Va18](#).[#] From [1984Pi03](#).[@] From [2024Ku24](#).& Assumed for extracting C^2S , unless otherwise noted.^a From PTOLEMY-DWBA analysis of the measured $\sigma(\theta)$ in [2024Ku24](#), unless otherwise noted. $\gamma(^{35}\text{S})$

E_γ	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
1572	1571.92	1/2 ⁺	0	3/2 ⁺	E_γ : from 2024Ku24 .
1991	1991.08	7/2 ⁻	0	3/2 ⁺	E_γ : from 2024Ku24 .

 $^{34}\text{S}(\text{d,p})$ 2024Ku24,1971Va18,1969Mo12

Level Scheme

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

1972Fr11: 3.50- and 4.00-MeV deuteron beams were produced from the 4-MV Van de Graaff accelerator at the Nuclear Research Center of Strasbourg, France. Targets were antimony sulphide enriched to 85.6%, $550\text{ }\mu\text{g}/\text{cm}^2$ on a 0.03 mm molybdenum layer and $150\text{ }\mu\text{g}/\text{cm}^2$ on a 0.05 mm silver layer. Protons were detected using a surface-barrier annular silicon detector at 180° . γ rays were detected using a 40 cm^3 Ge(Li) detector at 0° , 55° and 90° with FWHM=3.2 keV at $E_\gamma=1.332\text{ MeV}$ in the first experiment and a NaI crystal at 0° , 30° , 45° , 60° and 90° in the second experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 12 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 7 levels using the Doppler-Shift Attenuation Method (DSAM).

1975VaYG,1972Va07: A 4.72-MeV deuteron beam of 175 nA was provided from the Groningen 5 MV Van de Graaff generator. Targets were $80\text{-}\mu\text{g}/\text{cm}^2$ Zn^{34}S enriched to 89% evaporated onto $10\text{-}\mu\text{g}/\text{cm}^2$ Formvar plus $10\text{-}\mu\text{g}/\text{cm}^2$ carbon. Protons were detected using a 2-mm annular silicon detector. γ rays were detected using a $7.6\text{-cm}\times 7.6\text{-cm}$ NaI(Tl) detector. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced levels, J, γ -ray branching ratios, and mixing ratios. Also see **1971VaZG**.

1970Bu18: 2-3.25 MeV deuteron beams were produced from the University of Arizona 5.5 MV Van de Graaff accelerator. Targets were $480\text{-}\mu\text{g}/\text{cm}^2$ $^{120}\text{Ag}_2\text{S}$ made by heating 67% enriched ^{34}S powder and silver foil. Protons were detected using a 1-mm annular silicon surface barrier detector and γ rays were detected using a $7.62\text{ cm}\times 7.62\text{ cm}$ NaI(Tl) detector in the angular correlation experiment and a Ge(Li) detector in the lifetime experiment. Measured E_p , E_γ , I_γ , and $\text{p}\gamma(\theta)$ -coin. Deduced 3 levels, J, γ -ray branching ratios, and mixing ratios. Deduced $T_{1/2}$ for 3 levels using the Doppler-shift attenuation method (DSAM).

2024Co04: A 16-MeV deuteron beam from the John D. Fox Superconducting Linear Accelerator, FSU. Particles were momentum-analyzed by the Super-Enge Split-Pole Spectrograph and identified using a position-sensitive ionization chamber with two anode wires in the isobutane gas (ΔE) and a large plastic scintillator (E) at the focal plane. γ rays were detected using a CeBr_3 Array of five CeBr_3 detectors. Measured $\text{p}\gamma$ delayed coincidence time spectra and deduced $T_{1/2}$ for the 1991 level of ^{35}S .

1971Pr11: A 4.48-MeV deuteron beam was produced from the insulated-core transformer tandem accelerator at the Aerospace Research Laboratories (ARL). The target was Ag_2S enriched to 85.61% in ^{34}S . Protons were detected using a $500\text{ }\mu\text{m}$ -thick, 50-mm^2 silicon detector and γ rays were detected using a 1-in.-thick, 1.5-in. diameter NE111 scintillator. Measured $\text{p}\gamma$ -delayed coincidence time spectrum and deduced $T_{1/2}$ for the 1.99-MeV level.

1972Go31: A 3.3-MeV deuteron beam was produced from the BNL 3.5-MeV van de Graaff facility. The target was Sb_2S_3 enriched to 86% in ^{34}S . γ rays were detected using a Ge(Li) detector. Measured $E_\gamma=1572.24\text{ }17$ deexciting the first excited state.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		
1572.33 20	$1/2^+$	$>1.3\text{ ps}$	E(level): weighted average of 1574.2 10 (1970Bu18), 1572.3 3 (1972Fr11), and 1572.28 17 (1972Go31). J^π : spin= $1/2$ from isotropic $\gamma(\theta)$ in 1972Fr11 and 1972Va07 . $T_{1/2}$: from $\tau>1.8\text{ ps}$ in 1970Bu18 by DSAM. Other: $>1.1\text{ ps}$ (1972Fr11 , $\tau>1.6\text{ ps}$). E(level): unweighted average of 1994.6 10 (1970Bu18) and 1991.2 3 (1972Fr11). J^π : spin= $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Fr11 , 1972Va07 , and 1975VaYG . $T_{1/2}$: from $\tau=1.48\text{ ns}$ 7 from weighted average of 1.7 ns 3 (2024Co04 ,) and 1.47 ns 7 (1971Pr11), both from $\text{p}\gamma(\text{t})$. Other: $\tau>4.5\text{ ps}$ using DSAM (1970Bu18).
1992.9 17	$7/2^-$	$1.03\text{ ns } 5$	E(level): unweighted average of 2350.0 10 (1970Bu18) and 2347.5 2 (1972Fr11). J^π : spin from $\gamma(\theta)$ in 1972Va07 , 1975VaYG , and 1972Fr11 . $T_{1/2}$: from $\tau=1.03\text{ ns}$ 21 from weighted average of 0.8 ps 2 (1970Bu18) and 1.22 ps 18 (1972Fr11), both by DSAM.
2348.8 13	$3/2^-$	$0.71\text{ ps } 15$	J^π : spin= $3/2$, $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Va07 and 1972Fr11 . 2717.9 γ M1+E2 to $3/2^+$. $T_{1/2}$: from $\tau=100\text{ fs}$ 35 in 1972Fr11 by DSAM.
2718.0 10	$5/2^+$	$69\text{ fs } 24$	J^π : spin= $3/2$, $5/2$ from $\gamma(\theta)$ in 1972Va07 and 1975VaYG . $T_{1/2}$: from $\tau=100\text{ fs}$ in 1972Fr11 by DSAM.
2935 3	$3/2^+$		J^π : spin= $3/2$, $5/2$ from $\gamma(\theta)$ in 1972Va07 and 1975VaYG .
3421.0 10	$5/2^+$	$<69\text{ fs}$	J^π : spin= $3/2$, $5/2$, $7/2$ from $\gamma(\theta)$ in 1972Fr11 . $T_{1/2}$: from $\tau<100\text{ fs}$ in 1972Fr11 by DSAM.
3558	$(5/2^+)$		E(level): rounded to the nearest integer based on the Adopted Levels. 1972Fr11 quotes 3563 8 from 1969Mo12 without reporting their own E(level).
3592 5	$(7/2^+)$		
3802.0 10	$3/2^-$	$25\text{ fs } 18$	J^π : spin= $3/2$ from $\gamma(\theta)$ in 1972Fr11 . $T_{1/2}$: from $\tau=36\text{ fs}$ 26 in 1972Fr11 by DSAM.
3815 3	$(9/2)^-$		
3885.0 10	$(5/2)$		

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$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18** (continued) ^{35}S Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
4025.0 10	(3/2 ⁺ ,5/2 ⁺)		
4107 2	(5/2 ⁺)	<55 fs	$T_{1/2}$: from $\tau < 80$ fs in 1972Fr11 by DSAM.
4189 2	1/2 ⁻	<35 fs	$T_{1/2}$: from $\tau < 50$ fs in 1972Fr11 by DSAM.
4480 2	(5/2 ⁺)	<62 fs	$T_{1/2}$: from $\tau < 90$ fs in 1972Fr11 by DSAM.

[†] From the average of available measurements for the lowest three excited states, and from 1972Fr11 for higher-lying states.

[‡] From the Adopted Levels. Supporting arguments from this dataset are given under comments if available.

 $\gamma(^{35}\text{S})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.	δ	Comments
1572.33	1/2 ⁺	1572.24 17	100	0	3/2 ⁺			E_γ : from 1972Go31. $A_2 = -0.09$ 5, $A_4 = +0.07$ 6 (1972Fr11); $A_2 = -0.007$ 10, $A_4 = +0.004$ 20 (1975VaYG).
1992.9	7/2 ⁻	1992.8	100	0	3/2 ⁺	M2+E3	+0.17 6	Mult.: D+Q or Q+O from $\gamma(\theta)$ in 1970Bu18, 1972Fr11, 1972Va07, and 1975VaYG; E2+M3 is ruled out by RUL. D+Q is inconsistent with the $J^\pi = 7/2^-$. δ : unweighted average of $\delta(\text{E3/M2}) = 0.11$ 4 (1970Bu18), +0.11 4 (1972Fr11), and +0.28 3 (1972Va07,1975VaYG) for $J_i = 7/2$. Others: $\delta(\text{Q/D}) = -1.9$ 7 (1970Bu18) and -0.43 3 (1972Va07,1975VaYG) for $J_i = 5/2$. $A_2 = +0.35$ 4, $A_4 = -0.39$ 4 (1972Fr11); $A_2 = +0.16$ 3, $A_4 = -0.10$ 4 (1975VaYG).
2348.8	3/2 ⁻	776.5	27 2	1572.33	1/2 ⁺	(E1)		Mult., δ : D(+Q) from $\gamma(\theta)$, $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG, superseding $0 < \delta < +1.7$ from $\gamma(\theta)$ in 1972Va07. $A_2 = -0.37$ 3, $A_4 = -0.03$ 3 (1972Fr11); $A_2 = -0.38$ 2, $A_4 = +0.01$ 4 (1975VaYG).
		2348.7	73 2	0	3/2 ⁺	(E1)		Mult., δ : D(+Q) from $\gamma(\theta)$, $\Delta\pi = \text{yes}$ from level scheme. M2 component is less likely. $\delta < 0.04$ from $\gamma(\theta)$ and RUL in 1975VaYG, superseding -0.27 12 from $\gamma(\theta)$ in 1972Va07; +0.01 4 from $\gamma(\theta)$ in 1972Fr11 assuming pure E1 for 776.5 γ . $A_2 = +0.33$ 2, $A_4 = -0.05$ 2 (1972Fr11); $A_2 = +0.30$ 2, $A_4 = -0.02$ 4 (1975VaYG).
2718.0	5/2 ⁺	369.2 725.1 1145.7 2717.9	<5 <5 <5 100	2348.8 1992.9 1572.33 0	3/2 ⁻ 7/2 ⁻ 1/2 ⁺ 3/2 ⁺	M1+E2	-0.39 5	Mult., δ : D+Q from $\gamma(\theta)$ in 1975VaYG with $-6.9 < \delta < -0.11$ or $+0.42 < \delta < +6.8$ for $J_i = 3/2$, -0.39 5 or -15.0 34 for $J_i = 5/2$, 0.18 4 or 3.8 7 for $J_i = 7/2$. E1+M2 is ruled out by RUL. $\delta = -0.39$ 5 is adopted here based on $J^\pi = 5/2^+$ in the Adopted Levels. -15.0 34 is unlikely based on RUL. $A_2 = +0.47$ 6, $A_4 = -0.08$ 7 (1972Fr11); $A_2 = +0.22$ 2, $A_4 = -0.02$ 3 (1975VaYG).
2935	3/2 ⁺	586 942	<10 <20	2348.8 1992.9	3/2 ⁻ 7/2 ⁻			

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d},\text{p}\gamma)$ **1972Fr11,1975VaYG,1970Bu18** (continued) $\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Comments
2935	$3/2^+$	1363	<15	1572.33	$1/2^+$	
		2935	100	0	$3/2^+$	$A_2=-0.06$ 3, $A_4=-0.01$ 6 (1975VaYG).
3421.0	$5/2^+$	486	<4	2935	$3/2^+$	
		703.0	<6	2718.0	$5/2^+$	
		1072.2	<7	2348.8	$3/2^-$	
		1428.1	<30	1992.9	$7/2^-$	
		1848.6	<7	1572.33	$1/2^+$	
		3420.8	100	0	$3/2^+$	$A_2=-0.54$ 8, $A_4=-0.11$ 7 (1972Fr11).
3592	$(7/2^+)$	657	<10	2935	$3/2^+$	
		874	<11	2718.0	$5/2^+$	
		1243	<15	2348.8	$3/2^-$	
		1599	<17	1992.9	$7/2^-$	
		2020 ^{#@}	<17	1572.33	$1/2^+$	
		3592	100	0	$3/2^+$	
3802.0	$3/2^-$	867	<4	2935	$3/2^+$	
		1084.0	5 2	2718.0	$5/2^+$	
		1453.2	8 3	2348.8	$3/2^-$	
		1809.1	<4	1992.9	$7/2^-$	
		2229.6	42 4	1572.33	$1/2^+$	$A_2=-0.56$ 15, $A_4=0.0$ 1 (1972Fr11).
		3801.8	45 4	0	$3/2^+$	$A_2=+0.19$ 12, $A_4=0.0$ 1 (1972Fr11).
3815	$(9/2)^-$	1822	100	1992.9	$7/2^-$	
3885.0	$(5/2)$	464.0	<30	3421.0	$5/2^+$	
		950	<12	2935	$3/2^+$	
		1167.0	<15	2718.0	$5/2^+$	
		1536.2	<18	2348.8	$3/2^-$	
		1892.1	100	1992.9	$7/2^-$	
		2312.6	<16	1572.33	$1/2^+$	
		3884.8	<16	0	$3/2^+$	
4025.0	$(3/2^+, 5/2^+)$	467	<5	3558	$(5/2^+)$	
		604.0	<10	3421.0	$5/2^+$	
		1090	31 8	2935	$3/2^+$	
		1307.0	<8	2718.0	$5/2^+$	
		1676.2	33 12	2348.8	$3/2^-$	
		2032.0	<10	1992.9	$7/2^-$	
		2452.6	36 12	1572.33	$1/2^+$	
		4024.8	<13	0	$3/2^+$	
4107	$(5/2^+)$	549	<4	3558	$(5/2^+)$	
		686	<4	3421.0	$5/2^+$	
		1172	13 6	2935	$3/2^+$	
		1389	<10	2718.0	$5/2^+$	
		1758	<5	2348.8	$3/2^-$	
		2114	<7	1992.9	$7/2^-$	
		2535	<5	1572.33	$1/2^+$	
		4107	87 6	0	$3/2^+$	
4189	$1/2^-$	387 ^{#@}	<4	3802.0	$3/2^-$	
		597 ^{#@}	<4	3592	$(7/2^+)$	
		631	<4	3558	$(5/2^+)$	
		768 ^{#@}	<4	3421.0	$5/2^+$	
		1254	<5	2935	$3/2^+$	
		1471 ^{#@}	<5	2718.0	$5/2^+$	
		1840	48 5	2348.8	$3/2^-$	
		2196 ^{#@}	<8	1992.9	$7/2^-$	
		2617	<6	1572.33	$1/2^+$	
		4189	52 5	0	$3/2^+$	

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18 (continued) $\gamma(^{35}\text{S})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
4480	(5/2 ⁺)	595	<5	3885.0	(5/2)	4480	1762	61 7	2718.0	5/2 ⁺
		678	<5	3802.0	3/2 ⁻		2131	<8	2348.8	3/2 ⁻
		888	<5	3592	(7/2 ⁺)		2487	<10	1992.9	7/2 ⁻
		922	<5	3558	(5/2 ⁺)		2908	39 7	1572.33	1/2 ⁺
		1059	<10	3421.0	5/2 ⁺		4480	<12	0	3/2 ⁺
		1545	<7	2935	3/2 ⁺					

[†] Deduced by evaluators from level-energy difference, unless otherwise noted.

[‡] From 1972Fr11, unless otherwise noted.

[#] γ transitions are considered questionable and not included in the Adopted Levels. The transition strengths estimated by evaluators exceed RUL. $^{34}\text{S}(\text{n},\gamma)$ E=thermal observed other γ transitions from the same level but did not observe these questionable transitions.

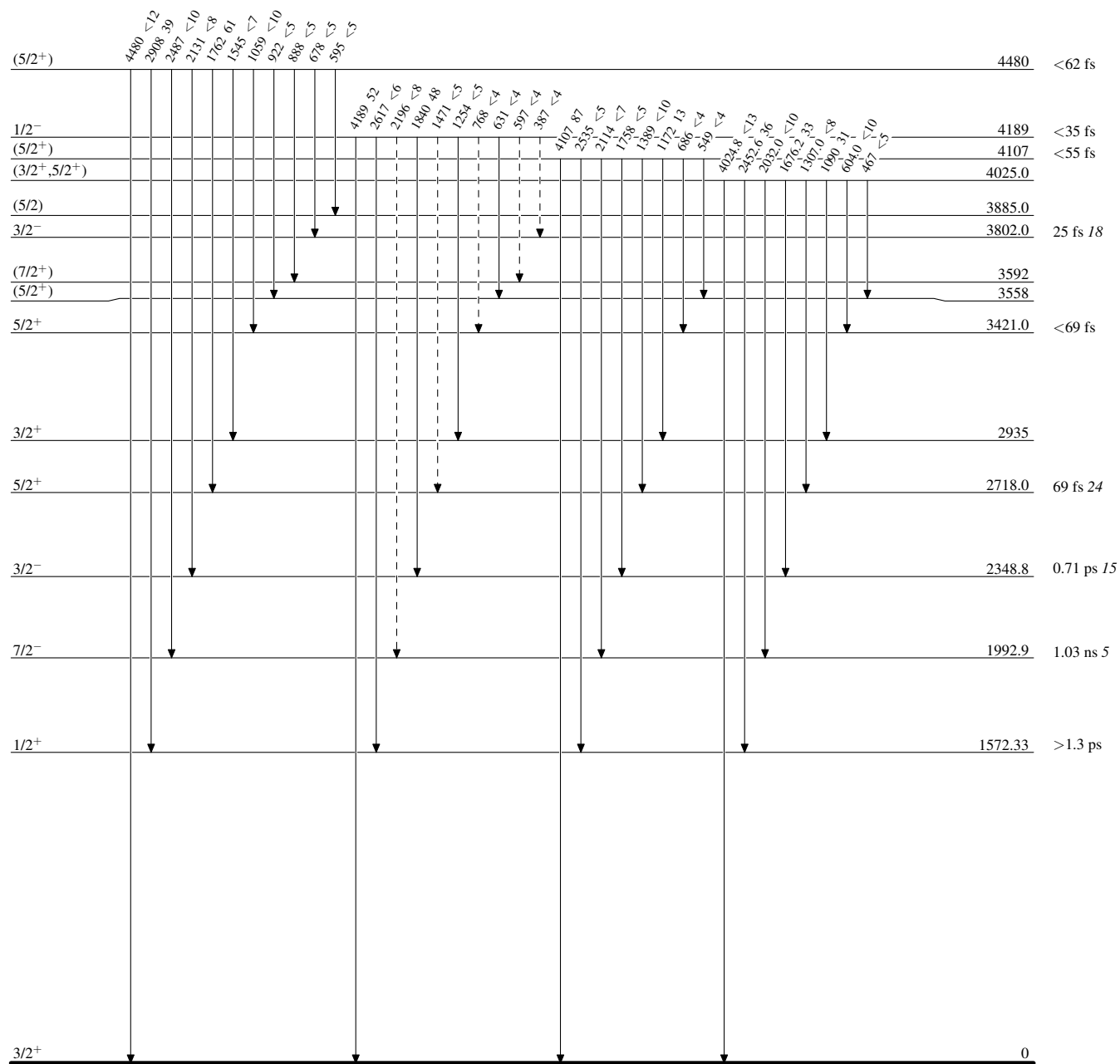
[@] Placement of transition in the level scheme is uncertain.

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

Legend

Level Scheme

Intensities: % photon branching from each level

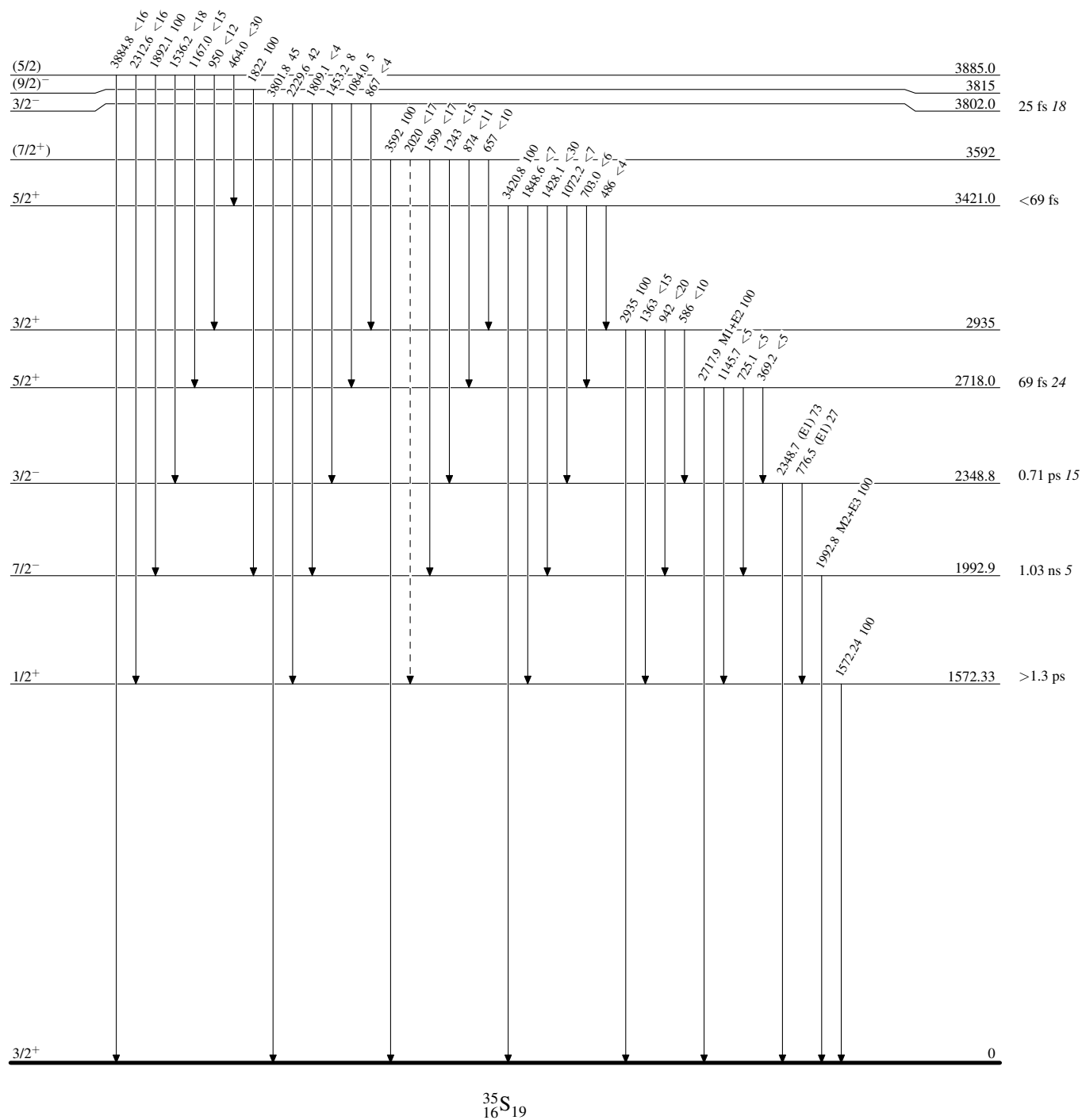
-----► γ Decay (Uncertain) $^{35}_{16}\text{S}_{19}$

$^{34}\text{S}(\text{d},\text{p}\gamma)$ 1972Fr11,1975VaYG,1970Bu18

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

-----► γ Decay (Uncertain)

$^{36}\text{S}(\text{p,d})$ 2026Jo01

$J^\pi=0^+$, $T=2$ for ^{36}S g.s.

2026Jo01: E=66 MeV dispersion-matched proton beam was produced from the Separated Sector Cyclotron facility of iThemba LABS, South Africa. The target was 1-mg/cm², 99.24%-enriched ^{36}S powder encapsulated between two 0.5- μm Mylar foils. The outgoing deuteron and triton reaction products were momentum-analyzed by the K=600 quadrupole-dipole-dipole magnetic spectrometer and detected by the focal-plane detectors consisting of two multiwire drift chambers (MWDC) and a downstream plastic scintillator. Measured $\sigma(E_d, \theta)$. Deduced 98 levels, L-transfers, and single-neutron spectroscopic factors for 81 levels from zero-range adiabatic distorted-wave approximation (ADWA) calculations using DWUCK4. Deduced J-dependence of L=2 angular distributions and reproduced with finite-range ADWA calculations. Comparisons with large-scale shell-model and ab initio calculations. Confirmed the N=20 shell closure in ^{36}S .

Also see Ph.D. Thesis 2021JoZZ.

 ^{35}S Levels

T: $T=3/2$ or $5/2$ for ^{35}S levels populated in the $^{36}\text{S}(\text{p,d})$ reaction based on $T=2$ for the 0^+ g.s. of ^{36}S . ^{35}S levels below 9 MeV are likely $T=3/2$ and those above 9 MeV are likely $T=5/2$ considering the binding, Coulomb, and proton-neutron mass differences suggest the IAS of ^{35}P ground state ($T=5/2$) should be around 9017 keV in ^{35}S . C^2S extracted for these ^{35}S levels show excellent agreement with proton-removal data from ^{35}P when scaled by expected Clebsch-Gordan ratio of 0.167 (2026Jo01).

E(level)	J^π ^{†‡}	L [†]	C^2S ^{†#}	Comments
0 5	$3/2^+$	2	3.5 4	J^π : L-1/2 transfer from J-dependence.
1569 5	$1/2^+$	0	1.28 14	
1990 5	$7/2^-$	3	0.170 19	
2348 5	$3/2^-$	1	0.033 4	
2718 5	$5/2^+$	2	0.53 6	J^π : L+1/2 transfer from J-dependence.
2937 5	$3/2^+$	2	0.034 4	J^π : L-1/2 transfer from J-dependence.
3420 5	$5/2^+$	2	0.50 6	J^π : L+1/2 transfer from J-dependence.
3558 5	($5/2^+$)	(2)	0.0300 34	
3592 5	($5/2^+$)	(2)	0.0040 5	
3800 5	$3/2^-$	1	0.0200 23	
3883 5	$7/2^-$	3	0.0180 20	
4023 5	($3/2^+$)	(2)	0.0056 6	
4107 7	($3/2^+$)	(2)	0.00230 26	
4182 5	$1/2^-$	1	0.00100 11	
4300 7	($1/2^-, 3/2^-$)	(1)	3.00×10^{-4} 34	
4486 7	$7/2^-$	3	0.045 5	
4574 5	$3/2^+$	2	0.080 9	J^π : L-1/2 transfer from J-dependence.
4614 5	$5/2^+$	2	0.110 12	
4838 5	($1/2^+$)	(0)	0.0035 4	
4907 5	$1/2^+$	0	0.0110 12	
4955 5	$5/2^+$	2	0.250 28	J^π : L+1/2 transfer from J-dependence.
5121 5	$1/2^+$	0	0.0092 10	
5286 5	($1/2^-, 3/2^-$)	(1)	6.5×10^{-4} 7	
5476 5	$3/2^+$	2	0.0090 10	
5630 5	$5/2^+$	2	0.110 12	
5766 5	$5/2^+$	2	0.76 9	J^π : L+1/2 transfer from J-dependence.
5844 5	($5/2^+$)	(2)	0.080 9	
6121 5	($3/2^+$)	(2)	0.035 4	
6218 5	($7/2^-$)	(3)	0.00300 34	
6338 5	$1/2^+$	0	0.044 5	
6439 5	($1/2^+$)	(0)	0.00260 29	
6550 5	$3/2^+$	2	0.0035 4	
6639 5	$5/2^+$	2	0.250 28	
6682 5	$1/2^+$	0	0.056 6	
6790 5	($5/2^+$)	(2)	0.037 4	
6962 5	($1/2^-, 3/2^-$)	(1)	0.0050 6	

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$^{36}\text{S}(\text{p,d})$ [2026Jo01](#) (continued) ^{35}S Levels (continued)

E(level)	J^π	$L^\dagger$	$C^2S^\dagger\#$	Comments
7019 5	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00150 17	
7102 5	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00200 23	
7151 5	(7/2 ⁻)	(3)	0.0090 10	
7215 5	3/2 ⁺	2	0.0120 14	
7249 5	(3/2 ⁺)	(2)	0.0120 14	
7279 5	1/2 ⁺	0	0.0068 8	
7326 5	1/2 ⁺	0	0.0073 8	
7349 5	(1/2 ⁺)	(0)	0.0095 11	
7441 5	1/2 ⁺	0	0.0094 11	
7490 5	1/2 ⁻	1	0.0060 7	
7753 5	5/2 ⁺	2	0.200 23	
7885 5	(5/2 ⁺)	(2)	0.0300 34	
7980 7	3/2 ⁻	1	0.00200 23	
8073 7	(5/2 ⁺)	(2)	0.0180 20	
8103 5	1/2 ⁻	1	0.00170 19	
8221 5	1/2 ⁻	1	0.0040 5	
8270 5	1/2 ⁻	1	0.0090 10	
8410 7	(5/2 ⁺)	(2)	0.0100 11	
8465 7	(5/2 ⁺)	(2)	0.0100 11	
8509 5	5/2 ⁺	2	0.0120 14	
8557 7	(7/2 ⁻)	(3)	0.00300 34	
8602 5	(7/2 ⁻)	(3)	0.0100 11	
8649 7	(7/2 ⁻)	(3)	0.0065 7	
8707 5	(5/2 ⁺)	(2)	0.0220 25	
8761 7	(5/2 ⁺)	(2)	0.0110 12	
8822 5	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0060 7	
8876 5	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00250 28	
8947 7	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00100 11	
9032 15	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0070 8	
9065 15	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0070 8	
9126 7	1/2 ⁺	0	0.35 4	T=5/2 J^π, T : isobaric analog state of 1/2 ⁺ , T=5/2, g.s. of ^{35}P (2026Jo01).
9180 5	1/2 ⁺	0	0.060 7	
9219 15	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00300 34	
9294 8	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0100 11	
9339 8	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00200 23	
9395 5	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0045 5	
9446 7	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00180 20	
9510 10	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0035 4	
9565 20	(1/2 ⁻ ,3/2 ⁻)	(1)	0.0033 4	
9615 10	(1/2 ⁻ ,3/2 ⁻)	(1)	0.00280 32	
9666 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
9716 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
9774 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
9849 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
9911 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
9989 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0049×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0166×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0225×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0277×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0350×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0421×10 ⁴ 10	(1/2 ⁻ ,3/2 ⁻)	(1)		
1.0493×10 ⁴ 10	(7/2 ⁻)	(3)	0.0200 23	

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$^{36}\text{S}(\text{p,d})$ [2026Jo01](#) (continued) ^{35}S Levels (continued)

E(level)	J^π ^{†‡}	L [†]	C^2S ^{†#}	Comments
1.0563×10^4 10	(1/2 ⁻ , 3/2 ⁻)	(1)		
1.0663×10^4 10	(1/2 ⁻ , 3/2 ⁻)	(1)		
1.0746×10^4 10	(1/2 ⁻ , 3/2 ⁻)	(1)		
1.0810×10^4 10	(1/2 ⁻ , 3/2 ⁻)	(1)		
1.147×10^4 2	3/2 ⁺	2	0.052 6	T=5/2 J^π, T : IAS of 3/2 ⁺ , T=5/2, 2387 level in ^{35}P (2026Jo01).
1.288×10^4 10	5/2 ⁺	2	0.49 6	T=5/2 J^π, T : IAS of 5/2 ⁺ , T=5/2, 3860 level in ^{35}P (2026Jo01).
1.368×10^4 10	5/2 ⁺	2	0.156 18	T=5/2 J^π, T : IAS of 5/2 ⁺ , T=5/2, 4666 level in ^{35}P (2026Jo01).
1.422×10^4 10	5/2 ⁺	2	0.247 28	T=5/2 J^π, T : IAS of 5/2 ⁺ , T=5/2, 5199 level in ^{35}P (2026Jo01).
1.470×10^4 10				

[†] From ADWA analysis of the measured $\sigma(\theta)$ in [2026Jo01](#). L-transfers are not specifically given but can be inferred from the reported J^π .

[‡] As given in [2026Jo01](#), which selected some $J=L-1/2$ or $L+1/2$ based on known J^π values ([2011Ch48](#)). When considered in the Adopted Levels, $J=L-1/2$ and $L+1/2$ are both possible unless other constraints are available.

[#] C^2S from [2026Jo01](#). For levels with $J^\pi=(1/2^-, 3/2^-)$, the C^2S values are given for 3/2⁻. C^2S is normalized to a ^{40}Ca benchmark where Woods-Saxon parameters ($r_0=1.27$ fm, $a_0=0.70$ fm) were adjusted to ensure the summed strength of the $1d_{3/2}$, $2s_{1/2}$, and $1f_{7/2}$ orbitals equals 6. A systematic uncertainty of 11.2% is associated with all spectroscopic factors due to the effective target-thickness variation (10%) and the nominal target thickness uncertainty (5%) ([2026Jo01](#)). This uncertainty is included by the evaluators.

$^{37}\text{Cl}(\text{p}, ^3\text{He})$ 1975Gu15

$^{37}\text{Cl} \rightarrow ^{35}\text{S} + \text{p} + \text{n}$ from $J^\pi = 3/2^+$ ^{37}Cl ground state.

1975Gu15: A 40.2-MeV proton beam was produced from the Michigan State University cyclotron. The target was $55 \mu\text{g}/\text{cm}^2$ NaCl with 97% enriched ^{37}Cl made by evaporation of the salt onto a $30\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. The reaction products were detected using a wire-counter plastic-scintillator in the focal plane of an Enge split-pole spectrograph with FWHM=30 keV. Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced levels, L from zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. The target was natural purified chlorine gas 100 mm in diameter and 25 mm in height. The reaction particles were detected using two separate counter telescopes with each consisting of a 200- μm phosphorous-drifted silicon ΔE detector, a 2-mm lithium-drifted silicon E detector and a 3-mm lithium-drifted silicon E-reject detector with FWHM=180 keV for ^3He . Measured $\sigma(\text{E}(^3\text{He}), \theta)$. Deduced L for ^{35}S ground state from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}S Levels

σ_{max} under comments are from 1975Gu15.

$\text{E}(\text{level})^\dagger$	$\text{L}^\ddagger$	Comments
0	0+2+4	$\sigma_{\text{max}} = 26.5 \mu\text{b}/\text{sr}$.
1575 10	0+2	$\sigma_{\text{max}} = 19 \mu\text{b}/\text{sr}$.
1992 10	3+5	$\sigma_{\text{max}} = 1.5 \mu\text{b}/\text{sr}$.
2717 10	0+2+4	$\sigma_{\text{max}} = 53.9 \mu\text{b}/\text{sr}$.
2938 10	0+2+4	$\sigma_{\text{max}} = 29.2 \mu\text{b}/\text{sr}$.
3421 10	0+2+4	$\sigma_{\text{max}} = 81.5 \mu\text{b}/\text{sr}$.
3598 10	2+4	$\sigma_{\text{max}} = 27.8 \mu\text{b}/\text{sr}$.
3811 10	3	$\sigma_{\text{max}} = 2.6 \mu\text{b}/\text{sr}$.
4027 10	2	$\sigma_{\text{max}} = 2.9 \mu\text{b}/\text{sr}$.
4114 10	0+2	$\sigma_{\text{max}} = 8.8 \mu\text{b}/\text{sr}$.
4186 10	(2,3)	$\sigma_{\text{max}} = 5.6 \mu\text{b}/\text{sr}$.
4290 10	(2)	$\sigma_{\text{max}} = 2.8 \mu\text{b}/\text{sr}$.
4489 10	2	$\sigma_{\text{max}} = 2.7 \mu\text{b}/\text{sr}$.
4577 10	0+2	$\sigma_{\text{max}} = 10.7 \mu\text{b}/\text{sr}$.
4617 10	(1,2)	$\sigma_{\text{max}} = 13.2 \mu\text{b}/\text{sr}$.
4843 10	2	$\sigma_{\text{max}} = 15.9 \mu\text{b}/\text{sr}$.
4963 10	(0+2)	$\sigma_{\text{max}} = 17.3 \mu\text{b}/\text{sr}$.
4990 10	0+2	$\sigma_{\text{max}} = 19.5 \mu\text{b}/\text{sr}$.
5127 10	2	$\sigma_{\text{max}} = 3.5 \mu\text{b}/\text{sr}$.
5345 10	3	$\sigma_{\text{max}} = 2.2 \mu\text{b}/\text{sr}$.
5550 10	(3)	$\sigma_{\text{max}} = 6.2 \mu\text{b}/\text{sr}$.
5771 10	2	$\sigma_{\text{max}} = 10.1 \mu\text{b}/\text{sr}$.
5915? 10	(2,3)	$\sigma_{\text{max}} = 3.4 \mu\text{b}/\text{sr}$.
6129 10	0+2	$\sigma_{\text{max}} = 7.8 \mu\text{b}/\text{sr}$.
6347 10	(2)	$\sigma_{\text{max}} = 4.8 \mu\text{b}/\text{sr}$.
6654 10	(3)	$\sigma_{\text{max}} = 7.7 \mu\text{b}/\text{sr}$.
6696 10	2	$\sigma_{\text{max}} = 6.7 \mu\text{b}/\text{sr}$.
7151? 10	(4)	$\sigma_{\text{max}} = 3.3 \mu\text{b}/\text{sr}$.
7375? 10		
7712? 10	4	$\sigma_{\text{max}} = 4.6 \mu\text{b}/\text{sr}$.
7770 10		$\sigma_{\text{max}} = 5.5 \mu\text{b}/\text{sr}$.
8103 10	(1+3)	$\sigma_{\text{max}} = 10.6 \mu\text{b}/\text{sr}$.
8160 10	(1)	$\sigma_{\text{max}} = 8.4 \mu\text{b}/\text{sr}$.
8430 10	2	$\sigma_{\text{max}} = 5.7 \mu\text{b}/\text{sr}$.
9155 10	2	$\sigma_{\text{max}} = 10 \mu\text{b}/\text{sr}$.

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$^{37}\text{Cl}(\text{p}, ^3\text{He})$ **1975Gu15** (continued)

^{35}S Levels (continued)

[†] From **1975Gu15**.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in **1975Gu15**. L is not considered firm by the evaluators when assigning J^π in the Adopted Levels because the theory of $(\text{p}, ^3\text{He})$ reactions is not robust.

$^{37}\text{Cl}(\text{d},\alpha\gamma)$ 1975VaYG,1972Va07

1975VaYG,1972Va07: A 4.25-MeV deuteron beam of 55 nA was provided from the Groningen 5 MV Van de Graaff generator.

The target was $100\text{ }\mu\text{g}/\text{cm}^2$ Co^{37}Cl enriched to 98% evaporated onto $10\text{-}\mu\text{g}/\text{cm}^2$ Formvar plus $10\text{-}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using a $60\text{-}\mu\text{m}$ annular silicon detector. γ rays were detected using a 120 cm^3 Ge(Li) detector at 90° . Measured $\sigma(E_\alpha)$, E_γ , I_γ , and $\alpha\gamma$ -coin. Deduced levels, γ -ray branching ratios. Also see 1971VaZG.

1968Te06: 3.1-4.6-MeV deuteron beams of 50 nA were provided from the Groningen 5-MV Van de Graaff generator. The target was a $100\text{-}\mu\text{g}/\text{cm}^2$ Co^{37}Cl both of natural ^{37}Cl abundance and enriched to 93% evaporated onto $10\text{-}\mu\text{g}/\text{cm}^2$ Formvar plus $10\text{-}\mu\text{g}/\text{cm}^2$ carbon. α particles were detected using an annular solid-state detector at $168\text{-}173^\circ$. γ rays were detected using a 3 in. by 3 in. NaI(Tl) scintillator at 55° . Measured $\sigma(E_\alpha)$, E_γ , and $\alpha\gamma$ -coin. Deduced levels.

1955Pa54: $^{37}\text{Cl}(\text{d},\alpha)^{35}\text{S}$ with 3.0, 5.6, 6.0, 7.0, and 7.5-MeV deuteron beams from the MIT-ONR electrostatic generator. Targets were 80 and $300\text{-}\mu\text{g}/\text{cm}^2$ barium chloride (75.4% ^{35}Cl , 24.6% ^{37}Cl) evaporated onto Formvar films on a gold layer. Charged reaction products emitted at 90° were magnetically analyzed by a broad-range spectrograph. Measured $\sigma(E_\alpha)$. Deduced levels.

 ^{35}S Levels

$E(\text{level})^\dagger$	Comments
0	
1572.2 12	
1990.0 10	E(level): Other: 1992 10 from 1955Pa54.
2348.2 10	E(level): Other: 2348 10 from 1955Pa54.
2716.7 10	E(level): Other: 2714 10 from 1955Pa54.
2939.2 13	
3423 5	
3560.8 19	
3598.4 21	
3803.6 19	
3818.1 11	
3889.0 19	
4022.2 22	
4027.7 10	E(level): Other: 4025 10 from 1955Pa54.
4108 3	
4180 3	
4187 3	
4302 4	
4480.0 16	

† From 1975VaYG based on measured E_γ , but the E_γ values are not provided in 1975VaYG.

 $\gamma(^{35}\text{S})$

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	Comments
1572.2	1572.2	100	0	
1990.0	1989.9	100	0	
2348.2	776.0	25 2	1572.2	
	2348.1	75 2	0	
2716.7	2716.6	100	0	
2939.2	2939.1	100	0	
3423	3423	100	0	
3560.8	1212.6	35 4	2348.2	
	1570.8	65 4	1990.0	
3598.4	3598.2	100	0	I_γ : >95 in 1975VaYG.
3803.6	2231.3	38 3	1572.2	
	3803.4	62 3	0	
3818.1	1828.1	100	1990.0	
3889.0	1540.8	40 4	2348.2	
	1898.9	45 5	1990.0	

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$^{37}\text{Cl}(\text{d},\alpha\gamma)$ [1975VaYG,1972Va07](#) (continued) $\gamma(^{35}\text{S})$ (continued)

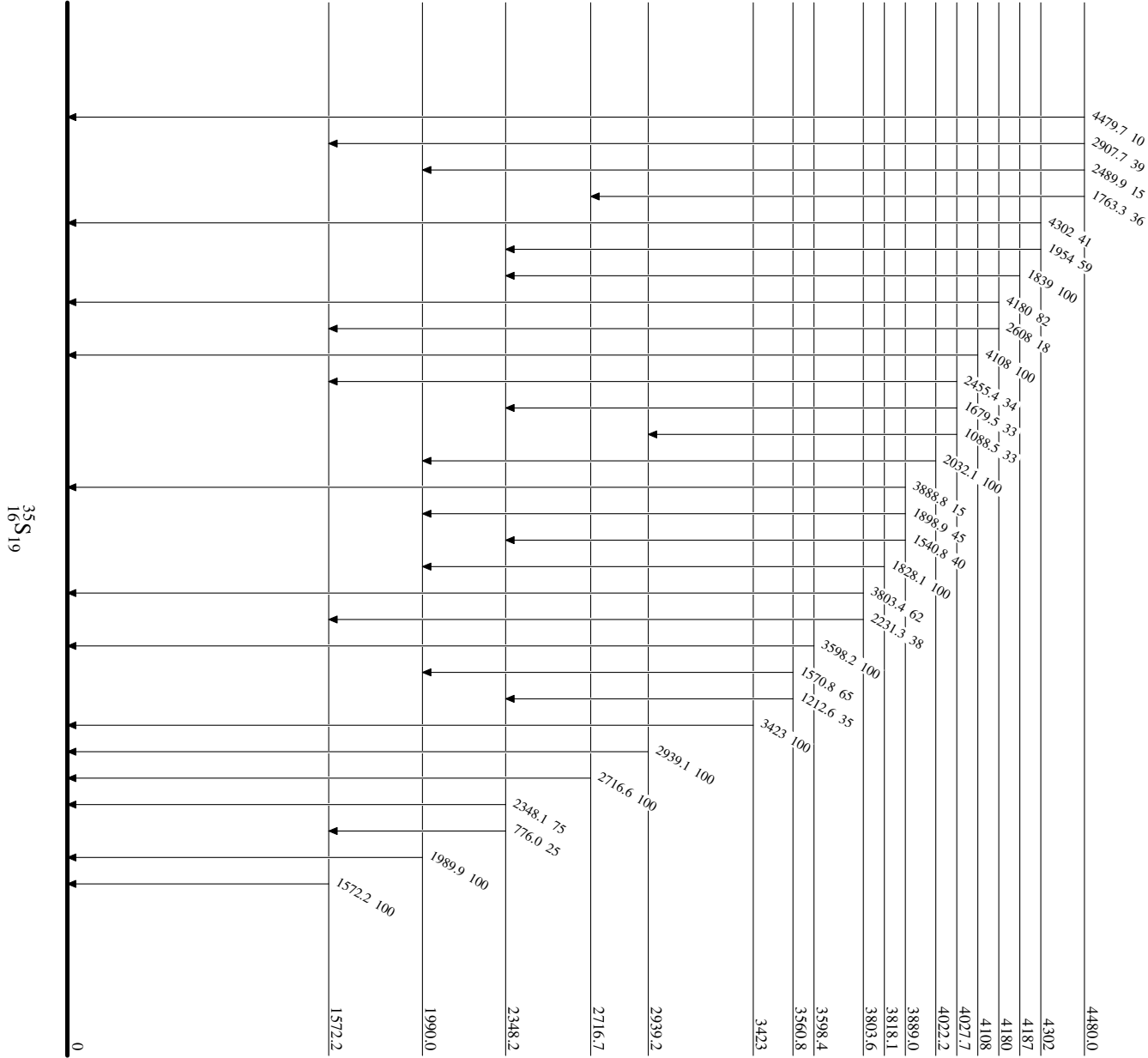
$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	Comments
3889.0	3888.8	15 3	0	
4022.2	2032.1	100	1990.0	
4027.7	1088.5	33 4	2939.2	
	1679.5	33 4	2348.2	
	2455.4	34 6	1572.2	
4108	4108	100	0	I_γ : >95 in 1975VaYG .
4180	2608	18 5	1572.2	
	4180	82 5	0	
4187	1839	100	2348.2	I_γ : >95 in 1975VaYG .
4302	1954	59 5	2348.2	
	4302	41 5	0	
4480.0	1763.3	36 4	2716.7	
	2489.9	15 4	1990.0	
	2907.7	39 4	1572.2	
	4479.7	10 2	0	

[†] Deduced by evaluators from level-energy difference in [1975VaYG](#).[‡] From [1975VaYG](#).

³⁷Cl(d,αγ) ¹⁹⁷5VaYG,1972Va07

Level Scheme

Intensities: % photon branching from each level



$^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04

1994Fo04: 159-MeV ^{34}S and 167-MeV ^{37}Cl beams were produced from the Argonne Tandem Linac Accelerator System (ATLAS). Targets were 1 mg/cm² 98.1% enriched ^{160}Gd backed by 15 mg/cm² gold. γ rays were detected using the Argonne-Notre Dame BGO γ -ray facility consisting of 12 Compton-suppressed Ge detectors and a 50-element bismuth germanate (BGO) array. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin. Deduced levels. $\gamma\gamma$ -coin gates were placed on known γ rays in specific $A\approx 160$ products to select individual reaction channels and identify coincident γ rays in light product partners. $\gamma\gamma$ -correlation matrices produced with the fold condition $4 \leq k \leq 9$ were found to include most of the events associated with deep inelastic processes.

 ^{35}S Levels

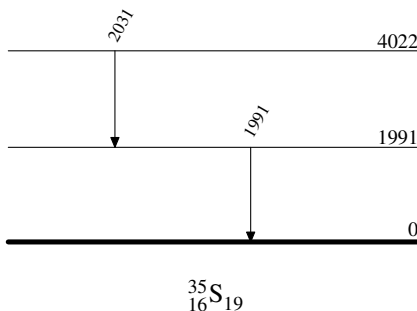
<u>$E(\text{level})^\dagger$</u>	<u>Comments</u>
0	
1991	
4022	J^π : 1994Fo04 suggests this level is a likely candidate to be the expected $9/2^-$ yrast excitation, while $(11/2)^-$ is adopted in the Adopted Levels.

† From $E\gamma$ data in 1994Fo04.

 $\gamma(^{35}\text{S})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>
1991	1991	0
2031	4022	1991

† From 1994Fo04.

 $^{160}\text{Gd}(^{34}\text{S},\text{X}\gamma),(^{37}\text{Cl},\text{X}\gamma)$ 1994Fo04Level Scheme

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2010WaZT,2022Gr07

2022Gr07: A 215-MeV ^{36}S beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 1 mg/cm² 99.7% enriched ^{208}Pb with 1 mg/cm² Nb backing and mounted onto the Cologne differential plunger. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the AGATA demonstrator array of five triple cluster modules of 36-fold segmented Ge crystals covering backward angles from 135° to 175°. Measured E_γ with Doppler corrections, (^{35}S) γ -coin, and level lifetimes using the differential recoil-distance method (DRDM). Compared with shell-model calculations. Measured E_γ , I_γ , fragment- γ -coin, recoil distance. Deduced levels, lifetimes. Compared with shell-model calculations with PSDPF, SDPF-U, and FSU effective interactions.

2010WaZT,2010Wa12: A 215-MeV $^{36}\text{S}^9$ beam was produced from the XTU-Tandem ALPI-superconducting linear accelerator complex at the INFN Legnaro National Laboratory, Italy. The target was 300 $\mu\text{g}/\text{cm}^2$ 99.7% enriched ^{208}Pb with 20 $\mu\text{g}/\text{cm}^2$ carbon backing. Projectile-like fragments produced in binary grazing reactions were separated and identified by the PRISMA spectrometer. γ rays were detected using the CLARA array of 22 escape-suppressed Ge clover detectors covering the azimuthal angles from 98° to 180°. Measured E_γ with Doppler corrections, I_γ , and (^{35}S) γ -coin. Deduced levels. Compared with shell-model calculations.

2010WaZT states that the observed γ -ray transitions are consistent with previous published work and no attempt has been made here to construct an independent level scheme.

 ^{35}S Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		2022Gr07 shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2 \otimes \nu(1d_{5/2})^6(2s_{1/2})^2(1d_{3/2})^3$ (77%).
1572.0 9	$1/2^+$	2.29 ps 14	2022Gr07 shell-model calculated configuration: $\pi(1d_{5/2})^6(2s_{1/2})^2 \otimes \nu(1d_{5/2})^6(2s_{1/2})^1(1d_{3/2})^4$ (44%). $T_{1/2}$: Lifetime=3.3 ps 2 from 2022Gr07 using the recoil-distance method with decay-curve analysis.
1991.1 9	$7/2^-$		
2347.1 13	$3/2^-$		
2717.1 14	$5/2^+$		
3421 2	$5/2^+$		
3558.1 14	$(5/2^+)$		
3593 2	$(7/2^+)$		
3818.1 14	$(9/2)^-$		
4023.1 22	$(11/2)^-$		

[†] From a least-squares fit to γ -ray energies.

[‡] From the Adopted Levels.

 $\gamma(^{35}\text{S})$

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
370 1	3.2 6	2717.1	$5/2^+$	2347.1	$3/2^-$	
1211 $\ddagger$ 2	1.7 7	3558.1	$(5/2^+)$	2347.1	$3/2^-$	
^x 1227 1	5.2 9					
1567 $\ddagger$ 2	9.6 9	3558.1	$(5/2^+)$	1991.1	$7/2^-$	
1572 1	100.0 24	1572.0	$1/2^+$	0	$3/2^+$	2022Gr07 shell-model calculated $B(E2; 1/2^+ \text{ to } 3/2^+) = 35.2 \text{ e}^2 \text{ fm}^4$ (5.2 W.u.), $B(M1; 1/2^+ \text{ to } 3/2^+) = 0.0204 \mu_N^2$ (1.14×10^{-2} W.u.), and mixing ratio $[\delta^2 = \lambda(E2)/\lambda(M1)] = 0.30$.
1827 1	12.9 11	3818.1	$(9/2)^-$	1991.1	$7/2^-$	
1986 $\ddagger$ 2	3.7 6	3558.1	$(5/2^+)$	1572.0	$1/2^+$	
1991 1	37.2 18	1991.1	$7/2^-$	0	$3/2^+$	
2032 2	13.1 12	4023.1	$(11/2)^-$	1991.1	$7/2^-$	

Continued on next page (footnotes at end of table)

²⁰⁸Pb(³⁶S, ³⁵Sγ) [2010WaZT,2022Gr07](#) (continued)

γ(³⁵S) (continued)

<u>E_γ[†]</u>	<u>I_γ[†]</u>	<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
2347 2	6.5 9	2347.1	3/2 ⁻	0	3/2 ⁺
2717 2	14.5 15	2717.1	5/2 ⁺	0	3/2 ⁺
^x 3034 3	7.2 9				
3421 2	8.6 10	3421	5/2 ⁺	0	3/2 ⁺
3593 2	8.8 11	3593	(7/2 ⁺)	0	3/2 ⁺

[†] From [2010WaZT](#).

[‡] Placement of transition in the level scheme is uncertain.

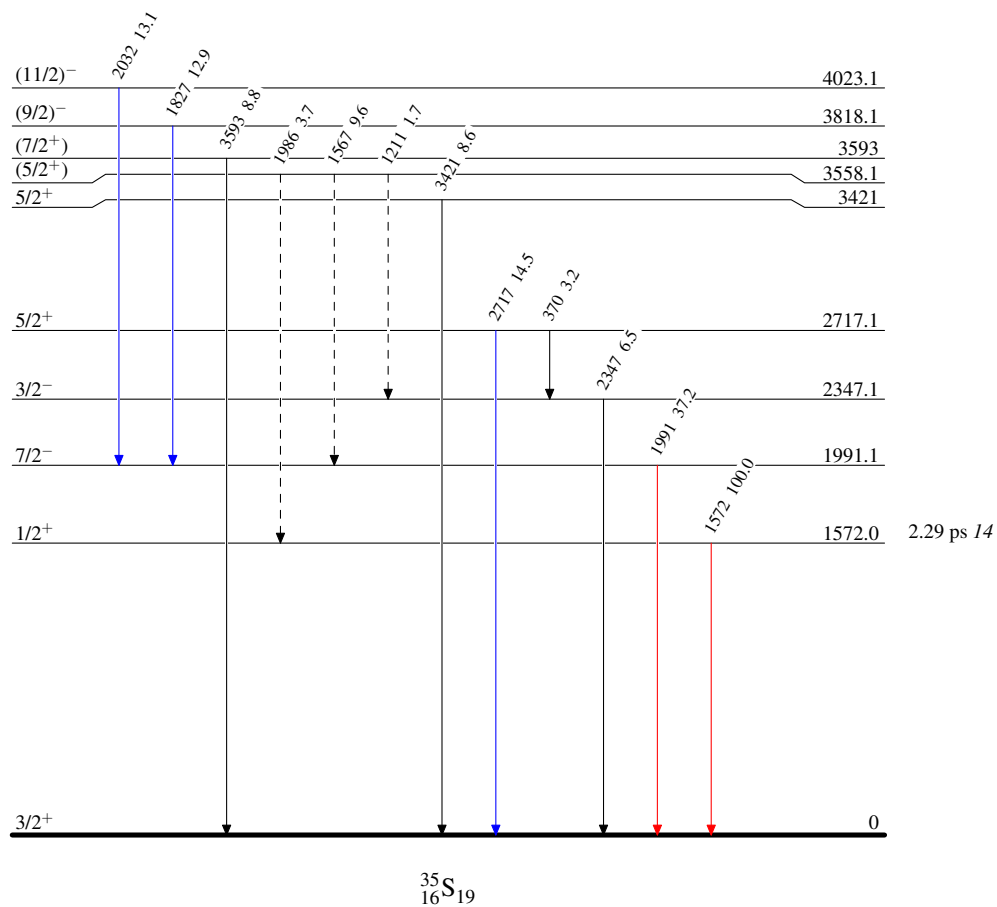
^x γ ray not placed in level scheme.

$^{208}\text{Pb}(^{36}\text{S}, ^{35}\text{S}\gamma)$ 2010WaZT,2022Gr07

Legend

Level Scheme
Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
- $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$
- $\cdots\cdots\cdots\longrightarrow$ γ Decay (Uncertain)



Adopted Levels, Gammas

$Q(\beta^-) = -5966.2$ 7; $S(n) = 12644.76$ 5; $S(p) = 6370.81$ 4; $Q(\alpha) = -6997.90$ 4 [2021Wa16](#)
 $S(2n) = 24152.8$ 4, $S(2p) = 17254.1$ 11 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): positive-ray mass spectrograph at Cambridge, UK ([1919As01](#)).

 ^{35}Cl Levels**Cross Reference (XREF) Flags**

A	^{35}S β^- decay (87.61 d)	K	^{31}P ($^6\text{Li}, d$)	U	^{35}Cl ($p, p'\gamma$)
B	^{35}Ar ε decay (1.7755 s)	L	^{32}S (α, p)	V	^{36}Ar (n, d)
C	^{36}K εp decay (341 ms)	M	^{32}S ($\alpha, p\gamma$)	W	^{36}Ar ($d, ^3\text{He}$)
D	^1H ($^{34}\text{S}, ^{35}\text{Cl}\gamma$)	N	^{33}S (α, d)	X	^{36}Ar ($\text{pol } d, ^3\text{He}$)
E	^2H ($^{34}\text{S}, n\gamma$)	O	^{34}S (p, γ)	Y	^{37}Cl (p, t)
F	^{12}C ($^{28}\text{Si}, \alpha p\gamma$)	P	^{34}S (p, p), ($p, p'\gamma$):resonances	Z	^{39}K (γ, α)
G	^{16}O ($^{24}\text{Mg}, \alpha p\gamma$)	Q	^{34}S (d, n)	Others:	
H	^{24}Mg ($^{16}\text{O}, \alpha p\gamma$)	R	^{34}S ($^3\text{He}, d$)	AA	^{40}Ca ($\mu^-, \nu\alpha n\gamma$)
I	^{27}Al ($^{14}\text{N}, \alpha p n\gamma$)	S	^{35}Cl (γ, γ')	AB	Coulomb excitation
J	^{31}P (α, p), (α, n):resonances	T	^{35}Cl ($n, n'\gamma$)		

E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
0.0	3/2 ⁺	stable	ABC EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB $\mu = +0.821870$ 2 (2013Ja17 , 2019StZV) $Q = -0.0817$ 8 (2008Py02 , 2021StZZ) J ^π : L=2 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10 , 1949Da14). μ : Nuclear Magnetic Resonance (2013Ja17). Q: Atomic beam magnetic resonance (2008Py02). Others: -0.08249 2 (1972St38), -0.076 5 (1986El09), -0.817 8 (1993Su36), 0.0819 11 (2000Ha64), 0.0850 11 (2004Al08). Evaluated rms nuclear charge radius $R = 3.365$ fm 19 (2013An02).
1219.38 7	1/2 ⁺	0.119 ps 14	B E H KLM O QRSTU VWXYZ	XREF: Others: AA , AB B(E2) [†] =0.00076 7 B(E2) [†] : from Coulomb excitation. E(level): weighted average of 1219.2 2 from ^{35}Ar ε decay, 1219.39 7 from (p,γ), and 1219.42 10 from (p,p'γ), 1219.0 10 from ($^{16}\text{O}, \alpha p\gamma$), and 1219.4 6 from ($\alpha, p\gamma$). J ^π : L=0 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He). T _{1/2} : from lifetime $\tau = 0.172$ ps 20: weighted average of 0.29 ps 4 from ($^{34}\text{S}, n\gamma$) (1973Wa10 , DSAM), 0.21 ps +10-8 from ($\alpha, p\gamma$) (1969In04 , DSAM), 80 fs 40 (1971Wi13 , DSAM), 175 fs 20 (1972Hu11 , DSAM), 150 fs 20 (1973Fa07 , DSAM), 145 fs 30 (1976Me12 , DSAM) from (p,γ), 0.13 ps +3-2 from (γ, γ') (1966Ho02 , resonance fluorescence), 200 fs 80 from (n,n'γ) (1989Ge09 , DSAM), 0.215 ps 75 (1972Va06 , DSAM), 0.21 ps +6-4 (1972Br45 , DSAM), 0.27 ps +19-10 (1969Du08 , DSAM) from (p,p'γ), 0.16 ps 5 (1969Ha17 , DSAM), and 0.230 ps 25 (1977Sc36 , DSAM) from Coulomb excitation.
1763.15 4	5/2 ⁺	0.36 ps 3	B EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
				T=1/2 B(E2)↑=0.0106 7 XREF: X(1785)Y(1750) B(E2)↑: from Coulomb excitation. E(level): weighted average of 1763.1 2 from ^{35}Ar ε decay, 1763.0 6 from ($^{28}\text{Si},\alpha\text{py}$), 1762.95 17 from ($^{24}\text{Mg},\alpha\text{py}$), 1763.3 2 from ($^{16}\text{O},\alpha\text{py}$), 1763.19 10 from ($^{14}\text{N},\alpha\text{pny}$), 1763.14 4 from (p,$\gamma$), 1763.26 10 from (p,p'$\gamma$). J^π: L=2 from 0⁺ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in (α,py), (p,γ), and (p,p'γ). T_{1/2}: lifetime τ=0.52 ps 5: weighted average of 0.54 ps 7 from ($^{34}\text{S},\text{n}\gamma$) (1973Wa10, DSAM), 0.55 ps 15 from (α,py) (1969In04, DSAM), 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), 0.53 ps 9 (1976Me12, DSAM) from (p,γ), 0.60 ps 21 from (γ,γ') (1962Bo17, resonance fluorescence), 0.205 ps 75 (1972Va06, DSAM), 0.49 ps +9-6 (1972Br45, DSAM), 1.15 ps +45-63 (1969Du08, DSAM) from (p,p'γ), 0.57 ps 7 (1969Ha17, DSAM), and 0.63 ps 5 (1977Sc36, DSAM) from Coulomb excitation.
2645.67 8	7/2 ⁺	0.159 ps 21	B FGHI LM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 18 from ($^{24}\text{Mg},\alpha\text{py}$), 2645.9 3 from ($^{16}\text{O},\alpha\text{py}$), 2645.45 12 from ($^{14}\text{N},\alpha\text{pny}$), 2645.70 8 from (p,$\gamma$), and 2645.74 17 from (p,p'γ). Others: 2645.1 10 from ^{35}Ar ε decay, 2645.8 6 from ($^{28}\text{Si},\alpha\text{py}$), and 2645.8 8 from ($\alpha,\text{py}$). J^π: spin=7/2 from p$\gamma$($\theta$) in ($\alpha,\text{py}$) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to 5/2⁺, 1763 level in (α,py). T_{1/2}: lifetime τ=0.229 ps 30: weighted average of 0.30 ps 9 from (α,py) (1969In04, DSAM), 0.255 ps 65 (1972Hu11, DSAM), 0.200 ps 30 (1976Me12, DSAM) in (p,γ), 0.27 ps 9 in (n,n'γ) (1989Ge09, DSAM), 0.185 ps 50 (1972Va06, DSAM), 0.26 ps 6 (1972Br45, DSAM), 0.35 ps +10-7 (1969Du08, DSAM) in (p,p'γ), and 0.26 ps 9 in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in ($^{28}\text{Si},\alpha\text{py}$) (2007Ks01, DSAM).
2694.00 20	3/2 ⁺	32 fs 6	B LM O QRSTU WXY	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 2 from ^{35}Ar ε decay, 2693.6 13 from (α,py), 2693.9 3 from (p,γ), 2694.50 28 from (p,p'γ). J^π: spin=3/2 from γ(θ) in (p,p'γ); L=2 from 0⁺ in ($^3\text{He},\text{d}$) and (d,^3He). T_{1/2}: lifetime τ=46 fs 9: unweighted average of 70 fs 30 in (α,py) from 1972Br33 with DSAM, 21 fs 3 in (α,py) from 1972Hu11 with DSAM, 19 fs 4 in (α,py) from 1973Fa07 with DSAM, 20 fs 4 in (p,γ) from 1976Me12 with DSAM, 65 fs 20 in (p,γ) from 1972Va06 with DSAM, 62 fs 16 in (p,p'γ) from 1969Du08 with DSAM, and 62 fs 8 in (n,n'γ) from 1989Ge09 with DSAM. Others: 115 fs +95-59 in

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
3002.64 11	5/2 ⁺	22 fs 6	B F HI LM O QRSTU WXY	(p,p'γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs 7 in (p,p'γ) from 1971Ca40 with DSAM is considered an outlier. E(level): weighted average of 3002.3 3 from ^{35}Ar ε decay, 3003.1 8 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3002.0 10 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3001.98 22 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3002.6 15 from ($\alpha,\text{p}\gamma$), 3002.75 8 from (p,γ), 3002.6 5 from (p,p'γ). J ^π : L=2 from 0 ⁺ in ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d, ^3He) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0 ⁺ in (d,n) and the angular distribution fit seems good. T _{1/2} : lifetime τ=32 fs 8: unweighted average of 48 fs 9 in (α,pγ) from 1972Br33 with DSAM, 22 fs 3 in (α,pγ) from 1972Hu11 with DSAM, 14 fs 2 in (α,pγ) from 1973Fa07 with DSAM, 16 fs 4 in (p,γ) from 1976Me12 with DSAM, 72 fs 12 in (n,n'γ) from 1989Ge09 with DSAM, 18 fs +26-15 in (p,p'γ) from 1972Va06 with DSAM, and 31 fs 13 in (p,p'γ) from 1969Du08 with DSAM.
3162.87 9	7/2 ⁻	29.2 ps 12	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 6 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3163.43 19 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 3162.9 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3162.64 9 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3163.1 11 from (α,pγ), 3162.99 7 from (p,γ), 3162.75 10 from (p,p'γ). J ^π : spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p'γ); L=3 from 0 ⁺ in (d,n), ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d, ^3He). T _{1/2} : lifetime τ=42.1 ps 17: weighted average of 42 ps 2 (1974Va13, RDM), 41.7 ps 17 (1976Ke02, RDM), 45 ps 6 (1991Ja11, RDM) from $^{24}\text{Mg}(\text{}^{16}\text{O},\alpha\text{p}\gamma)$, 41.8 ps 18 from $^{27}\text{Al}(\text{}^{14}\text{N},\alpha\text{p}\text{n}\gamma)$ (1976Wa11, RDM), 60 ps 7 from (α,pγ) (1969In04, DSAM), 37 ps 4 from (p,γ), (1971Ba23, delayed coincidence), and 46 ps 6 from (p,p'γ) (1974Hu09, RDM). Others: τ=140 ps 40 (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).
3918.47 12	3/2 ⁺	4.9 fs 14	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 5 from ^{35}Ar ε decay, 3915 2 from (α,pγ), 3918.48 10 from (p,γ). J ^π : spin=3/2 determined by the feeding 5162γ(θ) from 5/2 ⁺ , 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2 ⁺ ground state. L=2 from 0 ⁺ in (pol d, ^3He). T _{1/2} : lifetime τ=7 fs 2 from (p,γ), (1972Hu11, DSAM). Others: <22 fs from (α,pγ) (1972Br33, DSAM).
3942.9 2	9/2 ⁺	0.18 ps 3	F HI LM O	E(level): weighted average of 3942.0 7 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3943.4 10 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3942.3 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3943.4 12 from (α,pγ), and 3942.9 2 from (p,γ). J ^π : spin=9/2 from pγ(θ) in (α,pγ); 2180.1γ E2 to 5/2 ⁺ . T _{1/2} : lifetime τ=0.26 ps 5: weighted average of 0.33 ps 5 in (α,pγ) (1972Br33, DSAM), 0.39 ps 11 (1972Hu11,

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
3967.6 4	1/2 ⁺	11.8 fs 28	B M O R	DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in (²⁸ Si,αpγ) (2007Ks01, DSAM). E(level): weighted average of 3967.7 4 from ³⁵ Ar ε decay, 3968 2 from (α,pγ), and 3967.3 6 from (p,γ). J ^π : L=0 from 0 ⁺ in (³ He,d). T _{1/2} : lifetime τ=17 fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM) in (p,γ). Other: τ<54 fs from (α,pγ) (1972Br33, DSAM).
4059.1 4	(3/2) ⁻	13.9 fs 21	LM O QR	E(level): weighted average of 4056.9 27 from (α,p), 4058 3 from (α,pγ), 4059.2 4 from (p,γ). J ^π : L=1 from 0 ⁺ in (³ He,d); (3/2 ⁻) from FRESKO-DWBA analysis of measured σ(θ) of (α,p) (2019Se09). T _{1/2} : lifetime τ=20 fs 3: weighted average of 31 fs 13 from (α,pγ) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM) from (p,γ).
4112.4 16	7/2 ⁺	49 fs 11	LM O Q	E(level): weighted average of 4111.1 29 from (α,p), 4108 2 from (α,pγ), and 4113.7 10 from (p,γ). J ^π : 4113.4γ E2(+M3) to 3/2 ⁺ . T _{1/2} : lifetime τ=70 fs 16 from (α,pγ) (1972Br33, DSAM).
4173.43 23	(5/2 ⁺)	35 fs 6	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 2 from (α,pγ) and 4173.45 19 from (p,γ). J ^π : based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2 ⁺), 4456.4γ from 8630.1, 7/2 ⁻ level, 4243.0γ from 8416.7, 1/2 ⁺ level. Note that primary 3099.1γ from 7272.7, 1/2 ⁻ resonance disfavors 5/2 ⁺ assignment, but this γ branch is relatively weak. T _{1/2} : lifetime τ=51 fs 8: weighted average of 68 fs 24 in (α,pγ) (1972Br33, DSAM), 105 fs +35-30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM) in (p,γ).
4177.9 2	3/2 ⁻	27.0 fs 28	L O qR	XREF: q(4170)R(4167) J ^π : spin=3/2 from γ(θ) in (p,γ); L=1 from 0 ⁺ in (³ He,d). T _{1/2} : lifetime τ=39 fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM) from (p,γ).
4347.76 7	9/2 ⁻	1.0 ps 6	FGHI LM O	E(level): weighted average of 4347.8 6 from (²⁸ Si,αpγ), 4348.29 24 from (²⁴ Mg,αpγ), 4347.8 3 from (¹⁶ O,αpγ), 4347.43 17 from (¹⁴ N,αpγ), 4347.8 12 from (α,pγ), and 4347.76 6 from (p,γ). J ^π : spin from γ(θ) in (p,γ); parity from 1184.6γ, M1+E2 to 7/2 ⁻ , 3163 level in (p,γ) (1974Lo17). T _{1/2} : lifetime τ=1.5 ps 8: unweighted average of 1.31 ps +30-23 from (²⁸ Si,αpγ) (2007Ks01, DSAM), 2.9 ps +14-7 (1972Br33, DSAM), and 0.33 ps 6 (1976Me12, DSAM) from (p,γ).
4624.3 3	3/2,5/2	40 fs 17	B LM O	XREF: B(?) E(level): other: 4625.3 28 from (α,p); 4618 2 from

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
				($\alpha, \text{p}\gamma$) is discrepant. J ^π : from $\text{p}\gamma(\theta)$ in ($\alpha, \text{p}\gamma$). T _{1/2} : lifetime $\tau=58$ fs 24 from ($\alpha, \text{p}\gamma$) in 1972Br33 with DSAM. E(level): others: 4770.4 26 from (α, p), 4770 15 from ($\alpha, \text{p}\gamma$). J ^π : from primary transitions in (p, γ): 4059.2 γ from 8829.3, 1/2 ⁻ level, 2778.9 γ from 7548.9, 7/2 ⁻ level, 3231.0 γ from 8001.0, (7/2 ⁺) level. Other: 7/2, 9/2 from $\text{p}\gamma(\theta)$ for a group at E=4770 15 in ($\alpha, \text{p}\gamma$) is inconsistent and disfavored by γ rays to 1/2 ⁺ and 3/2 ⁺ . T _{1/2} : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 (1972Hu11 , DSAM) and 105 fs 17 (1976Me12 , DSAM) in (p, γ). XREF: x(4839) J ^π : from primary γ transitions in (p, γ): 2339.4 γ from 7178.6, 1/2 ⁽⁺⁾ level, 2355.4 γ from 7194.6, 1/2 ⁻ level, 2867.2 γ from 7706.4, 5/2 ⁺ level. T _{1/2} : lifetime $\tau=14$ fs 5 from (p, γ) (1976Me12 , DSAM). XREF: x(4839) E(level): other: 4859.2 28 from (α, p). J ^π : from primary γ transitions in (p, γ): 2340.1 γ from 7194.6, 1/2 ⁻ level, 3183.9 γ from 8038.4, 1/2 ⁺ , 3/2 ⁺ level; decay 3634.8 γ to 1219.38, 1/2 ⁺ . T _{1/2} : lifetime $\tau=7$ fs 2 from (p, γ) (1976Me12 , DSAM). E(level): others: 4881.8 29 from (α, p), 4880.8 21 from ($\alpha, \text{p}\gamma$). J ^π : spin=7/2 from $\text{p}\gamma(\theta)$ in ($\alpha, \text{p}\gamma$); 3417.1 primary γ from 8298.2, 3/2 ⁺ resonance in (p, γ). T _{1/2} : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 (1972Hu11 , DSAM), 7 fs 3 (1973Fa07 , DSAM), and 10 fs 3 (1976Me12 , DSAM) from (p, γ). Other: $\tau=0.28$ ps 6 from ($\alpha, \text{p}\gamma$) in 1972Br33 with DSAM. E(level): weighted average of 5006.9 27 from (α, p) and 5010.1 18 from (p, γ). Others: 5015 20 from ($\alpha, \text{p}\gamma$) and 5010 15 from ($^3\text{He}, \text{d}$). J ^π : primary 2185.4 γ from 7194.6, 1/2 ⁻ level and 3407.4 γ from 8416.7, 1/2 ⁺ level. T _{1/2} : lifetime $\tau=11$ fs 3 from (p, γ) (1976Me12 , DSAM). XREF: m(5175) E(level): weighted average of 5158.9 29 from (α, p), 5163 15 from ($^3\text{He}, \text{d}$), 5150 20 from $^{36}\text{Ar}(\text{d}, ^3\text{He})$, and 5155 11 from $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$. J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and (d, ^3He). XREF: m(5175) J ^π : spin=7/2 from the feeding 3466 $\gamma(\theta)$ from 8630, 7/2 ⁻ level and RUL in (p, γ); parity=- from 3466 γ , M1+E2 from 8630, 7/2 ⁻ level. XREF: M(5230) E(level): weighted average of 5209.5 28 from (α, p),
4769.86 23	(5/2 ⁻)	76 fs 20	LM 0	
4839.10 11	(1/2 ⁺ , 3/2)	9.7 fs 35	0 x	
4854.4 19	(1/2, 3/2)	4.9 fs 14	L 0 x	
4880.96 15	7/2 ⁽⁺⁾	5.6 fs 14	LM 0	
5009.1 18	(1/2, 3/2)	7.6 fs 21	LM 0 R	
5158.6 29	3/2 ⁺ , 5/2 ⁺		Lm R WX	
5163.34 22	7/2 ⁻		m 0	
5214.5 20	(5/2 ⁺) [#]	<4.85 fs	LM 0	

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Adopted Levels, Gammas (continued)

³⁵ Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
					5230 20 from (α,py), and 5215.8 15 from (p,γ). J ^π : other: 3/2,5/2 from py(θ) in (α,py). T _{1/2} : lifetime τ<7 fs from (p,γ) (1976Me12, DSAM).
5403.3 10	3/2 ⁻	11.8 fs 28	L	O R	E(level): weighted average of 5402.0 29 from (α,p) and 5403.5 10 from (p,γ). Other: 5409 12 from (³ He,d). J ^π : L=1 from 0 ⁺ in (³ He,d); 2400.6γ to 5/2 ⁺ rules out 1/2 ⁻ based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL. T _{1/2} : lifetime τ=17 fs 4 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 18 from (p,γ).
5407.07 ^a 21	11/2 ⁻	0.28 ps 7	FGHI	LM	E(level): weighted average of 5407.4 8 from (²⁸ Si,αpy), 5407.64 27 from (²⁴ Mg,αpy), 5407.3 3 from (¹⁶ O,αpy), 5406.68 19 from (¹⁴ N,αpny), and 5407.1 15 from (α,py). J ^π : spin=11/2 from py(θ) in (p,γ); parity from 1059γ, M1+E2 to 9/2 ⁻ , 4347 level. T _{1/2} : lifetime τ=0.4 ps 1 from (α,py) (1974Lo17, DSAM). Other: τ<1.1 ps from (²⁸ Si,αpy) (2007Ks01, DSAM).
5531 5			LM	O	E(level): weighted average of 5531 5 from (α,p) and 5535 20 from (α,py).
5586.0 20	5/2 ⁺		Lm	O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 26 from (α,p) and 5586 2 from (p,γ). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and L+1/2 transfer from analyzing power.
5598.4 22	3/2 ⁺ ,5/2 ⁺	2.1 fs 7	Lm	O wx	XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 26 from (α,p) and 5599.7 15 from (p,γ). J ^π : L(d, ³ He)=2 from 0 ⁺ . T _{1/2} : lifetime τ=3 fs 1 from (p,γ) (1976Me12, DSAM).
5633.6 30			Lm	r w y	XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p).
5646 2	(5/2,7/2,9/2 ⁺)	2.8 fs 7	Lm	O r y	XREF: m(5650)r(5660)y(5650) E(level): weighted average of 5645 4 from (α,p) and 5646 2 from (p,γ). J ^π : from primary transitions in (p,γ): 1903γ from 7548.9, 7/2 ⁻ level, 2355γ from 8001.0, (7/2 ⁺) level, 3261γ from 8906.8, 5/2 ⁺ level. T _{1/2} : lifetime τ=4 fs 1 from (p,γ) in 1976Me12 with DSAM.
5655 2	(3/2) ⁺	13.9 fs 35	Lm	O r w y	T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 4 from (α,p) and 5655 2 from (p,γ). J ^π : L=2 from 0 ⁺ in (³ He,d); primary 2142γ from 7797.2, 1/2 ⁻ level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2 ⁻ level in

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
				(p,γ) favors 5/2 ⁺ . T _{1/2} : lifetime τ=20 fs 5 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 18% of Iγ(2652)=100 6 from (p,γ). XREF: m(5650) E(level): weighted average of 5679.7 34 from (α,p), 5683 2 from (p,γ), and 5684 12 from (³ He,d). J ^π : L=1 from 0 ⁺ in (³ He,d). The contribution of Iγ to unknown levels adds up to 18% of Iγ(1504)=100 from (p,γ). E(level): others: 5731.2 25 from (α,p), 5720 20 from ³⁶ Ar(d, ³ He), and 5720 11 from ³⁶ Ar(pol d, ³ He). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and (d, ³ He) and L+1/2 transfer from analyzing power. The contribution of Iγ to unknown levels adds up to 100% of Iγ(3960)=100 11 from (p,γ). E(level): weighted average of 5757 4 from (α,p), 5758 3 from (p,γ), and 5760 12 from (³ He,d). J ^π : L(³ He,d)=(0,1) from 0 ⁺ . The contribution of Iγ to unknown levels adds up to 56% of Iγ(5758)=100 22 from (p,γ). E(level): weighted average of 5808.1 31 from (α,p) and 5806 2 from (p,γ). J ^π : primary 2890.2γ from 8696.9, 3/2 ⁻ level; 5806.1γ to 3/2 ⁺ g.s. T _{1/2} : lifetime τ=5 fs 1 from (p,γ) (1976Me12, DSAM). XREF: M(5850) E(level): weighted average of 5829 4 from (α,p) and 5850 25 from (α,pγ). J ^π : other: spin=5/2,9/2 from pγ(θ) in (α,pγ). E(level): weighted average of 5927.1 7 from (²⁸ Si,αpγ), 5926.9 5 from (¹⁶ O,αpγ), 5926.47 29 from (¹⁴ N,αpγ), 5928 3 from (α,p), and 5927.1 19 from (α,pγ). J ^π : 1579γ M1+E2, ΔJ=1 to 4348, 9/2 ⁻ in ¹² C(²⁸ Si,αpγ), also assuming spins increase with excitation energies in fusion-evaporation reactions. Spin=7/2,11/2 from pγ(θ) in (α,pγ). T _{1/2} : lifetime τ<0.4 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM). E(level): weighted average of 6087.6 8 from (²⁸ Si,αpγ), 6088.05 29 from (²⁴ Mg,αpγ), 6087.2 4 from (¹⁶ O,αpγ), 6086.89 23 from (¹⁴ N,αpγ), 6084.2 29 from (α,p), and 6087.0 18 from (α,pγ). J ^π : spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to 11/2 ⁻ , 5406 level in (α,pγ). T _{1/2} : lifetime τ=8.9 ps 8: weighted average of 9.3 ps 8 from (α,pγ) in 1974Lo17 with RDM and 7.7 ps 14 from (¹⁴ N,αpγ) in 1976Wa11. XREF: l(6104)w(6140)x(6136) E(level): weighted average of 6104 4 from (α,p)
5682.2 20	1/2 ⁻ ,3/2 ⁻		Lm O R	
5723.52 18	5/2 ⁺		L O WX	
5758 3	(1/2,3/2 ⁻)		L O R	
5806.6 20	(1/2,3/2,5/2)	3.5 fs 7	L O	
5830 4	(5/2 ⁻) [#]		LM	
5926.65 29	(11/2) ⁻	<0.28 ps	F HI LM	
6087.31 24	13/2 ⁻	6.2 ps 6	FGHI LM	
6106.2 21	(3/2,5/2 ⁺)	8.3 fs 21	1 O wx	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>XREF</u>		<u>Comments</u>
				and 6106.2 21 from (p,γ). J ^π : 4886.5γ to 1/2 ⁺ , 4342.8γ to 5/2 ⁺ ; primary 1675.5γ from 7781.7, (5/2 ⁻) level. T _{1/2} : lifetime τ=12 fs 3 from (p,γ) in 1976Me12 with DSAM. XREF: w(6140)x(6136) J ^π : L=2 from 0 ⁺ in (d, ³ He) and (pol d, ³ He), and L+1/2 transfer from analyzing power.
6141 4	5/2 ⁺	L	WX	XREF: n(6200) E(level): other: 6180 3 from (α,p). J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from 7618.7, 5/2 ⁺ level in (p,γ). Apart from the γ branch I _γ (6180)=100 observed in (p,γ), the contribution of I _γ to unknown levels adds up to 122% of I _γ (6180).
6181 3	(1/2 ⁺ to 7/2 ⁺)	L n0		XREF: n(6200) E(level): other: 6180 3 from (α,p). J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from 7618.7, 5/2 ⁺ level in (p,γ). Apart from the γ branch I _γ (6180)=100 observed in (p,γ), the contribution of I _γ to unknown levels adds up to 122% of I _γ (6180).
6224.4 30		L n		XREF: n(6200) J ^π : L(α,d)=6 for a group at 6200 10.
6283 3	(5/2 ⁺) [#]	L	R	E(level): weighted average of 6283 3 from (α,p) and 6284 4 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d). J ^π : L=(0,1) from 0 ⁺ in (³ He,d).
6329 4	(1/2,3/2 ⁻)		R	XREF: R(6377) E(level): weighted average of 6380.8 8 from (¹⁶ O,αpγ) and 6379.3 25 from (α,p). J ^π : other: L=(2,3) from 0 ⁺ in (³ He,d) for a group at E=6377 2 based on very few data points in a very narrow range of angles in the DWBA fit.
6380.2 9	(9/2 ⁻) [#]	H L	R	
6401.2 22	(1/2 ⁻) [#]	LM		
6428 2	(1/2 ⁺) [#]	L	R	E(level): weighted average of 6429 3 from (α,p) and 6427 2 from (³ He,d). J ^π : other: L=(3) from 0 ⁺ in (³ He,d) is inconsistent.
6469 3	(1/2 ⁻ ,3/2 ⁻)	L	R	E(level): weighted average of 6475 4 from (α,p) and 6468 2 from (³ He,d). J ^π : L=(1) from 0 ⁺ in (³ He,d).
6492 2	(3/2 ⁺) [#]	L O R		E(level): weighted average of 6492 3 from (α,p), 6492 3 from (p,γ), and 6491 2 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d).
6546 2	(1/2 ⁺) [#]	L	R	E(level): weighted average of 6549 3 from (α,p) and 6545 2 from (³ He,d). J ^π : other: L=(0,1) from 0 ⁺ in (³ He,d), indicating (1/2,3/2 ⁻). E(level): from (³ He,d). J ^π : from L=(1) from 0 ⁺ in (³ He,d).
6643 2	(1/2 ⁻ ,3/2 ⁻)	D	R	XREF: y(6677) J ^π : 2312.4γ to 9/2 ⁻ , 3497.1γ to 7/2 ⁻ in (²⁸ Si,αpγ), assuming spin increasing as excitation energy increases.
6659.1 30	(7/2 ⁺) [#]	L	y	XREF: y(6677)
6660.2 8	(11/2 ⁻)	F		E(level): weighted average of 6679.4 31 from (α,p) and 6674 2 from (³ He,d).
6675.6 25		D L	R y	XREF: X(6745) E(level): weighted average of 6761 2 from (³ He,d), 6750 20 from ³⁶ Ar(d, ³ He), and 6745 12 from ³⁶ Ar(pol d, ³ He). J ^π : L(d, ³ He)=2 from 0 ⁺ .
6761 2	3/2 ⁺ ,5/2 ⁺	D	R WX	E(level): weighted average of 6781.3 30 from (α,p) and 6778 2
6779.0 20		D L	R	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF			Comments
6800 4			D	L		from ($^3\text{He},d$). XREF: D(6802) E(level): from (α,p).
6823 2			D		R	E(level): from ($^3\text{He},d$).
6842 2			D	L	R	E(level): weighted average of 6842 4 from (α,p) and 6842 2 from ($^3\text{He},d$).
6863 3	(9/2 ⁺) [#]			L		
6866.7 6	(1/2,3/2 ⁻)		D	O	R	E(level): other: 6866 2 from ($^3\text{He},d$). J ^π : L($^3\text{He},d$)=(0,1) from 0 ⁺ .
6891.6 30	(9/2 ⁺) [#]			L		
6948.1 34	5/2 ⁺			L	WX	XREF: W(6930) E(level): weighted average of 6948.6 34 from (α,p) and 6930 20 from $^{36}\text{Ar}(d,^3\text{He})$. Other: 6953 43 from $^{36}\text{Ar}(\text{pol } d,^3\text{He})$. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
6986.3 34	(9/2 ⁺) [#]			L		
7066.2 7	(5/2 ⁺) [#]			L	O R	E(level): weighted average of 7066.2 7 from (p, γ), and 7066 2 from ($^3\text{He},d$).
7103.4 7	3/2 ⁽⁻⁾			L	O R	XREF: L(7105?) E(level): weighted average of 7103.4 7 from (p, γ), and 7103 2 from ($^3\text{He},d$). J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); parity from L($^3\text{He},d$)=(1,3) from 0 ⁺ .
7121.7 24				L		
7178.6 8	1/2 ⁽⁺⁾				O	T=3/2 J ^π : spin=1/2 from $\gamma(\theta)$ in (p, γ); parity from IAS in ^{35}S .
7179 4	(7/2 ⁺) [#]			L	N	E(level): weighted average of 7180 4 from $^{32}\text{S}(\alpha,p)$ and 7170 10 from $^{33}\text{S}(\alpha,d)$. J ^π : also L=6 from 3/2 ⁺ in $^{33}\text{S}(\alpha,d)$, indicating (7/2;17/2) ⁺ .
7185	5/2 ⁺				O R X	XREF: R(7178)X(7181) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=7181 60. Other: L($^3\text{He},d$)=(2) from 0 ⁺ for a group at E=7178 2 considered the same level.
7194.6 7	1/2 ⁻	0.027 keV 8		OP	R	J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances. Other: 1/2,3/2 from $\gamma(\theta)$ in (p, γ).
7213.7 25				L		y XREF: y(7250)
7225.5 8	5/2			O		y XREF: y(7250)
7228.2 20	(3/2 ⁺) [#]			L	R y	J ^π : from $\gamma(\theta)$ in (p, γ). XREF: R(7227)y(7250)
7234.0 8	5/2 ⁽⁺⁾			O		E(level): weighted average of 7230.9 30 from $^{32}\text{S}(\alpha,p)$, 7227 2 from $^{34}\text{S}(^3\text{He},d)$. J ^π : other: L=(0,1) from 0 ⁺ in ($^3\text{He},d$) is inconsistent.
7269.2 1				O		J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); primary 6014.1 γ to 1/2 ⁺ .
7272.7 9	1/2 ⁻	0.014 keV 4		OP	R y	XREF: P(7274)R(7273)y(7250)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^{π‡}	T _{1/2} or Γ ^{&}	XREF	Comments
				J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p, γ).
7348.3	(7/2) [#]		L	
7361.9 8	3/2 ⁽⁺⁾		L 0 R	E(level): weighted average of 7362.3 from (α ,p), 7362.0 8 from (p, γ), and 7361.2 from (^3He ,d). J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); 6141.9 γ (M1+E2) to 1/2 ⁺ .
7396.4 10	7/2 ⁽⁻⁾		0 R	E(level): weighted average of 7396.0 10 from (p, γ), and 7398.2 from (^3He ,d). J ^π : spin=7/2 from $\gamma(\theta)$ in (p, γ); 4233.3 γ (M1+E2) to 7/2 ⁻ .
7413.8 30			L	
7450.9 8	3/2		L 0	E(level): weighted average of 7447.3 from (α ,p) and 7451.1 6 from (p, γ). J ^π : from $\gamma(\theta)$ in (p, γ).
7496			0	
7501.1 8	(5/2 ⁺ , 7/2, 9/2 ⁺)		1 0	XREF: l(7503) J ^π : primary 3558.0 γ to 9/2 ⁺ , 4498.2 γ to 5/2 ⁺ .
7502.9 7			1 0	XREF: l(7503)
7518.8 7	7/2		0	J ^π : from $\gamma(\theta)$ in (p, γ).
7548.9 6	7/2 ⁻	<0.69 fs	OP	T=3/2 XREF: P(7550) J ^π : from resonance analysis of measured $\sigma(\theta)$ in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances. Spin=7/2 also from $\gamma(\theta)$ in (p, γ). T _{1/2} : lifetime $\tau < 1$ fs from (p, γ) (1973Fa07, DSAM). Other: $\Gamma_p = 4.6$ eV 10 and $\Gamma = 5$ eV +4-2 from $^{34}\text{S}(\text{p},\text{p})$ elastic scattering excitation function analysis (1971BrXT). Other Γ_p , Γ_γ , Γ values/ratios are mostly deduced from (p, γ) resonance strengths, which have large discrepancies. See comments in the (p, γ) dataset for details.
7561.3 6	1/2, 3/2		L 0	E(level): other: 7566.5 35 from (α ,p). J ^π : from $\gamma(\theta)$ in (p, γ).
7565 20	5/2 ⁺			X J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and L+1/2 transfer from analyzing power.
7590 4			L	
7600.8 8	5/2 ⁺	<13.9 fs	0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7599.9 γ M1+E2 to 3/2 ⁺ . T _{1/2} : lifetime $\tau < 20$ fs from (p, γ) (1971Wi13, DSAM), which is equivalent to $\Gamma > 0.033$ eV.
7618.7 5	5/2 ⁺		0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7617.8 γ M1+E2 to 3/2 ⁺ .
7648.1 30			L	
7656.6 8	(1/2, 3/2, 5/2 ⁺)		0	J ^π : primary 3688.8 γ to 1/2 ⁺ .
7670 10	(7/2 to 17/2) ⁺		N	J ^π : L(α ,d)=6 from 3/2 ⁺ .
7671.9 8	5/2 ⁺		1 0p	XREF: p(7686) J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 4668.8 γ , M1+E2 to 5/2 ⁺ ; 5908.2 γ , M1+E2 to 5/2 ⁺ ; other: 3324.0 γ to 9/2 ⁻ seems to favor 5/2 ⁻ .
7685.5 8	3/2 ⁻		OP	J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ) and 3/2 ⁻ from R-Matrix analysis in (p,p): resonances. $\Gamma_p = 446$ eV 15 from (p,p): resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
7693.9 8	(1/2 ⁺ , 3/2, 5/2)		0	J ^π : primary 7693.0γ to 3/2 ⁺ , 3634.6γ to (3/2) ⁻ , 4690.9γ to 5/2 ⁺ .
7706.4 8	5/2 ⁺		L OP	E(level): other: 7709.9 25 from (α,p). J ^π : spin=5/2 from γ(θ) in (p,γ) and 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances.
7744.8 9	7/2 ⁻		L 0	Γ _p =4 eV 1 in (p,p). E(level): other: 7744.1 30 from (α,p). J ^π : spin=7/2 from γ(θ) in (p,γ); 4581.6γ M1+E2 to 7/2 ⁻ .
7748	(7/2 to 17/2) ⁺		NO	XREF: N(7750) E(level): from (p,γ). J ^π : L(α,d)=6 from 3/2 ⁺ .
7777.1 9	(5/2 ⁺)		L 0	E(level): from (p,γ). Other: 7770.3 33 from (α,p). J ^π : based on primary transitions: 4613.9γ to 7/2 ⁻ , 5131.0γ to 7/2 ⁺ , 6557.1γ to 1/2 ⁺ .
7781.7 11	(5/2 ⁻)		0	J ^π : primary 3433.8γ to 9/2 ⁻ ; relatively weak 6561.7γ to 1/2 ⁺ .
7797.2 9	1/2 ⁻	0.031 keV 10	L OP	E(level): weighted average of 7799.8 30 from (α,p), 7797.1 9 from (p,γ), and 7796 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=1/2 from γ(θ) in (p,γ); (1/2) ⁻ from R-Matrix analysis of in (p,p):resonances.
7837.2 10	3/2 ⁻	3.7 keV 4	OP	T=3/2 E(level): weighted average of 7837.2 10 from (p,γ), and 7837 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and 3/2 ⁻ from R-Matrix analysis in (p,p):resonances. Γ: other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM).
7839.0 9	(5/2 ⁺)		0	J ^π : primary 6619.0γ to 1/2 ⁺ , 3895.9γ to 9/2 ⁺ .
7868.7 10	(1/2 ⁺ , 3/2, 5/2 ⁺)		L 0	E(level): other: 7868.5 35 from (α,p). J ^π : primary 6648.6γ to 1/2 ⁺ , 2654.1γ to (5/2 ⁺).
7873.03 ^b 30	13/2 ⁺		F HI NO	XREF: N(7870) E(level): weighted average of 7873.3 8 from (²⁸ Si,αpγ), 7873.4 4 from (¹⁶ O,αpγ), and 7872.78 30 from (¹⁴ N,αpnγ). J ^π : 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2) ⁻ ; 1786.2γ D(+Q) to 13/2 ⁻ ; band head.
7880.8 9	3/2 ⁺ , 5/2 ⁺	0.008 keV 4	OP	J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885			0	
7889.1 15			0	
7899.2 7	(5/2)		1 0	XREF: l(7901) J ^π : primary 2735.8γ to 7/2 ⁻ , 3018.1γ to

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	$J^{\pi\ddagger}$	$T_{1/2}$ or Γ ^{&}	XREF		Comments
7903			L	0	$7/2^{(+)}$, 3721.1 γ to $3/2^{-}$, 5204.9 γ to $3/2^{+}$. XREF: I(7901)
7923.3 9	($3/2^{+}$, $5/2^{+}$)			0	J^{π} : primary 5277.2 γ to $7/2^{+}$, 6703.2 γ to $1/2^{+}$.
7930				0	
7949				0	
7970.3 10	($5/2^{-}$)			0	J^{π} : primary 3622.3 γ to $9/2^{-}$, weak 6750.2 γ to $1/2^{+}$.
7978.6 35			L		
7987.8 10	$3/2^{+}$			0	J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ); 7986.8 γ M1+E2 to $3/2^{+}$.
7995.6 10	($5/2$)			0	J^{π} : primary 2592.2 γ to $3/2^{-}$, 7994.6 γ to $3/2^{+}$, 2832.1 γ to $7/2^{-}$, 5349.5 γ to $7/2^{+}$.
8001.0 9	($7/2^{+}$)		L	0	E(level): other: 8000.7 41 from (α ,p). J^{π} : from primary transitions in (p, γ): 3653.0 γ to $9/2^{-}$, 4057.9 to $9/2^{+}$, 3827.3 γ to $5/2^{-}$, 6237.4 γ to $5/2^{+}$, 8000.0 γ to $3/2^{+}$. XREF: P(8004)W(8010)X(8007)
8005.0 10	$5/2^{+}$	0.011 keV 5	OP	WX	J^{π} : L(d, ^3He)= 2 from 0^{+} ; spin= $5/2$ from $\gamma(\theta)$ in (p, γ). J^{π} : L(α ,d)= 6 from $3/2^{+}$.
8010 10	($7/2$ to $17/2$) ⁺		N		
8019.2 35			L		
8035.7 10	$1/2^{-}$, $3/2^{-}$	0.026 keV 7		OP	E(level): other: 8034 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances.
8038.4 11	$1/2^{+}$, $3/2^{+}$	0.30 keV 2		OP	E(level): weighted average of 8038.5 11 from (p, γ), and 8038 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances. J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ), but $1/2^{+}$ from R-Matrix analysis in (p,p); 8037.5 γ M1+E2 to $3/2^{+}$.
8075.9 10	($5/2$)		L	0	E(level): weighted average of 8075.1 46 from (α ,p) and 8075.9 10 from (p, γ). J^{π} : primary 3897.8 γ to $3/2^{-}$, 8074.9 γ to $3/2^{+}$; 2912.4 γ to $7/2^{-}$, 3962.0 γ to $7/2^{+}$.
8080				0	
8096.1 11	($5/2$)			0	J^{π} : primary 2932.6 γ to $7/2^{-}$, 5450.0 γ to $7/2^{+}$; 3918.0 γ to $3/2^{-}$, 4177.4 γ to $3/2^{+}$.
8100 10	($7/2$ to $17/2$) ⁺		N		J^{π} : L(α ,d)= 6 from $3/2^{+}$.
8106.4 11	$3/2^{+}$		L	0	E(level): other: 8103.8 42 from (α ,p). J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ); 8105.4 γ M1+E2 to $3/2^{+}$.
8113.3 11	($1/2^{+}$, $3/2$, $5/2^{+}$)		L	0	XREF: L(8121?) J^{π} : primary 2897.4 γ ($5/2^{+}$), 4145.4 γ to $1/2^{+}$.
8147.2 11	$3/2^{-}$	0.56 keV 6		OP	XREF: P(8150) J^{π} : from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p, γ) based on strengths of primary γ rays. Γ : from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances in 1977Ou01.
8148 3	$1/2^{-}$	2.66 keV 27		P	
8150 3	$3/2^{-}$	0.56 keV 6		P	
8156.8 11	($5/2^{+}$)			0	J^{π} : primary 3978.7 γ to $3/2^{-}$, 8155.8 γ to $3/2^{+}$, 4213.6 γ to $9/2^{+}$.
8171.7 42			L		

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
8179.4 12	(3/2 ⁻ , 5/2, 7/2 ⁺)		0		J ^π : primary 5016.1γ to 7/2 ⁻ , 5484.9γ to 3/2 ⁺ .
8208.2 10	5/2 ⁺	0.033 keV 10	OP	WX	T=3/2 XREF: W(8210)X(8204) E(level): other: 8209 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p}, \text{p}'\gamma)$:resonances. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
8211 3	1/2 ⁺	0.094 keV 15	L	OP	XREF: O(8209) E(level): from (p,p):resonances. Other: 8210.2 46 from (α,p).
8216.3 10	5/2 ⁺	0.014 keV 3	1	OP	WX XREF: w(8210)x(8204) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8242.4 12	3/2 ⁻	0.140 keV 15	1	OP	XREF: l(8246.0)P(8243) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8251 5	(3/2 ⁺ , 5/2 ⁺)		1	0	XREF: l(8246.0)
8269.0 5	5/2 ⁺	0.005 keV 3	L	OP	J ^π : primary 4138γ to 7/2 ⁺ , 7031γ to 1/2 ⁺ . XREF: L(8270.6) E(level): other: 8271 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p}, \text{p}'\gamma)$:resonances.
8277.2 5	(5/2 ⁺)	0.006 keV 3	OP		J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2 ⁻ rules out 3/2 ⁺ , since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8282.4 5	(3/2 ⁻ , 5/2)		0		XREF: P(8279) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2 ⁻ disfavors both 3/2 ⁺ and 5/2 ⁺ , but 3/2 ⁺ is much less likely.
8287.82 30	1/2 ⁻	0.04 keV 1	L	OP	J ^π : primary 3118.9γ to 7/2 ⁻ , 4104.2γ to 3/2 ⁻ , 8281.4γ to 3/2 ⁺ . XREF: L(8292)P(8290) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8297.2 9	3/2 ⁻	0.073 keV 15	OP		XREF: P(8300)
8298.2 5	3/2 ⁺		1	0	XREF: l(8292) J ^π : spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2 ⁺ .
8318.1 3	(5/2 ⁺)		L	0	XREF: L(8321.0) J ^π : primary 4350.2γ to 1/2 ⁺ , 5154.8γ to 7/2 ⁻ , 5671.9γ to 7/2 ⁺ .
8319.6 ^a 5	15/2 ⁻	12.5 fs 7	F	H	E(level): weighted average of 8319.4 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) and 8319.7 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$). J ^π : 2911.9γ E2, ΔJ=2 to 11/2 ⁻ , 2232.7γ to 13/2 ⁻ ; band assignment. T _{1/2} : lifetime τ=18 fs 1 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2013Bi10, DSAM).
8323.1 13	(3/2 ⁺ , 5/2 ⁺)		L	0	T _{1/2} : from (HI,xnγ). XREF: L(8321.0) J ^π : primary 4210.4γ to 7/2 ⁺ , 7103.0γ to 1/2 ⁺ .
8346 5			0		

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J^π [‡]	$T_{1/2}$ or Γ ^{&}	XREF		Comments
8381.8 8	5/2 ⁺	0.023 keV 7	OP		XREF: P(8383) J^π : from $\gamma(\theta)$ in (p, γ) and R-Matrix analysis in (p,p):resonances.
8389.4 16 8402.9 5	5/2 ⁻	0.002 keV 1	L 0 L OP		XREF: L(8398.8)P(8404) E(level): others: 8398.8 38 from (α ,p), and 8404 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; its primary 5708.5 γ to 3/2 ⁺ , 2694 level rules out 7/2 ⁻ based on RUL and $\Gamma=0.002$ keV.
8405.8 14	(5/2 ⁻)	0.001 keV 1	OP		E(level): weighted average of 8405.7 14 from (p, γ), and 8406 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : (5/2 ⁻ , 7/2 ⁻) from R-Matrix analysis in (p,p):resonances; primary 8404.6 γ to 3/2 ⁺ .
8409.4 18	1/2 ⁻	0.125 keV 15	OP		E(level): weighted average of 8409.6 18 from $^{34}\text{S}(\text{p},\gamma)$ and 8409 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8416.7 5	1/2 ⁺	0.026 keV 7	OP		E(level): weighted average of 8416.7 5 from (p, γ), and 8418 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 1/2 ⁺ from R-Matrix analysis in (p,p):resonances; but primary 4473.5 γ to 9/2 ⁺ and 7196.5 γ to 1/2 ⁺ in (p, γ) gives (5/2 ⁺), which may indicate two different resonances with close energies.
8430.3 5 8434.8 5 8464.3 5	(3/2) ⁺ (1/2 ⁺ , 3/2, 5/2 ⁺)	0.090 keV 15	L 0 OP 0		E(level): other: 8429.2 40 from (α ,p). XREF: P(8435) J^π : primary 4496.4 γ to 1/2 ⁺ , 3249.6 γ to (5/2 ⁺).
8484.4 5	(3/2) ⁺	0.012 keV 5	1 OP		E(level): weighted average of 8484.4 5 from (p, γ), and 8486 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; 3/2 ⁺ proposed by 2001Vo24 in (p, γ) based on proposed isobaric analog to 2940, 3/2 ⁺ level in ^{35}S .
8486.2 5	3/2 ⁻	0.150 keV 15	1 OP		E(level): weighted average of 8486.1 5 from (p, γ), and 8488 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8487.5 5	15/2 ⁻	<69 fs	F H		E(level): weighted average of 8487.8 10 from ($^{28}\text{Si},\alpha\text{p}\gamma$) and 8487.4 5 from ($^{16}\text{O},\alpha\text{p}\gamma$). J^π : 3080.0 γ E2, $\Delta J=2$ to 11/2 ⁻ , 2399.8 γ D(+Q) to 13/2 ⁻ . $T_{1/2}$: lifetime $\tau<0.1$ ps from ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01, DSAM).
8506.5 5 8514.3 5 8533.9 5	1/2 ⁻	0.150 keV 15	L 0 OP L 0		E(level): other: 8505.1 47 from (α ,p). XREF: P(8515) E(level): weighted average of 8532.4 43 from (α ,p) and 8533.9 5 from (p, γ).
8558.2? 13	(13/2 ⁻)	<0.14 ps	F		XREF: F(?) J^π : 2631 γ D(+Q), $\Delta J=1$ to 11/2 ⁻ in

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Adopted Levels, Gammas (continued)

³⁵ Cl Levels (continued)						
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF			Comments
8571.9 5	(5/2) ⁺	0.08 keV 1	L	OP	X	(²⁸ Si,αpγ). T _{1/2} : lifetime τ<0.2 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM). T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 5 from (p,γ), and 8573 3 from ³⁴ S(p,p),(p,p'γ):resonances. Other: 8567.9 37 from (α,p). XREF: P(8582)
8580.5 5	1/2 ⁺	0.75 keV 8		OP		XREF: P(8582)
8585.8 5				O		
8590.1 5	5/2 ⁺	0.003 keV 2		OP		XREF: P(8592) E(level): weighted average of 8590.0 5 from (p,γ), and 8592 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 3/2 ⁺ ,5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2 ⁻ ruled out 3/2 ⁺ based on RUL and Γ=0.003 keV.
8612.0 5	3/2 ⁺ ,5/2 ⁺		1	O	W	XREF: W(8610) J ^π : L(d, ³ He)=2 from 0 ⁺ .
8613.7 5	5/2 ⁺	0.175 keV 20	1	OP		XREF: P(8616)
8618.2 5	3/2 ⁺ ,5/2 ⁺	0.002 keV 1	1	OP		XREF: P(8620) E(level): weighted average of 8618.1 5 from (p,γ), and 8620 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 4282.1γ M1+E2 to 9/2 ⁻ , 6866.2γ D+Q to 5/2 ⁺ , spin=3/2,7/2 from γ(θ) in (p,γ).
8630.1 3	7/2 ⁻	0.001 keV 1		O		J ^π : spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2 ⁺ . XREF: P(8643)
8633 3	(3/2,5/2 ⁺)	0.001 keV 1		P		
8640	(5/2 ⁺)			O		
8641.4 5	3/2 ⁺ ,5/2 ⁺	0.003 keV 2		OP		
8654 4			L			
8686.2 5	5/2 ⁻	0.001 keV 1		OP		E(level): weighted average of 8686.2 5 from (p,γ), and 8687 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : 5/2 ⁻ ,7/2 ⁻ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2 ⁺ disfavors both 5/2 ⁻ and 7/2 ⁻ , but 7/2 ⁻ is much less likely. XREF: P(8689)
8688.3 5	1/2 ⁺	0.20 keV 2		OP		
8691 3	1/2 ⁻	6.4 keV 7		P		
8696.9 5	3/2 ⁻	0.8 keV 1	L	OP		XREF: L(8698)P(8698)
8700 10	(7/2 to 17/2) ⁺			N		J ^π : L(α,d)=6 from 3/2 ⁺ .
8706.3 5				O		
8717.8 5			L	O		E(level): weighted average of 8720.9 37 from (α,p) and 8717.7 5 from (p,γ).
8750.9 5	3/2 ⁻	0.30 keV 3		OP		E(level): weighted average of 8750.9 5 from (p,γ), and 8750 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8767.0 5				O		
8772.9 5	1/2 ⁻	0.57 keV 6		OP		XREF: P(8775)
8779.8 5	3/2 ⁻	0.214 keV 25		OP		XREF: P(8781) E(level): weighted average of 8779.8 5 from (p,γ), and 8781 3 from ³⁴ S(p,p),(p,p'γ):resonances.

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π_‡</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
8787.2 5	(5/2 ⁺)	0.001 keV 1	L OP	E(level): others: 8786.3 38 from (α,p), and 8788 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : (3/2 ⁺ , 5/2 ⁺) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2 ⁻ .
8788.3 ^b 6	(15/2 ⁺)	<0.28 ps	F H	XREF: F(?) J ^π : 915.4γ D, ΔJ=1 to 13/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2007Ks01, DSAM).
8798.4 5	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	OP	E(level): weighted average of 8798.4 5 from (p,γ), and 8799 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8820.9 5	(1/2 ⁺ to 7/2 ⁺)		0	J ^π : primary 5817.6γ and 7057.0γ to 5/2 ⁺ ; 6126.3γ and 8819.7γ to 3/2 ⁺ , which yields (1/2 ⁺ , 3/2 [±] , 5/2 [±] , 7/2 ⁺).
8824.2 5	1/2 ⁺	1.70 keV 17	OP	XREF: P(8825) E(level): weighted average of 8824.2 5 from (p,γ), and 8825 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8829.3 5	1/2 ⁻	12.2 keV 12	OP	E(level): weighted average of 8829.3 5 from (p,γ), and 8829 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 1/2 ⁻ from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2 ⁺ gives J≥3/2, which may indicate a different level.
8830 3	1/2 ⁺	0.080 keV 15	P	
8833.1 5	(5/2 ⁻ , 7/2)		0	J ^π : primary 3618.4γ to (5/2 ⁺), 4063.0γ to (5/2 ⁻); 4720.4γ to 7/2 ⁺ , 5669.6γ to 7/2 ⁻ .
8838.1 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	1 OP	XREF: P(8839)
8844.4 ^b 4	17/2 ⁺	5.9 ps 11	F HI N	XREF: N(8840) E(level): weighted average of 8845.1 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$), 8844.6 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$), and 8844.2 4 from ($^{14}\text{N}, \alpha\text{p}\gamma$). J ^π : 971.38γ E2, ΔJ=2 to 13/2 ⁺ , 524.8γ to 15/2 ⁻ ; band assignment. T _{1/2} : weighted average of 6.1 ps 11 from ($^{16}\text{O}, \alpha\text{p}\gamma$) (1991Ja11, DSAM) and 5.5 ps 14 from ($^{14}\text{N}, \alpha\text{p}\gamma$) (1976Wa11, DSAM).
8856.2 5	5/2 ⁺	0.010 keV 5	OP	E(level): weighted average of 8856.1 5 from (p,γ), and 8858 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.010 keV.
8868.6 5	3/2 ⁺ , 5/2 ⁺	0.027 keV 10	OP	E(level): weighted average of 8868.6 5 from (p,γ), and 8870 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8884.1 5			0	
8886.1 5	(5/2)		L 0	E(level): other: 8887.5 41 from (α,p). J ^π : primary 3722.6γ to 7/2 ⁻ , 6239.8γ to 7/2 ⁺ ; 4707.9γ to 3/2 ⁻ , 4967.2γ to 3/2 ⁺ .

Continued on next page (footnotes at end of table)

Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
8893.3 5	(5/2 ⁺ , 7/2 ⁺)		0		J ^π : primary 4950.0γ to 9/2 ⁺ , 6198.7γ to 3/2 ⁺ .
8904.8 5	(1/2, 3/2, 5/2 ⁺)		0		J ^π : primary 4726.6γ to 3/2 ⁻ , 6210.2γ to 3/2 ⁺ , 7684.5γ to 1/2 ⁺ .
8906.8 5	5/2 ⁺	0.002 keV 1	OP	x	XREF: P(8909)x(8921) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2 ⁺ rules out 3/2 ⁺ based on RUL and Γ=0.002 keV.
8919.8 5	(5/2 ⁻ , 7/2, 9/2 ⁺)		0		XREF: x(8921) J ^π : primary 4571.7γ to 9/2 ⁻ , 7155.9γ to 5/2 ⁺ .
8933.3 5			0		
8953.1 5	3/2 ⁺	0.075 keV 15	OP		XREF: P(8955) E(level): weighted average of 8953.0 5 from (p,γ), and 8955 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and (3/2) ⁺ from R-Matrix analysis in (p,p):resonances.
8957.9 5	(1/2 ⁻ , 3/2 ⁻)	0.04 keV 1	L	OP	XREF: P(8959) E(level): others: 8957.7 44 from (α,p) and 8959 3 from (p,p):resonances.
8981.9 5	5/2 ⁻ , 7/2 ⁻	0.003 keV 2	OP		XREF: P(8983) E(level): weighted average of 8981.9 5 from (p,γ), and 8983 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8984.2 5	5/2 ⁺	0.025 keV 10	OP		XREF: P(8985) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.025 keV.
8988.4 5			0		
8992.4 5			0		
8996.7 5	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	L	OP	XREF: L(8996.9)P(8998)
9001.1 5			0		
9019.3 5	3/2 ⁻	3.27 keV 30	OP		XREF: P(9021) Γ: weighted average of 3.1 keV 3 from (p,γ) and 3.50 keV 35 from (p,p):resonances.
9024.5 5			1	0	
9029.9 5	(5/2) ⁺	0.04 keV 1	1	OP	XREF: P(9031)
9033.1 5			0		
9038.4 5	1/2 ⁻	0.292 keV 30	OP		XREF: P(9040) E(level): weighted average of 9038.3 5 from (p,γ), and 9040 3 from ³⁴ S(p,p),(p,p'γ):resonances.
9048.4 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	OP		XREF: P(9048)
9050 3	(5/2) ⁺	0.095 keV 15	P		
9081.5 4	5/2 ⁺	59 eV 10	L	OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 4 from (p,γ), and 9084 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=5/2 from γ(θ) in (p,γ) and (5/2) ⁺ from R-Matrix analysis in (p,p):resonances.
					Γ: weighted average of 65 eV 20 from (p,γ) and 57 eV 10 from (p,p):resonances.
9088.5 5			L	OP	XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 5 from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
				(p,γ), and 9089 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9099.3 5			O	
9100.4 5	3/2 ⁻	0.20 keV 2	OP	XREF: P(9101)
9107.4 5			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 5 from (p,γ), and 9108 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9110.1 5			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 5 from (p,γ), and 9114 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9124.0 5			OP	XREF: P(9120)
9127 4	5/2 [@]		J P	E(level): from (p,p):resonances. Other: 9127 9 from $^{31}\text{P}(\alpha, \text{p})$:resonances.
9135.1 5		2.0 keV 4	Op	XREF: p(9136) Γ: from (p,γ).
9138.3 5			Op	XREF: p(9136)
9147 4	(7/2 to 17/2) ⁺		N P	XREF: N(9150)P(9147) J ^π : L(α,d)=6 from 3/2 ⁺ .
9155.7 5			O	
9157.1 5	5/2 ⁻		OP	XREF: P(9159) J ^π : spin=5/2 from γ(θ) in (p,γ); 5993.7γ M1+E2 to 7/2 ⁻ .
9161 4	1/2 [@]		J 1	XREF: l(9160.0)
9163.3 5	(3/2 ⁻)		1 OP	XREF: l(9160.0)P(9168) E(level): other: 9168 4 from (p,p):resonances.
9184.2 5			Op	XREF: p(9185)
9188.8 5			Op	XREF: p(9185)
9194.1 5			L OP	E(level): others: 9194.0 39 from (α,p), and 9196 4 from (p,p):resonances.
9207 4	9/2 ⁺		P	
9222 4	5/2 ⁺		P	X XREF: X(9220) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=9220 120.
9227 4	5/2 ⁻		P	
9241 4	7/2 ⁺		P	
9255 4	1/2 [@]		J P	E(level): from (p,p):resonances. Other: 9256 4 from $^{31}\text{P}(\alpha, \text{p})$:resonances.
9264.9 39			L	
9271 4			P	
9276 4			P	
9284 4	5/2 ⁻		P	
9294 4			P	
9299 4			P	
9316.2 39			L	
9325 4	5/2 ⁺		P	
9335 4	7/2 ⁺		L P	E(level): weighted average of 9336 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances and 9334 5 from $^{32}\text{S}(\alpha, \text{p})$.
9345 4			P	
9349 4			P	
9355 4			P	
9376 4			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
9400 9	1/2 [@]		J	
9450 10	(7/2 to 17/2) ⁺		N	J ^π : L(α,d)=6 from 3/2 ⁺ .
9456 4	3/2 [@]		J	
9476 4			L	
9481.0 18	3/2 [@]		J	
9508 4	(1/2 ⁻ , 3/2, 5/2 ⁺)		L X	XREF: X(9514) J ^π : L(pol d, ³ He)=(1,2) from 0 ⁺ .
9551 6	5/2 [@]		J	
9673 6	1/2 [@]		J	
9712.7 27	1/2 [@]		J L	E(level): weighted average of 9713.0 27 from ³¹ P(α,p),(α,n):resonances and 9711.8 44 from (α,p).
9740 4			L	
9751.1 27	7/2 [@]		J	
9799 5			L	
9814.0 27	5/2 ⁽⁺⁾		J X	XREF: X(9820) J ^π : spin=5/2 from angular distributions in ³¹ P(α,p):resonances; L=(1,2) from 0 ⁺ in (pol d, ³ He) gives (1/2 ⁻ , 3/2, 5/2 ⁺).
9836 5			L	
9869.8 27	1/2 [@]		J L	XREF: L(9870)
9900.8 27	3/2 [@]		J	
9923 4	3/2 ^{-@}		J	
9958 6	3/2 [@]		J	
9969 5	(1/2, 3/2) [@]		J L	XREF: J(9968) E(level): from (α,p) and (α,n):resonances.
10031 5	(1/2 ⁺ , 3/2 ⁻) [@]		J	
10058 5	3/2 [@]		J	
10075 5	(1/2) [@]		J L	E(level): from (α,p) and (α,n):resonances.
10090 5	5/2 ⁺ [@]		J	
10132 5	3/2 ^{-@}		J	
10163 5	[@]		L	
10172 5	3/2 ⁻ , 5/2 ^{-@}		J	
10179 5			L	
10181.1 ^a 5	19/2 ⁻	42.3 fs 28	F H	E(level): weighted average of 10181.1 10 from (²⁸ Si,αpγ) and 10181.1 5 from (¹⁶ O,αpγ). J ^π : 1693.5γ E2, ΔJ=2 to 15/2 ⁻ ; 1336.3γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=61 fs 4 from (²⁸ Si,αpγ) (2013Bi10, DSAM).
10215 5	3/2 [@]		J L	XREF: L(10218.0)
10221.9 11	(17/2 ⁻)	<0.28 ps	F H	XREF: F(?) E(level): from (¹⁶ O,αpγ). J ^π : 1377.8γ to 17/2 ⁺ consistent with ΔJ=0. T _{1/2} : lifetime τ<0.4 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM).
10236 5	3/2 ^{-@}		J	
10278 5	1/2 [@]		J	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
10295 5			J	
10322 5	5/2 ⁺ @		J	
10340 5	(3/2,5/2) @		J	
10359 5	3/2 @		J	
10385 5			J	
10395 5			L	
10399 5	3/2 ⁺ , 7/2 ⁺ @		J	
10431 5	5/2 ⁺ @		J	
10465 5	5/2 ⁺ @		J L	E(level): weighted average of 10467 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10463.2 49 from (α, p) .
10486 5	5/2 ⁺ @		J	
10517 6			L	
10534 5	1/2 ⁺ , 5/2 ⁺ @		J	
10548 6			L	
10584 5	3/2 ⁺ @		J L	E(level): weighted average of 10587 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10579.5 54 from (α, p) .
10642 5	5/2 ⁺ @		J L	E(level): weighted average of 10641 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10643.0 50 from (α, p) .
10678 5	3/2 ⁺ , 5/2 ⁺ @		J	
10696 5	7/2 ⁻ @		J	
10733 5	3/2 ⁺ @		J L	E(level): weighted average of 10734 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10732.1 50 from (α, p) .
10760 5	3/2 ⁺ , 7/2 ⁻ @		J L	E(level): weighted average of 10760 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10759.1 55 from (α, p) .
10800 5	5/2 ⁺ @		J	
10816 5	1/2 ⁺ , 3/2 ⁻ @		J	
10844 5	3/2 ⁻ @		J	
10858.5 ^b 8	19/2 ⁺	<0.28 ps	F H	XREF: F(?) J ^π : 2069.8γ E2, ΔJ=2 to 15/2 ⁺ , 2014.7γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha \text{py}$) (2007Ks01, DSAM).
10870 5	(5/2, 7/2) @		J	
10886 5			J	
10909 5	3/2 ⁺ , 5/2 ⁻ @		J	
10928 5	7/2 @		J	
10942 5			J	
10964 5	5/2 ⁺ @		J	
10973 5			J	
10998 5	3/2 ⁺ , 7/2 ⁻ @		J	
11034 5	3/2 ⁺ @		J	
11063 5	1/2 ⁺ , 9/2 ⁺ @		J	
11082 5	5/2 ⁺ , 9/2 ⁺ @		J	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
11105 5			J	
11123 5	3/2 [@]		J	
11142 5	5/2 ⁺ [@]		J	
11154 5			J	
11169 5			J	
11179 5	3/2 ⁻ , 5/2 ⁺ [@]		J	
11185 5			J	
11198 5	3/2 [@]		J	
11230 5			J	
11246 5	3/2 [@]		J	
11265 5	3/2 ⁻ [@]		J	
11287 5			J	
11307 5			J	
11314 5	1/2 ⁻ , 5/2 ⁺ [@]		J	
11327 5	1/2 ⁻ , 5/2 ⁺ [@]		J	
11357 5			J	
11375 5			J	
11390 5			J	
11410 5			J	
11421 5	3/2 ⁻ [@]		J	
11434 5			J	
11459.1 ^b 7	21/2 ⁺	43.0 fs 21	F H J	E(level): weighted average of 11459.2 12 from ($^{28}\text{Si}, \alpha\gamma$) and 11459.0 7 from ($^{16}\text{O}, \alpha\gamma$). Other: 11460 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances. J ^π : 2614.5γ E2, ΔJ=2 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=62 fs 3 from ($^{28}\text{Si}, \alpha\gamma$) (2013Bi10, DSAM).
11473 5	1/2 ⁻ , 5/2 ⁻ [@]		J	
11492 5			J	
11505 5	3/2 ⁻ , 5/2 ⁻ [@]		J	
11525 5			J	
11540 5			J	
11551 5	3/2 ⁻ , 5/2 ⁺ [@]		J	
11565 5			J	
11589 5			J	
11609 5	5/2 ⁺ , 7/2 ⁻ [@]		J	
11627 5	1/2 ⁺ , 5/2 ⁺ [@]		J	
11638 5			J	
11651 5			J	
11684 5	1/2 ⁺ , 5/2 ⁺ [@]		J	
11773 5	3/2 ⁻ , 7/2 ⁻ [@]		J	
11783 5	3/2 ⁻ [@]		J	
1.25×10 ⁴ 5			J	X T=3/2 XREF: J(12900) E(level): from (pol d, ^3He). E(level): weighted average of 12572.2 12 from ($^{28}\text{Si}, \alpha\gamma$) and 12572.2 6 from ($^{16}\text{O}, \alpha\gamma$). J ^π : 2390.8γ E2, ΔJ=2 to 19/2 ⁻ , 1113.3γ D, ΔJ=1 to 21/2 ⁺ ; band assignment.
12572.2 ^d 6	23/2 ⁻	33 fs 4	F H	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
13900				T _{1/2} : lifetime $\tau=48$ fs 6 from ($^{28}\text{Si},\alpha p\gamma$) (2013Bi10, DSAM).
16306.4 ^a 16	(27/2 ⁻)	<9.0 fs	F J	T=3/2 J ^π : 3734 γ to 23/2 ⁻ ; band assignment. T _{1/2} : lifetime $\tau<13$ fs from ($^{28}\text{Si},\alpha p\gamma$) (2013Bi10, DSAM).

[†] Most of Adopted Levels are reported in (p, γ) dataset with precise E(level) values, deduced based on measured γ -ray energies which however are not reported in references in (p, γ) dataset, while E(level) values from E γ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p, γ) datasets. Therefore, a least-squares fit to available E γ values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p, γ) and the evaluators have adopted E(level) values from (p, γ) or averages of values based on E γ data from all available datasets where applicable, as noted.

[‡] For proton-resonance levels populated in (p,p):resonances, J^π assignments are from R-Matrix analysis, unless otherwise noted.

[#] From FRESKO-DWBA analysis of measured $\sigma(\theta)$ of (α ,p) (2019Se09).

[@] From angular distributions in $^{31}\text{P}(\alpha,p)$:resonances.

[&] Sources of T_{1/2} values are listed under comments; Γ from $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances, unless otherwise noted.

^a Band(A): Band based on f_{7/2} orbital.

^b Band(B): Band based on 13/2⁺.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	
1219.38	1/2 ⁺	1219.4 1	100	0.0	3/2 ⁺	M1+E2	0.093 10	3.66×10^{-5} 5	B(M1)(W.u.)=0.101 +14-11; B(E2)(W.u.)=2.2 +6-5 E _γ : weighted average of 1219.3 2 from ^{35}Ar ε decay and 1219.4 1 from (p,p'γ), and 1219.4 6 from (α,pγ). Mult.,δ: ΔJ=1 from γ(θ) in (p,p'γ), E1+M2 ruled out by RUL. δ from adopted B(E2)†=0.00076 7 measured in Coulomb excitation and adopted lifetime τ=0.172 ps 20.
1763.15	5/2 ⁺	543.77 1763.16 10	<0.2 100	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[E2] M1+E2	-2.85 10	0.000323 5 0.0002124 30	I _γ : from (p,γ). Other: <0.5 from (α,pγ). B(M1)(W.u.)=0.00122 +14-12; B(E2)(W.u.)=12.1 +11-9 E _γ : weighted average of 1763.0 2 from ^{35}Ar ε decay, 1763.1 2 from (^{24}Mg ,αpγ), 1763.3 2 from (^{16}O ,αpγ), 1763.15 10 from (^{14}N ,αpnγ), and 1763.2 1 from (p,p'γ). Other: 1763.2 6 from (α,pγ). Mult.: from γγ(ADO) in (^{16}O ,αpγ) with ΔJ=1 and γ(lin pol) in (^{28}Si ,αpγ). δ: weighted average of -3.0 1 from (α,pγ), -2.66 12 from (p,p'γ), -2.95 45 from Coulomb excitation, -2.6 4 from (^{16}O ,αpγ), all from γ(θ) or pγ(θ). Other: -0.42 17 from γγ(θ) in (p,γ); 1.8 +10-4; deduced by the evaluators from adopted B(E2)†=0.0106 7 and T _{1/2} =0.36 ps 3.
2645.67	7/2 ⁺	882.80 11	11.4 10	1763.15	5/2 ⁺	M1+E2	+0.21 5	5.29×10^{-5} 9	B(M1)(W.u.)=0.0196 +35-27; B(E2)(W.u.)=4.2 +23-18 E _γ : weighted average of 882.9 1 from (^{24}Mg ,αpγ), 882.7 5 from (^{16}O ,αpγ), 882.32 35 from (^{14}N ,αpnγ), 882.4 10 from (α,pγ), and 882.3 3 from (p,p'γ). I _γ : unweighted average of 8.7 6 from (^{28}Si ,αpγ), 15.2 13 from (^{24}Mg ,αpγ), 11.08 28 from (^{16}O ,αpγ), 11.1 22 from (α,pγ), 9.5 11 from (p,γ), and 13 5 from (p,p'γ). Mult.: D+Q from pγ(θ) in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 5 from (α,pγ) and +0.17 5 from (p,p'γ). I _γ : from (α,pγ). Other: <2.2 from (p,γ). B(E2)(W.u.)=3.6 +6-4 E _γ : weighted average of 2645.5 4 from (^{24}Mg ,αpγ), 2645.8 5 from (^{16}O ,αpγ), 2645.79 30 from
		1426.26 2645.70 20	<1.2 100.0 11	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[M3] E2		7.52×10^{-5} 11 0.000633 9	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	
									($^{14}\text{N}, \alpha p n \gamma$), and 2645.7 2 from (p,p' γ). Other: 2646.2 15 from ($\alpha, p \gamma$).
									I_γ : others: 100.0 31 from ($^{24}\text{Mg}, \alpha p \gamma$), 100.0 22 from ($^{16}\text{O}, \alpha p \gamma$), 100.0 22 from ($\alpha, p \gamma$), and 100.0 20 from (p,p' γ). Other: 100 6 from ($^{28}\text{Si}, \alpha p \gamma$).
									Mult.: from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha p \gamma$); $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ).
2694.00	3/2 ⁺	930.9 2	16.3 13	1763.15	5/2 ⁺	M1+E2	+0.09 3	4.68×10 ⁻⁵ 7	B(M1)(W.u.)=0.110 +26-19; B(E2)(W.u.)=3.9 +33-22 E_γ : weighted average of 930.7 2 from ^{35}Ar ε decay, 930.5 15 from ($\alpha, p \gamma$), and 931.3 3 from (p,p' γ). I_γ : weighted average of 14.3 18 from ^{35}Ar ε decay, 18.2 26 from ($\alpha, p \gamma$), 16.3 13 from (p, γ), and 18.6 25 from (p,p' γ).
		1474.8 3	9.5 10	1219.38	1/2 ⁺	M1+E2	-0.63 21	8.4×10 ⁻⁵ 4	Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ). B(M1)(W.u.)=0.0116 +37-29; B(E2)(W.u.)=8.0 +45-39 E_γ : weighted average of 1474.8 3 from ^{35}Ar ε decay and 1475 2 from (p,p' γ). I_γ : weighted average of 9.3 10 from ^{35}Ar ε decay, 11.7 26 from ($\alpha, p \gamma$), 9.5 13 from (p, γ), and 8.3 25 from (p,p' γ).
		2693.7 2	100.0 17	0.0	3/2 ⁺	M1+E2	+0.26 2	0.000553 8	Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ). B(M1)(W.u.)=0.026 +6-4; B(E2)(W.u.)=0.92 +26-19 E_γ : weighted average of 2693.7 2 from ^{35}Ar ε decay and 2694.1 6 from (p,p' γ). Other: 2693.2 20 from ($\alpha, p \gamma$). I_γ : weighted average of 100.0 17 from ^{35}Ar ε decay, 100 4 from ($\alpha, p \gamma$), 100.0 25 from (p, γ), and 100.0 30 from (p,p' γ).
									Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ). δ : weighted average of +0.17 8 from ($\alpha, p \gamma$) and +0.26 2 from (p,p' γ).
3002.64	5/2 ⁺	308.64	<5	2694.00	3/2 ⁺	[M1,E2]		0.0014 9	I_γ : from ($\alpha, p \gamma$).
		356.97	<1	2645.67	7/2 ⁺	[M1,E2]		9	I_γ : from ($\alpha, p \gamma$).
		1239.47	2.2 11	1763.15	5/2 ⁺	[M1,E2]		4.4×10 ⁻⁵ 6	I_γ : from (p,p' γ). B(M1)(W.u.)=0.011 +8-5 if M1, B(E2)(W.u.)=26 +19-12 if E2.
		1783.21	3.3 11	1219.38	1/2 ⁺	[E2]		0.0002268 32	B(E2)(W.u.)=6.4 +35-22 I_γ : from (p,p' γ).
		3002.3 4	100.0 10	0.0	3/2 ⁺	M1+E2	+0.08 2	0.000671 9	B(M1)(W.u.)=0.034 +14-7; B(E2)(W.u.)=0.09 +7-4 E_γ : weighted average of 3002.1 4 from ^{35}Ar ε decay,

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Mult.[#]</u>	<u>$\delta^{\#}$</u>	<u>α^a</u>	<u>Comments</u>
3162.87	7/2 ⁻	160.67 20	1.5 4	3002.64	5/2 ⁺	[E1]		0.00330 5	3002.5 15 from (α ,p γ), and 3002.5 5 from (p,p' γ). I γ : weighted average of 100.0 19 from ³⁵ Ar ε decay and 100.0 10 from (p,p' γ). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,p' γ). δ : weighted average of +0.09 3 from (α ,p γ) and +0.07 2 from (p,p' γ). Other: -0.22 1 from (p, γ). B(E1)(W.u.)=6.6×10 ⁻⁵ 17 E γ : weighted average of 161 1 from (¹⁶ O, α p γ) and 160.66 20 from (¹⁴ N, α p γ). I γ : unweighted average of 1.10 30 from (²⁸ Si, α p γ) and 1.89 22 from (p, γ). B(E1)(W.u.)=1.61×10 ⁻⁵ 20
		517.47 11	12.3 16	2645.67	7/2 ⁺	(E1)		0.0001097 15	E γ : weighted average of 517.7 1 from (²⁴ Mg, α p γ), 517.3 5 from (¹⁶ O, α p γ), 517.26 10 from (¹⁴ N, α p γ), 517.4 15 from (α ,p γ), and 517.0 8 from (p,p' γ). I γ : unweighted average of 9.2 5 from (²⁸ Si, α p γ), 14.3 8 from (²⁴ Mg, α p γ), 11.1 13 from (¹⁶ O, α p γ), 11.1 33 from (α ,p γ), 8.9 11 from (p, γ), and 19 8 from (p,p' γ). Mult.: D from $\gamma\gamma(\theta)$ (DCO) in (²⁸ Si, α p γ); $\Delta\pi$ =yes from level scheme.
		1399.9 7	0.34 5	1763.15	5/2 ⁺	(E1+M2)	+0.44 12	0.000181 12	B(E1)(W.u.)=1.88×10 ⁻⁸ +31-34; B(M2)(W.u.)=0.0085 +43-38 I γ : weighted average of 0.50 20 from (¹⁶ O, α p γ) and 0.33 5 from (p, γ). Mult., δ : D+Q from $\gamma(\theta)$ in (p, γ) (1971Pr11); $\Delta\pi$ =yes from level scheme.
		1943.43	<0.22	1219.38	1/2 ⁺	[E3]		0.0001742 24	B(M2)(W.u.)=0.255 9; B(E3)(W.u.)=4.2 +11-9
		3162.53 10	100.0 11	0.0	3/2 ⁺	M2+E3	+0.17 2	0.000523 7	E γ : weighted average of 3162.7 4 from (²⁴ Mg, α p γ), 3162.6 3 from (¹⁶ O, α p γ), 3162.43 10 from (¹⁴ N, α p γ), and 3162.6 1 from (p,p' γ), and 3162.6 15 from (α ,p γ). I γ : weighted average of 100 5 from (²⁸ Si, α p γ), 100.0 30 from (²⁴ Mg, α p γ), 100.0 20 from (¹⁶ O, α p γ), 100.0 33 from (α ,p γ), 100.0 11 from (p, γ), and 100.0 20 from (p,p' γ). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p, γ). δ : weighted average of +0.16 1 from (²⁸ Si, α p γ), +0.18 7 from (¹⁶ O, α p γ), and +0.25 4 from (p, γ). Others: -0.14 2 from (α ,p γ) and -0.22 5 from (p,p' γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
4112.4	7/2 ⁺	2349.2	89 6	1763.15	5/2 ⁺	M1+E2	-0.16 2	0.000405 6	B(M1)(W.u.)=0.0159 +48-31; B(E2)(W.u.)=0.28 +12-8 I _γ : other: 16 5 from (α,pγ).
		4112.1	100 6	0.0	3/2 ⁺	E2(+M3)	0.00 10	1.21×10 ⁻³ 2	B(E2)(W.u.)=0.76 +22-15 I _γ : other: 100 5 from (α,pγ).
4173.43	(5/2 ⁺)	1479.40	45 14	2694.00	3/2 ⁺	[M1,E2]		9.1×10 ⁻⁵ 13	B(M1)(W.u.)=0.048 +20-12 if M1, B(E2)(W.u.)=82 +34-21 if E2. B(E2)(W.u.)=82 +34-21 upper bound exceeds RUL=100 if E2.
		1527.72	<17	2645.67	7/2 ⁺	[M1,E2]		0.000106 15	
		2410.19	28 9	1763.15	5/2 ⁺	[M1,E2]		0.00048 5	B(M1)(W.u.)=0.0068 +33-19 if M1, B(E2)(W.u.)=4.5 +21-13 if E2.
		2953.92	≤5.2	1219.38	1/2 ⁺	[E2]		0.000771 11	
		4173.16	100 17	0.0	3/2 ⁺	[M1,E2]		0.00116 7	B(M1)(W.u.)=0.0047 +15-7 if M1, B(E2)(W.u.)=1.02 +33-14 if E2.
4177.9	3/2 ⁻	1175.2	<0.82	3002.64	5/2 ⁺	[E1]		6.06×10 ⁻⁵ 8	
		1483.9	13.1 33	2694.00	3/2 ⁺	[E1]		0.000265 4	B(E1)(W.u.)=5.7×10 ⁻⁴ +16-14
		1532.2	<0.82	2645.67	7/2 ⁺	[M2]		6.18×10 ⁻⁵ 9	
		2414.7	<1.6	1763.15	5/2 ⁺	[E1]		0.000921 13	
		2958.4	51 8	1219.38	1/2 ⁺	(E1+M2)	+0.11 4	1.22×10 ⁻³ 2	B(E1)(W.u.)=2.8×10 ⁻⁴ +5-4; B(M2)(W.u.)=1.7 +16-11 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme. B(M2)(W.u.)=1.7 +16-11 upper bound exceeds RUL=3.
		4177.6	100 8	0.0	3/2 ⁺	(E1+M2)	+0.06 3	1.74×10 ⁻³ 3	B(E1)(W.u.)=1.93×10 ⁻⁴ +28-20; B(M2)(W.u.)=0.18 +24-14 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme.
4347.76	9/2 ⁻	1184.90 20	100.0 29	3162.87	7/2 ⁻	M1+E2	-0.36 3	3.57×10 ⁻⁵ 5	B(M1)(W.u.)=0.007 +7-3; B(E2)(W.u.)=2.5 +27-10 E _γ : weighted average of 1185.0 2 from (²⁴ Mg,αpγ), 1184.9 3 from (¹⁶ O,αpγ), and 1184.80 20 from (¹⁴ N,αpnγ). Other: 1184.6 10 from (α,pγ). I _γ : weighted average of 100 4 from (²⁸ Si,αpγ), 100.0 31 from (²⁴ Mg,αpγ), 100 4 from (¹⁶ O,αpγ), 100.0 31 from (α,pγ), and 100.0 29 from (p,γ). Mult.: from γγ(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=1. δ: weighted average of -0.36 3 from (α,pγ), -0.3 1 from (²⁸ Si,αpγ), and -0.65 30 from (¹⁶ O,αpγ). B(E1)(W.u.)=3.7×10 ⁻⁵ +39-14; B(M2)(W.u.)=0.019
		1701.88 25	46.8 24	2645.67	7/2 ⁺	E1+M2	-0.018 12	0.000434 6	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
								+47-17 E_γ : weighted average of 1702.0 3 from ($^{24}\text{Mg}, \alpha\text{py}$), 1701.7 5 from ($^{16}\text{O}, \alpha\text{py}$), and 1701.85 25 from ($^{14}\text{N}, \alpha\text{pny}$). Other: 1702.1 15 from (α, py). I_γ : unweighted average of 45.6 24 from ($^{28}\text{Si}, \alpha\text{py}$), 39.5 24 from ($^{24}\text{Mg}, \alpha\text{py}$), 52.0 14 from ($^{16}\text{O}, \alpha\text{py}$), 52 7 from (α, py), and 44.9 29 from (p, γ). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha\text{py}$) with $\Delta J=1$ and $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (α, py). δ : from (p, γ). Others: +0.5 +5-3 from ($^{28}\text{Si}, \alpha\text{py}$), +0.5 4 from ($^{16}\text{O}, \alpha\text{py}$).
4347.76	9/2 ⁻	2584.51 4347.8 8	≤ 4.4 12.4 11	1763.15 0.0	5/2 ⁺ 3/2 ⁺	[M2] [E3]	0.000329 5 0.001000 14	B(E3)(W.u.)=44 +46-17 E_γ, I_γ : from ($^{16}\text{O}, \alpha\text{py}$). E_γ : other: 4618 2 from (α, py). Mult., δ : from $\text{py}(\theta)$ with -0.08 14 for J=3/2; +0.36 15 for J=5/2 in (α, py). If J(4624=5/2), Mult would be M1+E2 based on RUL.
4624.3	3/2, 5/2	4624.0	100	0.0	3/2 ⁺	D+Q		
4769.86	(5/2 ⁻)	1606.95 1767.17 2075.79 3550.29 4769.51	100 16 54 16 <15 <15 <15	3162.87 3002.64 2694.00 1219.38 0.0	7/2 ⁻ 5/2 ⁺ 3/2 ⁺ 1/2 ⁺ 3/2 ⁺	[M1, E2] [E1] [E1] [M2] [E1]	0.000133 19 0.000483 7 0.000702 10 0.000645 9 1.95 $\times 10^{-3}$ 3	B(M1)(W.u.)=0.040 +23-5 if M1, B(E2)(W.u.)=58 +34-7 if E2. B(E1)(W.u.)=4.6 $\times 10^{-4}$ +30-9
4839.10	(1/2 ⁺ , 3/2)	1836.41 2145.03 2193.36 ^b 3075.81 3619.52 4838.74	<12 <6.8 <5.1 100 9 <12 70 9	3002.64 2694.00 2645.67 1763.15 1219.38 0.0	5/2 ⁺ 3/2 ⁺ 7/2 ⁺ 5/2 ⁺ 1/2 ⁺ 3/2 ⁺			
4854.4	(1/2, 3/2)	3634.8 4854.0	100 7 33 7	1219.38 0.0	1/2 ⁺ 3/2 ⁺			
4880.96	7/2 ⁽⁺⁾	1878.27 2186.89 2235.21	15 5 <16 47 8	3002.64 2694.00 2645.67	5/2 ⁺ 3/2 ⁺ 7/2 ⁺	[M1, E2] [E2] [M1, E2]	0.000240 30 0.000417 6 0.00040 4	B(M1)(W.u.)=0.051 +30-16 if M1, B(E2)(W.u.)=55 +32-17 if E2. I_γ : other: 52 9 from (α, py) for a doublet. B(M1)(W.u.)=0.095 +43-18 if M1, B(E2)(W.u.)=72 +33-14 if E2. B(E2)(W.u.)=72 +33-14 upper bound exceeds RUL=100 if E2. E_γ : from (α, py). I_γ : weighted average of 100 9 from (α, py) and 100 8 from (p, γ). B(M1)(W.u.)=0.075 +33-12 if M1, B(E2)(W.u.)=29 +13-5 if E2.
		3117.4 20	100 8	1763.15	5/2 ⁺	[M1, E2]	0.00078 6	
		4880.60	<8.1	0.0	3/2 ⁺	[E2]	1.46 $\times 10^{-3}$ 2	

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
5009.1	(1/2,3/2)	835.7	<25	4173.43	(5/2 ⁺)				
		950.0	<10	4059.1	(3/2) ⁻				
		1846.2 ^b	<50	3162.87	7/2 ⁻				
		2006.4	<20	3002.64	5/2 ⁺				
		2315.0	<25	2694.00	3/2 ⁺				
		2363.3 ^b	<5	2645.67	7/2 ⁺				
		3245.8	<10	1763.15	5/2 ⁺				
		3789.5	<7	1219.38	1/2 ⁺				
		5008.7	100	0.0	3/2 ⁺				
5163.34	7/2 ⁻	2000.41	93 12	3162.87	7/2 ⁻	D+Q	+0.44 20		
		2160.63	23 9	3002.64	5/2 ⁺				
		3400.01	100 12	1763.15	5/2 ⁺	D(+Q)	0.00 3		
		3943.72 ^b	<23	1219.38	1/2 ⁺				
		5162.93	<23	0.0	3/2 ⁺				
5214.5	(5/2 ⁺)	205.4	<23	5009.1	(1/2,3/2)				
		1041.1	<60	4173.43	(5/2 ⁺)				
		1155.4	<10	4059.1	(3/2) ⁻				
		2051.6	<15	3162.87	7/2 ⁻				
		2211.8	<25	3002.64	5/2 ⁺				
		2520.4	<15	2694.00	3/2 ⁺				
		2568.7	<15	2645.67	7/2 ⁺				
		3451.2	<6	1763.15	5/2 ⁺				
		3994.9	<4	1219.38	1/2 ⁺				
		5214.1	100	0.0	3/2 ⁺	[M1,E2]		0.00147 8	
5403.3	3/2 ⁻	2400.6	46 9	3002.64	5/2 ⁺	[E1]		0.000913 13	B(E1)(W.u.)=0.0011 +6-2
		3640.0	<18	1763.15	5/2 ⁺	[E1]		1.54×10 ⁻³ 2	
		4183.7	100 18	1219.38	1/2 ⁺	[E1]		1.75×10 ⁻³ 2	B(E1)(W.u.)=4.5×10 ⁻⁴ +20-6
		5402.9	<13	0.0	3/2 ⁺	[E1]		2.14×10 ⁻³ 3	
5407.07	11/2 ⁻	1059.32 20	14.5 22	4347.76	9/2 ⁻	M1+E2	+0.14 3	3.67×10 ⁻⁵ 5	B(M1)(W.u.)=0.0082 +30-20; B(E2)(W.u.)=0.54 +35-23
									E_γ : weighted average of 1059.3 2 from (²⁴ Mg, $\alpha\gamma$), 1059.5 2 from (¹⁶ O, $\alpha\gamma$), and 1059.17 20 from (¹⁴ N, $\alpha p n\gamma$). Other: 1059.3 10 from ($\alpha, p\gamma$).
									I_γ : unweighted average of 12.0 12 from (²⁸ Si, $\alpha\gamma$), 16.9 17 from (²⁴ Mg, $\alpha\gamma$), 19.2 5 from (¹⁶ O, $\alpha\gamma$), and 9.9 22 from ($\alpha, p\gamma$).
									Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in (²⁸ Si, $\alpha\gamma$), $\Delta J=1$.
									δ : weighted average of +0.12 3 from (¹⁶ O, $\alpha\gamma$),

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
5407.07	11/2 ⁻	2243.71 23	100.0 22	3162.87	7/2 ⁻	E2		0.000444 6	+0.2 1 from (²⁸ Si, α py), and +0.25 8 from (α ,py). B(E2)(W.u.)=4.6 +15-9 E _γ : weighted average of 2244.2 3 from (²⁴ Mg, α py), 2243.3 2 from (¹⁶ O, α py), 2244.00 25 from (¹⁴ N, α pny), and 2244 3 from (α ,py). I _γ : weighted average of 100.0 33 from (²⁸ Si, α py), 100.0 32 from (²⁴ Mg, α py), 100.0 23 from (¹⁶ O, α py), and 100.0 22 from (α ,py). Mult.: from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, α py), $\Delta J=2$.
5586.0	5/2 ⁺	2940.2 3822.6 4366.3 5585.5	67 17 100 17 <17 <17	2645.67 7/2 ⁺ 1763.15 5/2 ⁺ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5598.4	3/2 ⁺ , 5/2 ⁺	3835.0	37 14	1763.15	5/2 ⁺	[M1,E2]		0.00105 7	B(M1)(W.u.)=0.050 +29-19 if M1, B(E2)(W.u.)=13 +8-5 if E2.
		5597.9	100 14	0.0 3/2 ⁺		[M1,E2]		0.00157 8	B(M1)(W.u.)=0.044 +23-12 if M1, B(E2)(W.u.)=5.3 +28-14 if E2.
5646	(5/2, 7/2, 9/2 ⁺)	2483 2643 3883	100	3162.87 7/2 ⁻ 3002.64 5/2 ⁺ 1763.15 5/2 ⁺					
5655	(3/2) ⁺	5646 2652	<8 100 6	0.0 3/2 ⁺ 3002.64 5/2 ⁺		[M1,E2]		0.00058 5	B(M1)(W.u.)=0.076 +30-14 if M1, B(E2)(W.u.)=41 +16-7 if E2.
		3892	7.5 25	1763.15 5/2 ⁺		[M1,E2]		0.00107 7	B(M1)(W.u.)=0.0018 +10-6 if M1, B(E2)(W.u.)=0.45 +25-15 if E2.
5682.2	1/2 ⁻ , 3/2 ⁻	4435 1504.3	<7.5 100	1219.38 1/2 ⁺ 4177.9 3/2 ⁻		[M1,E2]		0.00125 8	
5723.52	5/2 ⁺	3029.38 3960.13	63 8 100 11	2694.00 3/2 ⁺ 1763.15 5/2 ⁺					
5758	(1/2, 3/2 ⁻)	1580 4538 5758	67 22 <13 100 22	4177.9 3/2 ⁻ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5806.6	(1/2, 3/2, 5/2)	5806.1	100	0.0 3/2 ⁺					
5830	(5/2 ⁻)	2667	100	3162.87 7/2 ⁻		D+Q			I _γ : from (α ,py). δ : -2.2 5 or -0.6 1 for J=5/2 from $\gamma(\theta)$ in (α ,py).
5926.65	(11/2) ⁻	1579.09 30	100 11	4347.76 9/2 ⁻		M1+E2	-0.2 1	0.0001074 22	E _γ : weighted average of 1578.9 5 from (¹⁶ O, α py), 1579.15 30 from (¹⁴ N, α pny), and 1579.3 15 from (α ,py).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	α^a	Comments
									Mult., δ : D+Q and δ from $\gamma\gamma(\theta)$ (DCO) in $(^{28}\text{Si}, \alpha\gamma)$ with $\Delta J=1$; E1+M2 ruled out by RUL. Other: -0.8 4 from $\gamma(\theta)$ in (α, py) . I_γ : from $(^{28}\text{Si}, \alpha\gamma)$.
5926.65	(11/2) ⁻	1983.7	11.1 22	3942.9	9/2 ⁺	[E1]		0.000639 9	
		2763.66		3162.87	7/2 ⁻				
6087.31	13/2 ⁻	680.35 10	100.0 21	5407.07	11/2 ⁻	M1+E2	+0.020 15	8.70×10^{-5} 12	B(M1)(W.u.)=0.0107 10 E_γ : weighted average of 680.4 1 from $(^{24}\text{Mg}, \alpha\text{py})$, 680.5 5 from $(^{16}\text{O}, \alpha\text{py})$, and 680.22 15 from $(^{14}\text{N}, \alpha\text{pny})$. Other: 679.9 10 from (α, py) . I_γ : weighted average of 100.0 31 from $(^{28}\text{Si}, \alpha\text{py})$, 100.0 21 from $(^{16}\text{O}, \alpha\text{py})$, and 100 6 from (α, py) . Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (α, py) . Other: $\delta=+0.1$ 1 from $\gamma\gamma(\text{DCO})$ in $(^{28}\text{Si}, \alpha\text{py})$, $\Delta J=1$.
		1739.4 4	6.9 9	4347.76	9/2 ⁻	E2		0.0002076 29	B(E2)(W.u.)=0.054 +9-8 E_γ : weighted average of 1739.6 5 from $(^{16}\text{O}, \alpha\text{py})$ and 1739.3 4 from $(^{14}\text{N}, \alpha\text{pny})$. I_γ : weighted average of 6.3 6 from $(^{28}\text{Si}, \alpha\text{py})$, 7.9 9 from $(^{16}\text{O}, \alpha\text{py})$, and 18 6 from (α, py) . Mult.: from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\text{lin pol})$ in $(^{28}\text{Si}, \alpha\text{py})$, $\Delta J=2$.
6106.2	(3/2, 5/2 ⁺)	4342.8	10.3 35	1763.15	5/2 ⁺				
		4886.5	62 17	1219.38	1/2 ⁺				
		6105.6	100 17	0.0	3/2 ⁺				
6181	(1/2 ⁺ to 7/2 ⁺)	6180	100	0.0	3/2 ⁺				
6380.2	(9/2 ⁻)	296 1		6087.31	13/2 ⁻				I_γ : from $(^{16}\text{O}, \alpha\text{py})$.
		971 1		5407.07	11/2 ⁻				I_γ : from $(^{16}\text{O}, \alpha\text{py})$.
6492	(3/2 ⁺)	6491	100	0.0	3/2 ⁺				
6660.2	(11/2 ⁻)	2312.4		4347.76	9/2 ⁻				
		3497.1		3162.87	7/2 ⁻				
7066.2	(5/2 ⁺)	1902.8	1.8	5163.34	7/2 ⁻				
		2892.6	1.0	4173.43	(5/2 ⁺)				
		2953.7	1.2	4112.4	7/2 ⁺				
		3007.0	2.9	4059.1	(3/2) ⁻				
		3903.1	12	3162.87	7/2 ⁻				
		4063.3	6.1	3002.64	5/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[‡]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[#]</u>	<u>δ[#]</u>
7066.2	(5/2 ⁺)	4371.9	10	2694.00	3/2 ⁺		
		4420.2	3.3	2645.67	7/2 ⁺		
		5302.6	29	1763.15	5/2 ⁺		
		5846.3	37	1219.38	1/2 ⁺		
		7065.4	100	0.0	3/2 ⁺		
7103.4	3/2 ⁽⁻⁾	2925.4	2.3	4177.9	3/2 ⁻		
		3184.8	2.3	3918.47	3/2 ⁺		
		4100.5	4.7	3002.64	5/2 ⁺		
		4409.1	17	2694.00	3/2 ⁺	D(+Q)	0.00 3
		5339.8	6.3	1763.15	5/2 ⁺		
		5883.5	100	1219.38	1/2 ⁺	D(+Q)	0.00 3
		7102.6	23	0.0	3/2 ⁺		
7178.6	1/2 ⁽⁺⁾	1072.4	7.9	6106.2	(3/2,5/2 ⁺)		
		1775.3	4.5	5403.3	3/2 ⁻		
		2339.4	4	4839.10	(1/2 ⁺ ,3/2)		
		3005.0	2.1	4173.43	(5/2 ⁺)		
		3119.4	58	4059.1	(3/2) ⁻		
		3210.8	24	3967.6	1/2 ⁺		
		3260.0	7.9	3918.47	3/2 ⁺		
		4484.3	11	2694.00	3/2 ⁺		
		5958.7	45	1219.38	1/2 ⁺		
		7177.8	100	0.0	3/2 ⁺		
7185	5/2 ⁺	1970	<1.1	5214.5	(5/2 ⁺)		
		2176	<1.1	5009.1	(1/2,3/2)		
		3011	<11	4173.43	(5/2 ⁺)		
		3126	57 13	4059.1	(3/2) ⁻		
		4022	<6.5	3162.87	7/2 ⁻		
		4182	<6.5	3002.64	5/2 ⁺		
		4491	4.4 22	2694.00	3/2 ⁺		
		4539	13 5	2645.67	7/2 ⁺		
		5421	9 7	1763.15	5/2 ⁺		
		5965	35 15	1219.38	1/2 ⁺		
		7184	100 22	0.0	3/2 ⁺		
7194.6	1/2 ⁻	2185.4	1.2	5009.1	(1/2,3/2)		
		2340.1	3.0	4854.4	(1/2,3/2)		
		2355.4	2.7	4839.10	(1/2 ⁺ ,3/2)		
		3016.6	24	4177.9	3/2 ⁻		
		3135.4	4.5	4059.1	(3/2) ⁻		
		3226.8	2.7	3967.6	1/2 ⁺		
		3276.0	6	3918.47	3/2 ⁺		
		4500.3	1.6	2694.00	3/2 ⁺		
		5431.0	0.75	1763.15	5/2 ⁺		

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>								
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult. #</u>	<u>$\delta^\#$</u>	<u>Comments</u>
7194.6	1/2 ⁻	5974.7	100	1219.38	1/2 ⁺			
		7193.8	3.0	0.0	3/2 ⁺			
7225.5	5/2	1119.3	3.1	6106.2	(3/2,5/2 ⁺)			
		1418.9		5806.6	(1/2,3/2,5/2)			
		1543.3	3.4	5682.2	1/2 ⁻ ,3/2 ⁻			
		1571	1.4	5655	(3/2 ⁺)			
		1822.2	1.5	5403.3	3/2 ⁻			
		2010.9	12	5214.5	(5/2 ⁺)			
		3047.5	14	4177.9	3/2 ⁻			
		3166.3	3.4	4059.1	(3/2 ⁻)			
		3306.9	2.5	3918.47	3/2 ⁺			
		4222.6		3002.64	5/2 ⁺			
		4531.2	17	2694.00	3/2 ⁺	D+Q	+0.36 4	
		5461.9	12	1763.15	5/2 ⁺	D+Q	-0.36 7	
		7224.7	100	0.0	3/2 ⁺	D+Q	+0.36 4	
7234.0	5/2 ⁽⁺⁾	1579	0.22	5655	(3/2 ⁺)			
		3056.0		4177.9	3/2 ⁻			
		4539.7	1.9	2694.00	3/2 ⁺			
		4588.0	3.2	2645.67	7/2 ⁺			
		5470.4	0.22	1763.15	5/2 ⁺			
		6014.1	1.9	1219.38	1/2 ⁺			
		7233.2	100	0.0	3/2 ⁺	D+Q	-0.10	
7272.7	1/2 ⁻	1515	1.7	5758	(1/2,3/2 ⁻)			
		1869.4	0.29	5403.3	3/2 ⁻			
		2263.5	2.0	5009.1	(1/2,3/2)			
		2433.5	1.5	4839.10	(1/2 ⁺ ,3/2)			
		3099.1	0.58	4173.43	(5/2 ⁺)			
		3213.4	1.3	4059.1	(3/2 ⁻)			
		3304.9	1.2	3967.6	1/2 ⁺			
		4578.4	1.6	2694.00	3/2 ⁺			
		6052.8	35	1219.38	1/2 ⁺			
		7271.9	100	0.0	3/2 ⁺			
7361.9	3/2 ⁽⁺⁾	1707	0.41	5655	(3/2 ⁺)			
		2352.7	0.82	5009.1	(1/2,3/2)			
		2522.7	1.4	4839.10	(1/2 ⁺ ,3/2)			
		3183.8	0.69	4177.9	3/2 ⁻			
		3302.6	2.7	4059.1	(3/2 ⁻)			
		3394.1	4.1	3967.6	1/2 ⁺			
		3443.3	1.1	3918.47	3/2 ⁺			
		4359.0	0.41	3002.64	5/2 ⁺			
		4667.6	0.69	2694.00	3/2 ⁺			
		5598.3	11	1763.15	5/2 ⁺	D+Q	δ : >+2.5 or -0.01 10 from $\gamma(\theta)$ in (p, γ).	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7361.9	3/2 ⁽⁺⁾	6141.9	100	1219.38	1/2 ⁺	(M1+E2)			Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition. δ : -0.05 3 or $+2.5$ 3 in (p, γ). δ : $-0.16 < \delta(Q/D) < -0.12$ or $+7.5 < \delta(Q/D) < +10$ in (p, γ).
		7361.1	14	0.0	3/2 ⁺	D+Q			
7396.4	7/2 ⁽⁻⁾	3048.5	23	4347.76	9/2 ⁻				Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		3283.8	17	4112.4	7/2 ⁺				
		3453.3	11	3942.9	9/2 ⁺				
		4233.3	100	3162.87	7/2 ⁻	(M1+E2)	+0.28 25	1.12×10^{-3} 3	
		4393.5	30	3002.64	5/2 ⁺	D(+Q)	+0.011 18		
		4750.4	28	2645.67	7/2 ⁺				
		5632.8	4.3	1763.15	5/2 ⁺				
7450.9	3/2	3532.2	6.9 34	3918.47	3/2 ⁺				
		4448.0	13.7 14	3002.64	5/2 ⁺				
		4756.6	11.0 11	2694.00	3/2 ⁺				
		5687.3	5.5 28	1763.15	5/2 ⁺				
		6230.9	100 10	1219.38	1/2 ⁺				
7501.1	(5/2 ⁺ , 7/2, 9/2 ⁺)	3558.0	46 6	3942.9	9/2 ⁺				
		4498.2	23.6 24	3002.64	5/2 ⁺				
		4855.1	100 11	2645.67	7/2 ⁺				
		5737.5	12.7 13	1763.15	5/2 ⁺				
7502.9		1396.7 @	14 @	6106.2	(3/2, 5/2 ⁺)				
		1904.4 @	4 @	5598.4	3/2 ⁺ , 5/2 ⁺				
		2493.7 @	4 @	5009.1	(1/2, 3/2)				
		3324.8 @	6 @	4177.9	3/2 ⁻				
		3443.6 @	100 @	4059.1	(3/2) ⁻				
		3559.8 @	4 @	3942.9	9/2 ⁺				
		4339.7 @	8 @	3162.87	7/2 ⁻				
		4500.0 @	8 @	3002.64	5/2 ⁺				
		4856.9 @	14 @	2645.67	7/2 ⁺				
		5739.2 @	1.4 @	1763.15	5/2 ⁺				
		6282.9 @	36 @	1219.38	1/2 ⁺				
		7502.0 @	0.6 @	0.0	3/2 ⁺				
7518.8	7/2	1920.3	2.9 15	5598.4	3/2 ⁺ , 5/2 ⁺				
		1932.7	4.4 22	5586.0	5/2 ⁺				
		3170.9	2.9 15	4347.76	9/2 ⁻				
		3575.7	2.9 15	3942.9	9/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7518.8	7/2	4355.6	100 10	3162.87	7/2 ⁻	D(+Q)	+0.1 2		
		4872.8	4.4 22	2645.67	7/2 ⁺				
		5755.1	29.4 30	1763.15	5/2 ⁺	D+Q	+0.098 22		
7548.9	7/2 ⁻	1903	0.59 7	5646	(5/2,7/2,9/2 ⁺)				
		1962.8	0.11	5586.0	5/2 ⁺	[E1]		0.000624 9	
		2778.9	1.49 11	4769.86	(5/2 ⁻)	[M1,E2]		0.00064 6	
		4385.7	100.0 11	3162.87	7/2 ⁻	M1+E2	-0.07 2	1.16×10 ⁻³ 2	
		4545.9	2.13 11	3002.64	5/2 ⁺	(E1+M2)	+0.6 4	0.00163 22	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4902.9	0.72 8	2645.67	7/2 ⁺	[E1]		1.99×10 ⁻³ 3	
		5785.2	0.35 4	1763.15	5/2 ⁺	[E1]		2.24×10 ⁻³ 3	
		7548.0	0.287 32	0.0	3/2 ⁺	(M2)			Mult.: Q(+O) with $\delta(\text{O/Q})=0.1$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; E3 component is less likely based on RUL.
7561.3	1/2,3/2	3383.2	2.6	4177.9	3/2 ⁻				
		3502.0	17	4059.1	(3/2 ⁻)				
		3593.5	1.4	3967.6	1/2 ⁺				
		3642.6	4.6	3918.47	3/2 ⁺				
		4866.9	63	2694.00	3/2 ⁺				
		6341.3	100	1219.38	1/2 ⁺				
		7560.4	97	0.0	3/2 ⁺				
7600.8	5/2 ⁺	2437.4	5.8	5163.34	7/2 ⁻	[E1]		0.000934 13	
		2719.7	6.1	4880.96	7/2 ⁽⁺⁾				
		2761.6	6.1	4839.10	(1/2 ⁺ ,3/2)				
		2830.8	3.9	4769.86	(5/2 ⁻)				
		2976.4	5.5	4624.3	3/2,5/2				
		3422.7	9.1	4177.9	3/2 ⁻	[E1]		1.44×10 ⁻³ 2	
		3427.2		4173.43	(5/2 ⁺)				
		3541.5	6.1	4059.1	(3/2 ⁻)	[E1]		1.50×10 ⁻³ 2	
		4437.6	15	3162.87	7/2 ⁻	[E1]		1.85×10 ⁻³ 3	
		4597.8	24	3002.64	5/2 ⁺	M1+E2	+0.50 22	1.25×10 ⁻³ 3	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	3/2 ⁺	(M1+E2)		0.00139 8	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =no from level scheme.
		4954.8	5.8	2645.67	7/2 ⁺	[M1,E2]		0.00140 8	
		5837.1	58	1763.15	5/2 ⁺	M1+E2	+0.31 7	1.57×10 ⁻³ 2	
		6380.8	3.3	1219.38	1/2 ⁺	[E2]			
		7599.9	100	0.0	3/2 ⁺	M1+E2	-0.19 4		
7618.7	5/2 ⁺	1438	0.77	6181	(1/2 ⁺ to 7/2 ⁺)				
		2404.1	2.6	5214.5	(5/2 ⁺)				
		2455.3	0.9	5163.34	7/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7618.7	$5/2^+$	2779.5	0.9	4839.10	$(1/2^+, 3/2)$				
		3440.6	3.9	4177.9	$3/2^-$				
		3559.4	2.6	4059.1	$(3/2)^-$				
		4455.5	3.9	3162.87	$7/2^-$				
		5855.0	13	1763.15	$5/2^+$	M1+E2	+0.48 11	1.59×10^{-3} 3	
		7617.8	100	0.0	$3/2^+$	M1+E2	+2.2 3		
7656.6	$(1/2, 3/2, 5/2^+)$	1550.4	5.3 26	6106.2	$(3/2, 5/2^+)$				
		1974.3	1.8 9	5682.2	$1/2^-, 3/2^-$				
		2002	1.8 9	5655	$(3/2)^+$				
		2817.4	5.3 26	4839.10	$(1/2^+, 3/2)$				
		3478.5	1.8 9	4177.9	$3/2^-$				
		3597.3	56 6	4059.1	$(3/2)^-$				
		3688.8	3.5 18	3967.6	$1/2^+$				
		7655.7	100 11	0.0	$3/2^+$				
		7671.9	$5/2^+$	2073.4	2.1	5598.4	$3/2^+, 5/2^+$		
2508.5	7			5163.34	$7/2^-$				
2790.8	2.1			4880.96	$7/2^{(+)}$				
2901.9	1.8			4769.86	$(5/2^-)$				
3324.0	11			4347.76	$9/2^-$				
3498.3	8.8			4173.43	$(5/2^+)$				
4508.7	3.2			3162.87	$7/2^-$				
4668.9	16			3002.64	$5/2^+$	M1+E2	-1.4 5	0.00134 4	
5025.8	23			2645.67	$7/2^+$	D(+Q)	+0.3 22		
5908.2	100			1763.15	$5/2^+$	M1+E2	+0.86 14	1.64×10^{-3} 3	
7685.5	$3/2^-$	7671.0	1.4	0.0	$3/2^+$				
		1927		5758	$(1/2, 3/2^-)$				
		2087.0	2.9	5598.4	$3/2^+, 5/2^+$				
		2282.1	2.9	5403.3	$3/2^-$				
		2470.9	1	5214.5	$(5/2^+)$				
		2676.3	0.86	5009.1	$(1/2, 3/2)$				
		3507.4	4.3	4177.9	$3/2^-$				
		3511.9	5.7	4173.43	$(5/2^+)$				
		3626.2	8.6	4059.1	$(3/2)^-$				
		4682.5	14	3002.64	$5/2^+$	D(+Q)	-0.01 5		
		5921.8	1	1763.15	$5/2^+$				
		6465.5	1.4	1219.38	$1/2^+$				
		7684.6	100	0.0	$3/2^+$	D+Q	+6.0 7		
		Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);							

Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);
 $\Delta\pi$ =yes from level scheme; but E1+M2 with such a large $\delta(M2/E1)$ is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p, γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
7693.9	(1/2 ⁺ ,3/2,5/2)	2095.4	1.0 5	5598.4	3/2 ⁺ ,5/2 ⁺				
		3634.6	1.0 5	4059.1	(3/2) ⁻				
		4690.9	2.1 11	3002.64	5/2 ⁺				
		7693.0	100 11	0.0	3/2 ⁺				
7706.4	5/2 ⁺	2051	4.8	5655	(3/2) ⁺	M1+E2		0.00032 4	δ : <-2 or ≥ 13 from $\gamma(\theta)$ in (p, γ).
		2825.3	0.48	4880.96	7/2 ⁽⁺⁾				
		2867.2	1.6	4839.10	(1/2 ⁺ ,3/2)				
		3082.0	0.96	4624.3	3/2,5/2				
		3528.3	2.4	4177.9	3/2 ⁻				
		3593.8	2.4	4112.4	7/2 ⁺				
		3787.7	0.72	3918.47	3/2 ⁺				
		4703.4	1.2	3002.64	5/2 ⁺				
		5060.3	4.8	2645.67	7/2 ⁺	M1+E2		0.00143 8	δ : -1.01 10 or ≥ 13 from (p, γ).
		5942.7	0.48	1763.15	5/2 ⁺				
		6486.4	0.6	1219.38	1/2 ⁺				
		7705.5	100	0.0	3/2 ⁺	D(+Q)	+0.1 1		
7744.8	7/2 ⁻	2863.7	47	4880.96	7/2 ⁽⁺⁾				
		2974.8	4.7	4769.86	(5/2 ⁻)				
		3632.2	14	4112.4	7/2 ⁺				
		3801.7	23	3942.9	9/2 ⁺	(E1)		1.60 $\times 10^{-3}$ 2	Mult.: D(+Q) with $\delta=-0.16$ 17 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2 ⁻	M1+E2	+0.18 8	1.23 $\times 10^{-3}$ 2	
		5098.7	100	2645.67	7/2 ⁺	(E1)		2.05 $\times 10^{-3}$ 3	Mult.: D(+Q) with $\delta=+0.1$ 30 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.15	5/2 ⁺				
		6524.8 ^b	1.4	1219.38	1/2 ⁺	[E3]			This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2 ⁺)	2122	1.3	5655	(3/2) ⁺				
		2613.7	2.8	5163.34	7/2 ⁻				
		2937.9	5	4839.10	(1/2 ⁺ ,3/2)				
		3007.1	7.5	4769.86	(5/2 ⁻)				
		3599.0	18	4177.9	3/2 ⁻				
		3717.8	2.8	4059.1	(3/2) ⁻				
		4613.9	5	3162.87	7/2 ⁻				
		4774.1	10	3002.64	5/2 ⁺				
		5082.7	23	2694.00	3/2 ⁺				
		5131.0	3.3	2645.67	7/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7777.1	(5/2 ⁺)	6013.4	100	1763.15	5/2 ⁺				
		6557.1	28	1219.38	1/2 ⁺				
		7776.2	45	0.0	3/2 ⁺				
7781.7	(5/2 ⁻)	1675.5	3.9	6106.2	(3/2,5/2 ⁺)				
		2183.2		5598.4	3/2 ⁺ ,5/2 ⁺				
		2378.3		5403.3	3/2 ⁻				
		2567.1	4.2	5214.5	(5/2 ⁺)				
		3011.7	19	4769.86	(5/2 ⁻)				
		3433.8	5.6	4347.76	9/2 ⁻				
		3608.1	100	4173.43	(5/2 ⁺)				
		3669.1	17	4112.4	7/2 ⁺				
		3722.4	3.3	4059.1	(3/2) ⁻				
		3863.0	31	3918.47	3/2 ⁺				
		4618.5	14	3162.87	7/2 ⁻				
		5135.6	22	2645.67	7/2 ⁺				
		6018.0	50	1763.15	5/2 ⁺				
		6561.7	3.3	1219.38	1/2 ⁺				
		7780.8	4.7	0.0	3/2 ⁺				
7797.2	1/2 ⁻	2039		5758	(1/2,3/2 ⁻)				
		2142	1.6	5655	(3/2) ⁺				
		2393.8	2.4	5403.3	3/2 ⁻				
		2788.0	2.4	5009.1	(1/2,3/2)				
		2942.7	1.6	4854.4	(1/2,3/2)				
		2958.0	1.3	4839.10	(1/2 ⁺ ,3/2)				
		3737.9	0.36	4059.1	(3/2) ⁻				
		6577.2	11	1219.38	1/2 ⁺				
		7796.3	100	0.0	3/2 ⁺				
7837.2	3/2 ⁻	1656 ^{&}	1.4 ^{&}	6181	(1/2 ⁺ to 7/2 ⁺)				
		2238.7 ^{&}	1.4 ^{&}	5598.4	3/2 ⁺ ,5/2 ⁺				
		2622.6 ^{&}	3.8 ^{&}	5214.5	(5/2 ⁺)				
		3659.1 ^{&}	76 ^{&}	4177.9	3/2 ⁻	(M1+E2)	-0.05 3	0.000917 13	Mult.: D+Q from (p, γ); $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=1.03 $\times 10^3$ 19 exceeds RUL=10. B(E2)(W.u.)=7 $\times 10^2$ +11-6 exceeds RUL=100.
		3663.6 ^{&}	8.1 ^{&}	4173.43	(5/2 ⁺)				
		3777.9 ^{&}	5.4 ^{&}	4059.1	(3/2) ⁻				
		3894.1 ^{&b}	1.4 ^{&}	3942.9	9/2 ⁺				This γ is a primary γ from the unresolved 7837+7839 doublet goes to 9/2 ⁺ , which is unlikely from a 3/2 ⁻ , 7837 level. It is likely from the (5/2 ⁺), 7839.0 level.
		3918.5 ^{&}	0.54 ^{&}	3918.47	3/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7837.2	3/2 ⁻	4834.2 &	11 &	3002.64	5/2 ⁺				
		6073.5 &	5.1 &	1763.15	5/2 ⁺				
		6617.2 &	100 &	1219.38	1/2 ⁺	(E1)			Mult.: D(+Q) with $\delta=+0.06$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 +10-8 exceeds RUL=0.1.
		7836.3 &	57 &	0.0	3/2 ⁺	(E1)			Mult.: D(+Q) with $\delta=+0.02$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 +47-44 exceeds RUL=0.1.
7839.0	(5/2 ⁺)	1658 &	1.4 &	6181	(1/2 ⁺ to 7/2 ⁺)				
		2240.5 &	1.4 &	5598.4	3/2 ⁺ , 5/2 ⁺				
		2624.4 &	3.8 &	5214.5	(5/2 ⁺)				
		3660.9 &	76 &	4177.9	3/2 ⁻				
		3665.4 &	8.1 &	4173.43	(5/2 ⁺)				
		3779.7 &	5.4 &	4059.1	(3/2 ⁻)				
		3895.9 &	1.4 &	3942.9	9/2 ⁺				
		3920.3 &	0.54 &	3918.47	3/2 ⁺				
		4836.0 &	11 &	3002.64	5/2 ⁺				
		6075.3 &	5.1 &	1763.15	5/2 ⁺				
		6619.0 &	100 &	1219.38	1/2 ⁺				
		7838.1 &	57 &	0.0	3/2 ⁺				
		2654.1	1.6	5214.5	(5/2 ⁺)				
		2859.5	4.9	5009.1	(1/2, 3/2)				
7868.7	(1/2 ⁺ , 3/2, 5/2 ⁺)	3695.1	8.2	4173.43	(5/2 ⁺)				
		6105.0	28	1763.15	5/2 ⁺				
		6648.6	21	1219.38	1/2 ⁺				
		7867.8	100	0.0	3/2 ⁺				
		1212.8		6660.2	(11/2 ⁻)				
		1786.2 5	9.6 4	6087.31	13/2 ⁻	D(+Q)	-0.6 6		E $_\gamma$: from (²⁸ Si, $\alpha\pi\gamma$) only. I $_\gamma$: weighted average of 8.2 18 from (²⁸ Si, $\alpha\pi\gamma$) and 9.7 4 from (¹⁶ O, $\alpha\pi\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) in (²⁸ Si, $\alpha\pi\gamma$). E $_\gamma$: weighted average of 1946.2 5 from (¹⁶ O, $\alpha\pi\gamma$) and 1946.40 30 from (¹⁴ N, $\alpha\pi\pi\gamma$). I $_\gamma$: weighted average of 46 5 from (²⁸ Si, $\alpha\pi\gamma$)
7873.03	13/2 ⁺	1946.35 30	48.0 26	5926.65	(11/2 ⁻)	E1+M2	+0.2 1	0.000594 22	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	
7873.03	13/2 ⁺	2465.94 30	100.0 20	5407.07	11/2 ⁻	E1+M2	-0.25 10	0.00091 4	and 48.5 26 from ($^{16}\text{O},\alpha p\gamma$). Mult., δ : from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si},\alpha p\gamma$), $\Delta J=1$. E_γ : weighted average of 2466.2 5 from ($^{16}\text{O},\alpha p\gamma$) and 2465.85 30 from ($^{14}\text{N},\alpha p n\gamma$). I_γ : weighted average of 100 9 from ($^{28}\text{Si},\alpha p\gamma$) and 100.0 20 from ($^{16}\text{O},\alpha p\gamma$). Mult., δ : from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si},\alpha p\gamma$), $\Delta J=1$.
7880.8	3/2 ⁺ ,5/2 ⁺	2226	40	5655	(3/2) ⁺				
		3041.6	5.7	4839.10	(1/2 ⁺ ,3/2)				
		3256.3	6.7	4624.3	3/2,5/2				
		3702.7	10	4177.9	3/2 ⁻				
		3768.2	4.3	4112.4	7/2 ⁺				
		3821.5	6.7	4059.1	(3/2) ⁻				
		3962.1	6.7	3918.47	3/2 ⁺				
		4877.8	37	3002.64	5/2 ⁺				
		5186.4	83	2694.00	3/2 ⁺				
		6117.1	100	1763.15	5/2 ⁺				
		7879.9	33	0.0	3/2 ⁺				
7899.2	(5/2)	2253	2.2	5646	(5/2,7/2,9/2 ⁺)				
		2735.8	1.3	5163.34	7/2 ⁻				
		3018.1	1.2	4880.96	7/2 ⁽⁺⁾				
		3274.7	0.74	4624.3	3/2,5/2				
		3721.1	12	4177.9	3/2 ⁻				
		3725.6	12	4173.43	(5/2 ⁺)				
		3839.9	4.4	4059.1	(3/2) ⁻				
		4736.0	5.9	3162.87	7/2 ⁻				
		5204.8	1.9	2694.00	3/2 ⁺				
		6135.5	100	1763.15	5/2 ⁺				
		7898.2	5.9	0.0	3/2 ⁺				
7923.3	(3/2 ⁺ ,5/2 ⁺)	3084.1	3.4	4839.10	(1/2 ⁺ ,3/2)				
		3153.3	4.9	4769.86	(5/2 ⁻)				
		3298.8	4.6	4624.3	3/2,5/2				
		3745.2	7.3	4177.9	3/2 ⁻				
		3749.7	12	4173.43	(5/2 ⁺)				
		3864.0	4.2	4059.1	(3/2) ⁻				
		3955.5	3.7	3967.6	1/2 ⁺				
		4004.6	2.9	3918.47	3/2 ⁺				
		4920.3	2.2	3002.64	5/2 ⁺				
		5228.9	27	2694.00	3/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	
7923.3	(3/2 ⁺ , 5/2 ⁺)	5277.2	2.2	2645.67	7/2 ⁺		
		6159.6	3.7	1763.15	5/2 ⁺		
		6703.2	66	1219.38	1/2 ⁺		
		7922.3	100	0.0	3/2 ⁺		
7970.3	(5/2 ⁻)	2324	15	5646	(5/2, 7/2, 9/2 ⁺)		
		2384.2	15	5586.0	5/2 ⁺		
		2755.7	15	5214.5	(5/2 ⁺)		
		3089.2	50	4880.96	7/2 ⁽⁺⁾		
		3200.3	25	4769.86	(5/2 ⁻)		
		3622.3	40	4347.76	9/2 ⁻		
		3792.2	30	4177.9	3/2 ⁻		
		3857.7	30	4112.4	7/2 ⁺		
		4807.1	50	3162.87	7/2 ⁻		
		4967.3	25	3002.64	5/2 ⁺		
		5324.2	100	2645.67	7/2 ⁺		
		6206.6	100	1763.15	5/2 ⁺		
		6750.2	5	1219.38	1/2 ⁺		
		7987.8	3/2 ⁺	1881.6	0.83	6106.2	(3/2, 5/2 ⁺)
		2333	2.8	5655	(3/2 ⁺)		
		3809.7	<1.3	4177.9	3/2 ⁻		I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels.
		3814.2	<1.3	4173.43	(5/2 ⁺)		I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels.
7995.6	(5/2)	3928.5	1.7	4059.1	(3/2 ⁻)		
		6767.7	100	1219.38	1/2 ⁺	M1+E2	δ : +0.21 4 or -3.1 4 from (p, γ).
		7986.8	60	0.0	3/2 ⁺	M1+E2	δ : -0.36 2 or -11 3 from (p, γ).
		2188.9	0.77	5806.6	(1/2, 3/2, 5/2)		
		2397.1	1.4	5598.4	3/2 ⁺ , 5/2 ⁺		
		2592.2	3.9	5403.3	3/2 ⁻		
		2832.1	1.9	5163.34	7/2 ⁻		
		3817.5	5.1	4177.9	3/2 ⁻		
		3936.3	1.4	4059.1	(3/2 ⁻)		
		4832.4	9	3162.87	7/2 ⁻		
8001.0	(7/2 ⁺)	4992.6	2.6	3002.64	5/2 ⁺		
		5301.2	0.26	2694.00	3/2 ⁺		
		5349.5	0.77	2645.67	7/2 ⁺		
		6231.9	1.2	1763.15	5/2 ⁺		
		7994.6	100	0.0	3/2 ⁺		
		2355	2.6	5646	(5/2, 7/2, 9/2 ⁺)		
		2786.4	1.1	5214.5	(5/2 ⁺)		
		3119.9	1	4880.96	7/2 ⁽⁺⁾		
		3231.0	11	4769.86	(5/2 ⁻)		
		3653.0	2.7	4347.76	9/2 ⁻		
		3827.4	14	4173.43	(5/2 ⁺)		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)											
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments		
8001.0	(7/2 ⁺)	4057.9	4.3	3942.9	9/2 ⁺						
		4837.8	1.1	3162.87	7/2 ⁻						
		5354.9	2.3	2645.67	7/2 ⁺						
		6237.3	100	1763.15	5/2 ⁺						
		8000.0	2	0.0	3/2 ⁺						
8005.0	5/2 ⁺	2350	3.3	5655	(3/2) ⁺						
		3123.9	1.1	4880.96	7/2 ⁽⁺⁾						
		3165.8	5.5	4839.10	(1/2 ⁺ , 3/2)						
		3826.9	<2.4	4177.9	3/2 ⁻				I_γ : total $I_\gamma=2.4$ for γ decays to 4173 and 4178 doublet final levels.		
		3831.3	<2.4	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=2.4$ for γ decays to 4173 and 4178 doublet final levels.		
		4841.8	27	3162.87	7/2 ⁻						
		5002.0	3.6	3002.64	5/2 ⁺						
		5310.6	3.1	2694.00	3/2 ⁺						
		5358.9	7.3	2645.67	7/2 ⁺						
		6241.3	27	1763.15	5/2 ⁺						
		6784.9	1.1	1219.38	1/2 ⁺						
		8004.0	100	0.0	3/2 ⁺						
		8035.7	1/2 ⁻ , 3/2 ⁻	2229.0	10.7 11	5806.6	(1/2, 3/2, 5/2)				
				3857.6	26.8 27	4177.9	3/2 ⁻				
3976.4	19.6 20			4059.1	(3/2) ⁻						
5341.3	9 5			2694.00	3/2 ⁺						
6272.0	9 5			1763.15	5/2 ⁺						
6815.6	100 11			1219.38	1/2 ⁺						
8034.7	3.6 18			0.0	3/2 ⁺						
3183.8	1.1			4854.4	(1/2, 3/2)						
3199.1	2.8			4839.10	(1/2 ⁺ , 3/2)						
3413.9	5.3			4624.3	3/2, 5/2						
8038.4	1/2 ⁺ , 3/2 ⁺	3860.3	2.8	4177.9	3/2 ⁻						
		3979.1	5.3	4059.1	(3/2) ⁻						
		4070.6	1.4	3967.6	1/2 ⁺						
		4119.7	2.5	3918.47	3/2 ⁺						
		5035.4	7	3002.64	5/2 ⁺						
		5344.0	100	2694.00	3/2 ⁺	M1+E2	-0.20 2	1.44×10 ⁻³ 2	Mult., δ : D+Q with $\delta=-0.20$ 2 or +19 8 from (p, γ); E1+M2 and the larger $\delta(E2/M1)$ is ruled out by RUL. B(M1)(W.u.)=52.1 +43-37 exceeds RUL=10. B(E2)(W.u.)=2.8×10 ² +6-5 exceeds RUL=100.		
		6274.7	26	1763.15	5/2 ⁺						
		6818.3	7	1219.38	1/2 ⁺						
		8037.4	14	0.0	3/2 ⁺	M1+E2	-0.13 4		B(M1)(W.u.)=2.20 44; B(E2)(W.u.)=2.2 +16-12		



Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a
Comments							
Mult., δ : D+Q with $\delta = -0.13$ 4 or $+8.1$ 26 from (p, γ); E1+M2 and the larger $\delta(\text{E2/M1})$ is ruled out by RUL.							
8075.9	(5/2)	2430	20	5646	(5/2,7/2,9/2 ⁺)		
		2489.8	20	5586.0	5/2 ⁺		
		2861.3	14	5214.5	(5/2 ⁺)		
		2912.4		5163.34	7/2 ⁻		
		3194.8		4880.96	7/2 ⁽⁺⁾		
		3305.9	7.1	4769.86	(5/2 ⁻)		
		3897.8	14	4177.9	3/2 ⁻		
		3963.3	3.6	4112.4	7/2 ⁺		
		5072.9	3.6	3002.64	5/2 ⁺		
		6312.1	100	1763.15	5/2 ⁺		
		8074.9		0.0	3/2 ⁺		
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2)		
		2450	13	5646	(5/2,7/2,9/2 ⁺)		
		2497.6	13	5598.4	3/2 ⁺ , 5/2 ⁺		
		2881.5	3.3	5214.5	(5/2 ⁺)		
		2932.6	5.1	5163.34	7/2 ⁻		
		3256.8	7.7	4839.10	(1/2 ⁺ , 3/2)		
		3326.1	5.1	4769.86	(5/2 ⁻)		
		3918.0	4.9	4177.9	3/2 ⁻		
		4036.8	5.1	4059.1	(3/2) ⁻		
		4177.4	4.6	3918.47	3/2 ⁺		
		4932.9	13	3162.87	7/2 ⁻		
		5401.7	13	2694.00	3/2 ⁺		
		5450.0	44	2645.67	7/2 ⁺		
		6332.3	21	1763.15	5/2 ⁺		
		8095.1	100	0.0	3/2 ⁺		
8106.4	3/2 ⁺	2507.9	1.4	5598.4	3/2 ⁺ , 5/2 ⁺		
		3251.8	1.6	4854.4	(1/2, 3/2)		
		3267.1	1.4	4839.10	(1/2 ⁺ , 3/2)		
		3928.3	4.3	4177.9	3/2 ⁻		
		4047.1	3.5	4059.1	(3/2) ⁻		
		4138.5	8.1	3967.6	1/2 ⁺		
		5103.4	24	3002.64	5/2 ⁺		
		5412.0	24	2694.00	3/2 ⁺	M1+E2	0.00153 8 δ : +4.9 10 or -0.05 4.
		6342.6	8.1	1763.15	5/2 ⁺		
		6886.3	95	1219.38	1/2 ⁺		
		8105.4	100	0.0	3/2 ⁺	M1+E2	δ : +1.2 1 or -0.46 4.
		1621	19	6492	(3/2 ⁺)		
8113.3	(1/2 ⁺ , 3/2, 5/2 ⁺)	2007.0	19	6106.2	(3/2, 5/2 ⁺)		
		2306.6	3.4	5806.6	(1/2, 3/2, 5/2)		

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8113.3	$(1/2^+, 3/2, 5/2^+)$	2458	2.8	5655	$(3/2)^+$
		2514.8		5598.4	$3/2^+, 5/2^+$
		2898.7	5.6	5214.5	$(5/2^+)$
		3939.6	22	4173.43	$(5/2^+)$
		4054.0	4.1	4059.1	$(3/2)^-$
		4145.4	16	3967.6	$1/2^+$
		5110.3	5.9	3002.64	$5/2^+$
		5418.9	41	2694.00	$3/2^+$
		6349.5	16	1763.15	$5/2^+$
		6893.2	100	1219.38	$1/2^+$
		8112.3	78	0.0	$3/2^+$
8147.2	$3/2^-$	2464.9	1.6	5682.2	$1/2^-, 3/2^-$
		3138.0	4.8	5009.1	$(1/2, 3/2)$
		3969.1	13	4177.9	$3/2^-$
		4087.8	4.8	4059.1	$(3/2)^-$
		4179.3	3.2	3967.6	$1/2^+$
		4228.5	4.8	3918.47	$3/2^+$
		5452.7	4.8	2694.00	$3/2^+$
		6927.1	24	1219.38	$1/2^+$
		8146.2	100	0.0	$3/2^+$
8156.8	$(5/2^+)$	2511	7.8	5646	$(5/2, 7/2, 9/2^+)$
		2570.7	16	5586.0	$5/2^+$
		2942.2	12	5214.5	$(5/2^+)$
		3386.8	2.0	4769.86	$(5/2^-)$
		3978.7	9.8	4177.9	$3/2^-$
		4044.2	14	4112.4	$7/2^+$
		4097.4	2.0	4059.1	$(3/2)^-$
		4213.6	3.9	3942.9	$9/2^+$
		4993.6	9.8	3162.87	$7/2^-$
		5153.8	12	3002.64	$5/2^+$
		5510.7	100	2645.67	$7/2^+$
		6393.0	2.0	1763.15	$5/2^+$
		8155.8	5.9	0.0	$3/2^+$
8179.4	$(3/2^-, 5/2, 7/2^+)$	3554.9	13.2 <i>I3</i>	4624.3	$3/2, 5/2$
		5016.1	11.8 <i>I2</i>	3162.87	$7/2^-$
		5176.4	5.9 <i>30</i>	3002.64	$5/2^+$
		5484.9	4.4 <i>22</i>	2694.00	$3/2^+$
		6415.6	100 <i>I0</i>	1763.15	$5/2^+$
		8178.4	11.8 <i>I2</i>	0.0	$3/2^+$
8208.2	$5/2^+$	2553	0.9	5655	$(3/2)^+$
		3327.1	1.0	4880.96	$7/2^{(+)}$
		3368.9	0.51	4839.10	$(1/2^+, 3/2)$

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8208.2	$5/2^+$	4148.8	1.9	4059.1	$(3/2)^-$				
		4289.5		3918.47	$3/2^+$				
		5205.1	1.4	3002.64	$5/2^+$				
		5513.7	0.64	2694.00	$3/2^+$				
		6444.4	18	1763.15	$5/2^+$	M1+E2	+0.33 4		B(M1)(W.u.)=0.75 +35-22; B(E2)(W.u.)=7.5 +41-25
		6988.1	3.9	1219.38	$1/2^+$				
		8207.2	100	0.0	$3/2^+$	M1+E2	-0.36 4		B(M1)(W.u.)=2.0 +9-5; B(E2)(W.u.)=15 +7-4
8216.3	$5/2^+$	2561	1.8	5655	$(3/2)^+$				
		3446.3	1.8	4769.86	$(5/2)^-$				
		4038.2	<6.7	4177.9	$3/2^-$				
		4042.6	<6.7	4173.43	$(5/2^+)$				
		5053.0	91	3162.87	$7/2^-$	(E1)		2.04×10^{-3} 3	B(E1)(W.u.)=0.062 +21-12 Mult.: D+Q with $\delta=+0.11$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi=\text{yes}$ from level scheme; but M2 component is unlikely for this strong primary transition.
		5213.2	11	3002.64	$5/2^+$				
		5521.8	6.7	2694.00	$3/2^+$				
8242.4	$3/2^-$	6452.5	3.1	1763.15	$5/2^+$				
		8215.3	100	0.0	$3/2^+$	M1+E2	+0.06 3		B(M1)(W.u.)=0.55 +18-9; B(E2)(W.u.)=0.11 +16-8
		4068.7	7.7	4173.43	$(5/2^+)$				
		5239.3	2.2	3002.64	$5/2^+$				
		7022.3	100	1219.38	$1/2^+$				
		4138	4.9	4112.4	$7/2^+$				
		5248	7.3	3002.64	$5/2^+$				
8251	$(3/2^+, 5/2^+)$	7031	100	1219.38	$1/2^+$				
		8250	9.8	0.0	$3/2^+$				
		3499.0	4.4	4769.86	$(5/2)^-$				
		4090.8	27	4177.9	$3/2^-$				
		4350.2	6.7	3918.47	$3/2^+$				
		5105.7	16	3162.87	$7/2^-$				
		5265.9	29	3002.64	$5/2^+$				
8269.0	$5/2^+$	5574.5	13	2694.00	$3/2^+$				
		5622.9	8.9	2645.67	$7/2^+$				
		6505.2	18	1763.15	$5/2^+$				
		8268.0	100	0.0	$3/2^+$				
		2622	5.2	5655	$(3/2)^+$				
		3396.1	17	4880.96	$7/2^{(+)}$				
		3437.9	38	4839.10	$(1/2^+, 3/2)$				
8277.2	$(5/2)^+$	3507.2	4.1	4769.86	$(5/2)^-$				
		3652.7	10	4624.3	$3/2, 5/2$				
		3929.2	6.2	4347.76	$9/2^-$				
		4103.5	6.9	4173.43	$(5/2^+)$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	
8277.2	$(5/2)^+$	4217.8	5.2	4059.1	$(3/2)^-$		
		4358.4	17	3918.47	$3/2^+$		
		5274.1	35	3002.64	$5/2^+$		
		5582.7	10	2694.00	$3/2^+$		
		5631.0	10	2645.67	$7/2^+$		
		6513.4	79	1763.15	$5/2^+$		
		8276.2	100	0.0	$3/2^+$		
8282.4	$(3/2^-, 5/2)$	2696.3	29	5586.0	$5/2^+$		
		3067.8	29	5214.5	$(5/2^+)$		
		3118.9	11	5163.34	$7/2^-$		
		4104.2	50	4177.9	$3/2^-$		
		4223.0	21	4059.1	$(3/2)^-$		
		5119.1	93	3162.87	$7/2^-$		
		6518.6	25	1763.15	$5/2^+$		
		8281.4	100	0.0	$3/2^+$		
8287.82	$1/2^-$	2633	4.6	5655	$(3/2)^+$		
		3278.6	6.8	5009.1	$(1/2, 3/2)$		
		4109.7	21	4177.9	$3/2^-$		
		4228.5	4.6	4059.1	$(3/2)^-$		
		4319.9	6.8	3967.6	$1/2^+$		
		4369.06	2.3	3918.47	$3/2^+$		
		5593.3	52	2694.00	$3/2^+$		
		7067.67	100	1219.38	$1/2^+$	(E1)	$B(E1)(\text{W.u.})=0.069 +24-14$ Mult.: D+Q with $\delta(Q/D)=+0.21$ 2 or -3.2 2 in (p, γ); but M2 component is unlikely for this primary transition based on resonance $\Gamma=0.04$ keV.
8298.2	$3/2^+$	8286.77	30	0.0	$3/2^+$		
		3288.9	13	5009.1	$(1/2, 3/2)$		
		3417.1	17	4880.96	$7/2^{(+)}$		
		3443.6	21	4854.4	$(1/2, 3/2)$		
		4124.5	25	4173.43	$(5/2^+)$		
		4238.8	42	4059.1	$(3/2)^-$		
		5134.9	4.2	3162.87	$7/2^-$		This γ is questionable if $J^\pi(8298)=3/2^+$, since it would require Mult=M2 which is probably unlikely; however $J^\pi(8298)=3/2^-$ would make the strong 7078.0 γ and 8297.1 γ less likely.
		5295.1	4.2	3002.64	$5/2^+$		
		5603.7	42	2694.00	$3/2^+$		
		5652.0	25	2645.67	$7/2^+$		
		6534.4	33	1763.15	$5/2^+$		
8318.1	$(5/2^+)$	7078.1	100	1219.38	$1/2^+$	M1+E2	$\delta: -1.7$ 1 or -0.02 2 from $\gamma(\theta)$ in (p, γ).
		8297.1	54	0.0	$3/2^+$	M1+E2	$\delta: -0.15$ 4 or $+9.5$ 36 from $\gamma(\theta)$ in (p, γ).
		2511.4	0.9	5806.6	$(1/2, 3/2, 5/2)$		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8318.1	$(5/2^+)$	2672	0.9	5646	$(5/2, 7/2, 9/2^+)$				
		2719.6	1.9	5598.4	$3/2^+, 5/2^+$				
		3548.1	0.39	4769.86	$(5/2^-)$				
		3693.6	2.3	4624.3	$3/2, 5/2$				
		4139.9	<3.9	4177.9	$3/2^-$				
		4144.4	<3.9	4173.43	$(5/2^+)$				
		4258.7	1.5	4059.1	$(3/2)^-$				
		4350.2	3.9	3967.6	$1/2^+$				
		5154.8	0.64	3162.87	$7/2^-$				
		5315.0	3.9	3002.64	$5/2^+$				
		5623.6	2.1	2694.00	$3/2^+$				
		5671.9	1.2	2645.67	$7/2^+$				
		6554.3	2.3	1763.15	$5/2^+$				
		7098.0	2.6	1219.38	$1/2^+$				
		8317.0	100	0.0	$3/2^+$				
8319.6	$15/2^-$	2232.7 6	57.2 28	6087.31	$13/2^-$	[M1,E2]		0.00040 4	E_γ, I_γ : from ($^{16}\text{O}, \alpha\gamma$). B(M1)(W.u.)=0.0577 4I if M1, B(E2)(W.u.)=43.8 3I if E2.
		2911.9 8	100 5	5407.07	$11/2^-$	E2		0.000753 1I	B(E2)(W.u.)=20.3 12 $E_\gamma, I_\gamma, \text{Mult.}$: from ($^{16}\text{O}, \alpha\gamma$). Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{DCO})$; M2 ruled out by RUL.
8323.1	$(3/2^+, 5/2^+)$	4210.4	1.2	4112.4	$7/2^+$				
		5320.0	3.7	3002.64	$5/2^+$				
		5676.9	4.9	2645.67	$7/2^+$				
		6559.3	<2.4	1763.15	$5/2^+$				
		7103.0	9.8	1219.38	$1/2^+$				
		8322.0	100	0.0	$3/2^+$				
8381.8	$5/2^+$	3611.7		4769.86	$(5/2^-)$				
		3757.3		4624.3	$3/2, 5/2$				
		4269.1	23.5 24	4112.4	$7/2^+$				
		4463.0	71 7	3918.47	$3/2^+$	M1+E2	+0.15 3	1.19×10^{-3} 2	B(M1)(W.u.)=2.9 +13-7; B(E2)(W.u.)=13 +8-5
		5378.7	74 8	3002.64	$5/2^+$	M1+E2	+0.02 3	1.44×10^{-3} 2	B(M1)(W.u.)=1.8 +14-7
		5687.3	2.9 15	2694.00	$3/2^+$	M1+E2	-0.72 3I	0.00157 4	B(M1)(W.u.)=0.039 +32-2I; B(E2)(W.u.)=2.4 +23-18
		5735.6	15 8	2645.67	$7/2^+$				
		6618.0	100 10	1763.15	$5/2^+$	M1+E2	+0.06 3		B(M1)(W.u.)=1.3 +6-3; B(E2)(W.u.)=0.4 +6-3
		8380.7	9 5	0.0	$3/2^+$	D+Q	1.65		
		3632.8	16 8	4769.86	$(5/2^-)$				
8402.9	$5/2^-$	4224.7	29 16	4177.9	$3/2^-$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8402.9	$5/2^-$	4343.5	74 20	4059.1	$(3/2)^-$
		5708.4	32 16	2694.00	$3/2^+$
		5756.7	23 13	2645.67	$7/2^+$
		6639.1	48 13	1763.15	$5/2^+$
		8401.8	100 26	0.0	$3/2^+$
8405.8	$(5/2^-)$	3524.7	35 9	4880.96	$7/2^{(+)}$
		4232.1	3.7 19	4173.43	$(5/2^+)$
		4293.1	3.3 16	4112.4	$7/2^+$
		5402.7	35 9	3002.64	$5/2^+$
		6642.0	100 26	1763.15	$5/2^+$
		8404.7	56 14	0.0	$3/2^+$
		3407.4	14	5009.1	$(1/2,3/2)$
8416.7	$1/2^+$	4243.0	31	4173.43	$(5/2^+)$
		4473.5	8.6	3942.9	$9/2^+$
		5413.6	20	3002.64	$5/2^+$
		5770.5	23	2645.67	$7/2^+$
		6652.9	63	1763.15	$5/2^+$
		7196.5	26	1219.38	$1/2^+$
		8415.6	100	0.0	$3/2^+$
		2809	47 24	5655	$(3/2)^+$
		3249.6	34 17	5214.5	$(5/2^+)$
8464.3	$(1/2^+, 3/2, 5/2^+)$	3455.0	47 24	5009.1	$(1/2, 3/2)$
		3609.7	19 10	4854.4	$(1/2, 3/2)$
		3839.8	25 13	4624.3	$3/2, 5/2$
		4286.1	31 15	4177.9	$3/2^-$
		4290.6	32 17	4173.43	$(5/2^+)$
		4496.4	31 15	3967.6	$1/2^+$
		4545.5	15 8	3918.47	$3/2^+$
		5461.2	100 24	3002.64	$5/2^+$
		5769.8	94 24	2694.00	$3/2^+$
		7244.1	53 24	1219.38	$1/2^+$
		8463.2	59 18	0.0	$3/2^+$
		2829	12 6	5655	$(3/2)^+$
		3603.2	15 8	4880.96	$7/2^{(+)}$
8484.4	$(3/2)^+$	3859.9	2.2 11	4624.3	$3/2, 5/2$
		4516.5	2.6 13	3967.6	$1/2^+$
		4565.6	11 6	3918.47	$3/2^+$
		5481.3	15 9	3002.64	$5/2^+$
		5789.9	44 11	2694.00	$3/2^+$
		5838.2	6.1 31	2645.67	$7/2^+$
		6720.6	100 26	1763.15	$5/2^+$
		8483.3	9 5	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8486.2	3/2 ⁻	2728	2.4 13	5758	(1/2,3/2 ⁻)				
		3861.7	3.9 20	4624.3	3/2,5/2				
		4308.0	<15	4177.9	3/2 ⁻				I_γ : total $I_\gamma=10$ 5 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4312.5	<15	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=10$ 5 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4426.8	4.1 22	4059.1	(3/2) ⁻				
		4518.3	15 9	3967.6	1/2 ⁺				
		5483.1	17 9	3002.64	5/2 ⁺				
		5791.7	41 11	2694.00	3/2 ⁺				
		6722.4	8 4	1763.15	5/2 ⁺				
		7266.0	14 7	1219.38	1/2 ⁺				
		8485.1	100 26	0.0	3/2 ⁺				
8487.5	15/2 ⁻	2399.8 8	100.0 34	6087.31	13/2 ⁻	(M1(+E2))	+0.2 +3-2	0.000427 17	Mult., δ : D+Q and δ from $\gamma\gamma(\text{DCO})$ in (²⁸ Si, $\alpha\gamma\gamma$); $\Delta\pi$ =no from level scheme.
		3080.0 7	31.8 20	5407.07	11/2 ⁻	E2		0.000825 12	$E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
8514.3	1/2 ⁻	3659.7	16 8	4854.4	(1/2,3/2)				
		4336.1	7 4	4177.9	3/2 ⁻				
		4454.9	100 26	4059.1	(3/2) ⁻				
		4546.4	85 22	3967.6	1/2 ⁺				
		4595.5	24 12	3918.47	3/2 ⁺				
		7294.1	37 11	1219.38	1/2 ⁺				
		8513.2	100 26	0.0	3/2 ⁺				
8558.2?	(13/2 ⁻)	2631.4 ^b	100	5926.65	(11/2) ⁻	D(+Q)	-0.2 +2-3		$E_\gamma, \text{Mult.}, \delta$: from (²⁸ Si, $\alpha\gamma\gamma$); $\Delta\pi$ =no from level scheme; Mult=Q, $\Delta J=1$.
8571.9	(5/2) ⁺	2848.3	3.1 16	5723.52	5/2 ⁺				
		2917	1.9 10	5655	(3/2) ⁺				
		2973.4	1.0 6	5598.4	3/2 ⁺ ,5/2 ⁺				
		3408.4	2.6 13	5163.34	7/2 ⁻				
		3690.7	1.3 7	4880.96	7/2 ⁽⁺⁾				
		4398.2	1.0 6	4173.43	(5/2 ⁺)				
		4459.2	1.6 9	4112.4	7/2 ⁺				
		4653.1	1.6 9	3918.47	3/2 ⁺				
		5408.6	2.3 12	3162.87	7/2 ⁻				
		5568.8	6.4 33	3002.64	5/2 ⁺				
		5877.4	0.9 4	2694.00	3/2 ⁺				
		5925.7	11 6	2645.67	7/2 ⁺				
		7351.7	8 4	1219.38	1/2 ⁺				
		8570.8	100 10	0.0	3/2 ⁺				
8580.5	1/2 ⁺	2088	8 4	6492	(3/2 ⁺)				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J_f^{π}</u>
8580.5	1/2 ⁺	3741.2	2.4 13	4839.10	(1/2 ⁺ ,3/2)
		4402.3	8 4	4177.9	3/2 ⁻
		4521.1	3.7 19	4059.1	(3/2) ⁻
		4661.7	4.3 22	3918.47	3/2 ⁺
		5886.0	10 5	2694.00	3/2 ⁺
		7360.3	100 11	1219.38	1/2 ⁺
		8579.4	48 13	0.0	3/2 ⁺
8590.1	5/2 ⁺	2832	22 11	5758	(1/2,3/2 ⁻)
		2935	17 9	5655	(3/2) ⁺
		3820.0	73 18	4769.86	(5/2 ⁻)
		4416.4	59 14	4173.43	(5/2 ⁺)
		4477.4	16 8	4112.4	7/2 ⁺
		5426.8	36 18	3162.87	7/2 ⁻
		5587.0	100 27	3002.64	5/2 ⁺
		5943.9	25 13	2645.67	7/2 ⁺
		6826.2	64 18	1763.15	5/2 ⁺
		8589.0	46 14	0.0	3/2 ⁺
		6848.1	100 10	1763.15	5/2 ⁺
8612.0	3/2 ⁺ ,5/2 ⁺	7391.8	34 8	1219.38	1/2 ⁺
		8610.9	66 16	0.0	3/2 ⁺
		3732.5	2.9 14	4880.96	7/2 ⁽⁺⁾
8613.7	5/2 ⁺	4554.3	10 5	4059.1	(3/2) ⁻
		4670.5	7.1	3942.9	9/2 ⁺
		4694.9	13 7	3918.47	3/2 ⁺
		5610.6	4.5 23	3002.64	5/2 ⁺
		5919.2	8 4	2694.00	3/2 ⁺
		5967.5	41 11	2645.67	7/2 ⁺
		6849.8	16	1763.15	5/2 ⁺
		7393.5	3.6	1219.38	1/2 ⁺
		8612.6	100 11	0.0	3/2 ⁺
		2894.6	6.4 33	5723.52	5/2 ⁺
		2963	5.2 27	5655	(3/2) ⁺
8618.2	3/2 ⁺ ,5/2 ⁺	3763.6	4.2 21	4854.4	(1/2,3/2)
		3993.7	4.6 24	4624.3	3/2,5/2
		4650.3	9 5	3967.6	1/2 ⁺
		4699.4	7 4	3918.47	3/2 ⁺
		5615.1	49 12	3002.64	5/2 ⁺
		5923.7	42 12	2694.00	3/2 ⁺
		5972.0	5.2 27	2645.67	7/2 ⁺
		6854.3	70 18	1763.15	5/2 ⁺
		8617.1	100 24	0.0	3/2 ⁺
		2906.5	0.6 4	5723.52	5/2 ⁺
8630.1	7/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8630.1	7/2 ⁻	3466.6	17 9	5163.34	7/2 ⁻	M1+E2	+0.60 10	0.000884 15	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		3860.0	3.0 15	4769.86	(5/2 ⁻)				
		4282.1	26 7	4347.76	9/2 ⁻	M1+E2	-0.184 14	1.13×10 ⁻³ 2	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		4456.4	8 4	4173.43	(5/2 ⁺)				
		4517.4	21 11	4112.4	7/2 ⁺	(E1+M2)	-0.06 2	1.86×10 ⁻³ 3	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4686.9	21 11	3942.9	9/2 ⁺				
		5466.8	7 4	3162.87	7/2 ⁻				
		5627.0	3.0 15	3002.64	5/2 ⁺				
		5983.9	5.5 28	2645.67	7/2 ⁺				
		6866.2	100 26	1763.15	5/2 ⁺	(E1+M2)	-0.011 3		Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
8640	(5/2 ⁺)	8629.0	1.1 7	0.0	3/2 ⁺				
		4697	61	3942.9	9/2 ⁺				
		5477	53	3162.87	7/2 ⁻				
		5637	16	3002.64	5/2 ⁺				
		5946	34	2694.00	3/2 ⁺				
		6876	100	1763.15	5/2 ⁺				
8641.4	3/2 ⁺ ,5/2 ⁺	2986	10 6	5655	(3/2 ⁺)				
		3055.3	9 5	5586.0	5/2 ⁺				
		3237.9	17 9	5403.3	3/2 ⁻				
		3802.1	9 5	4839.10	(1/2 ⁺ ,3/2)				
		3871.3	86 22	4769.86	(5/2 ⁻)				
		4016.9	7 4	4624.3	3/2,5/2				
		4463.2	<64	4177.9	3/2 ⁻				I_γ : total $I_\gamma=50$ 14 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4467.7	<64	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=50$ 14 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4582.0	32 18	4059.1	(3/2 ⁻)				
		4722.6	21 11	3918.47	3/2 ⁺				
		5946.9	19 9	2694.00	3/2 ⁺				
		8640.3	100 25	0.0	3/2 ⁺				
8686.2	5/2 ⁻	3100.1	2.8 15	5586.0	5/2 ⁺				
		3282.7	4.1 21	5403.3	3/2 ⁻				
		4508.0	2.6 13	4177.9	3/2 ⁻				
		4626.8	4.6 24	4059.1	(3/2 ⁻)				
		5522.9	4.9 25	3162.87	7/2 ⁻				
		5991.7	3.7 19	2694.00	3/2 ⁺				
		6922.3	6.3 32	1763.15	5/2 ⁺				
		7466.0	2.5 13	1219.38	1/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a	Comments
8686.2	5/2 ⁻	8685.0	100 11	0.0	3/2 ⁺			
8696.9	3/2 ⁻	2890.2	3.6 19	5806.6	(1/2,3/2,5/2)			
		3098.4	6.0 30	5598.4	3/2 ⁺ ,5/2 ⁺			
		7476.7	15 4	1219.38	1/2 ⁺			
		8695.7	100 10	0.0	3/2 ⁺			
8750.9	3/2 ⁻	2944.2	4.2 21	5806.6	(1/2,3/2,5/2)			
		3347.4	9 5	5403.3	3/2 ⁻			
		3536.2	4.5 23	5214.5	(5/2 ⁺)			
		4572.7	4.0 21	4177.9	3/2 ⁻			
		4577.2	4.3 23	4173.43	(5/2 ⁺)			
		6056.3	5.7 28	2694.00	3/2 ⁺			
		6987.0	57 15	1763.15	5/2 ⁺			
		7530.7	3.8	1219.38	1/2 ⁺			
		8749.7	100 10	0.0	3/2 ⁺			
8779.8	3/2 ⁻	3181.2	16 8	5598.4	3/2 ⁺ ,5/2 ⁺			
		3565.1	24 12	5214.5	(5/2 ⁺)			
		4601.6	<71	4177.9	3/2 ⁻			I_γ : total $I_\gamma=56$ 15 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4606.0	<71	4173.43	(5/2 ⁺)			I_γ : total $I_\gamma=56$ 15 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4861.0	16 8	3918.47	3/2 ⁺			
		5776.7	100 26	3002.64	5/2 ⁺			
		6085.2	44 11	2694.00	3/2 ⁺			
		6133.6	48 11	2645.67	7/2 ⁺			
		7015.9	25 13	1763.15	5/2 ⁺			
		8778.6	41 11	0.0	3/2 ⁺			
8787.2	(5/2 ⁺)	3063.5	27 14	5723.52	5/2 ⁺			
		3623.7	76 20	5163.34	7/2 ⁻			
		4017.1	27 14	4769.86	(5/2 ⁻)			
		4609.0	36 20	4177.9	3/2 ⁻			
		5623.8	23 12	3162.87	7/2 ⁻			
		5784.1	80 20	3002.64	5/2 ⁺			
		7023.3	100 24	1763.15	5/2 ⁺			
		8786.0	32 16	0.0	3/2 ⁺			
8788.3	(15/2 ⁺)	915.4 4	100	7873.03	13/2 ⁺	(M1)	4.82×10^{-5} 7	E_γ , Mult.: from ($^{16}\text{O}, \alpha p \gamma$), with Mult=D, $\Delta J=1$ from $\gamma\gamma(\text{ADO})$; $\Delta\pi=(\text{no})$ from level scheme.
8798.4	(3/2 ⁺ ,5/2 ⁺)	4739.0	13 7	4059.1	(3/2 ⁻)			
		5795.2	26 7	3002.64	5/2 ⁺			
		7034.5	11 6	1763.15	5/2 ⁺			
		7578.1	11 6	1219.38	1/2 ⁺			
		8797.2	100 10	0.0	3/2 ⁺			
8820.9	(1/2 ⁺ to 7/2 ⁺)	5817.7	17 10	3002.64	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a	Comments
8820.9	(1/2 ⁺ to 7/2 ⁺)	6126.3	35 10	2694.00	3/2 ⁺			
		7057.0	40 10	1763.15	5/2 ⁺			
		8819.7	100 10	0.0	3/2 ⁺			
8824.2	1/2 ⁺	5821.0	100 24	3002.64	5/2 ⁺			
		6129.6	16.2 8	2694.00	3/2 ⁺			
		7060.3	43 11	1763.15	5/2 ⁺			
		7603.9	60 16	1219.38	1/2 ⁺			
		8823.0	51 14	0.0	3/2 ⁺			
8829.3	1/2 ⁻	3174	30 15	5655	(3/2) ⁺			
		3990.0	3.9 21	4839.10	(1/2 ⁺ , 3/2)			
		4059.2	4.6 24	4769.86	(5/2 ⁻)			
		4910.5	14 7	3918.47	3/2 ⁺			
		5826.1	67 18	3002.64	5/2 ⁺			
		6134.7	39 9	2694.00	3/2 ⁺			
		6183.0	100 24	2645.67	7/2 ⁺			
		7065.4	27 15	1763.15	5/2 ⁺			
		7609.0	8 4	1219.38	1/2 ⁺			
		8828.1	12 6	0.0	3/2 ⁺			
		3618.4	30 15	5214.5	(5/2 ⁺)			
		4063.0	11 6	4769.86	(5/2 ⁻)			
		4659.3	61 18	4173.43	(5/2 ⁺)			
		4720.4	20 11	4112.4	7/2 ⁺			
8833.1	(5/2, 7/2)	4773.7	48 13	4059.1	(3/2) ⁻			
		4914.3	48 13	3918.47	3/2 ⁺			
		5669.7	16 8	3162.87	7/2 ⁻			
		5829.9	29 15	3002.64	5/2 ⁺			
		6138.5	20 10	2694.00	3/2 ⁺			
		6186.8	100 26	2645.67	7/2 ⁺			
		7069.2	39 22	1763.15	5/2 ⁺			
		8831.9	13 7	0.0	3/2 ⁺			
		3674.6		5163.34	7/2 ⁻			
		3956.9		4880.96	7/2 ⁽⁺⁾			
		4068.0	13 7	4769.86	(5/2 ⁻)			
		4490.0		4347.76	9/2 ⁻			
		4778.7		4059.1	(3/2) ⁻			
		5674.7	46 11	3162.87	7/2 ⁻			
8844.4	17/2 ⁺	7074.2	100 10	1763.15	5/2 ⁺			
		524.8		8319.6	15/2 ⁻			
		971.38 20	100.0 30	7873.03	13/2 ⁺	E2	6.16×10 ⁻⁵ 9	B(E2)(W.u.)=16.3 +37-26 E _γ : weighted average of 971.4 5 from (¹⁶ O,αpγ) and 971.38 20 from (¹⁴ N,αpγ).

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>						
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
8844.4	17/2 ⁺	2757.0		6087.31	13/2 ⁻	
8856.2	5/2 ⁺	3975.0	6.7 33	4880.96	7/2 ⁽⁺⁾	
		4231.6	10 5	4624.3	3/2,5/2	
		4678.0	7 4	4177.9	3/2 ⁻	
		4937.4	22 6	3918.47	3/2 ⁺	
		5692.8	100 10	3162.87	7/2 ⁻	
		6161.6	5.5 28	2694.00	3/2 ⁺	
		6209.9		2645.67	7/2 ⁺	
		7092.3	14 8	1763.15	5/2 ⁺	
		8855.0	31 8	0.0	3/2 ⁺	
8868.6	3/2 ⁺ ,5/2 ⁺	4690.4	5.3 27	4177.9	3/2 ⁻	
		5865.4	33 8	3002.64	5/2 ⁺	
		6174.0	25 6	2694.00	3/2 ⁺	
		7104.7	9 5	1763.15	5/2 ⁺	
		7648.3	33 8	1219.38	1/2 ⁺	
		8867.4	100 25	0.0	3/2 ⁺	
8886.1	(5/2)	3231	23 12	5655	(3/2) ⁺	
		3287.5	25 13	5598.4	3/2 ⁺ ,5/2 ⁺	
		3671.4	9 4	5214.5	(5/2 ⁺)	
		3722.6	35 16	5163.34	7/2 ⁻	
		4116.0	17 9	4769.86	(5/2 ⁻)	
		4707.9	31 16	4177.9	3/2 ⁻	
		4967.3	16 8	3918.47	3/2 ⁺	
		5882.9	31 16	3002.64	5/2 ⁺	
		6239.8	100 27	2645.67	7/2 ⁺	
		7122.2	46 12	1763.15	5/2 ⁺	
		8884.9	54 16	0.0	3/2 ⁺	
8893.3	(5/2 ⁺ ,7/2 ⁺)	3307.1	4.3 22	5586.0	5/2 ⁺	
		4780.6	24 11	4112.4	7/2 ⁺	
		4950.0	11 6	3942.9	9/2 ⁺	
		5729.9	78 19	3162.87	7/2 ⁻	
		6198.7	100 24	2694.00	3/2 ⁺	
		7129.4	51 14	1763.15	5/2 ⁺	
		8892.1	2.7 14	0.0	3/2 ⁺	
8904.8	(1/2,3/2,5/2 ⁺)	2724	15 8	6181	(1/2 ⁺ to 7/2 ⁺)	
		3250	12 6	5655	(3/2) ⁺	
		3895.5	13 7	5009.1	(1/2,3/2)	
		4050.2	21 11	4854.4	(1/2,3/2)	
		4065.5	16 8	4839.10	(1/2 ⁺ ,3/2)	
		4726.6	62 17	4177.9	3/2 ⁻	
		4845.3	20 10	4059.1	(3/2) ⁻	
		6210.2	52 14	2694.00	3/2 ⁺	

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	$\gamma(^{35}\text{Cl})$ (continued)		Comments
						Mult. #	$\delta^\#$	
8904.8	(1/2,3/2,5/2 ⁺)	7684.5	35 10	1219.38	1/2 ⁺			
		8903.6	100 24	0.0	3/2 ⁺			
8906.8	5/2 ⁺	3252		5655	(3/2) ⁺			
		3261	5.4 28	5646	(5/2,7/2,9/2 ⁺)			
		3308.2	3.2 16	5598.4	3/2 ⁺ ,5/2 ⁺			
		3320.6	5.8	5586.0	5/2 ⁺			
		3376	3.6 19	5531				
		3692.1	6.4 32	5214.5	(5/2 ⁺)			
		4136.7	8 4	4769.86	(5/2 ⁻)			
		4963.5	19 5	3942.9	9/2 ⁺			
		7142.9	100 10	1763.15	5/2 ⁺			
8919.8	(5/2 ⁻ ,7/2,9/2 ⁺)	3274	10 5	5646	(5/2,7/2,9/2 ⁺)			
		4571.7	14 7	4347.76	9/2 ⁻			
		5756.4	100 11	3162.87	7/2 ⁻			
		6273.5	46 12	2645.67	7/2 ⁺			
		7155.9	7 4	1763.15	5/2 ⁺			
8953.1	3/2 ⁺	7732.8		1219.38	1/2 ⁺	M1+E2	-0.34 2	Mult., δ : or $\delta=-5.0$ 4 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
		8951.9		0.0	3/2 ⁺	M1+E2	-0.39 5	Mult., δ : or $\delta=-8.2$ 7 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
8984.2	5/2 ⁺	3260.5	1.3 7	5723.52	5/2 ⁺			
		4144.8	2.0 10	4839.10	(1/2 ⁺ ,3/2)			
		4214.1	0.8 5	4769.86	(5/2 ⁻)			
		4359.6	0.8 5	4624.3	3/2,5/2			
		4806.0	2.5 13	4177.9	3/2 ⁻			
		4810.4	2.5 13	4173.43	(5/2 ⁺)			
		5065.3	15 8	3918.47	3/2 ⁺			
		5820.8	10 5	3162.87	7/2 ⁻			
		6289.6	3.0 15	2694.00	3/2 ⁺			
		6337.9	2.0 10	2645.67	7/2 ⁺			
		7220.3	27 7	1763.15	5/2 ⁺			
		8983.0	100 10	0.0	3/2 ⁺			
9081.5	5/2 ⁺	3357.8	1.7	5723.52	5/2 ⁺			
		4200.3	1.7	4880.96	7/2 ⁽⁺⁾			
		4242.1		4839.10	(1/2 ⁺ ,3/2)			
		4311.4		4769.86	(5/2 ⁻)			
		4456.9		4624.3	3/2,5/2			
		4903.2	3.3	4177.9	3/2 ⁻			
		4907.7	3.3	4173.43	(5/2 ⁺)			
		5162.6	15	3918.47	3/2 ⁺	D(+Q)	-0.002 9	
		5918.1	10	3162.87	7/2 ⁻			
		6386.9	3.3	2694.00	3/2 ⁺			
		6435.2	1.7	2645.67	7/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
9081.5	5/2 ⁺	7317.5	27	1763.15	5/2 ⁺	M1+E2	+0.11 2		B(M1)(W.u.)=1.16 +31-25; B(E2)(W.u.)=0.99 +50-38
		9080.2	100	0.0	3/2 ⁺	M1+E2	+0.089 9		B(M1)(W.u.)=2.26 +49-33; B(E2)(W.u.)=0.82 +27-19
9157.1	5/2 ⁻	3942.4	17 5	5214.5	(5/2 ⁺)				
		4386.9	16 4	4769.86	(5/2 ⁻)				
		4978.8	<37	4177.9	3/2 ⁻				I_γ : total $I_\gamma=30$ 7 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<37	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=30$ 7 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		5993.7	22 6	3162.87	7/2 ⁻	M1+E2	+0.10 2	1.59×10 ⁻³ 2	
		6153.9	20 5	3002.64	5/2 ⁺				
		6462.5	3 2	2694.00	3/2 ⁺				
		6510.8	7 4	2645.67	7/2 ⁺				
		7393.1	17 5	1763.15	5/2 ⁺				
		9155.8	100 9	0.0	3/2 ⁺	(E1)			Mult.: D+Q with $\delta=+0.09$ 1 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely.
10181.1	19/2 ⁻	1336.3 5	100.0 34	8844.4	17/2 ⁺	(E1)		0.0001637 23	B(E1)(W.u.)=0.00235 +21-18 I_γ : weighted average of 100 17 from ($^{28}\text{Si},\alpha\gamma$) and 100.0 34 from ($^{16}\text{O},\alpha\gamma$). E_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$; $\Delta\pi$ =(yes) from level scheme.
		1693.5 5	84 9	8487.5	15/2 ⁻	E2		0.0001880 26	B(E2)(W.u.)=44.4 46 E_γ , I_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
		1861.3 5	83 6	8319.6	15/2 ⁻	E2		0.000262 4	B(E2)(W.u.)=27.4 +26-24 E_γ , I_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
10221.9	(17/2 ⁻)	1377.8 10	100	8844.4	17/2 ⁺				
10858.5	19/2 ⁺	2014.7 9	100 5	8844.4	17/2 ⁺	(M1)		0.000265 4	E_γ , I_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$; $\Delta\pi$ =no from level scheme.
		2069.8 10	32.5 22	8788.3	(15/2 ⁺)	E2		0.000361 5	E_γ , I_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$) with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$; M2 ruled out by RUL.
11459.1	21/2 ⁺	2614.5 5	100	8844.4	17/2 ⁺	E2		0.000619 9	B(E2)(W.u.)=15.9 8 E_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$; M2 ruled out by RUL.
12572.2	23/2 ⁻	1113.3 5	24.4 18	11459.1	21/2 ⁺	(E1)		3.60×10 ⁻⁵ 5	B(E1)(W.u.)=0.00272 +41-35 E_γ , I_γ ,Mult.: from ($^{16}\text{O},\alpha\gamma$), with Mult=D from $\gamma\gamma(\text{ADO})$ with $\Delta J=1$; $\Delta\pi$ =yes from level scheme.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
12572.2	23/2 ⁻	2390.8 5	100 4	10181.1	19/2 ⁻	E2	0.000515 7	B(E2)(W.u.)=26.0 +36-29 E _γ , I _γ , Mult.: from (¹⁶ O, αpγ), with Mult=Q, ΔJ=2 from γγ(ADO); M2 ruled out by RUL.
16306.4	(27/2 ⁻)	3734.0		12572.2	23/2 ⁻			

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p,γ), unless otherwise noted.

[#] From γ(θ) and/or γγ(θ) in (p,γ) for γ rays originating from low-spin levels (J<9/2) and from γγ(θ)(DCO) and γ(lin pol) in (²⁸Si, αpγ) for γ rays originating from high-spin (J≥9/2) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ(lin pol) or RUL with measured T_{1/2}, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p,γ), a D+Q with definite δ(Q/D)>0 from γ(θ) is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ. The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the ³⁴S(p,γ) dataset.

@ Primary transitions probably from the unresolved 7501+7503 doublet initial levels in 1976Me12 in (p,γ).

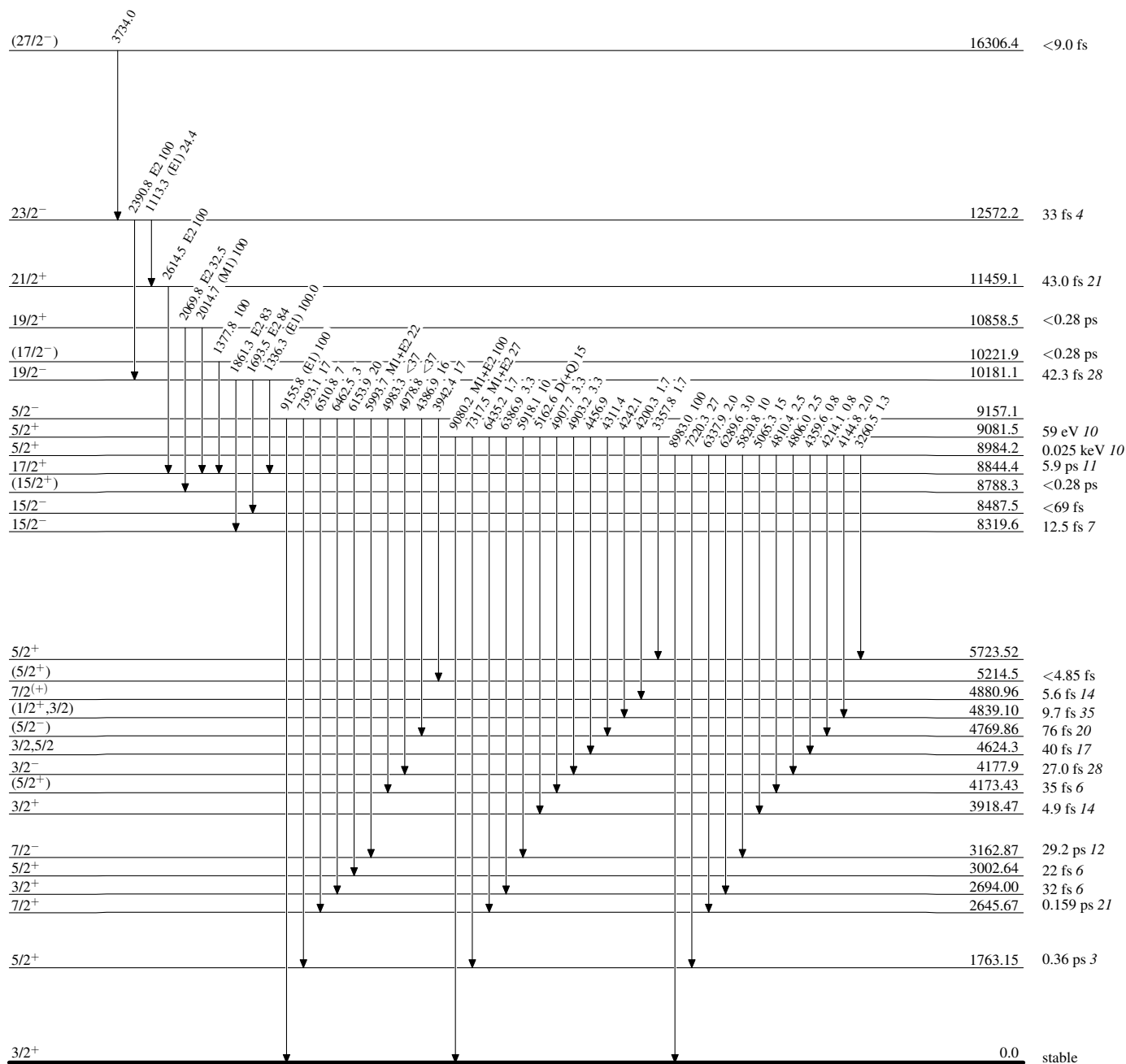
& Primary transitions probably from the unresolved 7837+7839 doublet initial levels in 1976Me12 in (p,γ).

^a Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^b Placement of transition in the level scheme is uncertain.

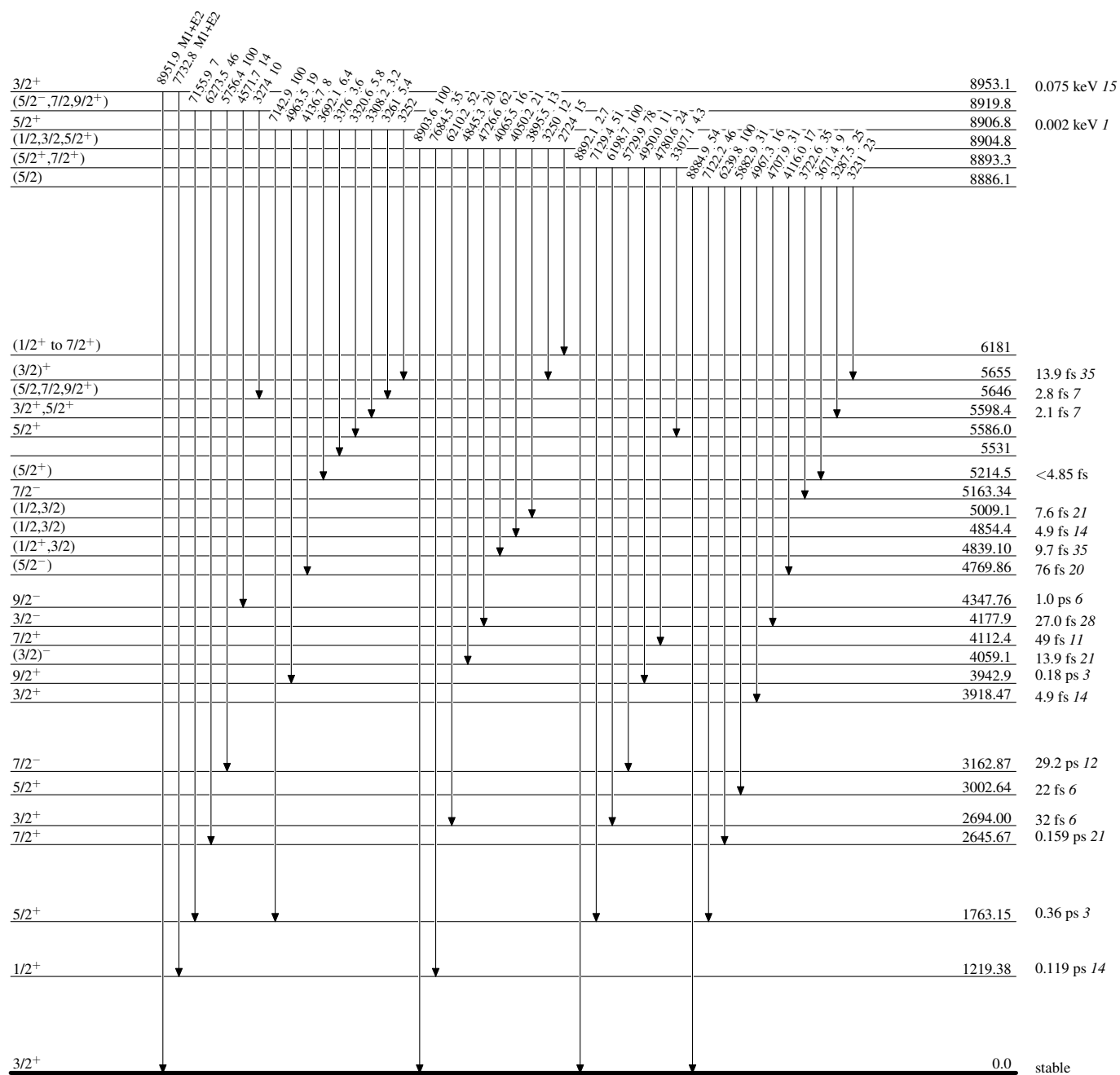
Adopted Levels, Gammas**Level Scheme**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

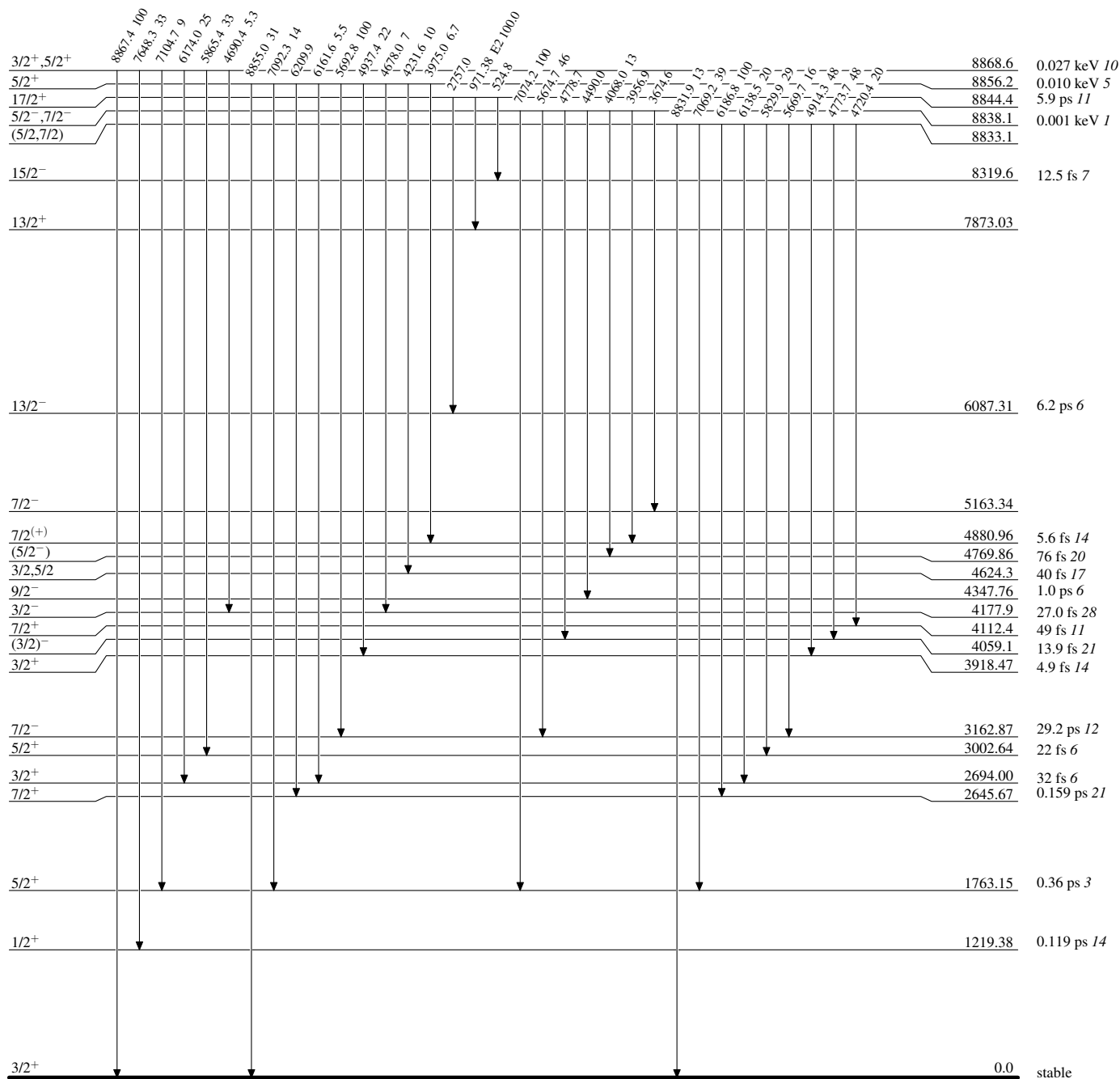
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

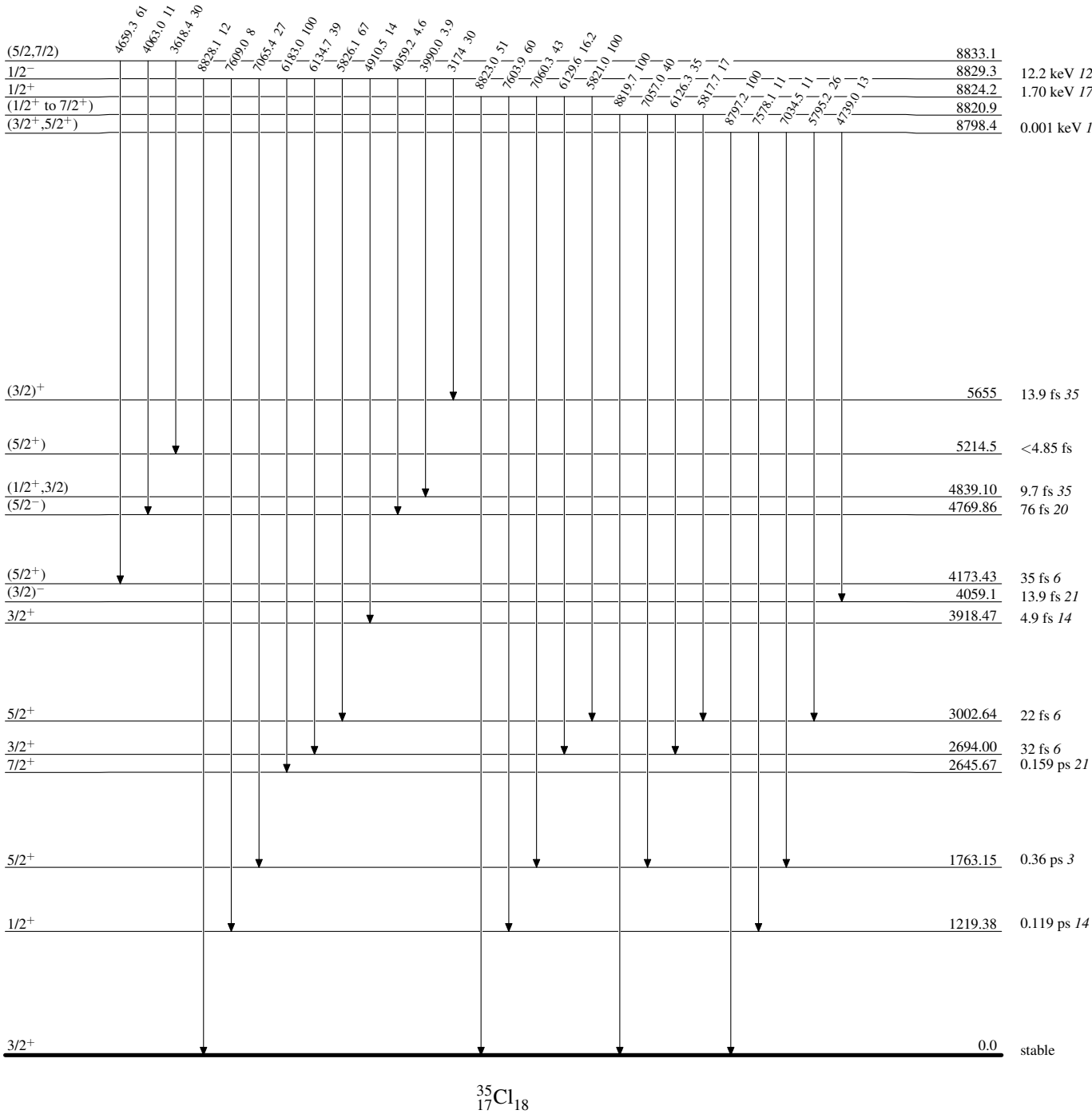
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

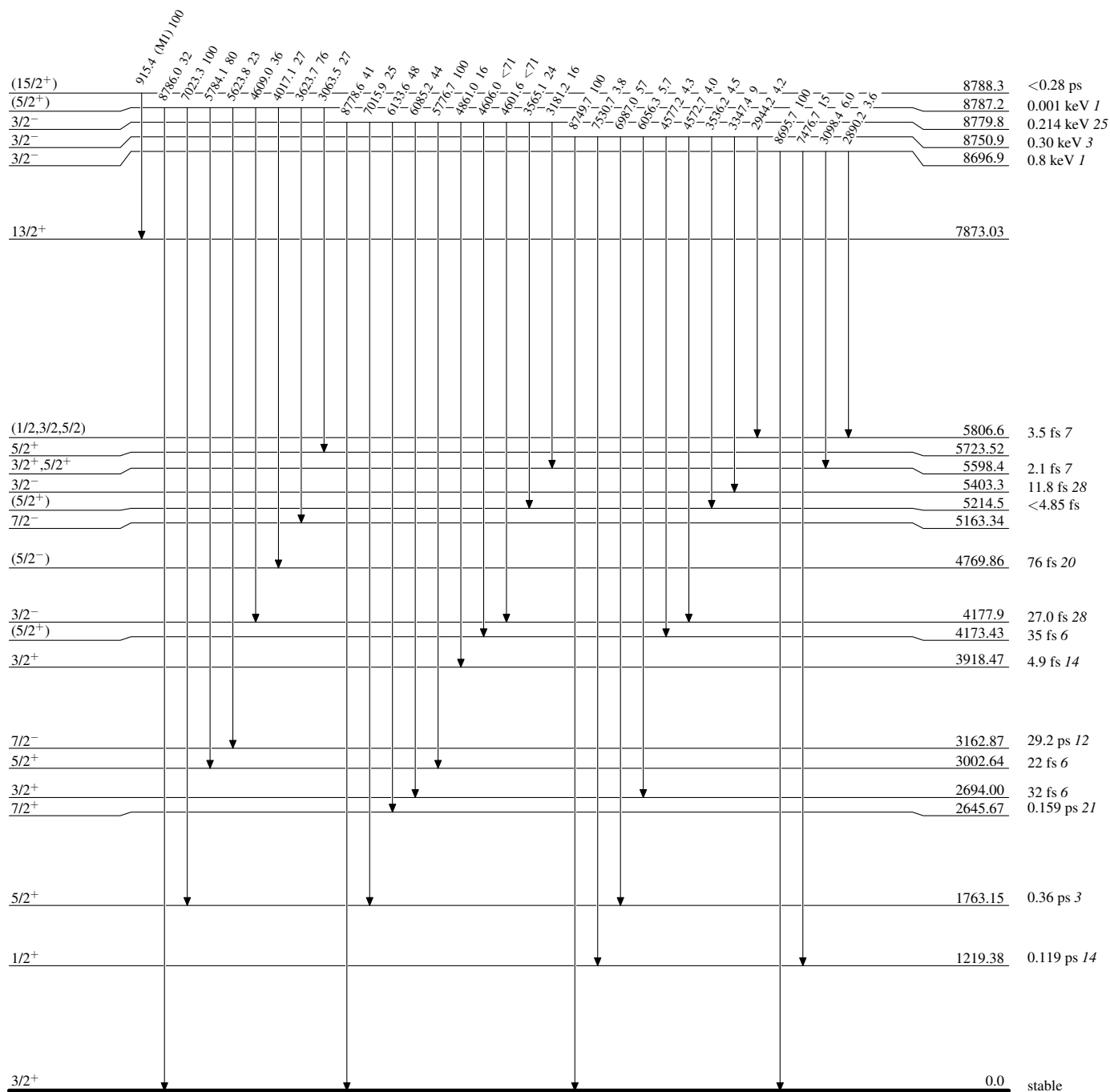
Level Scheme (continued)

Intensities: Relative photon branching from each level



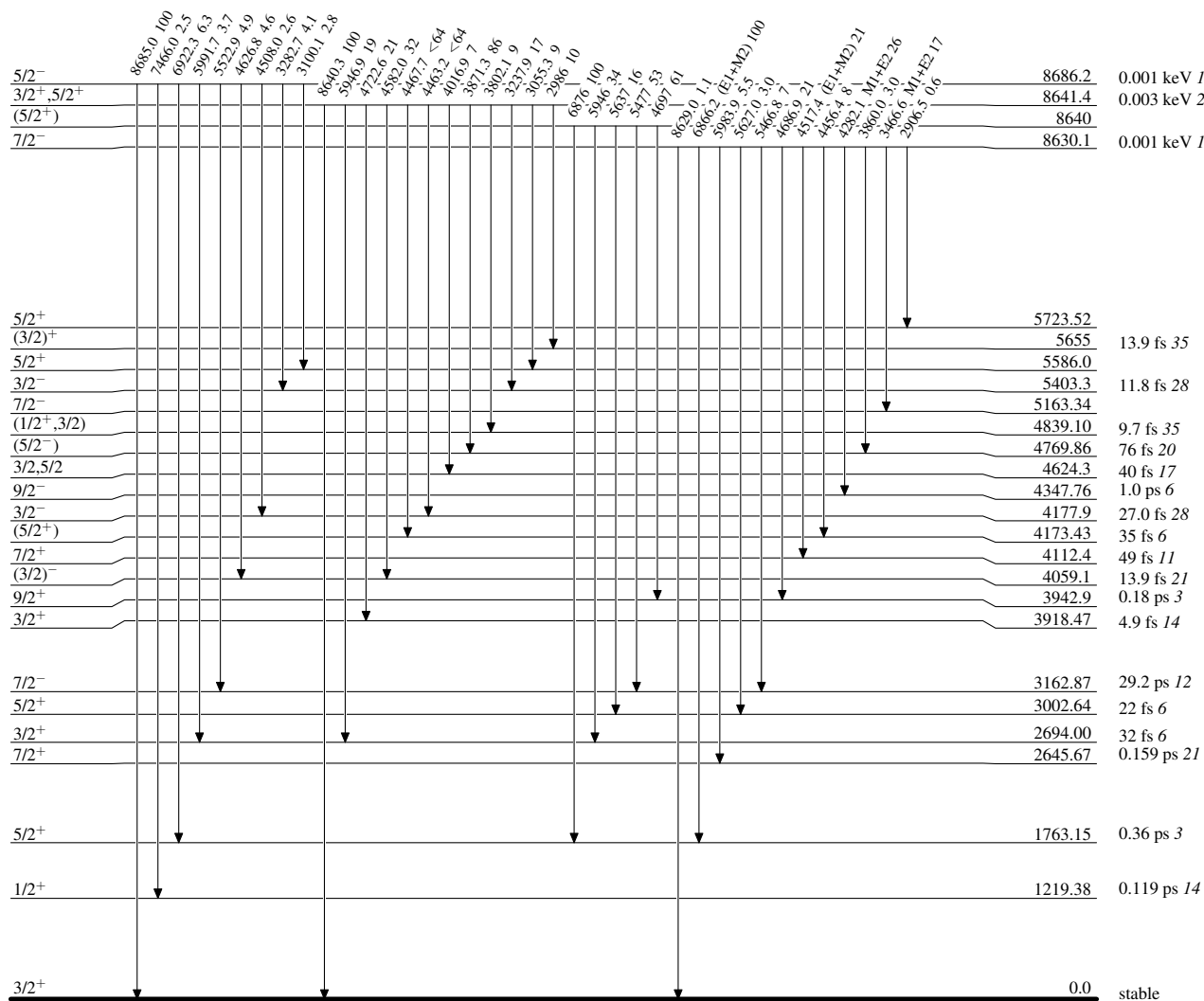
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

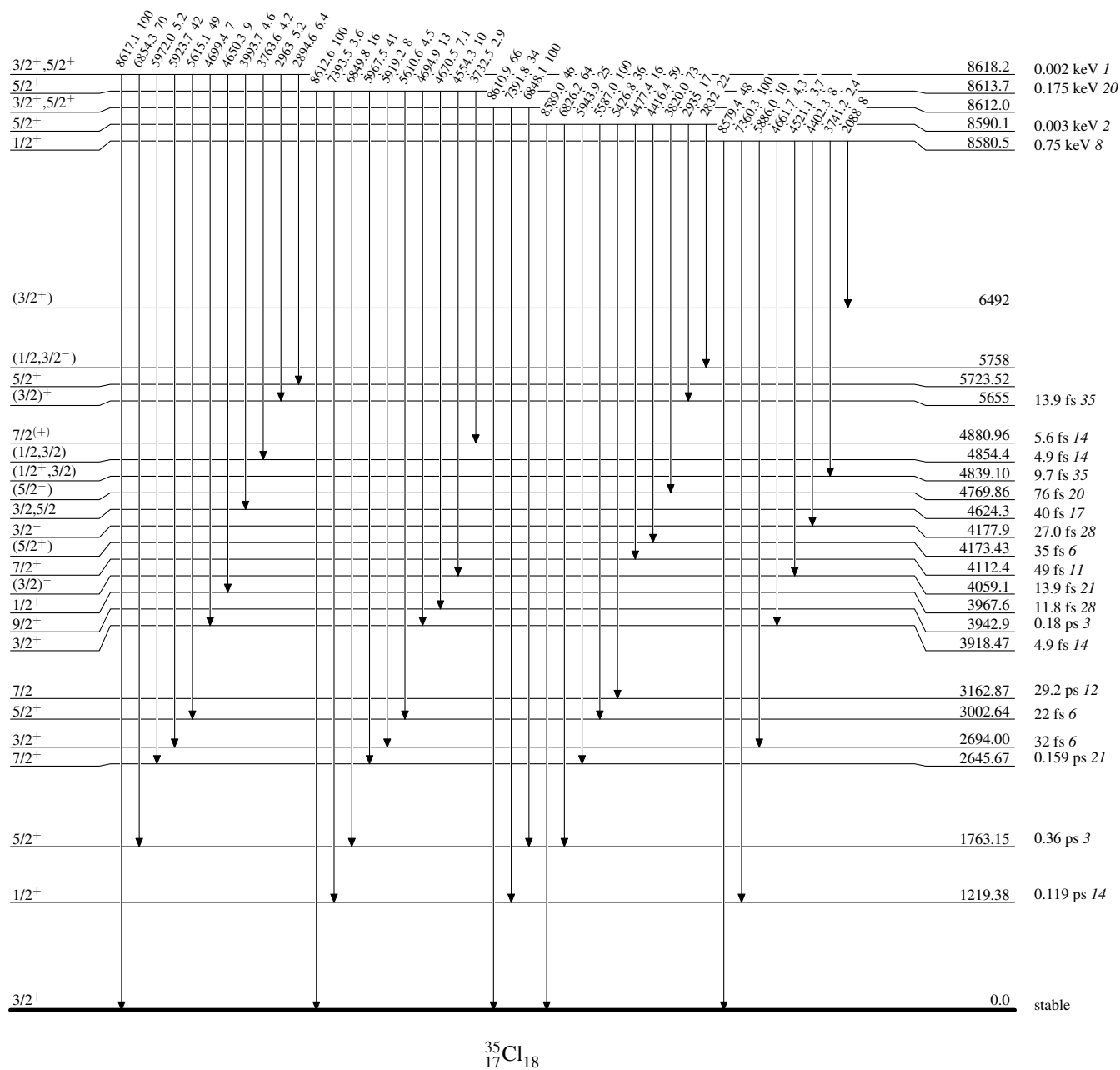
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

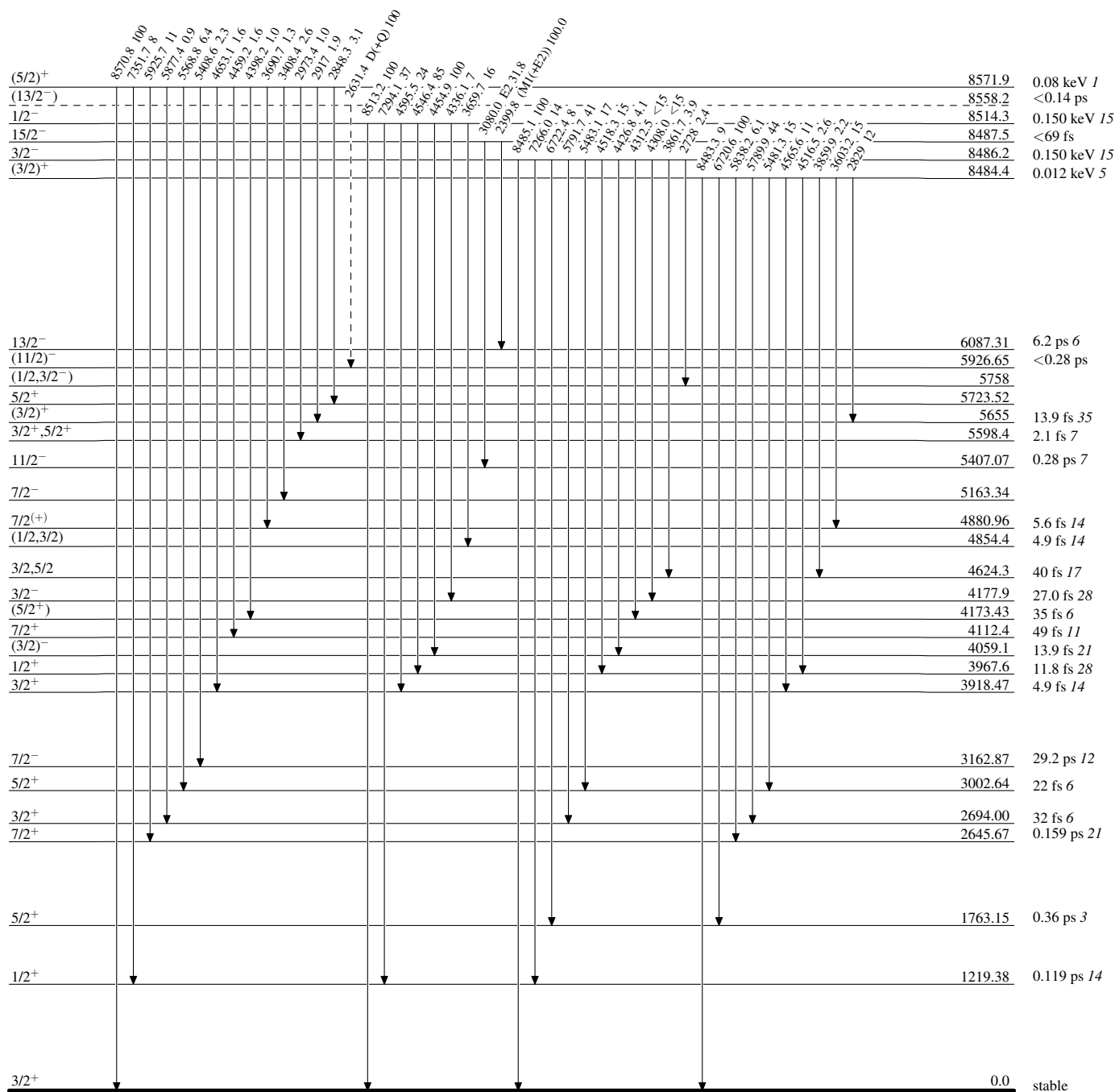


Adopted Levels, Gammas

Legend

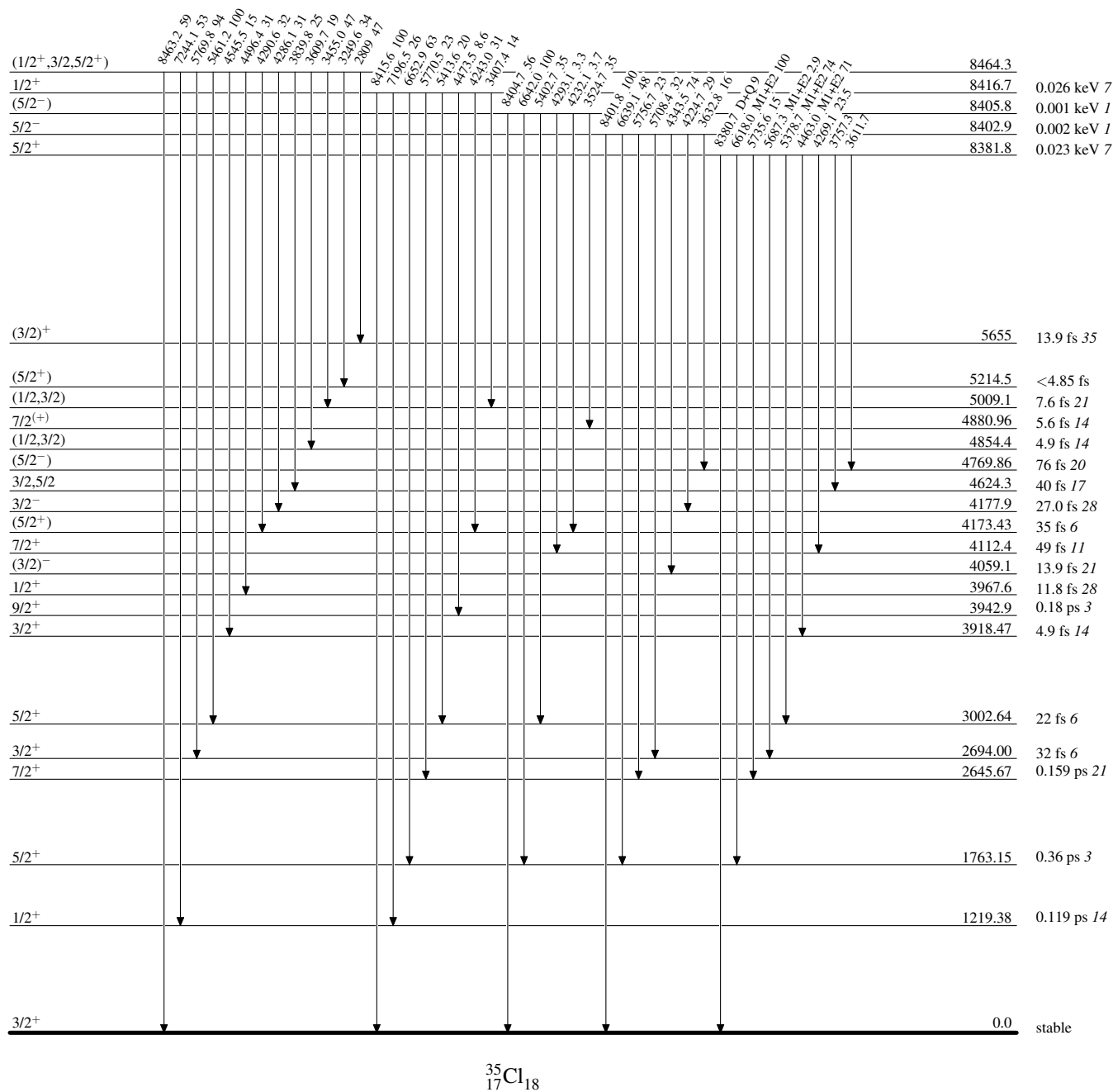
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

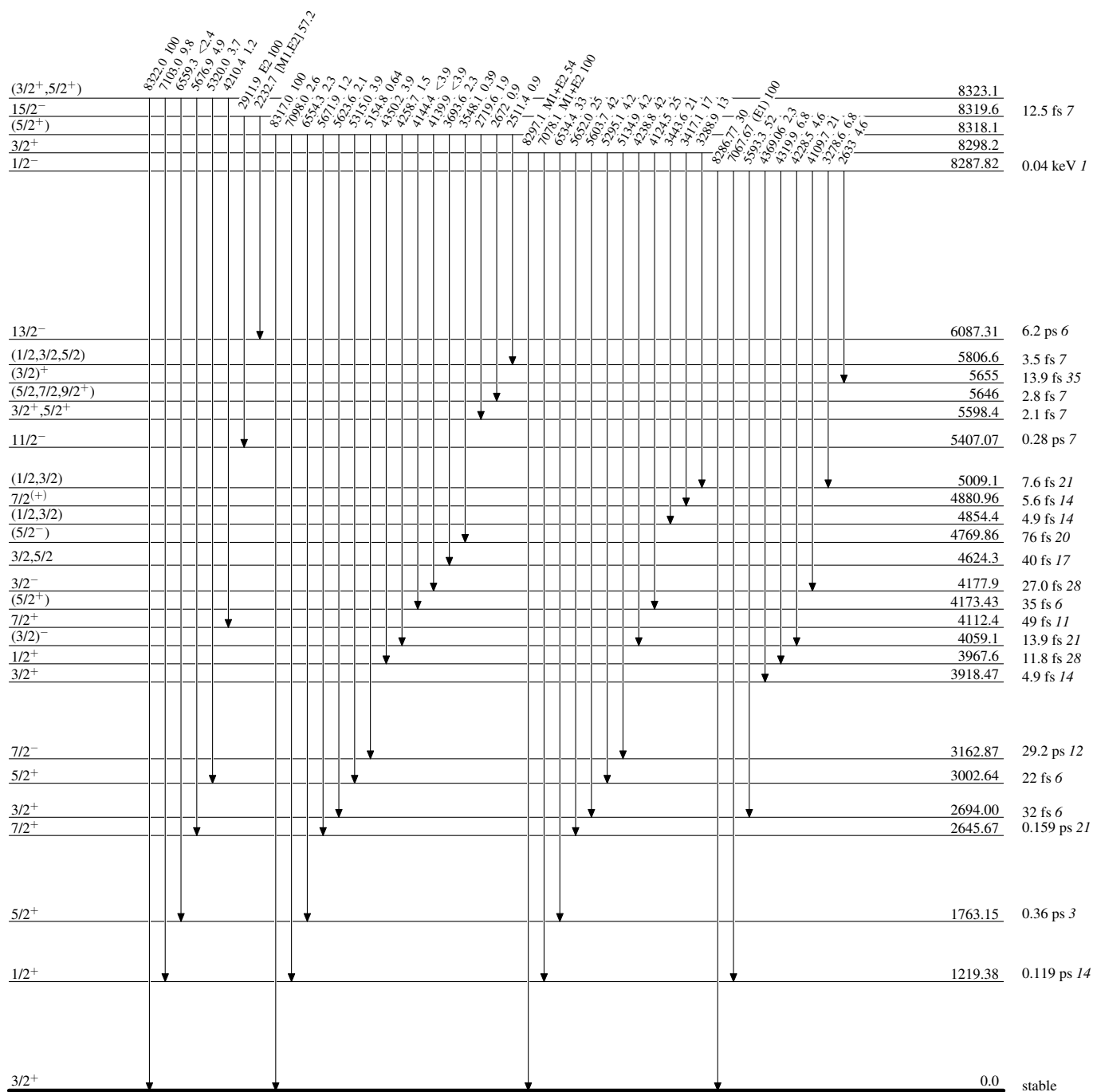
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

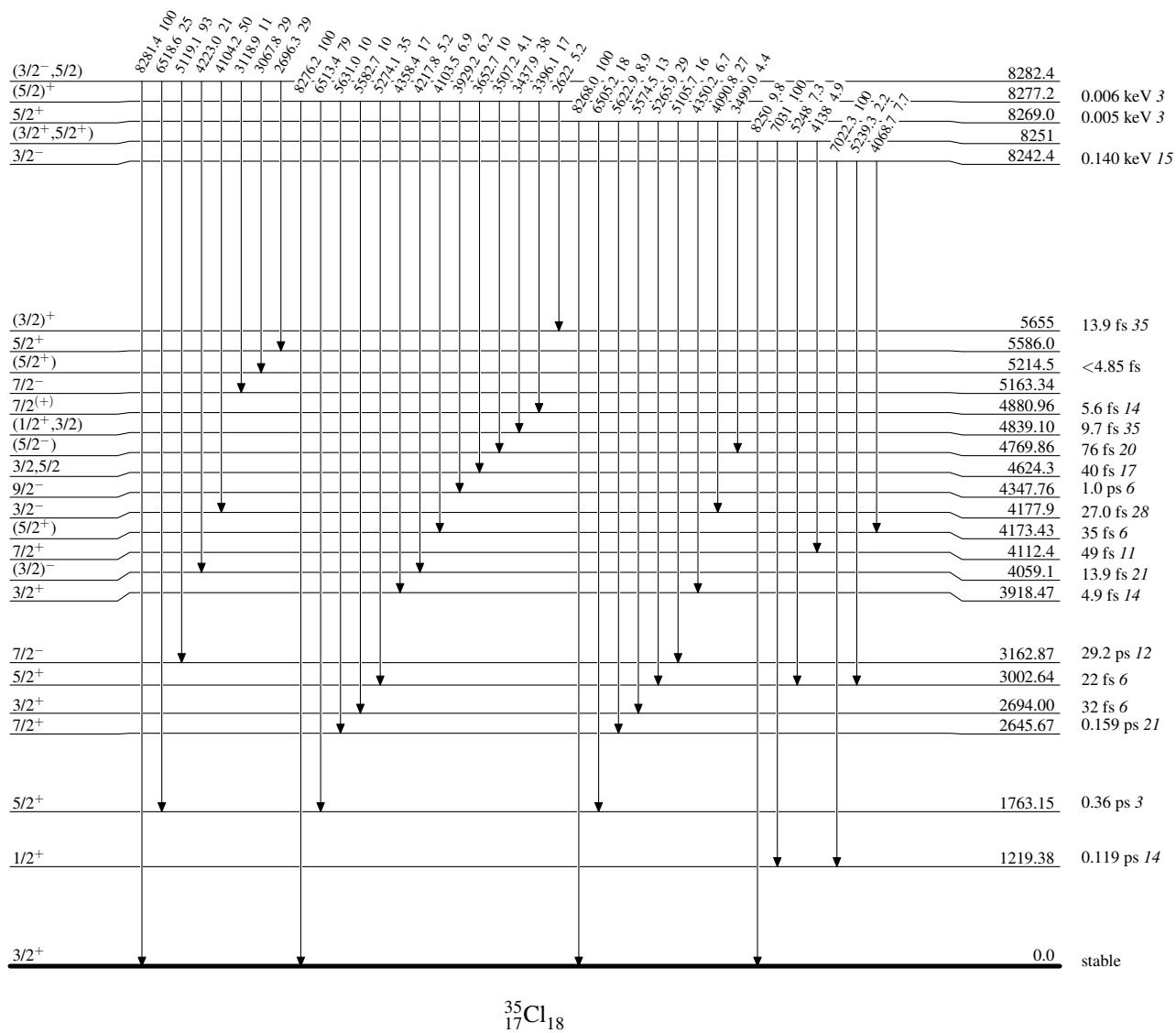
Level Scheme (continued)

Intensities: Relative photon branching from each level



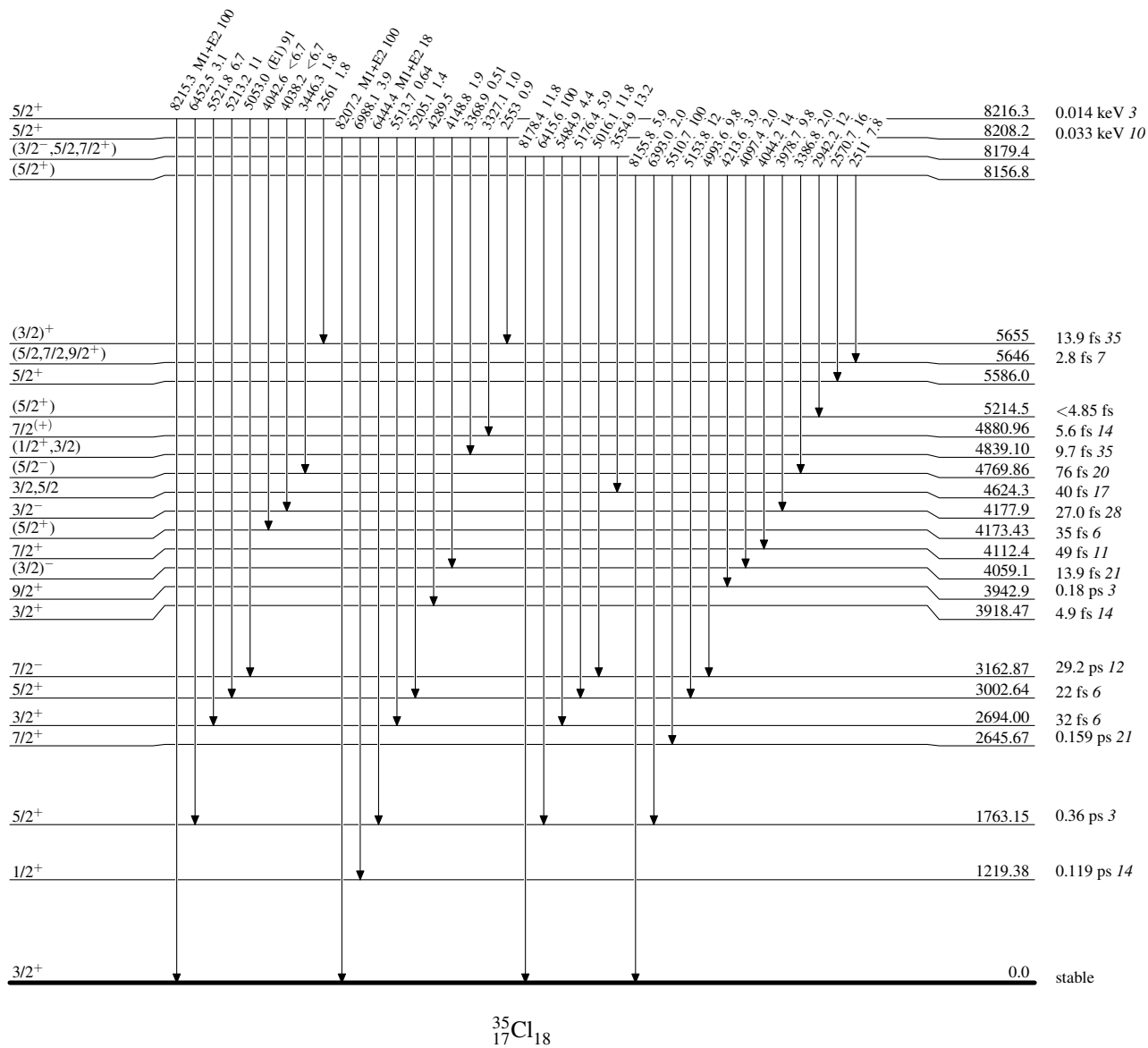
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, GammasLevel Scheme (continued)

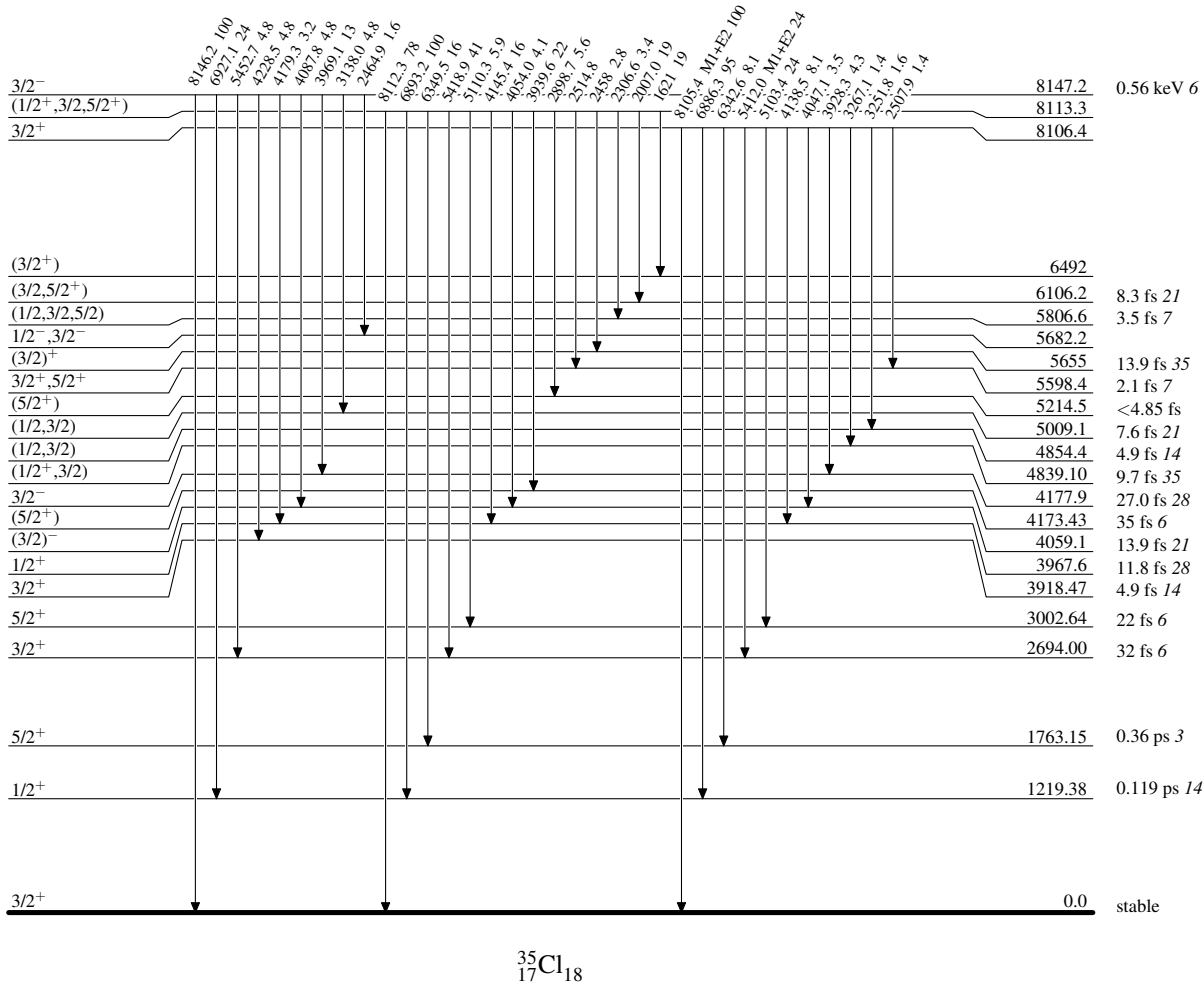
Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Level Scheme (continued)

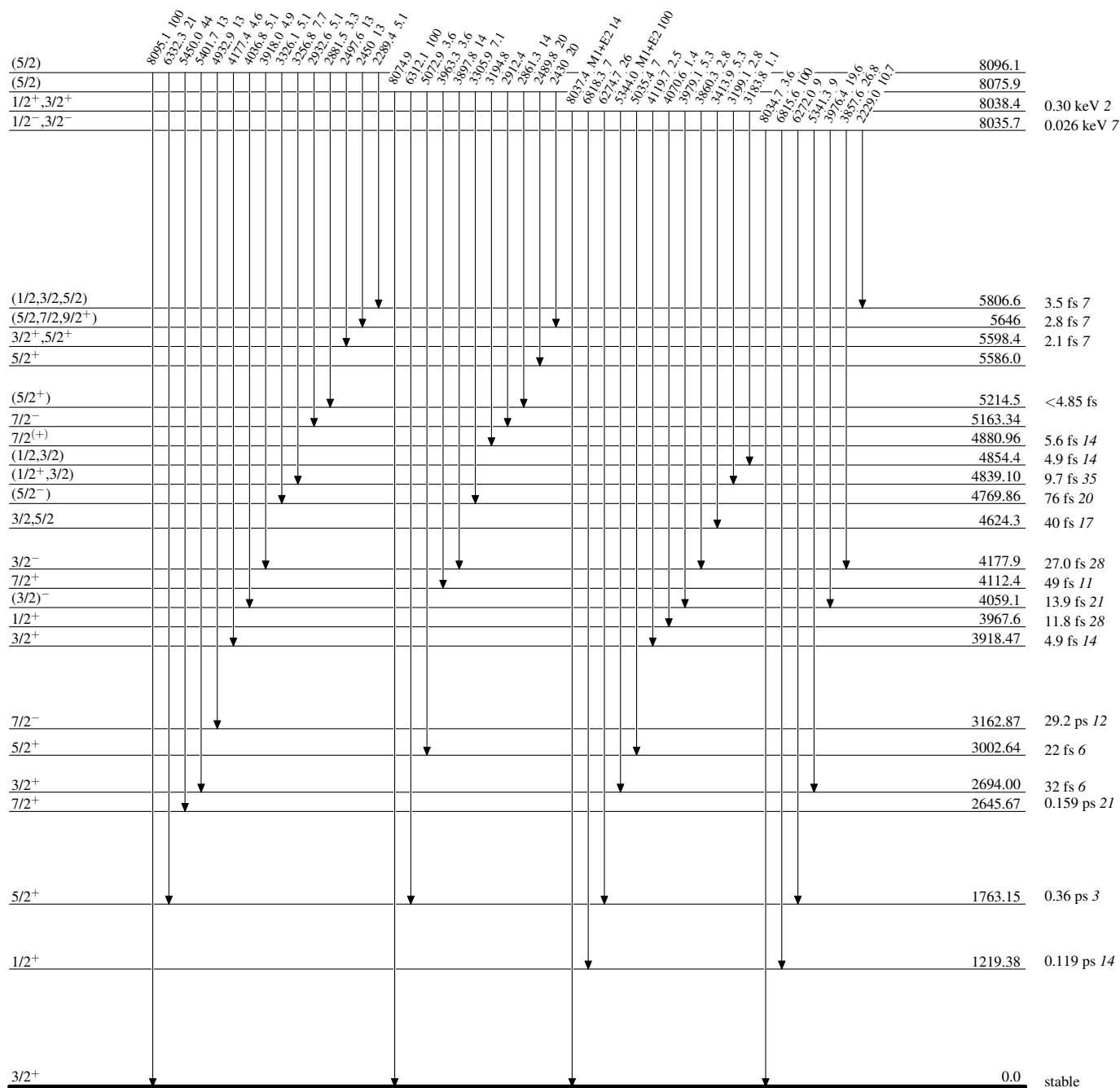
Intensities: Relative photon branching from each level



³⁵Cl₁₈

Adopted Levels, Gammas**Level Scheme (continued)**

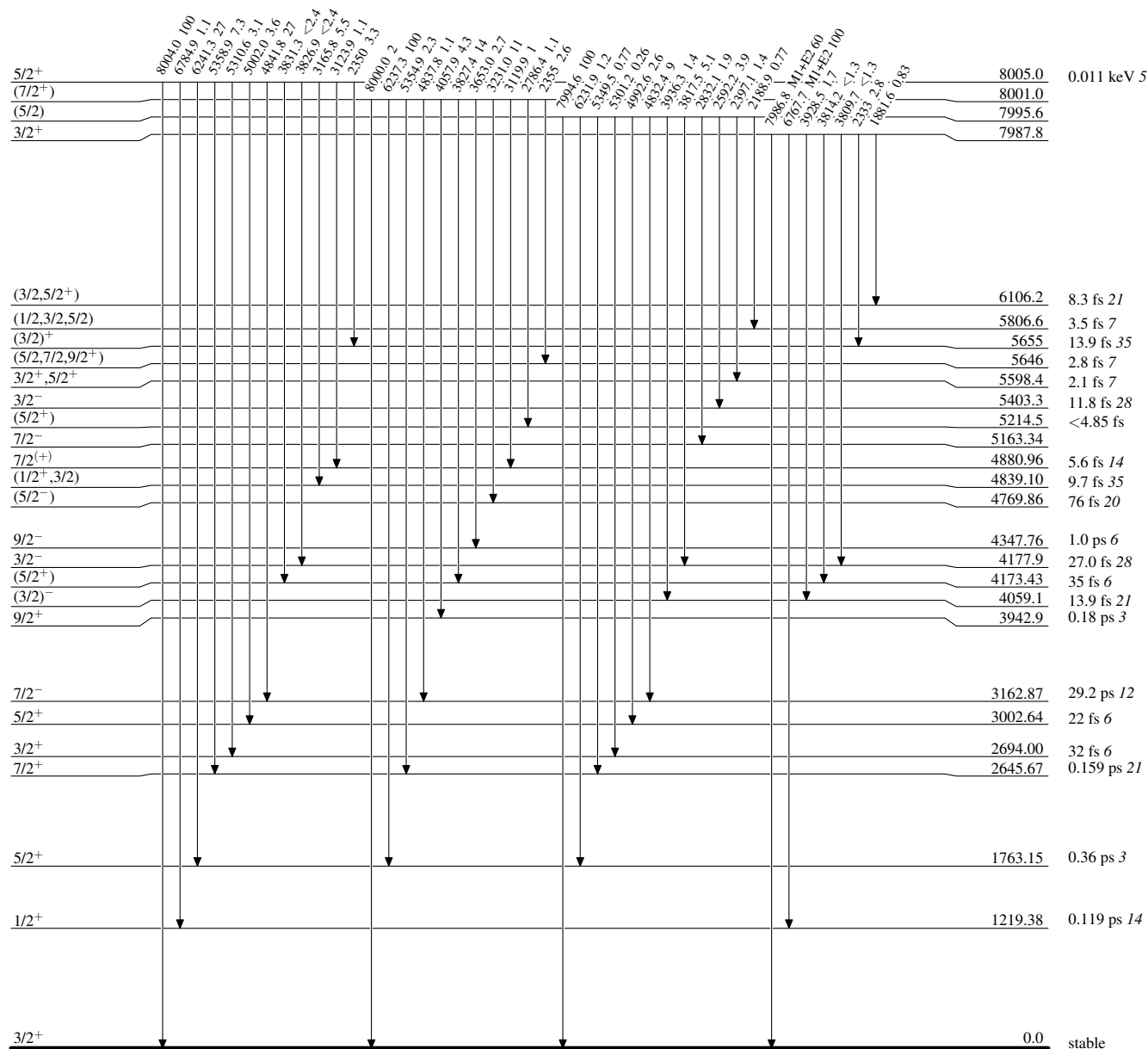
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

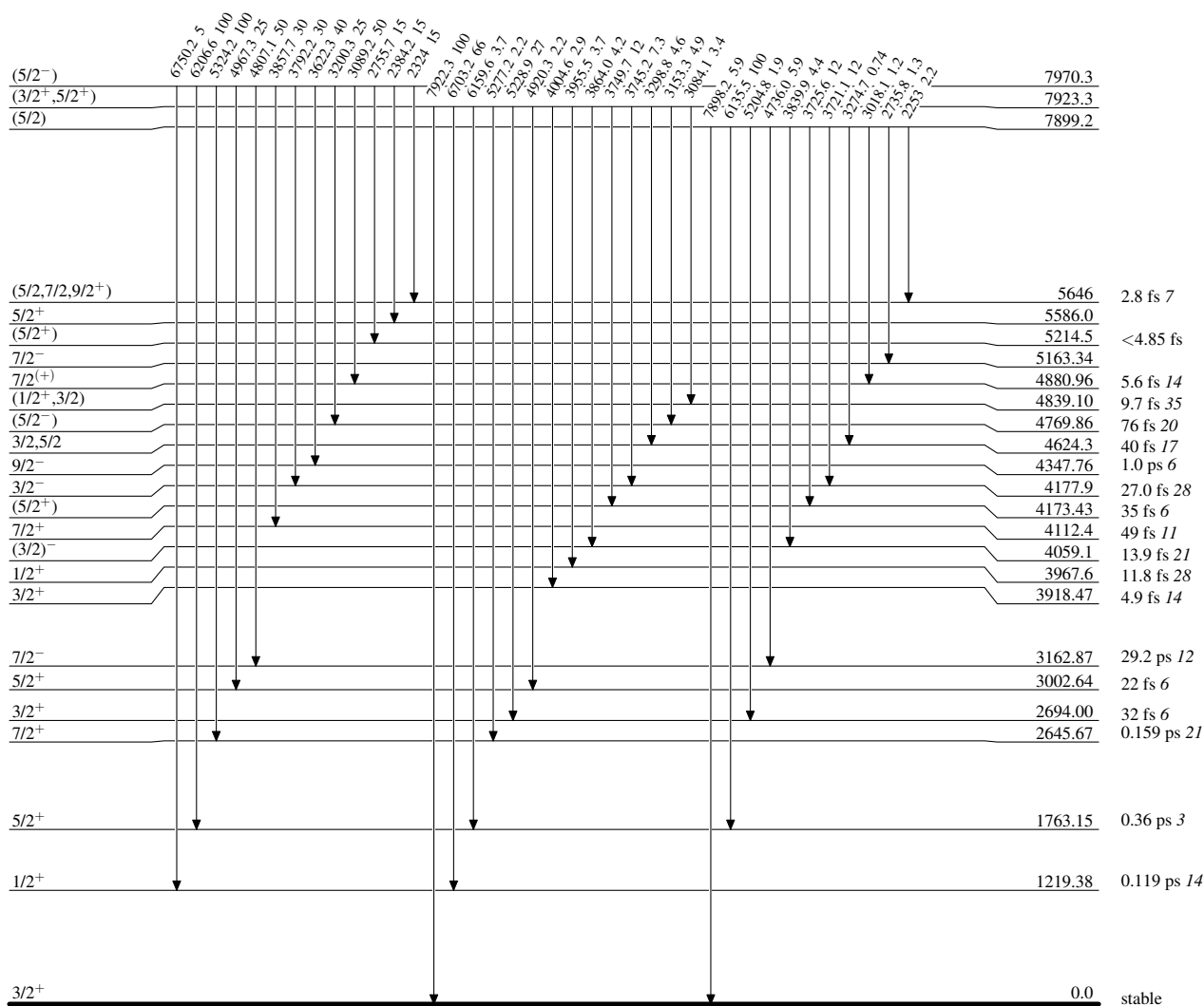
Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

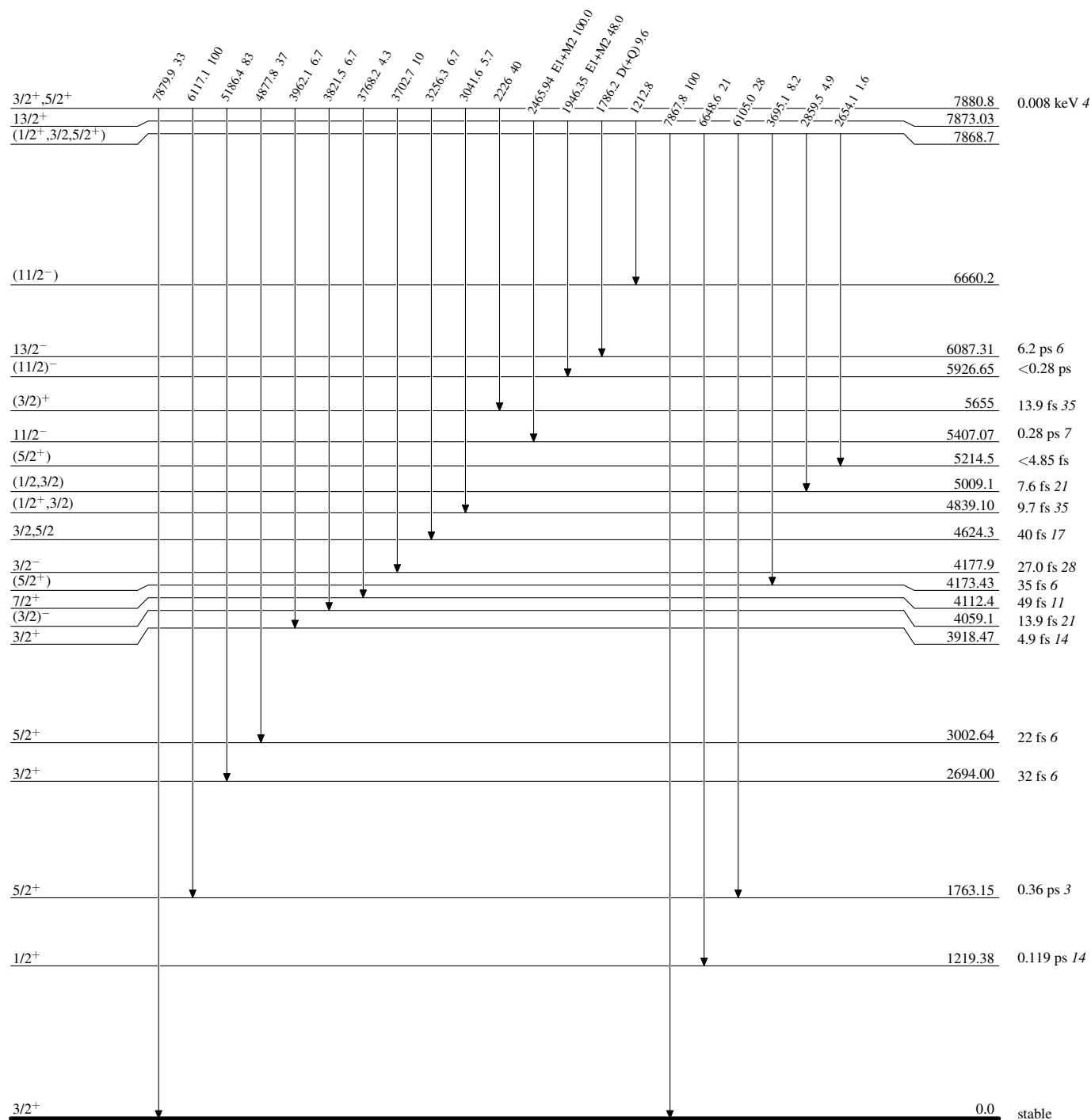
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

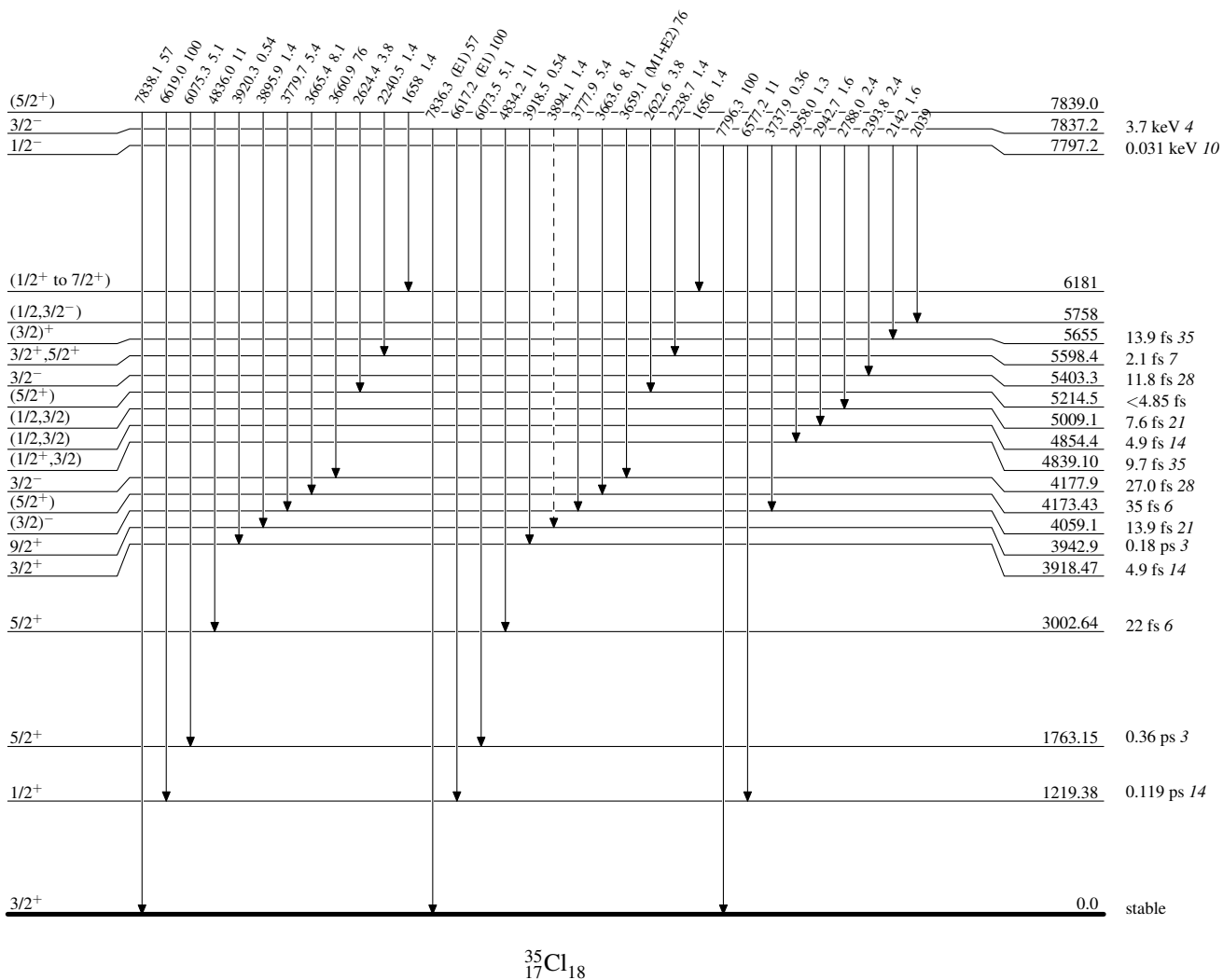


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

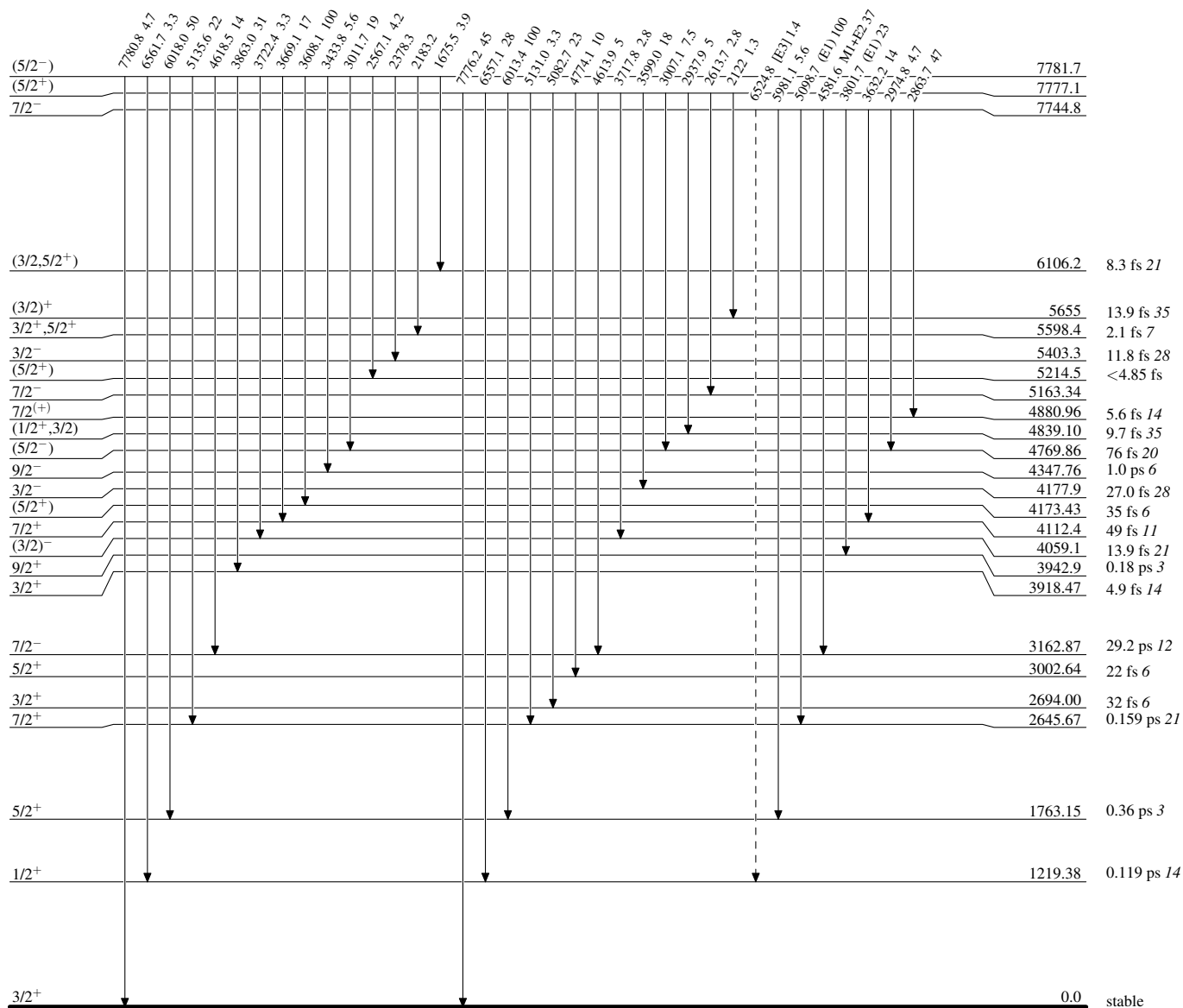
-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas

Legend

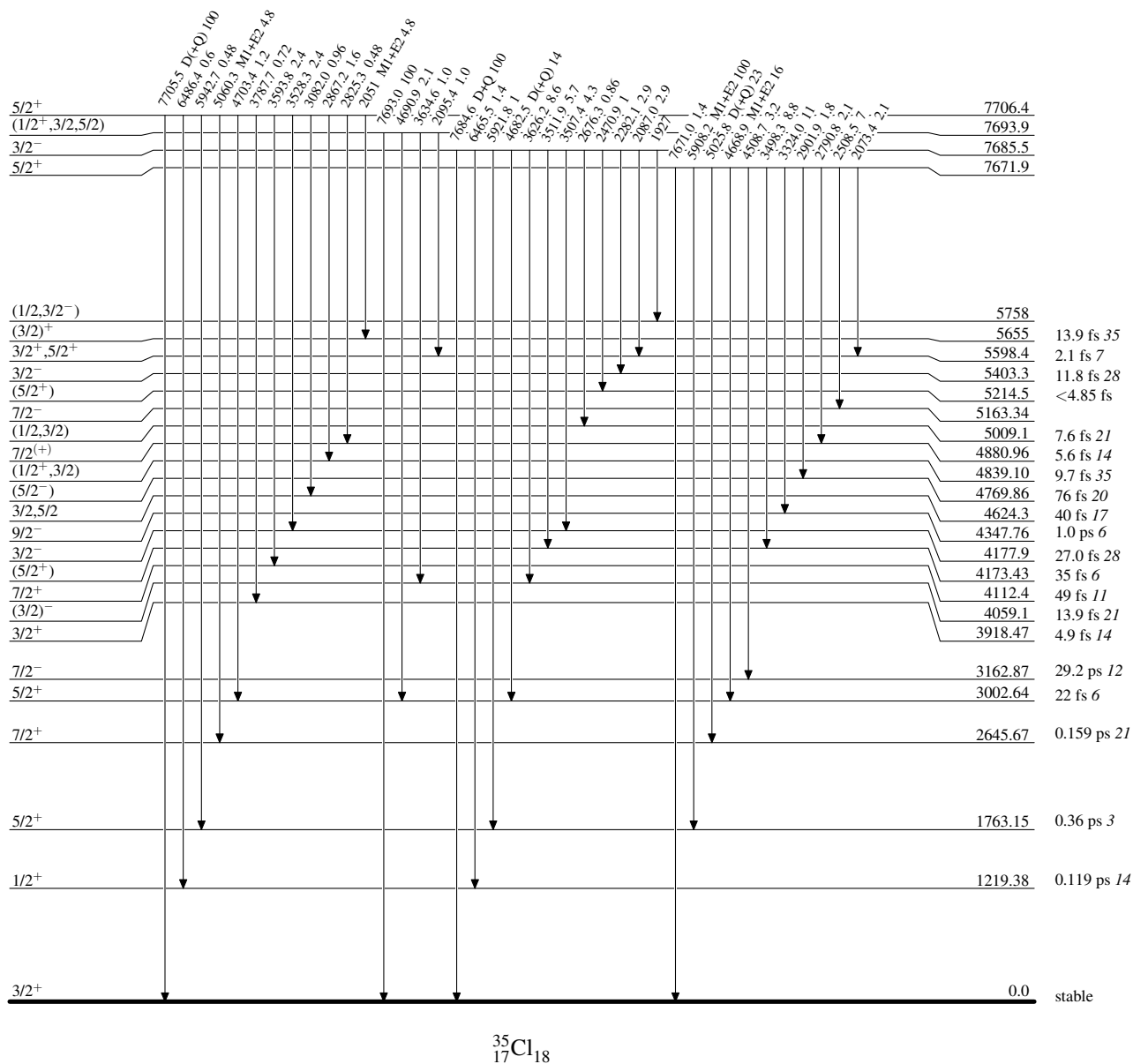
Level Scheme (continued)

Intensities: Relative photon branching from each level

 -----► γ Decay (Uncertain)

 $^{35}_{17}\text{Cl}_{18}$

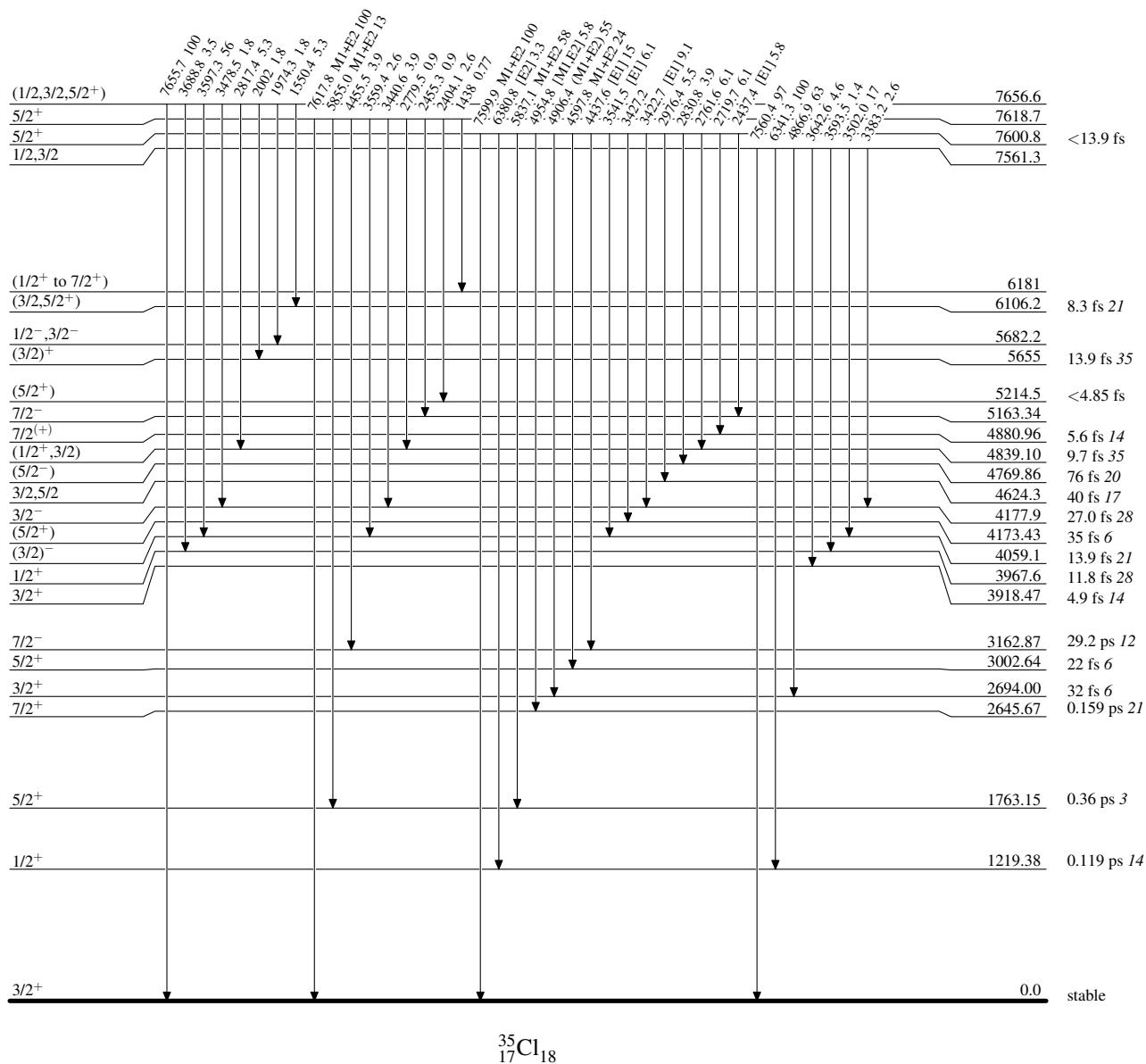
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

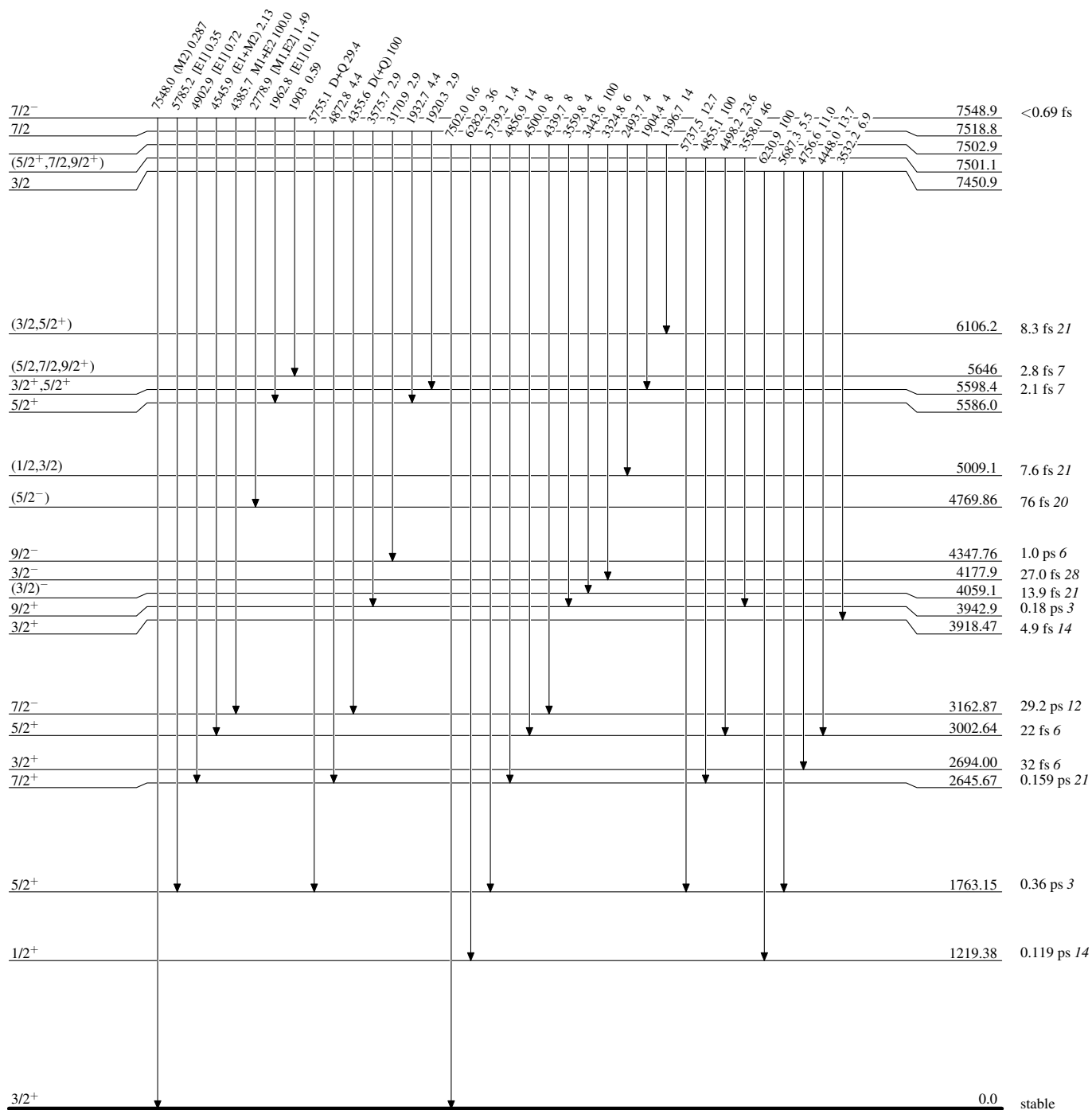
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



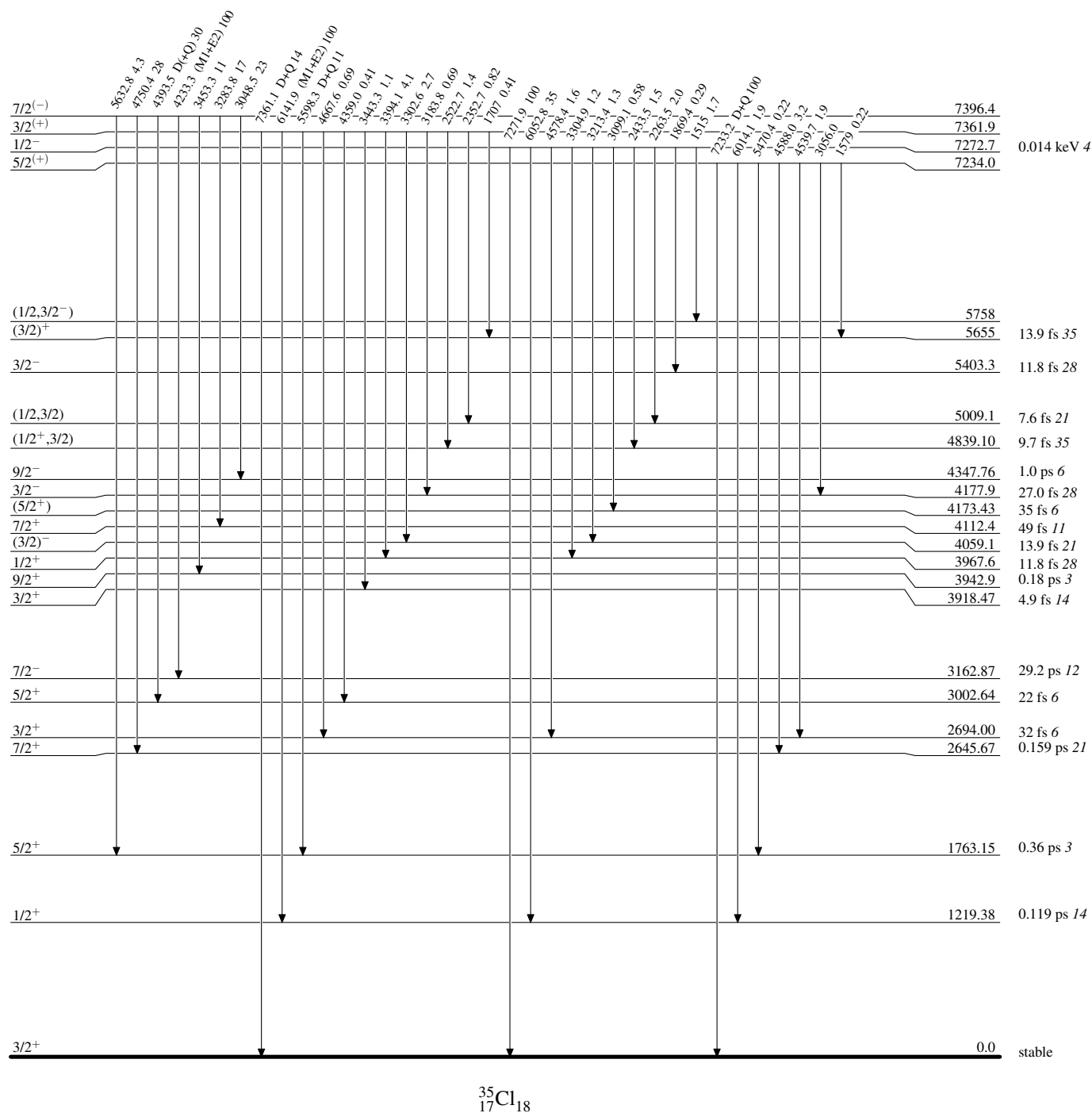
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



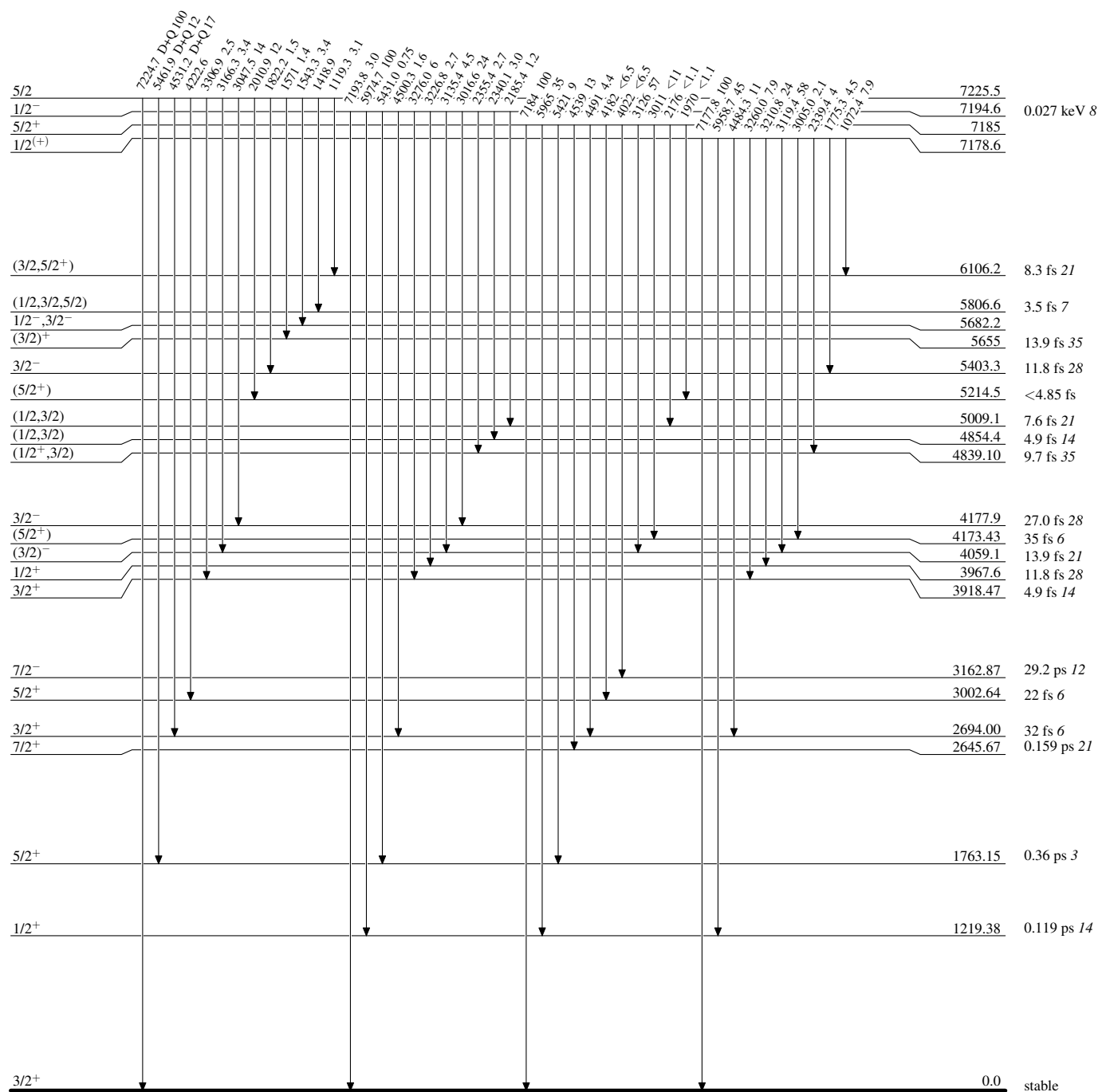
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



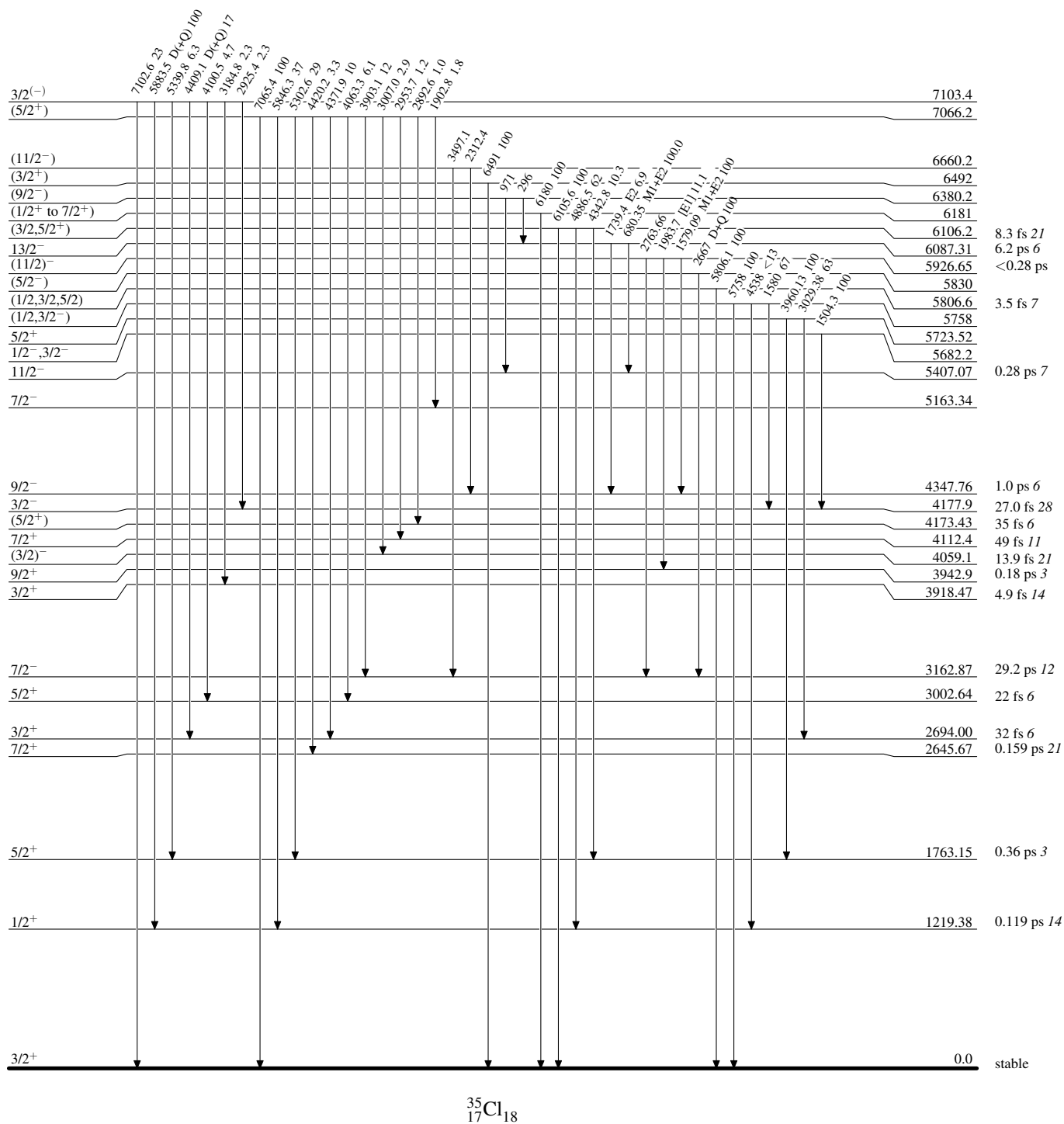
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

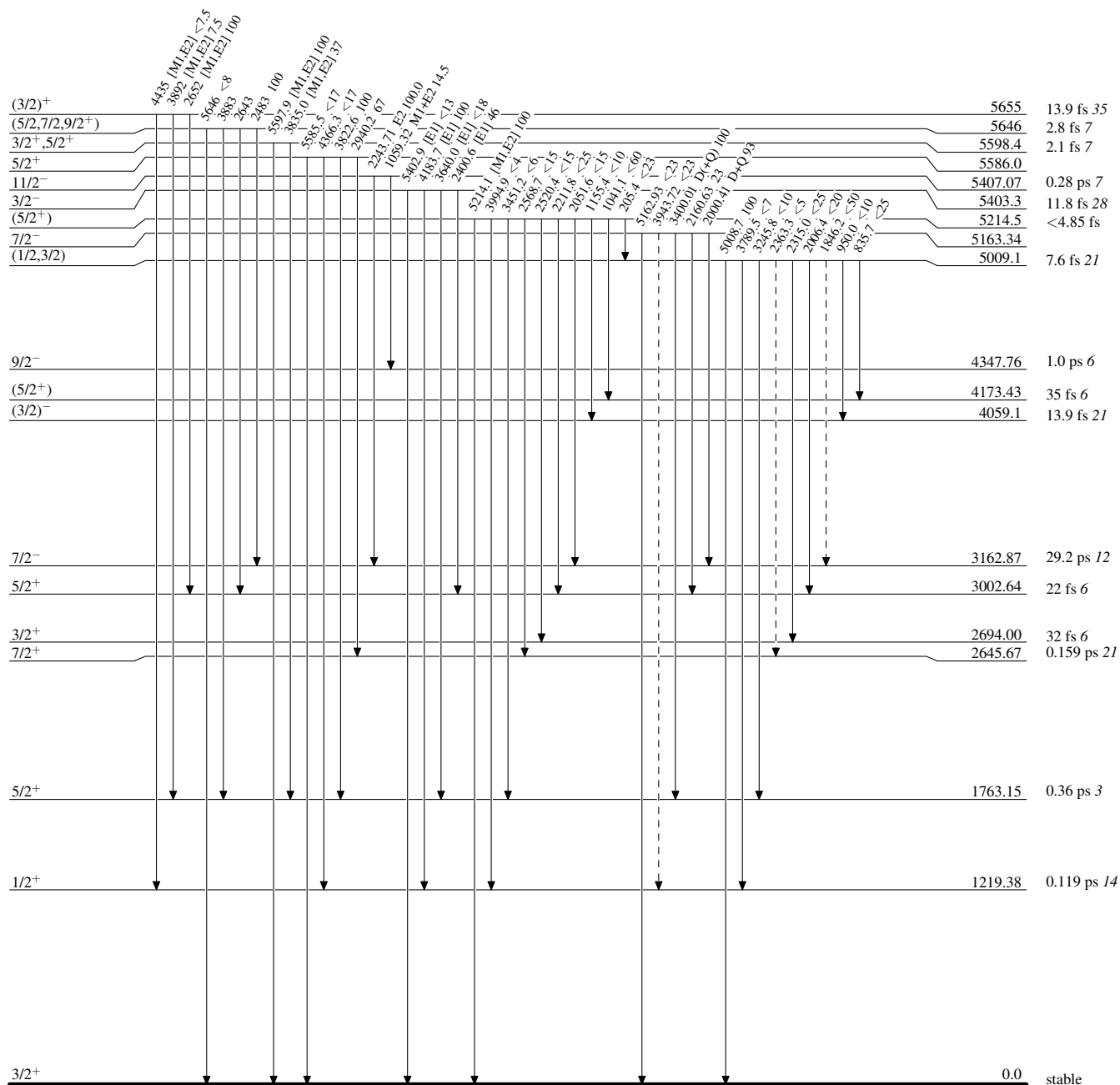


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

 -----► γ Decay (Uncertain)


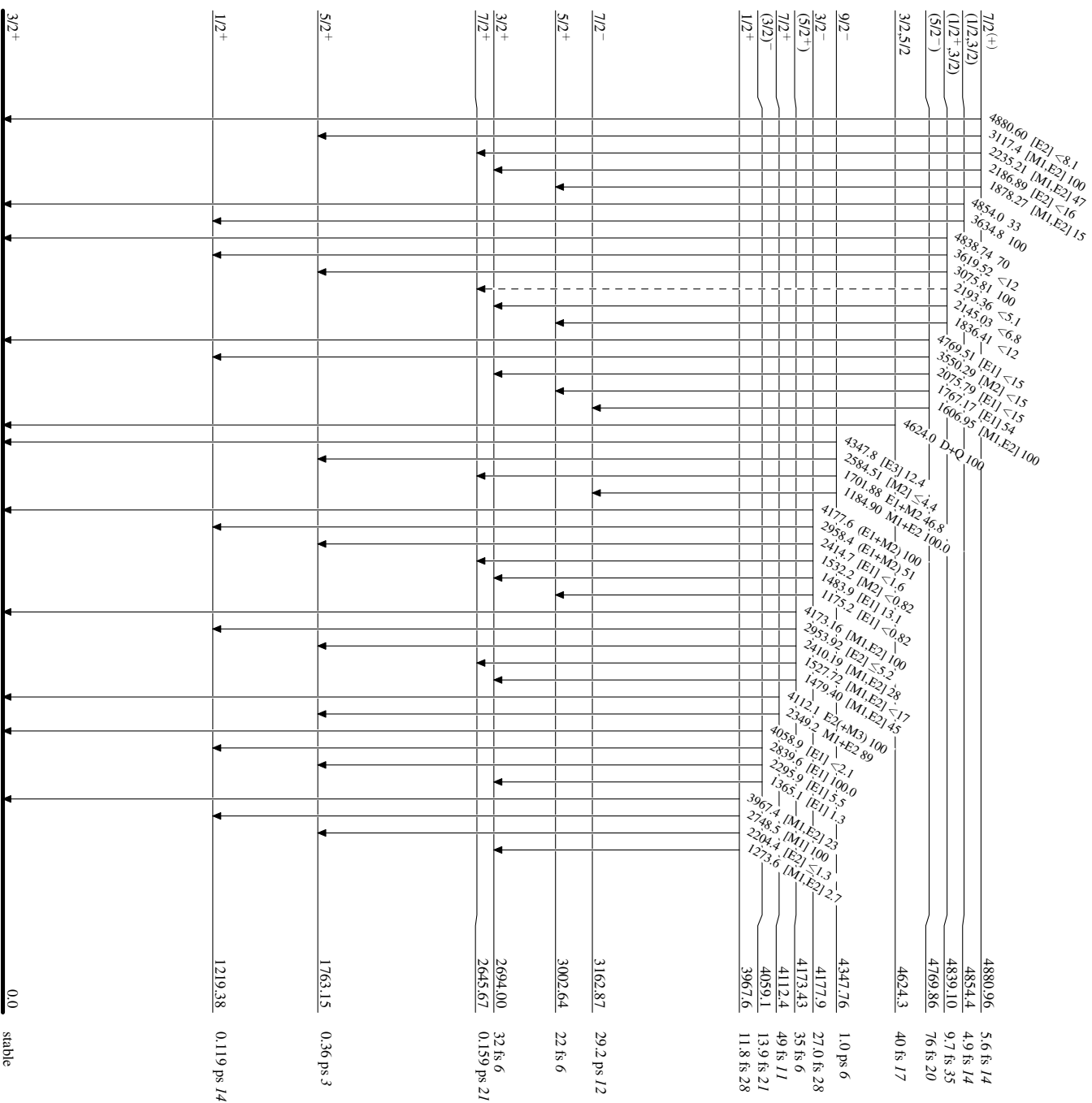
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

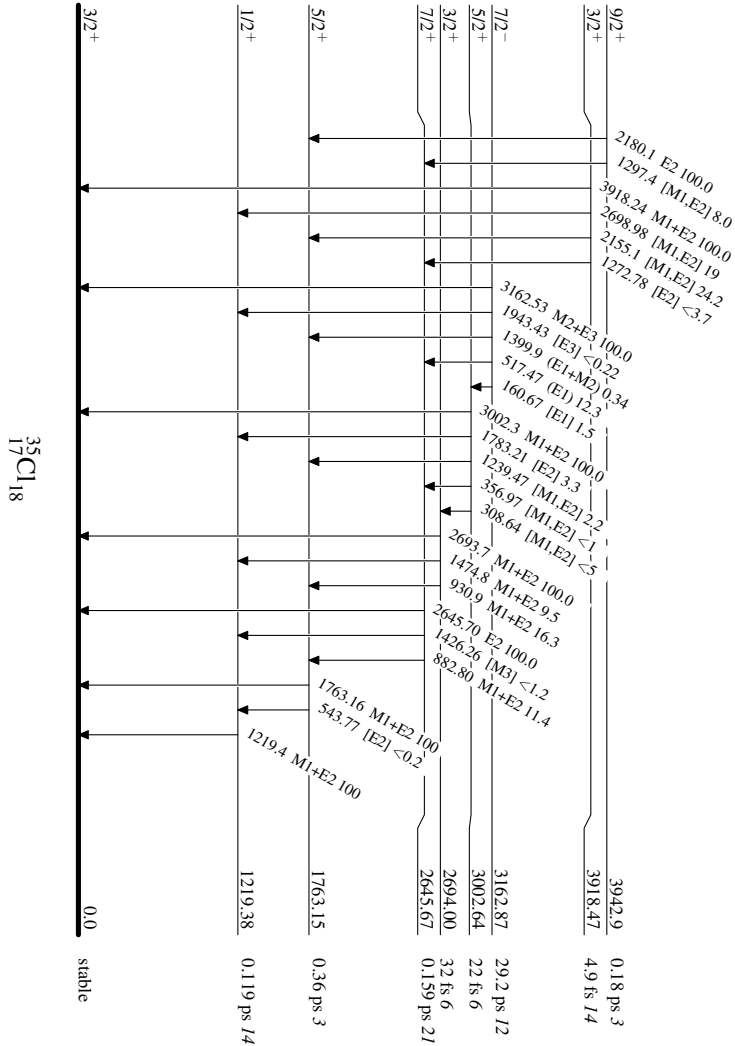
-----▶ γ Decay (Uncertain)



Adopted Levels, Gammas

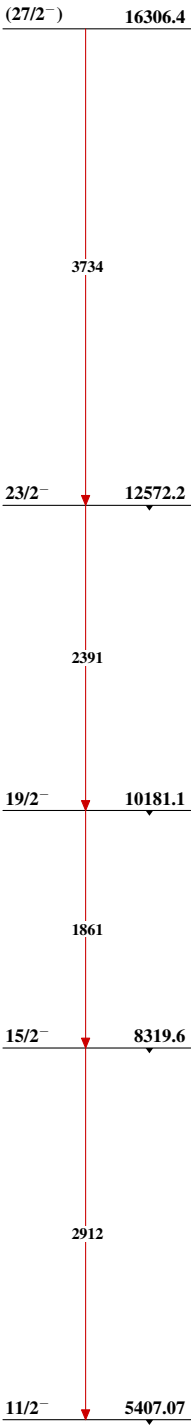
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Band(A): Band based on f_{7/2}
orbital



³⁵₁₇Cl₁₈

Adopted Levels, Gammas (continued)Band(B): Band based on $13/2^+$ $21/2^+$ 11459.1 $19/2^+$ 10858.5

2615

2070

2015

 $17/2^+$ 8844.4
 $(15/2^+)$ 8788.3

971

915

 $13/2^+$ 7873.03 $^{35}_{17}\text{Cl}_{18}$

^{35}S β^- decay (87.61 d)

Parent: ^{35}S : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=87.61$ d 24; $Q(\beta^-)=167.322$ 26; $\%\beta^-$ decay=100

^{35}S - $Q(\beta^-)$: From [2021Wa16](#).

^{35}S - $J^\pi, T_{1/2}$: From the Adopted Levels of ^{35}S .

[1983Ra04](#): $Q(\beta^-)=166.74$ keV 26.

[1985Ap01](#): $Q(\beta^-)=167.29$ keV 3.

[1985Oh06](#): $Q(\beta^-)=167.4$ keV 1.

[1985Al11](#): Upper limit for 17 keV neutrino: 0.4%.

[1985Oh06](#), [1992Hi06](#), [1993Ab11](#), [1993Be21](#), [1993Mo01](#): Deduced evidence against 17 keV neutrino.

[1994Bo33](#): Deduced spurious 17 keV neutrino signal origin.

[1995Bo43](#): $Q(\beta^-)=167.60$ keV 5 and upper limit for 17 keV neutrino is 1.8×10^{-3} .

[1995Mo17](#): Proposed 17 keV neutrino hypothesis exclusion.

Others: [2008Al39](#), [2000Ho13](#), [1999Oh09](#), [1996Je06](#), [1994Gr04](#), [1994Ho35](#), [1994Mu26](#), [1993Gi08](#), [1993Gr07](#), [1992Ch27](#), [1989Ba04](#), [1989Ta08](#), [1988Ch45](#), [1987Ge04](#), [1987Na22](#), [1985Ma59](#), [1999Pa18](#).

^{35}S decay leads only to the ground state of ^{35}Cl .

 ^{35}Cl Levels

<u>E(level)</u>	<u>J^π</u>	<u>Comments</u>
0	$3/2^+$	J^π : From the Adopted Levels.

 β^- radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>$I\beta^-^\dagger$</u>	<u>Log ft</u>
(167.3 14)	0	100	5.023 2

† Absolute intensity per 100 decays.

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01

Parent: ^{35}Ar : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=1.7755$ s 14; $Q(\varepsilon)=5966.2$ 7; $\% \varepsilon+\% \beta^+$ decay=100

^{35}Ar - J^π , $T_{1/2}$: From Adopted Levels of ^{35}Ar .

^{35}Ar - $Q(\varepsilon)$: From 2021Wa16.

1980Wi13: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on LiCl and PbCl_2 wrapped in 25- μm Ta foil. γ rays were detected using a 100- cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 20 γ -ray peaks. Deduced decay branching ratios for 9 levels, and log ft . Compared with shell-model calculations.

1984Ad01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick KCl target. γ rays were detected using a 28% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 8 γ -ray peaks. Deduced decay branching ratios for 6 levels, and log ft . Compared with shell-model calculations.

1985Da04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a NaCl target. γ rays were detected using a $\text{Ge}(\text{Li})$ detector with NaI anti-Compton shielding. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 3 levels.

1979Ha04: ^{35}Ar produced via the $^{51}\text{V}(p,p11n)$ spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38-g/ cm^2 vanadium target in the form of VC powder. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for the 1219 γ . Deduced the decay branching ratio for the 1219-keV level.

1977Az01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on CCl_4 and LiCl targets and 8-MeV protons from the University of Montreal impinging a thick LiCl pellet. Positrons were detected using an NE102 plastic scintillator coupled to an RCA 6342A photomultiplier tube. γ rays were detected using an 18% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 3 γ -ray peaks. Deduced decay branching ratios for 3 levels and parent ^{35}Ar $T_{1/2}$. Related: $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction threshold measurement in 1978Az01.

1971Ge04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on PbCl_2 targets. γ rays were detected using two coaxial $\text{Ge}(\text{Li})$ detectors. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 4 levels, decay branching ratios, log ft , and parent ^{35}Ar $T_{1/2}$.

1971De05: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction on KCl targets. γ rays were detected using a 25 cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ . Deduced 6 levels, decay branching ratios, and log ft .

1969Wi18,1969Fr09: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75-mg/ cm^2 KCl deposited on 0.13-mm Au backings and 160-mmHg gas CCl_4 targets. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and β end-point energy, log ft , and parent ^{35}Ar $T_{1/2}$.

1960Wa04: ^{35}Ar was produced at Crocker Laboratory and the Lawrence Radiation Laboratory. β spectra were measured with a uniform-field 180°-deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid CCl_4 target. β particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer. γ rays were detected using a $\text{NaI}(\text{Tl})$ crystal scintillation counter. Measured E_β and total I_β for two positron decays to ^{35}Cl excited states, E_γ and I_γ for two γ -ray peaks. Deduced two levels and decay branching ratios.

^{35}Ar β asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$3/2^+$	stable
1219.2 2	$1/2^+$	0.119 ps 14
1763.1 2	$5/2^+$	0.36 ps 3
2645.1 10	$7/2^+$	0.159 ps 21
2693.9 2	$3/2^+$	32 fs 6
3002.2 4	$5/2^+$	22 fs 6
3918.3 15	$3/2^+$	4.9 fs 14
3967.9 6	$1/2^+$	11.8 fs 28
4624.5? 5	$3/2, 5/2$	40 fs 17

† From a least-squares fit to γ -ray energies.

‡ From the Adopted Levels.

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01 (continued)

ε, β^+ radiations						Comments
E(decay)	E(level)	$I\beta^+$ ‡	$I\varepsilon$ ‡	Log ft	$I(\varepsilon+\beta^+)^{\dagger\ddagger}$	
(1341.7 [#] 13)	4624.5?	$<8.0\times 10^{-4}$	<0.0035	>3.8	<0.0043	$I(\varepsilon+\beta^+)$: deduced from $100-\Sigma I(\varepsilon+\beta^+)$ (excited states), where $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 5. Level scheme fed by β^+ decay may be complete considering the highest observed level is at 4.6 MeV. Observed $\Sigma I(\varepsilon)\approx 0.08$. Evaluators assume $\Sigma I(\varepsilon)\leq 0.08$ for all unobserved ε feedings to high-lying levels and include this limit in the uncertainty for $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 10 and $I(\varepsilon+\beta^+)$ (gs)=98.24 10.
(1998.3 14)	3967.9	0.015 10	0.0016 10	$4.5+5-2$	0.017 11	
(2047.9 19)	3918.3	0.0096 30	8.3×10^{-4} 26	$4.8+2-1$	0.0104 32	
(2964.0 13)	3002.2	0.089 14	0.0011 2	5.0 1	0.090 14	
(3272.3 13)	2693.9	0.164 9	0.00126 7	5.00 3	0.165 9	
(3321.1 16)	2645.1	<0.00146	$<2.82\times 10^{-5}$	>8.3	<0.00149	
(4203.1 13)	1763.1	0.256 13	7.0×10^{-4} 4	5.47 2	0.257 13	
(4747.0 13)	1219.2	1.22 4	0.00208 7	5.10 2	1.22 4	
(5966.2 16)	0	98.17 10	0.07309 8	3.759 1	98.24 10	

† Deduced from γ -ray intensity balance, relative to $I_\gamma(1219\gamma)=100$, normalized to absolute $\%I_\gamma(1219\gamma)=1.26$ 5.

‡ Absolute intensity per 100 decays.

Existence of this branch is questionable.



³⁵Ar ε+β⁺ decay (1.7755 s) [1980Wi13,1984Ad01](#) (continued)

γ(³⁵Cl)

I_γ normalization: absolute %I_γ(1219γ)=1.243 30 deduced from weighted average of 1.223 46 ([1969Wi18,1969Fr09](#)), 1.22 20 ([1971Ge04](#)), 1.55 15 ([1977Az01](#)), 1.34 8 ([1979Ha04](#)), 1.23 7 ([1984Ad01](#)), 1.228 30 ([1980Wi13](#)).

E _γ [†]	I _γ ^{‡&}	E _i (level)	J _i ^π	E _f	J _f ^π	Mult. [#]	δ [#]	α ^a	Comments
916 ^b		3918.3	3/2 ⁺	3002.2	5/2 ⁺				The intensity could not be determined due to being obscured by another peak (1980Wi13).
930.7 2	1.57 20	2693.9	3/2 ⁺	1763.1	5/2 ⁺	M1+E2	+0.09 3	4.69×10 ⁻⁵ 7	%I _γ =0.0195 25 I _γ : weighted average of 1.13 26 (1980Wi13) and 1.66 12 (1984Ad01).
965 ^b	<0.87 [@]	3967.9	1/2 ⁺	3002.2	5/2 ⁺				%I _γ <0.0111
1219.3 2	100.0 15	1219.2	1/2 ⁺	0	3/2 ⁺	M1+E2	0.093 10	3.66×10 ⁻⁵ 5	%I _γ =1.24 4 I _γ : others: 100.0 30 from 1984Ad01 , 100 10 from 1977Az01 , 100 from 1969Wi18 , 1971Ge04 , 1979Ha04 , and 1985Da04 .
1225 ^b		3918.3	3/2 ⁺	2693.9	3/2 ⁺				The intensity could not be determined due to being obscured by another peak (1980Wi13).
1239 ^b	<0.30 [@]	3002.2	5/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		4.4×10 ⁻⁵ 6	%I _γ <0.00382
1274 ^b	<0.42 [@]	3967.9	1/2 ⁺	2693.9	3/2 ⁺	[M1,E2]		4.9×10 ⁻⁵ 7	%I _γ <0.00535
1474.8 3	1.02 11	2693.9	3/2 ⁺	1219.2	1/2 ⁺	M1+E2	-0.63 21	8.4×10 ⁻⁵ 4	%I _γ =0.0127 14 I _γ : weighted average of 0.83 13 (1980Wi13) and 1.08 7 (1984Ad01).
1763.0 2	22.7 4	1763.1	5/2 ⁺	0	3/2 ⁺	M1+E2	-2.85 10	0.0002124 30	%I _γ =0.282 9 I _γ : weighted average of 23.4 13 (1969Wi18), 20.0 27 (1971Ge04), 22 6 (1977Az01), 23.08 37 (1980Wi13), 21.85 67 (1984Ad01), and 21.7 9 (1985Da04).
1783 ^b	<0.39 [@]	3002.2	5/2 ⁺	1219.2	1/2 ⁺	[E2]		0.0002267 32	%I _γ <0.00497
1931 ^b	<0.21 [@]	4624.5?	3/2,5/2	2693.9	3/2 ⁺				%I _γ <0.00267
2155.1 15	0.15 6	3918.3	3/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		0.00036 4	%I _γ =0.0019 8 I _γ : from 1984Ad01 . Other: <0.22 (1980Wi13).
2204 ^b	<0.27 [@]	3967.9	1/2 ⁺	1763.1	5/2 ⁺	[E2]		0.000425 6	%I _γ <0.00344
2645 ^b	<0.12 [@]	2645.1	7/2 ⁺	0	3/2 ⁺	E2		0.000633 9	%I _γ <0.00153
2693.7 2	10.96 18	2693.9	3/2 ⁺	0	3/2 ⁺	M1+E2	+0.26 2	0.000553 8	%I _γ =0.136 4 I _γ : weighted average of 10.2 15 (1971Ge04), 10.96 18 (1980Wi13) and 11.02 34 (1984Ad01).
2699 ^b	<0.33 [@]	3918.3	3/2 ⁺	1219.2	1/2 ⁺	[M1,E2]		0.00060 6	%I _γ <0.00420
2748.5 6	0.477 34	3967.9	1/2 ⁺	1219.2	1/2 ⁺	[M1]		0.000569 8	%I _γ =0.0059 5 I _γ : weighted average of 0.48 6 (1980Wi13) and 0.476 34 (1984Ad01).

³⁵Ar $\varepsilon+\beta^+$ decay (1.7755 s) [1980Wi13](#),[1984Ad01](#) (continued)

$\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [‡] &	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	α ^a	Comments
2861 ^b	<0.099 [@]	4624.5?	3/2,5/2	1763.1	5/2 ⁺				%I _γ <0.00126
3002.1 ⁴	7.29 ¹⁴	3002.2	5/2 ⁺	0	3/2 ⁺	M1+E2	+0.08 ²	0.000671 ⁹	%I _γ =0.0906 ²⁸ I _γ : weighted average of 9.0 ¹⁵ (1971Ge04), 7.24 ¹⁴ (1980Wi13), 7.45 ²³ (1984Ad01) and 6.9 ⁵ (1985Da04).
3405 ^b		4624.5?	3/2,5/2	1219.2	1/2 ⁺				I _γ : could not be determined due to being obscured by another peak (1980Wi13).
3918	0.52 ³	3918.3	3/2 ⁺	0	3/2 ⁺	M1+E2		0.00108 ⁷	%I _γ =0.0065 ⁴
3968	0.106 ²⁵	3967.9	1/2 ⁺	0	3/2 ⁺	[M1,E2]		0.00109 ⁷	%I _γ =0.00132 ³¹
4624 ^b	<0.035 [@]	4624.5?	3/2,5/2	0	3/2 ⁺	D+Q			%I _γ <4.46×10 ⁻⁴

[†] From [1984Ad01](#), unless otherwise noted.

[‡] from [1980Wi13](#), unless otherwise noted.

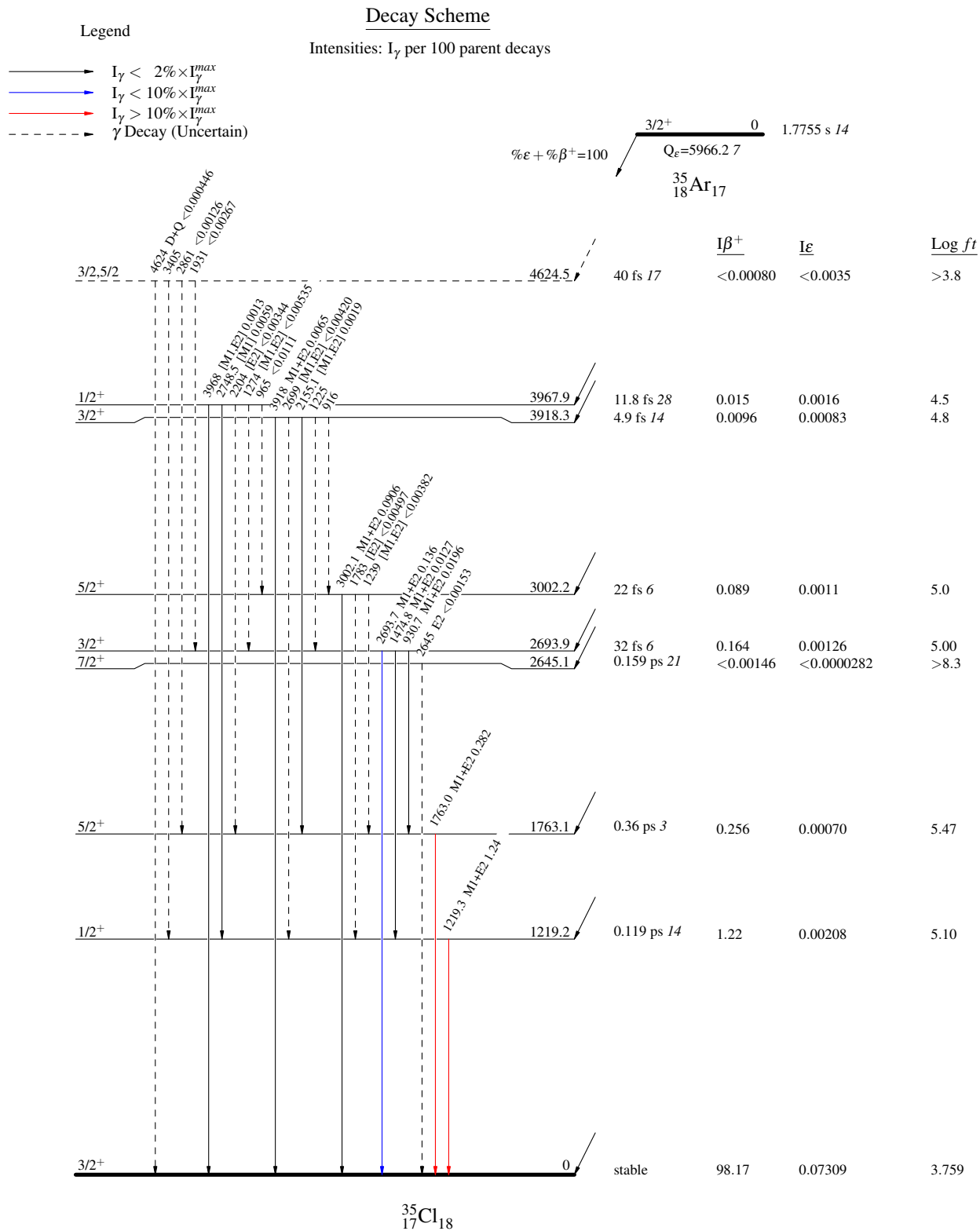
[#] From Adopted Gammas.

[@] [1980Wi13](#) gives upper limits of I_γ at 98% confidence level. These peaks are not observed in the spectra of [1980Wi13](#) or in [1984Ad01](#).

& For absolute intensity per 100 decays, multiply by 0.01243 ³⁰.

^a Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^b Placement of transition in the level scheme is uncertain.

^{35}Ar $\varepsilon + \beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01

 **^{36}K εp decay (341 ms) 1996II02,1980Es01**

Parent: ^{36}K : $E=0$; $J^\pi=2^+$; $T_{1/2}=341$ ms 3; $Q(\varepsilon\text{p})=4307.4$ 3; $\% \varepsilon\text{p}$ decay= 5.1×10^{-2}

^{36}K - J^π , $T_{1/2}$: From the Adopted Levels of ^{36}K (2012Ni01).

^{36}K - $Q(\varepsilon\text{p})$: From 2021Wa16.

^{36}K - $\% \varepsilon\text{p}$ decay: $\% \varepsilon+\beta^+\text{p}=5.1 \times 10^{-4}$ is scaled by evaluators from $\% \varepsilon+\beta^+\text{p}=4.9 \times 10^{-4}$ reported in 1996II02. $\% \varepsilon+\beta^+\text{p}=4.9 \times 10^{-4}$ in 1996II02 was normalized based on the strongest 970-keV $\varepsilon+\beta^+$ -delayed proton $I_p=3.3 \times 10^{-4}$ 9 measured by 1980Es01. 1980Es01 reported $\% \varepsilon+\beta^+\text{p}=4.8 \times 10^{-4}$ and their I_p was normalized based on the strongest 1970-keV $\varepsilon+\beta^+$ -delayed γ ray $\% I_\gamma=79$ 8 measured by 1972Mi13. Evaluators adopt a more precise $\% I_\gamma(1970)=81.8$ 16 from the evaluated ^{36}K ε decay dataset (2012Ni01,1976Fr03) as the new normalization reference, and therefore, all original I_p values from 1996II02 and 1980Es01 are scaled by 1.04 11.

1996II02,1997II03: a 500-MeV proton beam was produced from the TRIUMF cyclotron and bombarded a CaO powder target. The spallation reaction products diffused out of the target and were selectively ionized at a Re surface. A=36 singly charged ions were selected by a magnetic mass analyzer and implanted into a carbon foil at 7.0×10^5 pps. Protons and α particles were detected using a 30- μm silicon surface barrier detector with FWHM=23 keV. ^{32}S and ^{35}Cl recoils were detected using an MCP in a back-to-back geometry for timing stop signals. Measured $E_p(>0.5$ MeV) and I_p for 15 proton peaks, $E_\alpha(>1.5$ MeV) and I_α for 19 α peaks, p(recoil)-coin, and α (recoil)-coin. Deduced levels, J , π , decay branching ratios, $\log ft$, partial decay widths for ^{36}Ar , and ratios of $^{35}\text{Cl}(p,\alpha)^{32}\text{S}$ to $^{35}\text{Cl}(p,\gamma)^{36}\text{Ar}$ reaction rates.

1980Es01: a 20-MeV proton beam was produced from the University of Jyväskylä MC-20 cyclotron and bombarded a stack of four ^{36}Ar -implanted Al foils. The $^{36}\text{Ar}(p,n)$ reaction products were thermalized by helium, transported with NaCl-loaded helium, and deposited onto a fast-transport Al-coated mylar tape in a measurement chamber. Protons and α particles were detected using 14 and 26- μm fully depleted (FWHM=40 keV) and 100- μm partially depleted Si(Au) surface barrier detectors. A polyethylene absorber foil was placed in front of Si(Au) to cause an energy shift for distinguishing p/ α . Measured $E_p(>0.5$ MeV) and I_p for 11 proton peaks, $E_\alpha(>2.0$ MeV) and I_α for 12 α peaks. Deduced levels, J , π , decay branching ratios, $\log ft$, and partial decay widths for ^{36}Ar .

1980Ew01: a 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC_2 target. The $^{45}\text{Sc}(p,7n3p)$ spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a carbon foil for proton measurements. Protons and α particles were detected using 20- μm and 100- μm silicon surface barrier detectors with FWHM=35 and 23 keV. Measured $E_p(>0.5$ MeV) and I_p for 3 proton peaks, E_α and I_α for 4 α peaks. Deduced levels, J , π , and decay branching ratios for ^{36}Ar .

 ^{35}Cl Levels

<u>E(level)</u>	<u>$J^\pi^\dagger$</u>
0	$3/2^+$

† From the Adopted Levels.

Delayed Protons (^{35}Cl)

<u>E(p)†</u>	<u>E(^{35}Cl)</u>	<u>I(p)†@</u>	<u>E(^{36}Ar)$^\#$</u>	<u>Comments</u>
501.8 11	0	1.14×10^{-6} 33	9023	E(p): other: 501 10 (1980Es01). I(p): other: 3.6×10^{-6} 13 (1980Es01).
622.7 14	0	1.6×10^{-6} 5	9147	
693.5 8	0	7.9×10^{-5} 22	9220	E(p): other: 693 5 (1980Es01), 691 8 (1980Ew01). I(p): other: 7.8×10^{-5} 24 (1980Es01).
851.7 10	0	2.3×10^{-5} 7	9383	E(p): other: 849 5 (1980Es01). I(p): other: 2.0×10^{-5} 7 (1980Es01).
969.6 12	0	3.4×10^{-4} 10	9504	E(p): other: 970 5 (1980Es01), 959 7 (1980Ew01). I(p): from 1980Es01 and also adopted as the normalization reference by 1996II02.
1168.5 14	0	6.2×10^{-7} 7	9709	

Continued on next page (footnotes at end of table)

^{36}K ε p decay (341 ms) 1996II02,1980Es01 (continued)Delayed Protons (continued)

$E(p)^{\dagger}$	$E(^{35}\text{Cl})$	$I(p)^{\dagger@}$	$E(^{36}\text{Ar})^{\#}$	Comments
1198.5 11	0	2.5×10^{-6} 7	9740	
1272.0 20	0	1.6×10^{-6} 5	9815	
1334.2 7	0	4.5×10^{-5} 12	9879	E(p): other: 1333 5 (1980Es01), 1325 7 (1980Ew01). I(p): other: 3.6×10^{-5} 11 (1980Es01).
1409.3 22	0	1.04×10^{-6} 33	9957	
1874 ‡ 10	0	1.1×10^{-6} 5	10435	
1992 ‡ 10	0	5.0×10^{-6} 20	10556	
2048 ‡ 10	0	4.9×10^{-6} 19	10613	
2458 ‡ 10	0	3.0×10^{-6} 13	11035	
2640 ‡ 10	0	2.1×10^{-6} 10	11222	

† From 1996II02, unless otherwise noted. E(p) is in lab frame.

‡ From 1980Es01 and also adopted by 1996II02.

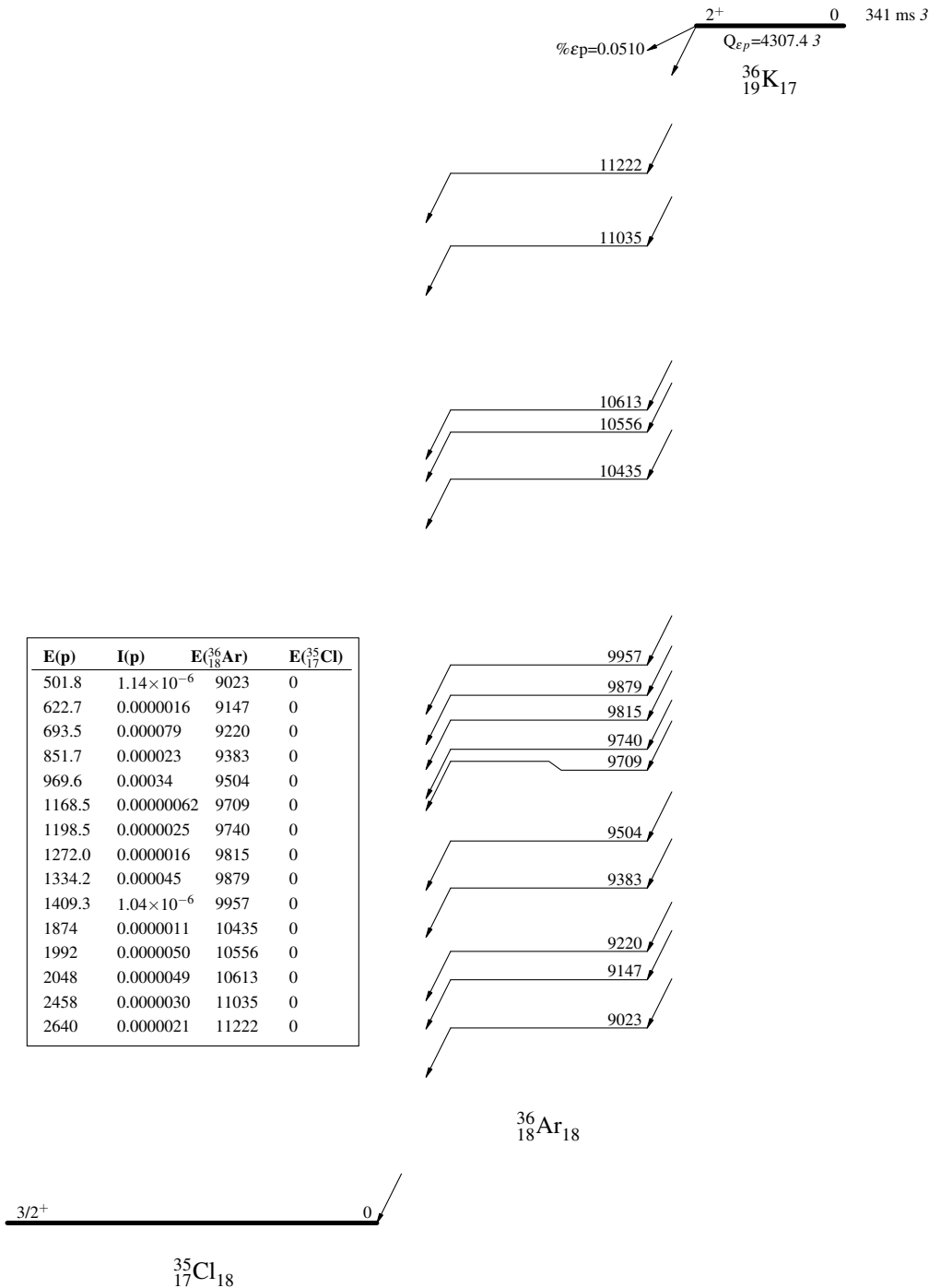
$^{\#}$ $E(\text{level})(^{36}\text{Ar}) = E(p)_{\text{lab}} \times [m(p) + m(^{35}\text{Cl})] / m(^{35}\text{Cl}) + S(p)(^{36}\text{Ar}) + E(\text{level})(^{35}\text{Cl})$, where $S(p)(^{36}\text{Ar}) = 8506.97$ 4 (2021Wa16). Proton decay is assumed to populate only the ground state of the daughter nucleus ^{35}Cl based on the arguments proposed in 1996II02: (A) unreasonably small $\log ft$ values if populating excited states; (B) no pairs of particle groups characteristic of the g.s./first excited state energy difference; (C) compatible level matching with resonance reaction studies.

$^@$ Absolute intensity per 100 decays.

³⁶K ϵ p decay (341 ms) 1996IN02,1980Es01

Decay Scheme

I(p) Intensities: I(p) per 100 parent decays



$^1\text{H}(^{34}\text{S}, ^{35}\text{Cl}\gamma)$ **2021Lo09**

$^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$ in inverse kinematics.

2021Lo09: ^{34}S was produced by the off-line ion source (OLIS) and accelerated by the ISAC accelerator at TRIUMF. 10^{10} – 10^{11} pps $^{34}\text{S}^{+7}$ ions were delivered to a windowless 2-10 Torr hydrogen gas target with an effective length of 12.3 cm. Elastically scattered protons were detected by two silicon surface barrier detectors inside the gas target. Prompt γ rays from the compound recoil nuclei were detected by a BGO array in coincidence with the heavy-ion recoils separated by the DRAGON separator and identified by two MCP detectors, a DSSD, and an ionization chamber at the focal plane. Deduced resonance strengths ($\omega\gamma$) for 8 resonances and the $^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$ astrophysical reaction rate.

 ^{35}Cl Levels

Values of resonance strength $\omega\gamma$ under comments are from direct measurements in **2021Lo09**.

<u>E(level)[†]</u>	<u>Comments</u>
6643	$E_{\text{R}}=272$ keV. $\omega\gamma=1.22\times 10^{-5}$ eV 36.
6674	$E_{\text{R}}=303$ keV. $\omega\gamma \leq 7.2\times 10^{-6}$ eV.
6761	$E_{\text{R}}=390$ keV. $\omega\gamma \leq 6.9\times 10^{-6}$ eV.
6778	$E_{\text{R}}=407$ keV. $\omega\gamma=8.0\times 10^{-4}$ eV 19.
6802	$E_{\text{R}}=431$ keV. $\omega\gamma \leq 3.9\times 10^{-5}$ eV.
6823	$E_{\text{R}}=452$ keV. $\omega\gamma \leq 3.1\times 10^{-5}$ eV.
6842	$E_{\text{R}}=471$ keV. $\omega\gamma=9.4\times 10^{-5}$ eV 29.
6866	$E_{\text{R}}=496$ keV. $\omega\gamma=1.37\times 10^{-2}$ eV 22.

[†] Excitation energies are deduced by evaluators from resonance energies $E_{\text{R}}+S_{\text{p}}(^{35}\text{Cl})$, where $S_{\text{p}}(^{35}\text{Cl})=6370.81$ 4 (**2021Wa16**).

$^2\text{H}(^{34}\text{S}, \text{n}\gamma)$ 1973Wa10

$^{34}\text{S}(\text{d}, \text{n})^{35}\text{Cl}$ in inverse kinematics.

1973Wa10: A 59.6-MeV ^{34}S beam was produced from the BNL MP-tandem Van de Graaff facility. Targets were 200- $\mu\text{g}/\text{cm}^2$ TiD prepared by evaporating titanium onto the Cu, Al, and Mg target backings in a deuterium atmosphere. γ rays were detected using a 35-cm³ Ge(Li) detector at 0° with FWHM=2 keV at 656 keV. Measured E_γ . Deduced $T_{1/2}$ for two ^{35}Cl levels using Doppler Shift Attenuation lineshape analysis.

 ^{35}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$T_{1/2}^\#$</u>	Comments
0	$3/2^+$		
1219	$1/2^+$	201 fs 28	$T_{1/2}$: Lifetime=0.29 ps 4 from 1973Wa10.
1762	$5/2^+$	374 fs 49	$T_{1/2}$: Lifetime=0.54 ps 7 from 1973Wa10.

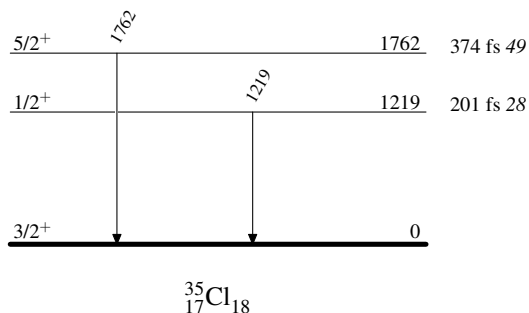
† From 1973Wa10.

‡ From the Adopted Levels.

$^\#$ From Doppler Shift Attenuation Method (DSAM) in 1973Wa10.

 $\gamma(^{35}\text{Cl})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1219	1219	$1/2^+$	0	$3/2^+$
1762	1762	$5/2^+$	0	$3/2^+$

 $^2\text{H}(^{34}\text{S}, \text{n}\gamma)$ 1973Wa10Level Scheme

$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ **2007Ks01, 2013Bi10, 2020Ra14**

2007Ks01: two different experiments: one with $E=70$ MeV ^{28}Si beam at BARC-TIFR facility and the second with $E=88$ MeV ^{28}Si beam at NSC facility, both using a target of about $50 \mu\text{g}/\text{cm}^2$ ^{12}C on a $\approx 10.5\text{-mg}/\text{cm}^2$ gold backing. γ rays were detected with almost identical (INGA) arrays of eight Compton-suppressed Clover HPGe detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)(\text{DCO})$, $\gamma\gamma(\text{lin pol})$, Doppler-shift attenuations. Deduced levels, J , π , $T_{1/2}$, γ -ray multipolarities and mixing ratios. Comparisons with available data and large-scale shell-model calculations.

2013Bi10: $E=110$ MeV ^{28}Si beam was produced from the IUAC-New Delhi accelerator facility. Target was $50 \mu\text{g}/\text{cm}^2$ thick ^{12}C on $18 \text{mg}/\text{cm}^2$ Au backing. γ rays were detected with the INGA array of thirteen Compton-suppressed Clover detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)(\text{DCO})$, Doppler-shift attenuations. Deduced levels, J , π , $T_{1/2}$, evidence for an SD band, γ -ray multipolarities. Comparison with large-scale shell-model calculations, two-level mixing calculation, and with structure of SD states in ^{36}Ar .

2020Ra14: $E=110$ MeV ^{28}Si beam was produced from the 15UD Pelletron accelerator at the Inter-University Accelerator Centre (IUAC), New Delhi. Target was $50 \mu\text{g}/\text{cm}^2$ ^{12}C evaporated onto $18 \text{mg}/\text{cm}^2$ Au backing. γ rays were detected with a multi-detector array (Indian National Gamma Array) consisting of 13 Compton-suppressed clovers. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\text{DCO})$, $\gamma\gamma(\text{ADO})$, $\gamma\gamma(\text{IPDCO})$. Deduced levels, J , π , γ -ray multipolarities. Comparisons with theoretical calculations.

Other reactions:

1982Le04: $^{28}\text{Si}(^{12}\text{C}, X\gamma) ^{12}\text{C}$ ($E_{\text{cm}}=23\text{-}35$ MeV) beam produced from the University of Washington FN Tandem Van de Graaff.

A target of $100 \mu\text{g}/\text{cm}^2$ ^{28}Si on a thick tantalum backing. A Ge(Li) detector. Measured $\sigma(E)$, $E\gamma$.

1985Kr21: $^{27}\text{Al}(^{12}\text{C}, X) E=85$ MeV/nucleon. Measured particle-particle angular correlations.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J\pi^\ddagger$	$T_{1/2}^\#$	Comments
0.0	$3/2^+$		
1763.0 6	$5/2^+$		
2645.8 6	$7/2^+$	<0.90 ps	
3003.1 8	$5/2^+$		
3163.1 6	$7/2^-$		
3942.0 7	$9/2^+$	<0.35 ps	
4347.8 6	$9/2^-$	0.91 ps $+21-16$	
5407.4 8	$11/2^-$	<0.76 ps	
5927.1 7	$(11/2)^-$	<0.28 ps	
6087.6 8	$13/2^-$		
6119.5? 13	$(11/2^-)$		
6660.2 8	$(11/2^-)$		
7873.3 8	$13/2^+$		
8319.4 ^a 9	$15/2^-$	12.5& fs 7	Tentative 8324, $(13/2^-)$ level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <70 fs (2007Ks01). $Q(\text{transition})=0.65$ 2, $\beta_2=0.30$ (2013Bi10).
8487.8 10	$15/2^-$	<0.07 ps	Tentative 8481, $(15/2^-)$ level in 2007Ks01 is considered the same level here by the evaluators.
8558.2? 13	$(13/2^-)$	<0.14 ps	
8787.3? 13	$(15/2^+)$	<0.28 ps	
8845.1 [@] 9	$17/2^+$		
10181.1 ^a 10	$19/2^-$	42.3& fs 28	Tentative 10181, $(19/2^+)$ level in 2007Ks01 is considered the same level here by the evaluators. $T_{1/2}$: other: <0.14 ps (2007Ks01). $Q(\text{transition})=0.82$ 6, $\beta_2=0.37$ (2013Bi10).
10223.2? 14	$(17/2^-)$	<0.28 ps	
10862.2? 14	$19/2^+$	<0.28 ps	
11459.2 [@] 12	$21/2^+$	43.0& fs 21	$Q(\text{transition})=0.56$ 2, $\beta_2=0.27$ (2013Bi10).
12572.2 ^a 12	$23/2^-$	33& fs 4	$Q(\text{transition})=0.71$ 5, $\beta_2=0.32$ (2013Bi10).
16306.4 ^a 16	$(27/2^-)$	$<9.0\text{&}$ fs	$Q(\text{transition})>0.5$, $\beta_2=0.24$ (2013Bi10).

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ **2007Ks01, 2013Bi10, 2020Ra14 (continued)** ^{35}Cl Levels (continued)[†] From a least-squares fit to γ -ray energies, assuming $\Delta E_{\gamma}=1$ keV in the fitting procedure.[‡] From the Adopted Levels, unless otherwise noted. Assignments for tentative levels not adopted in Adopted Levels are proposed by **2007Ks01**.# From DSAM in **2007Ks01**, unless otherwise noted.@ Existence of $17/2^{+}$ and $21/2^{+}$ levels provides indirect evidence in favor of SD band interpretation and parity-doublet cluster structure (**2013Bi10**).& From DSAM in **2013Bi10**.^a Band(A): SD band. Proposed configuration= $3p-3h$, $(sd)^{16}(pf)^3$. Negative-parity partner of SD band in ^{36}Ar (**2013Bi10**). $\gamma(^{35}\text{Cl})$ Δ_{IPDCO} corresponds to integrated polarization asymmetry measurement at $E=70$ MeV at TIFR facility (**2007Ks01**). The values are from e-mail reply of Dec 1, 2006 from one of the authors (M. Saha Sarkar) of **2007Ks01** to the previous evaluator B. Singh. Δ_{IPDCO} and DCO under comments are from **2007Ks01**, unless otherwise noted. Positive value of Δ_{IPDCO} indicates electric character and negative value indicates magnetic character. Expected value of DCO is ≈ 1.0 for a stretched transition gated by a transition of same multipolarity, ≈ 2.0 for dipole when gated by a stretched quadrupole (DCO(Q)) and ≈ 0.5 for stretched quadrupole when gated by a stretched dipole (DCO(D)) (**2007Ks01**). Values deviating from above values indicate a mixed transition. DCO values are also available in **2020Ra14**, but expected values are opposite to those in **2007Ks01** for stretched dipole and quadrupole transitions when gated by a transition of different multipolarity, that is DCO(Q) ≈ 0.5 for stretched dipole and DCO(D) ≈ 2.0 for stretched quadrupole (**2020Ra14**).For ADO ratios under comments from **2020Ra14**, expected values are 0.6 for pure dipole transitions and 1.6 for pure quadrupole transitions or $\Delta J=0$ transitions (**2020Ra14**).

$E_{\gamma}^{\dagger}$	$I_{\gamma}^{\ddagger}$	$E_i(\text{level})$	J_i^{π}	E_f	J_f^{π}	Mult. [#]	$\delta^{\#}$	Comments
160 [@]	1.1 3	3163.1	$7/2^{-}$	3003.1	$5/2^{+}$	D		I _{γ} : other: 0.6 2 at $E=70$ MeV. I _{γ} : other: 10.0 2 at $E=70$ MeV. DCO(Q)=0.8 1. DCO(D)=1.5 2, R _{ADO} =1.6 4 (2020Ra14).
518	9.2 5	3163.1	$7/2^{-}$	2645.8	$7/2^{+}$			
526		8845.1	$17/2^{+}$	8319.4	$15/2^{-}$	M1(+E2)	+0.1 1	I _{γ} : other: 8.6 2 at $E=70$ MeV. DCO(Q)=1.3 1 at $E=70$ MeV. $\Delta_{\text{IPDCO}}=-0.09$ 2.
680	32 1	6087.6	$13/2^{-}$	5407.4	$11/2^{-}$			
712 ^{@&}	<1	6119.5?	$(11/2^{-})$	5407.4	$11/2^{-}$	D(+Q) E2	-0.2 2	I _{γ} =4.6 2 at $E=70$ MeV. $\Delta_{\text{IPDCO}}=+0.01$ 5. DCO(Q)=2.2 6 at $E=88$ MeV. I _{γ} : other: 2.0 1 at $E=70$ MeV. DCO(Q)=1.0 2 at $E=70$ MeV, 0.9 1 at $E=88$ MeV. $\Delta_{\text{IPDCO}}=+0.08$ 4. DCO(D)=1.8 2, DCO(Q)=0.9 1, R _{ADO} =1.6 2 (2020Ra14).
779 ^{@&}	<1	3942.0	$9/2^{+}$	3163.1	$7/2^{-}$			
883	3.2 2	2645.8	$7/2^{+}$	1763.0	$5/2^{+}$			
914 ^{&}	2.4 5	8787.3?	$(15/2^{+})$	7873.3	$13/2^{+}$	E2		I _{γ} : other: 2.0 1 at $E=70$ MeV. DCO(Q)=1.0 2 at $E=70$ MeV, 0.9 1 at $E=88$ MeV. $\Delta_{\text{IPDCO}}=+0.08$ 4. DCO(D)=1.8 2, DCO(Q)=0.9 1, R _{ADO} =1.6 2 (2020Ra14).
972	33 1	8845.1	$17/2^{+}$	7873.3	$13/2^{+}$			
1060	7.3 7	5407.4	$11/2^{-}$	4347.8	$9/2^{-}$	M1+E2	+0.2 1	I _{γ} : other: 6.9 3 at $E=70$ MeV. DCO(D)=0.9 1. $\Delta_{\text{IPDCO}}=-0.10$ 4.
1113		12572.2	$23/2^{-}$	11459.2	$21/2^{+}$	M1+E2	-0.3 1	I _{γ} : other: 25 1 at $E=70$ MeV. DCO(Q)=2.0 2 at $E=70$ MeV, 2.8 3 at $E=88$ MeV. $\Delta_{\text{IPDCO}}=-0.08$ 3.
1185	25 1	4347.8	$9/2^{-}$	3163.1	$7/2^{-}$			

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$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14** (continued) $\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
1213		7873.3	13/2 ⁺	6660.2	(11/2 ⁻)			
1296	0.4 2	3942.0	9/2 ⁺	2645.8	7/2 ⁺			I_γ : other: 0.2 1 at E=70 MeV.
1336	6 1	10181.1	19/2 ⁻	8845.1	17/2 ⁺	D(+Q)	-0.2 2	DCO(Q)=2.0 4 at E=88 MeV.
1378 @&	2 1	10223.2?	(17/2 ⁻)	8845.1	17/2 ⁺			
1400	<1	3163.1	7/2 ⁻	1763.0	5/2 ⁺			
1579	18 2	5927.1	(11/2 ⁻)	4347.8	9/2 ⁻	(M1+E2)	-0.2 1	I_γ : other: 12 1 at E=70 MeV. DCO(D)=0.9 1 at E=70 MeV, 1.0 1 at E=88 MeV. $\Delta\text{IPDCO}=+0.02$ 3.
1693		10181.1	19/2 ⁻	8487.8	15/2 ⁻			
1702	11.4 6	4347.8	9/2 ⁻	2645.8	7/2 ⁺	E1+M2	+0.5 +5-3	I_γ : other: 10.8 5 at E=70 MeV. DCO(Q)=1.0 2 from 2007Ks01 indicating $\Delta J=2$ or 0 seems inconsistent with $\Delta J=1$ from level scheme. $\Delta\text{IPDCO}=+0.04$ 4. DCO(Q)=0.49 6, $R_{\text{ADO}}=0.60$ 6 (2020Ra14).
1740	2.0 2	6087.6	13/2 ⁻	4347.8	9/2 ⁻	E2		I_γ : other: 2.3 2 at E=70 MeV. $\Delta\text{IPDCO}=+0.12$ 7.
1763	>9.7	1763.0	5/2 ⁺	0.0	3/2 ⁺	E2+M1		I_γ : other: >15.7 at E=70 MeV. $\Delta\text{IPDCO}=+0.13$ 5.
1786	1.8 4	7873.3	13/2 ⁺	6087.6	13/2 ⁻	D(+Q)	-0.6 6	I_γ : other: 0.1 1 at E=70 MeV. DCO(Q)=1.2 3 at E=88 MeV.
1862		10181.1	19/2 ⁻	8319.4	15/2 ⁻			
1946	10 1	7873.3	13/2 ⁺	5927.1	(11/2 ⁻)	E1+M2	+0.2 1	I_γ : other: 0.5 1 at E=70 MeV. DCO(Q)=1.3 2 at E=88 MeV. $\Delta\text{IPDCO}=+0.11$ 6. DCO(Q)=0.45 8, $R_{\text{ADO}}=0.59$ 8 (2020Ra14).
1985 @	2.0 4	5927.1	(11/2 ⁻)	3942.0	9/2 ⁺			I_γ : other: 1.6 6 at E=70 MeV.
2017 @&	7 1	10862.2?	19/2 ⁺	8845.1	17/2 ⁺			
2179	6.5 7	3942.0	9/2 ⁺	1763.0	5/2 ⁺	E2		I_γ : other: 11.1 6 at E=70 MeV. $\Delta\text{IPDCO}=+0.07$ 5.
2232		8319.4	15/2 ⁻	6087.6	13/2 ⁻			E_γ : from 2013Bi10 only.
2244	61 2	5407.4	11/2 ⁻	3163.1	7/2 ⁻	E2		I_γ : other: 50 3 at E=70 MeV. DCO(Q)=1.0 1 at E=70 MeV, 1.2 1 at E=88 MeV. $\Delta\text{IPDCO}=+0.04$ 2.
2312		6660.2	(11/2 ⁻)	4347.8	9/2 ⁻			
2391		12572.2	23/2 ⁻	10181.1	19/2 ⁻			
2400	<1	8487.8	15/2 ⁻	6087.6	13/2 ⁻	D(+Q)	+0.2 +3-2	E_γ : other: 2394 (2007Ks01). DCO(Q)=1.4 4 at E=88 MeV.
2466	22 2	7873.3	13/2 ⁺	5407.4	11/2 ⁻	E1+M2	-0.25 10	I_γ : other: 1.2 1 at E=70 MeV. DCO(Q)=2.4 3. $\Delta\text{IPDCO}=+0.21$ 6. DCO(Q)=0.51 8, $R_{\text{ADO}}=0.65$ 9 (2020Ra14).
2614		11459.2	21/2 ⁺	8845.1	17/2 ⁺			
2631 @&	<1	8558.2?	(13/2 ⁻)	5927.1	(11/2 ⁻)	D(+Q)	-0.2 +2-3	DCO(Q)=2.4 7 at E=88 MeV.
2646	37 2	2645.8	7/2 ⁺	0.0	3/2 ⁺	E2		I_γ : other: 47 2 at E=70 MeV. $\Delta\text{IPDCO}=+0.08$ 5. DCO(D)=1.7 2, $R_{\text{ADO}}=1.6$ 2 (2020Ra14).
2757		8845.1	17/2 ⁺	6087.6	13/2 ⁻			
2764		5927.1	(11/2 ⁻)	3163.1	7/2 ⁻			E_γ : from 2013Bi10 only.

Continued on next page (footnotes at end of table)

$^{12}\text{C}(^{28}\text{Si},\alpha p\gamma)$ **2007Ks01,2013Bi10,2020Ra14 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ [‡]	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	Comments
2912	<1	8319.4	15/2 ⁻	5407.4	11/2 ⁻			E_γ : other: 2917 (2007Ks01, tentative). Mult., δ : D(+Q) with $\delta=-0.1$ +3-2 if J(8319)=13/2 in 2007Ks01. DCO(Q)=1.9 6 at E=88 MeV.
3003 [@]	>1.1	3003.1	5/2 ⁺	0.0	3/2 ⁺			I_γ : other: >0.6 at E=70 MeV.
3080	<1	8487.8	15/2 ⁻	5407.4	11/2 ⁻			E_γ : other: 3074 (2007Ks01).
3163	100 5	3163.1	7/2 ⁻	0.0	3/2 ⁺	M2+E3	+0.16 1	I_γ : other: 100 2 at E=70 MeV. DCO(Q)=0.9 1 at E=70 MeV, 0.7 1 at 88 MeV. $\Delta\text{IPDCO}=-0.06$ 2.
3497		6660.2	(11/2 ⁻)	3163.1	7/2 ⁻			
3734		16306.4	(27/2 ⁻)	12572.2	23/2 ⁻			
4347 [@]	<1	4347.8	9/2 ⁻	0.0	3/2 ⁺			I_γ : other: <1 at E=70 MeV.

[†] From 2013Bi10, unless otherwise noted.[‡] From 2007Ks01, measured at E(^{28}Si)=88 MeV at the NSC. Values are also available from E=70 MeV data of 2007Ks01 at TIFR facility, but less complete and are given under comments where available.[#] Q and D+Q with $\delta(Q/D)$ are from $\gamma\gamma(\theta)(\text{DCO})$ in 2007Ks01 and magnetic/electric characters are from $\gamma\gamma(\text{lin pol})$ in 2007Ks01 where available and/or RUL where level T_{1/2} is measured, unless otherwise noted.[@] From 2007Ks01 only.[&] Placement of transition in the level scheme is uncertain.

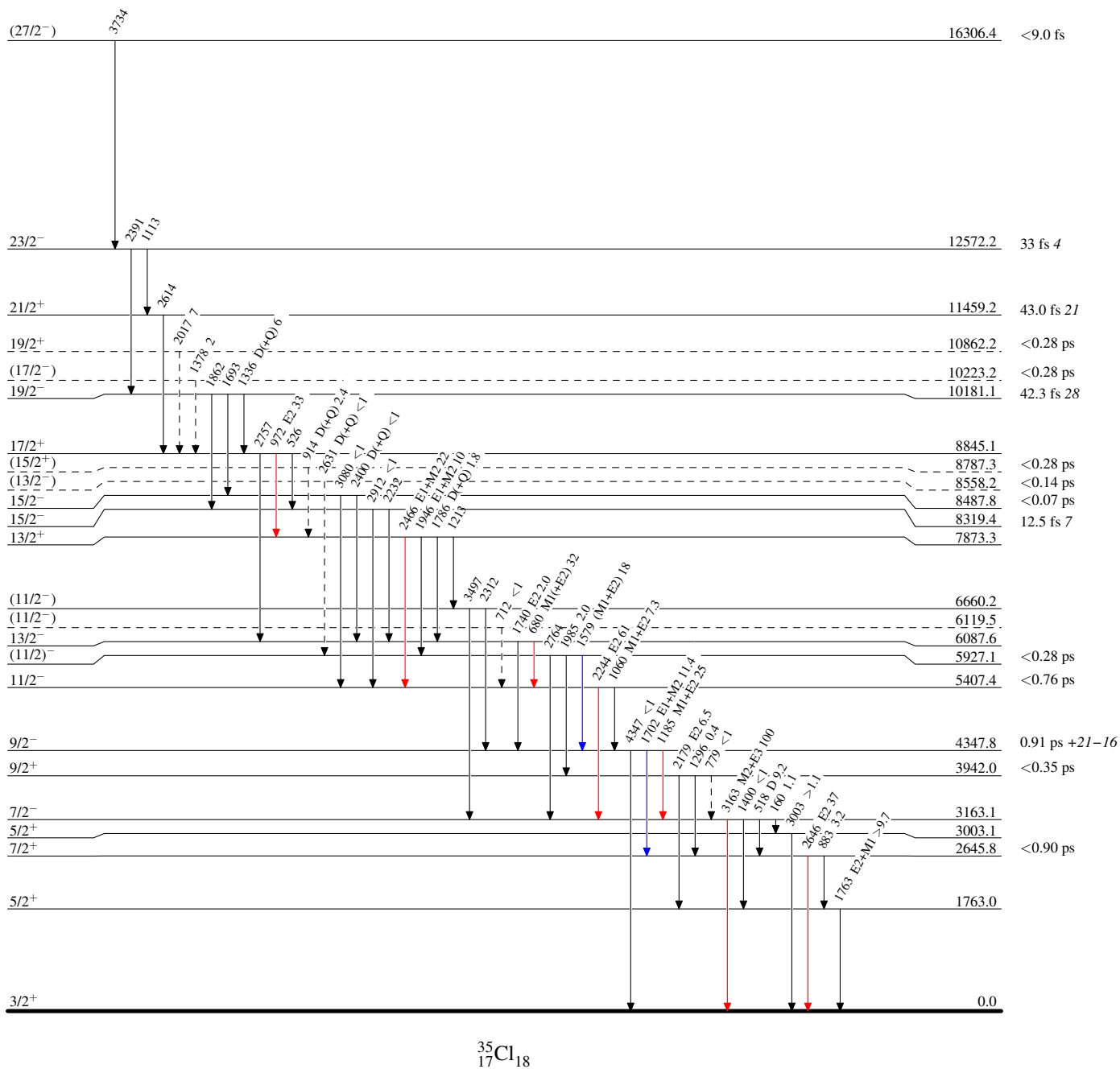
$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ 2007Ks01, 2013Bi10, 2020Ra14

Legend

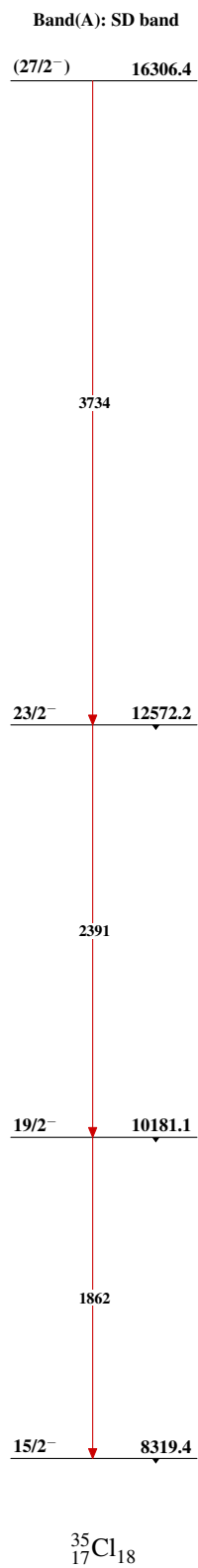
Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ γ Decay (Uncertain)

 $^{35}_{17}\text{Cl}_{18}$

$^{12}\text{C}(^{28}\text{Si}, \alpha p \gamma)$ 2007Ks01, 2013Bi10, 2020Ra14



$^{16}\text{O}(^{24}\text{Mg}, \alpha p \gamma)$ 2004Ek01, 2005Ek01

2004Ek01, 2005Ek01: E=60 MeV ^{24}Mg beam was produced at the Legnaro National Laboratory. Target was a 0.5 mg/cm² enriched ^{40}Ca target with a 7 mg/cm² tantalum backing and with oxygen inside. γ rays were detected with the GASP array of Ge detectors and charged particles were detected with the ISIS array of 40 $\Delta\text{E-E}$ Si telescopes. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma(\text{particle})$ -coin. Deduced levels, J, π from measured asymmetry ratios $R(\gamma(\theta))$.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0.0	$3/2^+$
1762.95 17	$5/2^+$
2645.81 18	$7/2^+$
3163.43 19	$7/2^-$
4348.29 24	$9/2^-$
5407.64 27	$11/2^-$
6088.05 29	$13/2^-$

[†] From a least-squares fit to γ -ray energies.

[‡] As given in **2004Ek01** based on measured $\gamma(\theta)$ asymmetry and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$

$R(\gamma(\theta))$ =Asymmetry ratio measured at 35° and 81° with respect to the beam axis. Value of ≈ 1.2 for stretched quadrupole and ≈ 0.7 for stretched dipole ($\Delta J=1$).

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	Comments
517.7 1	14.3 8	3163.43	$7/2^-$	2645.81	$7/2^+$		
680.4 1	39.1 12	6088.05	$13/2^-$	5407.64	$11/2^-$	D	$R(\gamma(\theta))=0.65$ 3.
882.9 1	4.9 4	2645.81	$7/2^+$	1762.95	$5/2^+$		
1059.3 2	9.1 9	5407.64	$11/2^-$	4348.29	$9/2^-$		
1185.0 2	32.9 10	4348.29	$9/2^-$	3163.43	$7/2^-$		
1702.0 3	13.0 8	4348.29	$9/2^-$	2645.81	$7/2^+$		
1763.1 2	22.9 7	1762.95	$5/2^+$	0.0	$3/2^+$		$R(\gamma(\theta))=1.43$ 7 indicating Q seems inconsistent with $\Delta J=1$ from level scheme.
2244.2 3	53.9 17	5407.64	$11/2^-$	3163.43	$7/2^-$	Q	$R(\gamma(\theta))=1.24$ 6.
2645.5 4	32.3 10	2645.81	$7/2^+$	0.0	$3/2^+$	Q	$R(\gamma(\theta))=1.21$ 6.
3162.7 4	100 3	3163.43	$7/2^-$	0.0	$3/2^+$	Q	$R(\gamma(\theta))=1.43$ 7.

[†] From **2004Ek01**.

[‡] Deduced from $\gamma(\theta)$ asymmetry in **2004Ek01** by the evaluators.

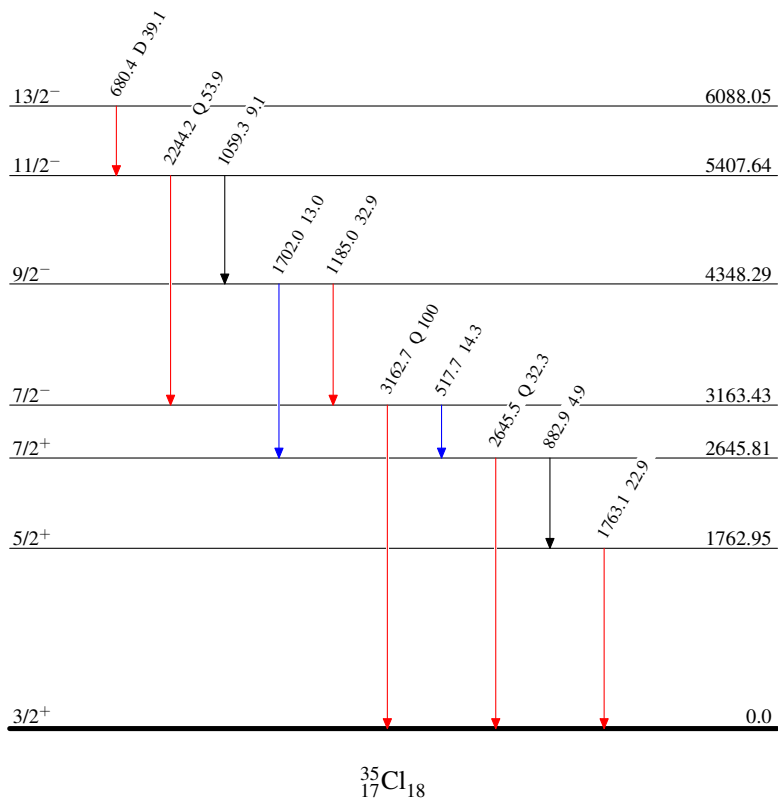
$^{16}\text{O}(^{24}\text{Mg},\alpha p\gamma)$ 2004Ek01,2005Ek01

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
—→ $I_\gamma < 10\% \times I_\gamma^{\max}$
—→ $I_\gamma > 10\% \times I_\gamma^{\max}$



$^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)$ **2007De14,1976Va24**

Also includes data from $^{25}\text{Mg}(^{16}\text{O},\alpha p n\gamma)$ in [1976Ke02](#) and $^{27}\text{Al}(^{16}\text{O},2\alpha\gamma)$ in [1991Ja11](#).

2007De14: E=70 MeV ^{16}O beam was produced from the XTU-Tandem accelerator at Legnaro National Laboratory. Target was a $400\text{ }\mu\text{g}/\text{cm}^2$ self-supporting target of ^{24}Mg . γ rays were detected with the GASP spectrometer of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators and charged particles were detected with the 4π array ISIS. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma\gamma(\theta)$ (ADO), particle- γ -coin. Deduced levels, J, π , band structures, γ -ray multipolarities.

1976Va24: E=38 and 45 MeV ^{16}O beams were produced at Fysisch laboratorium in the Netherlands. Targets were $300\text{ }\mu\text{g}/\text{cm}^2$ ^{24}Mg enriched to 99.94% evaporated onto $30\text{ }\mu\text{m}$ Au backings. γ rays were detected with a Ge(Li)-NaI(Tl) Compton-suppression spectrometer and a three-Ge(Li) Compton polarimeter. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, $\gamma(\theta)$, $\gamma(\text{lin pol})$. Deduced levels, J, π , γ -ray multipolarities and mixing ratios.

1974Va13: E=36 MeV ^{16}O beam was produced at Fysisch laboratorium in the Netherlands. Targets were 100 and $300\text{ }\mu\text{g}/\text{cm}^2$ Mg on $1\text{ }\mu\text{m}$ Ni backings. γ rays were detected with Ge(Li) detectors. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$ -coin, recoil distances. Deduced the half-life for the level at 3.16 MeV.

1982Ra25: E=40-51 MeV ^{16}O beams were produced at the Argonne National Laboratory. Target was $50\text{ }\mu\text{g}/\text{cm}^2$ layer of ^{24}Mg (>99.9%) evaporated onto a $50\text{ }\mu\text{g}$ Au backing. γ rays were detected with a Ge(Li) detector. Measured γ -ray yields.

Other reactions:

1991Ja11: $^{27}\text{Al}(^{16}\text{O},2\alpha\gamma)$ E=60 MeV ^{16}O beam was produced from the 14 UD-Pelletron at TIFR. Target was $350\text{ }\mu\text{g}/\text{cm}^2$ Al evaporated onto a $1.8\text{ mg}/\text{cm}^2$ thick Ta backing. γ rays were detected with Ge(Li) detector. Measured $E\gamma$, recoil distances. Deduced $T_{1/2}$ for the levels of 3162 and 8844 keV.

1976Ke02: $^{25}\text{Mg}(^{16}\text{O},\alpha p n\gamma)$ E=43 and 50 MeV ^{16}O beams were produced at the Institut fur Kernphysik in Germany. The target was $150\text{ }\mu\text{g}/\text{cm}^2$ enriched ^{25}Mg (99.2%) evaporated onto a Au backing. γ rays were detected with two true coaxial Ge(Li) detectors of 45 cm^3 . Measured $E\gamma$, $I\gamma$, Doppler-shift recoil distances. Deduced $T_{1/2}$ for the level of 3162 keV.

1989De02: $^{27}\text{Al}(^{16}\text{O},X)$ E=140 MeV ^{16}O from the Stony Brook LINAC facility. Measured particle-particle correlations.

1986Br26: $^{27}\text{Al}(^{16}\text{O},X)$ E=65 MeV ^{16}O from the Pelletron Laboratory at the Universidade de Sao Paulo, Brazil. Measured σ .

1979Ta03: $^{27}\text{Al}(^{16}\text{O},X)$ E=50, 60, and 70 MeV ^{16}O from the NBI super FN tandem. Measured particle-particle angular correlations.

2001Pi10: $^{12}\text{C}(^{32}\text{S},X)$ E=19.5 MeV/nucleon ^{32}S beam from the superconducting cyclotron of the Laboratori Nazionali del Sud in Catania. Measured σ .

 ^{35}Cl Levels

Band assignments are from [2007De14](#).

E(level) [†]	J π [‡]	$T_{1/2}$	Comments
0	3/2 ⁺		
1219.0 <i>10</i>			
1763.3 <i>2</i>	5/2 ⁺		
2645.9 <i>3</i>	7/2 ⁺		
3001.9 <i>10</i>	5/2 ⁺		
3162.9 [#] <i>3</i>	7/2 ⁻	31.0 ps <i>6</i>	$T_{1/2}$: weighted average of 29.1 ps <i>14</i> (1974Va13), 28.9 ps <i>12</i> (1976Ke02), and 31.4 ps <i>4</i> (1991Ja11), all by RDM.
3943.4 <i>10</i>	9/2 ⁺		
4347.8 <i>3</i>	9/2 ⁻		
5407.3 [#] <i>3</i>	11/2 ⁻		
5926.9 <i>5</i>	11/2 ⁻		
6087.2 <i>4</i>	13/2 ⁻		
6380.8 <i>8</i>			
7873.4 [@] <i>4</i>	13/2 ⁺		
8319.7 [#] <i>5</i>	15/2 ⁻		
8487.4 <i>5</i>	15/2 ⁻		
8788.9 [@] <i>6</i>	15/2 ⁺		
8844.6 [@] <i>5</i>	17/2 ⁺	6.1 ps <i>11</i>	$T_{1/2}$: from RDM in 1991Ja11 .

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)$ **2007De14,1976Va24 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$
10181.1 [#] 5	19/2 ⁻
10222.5 11	17/2 ⁻
10859.1 [@] 8	19/2 ⁺
11459.0 [@] 7	21/2 ⁺
12572.2 [#] 6	23/2 ⁻

[†] From a least-squares fit to γ -ray energies.

[‡] As proposed in **2007De14** based on measured $\gamma\gamma(\theta)(\text{ADO})$ and band assignments, unless otherwise noted. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

[#] Band(A): Band based on 3163, 7/2⁻ level. Configuration=f_{7/2}.

[@] Band(B): Band based on 7873, 13/2⁺ level.

 $\gamma(^{35}\text{Cl})$

R_{ADO} values under comments are from $\gamma\gamma(\theta)(\text{ADO})$ in **2007De14**. Expected values are ≈ 1.3 for $\Delta J=2$, quadrupole (and $\Delta J=0$, dipole in few cases) and ≈ 0.8 for $\Delta J=1$, dipole transitions (**2007De14**).

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [‡]	$\delta^\#$	Comments
161 [@] 1		3162.9	7/2 ⁻	3001.9	5/2 ⁺			
296 [@] 1		6380.8		6087.2	13/2 ⁻			
517.3 5	11.1 13	3162.9	7/2 ⁻	2645.9	7/2 ⁺			E_γ : weighted average of 517.4 5 (2007De14) and 517 1 (1976Va24).
680.5 5	33.0 7	6087.2	13/2 ⁻	5407.3	11/2 ⁻	D		$R_{\text{ADO}}=0.84$ 2.
882.7 5	4.0 1	2645.9	7/2 ⁺	1763.3	5/2 ⁺	D		E_γ : weighted average of 882.9 5 (2007De14) and 882 1 (1976Va24).
								$R_{\text{ADO}}=0.75$ 3.
915.4 4	7.2 5	8788.9	15/2 ⁺	7873.4	13/2 ⁺	D		$R_{\text{ADO}}=0.69$ 3.
971 [@] 1		6380.8		5407.3	11/2 ⁻			
971.4 5	75.3 19	8844.6	17/2 ⁺	7873.4	13/2 ⁺	Q		$R_{\text{ADO}}=1.38$ 3.
1059.5 2	17.1 4	5407.3	11/2 ⁻	4347.8	9/2 ⁻	M1+E2	+0.12 3	E_γ : weighted average of 1059.6 5 (2007De14) and 1059.5 2 (1976Va24).
								I_γ : other: 7.7 3 (1976Va24 , E=38 MeV).
								$R_{\text{ADO}}=1.05$ 4.
								$A_2=-0.08$ 6, $A_4=+0.049$ 46, POL=-0.11 12 (1976Va24).
1113.3 5	4.1 3	12572.2	23/2 ⁻	11459.0	21/2 ⁺	D		$R_{\text{ADO}}=0.86$ 9.
1184.9 3	44.4 16	4347.8	9/2 ⁻	3162.9	7/2 ⁻	M1+E2	-0.65 30	E_γ : weighted average of 1184.9 5 (2007De14) and 1184.9 3 (1976Va24).
								I_γ : other: 22.2 3 (1976Va24 , E=38 MeV).
								$R_{\text{ADO}}=0.54$ 2, giving $\Delta J=1$.
								$A_2=-0.663$ 25, $A_4=+0.004$ 25, POL=+0.07 7 (1976Va24).
1336.3 5	23.5 8	10181.1	19/2 ⁻	8844.6	17/2 ⁺	D		$R_{\text{ADO}}=0.75$ 2.
1377.8 10	5.4 4	10222.5	17/2 ⁻	8844.6	17/2 ⁺			$R_{\text{ADO}}=1.32$ 12 also consistent with $\Delta J=0$.
1399.9 7	0.5 2	3162.9	7/2 ⁻	1763.3	5/2 ⁺			
1578.9 5	31.8 13	5926.9	11/2 ⁻	4347.8	9/2 ⁻	D		E_γ : other: 1579 1 (1976Va24).
								$R_{\text{ADO}}=0.48$ 1.
1693.5 5	19.8 21	10181.1	19/2 ⁻	8487.4	15/2 ⁻	Q		$R_{\text{ADO}}=1.44$ 7.
1701.7 5	23.1 6	4347.8	9/2 ⁻	2645.9	7/2 ⁺	E1+M2	+0.5 4	E_γ : weighted average of 1701.6 5

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$^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)$ **2007De14,1976Va24 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. ‡	$\delta^\#$	Comments
1739.6 5	2.6 3	6087.2	13/2 ⁻	4347.8	9/2 ⁻	Q		(2007De14) and 1702 1 (1976Va24). R _{ADO} =0.76 6, giving $\Delta J=1$. E _γ : weighted average of 1739.5 5 (2007De14) and 1740 1 (1976Va24). R _{ADO} =1.38 10.
1763.3 2	21.2 7	1763.3	5/2 ⁺	0	3/2 ⁺	D+Q	-2.6 4	E _γ : weighted average of 1763.1 5 (2007De14) and 1763.3 2 (1976Va24). I _γ : other: 18.5 3 (1976Va24, E=38 MeV). A ₂ =-0.368 18, A ₄ =+0.13 2, POL=+0.06 20 (1976Va24). R _{ADO} =0.77 3, giving $\Delta J=1$. R _{ADO} =1.37 7, consistent with $\Delta J=0$ from level scheme.
1786.2 5	4.5 2	7873.4	13/2 ⁺	6087.2	13/2 ⁻			R _{ADO} =1.39 7.
1861.3 5	19.6 13	10181.1	19/2 ⁻	8319.7	15/2 ⁻	Q		R _{ADO} =0.80 4.
1946.2 5	22.4 12	7873.4	13/2 ⁺	5926.9	11/2 ⁻	D		R _{ADO} =0.50 3.
2014.7 9	23.1 12	10859.1	19/2 ⁺	8844.6	17/2 ⁺	D		R _{ADO} =1.57 11.
2069.8 10	7.5 5	10859.1	19/2 ⁺	8788.9	15/2 ⁺	Q		
2180 [@] 1		3943.4	9/2 ⁺	1763.3	5/2 ⁺			
2232.7 6	10.3 5	8319.7	15/2 ⁻	6087.2	13/2 ⁻			
2244.3 2	89 2	5407.3	11/2 ⁻	3162.9	7/2 ⁻	E2(+M3)	0.000 1	E _γ : weighted average of 2243.9 5 (2007De14) and 2244.3 2 (1976Va24). I _γ : other: 30.3 20 (1976Va24, E=38 MeV). R _{ADO} =1.26 3, giving $\Delta J=2$. A ₂ =+0.303 20, A ₄ =-0.119 22, POL=+0.26 28 (1976Va24). R _{ADO} =1.34 12.
2390.8 5	16.8 7	12572.2	23/2 ⁻	10181.1	19/2 ⁻	Q		
2399.8 8	14.8 5	8487.4	15/2 ⁻	6087.2	13/2 ⁻			
2466.2 5	46.2 9	7873.4	13/2 ⁺	5407.3	11/2 ⁻	D		E _γ : other: 2466 1 (1976Va24, unplaced). R _{ADO} =0.77 2.
2614.5 5	19.0 8	11459.0	21/2 ⁺	8844.6	17/2 ⁺	Q		R _{ADO} =1.33 4.
2645.8 5	36.1 8	2645.9	7/2 ⁺	0	3/2 ⁺	Q		E _γ : weighted average of 2645.7 5 (2007De14) and 2646 1 (1976Va24). R _{ADO} =1.35 4.
2911.9 8	18.0 8	8319.7	15/2 ⁻	5407.3	11/2 ⁻	Q		R _{ADO} =1.35 5.
3080.0 7	4.7 3	8487.4	15/2 ⁻	5407.3	11/2 ⁻	Q		R _{ADO} =1.20 12.
3162.6 3	100 2	3162.9	7/2 ⁻	0	3/2 ⁺	M2+E3	+0.18 7	E _γ : weighted average of 3163.1 5 (2007De14) and 3162.4 3 (1976Va24). I _γ : other: 100.0 15 (1976Va24, E=38 MeV). R _{ADO} =1.30 2, giving $\Delta J=2$. A ₂ =+0.416 6, A ₄ =-0.001 6, POL=-0.57 12 (1976Va24).
4347.8 8	5.5 5	4347.8	9/2 ⁻	0	3/2 ⁺			

[†] From 2007De14. Note that a few I_γ values in 2007De14 have very small uncertainties (<1%) which is unrealistic for an in-beam measurement with Ge detectors. It is very likely the reported uncertainties are statistical only and the evaluators have added in quadrature a typical 2% systematic uncertainty from efficiency calibration.

[‡] D with $\Delta J=1$ or Q with $\Delta J=2$ are deduced by the evaluators based on measured $\gamma\gamma(\theta)(\text{ADO})$ in 2007De14 and others are from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1976Va24. Assignments are not explicitly listed in 2007De14.

[#] From $\gamma(\theta)$ in 1976Va24.

[@] From 1976Va24 only.

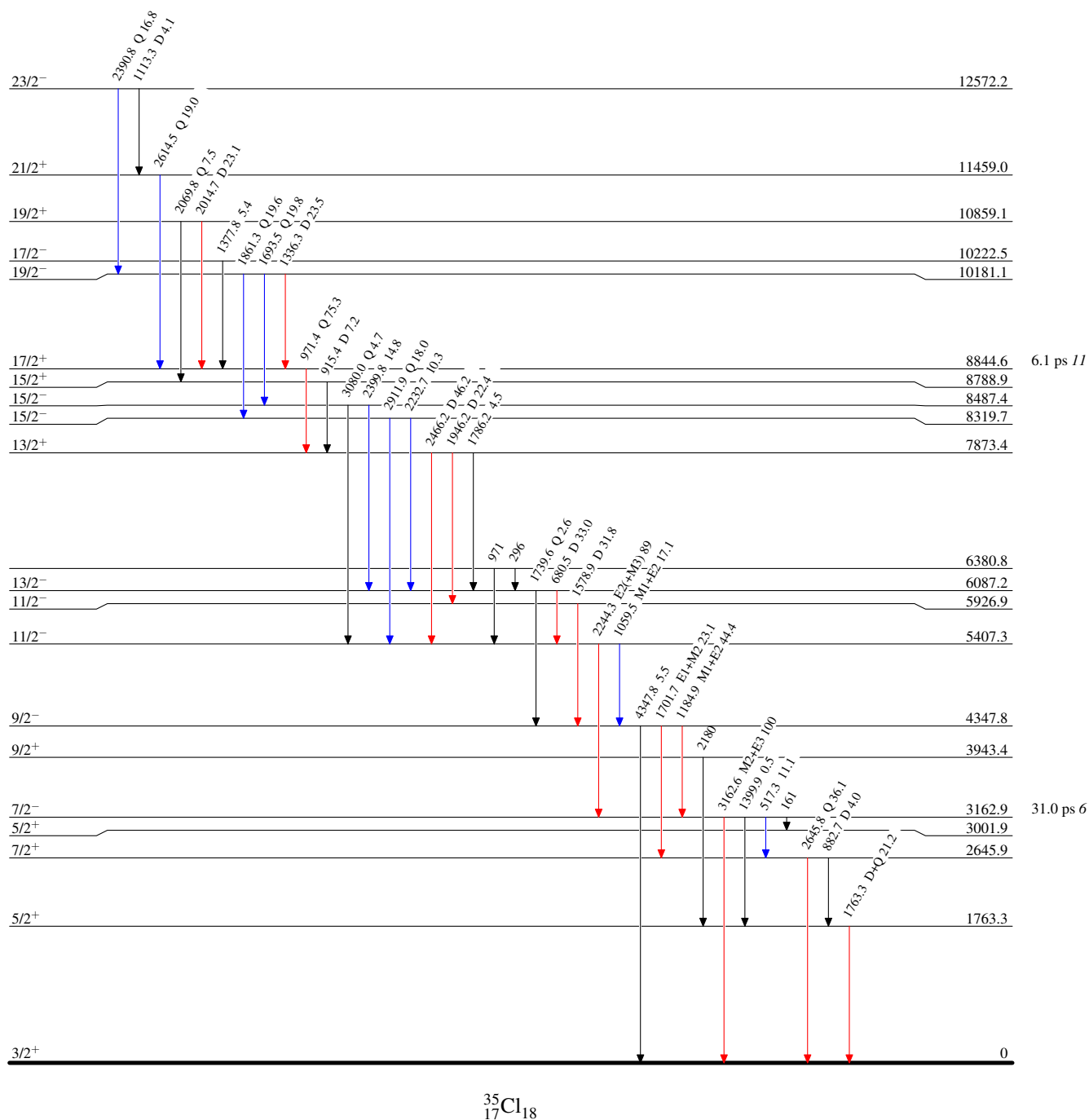
$^{24}\text{Mg}(^{16}\text{O}, \alpha p \gamma)$ 2007De14, 1976Va24

Level Scheme

Intensities: Relative I_γ

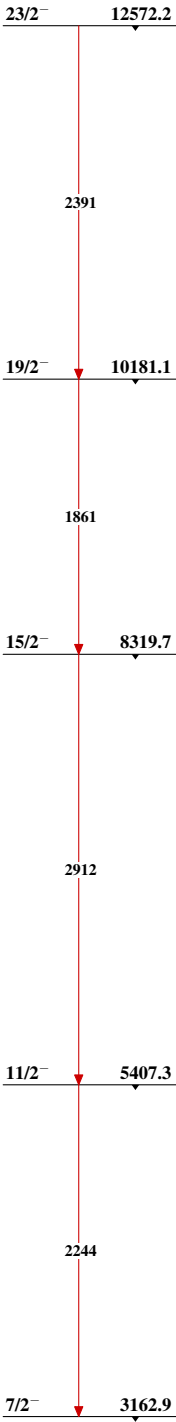
Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$



$^{24}\text{Mg}(^{16}\text{O},\alpha\text{p}\gamma)$ 2007De14,1976Va24

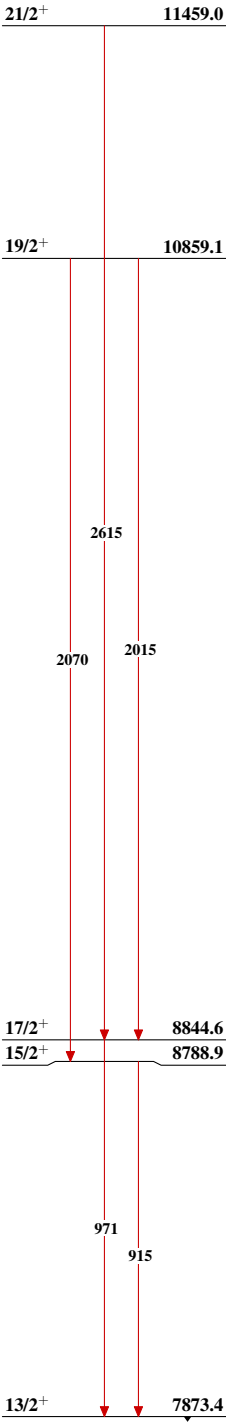
Band(A): Band based on
3163, 7/2⁻ level



$^{35}_{17}\text{Cl}_{18}$

²⁴Mg(¹⁶O, α p γ) 2007De14,1976Va24 (continued)

Band(B): Band based on 7873, 13/2⁺ level



³⁵₁₇Cl₁₈

$^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$ **1976Wa11**

1976Wa11: E=40 MeV ^{14}N beam was produced at the Brookhaven National Laboratory. Target was ^{27}Al . γ rays were detected with Ge(Li) detectors. Measured E_γ , I_γ , $\gamma(\theta)$, $\gamma\gamma$ -coin, recoils distances. Deduced levels, J, π , $T_{1/2}$, γ -ray branching ratios, multipolarities.

 ^{35}Cl Levels

E(level) [†]	J π [‡]	$T_{1/2}$	Comments
0	3/2 ⁺		
1763.19 10	5/2 ⁺		
2645.45 12	7/2 ⁺		
3001.98 22	5/2 ⁺		
3162.64 9	7/2 ⁻	29.0 ps 13	$T_{1/2}$: from $\tau=41.8$ ps 18 (1976Wa11).
3942.3 10	9/2 ⁺		
4347.43 17	9/2 ⁻		
5406.68 19	11/2 ⁻		
5926.47 29	11/2 ⁻		
6086.89 23	13/2 ⁻	5.3 ps 10	$T_{1/2}$: from $\tau=7.7$ ps 14 (1976Wa11).
7872.78 30	13/2 ⁺		$T_{1/2}$: 0.5 ps< τ <50 ps (1976Wa11).
8844.2 4	17/2 ⁺	5.5 ps 14	$T_{1/2}$: from $\tau=8$ ps 2 (1976Wa11).

[†] From a least-squares fit to γ -ray energies.

[‡] As given in **1976Wa11** based on measured $\gamma(\theta)$ and known assignments of low-lying levels. When considered in Adopted Levels, firm assignments here will be placed in parentheses if there are no strong supporting arguments.

 $\gamma(^{35}\text{Cl})$

A_2 and A_4 under comments are from $\gamma(\theta)$ in **1976Wa11**, with $A_4=0$ if not given.

E_γ [†]	I_γ [†]	$E_i(\text{level})$	J π_i	E_f	J π_f	Comments
160.66 [#] 20	2.3	3162.64	7/2 ⁻	3001.98	5/2 ⁺	$A_2=-0.24$ 15.
517.26 10	<2.2 [‡]	3162.64	7/2 ⁻	2645.45	7/2 ⁺	E_γ, I_γ : doublet with γ rays in ^{36}Cl (1976Wa11). $A_2=+0.47$ 25.
680.22 15	25	6086.89	13/2 ⁻	5406.68	11/2 ⁻	$A_2=-0.20$ 2.
882.32 35	2.3	2645.45	7/2 ⁺	1763.19	5/2 ⁺	$A_2=+0.31$ 40.
971.38 20	8.0	8844.2	17/2 ⁺	7872.78	13/2 ⁺	$A_2=+0.28$ 7, $A_4=-0.18$ 7.
1059.17 20	4.6	5406.68	11/2 ⁻	4347.43	9/2 ⁻	$A_2=-0.04$ 15. %Branching=13 4.
1184.80 20	21	4347.43	9/2 ⁻	3162.64	7/2 ⁻	$A_2=-0.66$ 4, $A_4=+0.08$ 4. %Branching=69 5.
1579.15 30	1.9 [‡]	5926.47	11/2 ⁻	4347.43	9/2 ⁻	$A_2=-0.6$ 4.
1701.85 25	9.5	4347.43	9/2 ⁻	2645.45	7/2 ⁺	$A_2=-0.20$ 3. %Branching=31 5.
1739.3 4	1.7	6086.89	13/2 ⁻	4347.43	9/2 ⁻	
1763.15 10	32	1763.19	5/2 ⁺	0	3/2 ⁺	$A_2=-0.28$ 2, $A_4=+0.05$ 3.
1786 [#]	0.63	7872.78	13/2 ⁺	6086.89	13/2 ⁻	E_γ, I_γ : inferred from the presence of 680 γ -971 γ coin (1976Wa11). %Branching=8 5.
1946.40 30	1.9	7872.78	13/2 ⁺	5926.47	11/2 ⁻	$A_2=-0.2$ 1. %Branching=22 5.
2179	5.0	3942.3	9/2 ⁺	1763.19	5/2 ⁺	E_γ, I_γ : observed in $\gamma\gamma$ -coin only (1976Wa11). $A_2=+0.23$ 2, $A_4=-0.08$ 2.
2244.00 25	32	5406.68	11/2 ⁻	3162.64	7/2 ⁻	%Branching=87 4.
2465.85 30	5.9	7872.78	13/2 ⁺	5406.68	11/2 ⁻	$A_2=-0.21$ 6. %Branching=70 5.

Continued on next page (footnotes at end of table)

²⁷Al(¹⁴N, α pn γ) **1976Wa11** (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_{γ}[†]</u>	<u>I_{γ}[†]</u>	<u>E_i(level)</u>	<u>J_i^{π}</u>	<u>E_f</u>	<u>J_f^{π}</u>	<u>Comments</u>
2645.79 30	<26	2645.45	7/2 ⁺	0	3/2 ⁺	E _{γ} , I _{γ} : doublet with γ rays in ³⁸ K (1976Wa11). A ₂ =+0.37 3, A ₄ =-0.04 3.
3162.43 10	100	3162.64	7/2 ⁻	0	3/2 ⁺	

[†] From **1976Wa11**. Original values of relative intensities have been renormalized by the evaluators to I(3162 γ)=100.

[‡] Estimated from $\gamma\gamma$ -coin data or known branching ratios (**1976Wa11**).

Placement of transition in the level scheme is uncertain.

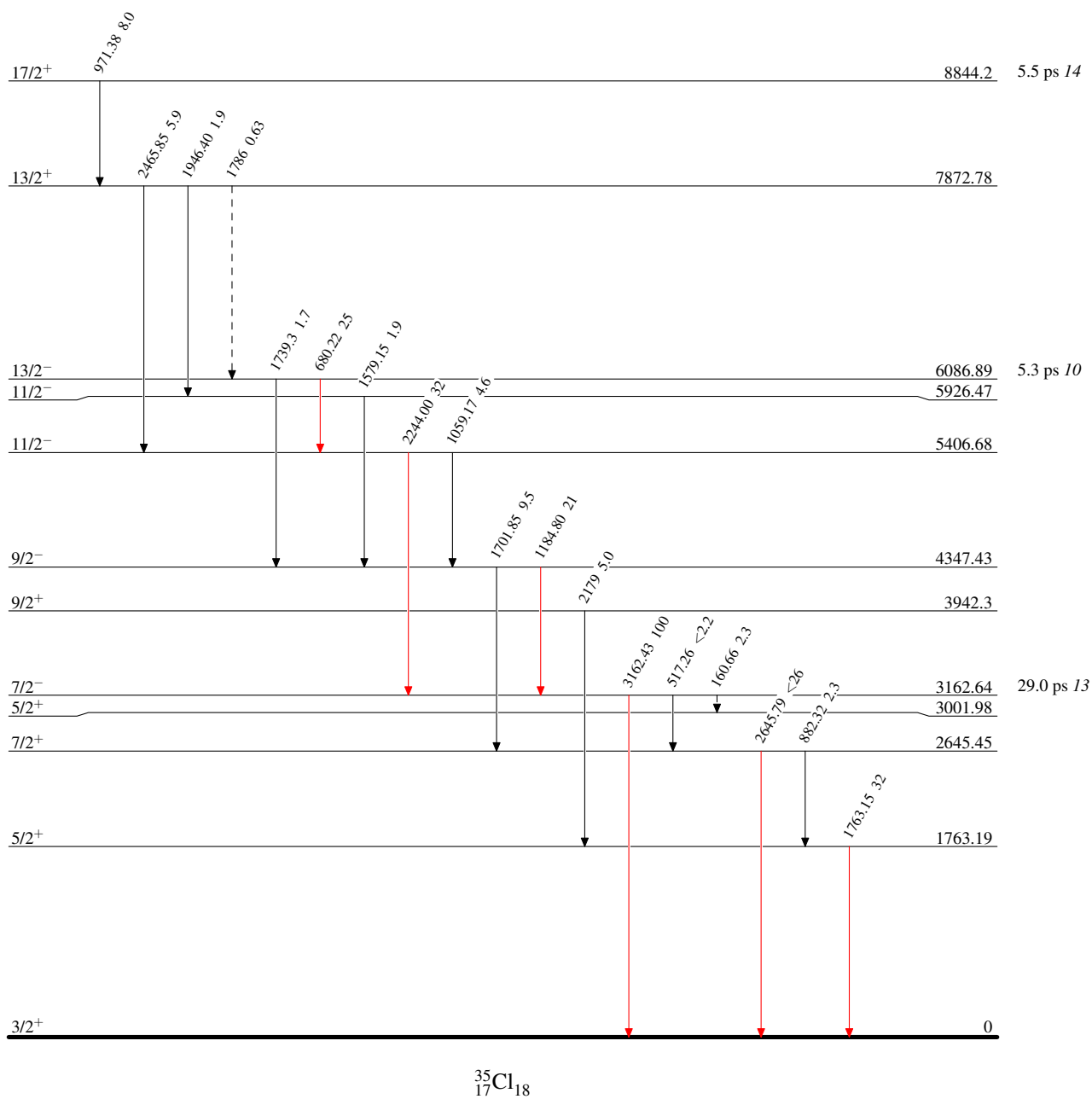
$^{27}\text{Al}(^{14}\text{N},\alpha\text{pn}\gamma)$ 1976Wa11

Legend

Level Scheme

Intensities: Relative I_γ

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\text{max}}$
 $\longrightarrow$ γ Decay (Uncertain)



$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n}): \text{resonances}$ [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) $J^\pi=1/2^+$ ^{31}P ground state.

[1975Sc40](#): $^{31}\text{P}(\alpha, \text{p})$ with 3.25-5.55-MeV α beams produced from the 5.5-MV Van de Graaff accelerator of the University of Lowell. The target was prepared by evaporating natural ^{31}P onto carbon backings. Protons were detected using four silicon surface-barrier detectors. Measured proton yields and angular distributions. Deduced resonance levels at $E_x=9.9$ -11.8 MeV, J, and π .

[1971Mc23](#): $^{31}\text{P}(\alpha, \text{p})$ with 3.25-5.25-MeV α beams produced from the 5.5-MeV Van de Graaff accelerator of the Southern Universities Nuclear Institute. The target was Zn_3P_2 evaporated onto carbon backings. Protons were detected using four solid state detectors. Measured proton yields and angular distributions. Deduced resonance levels at $E_x=9.9$ -11.7 MeV, J, and resonance strengths.

[1964Ku12](#): $^{31}\text{P}(\alpha, \text{p})$ with 1.7-3.3-MeV α beams produced from the Utrecht Van de Graaff accelerator. A zinc phosphide target was prepared by evaporation of 40-500-mg/cm² Zn_3P_2 onto solid copper backings. Protons were detected using five silicon surface barrier counters at 87°, 120°, 135°, 150°, and 172°. Measured resonance yields and proton angular distributions. Deduced ^{35}Cl resonance energies, J, and resonance strengths.

[1970Um01](#): $^{34}\text{S}(\text{p}, \text{n})$, $^{31}\text{P}(\alpha, \text{n})$, $^{31}\text{P}(\alpha, \alpha_0)$ reactions with proton and α beams produced from the Florida State University Tandem Van de Graaff accelerator. Targets for the $^{34}\text{S}(\text{p}, \text{n})$ experiment were a CdS target made by evaporating ^{34}S enriched CdS onto 0.12-mm gold plate and a AgS target with heating ^{34}S (67% enriched) in the presence of 0.025-mm Ag foil. Targets for the $^{31}\text{P}(\alpha, \text{n})$ experiment were prepared by evaporating natural Ca_3P_2 onto 0.075-mm Ta plate and 0.025-mm Al foil. Positrons from ^{34}Cl decay were detected using an NE102 plastic scintillator. γ rays in ^{34}S following the decay were detected using a Ge(Li) detector. Measured excitation functions. Deduced level energies and isospin T=3/2 for ^{35}Cl resonance states at 12.9 and 13.9 MeV.

 ^{35}Cl Levels

$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma$: from [1964Ku12](#) for $E_x<9950$ and from [1971Mc23](#) for $E_x>9950$.

E(level) [†]	J^π [‡]	$E_\alpha(\text{lab})$ (keV) ^{&}	Comments
9127 9	5/2 [#]	2404 [#] 10	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=0.16$ eV.
9161 4	1/2 [#]	2442 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.6$ eV.
9256 4	1/2 [#]	2550 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=2.6$ eV.
9400 9	1/2 [#]	2712 [#] 10	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=0.6$ eV.
9456 4	3/2 [#]	2776 [#] 4	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=2.8$ eV.
9481.0 18	3/2 [#]	2804 [#] 2	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=18$ eV.
9551 6	5/2 [#]	2883 [#] 7	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.9$ eV.
9673 6	1/2 [#]	3021 [#] 7	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=1.1$ eV.
9713.0 27	1/2 [#]	3066 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=41$ eV.
9751.1 27	7/2 [#]	3109 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=7.6$ eV.
9814.0 27	5/2 [#]	3180 [#] 3	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=20$ eV.
9869.8 27	1/2	3243 3	$E_\alpha(\text{lab})$ (keV): weighted average of 3243 3 (1964Ku12) and 3242 5 (1975Sc40). J^π : 1/2 from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=46$ eV.
9900.8 27	3/2	3278 3	$E_\alpha(\text{lab})$ (keV): weighted average of 3278 3 (1964Ku12) and 3278 5 (1975Sc40). J^π : 3/2 from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=38$ eV.
9923 4	3/2 ⁻	3303 4	$E_\alpha(\text{lab})$ (keV): weighted average of 3302 4 (1964Ku12) and 3305 5 (1975Sc40). J^π : (3/2) from 1964Ku12 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=80$ eV.
9958 6	3/2	3342 7	$E_\alpha(\text{lab})$ (keV): weighted average of 3348 5 (1971Mc23) and 3335 5 (1975Sc40). J^π : 3/2 from 1971Mc23 . $(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=34$ eV.
9968 5	(1/2, 3/2) [@]	3354 5	$(2J+1)\Gamma_\alpha\Gamma_p/\Gamma=17$ eV.
10031 5	1/2 ⁺ , 3/2 ⁻	3425 5	$E_\alpha(\text{lab})$ (keV): weighted average of 3424 5 (1971Mc23) and 3425 5 (1975Sc40). J^π : (1/2) from 1971Mc23 .

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) ^{35}Cl Levels (continued)

E(level) [†]	$J^{\pi\ddagger}$	$E_{\alpha}(\text{lab})$ (keV) ^{&}	Comments
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=124$ eV.
10058 5	3/2 [@]	3455 5	(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=40$ eV.
10075 5	(1/2) [@]	3475 5	(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=52$ eV.
10090 5	5/2 ⁺	3491 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3491 5 (1971Mc23) and 3490 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=63$ eV.
10132 5	3/2 ⁻	3539 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3538 5 (1971Mc23) and 3540 5 (1975Sc40). J^{π} : 3/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=165$ eV.
10172 5	3/2 ⁻ , 5/2 ⁻	3584 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3587 5 (1971Mc23) and 3580 5 (1975Sc40). J^{π} : (5/2) from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=26$ eV.
10215 5	3/2	3633 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3631 5 (1971Mc23) and 3635 5 (1975Sc40). J^{π} : 3/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=63$ eV.
10236 5	3/2 ⁻	3656 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3656 5 (1971Mc23) and 3655 5 (1975Sc40). J^{π} : 3/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=271$ eV.
10278 5	1/2 [@]	3704 5	(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=15$ eV.
10295 5		3723 5	
10322 5	5/2 ⁺ , 7/2 ⁻	3753 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3755 5 (1971Mc23) and 3750 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=63$ eV.
10340 5	(3/2, 5/2) [@]	3774 5	(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=50$ eV.
10359 5	3/2	3795 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3795 5 (1971Mc23) and 3795 5 (1975Sc40). J^{π} : (3/2) from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=155$ eV.
10385 5		3825 5	
10399 5	3/2 ⁺ , 7/2 ⁺	3841 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3842 5 (1971Mc23) and 3840 5 (1975Sc40). J^{π} : (1/2, 3/2) from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=61$ eV.
10431 5	5/2 ⁺	3877 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3876 5 (1971Mc23) and 3877 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=66$ eV.
10467 5	5/2 ⁺	3917 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3918 5 (1971Mc23) and 3915 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=211$ eV.
10486 5	5/2 ⁺	3939 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3939 5 (1971Mc23) and 3938 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=76$ eV.
10534 5	1/2 ⁺ , 5/2 ⁺	3993 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 3990 5 (1971Mc23) and 3995 5 (1975Sc40). J^{π} : 1/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=182$ eV.
10587 5	3/2 ⁺	4053 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 4050 5 (1971Mc23) and 4055 5 (1975Sc40). J^{π} : 3/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=71$ eV.
10641 5	5/2 ⁺	4114 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 4113 5 (1971Mc23) and 4114 5 (1975Sc40). J^{π} : 5/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=272$ eV.
10678 5	3/2 ⁺ , 5/2 ⁺	4156 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 4156 5 (1971Mc23) and 4155 5 (1975Sc40). J^{π} : 3/2 from 1971Mc23 .
			(2J+1) $\Gamma_{\alpha}\Gamma_{\text{p}}/\Gamma=132$ eV.
10696 5	7/2 ⁻	4176 5	$E_{\alpha}(\text{lab})$ (keV): weighted average of 4174 5 (1971Mc23) and 4178 5 (1975Sc40).

Continued on next page (footnotes at end of table)

$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances **1975Sc40, 1971Mc23, 1964Ku12** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _α (lab) (keV) ^{&}	Comments
10734 5	3/2 ⁺ , 9/2 ⁺	4219 5	J ^π : 7/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=100 eV. E _α (lab) (keV): weighted average of 4218 5 (1971Mc23) and 4220 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=269 eV.
10760 5	3/2 ⁺ , 7/2 ⁻	4248 5	E _α (lab) (keV): weighted average of 4246 5 (1971Mc23) and 4250 5 (1975Sc40).
10800 5	5/2 ⁺	4293 5	E _α (lab) (keV): weighted average of 4290 5 (1971Mc23) and 4295 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=238 eV.
10816 5	1/2 ⁺ , 3/2 ⁻	4311 5	E _α (lab) (keV): weighted average of 4308 5 (1971Mc23) and 4313 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=345 eV.
10844 5	3/2 ⁻	4343 5	E _α (lab) (keV): weighted average of 4342 5 (1971Mc23) and 4343 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=923 eV.
10870 5	(5/2, 7/2) [@]	4372 5	(2J+1)Γ _α Γ _p /Γ=137 eV.
10886 5		4390 5	
10909 5	3/2 ⁺ , 5/2 ⁻	4416 5	E _α (lab) (keV): weighted average of 4413 5 (1971Mc23) and 4418 5 (1975Sc40).
10928 5	7/2 [@]	4438 5	(2J+1)Γ _α Γ _p /Γ=231 eV.
10942 5		4454 5	
10964 5	5/2 ⁺	4478 5	E _α (lab) (keV): weighted average of 4478 5 (1971Mc23) and 4477 5 (1975Sc40). J ^π : 5/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=181 eV.
10973 5		4489 5	
10998 5	3/2 ⁺ , 7/2 ⁻	4517 5	E _α (lab) (keV): weighted average of 4517 5 (1971Mc23) and 4517 5 (1975Sc40). J ^π : (5/2) from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=172 eV.
11034 5	3/2 ⁺	4557 5	E _α (lab) (keV): weighted average of 4556 5 (1971Mc23) and 4558 5 (1975Sc40). J ^π : 3/2 from 1971Mc23 . (2J+1)Γ _α Γ _p /Γ=218 eV.
11063 5	1/2 ⁺ , 9/2 ⁺	4590 5	E _α (lab) (keV): weighted average of 4589 5 (1971Mc23) and 4590 5 (1975Sc40).
11082 5	5/2 ⁺ , 9/2 ⁺	4612 5	E _α (lab) (keV): weighted average of 4611 5 (1971Mc23) and 4613 5 (1975Sc40).
11105 5		4638 5	
11123 5	3/2	4658 5	E _α (lab) (keV): weighted average of 4657 5 (1971Mc23) and 4658 5 (1975Sc40).
11142 5	5/2 ⁺	4679 5	E _α (lab) (keV): weighted average of 4678 5 (1971Mc23) and 4680 5 (1975Sc40).
11154 5		4693 5	
11169 5		4710 5	
11179 5	3/2 ⁻ , 5/2 ⁺	4721 5	E _α (lab) (keV): weighted average of 4721 5 (1971Mc23) and 4720 5 (1975Sc40).
11185 5		4728 5	
11198 5	3/2	4743 5	E _α (lab) (keV): weighted average of 4745 5 (1971Mc23) and 4740 5 (1975Sc40).
11230 5		4779 5	
11246 5	3/2	4797 5	E _α (lab) (keV): weighted average of 4799 5 (1971Mc23) and 4795 5 (1975Sc40).
11265 5	3/2 ⁻	4818 5	E _α (lab) (keV): weighted average of 4820 5 (1971Mc23) and 4815 5 (1975Sc40).
11287 5		4843 5	
11307 5		4866 5	
11314 5	1/2 ⁻ , 5/2 ⁺	4874 5	E _α (lab) (keV): weighted average of 4875 5 (1971Mc23) and 4873 5 (1975Sc40).
11327 5	1/2 ⁻ , 5/2 ⁺	4888 5	E _α (lab) (keV): weighted average of 4886 5 (1971Mc23) and 4890 5 (1975Sc40).
11357 5		4922 5	
11375 5		4942 5	
11390 5		4960 5	
11410 5		4982 5	
11421 5	3/2 ⁻	4995 5	E _α (lab) (keV): weighted average of 4995 5 (1971Mc23) and 4995 5 (1975Sc40).
11434 5		5009 5	
11460 5		5038 5	
11473 5	1/2 ⁻ , 5/2 ⁻	5053 5	E _α (lab) (keV): weighted average of 5056 5 (1971Mc23) and 5050 5 (1975Sc40).

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$^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$: resonances [1975Sc40](#), [1971Mc23](#), [1964Ku12](#) (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _α (lab) (keV) ^{&}	Comments
11492 5		5075 5	
11505 5	3/2 ⁻ , 5/2 ⁻	5089 5	E _α (lab) (keV): weighted average of 5090 5 (1971Mc23) and 5088 5 (1975Sc40).
11525 5		5112 5	
11540 5		5129 5	
11551 5	3/2 ⁻ , 5/2 ⁺	5141 5	E _α (lab) (keV): weighted average of 5142 5 (1971Mc23) and 5140 5 (1975Sc40).
11565 5		5157 5	
11589 5		5184 5	
11609 5	5/2 ⁺ , 7/2 ⁻	5207 5	E _α (lab) (keV): weighted average of 5209 5 (1971Mc23) and 5205 5 (1975Sc40).
11627 5	1/2 ⁺ , 5/2 ⁺	5227 5	E _α (lab) (keV): weighted average of 5224 5 (1971Mc23) and 5230 5 (1975Sc40).
11638 5		5239 5	
11651 5		5254 5	
11684 5	1/2 ⁺ , 5/2 ⁺	5292 ^a 5	
11773 5	3/2 ⁻ , 7/2 ⁻	5392 ^a 5	
11783 5	3/2 ⁻	5403 ^a 5	
12900 ^b			T=3/2
13900 ^b			T=3/2

[†] From E_{c.m.}+S(α)(^{35}Cl), where S(α)(^{35}Cl)=6997.91 4 ([2021Wa16](#)). E_{c.m.}=E_α(lab)×m(^{31}P)/[m_α+m(^{31}P)].

[‡] From [1975Sc40](#) by analyzing measured angular distributions, unless otherwise noted.

From [1964Ku12](#).

@ From [1971Mc23](#).

& From [1971Mc23](#), unless otherwise noted.

^a From [1975Sc40](#).

^b From [1970Um01](#). Two ^{35}Cl resonances are found in $^{34}\text{S}(\text{p}, \text{n})$ and $^{31}\text{P}(\alpha, \text{n})$, but not in $^{31}\text{P}(\alpha, \alpha_0)$ reactions. T=1 ^{34}S g.s. and T=1/2 protons can populate either T=1/2 or T=3/2 states in ^{35}Cl . T=1/2 ^{31}P g.s. and T=0 α particles can populate T=1/2 states in ^{35}Cl . A small isospin impurity would allow formation of T=3/2 states in ^{35}Cl . The lack of resonances in $^{31}\text{P}(\alpha, \alpha_0)$ indicates that these two resonances are most likely T=3/2. A T=3/2 resonances in $^{31}\text{P}(\alpha, \alpha_0)$ scattering is doubly isospin forbidden (both entrance and exit channels) compared with singly forbidden (entrance channel only) for $^{31}\text{P}(\alpha, \text{n})$.

$^{31}\text{P}(^6\text{Li},\text{d})$ [1979Es05](#)

$^{31}\text{P}+\alpha\rightarrow^{35}\text{Cl}$ from $J^\pi=1/2^+$ ^{31}P ground state.

[1979Es05](#),[1978Es01](#): a 36-MeV $^6\text{Li}^{3+}$ beam at 150-400 nA was produced from the MP tandem accelerator at the University of Rochester and bombarded a $100\text{-}\mu\text{g}/\text{cm}^2$ cadmium phosphide Cd_3P_2 target. Outgoing deuterons were momentum analyzed by an Enge split-pole magnetic spectrograph and detected using Kodak NTB-50 nuclear emulsion plates with FWHM=50 keV. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, J, π , L-transfers, spectroscopic factors from finite-range, LOLA-DWBA analysis of the measured $\sigma(\theta)$. Other: [1982Ch10](#) (finite-range DWBA calculations).

 ^{35}Cl Levels

Spectroscopic factor $g^*\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}$, where $g=(2J_{\text{f}}+1)/(2J_{\text{i}}+1)$.

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>L[#]</u>	<u>S[#]</u>
0	$3/2^+$	2	1.00
1220	$1/2^+$	0	1.49
1760	$5/2^+$	2	0.24
3160	$7/2^-$	3	0.52

[†] From [1979Es05](#).

[‡] As given in [1979Es05](#) for extracting spectroscopic factors.

[#] From DWBA analysis of the measured $\sigma(\theta)$ in [1979Es05](#).

$^{32}\text{S}(\alpha, \text{p})$ 2019Se09, 1973Go16

$J^\pi=0^+$ for ^{32}S ground state.

2019Se09: a 21-MeV α beam was produced from the 10-MV FN tandem Van de Graaff accelerator at Triangle Universities Nuclear Laboratory (TUNL). Targets were $15.9\text{-}\mu\text{g}/\text{cm}^2$ natS in cadmium sulfide and $46.3\text{-}\mu\text{g}/\text{cm}^2$ natS in antimony sulfide. Reaction products were momentum-analyzed by the TUNL high-resolution Enge split-pole magnetic spectrograph and detected using a high-resolution position-sensitive focal-plane detector consisting of two position-sensitive gas avalanche counters: a gas proportionality energy loss section and a residual energy scintillator (FWHM=24 keV). Measured $\sigma(E_p, \theta)$, at $\theta=10^\circ$ to 50° . Deduced levels, J, π from one-step finite range triton transfer FRESKO-DWBA analysis. Deduced the thermonuclear $^{34}\text{S}(\text{p}, \gamma)^{35}\text{Cl}$ reaction rate via Monte Carlo methods.

1973Go16: 19.991-, 20.025- and 20.284-MeV α beams were produced from the University of Notre Dame FN tandem Van de Graaff accelerator. Natural sulfur targets were prepared by vacuum evaporation of CdS onto $20\text{-}\mu\text{g}/\text{cm}^2$ carbon backings. Protons were analyzed with broad-range magnetic spectrographs at 15° , 30° , 70° , 90° , and 120° and detected using nuclear track plates at the focal planes. Measured $\sigma(E_p, \theta)$. Deduced 102 levels.

1971Au07: a 28.8-MeV α beam of 0.1-0.5 μA was produced from the Naval Research Laboratory (NRL) isochronous cyclotron. The target was made by evaporating natural PbS onto a $30\text{-}\mu\text{g}/\text{cm}^2$ carbon backing, and the effective ^{32}S thickness was $163\text{ }\mu\text{g}/\text{cm}^2$. Protons were detected using two counter telescopes with FWHM=250 keV. Measured $\sigma(E_p, \theta)$. Deduced levels, J, π , L-transfers, and J-dependence for the g.s. and 1763-keV level from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

Other: **1996So21** (cross sections).

 ^{35}Cl Levels

E(level) [†]	$J^\pi \&$	L^a	Comments
0		2	J^π : $3/2^+$ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer.
1217.4 26			
1760.1 24		(2)	J^π : $5/2^+$ assumed in 1971Au07 for investigation of J-dependence of L=2 transfer.
			L: The parentheses are added by evaluators based on the DWBA fit in 1971Au07 .
2644.9 23			
2693?			
3001.5 31			
3158.2 25			
3945.2 23			E(level): weighted average of 3943.7 23 (1973Go16) and 3947.9 31 (2019Se09).
4056.9 27	$(3/2^-)^b$		E(level): 4059.12 quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
4111.1 29	$(7/2^+)^b$		E(level): weighted average of 4110.2 29 (1973Go16) and 4112.0 30 (2019Se09).
4177.9 24	$(3/2^-)^b$		E(level): weighted average of 4177.5 24 (1973Go16) and 4179 4 (2019Se09).
4346.6 24			E(level): 4347.82 quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
4625.3 28			E(level): weighted average of 4624.4 31 (1973Go16) and 4626.1 28 (2019Se09).
4770.4 26			E(level): weighted average of 4770.8 26 (1973Go16) and 4769.9 28 (2019Se09).
4859.2 [‡] 28			
4881.8 29			E(level): weighted average of 4883.1 29 (1973Go16) and 4878 5 (2019Se09).
5006.9 [‡] 27			
5158.9 29	$(5/2^+)^b$		E(level): weighted average of 5161.7 33 (1973Go16) and 5156.8 29 (2019Se09).
5209.5 28	$(5/2^+)$		E(level): weighted average of 5206.6 37 (1973Go16) and 5211.1 28 (2019Se09).
5402.0 29			
5407.2	$(11/2^-)^b$		E(level): quoted from 2011Ch48 and used for internal calibrations by 2019Se09 .
5531 [‡] 5			
5586.0 26			E(level): 5576 tentative level from 1973Go16 .
5594.6 26			E(level): weighted average of 5591.8 32 (1973Go16) and 5596.5 26 (2019Se09).
5633.6 30			E(level): weighted average of 5633.1 32 (1973Go16) and 5634 3 (2019Se09).
5645 [‡] 4			
5653 [‡] 4			
5679.7 34			E(level): weighted average of 5677.7 34 (1973Go16) and 5684 5 (2019Se09).
5731.2 [‡] 25			
5757 [‡] 4			

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$^{32}\text{S}(\alpha, p)$ **2019Se09, 1973Go16 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J^{π} &	Comments
5808.1 31		E(level): weighted average of 5809.2 34 (1973Go16) and 5807.1 31 (2019Se09).
5829 $\frac{1}{2}^{-}$ 4	(5/2 ⁻) ^b	E(level): 5823 tentative level from 1973Go16.
5928 $\frac{1}{2}^{-}$ 3		E(level): other: 5927.4 35 (1973Go16).
6084.2 29	(13/2 ⁻) ^b	E(level): 6087.4 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
6104 $\frac{1}{2}^{-}$ 4		
6141 4		E(level): weighted average of 6140.2 40 (1973Go16) and 6142 4 (2019Se09).
6180 $\frac{1}{2}^{-}$ 3		
6224.4 30		E(level): weighted average of 6224.9 36 (1973Go16) and 6224 3 (2019Se09).
6283 $\frac{1}{2}^{-}$ 3	(5/2 ⁺)	
6379.3 25	(9/2 ⁻)	E(level): from 2019Se09. Other: 6379.0 34 (1973Go16).
6401.2 22	(1/2 ⁻)	E(level): weighted average of 6402.4 41 (1973Go16) and 6400.9 22 (2019Se09).
6429 $\frac{1}{2}^{-}$ 3	(1/2 ⁺)	E(level): 6427 tentative level from 1973Go16.
6475 $\frac{1}{2}^{-}$ 4		
6492 $\frac{1}{2}^{-}$ 3	(3/2 ⁺)	E(level): other: 6491.9 34 (1973Go16).
6549 $\frac{1}{2}^{-}$ 3	(1/2 ⁺)	
6659.1 30	(7/2 ⁺)	E(level): weighted average of 6656.0 31 (1973Go16) and 6662 3 (2019Se09).
6679.4 31	(1/2 ⁺)	E(level): weighted average of 6680.8 31 (1973Go16) and 6677 4 (2019Se09).
6781.3 30	(3/2 ⁻)	E(level): weighted average of 6782.8 32 (1973Go16) and 6780 3 (2019Se09).
6800 4		E(level): weighted average of 6802.1 42 (1973Go16) and 6795 6 (2019Se09).
6842 $\frac{1}{2}^{-}$ 4	(3/2 ⁺)	
6863 $\frac{1}{2}^{-}$ 3	(9/2 ⁺)	E(level): 6867 tentative level from 1973Go16.
6891.6 30	(9/2 ⁺)	E(level): weighted average of 6893.5 32 (1973Go16) and 6890 3 (2019Se09).
6948.6 34	(5/2 ⁺)	E(level): weighted average of 6947.5 34 (1973Go16) and 6950 4 (2019Se09).
6986.3 34	(9/2 ⁺)	E(level): weighted average of 6985.7 34 (1973Go16) and 6987 4 (2019Se09).
7066.2	(5/2 ⁺)	E(level): 7066.2 quoted from 2011Ch48 and used for internal calibrations by 2019Se09.
7105?		
7121.7 [#] 24	(5/2 ⁻)	E(level): weighted average of 7120.8 34 (1973Go16) and 7122.1 24 (2019Se09).
7180 $\frac{1}{2}^{-}$ 4	(7/2 ⁺)	E(level): other: 7179.0 43 (1973Go16).
7213.7 25		E(level): weighted average of 7210.1 39 (1973Go16) and 7215.2 25 (2019Se09).
7230.9 30	(3/2 ⁺)	E(level): weighted average of 7229.5 34 (1973Go16) and 7232 3 (2019Se09).
7301?		
7348 $\frac{1}{2}^{-}$ 3	(7/2)	E(level): other: 7348.3 44 (1973Go16).
7362 $\frac{1}{2}^{-}$ 3		
7413.8 30		E(level): weighted average of 7417.5 44 (1973Go16) and 7412 3 (2019Se09).
7447 $\frac{1}{2}^{-}$ 3		
7503 4		E(level): weighted average of 7503.0 37 (1973Go16) and 7502 5 (2019Se09).
7566.4 35		E(level): weighted average of 7567.6 35 (1973Go16) and 7564 5 (2019Se09).
7590 4		E(level): weighted average of 7587.2 40 (1973Go16) and 7595 6 (2019Se09).
7648.1 30		E(level): weighted average of 7650.1 40 (1973Go16) and 7647 3 (2019Se09).
7674.6 $\frac{1}{2}^{-}$ 25		E(level): 7671 tentative level from 1973Go16.
7709.9 $\frac{1}{2}^{-}$ 25		
7730?		
7744.1 [#] 30		E(level): weighted average of 7742.7 38 (1973Go16) and 7745 3 (2019Se09).
7770.3 33		E(level): weighted average of 7776.3 45 (1973Go16) and 7768.5 25 (2019Se09).
7799.8 30		E(level): weighted average of 7800.8 35 (1973Go16) and 7799 3 (2019Se09).
7868.5 35		E(level): from 1973Go16. Other: 7867 7 (2019Se09).
7901 4		E(level): weighted average of 7900.6 38 (1973Go16) and 7899 12 (2019Se09).
7978.6 35		
8000.7 41		
8019.2 35		
8075.1 46		

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$^{32}\text{S}(\alpha, \text{p})$ **2019Se09,1973Go16** (continued) ^{35}Cl Levels (continued)

E(level) [†]	E(level) [†]	E(level) [†]	E(level) [†]
8103.8 [#] 42	8654.1 41	9334.0 46	10117?
8121?	8698?	9376.0 41	10162.7 53
8171.7 42	8720.9 [#] 37	9475.5 43	10179.1 53
8210.2 46	8786.3 38	9508.1 [#] 41	10218.0 46
8246.0 42	8840.3 38	9525?	10395.2 54
8270.6 [#] 42	8887.5 41	9711.8 44	10443?
8292?	8957.5 44	9740.4 40	10463.2 [#] 49
8321.0 38	8996.9 40	9799.2 47	10517.1 54
8398.8 38	9027.4 38	9836.1 44	10548.3 54
8429.2 40	9085.7 38	9870?	10579.5 54
8482.3 38	9109.3 49	9968.6 [@] 52	10643.0 50
8505.1 47	9160.0 42	9992?	10717?
8532.4 43	9194.0 39	10009?	10732.1 [#] 50
8567.9 37	9264.9 39	10075.2 [@] 48	10759.1 55
8616.9 37	9316.2 39	10098?	

[†] From **1973Go16**, unless otherwise noted.[‡] From **2019Se09**.[#] Doublet in **1973Go16**.[@] Triplet in **1973Go16**.[&] From **2019Se09**, based on FRESKO-DWBA analysis of measured $\sigma(\theta)$, unless otherwise noted.^a From **1971Au07**.^b Original firm assignments in **2019Se09** are placed in parentheses by the evaluators when DWBA fits to the measured $\sigma(\theta)$ were poor or when only a few data points were available for fitting.

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

$J^\pi=0^+$ for ^{32}S ground state.

1970Ho09: 10.975-, 11.960-, 12.120-MeV α beams were produced from the Chalk River MP tandem accelerator. Targets were $100\text{-}\mu\text{g}/\text{cm}^2$ Sb_2S_3 on a $10\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. γ rays were detected using 15.2-cm-long by 12.7-cm-diam NaI(Tl) detectors. Protons were detected using an annular silicon surface barrier detector close to 180° with FWHM=30 keV for protons and FWHM=50 keV for α . Measured E_p , E_γ , I_γ , $p\gamma(\theta)$. Deduced 25 level energies from E_p , J, branching ratios, multipole mixing ratios.

1972Br33: a 10-MeV α beam was produced from the Chalk River MP tandem accelerator. Targets were $340\text{-}\mu\text{g}/\text{cm}^2$ Sb_2S_3 on gold backings. γ rays were detected using a 44-cm³ Ge(Li) detector inside an anticoincidence split annular NaI(Tl) detector. Measured E_γ , I_γ . Deduced 11 level energies, 16 γ branching ratios, 11 lifetimes using the Doppler Shift Attenuation Method (DSAM). **1970In01** from the same experiment deduced the lifetime using DSAM for a level at 3942 keV and the mixing ratio for a $3942 \rightarrow 2646$ γ transition.

1973Al22: a 14-MeV α beam was produced from the 80-cm cyclotron at the Research Institute for Physics in Stockholm and 12-16-MeV α beams were produced from the Uppsala tandem accelerator. Targets were prepared by depositing pulverized sulphur enriched to 99.9% in ^{32}S on lead backings. γ rays were detected using an Ortec 60-cm³ Ge(Li) detector with FWHM=2.2 keV at 1333 keV and an additional Canberra 60-cm³ Ge(Li) detector for $\gamma\gamma$ -coincidence measurements. Measured E_γ , I_γ , $\gamma\gamma$ -coin, $\gamma(\theta)$ for 4 γ rays. Deduced 12 level energies from E_γ , J for 4 levels, branching ratios, and mixing ratios.

1969In04: 6-8-MeV α beams were produced from the Chalk River MP tandem accelerator. A thick target ($270\text{ }\mu\text{g}/\text{cm}^2$) of natural sulfur on a tantalum backing and a thin target ($50\text{ }\mu\text{g}/\text{cm}^2$) of Sb_2S_3 on a gold backing were used. γ rays were detected using two Ge(Li) detectors of 15 and 40 cm³ volume. Measured E_γ , I_γ . Deduced 6 level energies, branching ratios, 5 lifetimes using the Doppler Shift Attenuation Method (DSAM) and 1 lifetime for the 3163-keV level using the Recoil Distance Method (RDM), and transition strengths.

1974Lo17: 11.0, 12.0 and 14.7-MeV α beams were produced from the Oliver Lodge Laboratory at the University of Liverpool. Targets were natural CdS. Protons were detected using an annular ΔE -E telescope close to 180° with FWHM=80 keV. γ rays were detected using five NaI(Tl) detectors and one Ge(Li) detector. Measured E_γ , I_γ , $p\gamma$ -coin, $p\gamma(\theta)$, $\gamma\gamma$ -coin, γ -ray linear polarization. Deduced J and π for 3 levels, 6 γ branching ratios, 5 mixing ratios, and lifetimes for a 5408-keV level using DSAM and for a 6088-keV level using RDM.

1973Br26: a 11.2-MeV α beam was produced from the Legnaro (Padova) Van de Graaff generator. Targets were natural CdS. γ rays were detected using a Ge(Li) detector. Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

1973An01: 5.5- and 8.6-MeV α beams of 100-200 nA were produced from the Oxford Vertical Van de Graaff (Injector) or the Oxford EN Tandem. Targets were natural sulphur in AgS. γ rays were detected using a Ge(Li) detector at 0° . Deduced the lifetime using the Recoil Distance Method (RDM) for a level at 3163 keV.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
0	$3/2^+$		J^π : from the Adopted Levels.
1219.4 6	$1/2^+$	0.15 ps +7-6	J^π : from the Adopted Levels. $T_{1/2}$: Lifetime $\tau=0.21$ ps +10-8 (1969In04 DSAM).
1763.3 6	$5/2^+$	0.38 ps 11	J^π : spin=5/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity=+ from 1763.2 γ M1+E2 to $3/2^+$. $T_{1/2}$: Lifetime $\tau=0.55$ ps 15 from 1969In04 using DSAM.
2645.8 8	$7/2^+$	0.21 ps 6	J^π : spin=7/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 882 γ , M1+E2 to $5/2^+$ in 1970Ho09 . $T_{1/2}$: Lifetime $\tau=0.30$ ps 9 from 1969In04 using DSAM.
2693.6 13	$3/2^+, 5/2^+$	49 fs 21	$T_{1/2}$: Lifetime $\tau=70$ fs 30 from 1972Br33 using DSAM. Other: $\tau < 30$ fs (1969In04 , DSAM).
3002.6 15	$(5/2)^+$	33 fs 6	J^π : spin=3/2, 5/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 3002.5 γ , M1+E2 to $3/2^+$ in 1970Ho09 ; J=3/2 is disfavored since it would give a large $\delta(E2/M1)=1.2$ 5 for 3002.5 γ in 1970Ho09 which is less likely and also due to the absence of an expected γ to 1219.5, $1/2^+$ level if J=3/2. $T_{1/2}$: Lifetime $\tau=48$ fs 9 from 1972Br33 using DSAM. Other: $\tau < 50$ fs (1969In04 , DSAM).
3163.1 11	$7/2^-$	42 ps 5	J^π : spin=7/2 from $p\gamma(\theta)$ in 1970Ho09 ; parity from 3162.6 γ , M2+E3 to $3/2^+$ in 1970Ho09 .

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
3915 2		<15.3 fs	T _{1/2} : Lifetime $\tau=60$ ps 7 from 1969In04 . E(level): from 1972Br33 .
3943.4 12	9/2	0.229 ps 35	T _{1/2} : Lifetime $\tau<22$ fs from 1972Br33 using DSAM.
3968 2		<37 fs	T _{1/2} : Lifetime $\tau=0.33$ ps 5 from 1972Br33 using DSAM. E(level): from 1972Br33 .
4058 3		22 fs 9	T _{1/2} : Lifetime $\tau<54$ fs from 1972Br33 using DSAM.
4108 2	3/2, 7/2	49 fs 11	E(level): weighted average of 4045 10 (1970Ho09) and 4058 2 (1972Br33). T _{1/2} : Lifetime $\tau=31$ fs 13 from 1972Br33 using DSAM.
4171 2		47 fs 17	E(level): weighted average of 4110 10 (1970Ho09) and 4108 2 (1972Br33). T _{1/2} : Lifetime $\tau=70$ fs 16 from 1972Br33 using DSAM.
4347.8 12	9/2 ⁻	2.0 ps +10-5	E(level): weighted average of 4175 10 (1970Ho09) and 4171 2 (1972Br33). T _{1/2} : Lifetime $\tau=68$ fs 24 from 1972Br33 using DSAM.
4618 2	3/2, 5/2	40 fs 17	J ^π : spin from $p\gamma(\theta)$ in 1974Lo17 ; parity from 1184.6 γ , M1+E2 to 7/2 ⁻ in 1974Lo17 . Other: J=5/2, 9/2 from $p\gamma(\theta)$ in 1970Ho09 and $p\gamma(\theta)$ in 1973Al22 . T _{1/2} : Lifetime $\tau=2.9$ ps +14-7 from 1972Br33 using DSAM.
4770 [#] 15	7/2, 9/2		E(level): from 1972Br33 .
4880.8 21	7/2	0.19 ps 4	T _{1/2} : Lifetime $\tau=58$ fs 24 from 1972Br33 using DSAM.
5015 [#] 20			E(level): other: 4882 2 from 1972Br33 , 4885 20 from 1970Ho09 .
5175 [#] 20			T _{1/2} : Lifetime $\tau=0.28$ ps 6 from 1972Br33 using DSAM.
5230 [#] 20	1/2, 3/2, 5/2		
5407.1 15	11/2 ⁻	0.28 ps 7	E(level): other: 5415 20 from 1970Ho09 . J ^π : spin from $p\gamma(\theta)$ in 1974Lo17 and from $\gamma(\theta)$ and yield function in 1973Al22 ; parity from 1059.3 γ , M1+E2 to 9/2 ⁻ in 1974Lo17 . T _{1/2} : Lifetime $\tau=0.4$ ps 1 from 1974Lo17 using DSAM.
5535 [#] 20			
5600 [#] 20			
5650 [#] 20	5/2, 9/2		
5850 [#] 25	5/2, 9/2		
5927.1 19	7/2, 11/2		E(level): other: 5930 25 from 1970Ho09 . J ^π : spin from $\gamma(\theta)$ in 1973Al22 .
6087.0 18	13/2 ⁻	6.5 ps 6	J ^π : spin from $p\gamma(\theta)$ in 1974Lo17 ; parity from 679.9 γ , M1+E2 to 11/2 ⁻ in 1974Lo17 . Other: J=9/2, 13/2 from $\gamma(\theta)$ in 1973Al22 . T _{1/2} : Lifetime $\tau=9.3$ ps 8 from 1974Lo17 using RDM.
6400 [#] 25			

[†] From a least-squares fit to E_γ values with uncertainties, unless otherwise noted.[‡] Spin assignments are from $p\gamma(\theta)$ in [1970Ho09](#), unless otherwise noted.[#] From [1970Ho09](#), derived from measured E_p. $\gamma(^{35}\text{Cl})$

E _i (level)	J ^π _i	E _γ [†]	I _γ [‡]	E _f	J ^π _f	Mult. [#]	δ [#]	Comments
1219.4	1/2 ⁺	1219.4 6	100	0	3/2 ⁺			E _γ : weighted average of 1219.3 6 (1969In04) and 1219.5 10 (1973Al22).
1763.3	5/2 ⁺	543.9	<0.5	1219.4	1/2 ⁺			
		1763.2 6	100	0	3/2 ⁺	M1+E2	3.0 1	E _γ : from 1969In04 . Other: 1763.1 10 (1973Al22).
2645.8	7/2 ⁺	882.4 10	11.1 22	1763.3	5/2 ⁺	M1+E2	+0.25 5	I _γ : from 1970Ho09 . Other: 18 (1969In04), 10 (1973Al22).

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$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	Comments
2645.8	$7/2^+$	1426.4 2646.2 15	<1.2 100.0 22	1219.4 0	$1/2^+$ $3/2^+$			E_γ : from 1969In04. Other: 2646.0 20 (1973Al22). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). I_γ : from 1970Ho09. I_γ : from 1970Ho09. Other: 10 (1969In04). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : $\delta=+0.17$ 8 for $J=3/2$; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).
2693.6	$3/2^+, 5/2^+$	930.5 15 1474.2 2693.2 20	18.2 26 11.7 26 100 4	1763.3 1219.4 0	$5/2^+$ $1/2^+$ $3/2^+$	M1+E2		I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22). δ : $\delta=+0.17$ 8 for $J=3/2$; $\delta=-0.26$ 5 for $J=5/2$ (1970Ho09).
3002.6	$(5/2)^+$	309.0 356.8 1239.3 1783.2 3002.5 15	<5 <1 <2 <2 100	2693.6 2645.8 1763.3 1219.4 0	$3/2^+, 5/2^+$ $7/2^+$ $5/2^+$ $1/2^+$ $3/2^+$	M1+E2	+0.09 3	I_γ : other: <4 (1972Br33). I_γ : from 1969In04, 1972Br33. E_γ : weighted average of 3003 2 (1969In04), 3003 2 (1972Br33), and 3002.0 15 (1973Al22). δ : other: 1.2 5 if $J(3002.6)=3/2$ (1970Ho09).
3163.1	$7/2^-$	160.5 469.5 & 517.4 15 1399.8 3162.6 15	<0.54 <0.54 11.1 33 <1.1 100.0 33	3002.6 2693.6 2645.8 1763.3 0	$(5/2)^+$ $3/2^+, 5/2^+$ $7/2^+$ $5/2^+$ $3/2^+$	M2+E3	-0.14 2	I_γ : from 1970Ho09. Other: 8.7 (1969In04), 11 (1973Al22). I_γ : from 1970Ho09. Other: 100 (1969In04), 100 (1973Al22).
3915		2696 3915	19 5 100 5	1219.4 0	$1/2^+$ $3/2^+$			
3943.4	$9/2$	1297.4 15 2180.2 15	8.1 21 100.0 21	2645.8 1763.3	$7/2^+$ $5/2^+$			I_γ : weighted average of 9.9 22 (1970Ho09) and 6.4 21 (1972Br33). Other: 11 (1973Al22). I_γ : others: 100.0 22 (1970Ho09), 100 (1973Al22).
3968		2749 3968	100 <2	1219.4 0	$1/2^+$ $3/2^+$			
4058		2295 2839 4058	8 4 100 4 <2.2	1763.3 1219.4 0	$5/2^+$ $1/2^+$ $3/2^+$			
4108	$3/2, 7/2$	2345 4108	16 5 100 5	1763.3 0	$5/2^+$ $3/2^+$			
4347.8	$9/2^-$	1184.6 10 1702.1 15	100.0 31 52 7	3163.1 2645.8	$7/2^-$ $7/2^+$	M1+E2 E1(+M2)	-0.36 3 0.00 3	I_γ : from 1974Lo17. Others: 100 7 (1970Ho09) and 100 4 (1972Br33). δ : from 1974Lo17. Other: -0.40 +8-9 (1973Al22); +0.42 4 (1970Ho09). $A_2=-0.95$ 3, $A_4=+0.08$ 3 (1974Lo17). $\text{POL}=-0.005$ 40 (1974Lo17). I_γ : unweighted average of 61 7 (1970Ho09), 39 4 (1972Br33), and 56.3 31 (1974Lo17). Mult., δ : D(+Q) and δ from $p\gamma(\theta)$ and electric character from $\gamma(\text{pol})$ in 1974Lo17. $A_2=-0.34$ 5, $A_4=+0.04$ 5 (1974Lo17). $\text{POL}=+0.36$ 7 (1974Lo17).

Continued on next page (footnotes at end of table)

$^{32}\text{S}(\alpha, p\gamma)$ **1970Ho09, 1972Br33, 1973Al22 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
4618	3/2, 5/2	4618 2	100	0	3/2 ⁺			E_γ, I_γ : from 1972Br33. δ : +0.08 14 for J=3/2; -0.36 15 for J=5/2 (1970Ho09).
4880.8	7/2	2187.1 @	52 @ 9	2693.6	3/2 ⁺ , 5/2 ⁺			E_γ, I_γ : Final level is uncertain, either 2646 or 2694 (1970Ho09).
		2234.9 @	52 @ 9	2645.8	7/2 ⁺			E_γ, I_γ : Final level is uncertain, either 2646 or 2694 (1970Ho09).
		3117.4 20	100 9	1763.3	5/2 ⁺			I_γ : from 1970Ho09. Other: 100 (1973Al22).
5230	1/2, 3/2, 5/2	5230	100	0	3/2 ⁺			
5407.1	11/2 ⁻	1059.3 10	9.9 22	4347.8	9/2 ⁻	M1+E2	+0.25 8	Mult., δ : D+Q and δ from $\gamma(\theta)$ and magnetic character from $\gamma(\text{pol})$ (1974Lo17). $A_2=+0.19$ 11, $A_4=-0.01$ 13 (1974Lo17). POL=-0.40 14 (1974Lo17). Mult.: Q(+O) with $\delta(\text{O/Q})=0.00$ 3 from $\text{py}(\theta)$ in 1974Lo17; M2 ruled out by RUL. Other: Q(+O) with $\delta(\text{O/Q})=+0.02$ 9 (1973Al22). $A_2=+0.44$ 3, $A_4=-0.19$ 3 (1974Lo17).
		2244 3	100.0 22	3163.1	7/2 ⁻	E2		
5850	5/2, 9/2	2687	100	3163.1	7/2 ⁻	D+Q		Mult., δ : from $\text{py}(\theta)$ in 1970Ho09 with $\delta(\text{Q/D})=-2.2$ 5 or -0.6 1 for J=5/2; +0.28 4 for J=9/2 (1970Ho09).
5927.1	7/2, 11/2	1579.3 15	100	4347.8	9/2 ⁻	D+Q		Mult., δ : from $\gamma(\theta)$ in 1973Al22, with $\delta(\text{Q/D})=+1.1$ 5 for J=7/2; -0.8 4 for J=11/2.
6087.0	13/2 ⁻	679.9 10	100 6	5407.1	11/2 ⁻	M1+E2	+0.020 15	Mult., δ : D+Q and δ from $\text{py}(\theta)$ and magnetic character from $\gamma(\text{pol})$ in 1974Lo17. Other: $\delta=0.00$ 7 for $J_i=13/2$ (1973Al22). $A_2=-0.22$ 1, $A_4=0.005$ 15 (1974Lo17). POL=-0.45 5 (1974Lo17).
		1739.2	18 6	4347.8	9/2 ⁻			

[†] Values with uncertainties are from 1973Al22 and those without uncertainties are from level-energy differences, unless otherwise noted. 1973Al22 is the only literature in this dataset that reported measured E_γ with uncertainties. 1969In04 and 1972Br33 reported $E(\text{level})$ with uncertainties derived from their measured E_γ but did not report E_γ . Evaluators adopt the uncertainties of $E(\text{level})$ from 1969In04 and 1972Br33 as the uncertainties of an E_γ if this γ transition populates the ground state and is also the dominant decay branch of that level.

[‡] From 1969In04 for γ rays from level up to 3163, from 1972Br33 for γ rays from level above 3163 up to 4618 level, and from 1974Lo17 for γ rays from level above 4618, unless otherwise noted.

[#] Mult=D+Q, Q or Q+O and δ are determined from $\text{py}(\theta)$ in 1970Ho09; magnetic or electric characters are determined based on RUL and measured level $T_{1/2}$ where available, unless otherwise noted.

@ Multiply placed with undivided intensity.

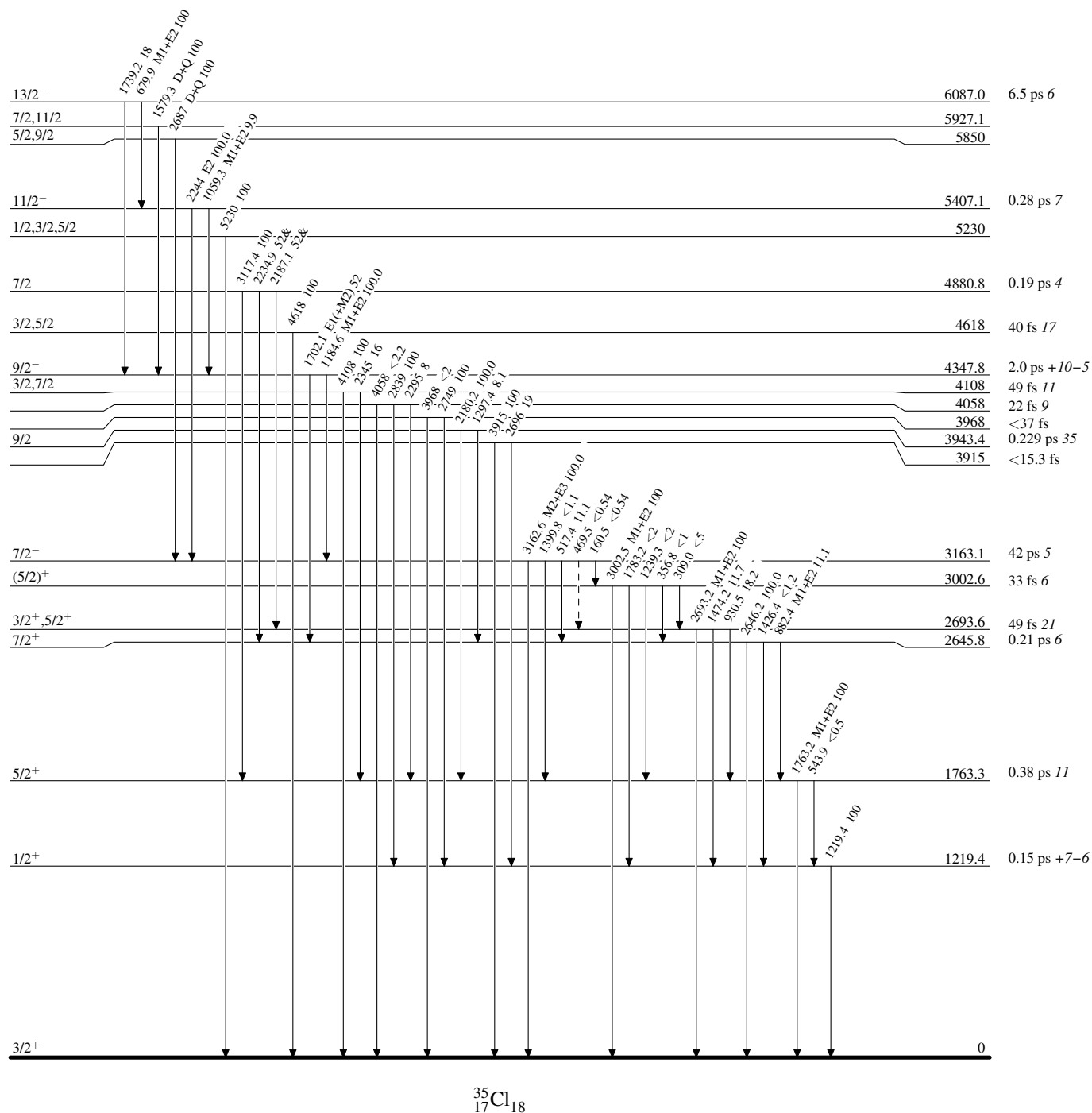
& Placement of transition in the level scheme is uncertain.

$^{32}\text{S}(\alpha, p\gamma)$ 1970Ho09, 1972Br33, 1973Al22

Legend

Level Scheme

Intensities: Relative photon branching from each level
& Multiply placed: undivided intensity given

-----► γ Decay (Uncertain)

$^{33}\text{S}(\alpha, \text{d})$ [1975Na18,1977Na10](#)

$J^\pi=3/2^+$ for ^{33}S ground state.

[1975Na18,1977Na10](#): a 40-MeV α beam was produced from the Michigan State University cyclotron. Targets were 76.8%-enriched ^{33}S of $15 \mu\text{g}/\text{cm}^2$ sandwiched between layers of Formvar and carbon foils. Reaction products were detected in the focal plane of an Enge split-pole magnetic spectrograph with a proportional-counter plastic-scintillator combination with FWHM=40-60 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, J, π , and L-transfers.

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>L[†]</u>	<u>Comments</u>
6200 <i>10</i>	6	
7170 <i>10</i>	6	
7670 <i>10</i>	6	
7750 <i>10</i>	6	
7870 <i>10</i>	4+6	L: 1977Na10 states that this is excited by a mixture of L=4 and 6 transfers. The L=6 admixture can be attributed to a $(f_{7/2})^2_{7,0}$ transfer whereas the L=4 component is associated with a mixture of $(f_{7/2})^2_{5,0}$ and $(f_{7/2}, p_{3/2})_{5,0}$ transfers.
8010 <i>10</i>	6	
8100 <i>10</i>	6	
8700 <i>10</i>	6	
8840 <i>10</i>	6	
9150 <i>10</i>	6	
9450 <i>10</i>	6	

[†] From [1977Na10](#). Experimental angular distributions of L=4, 5, 6 to ^{42}Sc and ^{34}Cl levels with known J^π are compared with the $\sigma(E_d, \theta)$ measured in [1977Na10](#) to assign L-transfers.

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

$J^\pi=0^+$ for ^{34}S ground state.

- 1976Me12:** 0.7-2.1-MeV proton beams were produced from the 1.1-MV Cockcroft-Walton generator and a 3-MV Van de Graaff accelerator. Targets were ZnS (46% ^{34}S) and CdS (37% ^{34}S) on tantalum backings. γ rays were detected using a 10-cm by 10-cm NaI detector and a 40-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced level energies for 33 bound states and 47 resonances, J, π , resonance strengths, γ branching ratios, mixing ratios, and lifetimes for 25 bound states using DSAM.
- 1976Sp08:** 1.21, 1.95-2.91-MeV protons were produced from the Utrecht University 3-MV Van de Graaff accelerator. Targets were Ag²S (95% ^{34}S) evaporated onto a tantalum backing. γ rays were detected using a 10 cm by 10 cm diameter NaI scintillator and a 120- or 60-cm³ Ge(Li) detector. Measured reaction yields, E_γ . Deduced Q value, 84 resonance energies, resonance strengths, widths, J, π , and γ -branching ratios.
- 1976Sp09:** similar setup as **1976Sp08** with 2.33-2.79-MeV protons. Measured reaction yields, E_γ , $\gamma(\theta)$. Deduced levels, J, π , and γ -branching ratios.
- 1972Hu10:** 0.7-2.1-MeV protons were produced from the 4-MV Van de Graaff at the Centre d'Etudes Nucleaires de Bordeaux-Gradignan. Targets were 100- $\mu\text{g}/\text{cm}^2$ Ag₂S (90% ^{34}S). γ rays were detected using a 12.7-cm by 12.7-cm NaI(Tl) detector and a 60-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced 59 resonances, 32 bound states, resonance strengths, and γ branching ratios.
- 1972Hu11:** similar setup as **1972Hu10**. Deduced lifetimes using DSAM.
- 1973Fa07:** 1-2-MeV protons were produced from the Helsinki 2-MV Van de Graaff accelerator. Targets were ^{34}S on tantalum backings. γ rays were detected using a 55-cm³ Ge(Li) detector. Measured reaction yields, E_γ , I_γ . Deduced 28 resonances, resonance strengths, γ -branching ratios, and 9 lifetimes using DSAM.
- 2001Vo24:** 1.183-2.820-MeV protons were produced from the 4-MeV electrostatic Van de Graaff accelerator (NSC KhPEI). Enriched ^{34}S target (100%). A 63 cm³ Ge(Li) detector and a 15-cm-diam by 10-cm-long NaI(Tl) detector. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced 12 resonances and resonance strengths.
- 2002Vo17:** 2.187-MeV protons were produced from the ESU-4 accelerator (NSC KhPEI). Enriched ^{34}S target. A 63 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, resonance strengths, widths.
- 1981Bi05:** 1.4-2.8-MeV protons were produced from the Groningen 5-MV Van de Graaff accelerator. Targets were 4 and 1- $\mu\text{g}/\text{cm}^2$ Sb₂S₃ (90% in ^{34}S) evaporated onto 10- $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected using three silicon surface barrier detectors. γ rays were detected using a 10-cm by 10-cm NaI(Tl) detector and a Ge(Li) detector. Measured $\sigma(E_p, \theta)$, E_γ , $\gamma(\theta)$ for two γ rays. Deduced levels, J, π , widths, and resonance strengths.
- 1971Wi13:** protons were produced from the Ontario Cancer Institute. Targets were CdS. γ rays were detected using NaI(Tl) and Ge(Li) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -linear polarization. Deduced 1.020-, 1.214-, 1.267-, and 1.510-MeV resonances, J^π , branchings, mixing ratios, and half-lives using DSAM.
- 1971Pr11:** 1.211-MeV protons were produced from the 2-MeV ARL Van de Graaff at the Aerospace Research Laboratories (ARL). Targets were Ag₂S (85.6% ^{34}S). γ rays were detected using an 80-cm³ Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, $p\gamma$ -delay. Deduced resonances, levels, J, π , branchings.
- 1963Ha32:** 0.7-1.2-MeV protons were produced from the Laboratorium Technische Fysika. Target was ^{34}S . γ rays were detected using NaI(Tl) scintillation spectrometers. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced levels, branchings.
- 1967Da12:** 0.8-2.5-MeV protons were produced from the 2.5 MeV Van de Graaff generator of the U.A.R. Atomic Energy Establishment. Targets were ^{34}S on tantalum backings. γ rays were detected using two 15.2-cm-diam by 12.7-cm-thick NaI(Tl) crystals. Measured $\sigma(E_p)$, E_γ , I_γ . Deduced resonances, levels, branchings.
- 1960An06:** 0.7-1.9-MeV protons were produced from the electrostatic generator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were isotopic ^{34}S (4.2%). γ rays were detected using a 2-cm-thick by 3-cm-diam CsI(Tl) crystal. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced levels, J, branchings.
- 1966Wa09:** 1.214 MeV proton beam. Measured E_γ , γ -polarization. Deduced level energy and J, π for the level of 3163 and 7540 level.
- 1966En04:** 0.3-2.1-MeV protons were produced from the Utrecht 850-keV Cockcroft-Walton generator and the 3-MV Van de Graaff generator. Targets were natural sulphur compound on tantalum backings. γ rays were detected using a cylindrical 10-cm-by-10-cm NaI crystal. Measured $\sigma(E_p)$, E_γ . Deduced the resonance strength for $E_p=1214$ keV.
- 1966Az01:** 1.214-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS with enriched ^{34}S (46%). γ rays were detected using a Compton polarimeter. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J, π , branchings.
- 1967Wa22:** 1.214- and 1.512-MeV protons were produced from the Aerospace Research Laboratories. Targets were enriched CdS (40% ^{34}S) and Ag₂S (86% ^{34}S). γ rays were detected using two large volume NaI(Tl) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -polarization. Deduced levels, J, π , branchings.

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10 (continued)

- 1967Ta08:** 1.020-, 1.213- and 1.513-MeV protons were produced from the 3-MV Van de Graaff accelerator of the Ontario Cancer Institute. Targets were CdS (46% ^{34}S) on thick metal foils. γ rays were detected using a 7.6-cm-long by 7.6-cm-diam NaI(Tl) detector. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$, γ -polarization. Deduced levels, J , π , branchings, mixing ratios.
- 1977Ko35:** 1.055, 1.183, 1.214, and 1.416 MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Target of ^{34}S . A 40 cm³ coaxial Ge(Li) detector. Measured $\sigma(E_p)$, E_γ , $\gamma(\theta)$. Deduced levels, branchings, mixing ratios.
- 1977Ko34:** 1.0-1.4 MeV protons. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π , branchings, mixing ratios.
- 1970Ko33:** 2.317-, 2.223-, and 2.336-MeV protons were produced from the 4-MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were ^{34}S on tantalum backings. γ rays were detected using a 12-cm³ coaxial Ge(Li) detector. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced 3 levels, J , and branching ratios.
- 1967Ko19,1967Ko20,1967Ko23:** 2.079-MeV protons were produced from the 4 MV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences. Targets were made by implanting ^{34}S ions in a tantalum substrate. γ rays were detected using a 70-mm by 50-mm and a 70-mm by 60-mm NaI(Tl) detectors. Measured E_γ , I_γ , $p\gamma\gamma(\theta)$. Deduced levels, J , π , branchings, mixing ratios.
- 1968Az02:** 1.214- and 1.513-MeV protons were produced from the Ontario Cancer Institute 3-MeV Van de Graaff accelerator. Targets were CdS (37% ^{34}S) evaporated onto gold backings. γ rays were detected using a 5-cm-diam by 5-cm-long NaI(Tl) crystal. Measured E_γ , $\gamma\gamma$ -delay. Deduced lifetimes of the level of 3163 keV using the delayed coincidence method and DSAM.
- 1969Mi23:** 2.079-MeV protons. ^{34}S target. Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, branchings.
- 1971Ko32,1971Ko46:** 0.8-2.6-MeV protons. Targets were made by evaporating ^{34}S ions onto tantalum backings. γ rays were detected using Ge(Li) and NaI(Tl) detectors. Measured $\sigma(E_p)$, E_γ , I_γ , and $p\gamma\gamma(\theta)$. Deduced levels, J , π , level widths, and branchings.
- 1971Ba23:** 1.214-MeV protons were produced from the 3 MeV Dynamitron of Carleton and Ottawa Universities. Target of CdS (93% ^{34}S) on a gold backing. A 28 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced lifetime of the 3163 keV level using the delayed coincidence method.
- 1974Ai04:** 1.214-MeV protons. Targets were Ag₂S, CdS, and ZnS of 300- $\mu\text{g}/\text{cm}^2$. Ge(Li) detector. Measured $\sigma(E_\gamma)$. Deduced resonance strengths.
- 1975Ke11:** 1.214-MeV protons were produced from the Helsinki University 2.5-MV van de Graaff. Targets of CdS and ZnS on tantalum backings. γ rays were detected using a 120 cm³ Ge(Li) detector with FWHM=2.9 keV at 2.6 MeV. Measured $\sigma(E_\gamma)$. Deduced resonance strengths.
- 1979Pa16:** 1.211-MeV protons were produced from the University of Melbourne 5U Pelletron accelerator. Natural sulphur target. NaI(Tl) and Ge(Li) detector. Measured $\sigma(E_p)$. Deduced resonance strength.
- 1988Va06:** 0.4-0.7 and 1.47 MeV protons were produced from the Utrecht 3 MV Van de Graaff accelerator. Targets of Ag₂S (90% ^{34}S) of 10 and 34 $\mu\text{g}/\text{cm}^2$ on Ta backings. A 80-cm³ coaxial Philips Ge(Li) or an γ -X intrinsic Ge detector, FWHM=1.85 and 1.75 keV at 1.33 MeV. Measured $\sigma(E_p)$, E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π , branchings, mixing ratios.
- 1988Co03:** 1.211-MeV protons were produced from the Geel 7-MV CN-type Van de Graaff. Targets of ^{34}S . Ge(Li) detector. Measured $\gamma\gamma$ -coin. Deduced weighting function.
- 1996Vo20:** 1.891, 1.900 and 2.070-MeV protons were produced by an electrostatic accelerator ESU-4 at the Physical and Technical Institute. Target made by implanting ^{34}S ions in a tantalum backing. A 63 cm³ Ge(Li) detector. Measured E_γ , I_γ . Deduced levels, gamma widths.
- 1996Ka21:** 1-3 MeV protons were produced from the electrostatic accelerator of the Institute of Physica and Technology (Kharkov). Targets were 20- $\mu\text{g}/\text{cm}^2$ Ag₂S. γ rays were detected using a 60 cm³ Ge(Li) detector and a 150×100-mm² NaI(Tl) as a monitor. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced J^π for 3 resonances, resonance strengths.
- 2000Vo21:** 1.794 and 2.275 MeV protons. Measured E_γ , I_γ . Deduced gamma widths, transition strengths.
- 2001Ka69:** 0.8-3 MeV protons produced from the ESU-4 accelerator (NSC KhPEI). Target of ^{34}S . A 15-cm-diam by 19-cm-long NaI(Tl) detector. Measured E_γ , I_γ . Deduced levels, transition strengths.
- Others: **1962Du05**, **1963Du08**, **1964Hy01**, **1966Ko22**, **1966Ko15**, **1967Ma48**, **1969Ko26**, **1969Ah02**, **1969KrZW**, **1972LeYW**, **1974Ii01**, **1974Bi16**, **1981Va02**.
- Theoretical studies: **2020Ri06** (shell model).
- Standard resonance strength $\omega\gamma=(2J_r+1)/[(2J_p+1)(2J_T+1)]\times\Gamma_\gamma\Gamma_p/\Gamma$, where J_r is the spin of the resonance, $J_p=1/2$ is the spin of the proton, and $J_T=0$ is the spin of the target ^{34}S ground state. The resonance strengths S from the original papers in this dataset are mainly obtained from resonance γ -ray yields and are defined without the $(2J_p+1)$ factor. Therefore, to obtain the standard resonance strength, all the resonance strengths provided in the comments below should be divided by 2.
- $S=21$ eV 3 at $E_p=1213$ measured by **1966En04** has been used as a normalization standard by **1971BrXT**, **1972Hu10**, **1973Fa07**,

³⁴S(p,γ) 1976Me12,1976Sp08,1972Hu10 (continued)

1974Bi16, and 1976Me12. 1971BrXT measured ³⁴S(p,p) elastic scattering and determined $\Gamma_p=4.6$ eV 10 and $\Gamma=5$ eV +4-2 from a Q^2 minimization analysis of the excitation function. 1971BrXT added $\Gamma_p\Gamma_\gamma/\Gamma=2.6$ eV 4 converted from the S measured by 1966En04 to constrain their Q^2 analysis and reported a final $\Gamma_p=5.8$ eV 7, $\Gamma_\gamma=4.7$ eV 17, and $\Gamma=10.5$ eV 15. 1974Bi16 measured ³⁴S(p,p) and determined $\Gamma_p=7.2$ eV 15 from a Q^2 minimization analysis of the excitation function and quoted $S=21$ eV 3 (1966En04) to deduce $\Gamma_\gamma=3.8$ eV 15 and $\Gamma=11.0$ eV 15. 1974AI04 determined an absolute $S_{1213}=9.7$ eV 9. 1975Ke11 determined an absolute $S_{1213}=9.8$ eV 10 and quoted 1974Bi16's relative ratio of $\Gamma_p/\Gamma_\gamma=7.2/3.9=1.9$ to deduce $\Gamma_p=3.8$ eV 13, $\Gamma_\gamma=2.0$ eV 13, and $\Gamma=5.8$ eV 13. 1976Sp09 measured the resonant absorption of the $E_p=2791$ resonance, which indicates that normalization using $S_{1213}=9.7$ eV gives a better agreement than that using $S_{1213}=21$ eV. 1981Bi05 determined an absolute $S=4.1$ eV 5 for the $E_p=1892$ resonance and quoted the relative ratio of S_{1213}/S_{1892} from 1978En02 to deduce $S=9.5$ eV 12 for $E_p=1213$ resonance. The $S_{1213}/S_{1892}=2.3$ quoted by 1981Bi05 was based on two S values: first, 1978En02 quoted $S=4.2$ eV for a 1892+1893 resonance doublet, which was scaled down by a factor of 21/9.7 from 1976Me12's original $S_{1892}=9.0$ eV; second, 1978En02 quoted $S_{1213}=9.7$ eV 7 from 1976Sp08, 1974AI04, and 1975Ke11. 1979Pa16 determined $S=9.0$ eV 10 using the ²⁷Al(p,γ)²⁸Si resonances as normalization standards. 1996Ka21 normalized all their S to $S_{1213}=9.7$ eV 7 and cited 1976Sp08 but listed this value in the "this study" column. 2001Vo24 normalized all S to $S_{1213}=9.7$ eV 7 and cited 1976Sp08 and 1975Ke11. However, 2001Vo24 also reported a different $S_{1213}=9.9$ eV 7 without giving details. Although 1976Sp08 is frequently cited as the source of $S_{1213}=9.7$ eV 7, they did not measure the absolute S value; instead, they calculated a weighted average of 1974AI04 and 1975Ke11.

In summary, the evaluators adopt $S_{1213}=9.5$ eV 9: weighted average of all independently measured S_{1213} : 9.7 eV 9 (1974AI04), 9.8 eV 10 (1975Ke11), and 9.0 eV 10 (1979Pa16). All resonance strengths given in the comments below have been normalized to this value by the evaluators. $\Gamma_p=4.6$ eV 10 and $\Gamma=5$ eV +4-2 from (p,p) excitation function analysis in 1971BrXT are likely the only width data that are not influenced by the questionable $S_{1213}=21$ eV 3.

³⁵Cl Levels

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
0	3/2 ⁺		
1219.39 7	1/2 ⁺	105 fs 14	E(level): weighted average of 1219.1 5 (1972Hu10) and 1219.3 1 (1976Me12), and 1219.44 7 (1981Bi05). T _{1/2} : lifetime $\tau=152$ fs 20: weighted average of 80 fs 40 (1971Wi13, DSAM), 175 fs 20 (1972Hu11, DSAM), 150 fs 20 (1973Fa07, DSAM), and 145 fs 30 (1976Me12, DSAM).
1763.14 4	5/2 ⁺	0.347 ps 63	E(level): weighted average of 1762.8 5 (1972Hu10), 1763.4 7 (1976Me12), and 1763.14 4 (1981Bi05). J ^π : from $\gamma(\theta)$ in 1971Wi13. T _{1/2} : lifetime $\tau=0.50$ ps 9: weighted average of 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), and 0.53 ps 9 (1976Me12, DSAM).
2645.70 8	(5/2,7/2)	0.146 ps 21	E(level): weighted average of 2645.6 5 (1972Hu10), 2644.7 13 (1976Me12), and 2645.71 8 (1981Bi05). J ^π : from $\gamma(\theta)$ in 1971Wi13. Other: 3/2 from $\gamma(\theta)$ in 1967Ko20. T _{1/2} : lifetime $\tau=0.210$ ps 30: weighted average of 0.255 ps 65 (1972Hu11, DSAM) and 0.200 ps 30 (1976Me12, DSAM).
2693.9 3	(3/2,5/2)	13.9 fs 21	E(level): weighted average of 2694.0 3 (1972Hu10), 2694.7 12 (1976Me12), and 2693.8 3 (1981Bi05). J ^π : from $\gamma(\theta)$ in 1971Wi13. T _{1/2} : lifetime $\tau=20$ fs 3: weighted average of 21 fs 3 (1972Hu11, DSAM), 19 fs 4 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM).
3002.75 8	5/2	11.1 fs 21	E(level): weighted average of 3002.7 7 (1972Hu10), 3003.7 8 (1976Me12), and 3002.74 8 (1981Bi05). J ^π : 5/2 from $\gamma(\theta)$ in 1967Ko20 and 1971Wi13. T _{1/2} : lifetime $\tau=16$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 14 fs 2 (1973Fa07, DSAM), and 16 fs 4 (1976Me12, DSAM).
3162.99 7	7/2 ⁻	25.7 ps 28	T=1/2 E(level): weighted average of 3162.5 5 (1972Hu10), 3163.9 7 (1976Me12), and 3162.99 6 (1981Bi05). J ^π : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, 1967Ta08, and 1971Wi13; parity from 3163γ, M2+E3 to 3/2 ⁺ ground state in 1966Wa09 and 1967Wa22.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	Comments
			T _{1/2} : lifetime $\tau=37$ ps 4 (1971Ba23, delayed coincidence). Others: 140 ps 40 (1968Az02, delayed coincidence), >0.6 ps (1968Az02, DSAM), >0.2 ps (1971Wi13, DSAM), >2 ps (1972Hu11, DSAM), >0.9 ps (1973Fa07, DSAM), and >1 ps (1976Me12). 1971Ba23 states that the earlier 1968Az02 value was obtained using NaI(Tl) detectors with broad time resolution and the discrepancy between 1968Az02 and 1971Ba23 is not unreasonable.
3918.48 10	3/2 ⁺	4.9 fs 14	E(level): weighted average of 3917.9 6 (1972Hu10), 3920.7 13 (1976Me12), and 3918.48 7 (1981Bi05). J ^π : spin=3/2 determined by the feeding 5162 γ (θ) from 5/2 ⁺ , 9081 level in 1976Sp09; parity=+ determined by 3918 γ , M1+E2 to 3/2 ⁺ ground state.
3942.9 2		0.159 ps 42	T _{1/2} : lifetime $\tau=7$ fs 2 (1972Hu11, DSAM). Other: <6 fs (1976Me12, DSAM). E(level): weighted average of 3942.6 9 (1972Hu10), 3944.1 11 (1976Me12), and 3942.9 2 (1981Bi05).
3967.3 6		11.8 fs 28	T _{1/2} : lifetime $\tau=0.23$ ps 6: weighted average of 0.39 ps 11 (1972Hu11, DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM). E(level): from 1972Hu10. Other: 3979.0 15 (1976Me12). 1976Me12 matched their 3979 level with the 3967 level observed in other work. No other work has observed a level at 3979 keV.
4059.2 4		13.9 fs 21	T _{1/2} : lifetime $\tau=17$ fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM). E(level): weighted average of 4058.4 8 (1972Hu10), 4059.4 4 (1976Me12), and 4059.2 3 (1981Bi05).
4113.7 10	7/2		T _{1/2} : lifetime $\tau=20$ fs 3: weighted average of 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM). E(level): weighted average of 4111 3 (1972Hu10) and 4114.0 10 (1976Me12). J ^π : spin=7/2 from feeding 4516 γ (θ) from 7/2, 8630 level in 1976Sp09.
4173.45 19		34.7 fs 63	T=1/2 E(level): weighted average of 4173.0 6 (1972Hu10), 4174.7 10 (1976Me12), and 4173.45 19 (1981Bi05). T _{1/2} : lifetime $\tau=50$ fs 9: weighted average of 105 fs +35–30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM).
4177.9 2	3/2	27.0 fs 28	E(level): weighted average of 4177.2 9 (1972Hu10), 4180.1 15 (1976Me12), and 4177.9 2 (1981Bi05). J ^π : from γ (θ) in 1967Wa22.
4347.76 6	5/2,9/2	0.229 ps 42	T _{1/2} : lifetime $\tau=39$ fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM). E(level): weighted average of 4347.5 8 (1972Hu10), 4347.2 12 (1976Me12), and 4347.76 6 (1981Bi05). J ^π : spin=5/2,9/2 based on feeding 4282 γ (θ) from 7/2, 8630 level in 1976Sp09.
4624.4 3			T _{1/2} : lifetime $\tau=0.33$ ps 6 (1976Me12, DSAM). Other: >0.6 ps (1972Hu11, DSAM). E(level): weighted average of 4624.2 20 (1976Me12) and 4624.4 3 (1981Bi05). Other: 4625 5 (1972Hu10). 15% of I γ from this level decays to unknown final levels (1976Me12).
4769.86 23		76 fs 20	E(level): weighted average of 4768.9 12 (1972Hu10), 4766.9 15 (1976Me12), and 4769.90 15 (1981Bi05). T _{1/2} : lifetime $\tau=110$ fs 29: weighted average of 270 fs 95 (1972Hu11, DSAM) and 105 fs 17 (1976Me12, DSAM).
4839.10 11		9.7 fs 35	E(level): weighted average of 4838 3 (1972Hu10), 4841.7 19 (1976Me12), and 4839.09 11 (1981Bi05). T _{1/2} : lifetime $\tau=14$ fs 5 (1976Me12, DSAM).
4854.4 19		4.9 fs 14	E(level): weighted average of 4853 2 (1972Hu10) and 4855.7 19 (1976Me12). T _{1/2} : lifetime $\tau=7$ fs 2 (1976Me12, DSAM). Other: <6 fs (1972Hu11, DSAM).
4880.96 15		5.6 fs 14	E(level): weighted average of 4880.9 1 (1972Hu10), 4885 2 (1976Me12), and 4881.14 20 (1981Bi05). T _{1/2} : lifetime $\tau=8$ fs 2: weighted average of 7 fs 2 (1972Hu11, DSAM), 7 fs 3 (1973Fa07, DSAM), and 10 fs 3 (1976Me12, DSAM).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	E _p (lab) (keV)	Comments
5010.1 18		7.6 fs 21		E(level): weighted average of 5008 5 (1972Hu10) and 5010.4 18 (1976Me12). T _{1/2} : lifetime $\tau=11$ fs 3 (1976Me12, DSAM).
5163.34 22	7/2			E(level): weighted average of 5163 5 (1972Hu10), 5166.7 15 (1976Me12), and 5163.31 14 (1981Bi05). J ^π : spin=3/2, 7/2 from feeding 3466 γ (θ) from 7/2, 8630 level in 1976Sp09. Spin=3/2 and $\delta(\text{O}/\text{Q})=0.12$ 1 would require a large E3 or M3 component, which is ruled out by RUL. 10% of I γ transitions to unknown levels (1976Me12).
5215.8 15		<4.85 fs		E(level): weighted average of 5214 3 (1972Hu10) and 5216.2 15 (1976Me12). T _{1/2} : lifetime $\tau<7$ fs (1976Me12, DSAM).
5403.5 10		11.8 fs 28		E(level): weighted average of 5400 5 (1972Hu10) and 5403.6 10 (1976Me12). T _{1/2} : lifetime $\tau=17$ fs 4 (1976Me12, DSAM). 20% of I γ from this level decays (1976Me12). 50% of I γ from this level decays to unknown levels (1972Hu10).
5520				E(level): from 1976Sp08.
5586 2				E(level): weighted average of 5587 3 (1972Hu10) and 5586 2 (1976Me12).
5599.7 15		2.08 fs 69		E(level): weighted average of 5598 3 (1972Hu10) and 5600.1 15 (1976Me12). T _{1/2} : lifetime $\tau=3$ fs 1 (1976Me12, DSAM).
5646 2		2.77 fs 69		E(level): weighted average of 5645 3 (1972Hu10) and 5646 2 (1976Me12). T _{1/2} : lifetime $\tau=4$ fs 1 (1976Me12, DSAM).
5655 2		13.9 fs 35		E(level): weighted average of 5651 3 (1972Hu10) and 5656 2 (1976Me12). J ^π : Mirror level of ^{35}S ground state and thus 3/2 ⁺ (2001Vo24). Other: spin=5/2 based on the feeding 2051g(θ) from the 5/2, 7706 level in 1977Ko34. T _{1/2} : lifetime $\tau=20$ fs 5 (1976Me12, DSAM). $\Gamma_{\gamma}=0.033$ eV 10 (2001Vo24). 14% of I γ from this level decays to unknown final levels (1976Me12).
5683 2				E(level): weighted average of 5682 5 (1972Hu10) and 5683 2 (1976Me12). 40% of I γ from this level decays to unknown final levels (1976Me12).
5723.52 18				E(level): from 1981Bi05. 38% of I γ from this level decays to unknown final levels (1976Sp09).
5758 3				E(level): weighted average of 5754 5 (1972Hu10) and 5759 3 (1976Me12). 25% of γ from this level decays to unknown final levels and the other 75% is attributed to two γ transitions (1976Me12). Other: 100% of γ from this level decays to unknown final levels (1972Hu10).
5806 2		3.47 fs 69		E(level): weighted average of 5802 5 (1972Hu10) and 5806 2 (1976Me12). T _{1/2} : lifetime $\tau=5$ fs 1 (1976Me12, DSAM).
6106.2 21		8.3 fs 21		E(level): weighted average of 6102 3 (1972Hu10) and 6107.2 15 (1976Me12). T _{1/2} : lifetime $\tau=12$ fs 3 (1976Me12, DSAM).
6181 3				E(level): from 1976Me12. 55% of I γ from this level decays to unknown final levels (1976Me12).
6492 3				E(level): weighted average of 6489 5 (1972Hu10) and 6493 3 (1976Me12).
6866.7 6			510.6 6	E _p (lab) (keV): from 1988Va06.
7066.2 7			716.0 7	E _p (lab) (keV): weighted average of 716.0 7 (1972Hu10) and 716.0 10 (1976Me12). $\omega\gamma=0.09$ eV 5 (1972Hu10). $\omega\gamma=0.14$ eV 4 (1976Me12).
7103.4 7	3/2		754.3 7	J ^π : from γ (θ) in 1976Me12. E _p (lab) (keV): weighted average of 754.1 7 (1972Hu10) and 754.6 10 (1976Me12).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
7178.6 8	1/2	831.8 8	$\omega\gamma=0.04$ eV (1963Ha32). $\omega\gamma=0.23$ eV 11 (1972Hu10). $\omega\gamma=0.23$ eV 7 (1976Me12). J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 831.5 8 (1972Hu10), 832.2 15 (1976Me12), and 832 1 (2001Vo24). $\omega\gamma=0.13$ eV 2 (2001Vo24). $\Gamma_{\gamma} \geq 0.07$ eV (2001Vo24). $\omega\gamma=0.18$ eV 9 (1972Hu10). $\omega\gamma=0.14$ eV 4 (1976Me12).
7185		838	E _p (lab) (keV): from 1963Ha32.
7194.6 7	1/2,3/2	848.2 7	$\omega\gamma=0.056$ eV (1963Ha32). J ^π : from $\gamma(\theta)$ in 1960An06. E _p (lab) (keV): weighted average of 848.2 7 (1972Hu10) and 848.1 10 (1976Me12). $\omega\gamma=0.17$ eV (1963Ha32). $\omega\gamma=0.50$ eV 15 (1972Hu10). $\omega\gamma=0.54$ eV 16 (1976Me12).
7225.5 8	5/2	880.0 8	J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 880.5 8 (1972Hu10) and 879.3 10 (1976Me12). $\omega\gamma=0.073$ eV (1963Ha32). $\omega\gamma=0.18$ eV 9 (1972Hu10). $\omega\gamma=0.14$ eV 4 (1976Me12).
7234.0 8	5/2 ⁺	888.8 8	J ^π : spin=5/2 from $\gamma(\theta)$ in 1960An06 and 6014 γ to 1/2 ⁺ , 1219 level in 1972Hu10 and 1976Me12. E _p (lab) (keV): weighted average of 889.0 8 (1972Hu10) and 888.5 10 (1976Me12). $\omega\gamma=0.23$ eV (1963Ha32). $\omega\gamma=0.59$ eV 18 (1972Hu10). $\omega\gamma=0.86$ eV 26 (1976Me12).
7269.2 1		925.1 1	E _p (lab) (keV): from 1976Sp08.
7272.7 9	(1/2,3/2)	928.6 9	J ^π : from $\gamma(\theta)$ in 1960An06. E _p (lab) (keV): weighted average of 928.2 9 (1972Hu10) and 929.0 10 (1976Me12). $\omega\gamma=0.31$ eV (1963Ha32). $\omega\gamma=0.63$ eV 19 (1972Hu10). $\omega\gamma=1.00$ eV 30 (1976Me12).
7362.0 8	3/2	1020.6 8	J ^π : from $\gamma(\theta)$ in 1971Wi13 and 1977Ko34. E _p (lab) (keV): weighted average of 1020.4 8 (1972Hu10), 1020.5 11 (1973Fa07), and 1021.0 10 (1976Me12). $\omega\gamma=0.48$ eV (1963Ha32). $\omega\gamma=1.4$ eV 4 (1972Hu10). $\omega\gamma=1.13$ eV (1973Fa07). $\omega\gamma=1.4$ eV 4 (1976Me12).
7396.0 10	7/2	1055.6 10	J ^π : from $\gamma(\theta)$ in 1977Ko35. E _p (lab) (keV): weighted average of 1054.9 10 (1972Hu10), 1055.8 12 (1973Fa07), and 1056.9 16 (1976Me12). $\omega\gamma=0.67$ eV (1963Ha32). $\omega\gamma=0.18$ eV 9 (1972Hu10). $\omega\gamma=0.18$ eV (1973Fa07). $\omega\gamma=0.23$ eV 7 (1976Me12).
7451.1 6	3/2	1112.3 6	J ^π : from $\gamma(\theta)$ in 1972LeYW. E _p (lab) (keV): weighted average of 1112.2 10 (1972Hu10), 1112.8 13 (1973Fa07), and 1112.2 6 (1976Me12). $\omega\gamma=0.14$ eV 7 (1972Hu10). $\omega\gamma=0.14$ eV (1973Fa07). $\omega\gamma=0.18$ eV 5 (1976Me12).
7496?		1158	E(level): from 1960An06.
7501.1 8		1163.8 8	E _p (lab) (keV): from 1973Fa07. Other: 1165 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^{\dagger}$	$J^{\pi\dagger}$	$T_{1/2}$	$E_{\text{p}}(\text{lab})$ (keV)	Comments
7502.9 7			1165.7 7	$\omega\gamma=0.05$ eV (1973Fa07). $\omega\gamma=0.27$ eV 14 for $E_{\text{p}}=1165+1166$ doublet (1972Hu10). $E_{\text{p}}(\text{lab})$ (keV): from 1973Fa07. Other: 1165.4 7 (1976Me12), 1166 (1972Hu10). $\omega\gamma=0.27$ eV 14 for $E_{\text{p}}=1165+1166$ doublet (1972Hu10). $\omega\gamma=0.23$ eV (1973Fa07). $\omega\gamma=0.36$ eV 11 (1976Me12), possible doublet. J^{π} : from $\gamma(\theta)$ in 1977Ko35. $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1183.2 11 (1972Hu10), 1182.2 12 (1973Fa07), and 1181.4 7 (1976Me12). $\omega\gamma=0.09$ eV 5 (1972Hu10). $\omega\gamma=0.14$ eV (1973Fa07). $\omega\gamma=0.18$ eV 5 (1976Me12). $T=3/2$ J^{π} : spin from $\gamma(\theta)$ in 1966Wa09, 1967Wa22, and 1977Ko35; parity from 4386 γ , M1+E2 to 7/2 $^{-}$, 3163 level in 1966Wa09 and 1967Wa22. $T_{1/2}$: lifetime $\tau<1$ fs (1973Fa07, DSAM). Others: $\tau<13$ fs (1971Wi13, DSAM). $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1213.7 10 (1972Hu10), 1213.6 6 (1973Fa07), 1212.4 7 (1976Me12), and 1212 1 (2001Vo24). The resonance strength of this level has been intensively studied. See the general comments at the beginning of this dataset for details. 1971Pr11 reports a 3166 γ with $I_{\gamma}=0.37$ 13 to the 4382 final level. J^{π} : from $\gamma(\theta)$ in 1977Ko35. $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1225.6 10 (1972Hu10), 1226.3 6 (1973Fa07), and 1225.3 7 (1976Me12). $\omega\gamma=0.68$ eV 20 (1972Hu10). $\omega\gamma=0.50$ eV (1973Fa07). $\omega\gamma=0.68$ eV 20 (1976Me12). $\omega\gamma=0.7$ eV 2 (2001Vo24). $A_2=2$ (2001Vo24). J^{π} : spin=5/2 from $\gamma(\theta)$ in 1971Wi13 and 1976Me12. Other: spin=3/2 from 1977Ko34. $T_{1/2}$: lifetime $\tau<20$ fs (1971Wi13, DSAM). $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1266.5 10 (1972Hu10), 1267.9 11 (1973Fa07), and 1265.7 8 (1976Me12). $\omega\gamma=3.4$ eV, $\Gamma_{\gamma}\approx 0.57$ eV (1971Wi13). $\omega\gamma=1.4$ eV 4 (1972Hu10). $\omega\gamma=1.31$ eV (1973Fa07). $\omega\gamma=1.3$ eV 4 (1976Me12). J^{π} : from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12. $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1285.5 11 (1972Hu10), 1286.1 11 (1973Fa07), and 1284.5 5 (1976Me12). $\omega\gamma=0.54$ eV 16 (1972Hu10). $\omega\gamma=0.50$ eV (1973Fa07). $\omega\gamma=0.77$ eV 23 (1976Me12). $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1323.7 12 (1972Hu10), 1323.5 12 (1973Fa07), and 1324.2 8 (1976Me12). $\omega\gamma=0.23$ eV 11 (1972Hu10). $\omega\gamma=0.23$ eV (1973Fa07). $\omega\gamma=0.36$ eV 11 (1976Me12). J^{π} : spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=5/2,7/2 from $\gamma(\theta)$ in 1976Me12. $E_{\text{p}}(\text{lab})$ (keV): weighted average of 1339.8 11 (1972Hu10), 1339.4 11 (1973Fa07), and 1339.7 8 (1976Me12). $\omega\gamma=0.68$ eV 20 (1972Hu10). $\omega\gamma=0.63$ eV (1973Fa07).
7518.8 7	7/2		1182.0 7	
7548.9 6	7/2 $^{-}$	<0.69 fs	1213.0 6	
7561.3 6	1/2,3/2		1225.8 6	
7600.8 8	5/2	<13.9 fs	1266.5 8	
7618.7 5	5/2		1284.9 5	
7656.6 8			1323.9 8	
7671.9 8	5/2		1339.7 8	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2}	E _p (lab) (keV)	Comments
7685.5 8	3/2 ⁻		1353.7 8	$\omega\gamma=0.59$ eV 18 (1976Me12). J ^π : 3/2 ⁻ from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=3/2 from $\gamma(\theta)$ in 1976Me12. Spin=3/2,5/2 from $\gamma(\theta)$ in 1977Ko34. E _p (lab) (keV): weighted average of 1353.7 11 (1972Hu10), 1353.4 11 (1973Fa07), and 1353.8 8 (1976Me12). $\Gamma_p=445$ eV 11 (1981Bi05). $\omega\gamma=1.13$ eV 34 (1972Hu10). $\omega\gamma=1.09$ eV (1973Fa07). $\omega\gamma=1.3$ eV 4 (1976Me12).
7693.9 8			1362.3 8	E _p (lab) (keV): weighted average of 1362.4 12 (1972Hu10) and 1362.3 8 (1976Me12). $\omega\gamma=0.18$ eV 9 (1972Hu10). $\omega\gamma=0.27$ eV 8 (1976Me12).
7706.4 8	5/2 ⁺		1375.2 8	J ^π : 5/2 ⁺ from $\sigma(E_p,\theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1977Ko34. Spin=3/2,5/2 from $\gamma(\theta)$ in 1976Me12 and 5/2 is favored. E _p (lab) (keV): weighted average of 1375.7 11 (1972Hu10), 1375.0 11 (1973Fa07), and 1375.1 8 (1976Me12). $\omega\gamma=1.7$ eV 5 (1972Hu10). $\omega\gamma=1.40$ eV (1973Fa07). $\omega\gamma=2.0$ eV 6 (1976Me12). $\Gamma_p=4$ eV 1 (1981Bi05).
7744.8 9	7/2		1414.8 9	J ^π : from $\gamma(\theta)$ for J _f (2645.7)=7/2, J _f (3163)=7/2, J _f (3942.9)=9/2 (1977Ko35). E _p (lab) (keV): weighted average of 1415.6 12 (1972Hu10), 1414.2 12 (1973Fa07), and 1414.7 9 (1976Me12). $\omega\gamma=0.59$ eV 18 (1972Hu10). $\omega\gamma=0.54$ eV (1973Fa07). $\omega\gamma=0.77$ eV 23 (1976Me12).
7748			1418	E(level): from 1960An06.
7777.1 9			1448.0 9	E _p (lab) (keV): weighted average of 1449.0 11 (1972Hu10), 1447.5 12 (1973Fa07), and 1447.5 9 (1976Me12). $\omega\gamma=0.68$ eV 20 (1972Hu10). $\omega\gamma=(0.63)$ eV (1973Fa07). $\omega\gamma=0.68$ eV 20 (1976Me12).
7781.7 11	5/2 ⁻		1452.8 11	J ^π : proposed by 1976Me12 based on strengths of primary transitions. E _p (lab) (keV): weighted average of 1453.6 11 (1972Hu10), 1451.8 15 (1973Fa07), and 1452.3 13 (1976Me12). $\omega\gamma=0.36$ eV 18 (1972Hu10). $\omega\gamma=(0.41)$ eV (1973Fa07). $\omega\gamma=0.41$ eV 12 (1976Me12).
7797.1 9	1/2		1468.6 9	J ^π : from $\gamma(\theta)$ in 1988Va06, which rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1469.6 12 (1972Hu10), 1468.2 12 (1973Fa07), and 1468.2 9 (1976Me12). $\omega\gamma=0.68$ eV 20 (1972Hu10). $\omega\gamma=0.72$ eV (1973Fa07). $\omega\gamma=0.63$ eV 19 (1976Me12).
7837.2 10	3/2	<3.47 fs	1509.9 10	T=3/2 J ^π : spin from $\gamma(\theta)$ in 1967Wa22. T _{1/2} : lifetime $\tau<5$ fs (1971Wi13, DSAM). E _p (lab) (keV): weighted average of 1509.1 15 (1973Fa07), 1510.2 10 (1976Me12), and 1510 1 (2001Vo24). $\omega\gamma=3.4$ eV 6 (1967Wa22). $\omega\gamma\approx 7.5$ eV, $\Gamma_\gamma=1.9$ eV (1971Wi13). $\omega\gamma=5.0$ eV 15 for E _p =1510 doublet (1972Hu10). $\omega\gamma=0.59$ eV (1973Fa07). $\omega\gamma=2.2$ eV 7 (1976Me12). $\omega\gamma=2.1$ eV 3 (2001Vo24).

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³⁴S(p,γ) **1976Me12,1976Sp08,1972Hu10 (continued)**³⁵Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
7839.0 9		1511.8 9	Γ _γ =0.55 eV 6 (2001Vo24). E _p (lab) (keV): weighted average of 1512.3 15 (1973Fa07) and 1511.6 9 (1976Me12). ωγ=5.0 eV 15 for E _p =1510 doublet (1972Hu10). ωγ=3.94 eV (1973Fa07). ωγ=2.8 eV 8 (1976Me12).
7868.7 10		1542.3 10	E _p (lab) (keV): weighted average of 1542.3 13 (1972Hu10), 1542.3 12 (1973Fa07), and 1542.2 10 (1976Me12). ωγ=0.23 eV 11 (1972Hu10). ωγ=(0.23) eV (1973Fa07). ωγ=0.45 eV 14 (1976Me12).
7873?		1547	E(level): from 1960An06.
7880.8 9	3/2 ⁻ ,5/2	1554.8 9	J ^π : from γ(θ) in 1976Me12. E _p (lab) (keV): weighted average of 1554.9 12 (1972Hu10), 1554.8 11 (1973Fa07), and 1554.7 9 (1976Me12). ωγ=0.72 eV 22 (1972Hu10). ωγ=0.59 eV (1973Fa07). ωγ=1.00 eV 30 (1976Me12).
7885?		1559	E(level): from 1960An06.
7889.1 15		1563.3 15	E _p (lab) (keV): from 1973Fa07. ωγ=(0.05) eV (1973Fa07).
7899.2 7	(3/2 ⁻ ,5/2)	1573.7 7	E _p (lab) (keV): weighted average of 1574.4 12 (1972Hu10), 1573.2 11 (1973Fa07), and 1573.7 7 (1976Me12). ωγ=0.63 eV 19 (1972Hu10). ωγ=0.45 eV (1973Fa07). ωγ=1.00 eV 30 (1976Me12).
7903?		1578	E(level): from 1960An06.
7923.3 9		1598.6 9	E _p (lab) (keV): weighted average of 1598.9 13 (1972Hu10), 1598.7 12 (1973Fa07), and 1598.4 9 (1976Me12). ωγ=0.36 eV 18 (1972Hu10). ωγ=(0.32) eV (1973Fa07). ωγ=0.45 eV 14 (1976Me12).
7930?		1605	E(level): from 1960An06.
7949?		1625	E(level): from 1960An06.
7970.3 10		1646.9 10	E _p (lab) (keV): weighted average of 1647.3 12 (1972Hu10), 1645.9 13 (1973Fa07), and 1647.1 10 (1976Me12). ωγ=0.54 eV 16 (1972Hu10). ωγ=(0.45) eV (1973Fa07). ωγ=0.86 eV 26 (1976Me12).
7987.8 10	3/2	1665.0 10	J ^π : from γ(θ) in 1976Me12. E _p (lab) (keV): weighted average of 1665.2 12 (1972Hu10), 1663.3 12 (1973Fa07), and 1666.1 10 (1976Me12). ωγ=0.68 eV 20 (1972Hu10). ωγ=(0.59) eV (1973Fa07). ωγ=0.54 eV 16 (1976Me12).
7995.6 10	5/2	1673.0 10	E _p (lab) (keV): weighted average of 1673.7 11 (1972Hu10), 1672.4 13 (1973Fa07), and 1672.7 10 (1976Me12). ωγ=1.2 eV 4 (1972Hu10). ωγ=(1.04) eV (1973Fa07). ωγ=0.95 eV 28 (1976Me12).
8001.0 9	(7/2,9/2 ⁺)	1678.6 9	E _p (lab) (keV): weighted average of 1679.7 11 (1972Hu10), 1678.2 13 (1973Fa07), and 1678.1 9 (1976Me12). ωγ=1.9 eV 6 (1972Hu10). ωγ=(1.18) eV (1973Fa07). ωγ=1.3 eV 4 (1976Me12).
8005.0 10	5/2 ⁺	1682.7 10	J ^π : from σ(E _p ,θ) in 1981Bi05. Spin=5/2 from γ(θ) in 1981Bi05. Spin=3/2,5/2 from

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
			$\gamma(\theta)$ in 1976Me12 . E _p (lab) (keV): weighted average of 1684.3 <i>I3</i> (1972Hu10), 1681.9 <i>I1</i> (1973Fa07), and 1682.5 <i>I0</i> (1976Me12). $\omega\gamma=(1.54)$ eV (1973Fa07). $\omega\gamma=2.2$ eV <i>7</i> (1976Me12). $\omega\gamma=2.0$ eV <i>6</i> (1972Hu10). $\Gamma_p=21$ eV <i>3</i> (1981Bi05). E _p (lab) (keV): weighted average of 1714.0 <i>I1</i> (1972Hu10), 1713.5 <i>I4</i> (1973Fa07), and 1714.9 <i>I0</i> (1976Me12). $\omega\gamma=0.41$ eV <i>20</i> (1972Hu10). $\omega\gamma=(0.41)$ eV (1973Fa07). $\omega\gamma=0.36$ eV <i>11</i> (1976Me12). J ^π : from $\gamma(\theta)$ in 1976Me12 . E _p (lab) (keV): weighted average of 1717.9 <i>I1</i> (1972Hu10), 1716.3 <i>I1</i> (1973Fa07), and 1717.3 <i>I1</i> (1976Me12). $\omega\gamma=1.6$ eV <i>5</i> (1972Hu10). $\omega\gamma=1.36$ eV (1973Fa07). $\omega\gamma=1.3$ eV <i>4</i> (1976Me12). E _p (lab) (keV): weighted average of 1756.0 <i>I0</i> (1972Hu10), 1755.4 <i>I1</i> (1973Fa07), and 1755.7 <i>I1</i> (1976Me12). $\omega\gamma=0.72$ eV <i>22</i> (1972Hu10). $\omega\gamma=(0.59)$ eV (1973Fa07). $\omega\gamma=0.72$ eV <i>22</i> (1976Me12). E _p (lab) (keV): from 1960An06 . E _p (lab) (keV): weighted average of 1776.7 <i>I1</i> (1972Hu10), 1776.9 <i>I1</i> (1973Fa07), and 1776.0 <i>I1</i> (1976Me12). $\omega\gamma=1.13$ eV <i>34</i> (1972Hu10). $\omega\gamma=1.13$ eV (1973Fa07). $\omega\gamma=1.1$ eV <i>3</i> (1976Me12). J ^π : from $\gamma(\theta)$ in 1976Me12 . E _p (lab) (keV): weighted average of 1787.7 <i>I1</i> (1972Hu10), 1786.6 <i>I1</i> (1973Fa07), and 1786.9 <i>I1</i> (1976Me12). $\omega\gamma=1.9$ eV <i>6</i> (1972Hu10). $\omega\gamma=1.63$ eV (1973Fa07). $\omega\gamma=1.7$ eV <i>5</i> (1976Me12). E _p (lab) (keV): weighted average of 1794.8 <i>I3</i> (1972Hu10), 1793.3 <i>I4</i> (1973Fa07), and 1794.2 <i>I1</i> (1976Me12). $\omega\gamma=0.54$ eV <i>16</i> (1972Hu10). $\omega\gamma=(0.50)$ eV (1973Fa07). $\omega\gamma=0.72$ eV <i>22</i> (1976Me12). E _p (lab) (keV): weighted average of 1829.1 <i>I3</i> (1972Hu10), 1828.6 <i>I2</i> (1973Fa07), and 1829.5 <i>I1</i> (1976Me12). $\omega\gamma=1.09$ eV <i>33</i> (1972Hu10). $\omega\gamma=0.68$ eV (1973Fa07). $\omega\gamma=0.77$ eV <i>23</i> (1976Me12). E _p (lab) (keV): weighted average of 1839.0 <i>I2</i> (1972Hu10), 1838.8 <i>I2</i> (1973Fa07), and 1839.1 <i>I1</i> (1976Me12). $\omega\gamma=0.95$ eV <i>28</i> (1972Hu10). $\omega\gamma=0.95$ eV (1973Fa07). $\omega\gamma=0.86$ eV <i>26</i> (1976Me12). E _p (lab) (keV): weighted average of 1862.7 <i>I3</i> (1972Hu10), 1862.1 <i>I3</i> (1973Fa07), and 1862.1 <i>I2</i> (1976Me12). $\omega\gamma=0.36$ eV <i>18</i> (1972Hu10). $\omega\gamma=(0.32)$ eV (1973Fa07). $\omega\gamma=0.41$ eV <i>12</i> (1976Me12). J ^π : from $\sigma(E_p,\theta)$ in 1981Bi05 . Spin=5/2 from $\gamma(\theta)$ in 1996Ka21 and 2001Vo24 .
8035.7 <i>I0</i>		1714.3 <i>I0</i>	
8038.5 <i>I1</i>	3/2	1717.2 <i>I1</i>	
8075.9 <i>I0</i>		1755.7 <i>I0</i>	
8080?		1760	
8096.1 <i>I1</i>	(5/2, 7/2 ⁺)	1776.5 <i>I1</i>	
8106.4 <i>I1</i>	3/2	1787.1 <i>I1</i>	
8113.3 <i>I1</i>	(1/2 ⁺ , 3/2, 5/2 ⁺)	1794.2 <i>I1</i>	
8147.2 <i>I1</i>	3/2 ⁻	1829.1 <i>I1</i>	
8156.8 <i>I1</i>	(5/2 ⁺ , 7/2 ⁻)	1839.0 <i>I1</i>	
8179.4 <i>I2</i>		1862.3 <i>I2</i>	
8208.2 <i>I0</i>	5/2 ⁺	1891.9 <i>I0</i>	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π _z	E _p (lab) (keV)	Comments
			E _p (lab) (keV): weighted average of 1893.2 11 (1972Hu10), 1891.3 12 (1976Me12), 1892.0 11 (1973Fa07), and 1891 1 (2001Vo24). $\omega\gamma=3.6$ eV 11 (1972Hu10). $\omega\gamma=2.71$ eV (1973Fa07). $\omega\gamma=4.1$ eV 12 (1976Me12). $\omega\gamma=4.1$ eV 5, $\Gamma_p=39$ eV 5 (1981Bi05). $\omega\gamma=2.5$ eV 7 (1996Ka21). $\omega\gamma=4.0$ eV 8 (2001Vo24), sum of $\omega\gamma$ for the resonances at E _p =1891 and 1893, $\Gamma_\gamma=0.70$ eV 14 (2001Vo24).
8209	1/2 ⁺	1893	E(level): from 1974Bi16; a weak resonance in the tail of the 1892 resonance. J ^π : from $\sigma(E_p, \theta)$ in 1981Bi05. $\Gamma_p=120$ eV 11 (1981Bi05).
8216.3 10	5/2	1900.3 10	J ^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21. E _p (lab) (keV): weighted average of 1901.8 11 (1972Hu10), 1899.7 12 (1973Fa07), 1899.6 11 (1976Me12), and 1900 1 (2001Vo24). $\omega\gamma=2.8$ eV 8 (1972Hu10). $\omega\gamma=2.13$ eV (1973Fa07). $\omega\gamma=3.2$ eV 10 (1976Me12). $\omega\gamma=1.7$ eV 5 (1996Ka21). $\omega\gamma=3.2$ eV 5 (2001Vo24). $\Gamma_\gamma=0.55$ eV 8 (2001Vo24).
8242.4 12		1927.1 12	E _p (lab) (keV): weighted average of 1928.2 13 (1972Hu10), 1926.8 13 (1973Fa07), and 1926.5 12 (1976Me12). $\omega\gamma=0.86$ eV 26 (1972Hu10). $\omega\gamma=0.86$ eV (1973Fa07). $\omega\gamma=1.00$ eV 30 (1976Me12). $\omega\gamma=1.0$ eV 3 (2001Vo24). $\Gamma_\gamma=0.25$ eV 8 (2001Vo24). 20% of I γ transitions to unknown levels (1973Fa07).
8251? 5 8268.9 5		1936 5 1954.4 5	E _p (lab) (keV): from 1967Da12. E _p (lab) (keV): weighted average of 1955.9 13 (1972Hu10), 1952 2 (1973Fa07), 1954.3 12 (1976Me12), and 1954.4 5 (1976Sp08). $\omega\gamma=0.81$ eV 24 (1972Hu10). $\omega\gamma=(0.86)$ eV (1973Fa07). $\omega\gamma=0.90$ eV 27 (1976Me12). $\omega\gamma=0.6$ eV 3 (1976Sp08).
8277.2 5		1963.0 5	E _p (lab) (keV): weighted average of 1964.1 14 (1972Hu10), 1962 2 (1973Fa07), 1962.8 12 (1976Me12), and 1962.9 5 (1976Sp08). $\omega\gamma=0.72$ eV 22 (1972Hu10). $\omega\gamma=(0.77)$ eV (1973Fa07). $\omega\gamma=0.68$ eV 20 (1976Me12). $\omega\gamma=0.7$ eV 4 (1976Sp08).
8282.4 5	(3/2 ⁻ , 5/2)	1968.3 5	E _p (lab) (keV): weighted average of 1970.0 13 (1972Hu10), 1971 2 (1973Fa07), 1967.8 12 (1976Me12), and 1967.9 5 (1976Sp08). $\omega\gamma=1.09$ eV 33 (1972Hu10). $\omega\gamma=(1.09)$ eV (1973Fa07). $\omega\gamma=0.59$ eV 18 (1976Me12). $\omega\gamma=0.4$ eV 2 (1976Sp08).
8287.82 30	1/2 ⁻	1973.89 30	J ^π : from $\sigma(E_p, \theta)$ in 1981Bi05, which also rejects spin=3/2 from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1975.6 13 (1972Hu10), 1973.8 12 (1976Me12), and 1973.80 30 (1976Sp08). $\omega\gamma=1.3$ eV 4 (1972Hu10). $\omega\gamma=0.72$ eV 22 (1976Me12). $\omega\gamma=0.9$ eV 5 (1976Sp08). $\Gamma_p=48$ eV 6 (1981Bi05).
8297.2 9		1983.5 9	E _p (lab) (keV): weighted average of 1983.7 5 (1976Sp08) and 1980 2 (1973Fa07).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
8298.2 5	3/2	1984.6 5	$\omega\gamma=(1.36)$ eV (1973Fa07). $\omega\gamma=0.6$ eV 3 (1976Sp08). J ^π : from $\gamma(\theta)$ in 1976Me12. E _p (lab) (keV): weighted average of 1986.0 13 (1972Hu10), 1983.0 20 (1973Fa07), 1984.5 12 (1976Me12), and 1984.5 5 (1976Sp08). $\omega\gamma=1.8$ eV 5 (1972Hu10). $\omega\gamma=(0.77)$ eV (1973Fa07). $\omega\gamma=1.6$ eV 5 (1976Me12). $\omega\gamma=0.9$ eV 5 (1976Sp08).
8318.1 3		2005.1 3	E _p (lab) (keV): weighted average of 2006.7 12 (1972Hu10), 2004.6 13 (1973Fa07), 2006.0 13 (1976Me12), and 2005.0 3 (1976Sp08). $\omega\gamma=1.9$ eV 6 (1972Hu10). $\omega\gamma=2.08$ eV (1973Fa07). $\omega\gamma=1.9$ eV 6 (1976Me12). $\omega\gamma=2.6$ eV 5 (1976Sp08).
8323.1 13		2010.2 13	E _p (lab) (keV): weighted average of 2010.5 13 (1972Hu10) and 2009.9 13 (1976Me12). $\omega\gamma=0.45$ eV 14 (1972Hu10). $\omega\gamma=0.72$ eV 22 (1976Me12).
8346? 5		2034 5	E _p (lab) (keV): from 1967Da12.
8381.8 8	5/2 ⁺	2070.7 8	J ^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1967Ko23. E _p (lab) (keV): weighted average of 2073.5 12 (1972Hu10), 2070.4 5 (1976Sp08), and 2070 1 (2001Vo24). $\omega\gamma\approx 4$ eV (1967Ko23).  $\omega\gamma=6.4$ eV 19 (1972Hu10). $\omega\gamma=6.0$ eV 12 (1976Sp08). $\omega\gamma=5.9$ eV 6 (2001Vo24). $\Gamma_\gamma=1.02$ eV 10 (2001Vo24).
8389.4 16		2078.5 16	E _p (lab) (keV): unweighted average of 2080.0 14 (1972Hu10) and 2076.9 5 (1976Sp08). $\omega\gamma=0.27$ eV 14 (1972Hu10). $\omega\gamma=0.4$ eV 2 (1976Sp08).
8402.9 5		2092.4 5	E _p (lab) (keV): from 1976Sp08. $\omega\gamma=0.7$ eV 4 (1976Sp08).
8405.7 14		2095.3 14	E _p (lab) (keV): unweighted average of 2096.6 12 (1972Hu10) and 2093.9 5 (1976Sp08). $\omega\gamma=2.9$ eV 9 (1972Hu10). $\omega\gamma=2.3$ eV 5 (1976Sp08).
8409.6 18		2099.3 18	E _p (lab) (keV): unweighted average of 2101.0 14 (1972Hu10) and 2097.5 5 (1976Sp08). $\omega\gamma=1.6$ eV 5 (1972Hu10). $\omega\gamma=1.4$ eV 3 (1976Sp08). $\omega\gamma=0.5$ eV 3 (1976Sp08).
8416.7 5		2106.6 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8430.3 5		2120.6 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8434.8 5		2125.2 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
8464.3 5		2155.6 5	J ^π : from 2001Vo24 based on proposed isobaric analog to 3/2 ⁺ in ^{35}S .
8484.4 5	3/2 ⁺	2176.3 5	E _p (lab) (keV): weighted average of 2176.4 5 (1976Sp08) and 2176 1 (2001Vo24). $\omega\gamma=2.1$ eV 4 (1976Sp08). $\omega\gamma=2.1$ eV 4 (2001Vo24). $\Gamma_\gamma=0.36$ eV 6 (2001Vo24).
8486.1 5		2178.0 5	$\omega\gamma=1.9$ eV 4 (1976Sp08).
8506.5 5		2199.1 5	$\omega\gamma=0.3$ eV 2 (1976Sp08).
8514.3 5		2207.1 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8533.9 5		2227.3 5	$\omega\gamma=0.10$ eV 5 (1976Sp08).
8571.9 5		2266.4 5	$\omega\gamma=2.8$ eV 6 (1976Sp08).
8580.5 5		2275.3 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8585.8 5		2280.7 5	$\omega\gamma=0.10$ eV 5 (1976Sp08).
8590.0 5		2285.0 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8612.0 5		2307.7 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab) (keV)	Comments
8613.7 5		2309.4 5	$\omega\gamma=2.4$ eV 5 (1976Sp08).
8618.1 5	(3/2 ⁺ ,5/2 ⁺)	2314.0 5	J ^π : most probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.
			$\omega\gamma=3.7$ eV 8 (1976Sp08).
8630.1 3	7/2	2326.3 3	J ^π : spin=3/2,7/2 from $\gamma(\theta)$ in 1976Sp09; 1976Sp09 ruled out 3/2 based on 4687 γ to 9/2 ⁺ , 3943 level. Other: most probable spin=3/2 from $\gamma(\theta)$ in 1970Ko33.
			$\omega\gamma=2.6$ eV 5 (1976Sp08).
8640	5/2	2336	J ^π : probable spin=5/2 from $\gamma(\theta)$ in 1970Ko33.
			E _p (lab) (keV): from 1970Ko33.
8641.4 5	3/2,5/2	2338.0 5	J ^π : spin=3/2 from $\gamma(\theta)$ in 1976Sp09; spin=5/2 is also possible.
			$\omega\gamma=0.9$ eV 5 (1976Sp08).
8686.2 5		2384.1 5	$\omega\gamma=2.2$ eV 4 (1976Sp08).
8688.3 5		2386.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8696.9 5		2395.1 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8706.3 5		2404.8 5	$\omega\gamma=0.2$ eV 1 (1976Sp08).
8717.7 5		2416.5 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
8750.9 5		2450.7 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8767.0 5		2467.3 5	$\omega\gamma=0.3$ eV 2 (1976Sp08).
8772.9 5		2473.4 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8779.8 5		2480.5 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8787.2 5		2488.1 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
8798.4 5		2499.6 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
8820.9 5		2522.8 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8824.2 5		2526.2 5	$\omega\gamma=3.6$ eV 7 (1976Sp08).
8829.3 5		2531.4 5	$\omega\gamma=2.1$ eV 4 (1976Sp08).
8833.1 5		2535.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8838.1 5	7/2 ⁻	2540.5 5	J ^π : 5/2 ⁻ ,7/2 ⁻ from $\sigma(\text{E}_p,\theta)$ in 1981Bi05. Spin=7/2 from $\gamma(\theta)$ in 1981Bi05.
			$\omega\gamma=1.5$ eV 3 (1976Sp08).
			$\Gamma_p=4$ eV 1 (1981Bi05).
8856.1 5		2559.0 5	$\omega\gamma=4.5$ eV 9 (1976Sp08).
8868.6 5		2571.9 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8884.1 5		2587.9 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
8886.1 5		2589.9 5	$\omega\gamma=1.4$ eV 3 (1976Sp08).
8893.3 5	(5/2,7/2) ⁺	2597.3 5	J ^π : from 2001Vo24 with no supporting arguments.
			E _p (lab) (keV): weighted average of 2597.4 5 (1976Sp08) and 2597 1 (2001Vo24).
			$\omega\gamma=1.2$ eV 2 (1976Sp08).
			$\omega\gamma=1.2$ eV 4 (2001Vo24).
			$\Gamma_\gamma=0.20$ eV 6 (2001Vo24).
8904.8 5		2609.2 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8906.8 5	5/2 ⁺	2611.2 5	J ^π : from 2001Vo24 with no supporting arguments.
			E _p (lab) (keV): weighted average of 2611.3 5 (1976Sp08) and 2611 1 (2001Vo24).
			$\omega\gamma=1.5$ eV 3 (1976Sp08).
			$\omega\gamma=1.5$ eV 3 (2001Vo24).
			$\Gamma_\gamma=0.25$ eV 5 (2001Vo24).
8919.8 5		2624.6 5	$\omega\gamma=1.1$ eV 2 (1976Sp08).
8933.3 5		2638.5 5	$\omega\gamma=1.0$ eV 2 (1976Sp08).
8953.0 5	3/2	2658.8 5	J ^π : from $\gamma(\theta)$ in 1976Sp09.
			$\omega\gamma=2.4$ eV 5 (1976Sp08).
8957.9 5		2663.8 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
8981.9 5		2688.6 5	$\omega\gamma=2.2$ eV 4 (1976Sp08).
8984.2 5		2690.9 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).
8988.4 5		2695.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
8992.4 5		2699.4 5	$\omega\gamma=2.3$ eV 5 (1976Sp08).
8996.7 5		2703.8 5	$\omega\gamma=1.7$ eV 3 (1976Sp08).
9001.1 5		2708.3 5	$\omega\gamma=0.6$ eV 3 (1976Sp08).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	$E_p(\text{lab})$ (keV)	Comments
9019.3 5		3.1 keV 3	2727.1 5	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma=0.8$ eV 4 (1976Sp08).
9024.5 5			2732.4 5	$\omega\gamma=1.1$ eV 2 (1976Sp08).
9029.9 5			2738.0 5	$\omega\gamma=2.5$ eV 5 (1976Sp08).
9033.1 5			2741.3 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
9038.3 5			2746.6 5	$\omega\gamma=0.8$ eV 4 (1976Sp08).
9048.4 5			2757.0 5	$\omega\gamma=3.1$ eV 6 (1976Sp08).
9081.4 4	5/2 ⁺	65 eV 20	2791.0 4	J^π : from $\sigma(E_p, \theta)$ in 1981Bi05. Spin=5/2 from $\gamma(\theta)$ in 1976Sp09. $T_{1/2}$: width $\Gamma_p=62$ eV 20, $\Gamma_\gamma=2.3$ eV 4 (1976Sp09, resonant absorption). $E_p(\text{lab})$ (keV): weighted average of 2791.0 4 (1976Sp08) and 2791 1 (2001Vo24). $\omega\gamma=15.7$ eV 32 (1976Sp08). $\omega\gamma=16.0$ eV 4 (1996Ka21). $\omega\gamma=16.0$ eV 3 (2001Vo24). $\omega\gamma=13.4$ eV 22 (1976Sp09). $\Gamma_p=59$ eV 7 (1981Bi05). $\Gamma_\gamma=2.7$ eV 5 (2001Vo24).
9088.5 5			2798.3 5	$\omega\gamma=0.7$ eV 4 (1976Sp08).
9099.3 5			2809.4 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
9100.4 5			2810.6 5	$\omega\gamma=2.1$ eV 4 (1976Sp08).
9107.4 5			2817.8 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
9110.0 5			2820.5 5	$\omega\gamma=0.5$ eV 3 (1976Sp08).
9124.0 5			2834.9 5	$\omega\gamma=1.2$ eV 2 (1976Sp08).
9135.1 5		2.0 keV 4	2846.3 5	$T_{1/2}$: width measured from the slope of the low-energy side of the resonance peak (1976Sp08). $\omega\gamma=0.2$ eV 1 (1976Sp08).
9138.3 5			2849.6 5	$\omega\gamma=1.9$ eV 4 (1976Sp08).
9155.7 5			2867.5 5	$\omega\gamma=2.0$ eV 4 (1976Sp08).
9157.1 5	5/2		2869.0 5	J^π : spin=5/2 from $\gamma(\theta)$ in 2001Vo24 and 1996Ka21. $\omega\gamma=6.6$ eV 13 (1976Sp08). $\omega\gamma=6.3$ eV 1 (1996Ka21).
9163.3 5			2875.3 5	$\omega\gamma=0.4$ eV 2 (1976Sp08).
9184.2 5			2896.9 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
9188.8 5			2901.6 5	$\omega\gamma=0.9$ eV 5 (1976Sp08).
9194.1 5			2907.1 5	$\omega\gamma=3.9$ eV 8 (1976Sp08).

[†] For resonances above $S(\text{p})(^{35}\text{Cl})=6370.81$ 4 (2021Wa16), $E(\text{level})=E_{\text{c.m.}}+S(\text{p})(^{35}\text{Cl})$, where $E_{\text{c.m.}}=E_p(\text{lab})\times m(^{34}\text{S})/[m_p+m(^{34}\text{S})]$. Weighted average is taken when multiple $E_p(\text{lab})$ measurements are available. For levels below 8.27 MeV ($E_p<1.95$ MeV), $E_p(\text{lab})$ is mainly from 1972Hu10, 1973Fa07, and 1976Me12; for levels above 8.27 MeV ($E_p>1.95$ MeV), $E_p(\text{lab})$ is mainly from 1976Sp08, unless otherwise noted.

[‡] From the comparison of experimental angular distributions of γ -rays with theoretical predictions or deduced from γ -feedings, unless otherwise noted.

$\gamma(^{35}\text{Cl})$								Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	
1219.39	1/2 ⁺	1219.37	100	0	3/2 ⁺			E_γ : other: 1220 40 (1963Ha32). $A_2=-0.03$ 3, $A_4=-0.002$ 50 (1971Wi13). $A_2=-0.01$ 2, $A_4=+0.005$ 24 (1967Ta08).
1763.14	5/2 ⁺	543.75 1763.09	<0.2 100	1219.39 0	1/2 ⁺ 3/2 ⁺	D+Q	-0.42 17	I_γ : from 1972Hu10 and 1976Me12. I_γ : from 1972Hu10 and 1976Me12. Mult., δ : from $\gamma\gamma(\theta)$ in 1967Ko20 and 1967Ko23.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								Other: $+2.8 < \delta < +4.0$ (1971Wi13). $A_2 = -0.24$ 6, $A_4 = -0.08$ 7 (1971Wi13). $A_2 = +0.25$ 3, $A_4 = -0.49$ 8 (1967Ko20,1967Ko23,1970Ko33). I_γ : weighted average of 8 1 (1972Hu10) and 9.4 10 (1976Me12).
2645.70	(5/2,7/2)	882.55	8.7 10	1763.14	5/2 ⁺			I_γ : weighted average of 92 1 (1972Hu10) and 90.6 10 (1976Me12). Mult., δ : D+Q with $\delta = -0.62 + 38-36$ from $\gamma(\theta)$ for $J_1=3/2$ (1967Ko20). $A_2 = +0.124$ 17, $A_4 = -0.230$ 24 (1967Ko20,1967Ko23).
		1426.28	<2	1219.39	1/2 ⁺			E_γ : other: 933 7 (1970Ko33). I_γ : weighted average of 13 2 (1972Hu10) and 13.0 10 (1976Me12). $A_2 = -0.05$ 38 (1971Wi13). E_γ : other: 1475 7 (1970Ko33). I_γ : weighted average of 6 2 (1972Hu10) and 8.0 10 (1976Me12). Other: 66 14 from 1963Ha32.
		2645.59	91.3 10	0	3/2 ⁺			I_γ : weighted average of 81 3 (1972Hu10) and 79 2 (1976Me12). Other: 34 9 from 1963Ha32.
2693.9	(3/2,5/2)	930.8	13.0 10	1763.14	5/2 ⁺			δ : $+0.1 < \delta < +0.26$ or $-35 < \delta < -7$ for $J=3/2$, $\delta < -5$ for $J=5/2$ (1971Wi13). $A_2 = +0.20$ 10, $A_4 = -0.16$ 11 (1971Wi13). I_γ : from 1972Hu10. Other: $I_\gamma < 2$ (1976Me12).
		1474.5	7.6 10	1219.39	1/2 ⁺			I_γ : other: $I_\gamma \leq 5$ (1972Hu10). E_γ : other: 3006 7 (1970Ko33). I_γ : from 1972Hu10 and 1976Me12.
		2693.8	80 2	0	3/2 ⁺	D+Q		Mult., δ : from $\gamma\gamma(\theta)$ in 1967Ko20. $A_2 = +0.45$ 18, $A_4 = +0.05$ 19 (1971Wi13). $A_2 = -0.025$ 11, $A_4 = -0.084$ 16 (1967Ko20).
3002.75	5/2	1239.59	≤ 1	1763.14	5/2 ⁺			I_γ : from 1971Pr11. Other: <3 (1976Me12).
		1783.31	<3	1219.39	1/2 ⁺			I_γ : from 1971Pr11. Others: 11 2 (1972Hu10), 8.9 20 (1976Me12). $A_2 = +0.31$ 8, $A_4 = -0.06$ 8 (1971Wi13).
		3002.61	100	0	3/2 ⁺	D+Q	-0.22 1	I_γ : from 1971Pr11. Others: ≤ 0.2 (1972Hu10), <0.2 (1976Me12). Mult., δ : from $\gamma(\theta)$ in 1971Pr11, which states that the anomalous $\gamma(\theta)$ can be understood if it occurs by a mixture of M2 and E1 radiation. In view of the weakness of this transition, such a mixture is not unreasonable.
3162.99	7/2 ⁻	160.24	1.7 2	3002.75	5/2			E_γ : other: 3163 7 (1970Ko33). I_γ : from 1971Pr11. Others: 89 2 (1972Hu10), 91 2 (1976Me12). Mult.: Q+O from $\gamma(\theta)$ in 1966Az01, 1966Wa09, 1967Wa22, and 1967Ta08;
		517.29	8 1	2645.70	(5/2,7/2)			
		1399.82	0.30 4	1763.14	5/2 ⁺	E1+M2	+0.44 12	
		1943.54	<0.2	1219.39	1/2 ⁺			
		3162.84	90 1	0	3/2 ⁺	M2+E3	+0.25 4	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								magnetic character from $\gamma(\text{pol})$ in 1966Wa09 and 1967Wa22 . δ : from 1966Az01 . Others: +0.25 5 (1967Ta08); +0.16 1 (1966Wa09,1967Wa22). Asymmetry ratio $R=1.426$ 7, $N_{90}/N_0=1.25$ 11 (1971Wi13). $\text{POL}=-0.75$ 20, $\Delta\pi=\text{yes}$ (1967Wa22). $\text{POL}=-1.27$ 10, $\Delta\pi=\text{yes}$ (1966Wa09). $0.00<\text{POL}<0.384$ (1966Az01,1967Ta08). $A_2=+0.40$ 8, $A_4=-0.23$ 8 (1971Wi13). $A_2=+0.25$ 3, $A_4=+0.02$ 5 (1967Ta08). $A_2=+0.55$ 3, $A_4=-0.008$ 38 (1966Az01).
3918.48	3/2 ⁺	1272.76 2155.27	<3 19.5 23	2645.70 (5/2,7/2) 1763.14 5/2 ⁺				I_γ : weighted average of 17 3 (1972Hu10), 25 3 (1976Me12), and 18.0 21 (1976Sp09). I_γ : weighted average of 83 3 (1972Hu10), 75 3 (1976Me12), and 82.0 10 (1976Sp09). Mult.: D+Q from $\gamma(\theta)$ in 1976Sp09 ; $\delta=-0.21$ 2 or +20 +10-5 (1976Sp09). E1+M2 ruled by RUL.
		3918.25	81.5 15	0 3/2 ⁺		M1+E2		
3942.9		1297.2	8 2	2645.70 (5/2,7/2)				$A_2=+0.07$ 2, $A_4=-0.01$ 3 (1976Sp09). I_γ : weighted average of 7 3 (1972Hu10) and 8 2 (1976Me12). I_γ : weighted average of 93 3 (1972Hu10) and 92 2 (1976Me12).
		2179.7	92 2	1763.14 5/2 ⁺				
		2723.4 ^c 3942.7 ^c	<6 <10	1219.39 1/2 ⁺ 0 3/2 ⁺				
3967.3		1273.4	2.1 5	2693.9 (3/2,5/2)				I_γ : weighted average of 5 3 (1972Hu10) and 2.0 5 (1976Me12).
		1321.6 ^c 2204.1 2747.8	<2 ≤ 1 79 5	2645.70 (5/2,7/2) 1763.14 5/2 ⁺ 1219.39 1/2 ⁺				I_γ : from 1972Hu10 . Other: <2 (1976Me12). I_γ : weighted average of 74 5 (1972Hu10) and 83 5 (1976Me12). I_γ : weighted average of 21 3 (1972Hu10) and 15 5 (1976Me12).
		3967.1	19 3	0 3/2 ⁺				
4059.2		1365.3 1413.5 ^c 2296.0	1.2 4 <1 4.8 15	2693.9 (3/2,5/2) 2645.70 (5/2,7/2) 1763.14 5/2 ⁺				E_γ : other: 2296 7 (1970Ko33). I_γ : other: 3 2 (1972Hu10). E_γ : other: 2838 7 (1970Ko33). I_γ : other: 97 2 (1972Hu10). 45 5 (1971Wi13). E_γ : other: 4058 7 (1970Ko33). I_γ : other: 20 2 (1971Wi13).
		2839.7	94 2	1219.39 1/2 ⁺				I_γ : weighted average of 50 20 (1972Hu10), 44 7 (1976Me12), and 48 3 (1976Sp09). Mult., δ : from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.56$ 3, $A_4=+0.05$ 4 (1976Sp09).
		4059.0	<2	0 3/2 ⁺				
4113.7	7/2	2350.5	47 3	1763.14 5/2 ⁺		D+Q	-0.16 2	I_γ : weighted average of 50 20 (1972Hu10), 44 7 (1976Me12), and 48 3 (1976Sp09). Mult., δ : from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.56$ 3, $A_4=+0.05$ 4 (1976Sp09).
		2894.2 ^c 4113.4	<10 53 3	1219.39 1/2 ⁺ 0 3/2 ⁺		Q(+O)	0.00 10	I_γ : weighted average of 50 20 (1972Hu10) and 56 7 (1976Me12), and 52 3 (1976Sp09). Mult., δ : Q(+O) from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.46$ 25, $A_4=-0.23$ 25 (1976Sp09).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
4173.45		1479.5	26 8	2693.9	(3/2,5/2)			I_γ : weighted average of 27 8 (1972Hu10) and 25 8 (1976Me12).
		1527.71	<10	2645.70	(5/2,7/2)			
		2410.22	16 5	1763.14	5/2 ⁺			I_γ : weighted average of 18 6 (1972Hu10) and 15 5 (1976Me12).
		2953.93	≤ 3	1219.39	1/2 ⁺			I_γ : from 1972Hu10. Other: <4 (1976Me12); 32 3 (1971Wi13).
		4173.18	58 10	0	3/2 ⁺			I_γ : weighted average of 55 10 (1972Hu10) and 60 10 (1976Me12). $A_2=-0.01$ 3, $A_4=-0.006$ 40 (1967Ta08).
4177.9	3/2	1175.1	<0.5	3002.75	5/2			
		1484.0	8 2	2693.9	(3/2,5/2)			I_γ : weighted average of 7 2 (1972Hu10) and 10 3 (1976Me12).
		1532.2	<0.5	2645.70	(5/2,7/2)			
		2414.7	<1	1763.14	5/2 ⁺			I_γ : other ≤ 1 (1972Hu10).
		2958.4	31 5	1219.39	1/2 ⁺	D+Q	+0.11 4	I_γ : weighted average of 31 5 (1972Hu10) and 30 5 (1976Me12).
		4177.6	61 5	0	3/2 ⁺	D+Q	+0.06 3	Mult., δ : from $\gamma(\theta)$ in 1967Wa22. I_γ : weighted average of 62 6 (1972Hu10) and 60 5 (1976Me12).
								Mult., δ : from $\gamma(\theta)$ in 1967Wa22. POL=+0.12 31 (1967Wa22).
4347.76	5/2,9/2	1184.75	70 2	3162.99	7/2 ⁻			I_γ : weighted average of 70 5 (1972Hu10), 72 5 (1976Me12), and 69 2 (1976Sp09).
		1344.98 ^c	<4	3002.75	5/2			
		1653.8 ^c	<4	2693.9	(3/2,5/2)			
		1702.02	31 2	2645.70	(5/2,7/2)	D+Q	-0.018 12	I_γ : weighted average of 30 5 (1972Hu10), 28 5 (1976Me12), and 31 2 (1976Sp09).
								Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09. $A_2=-0.34$ 3, $A_4=-0.00$ 4 (1976Sp09).
		2584.52	≤ 3	1763.14	5/2 ⁺			I_γ : from 1972Hu10. Other: <5 (1976Me12).
		3128.22 ^c	<4	1219.39	1/2 ⁺			
		4347.47	≤ 5	0	3/2 ⁺			I_γ : from 1972Hu10. Other: <10 (1976Me12).
4624.4		4624.1	85 10	0	3/2 ⁺			I_γ : other: 100 (1972Hu10).
4769.86		1606.83	65 10	3162.99	7/2 ⁻			I_γ : other: 53 10 (1972Hu10) and 47% of I_γ to unknown levels (1972Hu10).
		1767.06	35 10	3002.75	5/2			
		2075.9	<10	2693.9	(3/2,5/2)			
		3550.28	<10	1219.39	1/2 ⁺			
		4769.51	<10	0	3/2 ⁺			
4839.10		1836.30	<7	3002.75	5/2			
		2145.1	<4	2693.9	(3/2,5/2)			
		2193.33	<3	2645.70	(5/2,7/2)			
		3075.82	59 5	1763.14	5/2 ⁺			I_γ : weighted average of 50 20 (1972Hu10) and 60 5 (1976Me12). Other: 48 4 (1976Sp09).
		3619.51	<7	1219.39	1/2 ⁺			
		4838.74	41 5	0	3/2 ⁺			I_γ : weighted average of 50 20 (1972Hu10) and 40 5 (1976Me12). Other: 18 3 (1976Sp09).
4854.4		3634.8	75 5	1219.39	1/2 ⁺			
		4854.0	25 5	0	3/2 ⁺			
4880.96		1878.16	9 3	3002.75	5/2			
		2187.0	<10	2693.9	(3/2,5/2)			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
4880.96		2235.18	29 5	2645.70	(5/2,7/2)			I_γ : other: 30 10 (1972Hu10).
		3117.67	62 5	1763.14	5/2 ⁺			I_γ : other: 70 10 (1972Hu10).
		3661.36 ^c	<4	1219.39	1/2 ⁺			
		4880.60	<5	0	3/2 ⁺			
5010.1		836.6	<25 ^a	4173.45				
		950.9	<10 ^a	4059.2				
		1847.1	<50 ^a	3162.99	7/2 ⁻			
		2007.3	<20 ^a	3002.75	5/2			
		2316.1	<25 ^a	2693.9	(3/2,5/2)			
		2364.3	<5	2645.70	(5/2,7/2)			
		3246.8	<10	1763.14	5/2 ⁺			
		3790.5	<7	1219.39	1/2 ⁺			
		5009.7	100	0	3/2 ⁺			I_γ : from 1976Me12 and 1972Hu10.
5163.34	7/2	2000.29	40 5	3162.99	7/2 ⁻	D+Q	+0.44 20	I_γ : weighted average of 40 20 (1972Hu10), 36 5 (1976Me12), and 54 10 (1976Sp09). Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09. $A_2=+0.52$ 5, $A_4=-0.02$ 6 (1976Sp09).
		2160.52	10 4	3002.75	5/2			I_γ : weighted average of 10 5 (1972Hu10), 10 4 (1976Me12), and 10 5 (1976Sp09).
		3400.02	43 5	1763.14	5/2 ⁺	D(+Q)	0.00 3	I_γ : weighted average of 50 20 (1972Hu10), 44 5 (1976Me12), and 36 10 (1976Sp09). Mult., δ : D(+Q) from $\gamma(\theta)$ in 1976Sp09. $A_2=-0.23$ 6, $A_4=-0.08$ 7 (1976Sp09).
		3943.71	<10	1219.39	1/2 ⁺			
		5162.93	<10	0	3/2 ⁺			
5215.8		205.7	<23 ^a	5010.1				
		1042.3	<60 ^a	4173.45				
		1156.6	<10 ^a	4059.2				
		2052.8	<15 ^a	3162.99	7/2 ⁻			
		2213.0	<25 ^a	3002.75	5/2			
		2521.8	<15 ^a	2693.9	(3/2,5/2)			
		2570.0	<15 ^a	2645.70	(5/2,7/2)			
		3452.5	<6	1763.14	5/2 ⁺			
		3996.2	<4	1219.39	1/2 ⁺			
		5215.4	100	0	3/2 ⁺			
5403.5		2400.7	25 5	3002.75	5/2			I_γ : other: 25 10 (1972Hu10).
		3640.2	<10	1763.14	5/2 ⁺			I_γ : other: 25 10 from 1972Hu10.
		4183.8	55 10	1219.39	1/2 ⁺			
		5403.1	<7	0	3/2 ⁺			
5586		2940	40 10	2645.70	(5/2,7/2)			I_γ : weighted average of 40 20 (1972Hu10) and 40 10 (1976Me12).
		3823	60 10	1763.14	5/2 ⁺			I_γ : weighted average of 60 20 (1972Hu10) and 60 10 (1976Me12).
		4366	<10	1219.39	1/2 ⁺			
		5586	<10	0	3/2 ⁺			
5599.7		3836.3	27 10	1763.14	5/2 ⁺			I_γ : weighted average of 20 10 (1972Hu10) and 33 10 (1976Me12).
		5599.2	74 10	0	3/2 ⁺			I_γ : weighted average of 80 10 (1972Hu10) and 67 10 (1976Me12).
5646		2483	100	3162.99	7/2 ⁻			I_γ : from 1972Hu10, 1976Me12, 1971Pr11.
		2643		3002.75	5/2			I_γ : 80 from 2001Vo24 only; not observed by others.
		3883		1763.14	5/2 ⁺			I_γ : 6 from 2001Vo24 only; not observed by others.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
5646		5646	<8	0	3/2 ⁺			
5655		2652	80 5	3002.75	5/2			
		3892	6 2	1763.14	5/2 ⁺			
		4435	<6	1219.39	1/2 ⁺			
5683		1505	60 10	4177.9	3/2			
5723.52		3029.5	24 3	2693.9	(3/2,5/2)			I_γ : from 1976Sp09.
		3960.14	38 4	1763.14	5/2 ⁺			I_γ : from 1976Sp09.
5758		1580	30 10	4177.9	3/2			
		4538	<6	1219.39	1/2 ⁺			
		5758	45 10	0	3/2 ⁺			
5806		5806	100	0	3/2 ⁺			I_γ : from 1972Hu10 and 1976Me12.
6106.2		4342.8	6 2	1763.14	5/2 ⁺			
		4886.4	36 10	1219.39	1/2 ⁺			I_γ : other: 40 10 (1972Hu10).
		6105.6	58 10	0	3/2 ⁺			I_γ : other: 60 10 (1972Hu10).
6181		6180	45 10	0	3/2 ⁺			
6492		6491	100	0	3/2 ⁺			
7066.2		1902.8	0.9	5163.34	7/2			
		2892.6	0.5	4173.45				
		2952.4	0.6	4113.7	7/2			
		3006.9	1.4	4059.2				I_γ : other: 1.0 5 from 1972Hu10.
		3903.0	6	3162.99	7/2 ⁻			I_γ : other: 6.0 6 from 1972Hu10.
		4063.2	3	3002.75	5/2			I_γ : other: 2 1 from 1972Hu10.
		4372.0	5	2693.9	(3/2,5/2)			I_γ : other: 6.0 6 from 1972Hu10.
		4420.2	1.6	2645.70	(5/2,7/2)			I_γ : other: 3.0 15 from 1972Hu10.
		5302.6	14	1763.14	5/2 ⁺			I_γ : other: 16.0 16 from 1972Hu10.
		5846.3	18	1219.39	1/2 ⁺			I_γ : other: 18.0 18 from 1972Hu10.
		7065.4	49	0	3/2 ⁺			I_γ : other: 48.0 48 from 1972Hu10.
7103.4	3/2	2925.4	1.5	4177.9	3/2			
		3184.8	1.5	3918.48	3/2 ⁺			I_γ : other: 2 1 (1972Hu10).
		4100.4	3	3002.75	5/2			I_γ : other: 3.0 15 from 1972Hu10.
		4409.2	11	2693.9	(3/2,5/2)	D(+Q)	0.00 3	I_γ , Mult., δ : from 1976Me12. Other: 13.0 13 (1972Hu10).
								$A_2=+0.34$ 7, $A_4=+0.03$ 11 (1976Me12).
		5339.8	4	1763.14	5/2 ⁺			I_γ : other: 4 2 from 1972Hu10.
		5883.5	64	1219.39	1/2 ⁺	D(+Q)	0.00 3	I_γ , Mult., δ : from 1976Me12. Other: 67.0 67 (1972Hu10).
								$A_2=-0.53$ 8, $A_4=+0.01$ 11 (1976Me12).
		7102.6	15	0	3/2 ⁺			I_γ : other: 11.0 11 (1972Hu10).
								$A_2=+0.54$ 5, $A_4=-0.01$ 8 (1976Me12).
7178.6	1/2	1072.4	3	6106.2				I_γ : other: 4 2 (1972Hu10).
		1775.1	1.7	5403.5				
		2339.4	1.5	4839.10				I_γ : other: 2 (2001Vo24).
		3005.0	0.8	4173.45				I_γ : other: 1 (2001Vo24).
		3119.3	22	4059.2				I_γ : others: 22 (2001Vo24), 24.0 24 (1972Hu10).
								$A_2=-0.02$ 4, $A_4=+0.10$ 66 (1976Me12).
								$A_2=-0.01$ 4, $A_4=+0.09$ 24 (2001Vo24).
		3211.1	9	3967.3				I_γ : others: 9 (2001Vo24), 10.0 10 (1972Hu10).
		3260.0	3	3918.48	3/2 ⁺			I_γ : others: 3 (2001Vo24), 2 1 (1972Hu10).
		4484.4	4	2693.9	(3/2,5/2)			I_γ : others: 4 (2001Vo24), 4 2 (1972Hu10).
		5958.7	17	1219.39	1/2 ⁺			I_γ : others: 17 (2001Vo24), 16.0 16 (1972Hu10).
								$A_2=-0.01$ 3, $A_4=+0.06$ 7 (2001Vo24).
								$A_2=-0.01$ 5, $A_4=+0.07$ 8 (1976Me12).
		7177.8	38	0	3/2 ⁺			I_γ : others: 38 (2001Vo24), 40.0 40 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
								$A_2=-0.01$ 6, $A_4=-0.02$ 7 (1976Me12). $A_2=-0.02$ 4, $A_4=-0.01$ 5 (2001Vo24).
7185		1969	<0.5 ^a	5215.8				
		2175	<0.5 ^a	5010.1				
		3011	<5 ^a	4173.45				
		3126	26 ^a 6	4059.2				
		4022	<3 ^a	3162.99	7/2 ⁻			
		4182	<3 ^a	3002.75	5/2			
		4491	2 ^a 1	2693.9	(3/2,5/2)			
		4539	6 ^a 2	2645.70	(5/2,7/2)			
		5421	4 ^a 3	1763.14	5/2 ⁺			
		5965	16 ^a 7	1219.39	1/2 ⁺			
7194.6	1/2,3/2	7184	46 ^a 10	0	3/2 ⁺			
		2184.4	0.8	5010.1				I_γ : other: 1.0 5 (1972Hu10).
		2340.1	2	4854.4				I_γ : other: 2 1 (1972Hu10).
		2355.4	1.8	4839.10				
		3016.6	16	4177.9	3/2			I_γ : other: 15.0 15 (1972Hu10).
		3135.3	3	4059.2				I_γ : other: 3.0 15 (1972Hu10).
		3227.1	1.8	3967.3				I_γ : other: 2 1 (1972Hu10).
		3276.0	4	3918.48	3/2 ⁺			I_γ : other: 5.0 25 (1972Hu10).
		4500.4	1.1	2693.9	(3/2,5/2)			
		5431.0	0.5	1763.14	5/2 ⁺			
7225.5	5/2	5974.7	67	1219.39	1/2 ⁺			I_γ : other: 72.0 72 (1972Hu10).
		7193.8	2	0	3/2 ⁺			
		1119.3	1.8	6106.2				I_γ : other: 3.0 15 (1972Hu10).
		1420		5806				I_γ : 1.0 5 (1972Hu10).
		1543	2.0	5683				I_γ : other: 2 1 (1972Hu10).
		1571	0.8	5655				
		1822.0	0.9	5403.5				I_γ : other: 1.0 5 (1972Hu10).
		2009.6	7	5215.8				I_γ : other: 6.0 30 (1972Hu10).
		3047.5	8	4177.9	3/2			I_γ : other: 7.0 35 (1972Hu10).
		3166.2	2	4059.2				
7234.0	5/2 ⁺	3306.9	1.5	3918.48	3/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		4222.5		3002.75	5/2			I_γ : 3.0 15 (1972Hu10).
		4531.3	10	2693.9	(3/2,5/2)	D+Q	+0.36 4	I_γ , Mult., δ : from 1976Me12. Other: 10.0 10 (1972Hu10).
								$A_2=+0.43$ 8, $A_4=-0.21$ 9 (1976Me12).
		5461.9	7	1763.14	5/2 ⁺	D+Q	-0.36 7	I_γ , Mult., δ : from 1976Me12. Other: 8.0 8 (1972Hu10).
								$A_2=+0.02$ 10, $A_4=-0.09$ 17 (1976Me12).
		7224.7	59	0	3/2 ⁺	D+Q	+0.36 4	I_γ , Mult., δ : from 1976Me12. Other: 58.0 58 (1972Hu10).
								$A_2=+0.40$ 7, $A_4=+0.04$ 12 (1976Me12).
		1579	0.2	5655				
		3056.0		4177.9	3/2			I_γ : 1.0 5 (1972Hu10).
7272.7	(1/2,3/2)	4539.8	1.8	2693.9	(3/2,5/2)			I_γ : other: 2 1 (1972Hu10).
		4588.0	3	2645.70	(5/2,7/2)			I_γ : other: 2 1 (1972Hu10).
		5470.4	0.2	1763.14	5/2 ⁺			
		6014.1	1.8	1219.39	1/2 ⁺			I_γ : other: 2 1 (1972Hu10).
		7233.2	93	0	3/2 ⁺	D+Q	-0.10	I_γ : other: 93.0 93 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1960An06.
		1515	1.2	5758				I_γ : other: 1.5 8 (1972Hu10).
		1869.2	0.2	5403.5				I_γ : other: 0.5 3 (1972Hu10).
		2262.5	1.4	5010.1				I_γ : other: 1.5 8 (1972Hu10).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7272.7	(1/2,3/2)	2433.5	1.0	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3099.1	0.4	4173.45				I_γ : other: 0.5 3 (1972Hu10).
		3213.3	0.9	4059.2				I_γ : other: 1.0 5 (1972Hu10).
		3305.2	0.8	3967.3				I_γ : other: 1.0 5 from 1972Hu10.
		4578.5	1.1	2693.9	(3/2,5/2)			I_γ : other: 1.0 5 (1972Hu10).
		6052.8	24	1219.39	1/2 ⁺			I_γ : other: 23.0 23 (1972Hu10).
		7271.9	69	0	3/2 ⁺			I_γ : other: 69.0 69 (1972Hu10).
7362.0	3/2	1707	0.3	5655				I_γ : other: 0.5 3 (1972Hu10).
		2351.8	0.6	5010.1				I_γ : other: 0.5 3 (1972Hu10).
		2522.8	1.0	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3183.9	0.5	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		3302.6	2	4059.2				I_γ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).
		3394.5	3	3967.3				I_γ : others: 3.0 15 (1972Hu10); 4 (1971Wi13).
		3443.3	0.8	3918.48	3/2 ⁺			I_γ : other: 0.5 3 (1972Hu10).
		4359.0	0.3	3002.75	5/2			I_γ : other: 0.5 3 (1972Hu10).
		4667.8	0.5	2693.9	(3/2,5/2)			
		5598.4	8	1763.14	5/2 ⁺	D(+Q)		I_γ : others: 10.0 10 (1972Hu10); 9 (1971Wi13).
								Mult., δ : $\delta=-0.01$ 10 or $\delta>+2.5$ (1977Ko34).
		6142.0	73	1219.39	1/2 ⁺	D+Q		$A_2=-0.20$ 24, $A_4=+0.92$ 25 (1977Ko34).
								I_γ : others: 70.0 70 (1972Hu10); 72 (1971Wi13).
								Mult., δ : $\delta=-0.05$ 3 or $+2.5$ 3 (1977Ko34); $\delta=-20$ or $+0.22<\delta<+0.26$ (1971Wi13).
								$A_2=-0.05$ 3, $A_4=-0.04$ 5 (1971Wi13).
								$A_2=-0.25$ 11, $A_4=+0.31$ 11 (1977Ko34).
		7361.2	10	0	3/2 ⁺	D+Q		I_γ : others: 10.0 10 (1972Hu10); 11 (1971Wi13).
								Mult., δ : $-0.16<\delta<-0.12$ or $+7.5<\delta<+10$ (1971Wi13).
								$A_2=+0.19$ 3, $A_4=-0.08$ 5 (1971Wi13).
7396.0	7/2	3048.1	11	4347.76	5/2,9/2			I_γ : other: 10.0 10 (1972Hu10).
		3282.1	8	4113.7	7/2			I_γ : other: 9.0 9 (1972Hu10).
		3452.9	5	3942.9				I_γ : other: 8.0 8 (1972Hu10).
		4232.7	47	3162.99	7/2 ⁻	D+Q	+0.28 25	I_γ : other: 49.0 49 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		4393.0	14	3002.75	5/2	D(+Q)	+0.011 18	I_γ : other: 14.0 14 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
								I_γ : other: 10.0 10 (1972Hu10).
		4750.0	13	2645.70	(5/2,7/2)			
		5632.4	2	1763.14	5/2 ⁺			
7451.1	3/2	3532.4	5.0 25	3918.48	3/2 ⁺			I_γ : from 1972Hu10.
		4448.1	10 1	3002.75	5/2			I_γ : from 1972Hu10.
		4756.9	8.0 8	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.
		5687.5	4 2	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
		6231.1	73 7	1219.39	1/2 ⁺			I_γ : from 1972Hu10.
7501.1		3558.0	25 3	3942.9				I_γ : from 1972Hu10.
		4498.0	13.0 13	3002.75	5/2			I_γ : from 1972Hu10.
		4855.0	55 6	2645.70	(5/2,7/2)			I_γ : from 1972Hu10.
		5737.5	7.0 7	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
7502.9		1396.7	7	6106.2				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
								Other: 10.0 10 for 7503 level only (1972Hu10).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7502.9		1903.1	2	5599.7				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		2492.7	2	5010.1				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3324.8	3	4177.9	3/2			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		3443.5	50	4059.2				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 70.0 70 for 7503 level only (1972Hu10).
		3559.8	2	3942.9				I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4339.6	4	3162.99	7/2 ⁻			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4499.8	4	3002.75	5/2			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		4856.8	7	2645.70	(5/2,7/2)			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		5739.3	0.7	1763.14	5/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
		6282.9	18	1219.39	1/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels. Other: 20.0 20 for 7503 level only (1972Hu10).
		7502.0	0.3	0	3/2 ⁺			I_γ : from 1976Me12 containing γ rays of 7501 and 7503 doublet initial levels.
7518.8	7/2	1919.0	2 1	5599.7				I_γ : from 1972Hu10 .
		1933	3.0 15	5586				I_γ : from 1972Hu10 .
		3170.9	2 1	4347.76	5/2,9/2			I_γ : from 1972Hu10 .
		3575.7	2 1	3942.9				I_γ : from 1972Hu10 .
		4355.5	68 7	3162.99	7/2 ⁻	D(+Q)	+0.1 2	I_γ : from 1972Hu10 . Mult., δ : from 1977Ko35 .
		4872.7	3.0 15	2645.70	(5/2,7/2)			I_γ : from 1972Hu10 .
		5755.2	20 2	1763.14	5/2 ⁺	D+Q	+0.098 22	I_γ : from 1972Hu10 . Mult., δ : from $\gamma(\theta)$ in 1977Ko35 .
7548.9	7/2 ⁻	1903	0.55 6	5646				I_γ : from 1971Pr11 . Others: 0.6 (1976Me12); 0.5 3 (1972Hu10); 1 (2001Vo24 , 2002Vo17).
		1963	0.1	5586				I_γ : other: 0.2 1 (1972Hu10).
		2778.9	1.4 1	4769.86				I_γ : from 1971Pr11 . Others: 1.3 (1976Me12); 1.5 8 (1972Hu10).
		4385.6	94 1	3162.99	7/2 ⁻	M1+E2	-0.07 2	I_γ : from 1971Pr11 . Others: 95 (1976Me12); 95.0 95 (1972Hu10); ≥ 93 (1971Wi13). Mult.: D+Q from $\gamma(\theta)$ in 1967Wa22 and 1966Wa09 ; magnetic character from $\gamma(\text{pol})$ in 1967Wa22 . δ : from 1966Wa09 and 1967Wa22 . Others: $\delta=+0.08$ 10 (1977Ko35), $\delta=-0.03$ 3 from 2001Vo24 and 2002Vo17 . $N_{90}/N_0=0.72$ 6 (1971Wi13). POL=+0.76 6, $\Delta\pi=\text{no}$ (1966Wa09). POL=+0.93 10, $\Delta\pi=\text{no}$ (1967Wa22). 0.021<POL<0.813 (1966Az01). $A_2=+0.42$ 3, $A_4=0.00$ 3 (2001Vo24 , 2002Vo17). $A_2=+0.42$ 3, $A_4=-0.001$ 27 (1966Az01). $A_2=+0.42$ 3, $A_4=+0.00$ 3 (1967Ta08). I_γ : from 1971Pr11 . Others: 2 (1976Me12); 1.8
		4545.8	2.0 1	3002.75	5/2	D(+Q)	+0.6 4	

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7548.9	7/2 ⁻	4902.8	0.68 7	2645.70	(5/2,7/2)			9 (1972Hu10); 1.3 (1971Wi13). Mult., δ : from $\gamma(\theta)$ in 1977Ko35. I $_\gamma$: from 1971Pr11. Others: 0.5 (1976Me12); 0.5 3 (1972Hu10); ≤ 5 (1971Wi13).
		5785.3	0.33 4	1763.14	5/2 ⁺			I $_\gamma$: from 1971Pr11. Others: 0.3 (1976Me12); 0.3 2 (1972Hu10); ≈ 0.4 (1971Wi13).
		7548.0	0.27 3	0	3/2 ⁺	Q(+O)	+0.1 2	I $_\gamma$: from 1971Pr11. Others: 0.2 (1976Me12); 0.2 1 (1972Hu10); ≈ 0.3 (1971Wi13). Mult., δ : from $\gamma(\theta)$ in 1977Ko35. I $_\gamma$: other: 1.0 5 (1972Hu10). I $_\gamma$: other: 5.0 25 (1972Hu10).
7561.3	1/2,3/2	3383.2	0.9	4177.9	3/2			I $_\gamma$: other: 1.0 5 (1972Hu10). I $_\gamma$: other: 21.0 21 (1972Hu10). A $_2$ =+0.04 3, A $_4$ =-0.05 3 (1976Me12). I $_\gamma$: other: 36.0 36 (1972Hu10). A $_2$ =-0.02 2, A $_4$ =-0.05 3 (1976Me12). I $_\gamma$: other: 36.0 36 (1972Hu10). A $_2$ =+0.01 2, A $_4$ =-0.03 3 (1976Me12).
		3501.9	6	4059.2				
		3593.8	0.5	3967.3				
		3642.6	1.6	3918.48	3/2 ⁺			
		4867.0	22	2693.9	(3/2,5/2)			
		6341.3	35	1219.39	1/2 ⁺			
		7560.4	34	0	3/2 ⁺			
7600.8	5/2	2437.4	1.9	5163.34	7/2			
		2719.7	2	4880.96				
		2761.6	2	4839.10				
		2830.8	1.3	4769.86				
		2976.3	1.8	4624.4				
		3422.7	3	4177.9	3/2			I $_\gamma$: other: 2 1 (1972Hu10). I $_\gamma$: 2 1 from 1972Hu10. I $_\gamma$: other: 1.5 8 (1972Hu10). I $_\gamma$: others: 4 2 (1972Hu10); 8 (1971Wi13). Mult., δ : Q+O with $\delta=-0.25$ 15 for J $_i$ =3/2 (1977Ko34). A $_2$ =+0.19 2, A $_4$ =+0.08 2 (1977Ko34). A $_2$ =+0.35 32, A $_4$ =-0.44 32 (1971Wi13). I $_\gamma$: others: 9.0 9 (1972Hu10); 12 (1971Wi13). Mult., δ : from 1971Wi13; δ =+0.85 42 for J $_i$ =3/2 (1977Ko34). A $_2$ =+0.75 7, A $_4$ =-0.06 7 (1971Wi13). A $_2$ =+0.53 21, A $_4$ =-0.04 25 (1976Me12). A $_2$ =+0.42 3, A $_4$ =-0.10 3 (1977Ko34). I $_\gamma$: others: 19.0 19 (1972Hu10); 24 (1971Wi13). δ : from 1971Wi13; other: $\delta=-5.3$ 9 or +0.01 5 for J $_i$ =3/2 (1977Ko34). A $_2$ =+0.33 1, A $_4$ =-0.05 1 (1977Ko34). A $_2$ =+0.18 4, A $_4$ =-0.02 5 (1976Me12). A $_2$ =+0.26 10, A $_4$ =-0.05 10 (1971Wi13). I $_\gamma$: other: 1.5 8 (1972Hu10). I $_\gamma$: others: 20 2 (1971Wi13), 2 1 (1972Hu10); 25 (1971Wi13). Mult., δ : from 1971Wi13. δ =+0.2 22 for J $_i$ =3/2 (1977Ko34). A $_2$ =+0.10 7, A $_4$ =-0.10 8 (1971Wi13). A $_2$ =+0.12 2, A $_4$ =-0.10 3 (1976Me12). A $_2$ =+0.30 9, A $_4$ =-0.17 9 (1977Ko34). I $_\gamma$: others: 1.0 5 (1972Hu10); ≤ 1.5 (1971Wi13).
		3427.2		4173.45				
		3541.4	2	4059.2				
		4437.5	5	3162.99	7/2 ⁻			
		4597.7	8	3002.75	5/2	D+Q	+0.50 22	
		4906.5	18	2693.9	(3/2,5/2)	D+Q	+0.32 7	
		4954.7	1.9	2645.70	(5/2,7/2)			
		5837.1	19	1763.14	5/2 ⁺	D+Q	+0.31 7	
		6380.8	1.1	1219.39	1/2 ⁺			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued)

$\gamma(^{35}\text{Cl})$ (continued)								Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	
7600.8	5/2	7599.9	33	0	3/2 ⁺	D+Q	-0.19 4	I_γ : others: 31.0 3I (1972Hu10); 31 (1971Wi13). Mult., δ : from 1976Me12. Others: $\delta=+0.17$ 3 (1971Wi13). $A_2=-0.73$ 7, $A_4=+0.04$ 6 (1971Wi13). $A_2=-0.62$ 5, $A_4=-0.06$ 5 (1976Me12). $A_2=-0.56$ 3, $A_4=-0.05$ 2 (1977Ko34).
7618.7	5/2	1438	0.6	6181				I_γ : other: 3.0 15 (1972Hu10).
		2402.8	2	5215.8				
		2455.3	0.7	5163.34	7/2			
		2779.5	0.7	4839.10				
		3440.6	3	4177.9	3/2			
		3559.3	2	4059.2				
		4455.4	3	3162.99	7/2 ⁻			
		5855.0	10	1763.14	5/2 ⁺	D+Q	+0.48 11	
		7617.8	78	0	3/2 ⁺	D+Q	+2.2 3	I_γ : other: 80 8 (1972Hu10). Mult., δ : from $\gamma(\theta)$ in 1977Ko34. $A_2=+0.38$ 5, $A_4=-0.01$ 6 (1976Me12). $A_2=-0.11$ 3, $A_4=-0.06$ 3 (1977Ko34).
7656.6		1550.4	3.0 15	6106.2				I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10. I_γ : from 1972Hu10.
		1974	1.0 5	5683				
		2002	1.0 5	5655				
		2817.4	3.0 15	4839.10				
		3478.5	1.0 5	4177.9	3/2			
		3597.2	32.0 32	4059.2				
		3689.1	2.0 10	3967.3				
		7655.7	57 6	0	3/2 ⁺			
7671.9	5/2	2072.1	1.2	5599.7				I_γ : other: 4 2 (1972Hu10). I_γ : other: 1.0 5 (1972Hu10). I_γ : other: 6.0 6 (1972Hu10). I_γ : other: 4 2 (1972Hu10). I_γ : other: 4 2 (1972Hu10). I_γ : other: 8.0 8 (1972Hu10). Mult., δ : from $\gamma(\theta)$ in 1977Ko34. $A_2=-0.35$ 4, $A_4=+0.01$ 5 (1976Me12). $A_2=+0.52$ 1, $A_4=-0.18$ 1 (1977Ko34). I_γ : other: 14.0 14 (1972Hu10). Mult., δ : from $\gamma(\theta)$ in 1977Ko34. $A_2=+0.41$ 3, $A_4=-0.03$ 4 (1976Me12). $A_2=+0.33$ 8, $A_4=-0.12$ 8 (1977Ko34). I_γ : other: 57.0 57 (1972Hu10). Mult., δ : from $\gamma(\theta)$ in 1977Ko34. $A_2=-0.38$ 4, $A_4=-0.06$ 5 (1976Me12). $A_2=-0.40$ 9, $A_4=+0.07$ 8 (1977Ko34). I_γ : other: 2 1 (1972Hu10). I_γ : 0.5 3 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 0.5 3 (1972Hu10).
		2508.5	4	5163.34	7/2			
		2790.8	1.2	4880.96				
		2901.9	1.0	4769.86				
		3324.0	6	4347.76	5/2,9/2			
		3498.3	5	4173.45				
		4508.6	1.8	3162.99	7/2 ⁻			
		4668.8	9	3002.75	5/2	D+Q	-1.4 5	
		5025.8	13	2645.70	(5/2,7/2)	D(+Q)	+0.3 22	I_γ : other: 2 1 (1972Hu10). I_γ : 0.5 3 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 0.5 3 (1972Hu10). I_γ : other: 2 1 (1972Hu10).
		5908.2	57	1763.14	5/2 ⁺	D+Q	+0.86 14	
7685.5	3/2 ⁻	7671.0	0.8	0	3/2 ⁺			I_γ : other: 2 1 (1972Hu10). I_γ : 0.5 3 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 2 1 (1972Hu10). I_γ : other: 0.5 3 (1972Hu10). I_γ : other: 2 1 (1972Hu10).
		1927		5758				
		2085.7	2	5599.7				
		2281.9	2	5403.5				
		2469.6	0.7	5215.8				
		2675.3	0.6	5010.1				
		3507.4	3	4177.9	3/2			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)**

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7685.5	3/2 ⁻	3511.9	4	4173.45				I_γ : other: 5.0 25 (1972Hu10). $A_2=+0.01$ 14, $A_4=+0.23$ 16 (1977Ko34).
		3626.1	6	4059.2				I_γ : other: 6.0 6 (1972Hu10).
		4682.4	10	3002.75	5/2	D(+Q)	-0.01 5	I_γ : other: 9.0 9 (1972Hu10). Mult., δ : other: $\delta=-2.4$ +20-4 (1976Me12); $\delta=+0.01$ 7 (1977Ko34). $A_2=-0.04$ 6, $A_4=-0.09$ 7 (1976Me12). $A_2=-0.06$ 12, $A_4=+0.08$ 13 (1977Ko34).
		5921.8	0.7	1763.14	5/2 ⁺			
		6465.5	1	1219.39	1/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		7684.6	70	0	3/2 ⁺	D+Q	+6.0 7	I_γ , Mult., δ : from 1976Me12. Other: $I_\gamma=72.0$ 72 (1972Hu10); $\delta<-3.5$ for $J_i=3/2$ (1977Ko34). $A_2=+0.25$ 1, $A_4=-0.03$ 1 (1976Me12). $A_2=+0.18$ 11, $A_4=-0.04$ 11 (1977Ko34).
7693.9		2094.1	1.0 5	5599.7				I_γ : from 1972Hu10.
		3634.5	1.0 5	4059.2				I_γ : from 1972Hu10.
		4690.8	2.0 10	3002.75	5/2			I_γ : from 1972Hu10.
		7693.0	96 10	0	3/2 ⁺			I_γ : from 1972Hu10.
7706.4	5/2 ⁺	2051	4	5655				I_γ : other: 4 2 (1972Hu10). Mult., δ : D+Q with $\delta\leq-2$ or ≥ 13 for $J_f(5655)=5/2$ (1977Ko34), but $J_f(5655)=(3/2)^+$ from the Adopted Levels. $A_2=+0.14$ 15, $A_4=-0.39$ 17 (1977Ko34).
		2825.3	0.4	4880.96				
		2867.2	1.3	4839.10				I_γ : other: 1.0 5 (1972Hu10).
		3081.9	0.8	4624.4				
		3528.3	2	4177.9	3/2			
		3592.5	2	4113.7	7/2			I_γ : other: 1.0 5 (1972Hu10).
		3787.7	0.6	3918.48	3/2 ⁺			
		4703.3	1	3002.75	5/2			I_γ : other: 1.0 5 (1972Hu10).
		5060.3	4	2645.70	(5/2,7/2)	D+Q	-1.01 10	I_γ : other: 4 2 (1972Hu10). Mult., δ : or ≥ 13 (1977Ko34). $A_2=-0.03$ 6, $A_4=+0.08$ 8 (1976Me12). $A_2=-0.18$ 17, $A_4=+0.41$ 18 (1977Ko34).
		5942.7	0.4	1763.14	5/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		6486.4	0.5	1219.39	1/2 ⁺			I_γ : other: 88.0 88 (1972Hu10).
		7705.5	83	0	3/2 ⁺	D(+Q)	+0.1 1	Mult., δ : from $\gamma(\theta)$ in 1977Ko34. $A_2=-0.54$ 3, $A_4=-0.07$ 3 (1976Me12). $A_2=-0.63$ 4, $A_4=+0.15$ 3 (1977Ko34).
7744.8	7/2	2863.7	20	4880.96				I_γ : other: 19.0 19 (1972Hu10).
		2974.8	2	4769.86				I_γ : other: 1.0 5 (1972Hu10).
		3630.9	6	4113.7	7/2			I_γ : other: 5.0 25 (1972Hu10).
		3801.7	10	3942.9		D(+Q)	-0.16 17	I_γ : other: 12.0 12 (1972Hu10). Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		4581.5	16	3162.99	7/2 ⁻	D+Q	+0.18 8	Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		5098.7	43	2645.70	(5/2,7/2)	D(+Q)	+0.1 30	Mult., δ : from $\gamma(\theta)$ in 1977Ko35.
		5981.1	2.4	1763.14	5/2 ⁺			
		6524.8	0.6	1219.39	1/2 ⁺			
7777.1		2122	0.5	5655				
		2613.7	1.1	5163.34	7/2			
		2937.9	2	4839.10				I_γ : other: 2 1 (1972Hu10).
		3007.1	3	4769.86				I_γ : other: 2 1 (1972Hu10).
		3599.0	7	4177.9	3/2			I_γ : other: 7.0 7 (1972Hu10).
		3717.7	1.1	4059.2				I_γ : other: 1.0 5 (1972Hu10).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7777.1		4613.8	2	3162.99	7/2 ⁻			
		4774.0	4	3002.75	5/2			I_γ : other: 4 2 (1972Hu10).
		5082.8	9	2693.9	(3/2,5/2)			I_γ : other: 7.0 7 (1972Hu10).
		5131.0	1.3	2645.70	(5/2,7/2)			I_γ : other: 2 1 (1972Hu10).
		6013.4	40	1763.14	5/2 ⁺			I_γ : other: 46.0 46 (1972Hu10).
		6557.1	11	1219.39	1/2 ⁺			I_γ : other: 13.0 13 (1972Hu10).
		7776.2	18	0	3/2 ⁺			I_γ : other: 16.0 16 (1972Hu10).
7781.7	5/2 ⁻	1675.5	1.4	6106.2				
		2181.9		5599.7				I_γ : 1.0 5 from 1972Hu10.
		2378.1		5403.5				I_γ : 1.0 5 from 1972Hu10.
		2565.8	1.5	5215.8				I_γ : other: 1.0 5 (1972Hu10).
		3011.7	7	4769.86				I_γ : other: 7.0 7 (1972Hu10).
		3433.8	2	4347.76	5/2,9/2			I_γ : other: 2 1 (1972Hu10).
		3608.1	36	4173.45				I_γ : other: 40.0 40 (1972Hu10).
		3667.8	6	4113.7	7/2			I_γ : other: 3.0 15 (1972Hu10).
		3722.3	1.2	4059.2				I_γ : other: 1.0 5 (1972Hu10).
		3863.0	11	3918.48	3/2 ⁺			I_γ : other: 10.0 10 (1972Hu10).
		4618.4	5	3162.99	7/2 ⁻			I_γ : other: 5.0 25 (1972Hu10).
		5135.6	8	2645.70	(5/2,7/2)			I_γ : other: 9.0 9 (1972Hu10).
		6018.0	18	1763.14	5/2 ⁺			I_γ : other: 20.0 20 (1972Hu10).
		6561.7	1.2	1219.39	1/2 ⁺			
		7780.8	1.7	0	3/2 ⁺			
7797.1	1/2	2039		5758				I_γ : 0.5 3 from 1972Hu10.
		2142	1.3	5655				I_γ : other: 1.0 5 (1972Hu10).
								$A_2=+0.03$ 8 (1988Va06).
		2393.5	2	5403.5				I_γ : other: 2 1 (1972Hu10).
								$A_2=+0.04$ 5 (1988Va06).
		2786.9	2	5010.1				I_γ : other: 2 1 (1972Hu10).
								$A_2=-0.07$ 5 (1988Va06).
		2942.6	1.3	4854.4				I_γ : other: 0.5 3 (1972Hu10).
		2957.9	1.1	4839.10				
		3737.7	0.3	4059.2				
		6577.1	9	1219.39	1/2 ⁺			I_γ : other: 9.0 9 (1972Hu10).
								δ : $\delta(Q/D)=+0.21$ 4 or -1.9 2 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.
								$A_2=-0.02$ 6, $A_4=-0.14$ 7 (1976Me12).
								$A_2=-0.03$ 3 (1988Va06).
		7796.2	83	0	3/2 ⁺			I_γ : other: 85.0 85 (1972Hu10).
								δ : $\delta(Q/D)=-0.33$ 2 or -19 +5-10 for $J=3/2$ (1976Me12), which is invalidated by the $J^\pi=1/2^-$ assignment for the 7797 level in 1988Va06.
								$A_2=-0.06$ 2, $A_4=-0.03$ 2 (1976Me12).
								$A_2=+0.006$ 14 (1988Va06).
7837.2	3/2	1656	0.5	6181				I_γ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 from 2001Vo24.
		2237.4	0.5	5599.7				I_γ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).
		2621.3	1.4	5215.8				I_γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 1 (2001Vo24); 1.0 5 (1972Hu10).
		3659.1	28	4177.9	3/2	M1+E2	-0.05 3	I_γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 28 (2001Vo24); 35.0 35 (1972Hu10).
								Mult., δ : D+Q from $\gamma(\theta)$ in 1967Wa22; magnetic

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$\gamma(^{35}\text{Cl})$ (continued)

E _i (level)	E _γ [†]	I _γ [#]	E _f	J _f ^π	Mult. ^b	δ ^b	Comments
							character from γ(pol) in 1967Wa22. Other: δ=−0.04 3 or +0.27 5 from 2001Vo24.
							A ₂ =+0.17 1, A ₄ =−0.01 2 (2001Vo24).
							POL=+0.66 18, Δπ=no (1967Wa22).
7837.2	3663.5	3	4173.45				I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 3 (2001Vo24); 3.0 15 (1972Hu10); 34 (1971Wi13).
	3777.8	2	4059.2				A ₂ =+0.17 1, A ₄ =−0.005 13 (1967Ta08).
							I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 2 (2001Vo24); 2 1 (1972Hu10); 4 (1971Wi13).
	3894.1	0.5	3942.9				I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Other: 1 (2001Vo24).
	3918.5	0.2	3918.48	3/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	4834.1	4	3002.75	5/2			I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 4 (2001Vo24); 4 2 (1972Hu10); 4 (1971Wi13).
	6073.5	1.9	1763.14	5/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 2 (2001Vo24); ≤1 (1971Wi13).
	6617.1	37	1219.39	1/2 ⁺	D+Q	+0.06 3	I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 37 (2001Vo24); 27.0 27 (1972Hu10); 33 (1971Wi13).
							Mult.,δ: from γ(θ) in 1967Wa22.
							POL=+0.17 45 (1967Wa22).
							A ₂ =−0.46 4, A ₄ =+0.04 6 (1967Ta08).
	7836.3	21	0	3/2 ⁺	D(+Q)	+0.02 3	I _γ : for 7837 and 7839 doublet initial levels (1976Me12). Others: 21 (2001Vo24); 27.0 27 (1972Hu10); 25 (1971Wi13).
							Mult.,δ: from γ(θ) in 1967Wa22. Other: 0.00 3 from 2001Vo24.
							POL=−1.1 8 (1967Wa22).
							A ₂ =+0.41 3, A ₄ =+0.012 40 (1967Ta08).
							A ₂ =+0.40 2, A ₄ =+0.01 4 (2001Vo24).
7839.0	1658	0.5	6181				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	2239.2	0.5	5599.7				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	2623.1	1.4	5215.8				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	3660.9	28	4177.9	3/2			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	3665.3	3	4173.45				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	3779.6	2	4059.2				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	3895.9	0.5	3942.9				I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	3920.3	0.2	3918.48	3/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	4835.9	4	3002.75	5/2			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	6075.3	1.9	1763.14	5/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	6618.9	37	1219.39	1/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
	7838.1	21	0	3/2 ⁺			I _γ : for 7837 and 7839 doublet initial levels (1976Me12).
7868.7	2652.8	1.0	5215.8				
	2858.5	3	5010.1				I _γ : other: 4 2 (1972Hu10).
	3695.0	5	4173.45				I _γ : other: 4 2 (1972Hu10).
	6105.0	17	1763.14	5/2 ⁺			I _γ : other: 18.0 18 (1972Hu10).
	6648.6	13	1219.39	1/2 ⁺			I _γ : other: 12.0 12 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
7868.7		7867.8	61	0	$3/2^+$	I_γ : other: 62.0 62 (1972Hu10).
7880.8	$3/2,5/2$	2226	12	5655		I_γ : other: 9.0 9 (1972Hu10).
		3041.6	1.7	4839.10		I_γ : other: 3.0 15 (1972Hu10).
		3256.2	2	4624.4		I_γ : other: 1.0 5 (1972Hu10).
		3702.7	3	4177.9	$3/2$	
		3766.9	1.3	4113.7	$7/2$	I_γ : other: 4 2 (1972Hu10).
		3821.4	2	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		3962.1	2	3918.48	$3/2^+$	I_γ : other: 2 1 (1972Hu10).
		4877.7	11	3002.75	$5/2$	I_γ : other: 10.0 10 (1972Hu10).
		5186.5	25	2693.9	$(3/2,5/2)$	I_γ : other: 30.0 30 (1972Hu10).
						$A_2=+0.50$ 3, $A_4=+0.09$ 4 (1976Me12).
		6117.1	30	1763.14	$5/2^+$	I_γ : other: 31.0 31 (1972Hu10).
						$A_2=+0.08$ 3, $A_4=-0.03$ 4 (1976Me12).
		7879.9	10	0	$3/2^+$	I_γ : other: 9.0 9 (1972Hu10).
						$A_2=-0.06$ 4, $A_4=+0.04$ 5 (1976Me12).
7899.2	$(3/2^-,5/2)$	2253	1.5	5646		
		2735.8	0.9	5163.34	$7/2$	
		3018.1	0.8	4880.96		
		3274.6	0.5	4624.4		
		3721.1	8	4177.9	$3/2$	I_γ : other: 8.0 8 (1972Hu10).
		3725.5	8	4173.45		I_γ : other: 6.0 6 (1972Hu10).
		3839.8	3	4059.2		I_γ : other: 2 1 (1972Hu10).
		4735.9	4	3162.99	$7/2^-$	I_γ : other: 4 2 (1972Hu10).
		5204.9	1.3	2693.9	$(3/2,5/2)$	
		6135.5	68	1763.14	$5/2^+$	I_γ : other: 77.0 77 (1972Hu10).
		7898.2	4	0	$3/2^+$	I_γ : other: 3.0 15 (1972Hu10).
7923.3		3084.1	1.4	4839.10		
		3153.3	2	4769.86		
		3298.7	1.9	4624.4		
		3745.2	3	4177.9	$3/2$	I_γ : other: 2 1 (1972Hu10).
		3749.6	5	4173.45		I_γ : other: 6.0 6 (1972Hu10).
		3863.9	1.7	4059.2		I_γ : other: 1.0 5 (1972Hu10).
		3955.8	1.5	3967.3		I_γ : other: 1.0 5 (1972Hu10).
		4004.6	1.2	3918.48	$3/2^+$	I_γ : other: 1.0 5 (1972Hu10).
		4920.2	0.9	3002.75	$5/2$	
		5229.0	11	2693.9	$(3/2,5/2)$	I_γ : other: 12.0 12 (1972Hu10).
		5277.2	0.9	2645.70	$(5/2,7/2)$	
		6159.6	1.5	1763.14	$5/2^+$	I_γ : other: 1.0 5 (1972Hu10).
		6703.2	27	1219.39	$1/2^+$	I_γ : other: 28.0 28 (1972Hu10).
		7922.3	41	0	$3/2^+$	I_γ : other: 48.0 48 (1972Hu10).
7970.3		2324	3	5646		I_γ : other: 3.0 15 (1972Hu10).
		2384	3	5586		
		2754.4	3	5215.8		I_γ : other: 3.0 15 (1972Hu10).
		3089.2	10	4880.96		I_γ : other: 9.0 9 (1972Hu10).
		3200.3	5	4769.86		I_γ : other: 6.0 6 (1972Hu10).
		3622.3	8	4347.76	$5/2,9/2$	I_γ : other: 8.0 8 (1972Hu10).
		3792.2	6	4177.9	$3/2$	I_γ : other: 6.0 6 (1972Hu10).
		3856.4	6	4113.7	$7/2$	I_γ : other: 2 1 (1972Hu10).
		4807.0	10	3162.99	$7/2^-$	I_γ : other: 13.0 13 (1972Hu10).
		4967.2	5	3002.75	$5/2$	I_γ : other: 5.0 25 (1972Hu10).
		5324.2	20	2645.70	$(5/2,7/2)$	I_γ : other: 22.0 22 (1972Hu10).
		6206.6	20	1763.14	$5/2^+$	I_γ : other: 21.0 21 (1972Hu10).
		6750.2	1	1219.39	$1/2^+$	
7987.8	$3/2$	1881.6	0.5	6106.2		I_γ : other: 1.0 5 (1972Hu10).
		2333	1.7	5655		I_γ : other: 1.0 5 (1972Hu10).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
7987.8	3/2	3809.7	<0.8	4177.9	3/2			I_γ : total $I_\gamma=0.8$ for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3814.1	<0.8	4173.45				I_γ : total $I_\gamma=0.8$ for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3928.4	1	4059.2				
		6767.7	60	1219.39	1/2 ⁺	D+Q	+0.21 4	I_γ : other: 62.0 62 (1972Hu10). Mult., δ : or -3.1 4 (1976Me12). $A_2=-0.07$ 2, $A_4=-0.05$ 3 (1976Me12).
		7986.8	36	0	3/2 ⁺	D+Q	-0.36 2	I_γ : other: 36.0 36 (1972Hu10). Mult., δ : or -11 3 (1976Me12). $A_2=-0.10$ 2, $A_4=-0.06$ 2 (1976Me12).
7995.6	5/2	2190	0.6	5806				
		2395.8	1.1	5599.7				
		2592.0	3	5403.5				I_γ : other: 3.0 15 (1972Hu10).
		2832.1	1.5	5163.34	7/2			
		3817.5	4	4177.9	3/2			I_γ : other: 3.0 15 (1972Hu10).
		3936.2	1.1	4059.2				
		4832.3	7	3162.99	7/2 ⁻			I_γ : other: 6.0 6 (1972Hu10).
		4992.5	2	3002.75	5/2			
		5301.3	0.2	2693.9	(3/2,5/2)			
		5349.5	0.6	2645.70	(5/2,7/2)			
		6231.9	0.9	1763.14	5/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		7994.6	78	0	3/2 ⁺			I_γ : other: 87.0 87 (1972Hu10).
		8001.0	(7/2,9/2 ⁺)	2355	1.8	5646		I_γ : other: 3.0 15 (1972Hu10).
		2785.1	0.8	5215.8				
		3119.9	0.7	4880.96				
8001.0	(7/2,9/2 ⁺)	3231.0	8	4769.86				I_γ : other: 8.0 8 (1972Hu10).
		3653.0	1.9	4347.76	5/2,9/2			
		3827.3	10	4173.45				I_γ : other: 8.0 8 (1972Hu10).
		4057.9	3	3942.9				I_γ : other: 2 1 (1972Hu10).
		4837.7	0.8	3162.99	7/2 ⁻			
		5354.9	1.6	2645.70	(5/2,7/2)			I_γ : other: 2 1 (1972Hu10).
		6237.3	70	1763.14	5/2 ⁺			I_γ : other: 76.0 76 (1972Hu10).
		8000.0	1.4	0	3/2 ⁺			I_γ : other: 1.0 5 (1972Hu10).
		2350	1.8	5655				
		3123.9	0.6	4880.96				
		3165.8	3	4839.10				
		3826.9	<1.3	4177.9	3/2			I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		3831.3	<1.3	4173.45				I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels (1976Me12).
		4841.7	15	3162.99	7/2 ⁻			I_γ : other: 17.0 17 (1972Hu10). $A_2=-0.084$ 14, $A_4=-0.0041$ 15 (1981Bi05).
		5001.9	2	3002.75	5/2			I_γ : other: 3.0 15 (1972Hu10).
8005.0	5/2 ⁺	5310.7	1.7	2693.9	(3/2,5/2)			
		5358.9	4	2645.70	(5/2,7/2)			
		6241.3	15	1763.14	5/2 ⁺			I_γ : other: 20.0 20 (1972Hu10). $A_2=-0.11$ 2, $A_4=-0.09$ 3 (1976Me12).
		6784.9	0.6	1219.39	1/2 ⁺			
		8004.0	55	0	3/2 ⁺			I_γ : other: 60.0 60 (1972Hu10). $A_2=-0.03$ 1, $A_4=-0.02$ 2 (1976Me12).
		2230	6.0 6	5806				I_γ : from 1972Hu10.
		3857.6	15.0 15	4177.9	3/2			I_γ : from 1972Hu10.
		3976.3	11.0 11	4059.2				I_γ : from 1972Hu10.
		5341.4	5.0 25	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.
8035.7		2230	6.0 6	5806				I_γ : from 1972Hu10.
		3857.6	15.0 15	4177.9	3/2			I_γ : from 1972Hu10.
		3976.3	11.0 11	4059.2				I_γ : from 1972Hu10.
		5341.4	5.0 25	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8035.7		6272.0	5.0 25	1763.14	5/2 ⁺			I_γ : from 1972Hu10.
		6815.6	56 6	1219.39	1/2 ⁺			I_γ : from 1972Hu10.
		8034.7	2.0 10	0	3/2 ⁺			I_γ : from 1972Hu10.
8038.5	3/2	3183.9	0.6	4854.4				
		3199.2	1.6	4839.10				
		3413.9	3	4624.4				
		3860.4	1.6	4177.9	3/2			
		3979.1	3	4059.2				I_γ : other: 4 2 (1972Hu10).
		4071.0	0.8	3967.3				
		4119.8	1.4	3918.48	3/2 ⁺			
		5035.4	4	3002.75	5/2			I_γ : other: 3.0 15 (1972Hu10).
		5344.2	57	2693.9	(3/2,5/2)	D+Q	-0.20 2	I_γ : other: 61.0 61 (1972Hu10). Mult., δ : or +19 +10-5 (1976Me12). $A_2=+0.08$ 2, $A_4=-0.03$ 2 (1976Me12). I_γ : other: 17.0 17 (1972Hu10). $A_2=+0.24$ 2, $A_4=+0.01$ 3 (1976Me12).
		6274.8	15	1763.14	5/2 ⁺			I_γ : other: 6.0 6 (1972Hu10).
		6818.4	4	1219.39	1/2 ⁺			I_γ : other: 9.0 9 (1972Hu10).
		8037.5	8	0	3/2 ⁺	D+Q	-0.13 4	Mult., δ : or +8.1 +33-18 (1976Me12). $A_2=+0.20$ 3, $A_4=-0.08$ 3 (1976Me12). I_γ : other: 10.0 10 (1972Hu10). I_γ : other: 11.0 11 (1972Hu10). I_γ : other: 7.0 7 (1972Hu10). I_γ : 1.0 5 from 1972Hu10. I_γ : 2 1 from 1972Hu10.
8075.9		2430	11	5646				I_γ : other: 4 2 (1972Hu10).
		2490	11	5586				I_γ : other: 7.0 7 (1972Hu10).
		2860.0	8	5215.8				I_γ : other: 2 1 (1972Hu10).
		2912.4		5163.34	7/2			I_γ : other: 2 1 (1972Hu10).
		3194.8		4880.96				I_γ : other: 53.0 53 (1972Hu10).
		3305.9	4	4769.86				I_γ : 1.0 5 from 1972Hu10.
		3897.8	8	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		3962.0	2	4113.7	7/2			
		5072.8	2	3002.75	5/2			
		6312.2	56	1763.14	5/2 ⁺			
		8074.9		0	3/2 ⁺			
8096.1	(5/2,7/2 ⁺)	2290	2	5806				I_γ : other: 1.0 5 (1972Hu10).
		2450	5	5646				I_γ : other: 5.0 25 (1972Hu10).
		2496.3	5	5599.7				I_γ : other: 6.0 6 (1972Hu10).
		2880.2	1.3	5215.8				I_γ : other: 1.0 5 (1972Hu10).
		2932.6	2	5163.34	7/2			I_γ : other: 2 1 (1972Hu10).
		3256.8	3	4839.10				I_γ : other: 3.0 15 (1972Hu10).
		3326.1	2	4769.86				I_γ : other: 1.0 5 (1972Hu10).
		3918.0	1.9	4177.9	3/2			
		4036.7	2	4059.2				I_γ : other: 2 1 (1972Hu10).
		4177.4	1.8	3918.48	3/2 ⁺			I_γ : other: 3.0 15 (1972Hu10).
		4932.7	5	3162.99	7/2 ⁻			I_γ : other: 5.0 25 (1972Hu10).
		5401.8	5	2693.9	(3/2,5/2)			I_γ : other: 5.0 25 (1972Hu10).
		5449.9	17	2645.70	(5/2,7/2)			I_γ : other: 17.0 17 (1972Hu10).
		6332.4	8	1763.14	5/2 ⁺			I_γ : other: 9.0 9 (1972Hu10).
		8095.1	39	0	3/2 ⁺			I_γ : other: 40.0 40 (1972Hu10).
8106.4	3/2	2506.6	0.5	5599.7				I_γ : other: 0.5 3 (1972Hu10).
		3251.8	0.6	4854.4				I_γ : other: 0.5 3 (1972Hu10).
		3267.1	0.5	4839.10				I_γ : other: 0.5 3 (1972Hu10).
		3928.3	1.6	4177.9	3/2			I_γ : other: 1.0 5 (1972Hu10).
		4047.0	1.3	4059.2				I_γ : other: 0.5 3 (1972Hu10).
		4138.8	3	3967.3				I_γ : other: 3.0 15 (1972Hu10).
		5103.3	9	3002.75	5/2			I_γ : other: 8.0 8 (1972Hu10).
		5412.1	9	2693.9	(3/2,5/2)	D+Q	+4.9 10	I_γ : other: 8.0 8 (1972Hu10).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)**

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^{<i>b</i>}	δ^b	Comments	
								Mult., δ : -0.05 4 (1976Me12).	
8106.4	3/2	6342.6	3	1763.14	5/2 ⁺			$A_2=+0.32$ 4, $A_4=-0.08$ 5 (1976Me12).	
		6886.3	35	1219.39	1/2 ⁺			I_γ : other: 3.0 15 (1972Hu10).	
								I_γ : other: 35.0 35 (1972Hu10).	
		8105.4	37	0	3/2 ⁺	D+Q	+1.2 1	$A_2=-0.46$ 2, $A_4=-0.05$ 3 (1976Me12).	
								I_γ : other: 40.0 40 (1972Hu10).	
								Mult., δ : or -0.46 4 (1976Me12).	
								$A_2=+0.83$ 3, $A_4=+0.01$ 2 (1976Me12).	
8113.3	(1/2 ⁺ ,3/2,5/2 ⁺)	1621	6	6492				I_γ : other: 6.0 6 (1972Hu10).	
		2007.0	6	6106.2					
		2307	1.1	5806					
		2458	0.9	5655					
		2513.5		5599.7					I_γ : 3.0 15 (1972Hu10).
		2897.4	1.8	5215.8					
		3939.6	7	4173.45					I_γ : other: 5.0 25 (1972Hu10).
		4053.9	1.3	4059.2					
		4145.7	5	3967.3					I_γ : other: 5.0 25 (1972Hu10).
		5110.2	1.9	3002.75	5/2				
		5419.0	13	2693.9	(3/2,5/2)				I_γ : other: 17.0 17 (1972Hu10).
		6349.5	5	1763.14	5/2 ⁺				I_γ : other: 6.0 6 (1972Hu10).
		6893.2	32	1219.39	1/2 ⁺				I_γ : other: 32.0 32 (1972Hu10).
		8112.3	25	0	3/2 ⁺				I_γ : other: 26.0 26 (1972Hu10).
8147.2	3/2 ⁻	2464	1.0	5683				I_γ : other: 1.0 5 (1972Hu10).	
		3137.0	3	5010.1					
		3969.1	8	4177.9	3/2				
		4087.7	3	4059.2					
		4179.6	2	3967.3					
		4228.5	3	3918.48	3/2 ⁺				I_γ : other: 3.0 15 (1972Hu10).
		5452.8	3	2693.9	(3/2,5/2)				I_γ : other: 4 2 (1972Hu10).
		6927.1	15	1219.39	1/2 ⁺				I_γ : other: 20.0 20 (1972Hu10).
		8146.2	62	0	3/2 ⁺				I_γ : other: 75.0 75 (1972Hu10).
		2511	4	5646					I_γ : other: 3.0 15 (1972Hu10).
		2571	8	5586					I_γ : other: 7.0 7 (1972Hu10).
		2940.9	6	5215.8					I_γ : other: 8.0 8 (1972Hu10).
		3386.8	1.0	4769.86					
		3978.7	5	4177.9	3/2				I_γ : other: 8.0 8 (1972Hu10).
4042.9	7	4113.7	7/2				I_γ : other: 7.0 7 (1972Hu10).		
4097.3	1	4059.2							
4213.6	2	3942.9							
4993.4	5	3162.99	7/2 ⁻				I_γ : other: 9.0 9 (1972Hu10).		
5153.6	6	3002.75	5/2				I_γ : other: 4 2 (1972Hu10).		
5510.6	51	2645.70	(5/2,7/2)				I_γ : other: 50.0 50 (1972Hu10).		
6393.0	1	1763.14	5/2 ⁺						
8155.8	3	0	3/2 ⁺				I_γ : other: 4 2 (1972Hu10).		
8179.4		3554.8	9.0 9	4624.4				I_γ : from 1972Hu10.	
		5016.0	8.0 8	3162.99	7/2 ⁻			I_γ : from 1972Hu10.	
		5176.2	4.0 20	3002.75	5/2			I_γ : from 1972Hu10.	
		5485.0	3.0 15	2693.9	(3/2,5/2)			I_γ : from 1972Hu10.	
		6415.6	68 7	1763.14	5/2 ⁺			I_γ : from 1972Hu10.	
		8178.4	8.0 8	0	3/2 ⁺			I_γ : from 1972Hu10.	
8208.2	5/2 ⁺	2553	0.7	5655				I_γ : other: 1 (2001Vo24).	
		3327.1	0.8	4880.96				I_γ : other: 1 (2001Vo24).	
		3368.9	0.4	4839.10				I_γ : other: 1 (2001Vo24).	
		4148.7	1.5	4059.2				I_γ : others: 2 (2001Vo24); 2 1 (1972Hu10).	

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8208.2	$5/2^+$	4289.4		3918.48	$3/2^+$			I_γ : 1.0 5 (1972Hu10).
		5205.0	1.1	3002.75	$5/2$			I_γ : others: 1 (2001Vo24); 1.0 5 (1972Hu10).
		5513.8	0.5	2693.9	$(3/2, 5/2)$			I_γ : other: 1 (2001Vo24); 1.0 5 (1972Hu10).
		6444.4	14	1763.14	$5/2^+$	D+Q	+0.33 4	I_γ : other: 14 (2001Vo24); 16.0 16 (1972Hu10).
								Mult., δ : from 2001Vo24.
								$A_2=+0.51$ 13, $A_4=-0.21$ 11, $A_6=+0.0004$ (1996Ka21).
								$A_2=+0.51$ 13, $A_4=-0.21$ 11 (2001Vo24).
		6988.1	3	1219.39	$1/2^+$			I_γ : other: 3 (2001Vo24); 3.0 15 (1972Hu10).
		8207.2	78	0	$3/2^+$	D+Q	-0.36 4	I_γ : other: 78 (2001Vo24); 76.0 76 (1972Hu10).
								Mult., δ : from 2001Vo24. Other: $\delta(\text{Q/D})=0.52$ 42 from 1996Ka21.
8216.3	$5/2$							$A_2=-0.55$ 8, $A_4=+0.004$ 78, $A_6=+0.006$ 72 (1996Ka21).
								$A_2=-0.55$ 8, $A_4=0.00$ 8 (2001Vo24).
		2561	0.8	5655				I_γ : other: 1 (2001Vo24).
		3446.3	0.8	4769.86				I_γ : other: 1 (2001Vo24).
		4038.2	<3	4177.9	$3/2$			I_γ : other: 3 (2001Vo24).
		4042.6	<3	4173.45				
		5052.9	41	3162.99	$7/2^-$	D+Q	+0.11 2	I_γ : other: 41 (2001Vo24); 44.0 44 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24.
								$A_2=-0.32$ 7, $A_4=-0.11$ 7, $A_6=+0.05$ 6 (1996Ka21).
								$A_2=-0.32$ 7, $A_4=-0.11$ 7 (2001Vo24).
		5213.1	5	3002.75	$5/2$			I_γ : other: 5 (2001Vo24); 5.0 25 (1972Hu10).
		5521.9	3	2693.9	$(3/2, 5/2)$			I_γ : other: 3 (2001Vo24); 4 2 (1972Hu10).
		6452.5	1.4	1763.14	$5/2^+$			I_γ : other: 1 (2001Vo24).
		8215.3	45	0	$3/2^+$	D+Q	+0.06 3	I_γ : other: 45 (2001Vo24); 47.0 47 (1972Hu10).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24. Other: $\delta(\text{Q/D})=0.59$ 56 from 1996Ka21.
								$A_2=-0.560$ 8, $A_4=+0.017$ 71, $A_6=+0.066$ 75 (1996Ka21).
8242.4		4068.7	7	4173.45				$A_2=-0.56$ 1, $A_4=+0.02$ 7 (2001Vo24).
		5239.2	2	3002.75	$5/2$			
		7022.3	91	1219.39	$1/2^+$			
8251?		4137	4&	4113.7	$7/2$			
		5248	6&	3002.75	$5/2$			
		7031	82&	1219.39	$1/2^+$			
		8250	8&	0	$3/2^+$			
8268.9		3498.9	2	4769.86				I_γ : from 1976Sp08.
		4090.7	12	4177.9	$3/2$			
		4350.1	3	3918.48	$3/2^+$			
		5105.5	7	3162.99	$7/2^-$			
		5265.7	13	3002.75	$5/2$			
		5574.5	6	2693.9	$(3/2, 5/2)$			
		5622.7	4	2645.70	$(5/2, 7/2)$			
		6505.1	8	1763.14	$5/2^+$			
		8267.9	45	0	$3/2^+$			
8277.2		2622	1.5	5655				
		3396.1	5	4880.96				
		3437.9	11	4839.10				
		3507.2	1.2	4769.86				
		3652.6	3	4624.4				
		3929.2	1.8	4347.76	$5/2, 9/2$			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	Comments
8277.2		4103.5	2	4173.45			
		4217.7	1.5	4059.2			
		4358.4	5	3918.48	3/2 ⁺		
		5274.0	10	3002.75	5/2		
		5582.8	3	2693.9	(3/2,5/2)		
		5631.0	3	2645.70	(5/2,7/2)		
		6513.4	23	1763.14	5/2 ⁺		
		8276.2	29	0	3/2 ⁺		
8282.4	(3/2 ⁻ ,5/2)	2696	8	5586			
		3066.5	8	5215.8			
		3118.9	3	5163.34	7/2		
		4104.2	14	4177.9	3/2		
		4222.9	6	4059.2			
		5119.0	26	3162.99	7/2 ⁻		
		6518.6	7	1763.14	5/2 ⁺		
		8281.4	28	0	3/2 ⁺		
8287.82	1/2 ⁻	2633	2	5655			
		3277.6	3	5010.1			
		4109.7	9	4177.9	3/2		
		4228.4	2	4059.2			
		4320.2	3	3967.3			
		4369.05	1	3918.48	3/2 ⁺		
		5593.4	23	2693.9	(3/2,5/2)		$A_2=-0.01$ 4, $A_4=-0.02$ 4 (1976Me12).
		7067.66	44	1219.39	1/2 ⁺		$\delta: \delta(\text{Q/D})=+0.21$ 2 or -3.1 2 for $J_i=3/2$ (1976Me12).
							$A_2=-0.08$ 2, $A_4=-0.01$ 3 (1976Me12).
		8286.77	13	0	3/2 ⁺		$\delta: \delta(\text{Q/D})=-0.35$ 4 or -11 3 for $J_i=3/2$ (1976Me12).
							$A_2=-0.11$ 3, $A_4=-0.01$ 4 (1976Me12).
8298.2	3/2	3287.9	3	5010.1			
		3417.1	4	4880.96			
		3443.6	5	4854.4			
		4124.5	6	4173.45			
		4238.7	10	4059.2			
		5134.8	1	3162.99	7/2 ⁻		
		5295.0	1	3002.75	5/2		
		5603.8	10	2693.9	(3/2,5/2)		$A_2=+0.09$ 6, $A_4=-0.07$ 7 (1976Me12).
		5652.0	6	2645.70	(5/2,7/2)		
		6534.4	8	1763.14	5/2 ⁺		$A_2=+0.43$ 11, $A_4=-0.51$ 12 (1976Me12).
		7078.0	24	1219.39	1/2 ⁺	D+Q	Mult., δ : -1.7 1 or -0.02 2 (1976Me12).
							$A_2=-0.53$ 3, $A_4=+0.11$ 3 (1976Me12).
		8297.1	13	0	3/2 ⁺	D+Q	Mult., δ : -0.15 4 or $+9.5$ +48-24 (1976Me12).
							$A_2=+0.18$ 3, $A_4=-0.10$ 4 (1976Me12).
8318.1		2512	0.7	5806			
		2672	0.7	5646			
		2718.3	1.5	5599.7			
		3548.1	0.3	4769.86			
		3693.5	1.8	4624.4			
		4139.9	<3	4177.9	3/2		
		4144.4	<3	4173.45			
		4258.6	1.2	4059.2			
		4350.5	3	3967.3			
		5154.7	0.5	3162.99	7/2 ⁻		
		5314.9	3	3002.75	5/2		
		5623.7	1.6	2693.9	(3/2,5/2)		
		5671.9	0.9	2645.70	(5/2,7/2)		
		6554.3	1.8	1763.14	5/2 ⁺		
		7097.9	2	1219.39	1/2 ⁺		

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8318.1		8317.0	78	0	3/2 ⁺			
8323.1		4209.1	1&	4113.7	7/2			
		5319.9	3&	3002.75	5/2			
		5676.9	4&	2645.70	(5/2,7/2)			
		6559.3	<2&	1763.14	5/2 ⁺			
		7102.9	8&	1219.39	1/2 ⁺			
		8322.0	82&	0	3/2 ⁺			
8381.8	5/2 ⁺	3611.7		4769.86				I_γ : 1 (2001Vo24).
		3757.2		4624.4				I_γ : 1 (2001Vo24).
		4267.8	8.0 8	4113.7	7/2			I_γ : from 1972Hu10. Other: 7 (2001Vo24).
		4463.0	24.0 24	3918.48	3/2 ⁺	D+Q	+0.15 3	I_γ : from 1972Hu10. Other: 24 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24.
								$A_2=+0.25$ 3, $A_4=-0.04$ 2 (2001Vo24).
		5378.6	25.0 25	3002.75	5/2	D(+Q)	+0.02 3	I_γ : from 1972Hu10. Other: 25 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24. Other:
								$\delta=-0.11$ 1 from 1967Ko20, 1967Ko23.
								$A_2=+0.60$ 7, $A_4=-0.02$ 7 (1967Ko20,1967Ko23).
		5687.4	1.0 5	2693.9	(3/2,5/2)			$A_2=+0.51$ 23, $A_4=-0.05$ 24 (2001Vo24).
								I_γ : from 1972Hu10. Other: 1 (2001Vo24).
		5735.6	5.0 25	2645.70	(5/2,7/2)			$A_2=+0.22$ 9, $A_4=-0.02$ 11 (1967Ko20,1967Ko23).
								I_γ : from 1972Hu10. Others: 5 (2001Vo24).
								Mult., δ : D+Q with $\delta=+0.72$ +42-20 and $J_f=3/2$
								for level 2645 (1967Ko20).
		6618.0	34.0 34	1763.14	5/2 ⁺	D+Q	+0.06 3	I_γ : from 1972Hu10. Other: 34 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 2001Vo24. Other:
								$\delta=-0.12$ 10 from $\gamma(\theta)$ in 1967Ko23.
								$A_2=+0.79$ 15, $A_4=-0.08$ 15
								(1967Ko20,1967Ko23).
								$A_2=+0.02$ 13, $A_4=-0.08$ 13 (2001Vo24).
		8380.7	3.0 15	0	3/2 ⁺	D+Q	1.65	I_γ : from 1972Hu10. Other: 2 (2001Vo24).
								Mult., δ : from $\gamma(\theta)$ in 1967Ko23.
								$A_2=-0.83$ 22, $A_4=+0.72$ 33
								(1967Ko20,1967Ko23).
8402.9		3632.8	4.8 24	4769.86				
		4224.7	9 5	4177.9	3/2			
		4343.4	23 6	4059.2				
		5708.5	10 5	2693.9	(3/2,5/2)			
		5756.7	7 4	2645.70	(5/2,7/2)			
		6639.1	15 4	1763.14	5/2 ⁺			
		8401.8	31 8	0	3/2 ⁺			
8405.7		3524.6	15 4	4880.96				
		4232.0	1.6 8	4173.45				
		4291.7	1.4 7	4113.7	7/2			
		5402.5	15 4	3002.75	5/2			
		6641.9	43 11	1763.14	5/2 ⁺			
		8404.6	24 6	0	3/2 ⁺			
8416.7		3406.4	5&	5010.1				
		4243.0	11&	4173.45				
		4473.5	3&	3942.9				
		5413.5	7&	3002.75	5/2			
		5770.5	8&	2645.70	(5/2,7/2)			
		6652.9	22&	1763.14	5/2 ⁺			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ †	$I_\gamma^\#$	E_f	J_f^π	Comments
8416.7		7196.5	9&	1219.39	1/2 ⁺	
		8415.6	35&	0	3/2 ⁺	
8464.3		2809	8 4	5655		
		3248.3	5.8 29	5215.8		
		3454.0	8 4	5010.1		
		3609.7	3.3 17	4854.4		
		3839.7	4.3 22	4624.4		
		4286.1	5.2 26	4177.9	3/2	
		4290.6	5.5 28	4173.45		
		4496.7	5.2 26	3967.3		
		4545.5	2.6 13	3918.48	3/2 ⁺	
		5461.1	17 4	3002.75	5/2	
		5769.9	16 4	2693.9	(3/2,5/2)	
		7244.1	9 4	1219.39	1/2 ⁺	
		8463.2	10 3	0	3/2 ⁺	
8484.4	3/2 ⁺	2829	5.6 28	5655		I_γ : other: 6 2001Vo24 .
		3603.2	7.0 35	4880.96		I_γ : other: 7 2001Vo24 .
		3859.8	1.0 5	4624.4		I_γ : other: 1 2001Vo24 .
		4516.8	1.2 6	3967.3		I_γ : other: 1 2001Vo24 .
		4565.6	5.2 26	3918.48	3/2 ⁺	I_γ : other: 5 2001Vo24 .
		5481.2	7 4	3002.75	5/2	I_γ : other: 7 2001Vo24 .
		5790.0	20 5	2693.9	(3/2,5/2)	I_γ : other: 20 2001Vo24 .
		5838.2	2.8 14	2645.70	(5/2,7/2)	I_γ : other: 3 2001Vo24 .
		6720.6	46 12	1763.14	5/2 ⁺	I_γ : other: 46 2001Vo24 .
		8483.3	4 2	0	3/2 ⁺	I_γ : other: 4 2001Vo24 .
8486.1		2728	1.1 6	5758		
		3861.5	1.8 9	4624.4		
		4307.9	<7.2	4177.9	3/2	I_γ : total $I_\gamma=4.8$ 24 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4312.4	<7.2	4173.45		I_γ : total $I_\gamma=4.8$ 24 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4426.6	1.9 10	4059.2		
		4518.5	7 4	3967.3		
		5482.9	8 4	3002.75	5/2	
		5791.7	19 5	2693.9	(3/2,5/2)	
		6722.3	3.6 18	1763.14	5/2 ⁺	
		7265.9	6.6 33	1219.39	1/2 ⁺	
		8485.0	46 12	0	3/2 ⁺	
8514.3		3659.7	4.4 22	4854.4		
		4336.1	2 1	4177.9	3/2	
		4454.8	27 7	4059.2		
		4546.7	23 6	3967.3		
		4595.5	6.6 33	3918.48	3/2 ⁺	
		7294.1	10 3	1219.39	1/2 ⁺	
		8513.2	27 7	0	3/2 ⁺	
8571.9		2848.3	2.2 11	5723.52		
		2917	1.3 7	5655		
		2972.1	0.7 4	5599.7		
		3408.4	1.8 9	5163.34	7/2	
		3690.7	0.9 5	4880.96		
		4398.2	0.7 4	4173.45		
		4457.9	1.1 6	4113.7	7/2	
		4653.1	1.1 6	3918.48	3/2 ⁺	
		5408.5	1.6 8	3162.99	7/2 ⁻	
		5568.7	4.5 23	3002.75	5/2	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	E_γ ^{$\dagger$}	I_γ ^{$\#$}	E_f	J_f^π	Mult. ^{b}	δ ^{b}	Comments
8571.9		5877.5	0.6 3	2693.9	(3/2,5/2)			
		5925.7	8 4	2645.70	(5/2,7/2)			
		7351.7	5.4 27	1219.39	1/2 ⁺			
		8570.8	70 7	0	3/2 ⁺			
8580.5		2088	4.4 22	6492				
		3741.2	1.3 7	4839.10				
		4402.3	4.4 22	4177.9	3/2			
		4521.0	2 1	4059.2				
		4661.7	2.3 12	3918.48	3/2 ⁺			
		5886.1	5.6 28	2693.9	(3/2,5/2)			
		7360.3	54 6	1219.39	1/2 ⁺			
		8579.4	26 7	0	3/2 ⁺			
8590.0		2832	4.8 24	5758				
		2935	3.7 19	5655				
		3819.9	16 4	4769.86				
		4416.3	13 3	4173.45				
		4476.0	3.6 18	4113.7	7/2			
		5426.6	8 4	3162.99	7/2 ⁻			
		5586.8	22 6	3002.75	5/2			
		5943.8	5.5 28	2645.70	(5/2,7/2)			
		6826.2	14 4	1763.14	5/2 ⁺			
		8588.9	10 3	0	3/2 ⁺			
8612.0		6848.1	50 5	1763.14	5/2 ⁺			
		7391.8	17 4	1219.39	1/2 ⁺			
		8610.9	33 8	0	3/2 ⁺			
8613.7		3732.5	1.6 8	4880.96				
		4554.2	5.4 27	4059.2				
		4670.5 ^c	4 &	3942.9				I_γ : from 1967Da12 only; not observed by others.
		4694.9	7 4	3918.48	3/2 ⁺			
		5610.5	2.5 13	3002.75	5/2			
		5919.3	4.3 22	2693.9	(3/2,5/2)			
		5967.5	23 6	2645.70	(5/2,7/2)			
		6849.8 ^c	9 &	1763.14	5/2 ⁺			I_γ : from 1967Da12 only; not observed by others.
		7393.5 ^c	2 &	1219.39	1/2 ⁺			I_γ : from 1967Da12 only; not observed by others.
8618.1	(3/2 ⁺ ,5/2 ⁺)	8612.6	56 6	0	3/2 ⁺			
		2894.5	2.1 11	5723.52				
		2963	1.7 9	5655				
		3763.5	1.4 7	4854.4				
		3993.5	1.5 8	4624.4				
		4650.5	3.0 15	3967.3				
		4699.3	2.3 12	3918.48	3/2 ⁺			
		5614.9	16 4	3002.75	5/2			
		5923.7	14 4	2693.9	(3/2,5/2)			$A_2=+0.147$ 33, $A_4=+0.114$ 52 (1970Ko33).
		5971.9	1.7 9	2645.70	(5/2,7/2)			
		6854.2	23 6	1763.14	5/2 ⁺			$A_2=+0.662$ 23, $A_4=+0.319$ 39 (1970Ko33 reported for 7855 γ , 7855 is likely a misprint).
		8617.0	33 8	0	3/2 ⁺			$A_2=-0.293$ 29, $A_4=-0.028$ 47 (1970Ko33).
8630.1	7/2	2906.5	0.3 2	5723.52				
		3466.6	8 4	5163.34	7/2	D+Q	+0.60 10	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.60$ 3, $A_4=-0.08$ 4 (1976Sp09).

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8630.1	7/2	3860.0	1.4 7	4769.86				
		4282.1	12 3	4347.76	5/2,9/2	D+Q	-0.184 14	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.11$ 2, $A_4=+0.02$ 2 (1976Sp09).
		4456.4	3.6 18	4173.45				
		4516.1	10 5	4113.7	7/2	D+Q	-0.06 2	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.43$ 2, $A_4=-0.01$ 2 (1976Sp09).
		4686.9	10 5	3942.9				
		5466.7	3.5 18	3162.99	7/2 ⁻			
		5626.9	1.4 7	3002.75	5/2			
		5983.9	2.6 13	2645.70	(5/2,7/2)			
		6866.2	47 12	1763.14	5/2 ⁺	D+Q	-0.011 3	Mult., δ : D+Q from $\gamma(\theta)$ in 1976Sp09 . $A_2=+0.10$ 33, $A_4=-0.65$ 46 (1970Ko33). $A_2=-0.387$ 7, $A_4=+0.012$ 8 (1976Sp09). $A_2=+0.398$ 46, $A_4=+0.019$ 78 (1970Ko33).
		8629.0	0.5 3	0	3/2 ⁺			E_γ : other: 4689 7 (1970Ko33).
8640	5/2	4697	23 @	3942.9				E_γ : other: 5473 7 (1970Ko33).
		5477	20 @	3162.99	7/2 ⁻			E_γ : other: 5630 7 (1970Ko33).
		5637	6 @	3002.75	5/2			E_γ : other: 5941 7 (1970Ko33).
		5946	13 @	2693.9	(3/2,5/2)			E_γ : other: 6874 7 (1970Ko33).
		6876	38 @	1763.14	5/2 ⁺			$A_2=-0.52$ 12, $A_4=-0.29$ 19 (1970Ko33).
8641.4	3/2,5/2	2986	2.9 15	5655				
		3055	2.5 13	5586				
		3237.7	4.7 24	5403.5				
		3802.1	2.6 13	4839.10				
		3871.3	24 6	4769.86				
		4016.8	1.9 10	4624.4				
		4463.2	<18	4177.9	3/2			I_γ : total $I_\gamma=14$ 4 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4467.6	<18	4173.45				I_γ : total $I_\gamma=14$ 4 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4581.9	9 5	4059.2				
		4722.6	5.9 30	3918.48	3/2 ⁺			
8686.2		5947.0	5.2 26	2693.9	(3/2,5/2)			
		8640.3	28 7	0	3/2 ⁺			$A_2=-0.12$ 2, $A_4=-0.01$ 2 (1976Sp09).
		3100	2.1 11	5586				
		3282.5	3.1 16	5403.5				
		4508.0	2 1	4177.9	3/2			
		4626.7	3.5 18	4059.2				
		5522.7	3.7 19	3162.99	7/2 ⁻			
		5991.8	2.8 14	2693.9	(3/2,5/2)			
		6922.3	4.8 24	1763.14	5/2 ⁺			
		7466.0	1.9 10	1219.39	1/2 ⁺			
8696.9		8685.0	76 8	0	3/2 ⁺			
		2891	2.9 15	5806				
		3097.1	4.8 24	5599.7				
		7476.7	12 3	1219.39	1/2 ⁺			
		8695.7	80 8	0	3/2 ⁺			
8750.9		2945	2.2 11	5806				
		3347.2	4.9 25	5403.5				
		3534.9	2.4 12	5215.8				
		4572.7	2.1 11	4177.9	3/2			
		4577.1	2.3 12	4173.45				
		6056.4	3.0 15	2693.9	(3/2,5/2)			
		6987.0	30 8	1763.14	5/2 ⁺			

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
8750.9	7530.6 ^c	2	1219.39	1/2 ⁺	I_γ : from 1967Da12 only; not observed by others.
	8749.7	53 5	0	3/2 ⁺	
8779.8	3179.9	4.3 22	5599.7		
	3563.8	6.6 33	5215.8		
	4601.6	<19	4177.9	3/2	I_γ : total $I_\gamma=15$ 4 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
	4606.0	<19	4173.45		I_γ : total $I_\gamma=15$ 4 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
	4861.0	4.2 21	3918.48	3/2 ⁺	
	5776.5	27 7	3002.75	5/2	
	6085.3	12 3	2693.9	(3/2,5/2)	
	6133.5	13 3	2645.70	(5/2,7/2)	
	7015.9	6.7 34	1763.14	5/2 ⁺	
	8778.6	11 3	0	3/2 ⁺	
8787.2	3063.5	6.8 34	5723.52		
	3623.7	19 5	5163.34	7/2	
	4017.1	6.7 34	4769.86		
	4609.0	9 5	4177.9	3/2	
	5623.7	5.8 29	3162.99	7/2 ⁻	
	5783.9	20 5	3002.75	5/2	
	7023.3	25 6	1763.14	5/2 ⁺	
	8786.0	8 4	0	3/2 ⁺	
8798.4	4738.9	8 4	4059.2		
	5795.1	16 4	3002.75	5/2	
	7034.5	7.0 35	1763.14	5/2 ⁺	
	7578.1	7.0 35	1219.39	1/2 ⁺	
	8797.2	62 6	0	3/2 ⁺	
8820.9	5817.6	9 5	3002.75	5/2	
	6126.4	18 5	2693.9	(3/2,5/2)	
	7057.0	21 5	1763.14	5/2 ⁺	
	8819.7	52 5	0	3/2 ⁺	
8824.2	5820.9	37 9	3002.75	5/2	
	6129.7	6.0 3	2693.9	(3/2,5/2)	
	7060.3	16 4	1763.14	5/2 ⁺	
	7603.9	22 6	1219.39	1/2 ⁺	
	8823.0	19 5	0	3/2 ⁺	
8829.3	3174	10 5	5655		
	3990.0	1.3 7	4839.10		
	4059.2	1.5 8	4769.86		
	4910.5	4.5 23	3918.48	3/2 ⁺	
	5826.0	22 6	3002.75	5/2	
	6134.8	13 3	2693.9	(3/2,5/2)	
	6183.0	33 8	2645.70	(5/2,7/2)	
	7065.4	9 5	1763.14	5/2 ⁺	
	7609.0	2.6 13	1219.39	1/2 ⁺	
	8828.1	3.8 19	0	3/2 ⁺	
8833.1	3617.1	6.9 35	5215.8		
	4063.0	2.6 13	4769.86		
	4659.3	14 4	4173.45		
	4719.1	4.7 24	4113.7	7/2	
	4773.6	11 3	4059.2		
	4914.3	11 3	3918.48	3/2 ⁺	
	5669.6	3.7 19	3162.99	7/2 ⁻	
	5829.8	6.7 34	3002.75	5/2	
	6138.6	4.5 23	2693.9	(3/2,5/2)	
	6186.8	23 6	2645.70	(5/2,7/2)	

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Comments
8833.1		7069.2	9 5	1763.14	5/2 ⁺	
		8831.9	2.9 15	0	3/2 ⁺	
8838.1	7/2 ⁻	3674.6 [‡]		5163.34	7/2	
		3956.9 [‡]		4880.96		
		4068.0	8 4	4769.86		
		4490.0 [‡]		4347.76	5/2,9/2	
		4778.6 [‡]		4059.2		
		5674.6	29 7	3162.99	7/2 ⁻	
		7074.2	63 6	1763.14	5/2 ⁺	$A_2=-0.20$ 2, $A_4=-0.00$ 2 (1981Bi05).
8856.1		3974.9	3.4 17	4880.96		
		4231.4	5.1 26	4624.4		
		4677.9	3.5 18	4177.9	3/2	
		4937.3	11 3	3918.48	3/2 ⁺	
		5692.6	51 5	3162.99	7/2 ⁻	
		6161.6	2.8 14	2693.9	(3/2,5/2)	
		6209.8		2645.70	(5/2,7/2)	
		7092.2	7 4	1763.14	5/2 ⁺	
		8854.9	16 4	0	3/2 ⁺	
8868.6		4690.4	2.6 13	4177.9	3/2	
		5865.3	16 4	3002.75	5/2	
		6174.1	12 3	2693.9	(3/2,5/2)	
		7104.7	4.4 22	1763.14	5/2 ⁺	
		7648.3	16 4	1219.39	1/2 ⁺	
		8867.4	49 12	0	3/2 ⁺	
8886.1		3231	6 3	5655		
		3286.2	6.6 33	5599.7		
		3670.1	2.2 11	5215.8		
		3722.6	9 4	5163.34	7/2	
		4116.0	4.5 23	4769.86		
		4707.9	8 4	4177.9	3/2	
		4967.2	4.1 21	3918.48	3/2 ⁺	
		5882.8	8 4	3002.75	5/2	
		6239.8	26 7	2645.70	(5/2,7/2)	
		7122.2	12 3	1763.14	5/2 ⁺	
		8884.9	14 4	0	3/2 ⁺	
8893.3	(5/2,7/2) ⁺	3293.4 ^c		5599.7		I_γ : 1 from 2001Vo24 only; not observed by others.
		3307	1.6 8	5586		
		4779.3	9 4	4113.7	7/2	I_γ : other: 9 2001Vo24 .
		4950.0	3.9 20	3942.9		I_γ : other: 4 2001Vo24 .
		5729.8	29 7	3162.99	7/2 ⁻	I_γ : other: 29 2001Vo24 .
		6198.8	37 9	2693.9	(3/2,5/2)	I_γ : other: 37 2001Vo24 .
		7129.4	19 5	1763.14	5/2 ⁺	I_γ : other: 19 2001Vo24 .
		8892.1	1.0 5	0	3/2 ⁺	I_γ : other: 1 2001Vo24 .
8904.8		2724	4.3 22	6181		
		3250	3.4 17	5655		
		3894.5	3.9 20	5010.1		
		4050.2	6.2 31	4854.4		
		4065.5	4.5 23	4839.10		
		4726.6	18 5	4177.9	3/2	
		4845.2	5.7 29	4059.2		
		6210.3	15 4	2693.9	(3/2,5/2)	
		7684.5	10 3	1219.39	1/2 ⁺	
		8903.6	29 7	0	3/2 ⁺	
8906.8	5/2 ⁺	3252		5655		I_γ : 4 from 2001Vo24 only; not observed by others.
		3261	3.7 19	5646		I_γ : other: 2 2001Vo24 .

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$^{34}\text{S}(\text{p},\gamma)$ **1976Me12,1976Sp08,1972Hu10 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
8906.8	5/2 ⁺	3306.9	2.2 11	5599.7				
		3321	4	5586				I_γ : other: 4 2001Vo24.
		3387	2.5 13	5520				
		3690.8	4.4 22	5215.8				
		4136.7	5.6 28	4769.86				I_γ : other: 6 2001Vo24.
		4963.5	13 3	3942.9				I_γ : other: 15 2001Vo24.
		7142.9	69 7	1763.14	5/2 ⁺			I_γ : other: 69 2001Vo24.
8919.8		3274	5.6 28	5646				
		4571.7	8 4	4347.76	5/2,9/2			
		5756.3	57 6	3162.99	7/2 ⁻			
		6273.5	26 7	2645.70	(5/2,7/2)			
8953.0	3/2	7155.9	4 2	1763.14	5/2 ⁺			
		7732.7		1219.39	1/2 ⁺	D+Q		Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.34$ 2 or $\delta=-5.0$ 4 (1976Sp09).
		8951.8		0	3/2 ⁺	D+Q		$A_2=+0.12$ 3, $A_4=-0.02$ 4 (1976Sp09). Mult.: from $\gamma(\theta)$ in 1976Sp08; $\delta=-0.39$ 5 or $\delta=-8.2$ 7 (1976Sp09).
8984.2		3260.5	0.8 4	5723.52				$A_2=-0.19$ 2, $A_4=+0.02$ 2 (1976Sp09).
		4144.8	1.2 6	4839.10				
		4214.1	0.5 3	4769.86				
		4359.5	0.5 3	4624.4				
		4806.0	1.5 8	4177.9	3/2			
		4810.4	1.5 8	4173.45				
		5065.3	9 5	3918.48	3/2 ⁺			
		5820.7	6 3	3162.99	7/2 ⁻			
		6289.7	1.8 9	2693.9	(3/2,5/2)			
		6337.9	1.2 6	2645.70	(5/2,7/2)			
		7220.3	16 4	1763.14	5/2 ⁺			
		8983.0	60 6	0	3/2 ⁺			
9081.4	5/2 ⁺	3357.7	1	5723.52				I_γ : from 2001Vo24.
		4200.2	1	4880.96				I_γ : from 2001Vo24.
		4242.0 [‡]		4839.10				
		4311.3 [‡]		4769.86				
		4456.7 [‡]		4624.4				
		4903.1	2	4177.9	3/2			I_γ : from 2001Vo24.
		4907.6	2	4173.45				I_γ : from 2001Vo24.
		5162.5	9	3918.48	3/2 ⁺	D(+Q)	-0.002 9	I_γ : from 2001Vo24. Mult., δ : from $\gamma(\theta)$ in 1976Sp09. $A_2=-0.43$ 3, $A_4=+0.05$ 3 (1976Sp09).
		5917.9	6	3162.99	7/2 ⁻			I_γ : from 2001Vo24.
		6386.9	2	2693.9	(3/2,5/2)			I_γ : from 2001Vo24.
		6435.1	1	2645.70	(5/2,7/2)			I_γ : from 2001Vo24.
		7317.4	16	1763.14	5/2 ⁺	D+Q	+0.11 2	I_γ : from 2001Vo24. Mult., δ : from $\gamma(\theta)$ in 1976Sp09. $A_2=+0.555$ 14, $A_4=+0.03$ 2 (1976Sp09).
		9080.1	60	0	3/2 ⁺	D+Q	+0.089 9	I_γ : from 2001Vo24. Mult., δ : from $\gamma(\theta)$ in 1976Sp09. $A_2=-0.243$ 11, $A_4=+0.05$ 2 (1976Sp09).
9157.1	5/2	3941.1	7.4 19	5215.8				
		4386.9	6.9 17	4769.86				
		4978.8	<16	4177.9	3/2			I_γ : total $I_\gamma=13$ 3 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<16	4173.45				I_γ : total $I_\gamma=13$ 3 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p},\gamma)$ [1976Me12](#),[1976Sp08](#),[1972Hu10](#) (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\#$	E_f	J_f^π	Mult. ^b	δ^b	Comments
9157.1	5993.6	9.6 24	3162.99	7/2 ⁻	D+Q	+0.10 2	Mult., δ : from $\gamma(\theta)$ in 2001Vo24 . A ₂ =-0.15 8, A ₄ =-0.31 9, A ₆ =-0.05 8 (1996Ka21).
	6153.8	8.4 21	3002.75	5/2			
	6462.6	1.3 7	2693.9	(3/2,5/2)			
	6510.8	2.9 15	2645.70	(5/2,7/2)			
	7393.1	7.5 19	1763.14	5/2 ⁺			
	9155.8	43 4	0	3/2 ⁺	D+Q	+0.09 1	Mult., δ : from $\gamma(\theta)$ in 2001Vo24 . Other: $\delta(\text{Q/D})=0.47$ 50 (1996Ka21). A ₂ =-0.29 6, A ₄ =-0.38 8, A ₆ =-0.10 6 (1996Ka21).

[†] From level-energy differences, unless otherwise noted.

[‡] γ branches observed in [1981Bi05](#). Branching ratios are not given.

[#] For levels below 8.27 MeV ($E_p < 1.95$ MeV), I_γ is mainly from [1976Me12](#) and for levels above 8.27 MeV ($E_p > 1.95$ MeV), I_γ is mainly from [1976Sp08](#), unless otherwise noted.

@ From [1970Ko33](#).

& From [1967Da12](#).

^a From [1963Ha32](#).

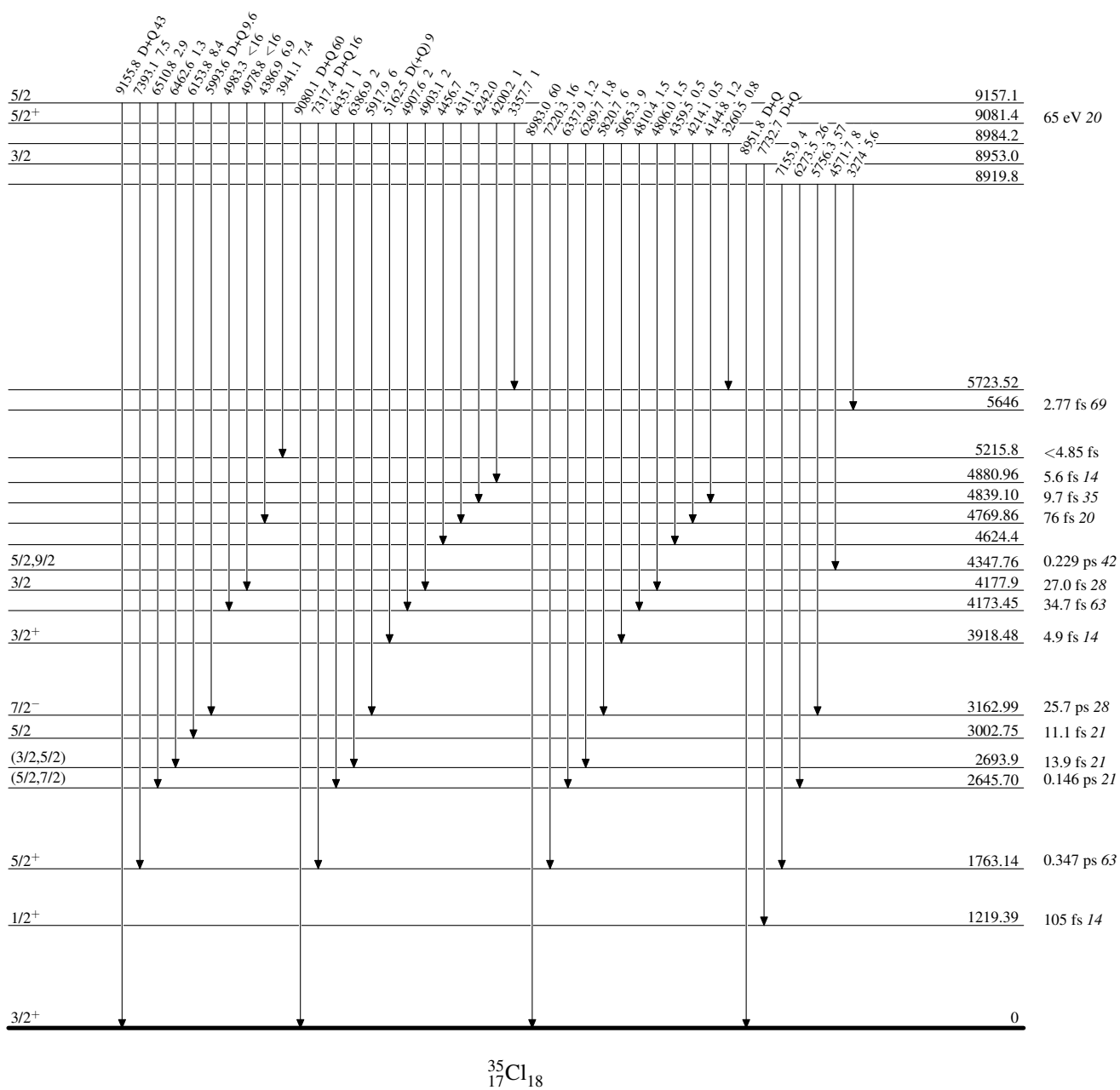
^b From $\gamma(\theta)$ and/or $\gamma\gamma(\theta)$ data as given under comments.

^c Placement of transition in the level scheme is uncertain.

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme

Intensities: % photon branching from each level

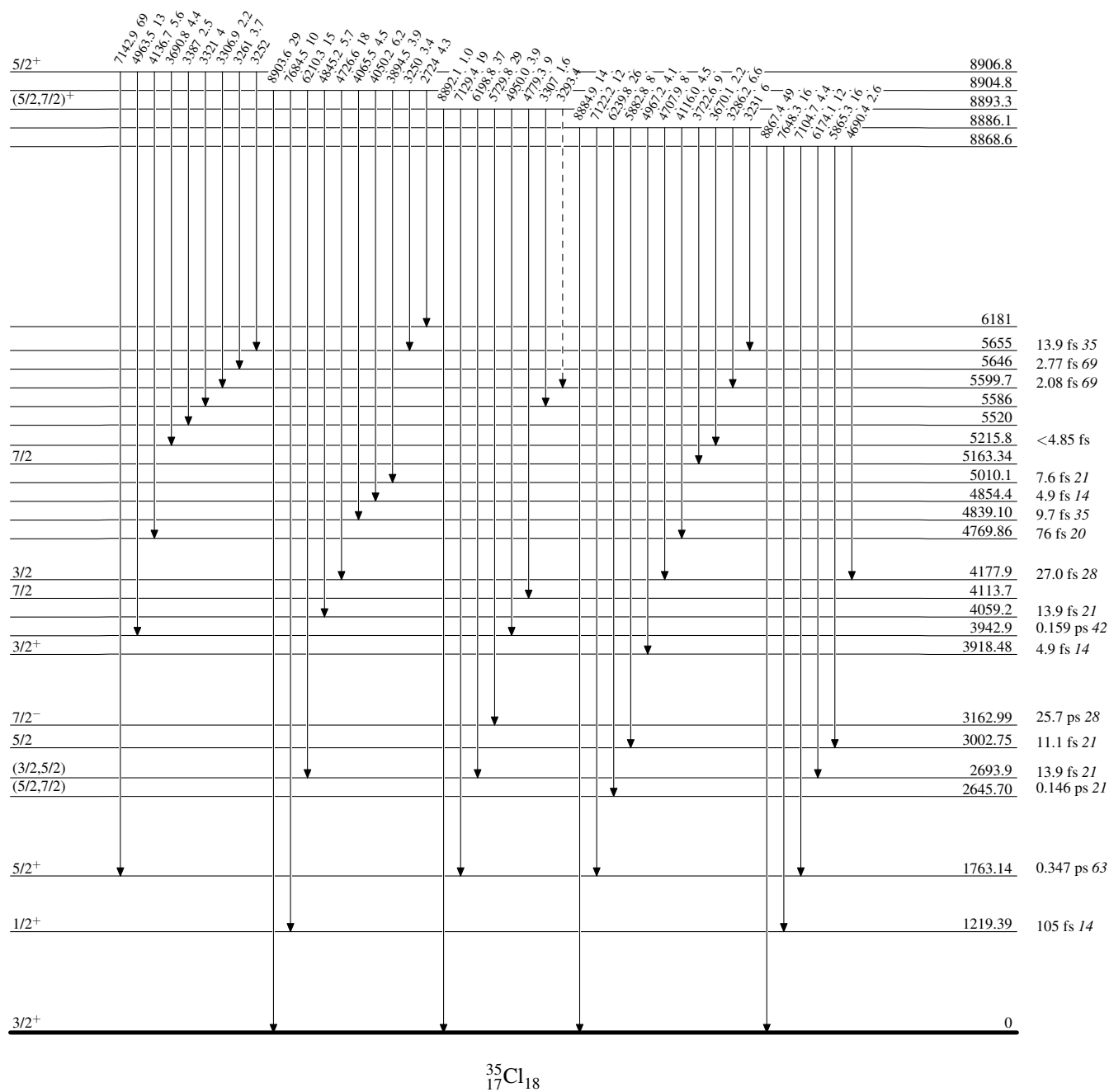


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

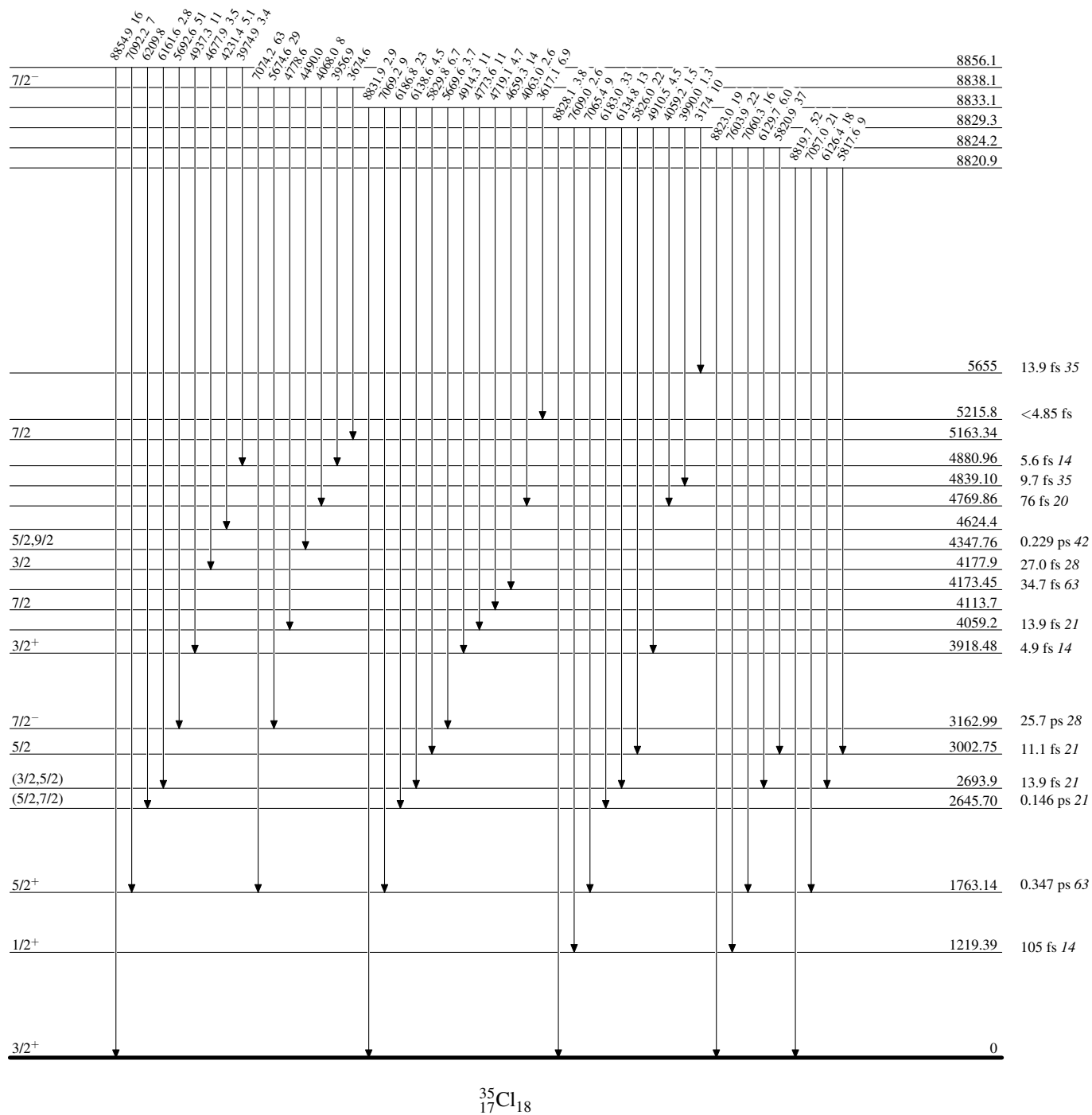
Intensities: % photon branching from each level

-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

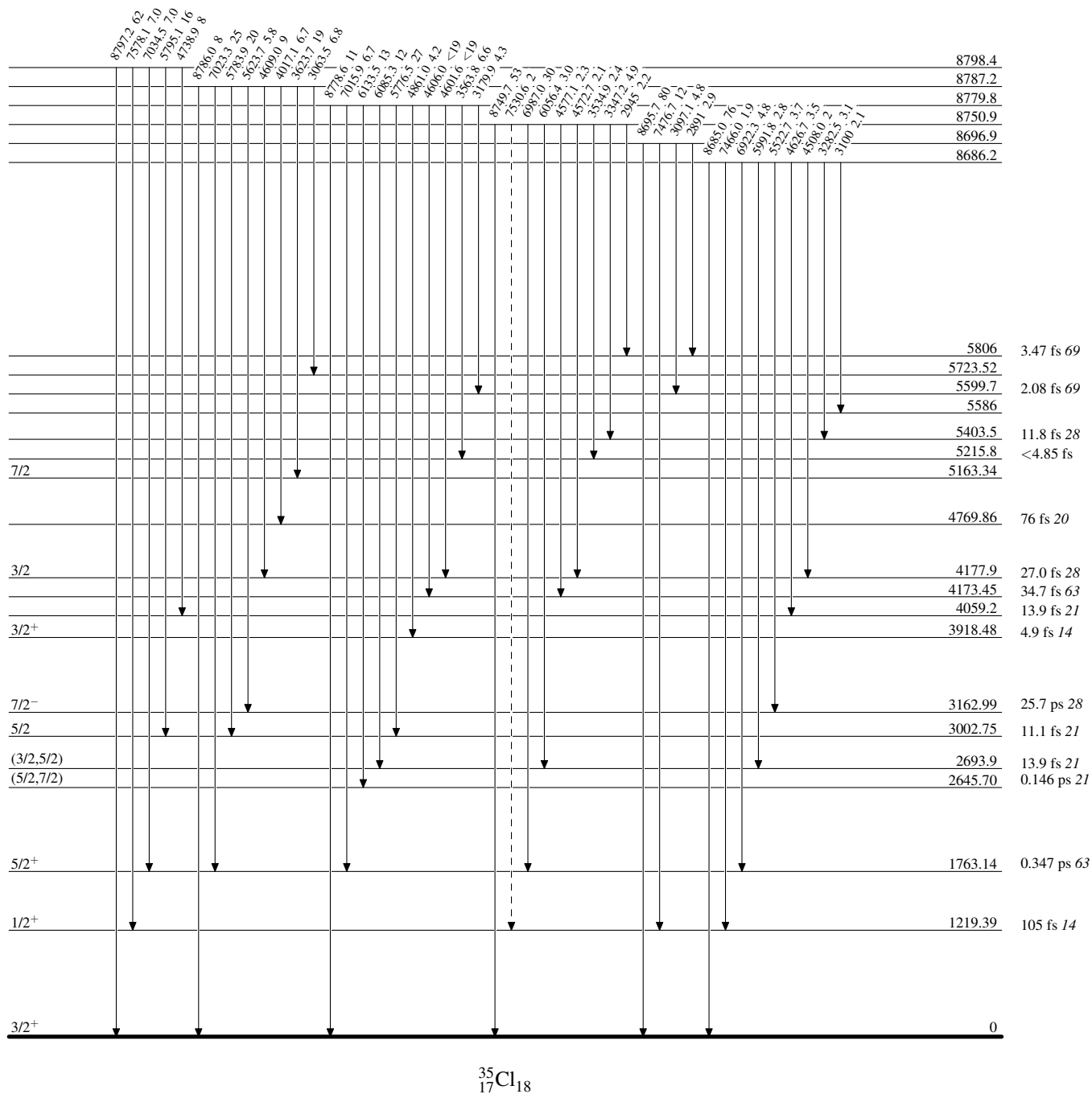


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

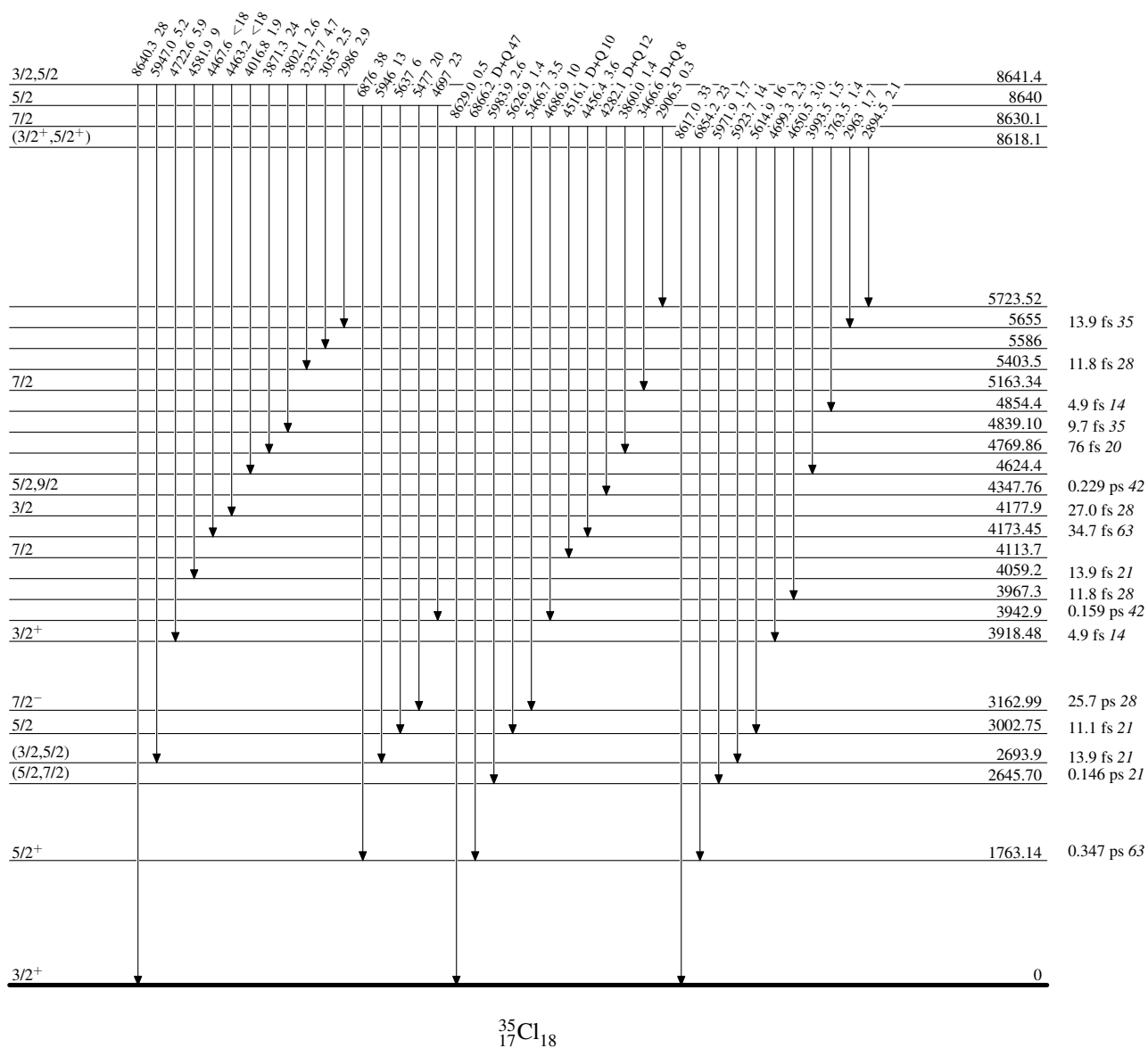
Intensities: % photon branching from each level

-----► γ Decay (Uncertain)

³⁴S(p,γ) 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

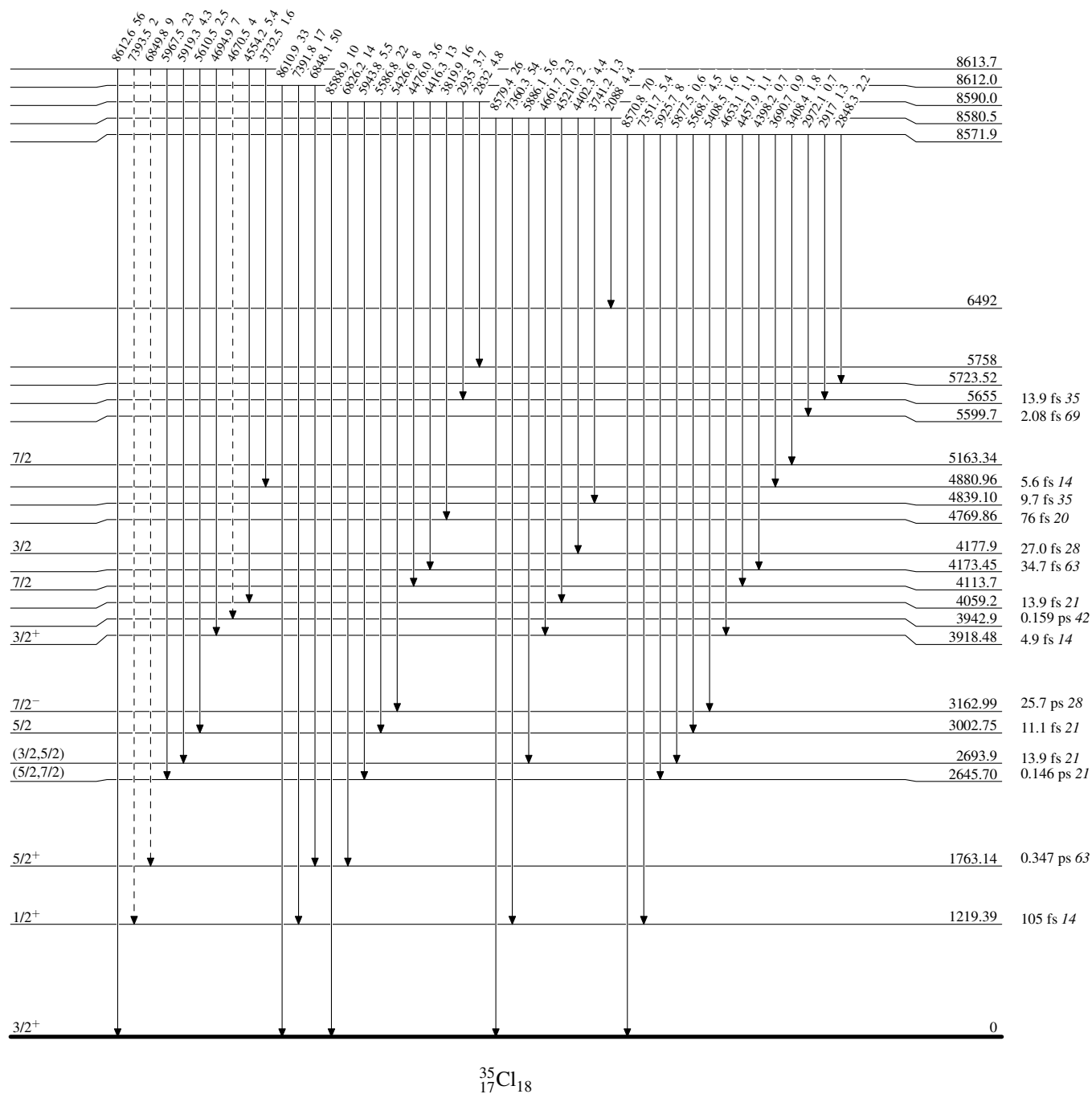


$^{34}\text{S}(p,\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

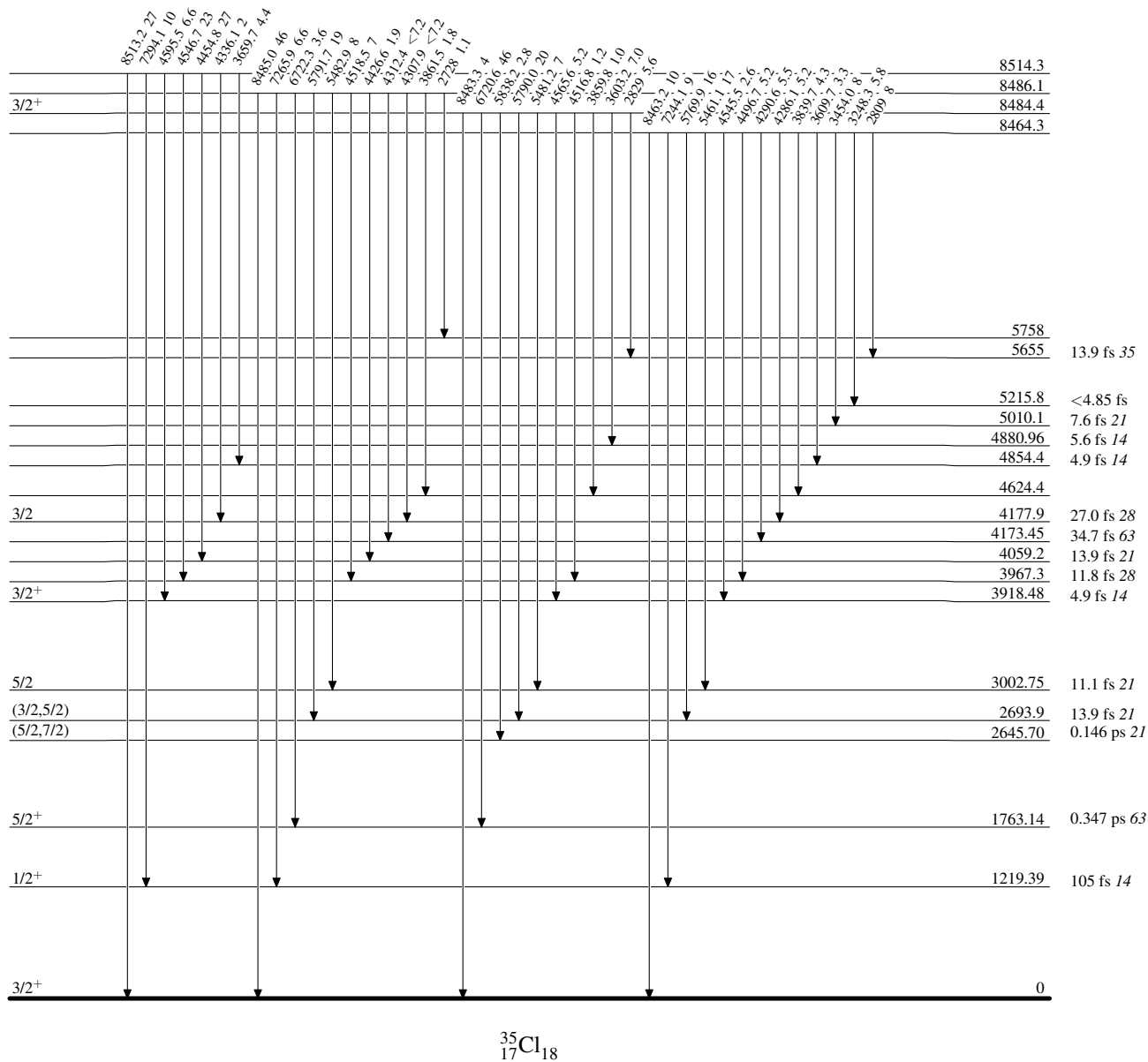
Intensities: % photon branching from each level

-----► γ Decay (Uncertain)


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

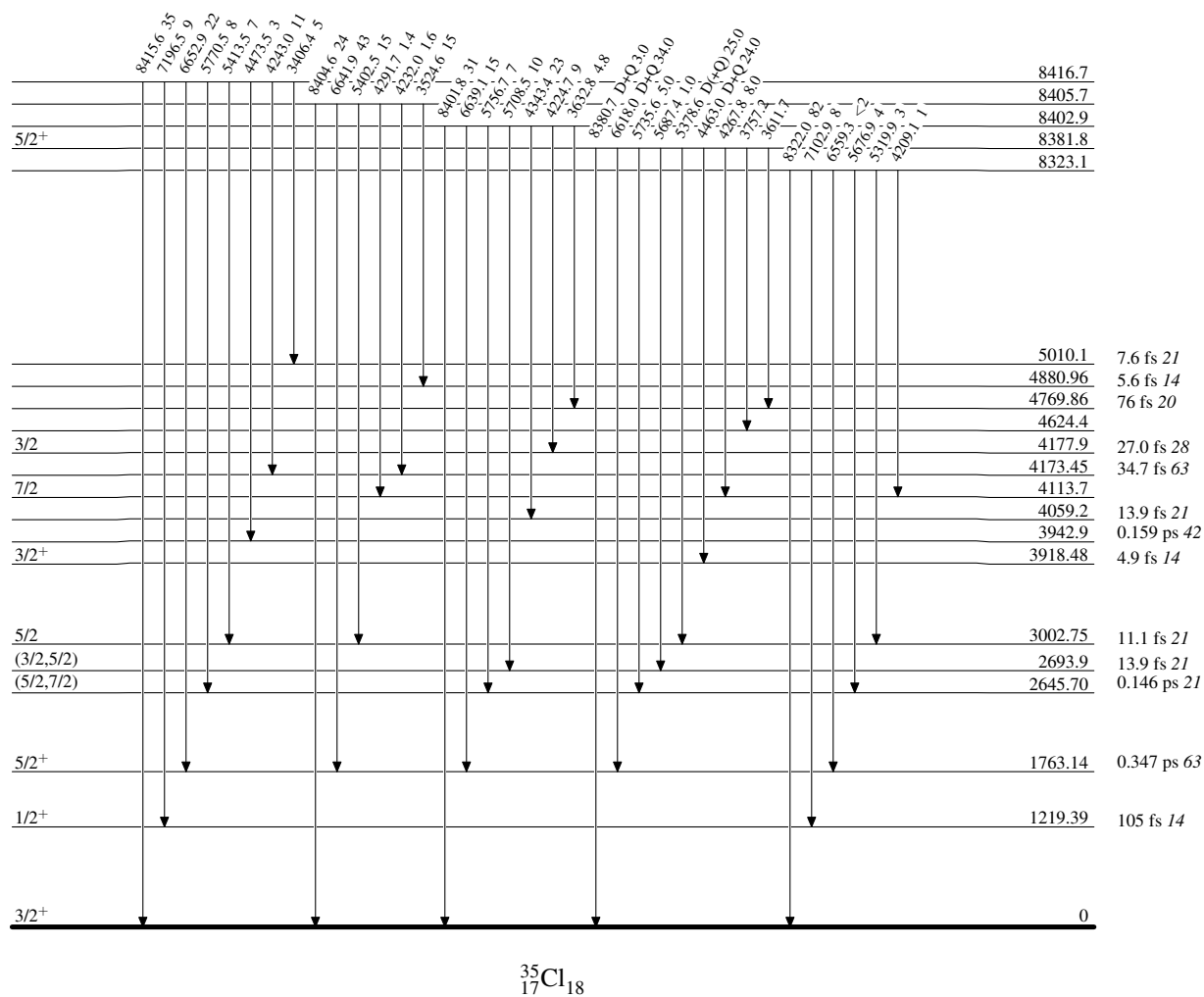
Intensities: % photon branching from each level


 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

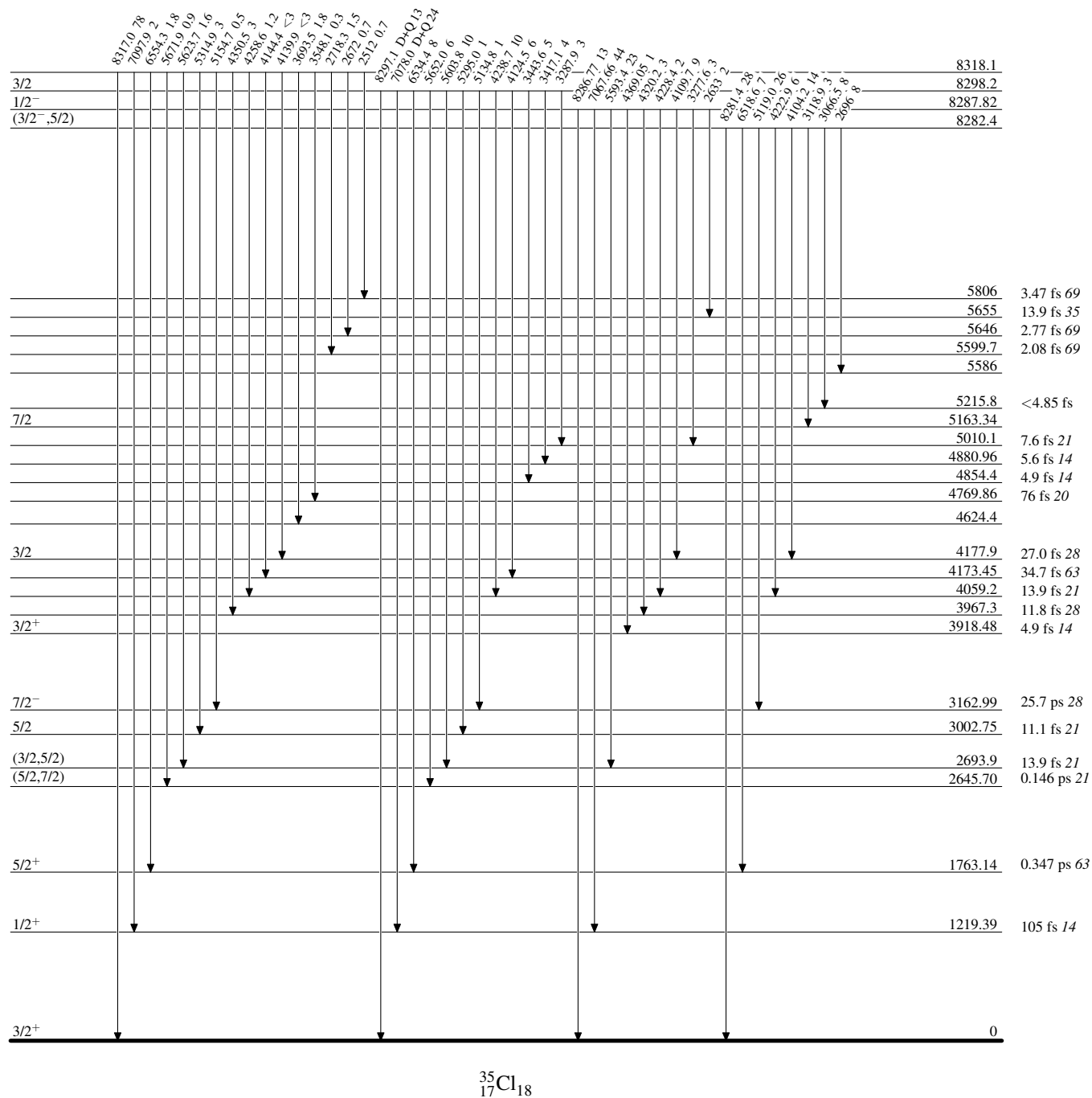
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

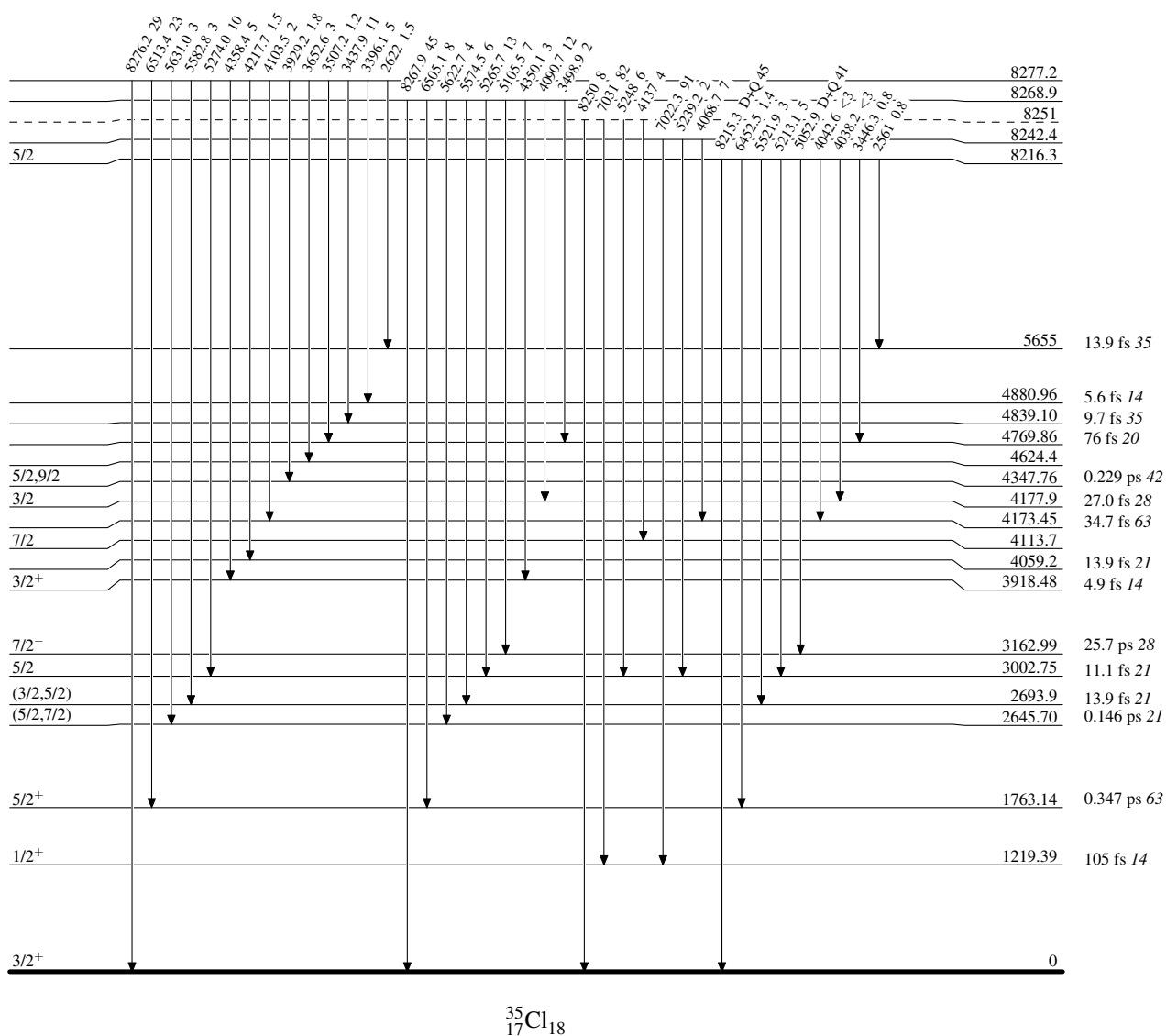
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

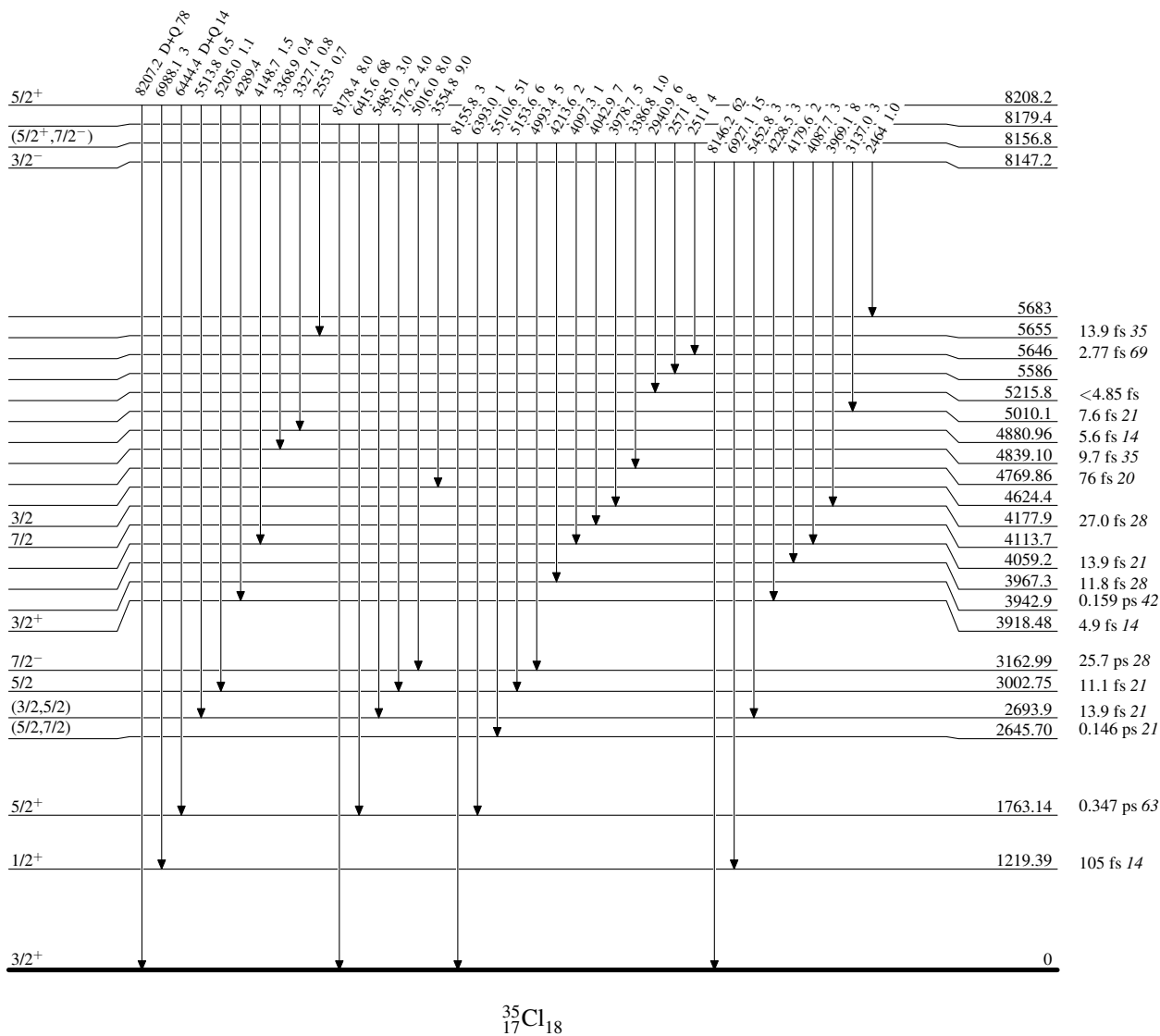
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

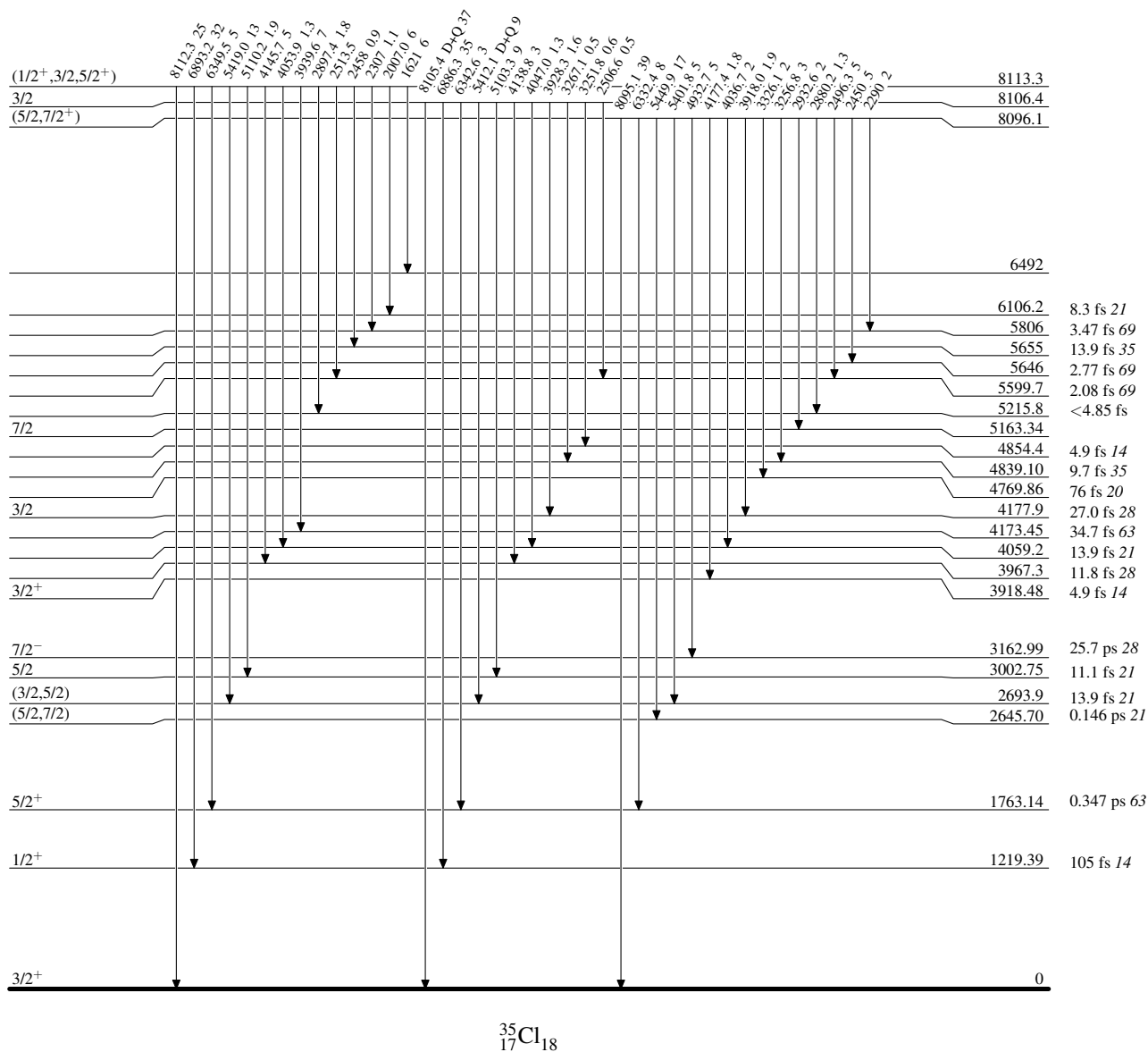
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

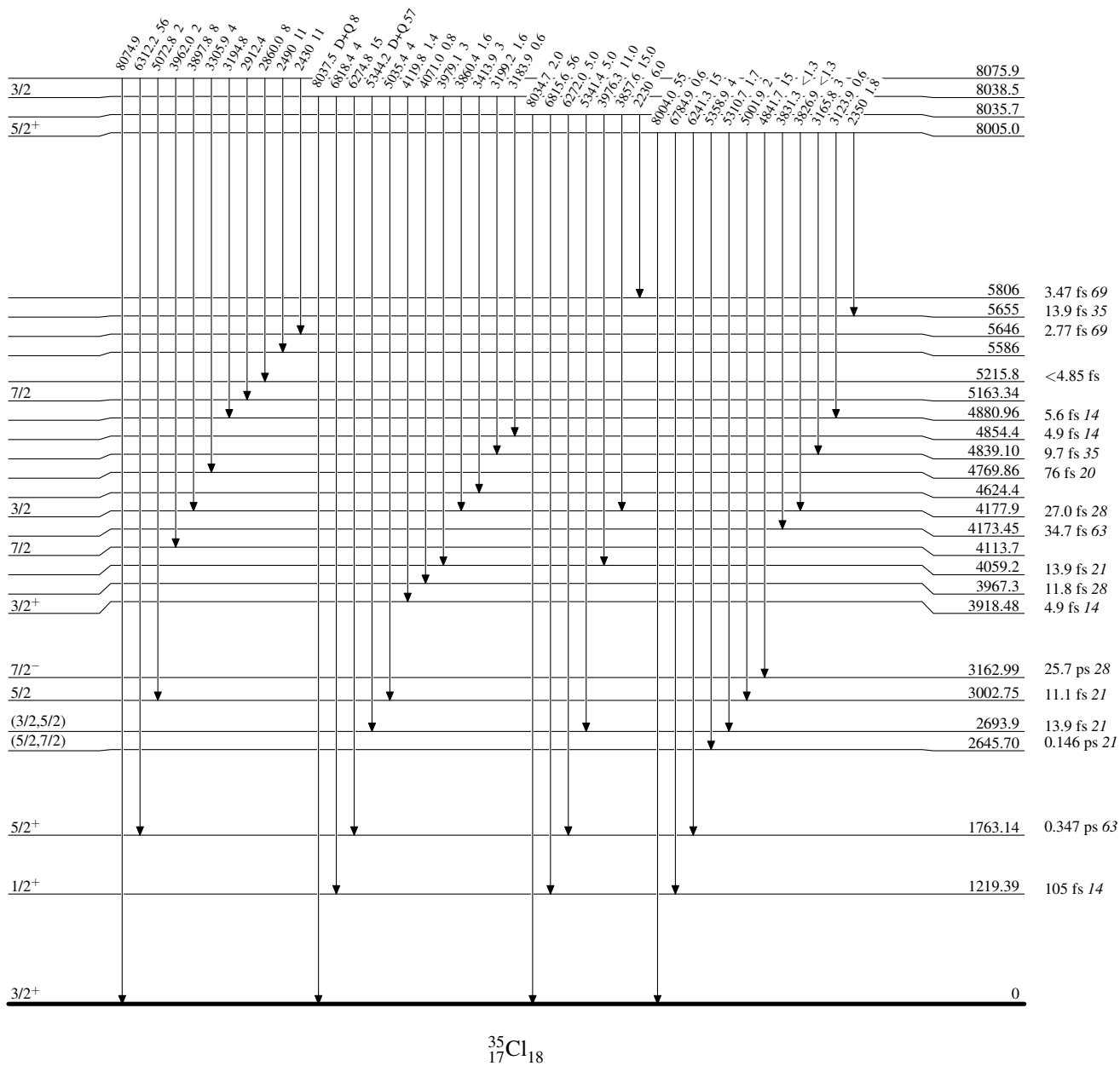
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

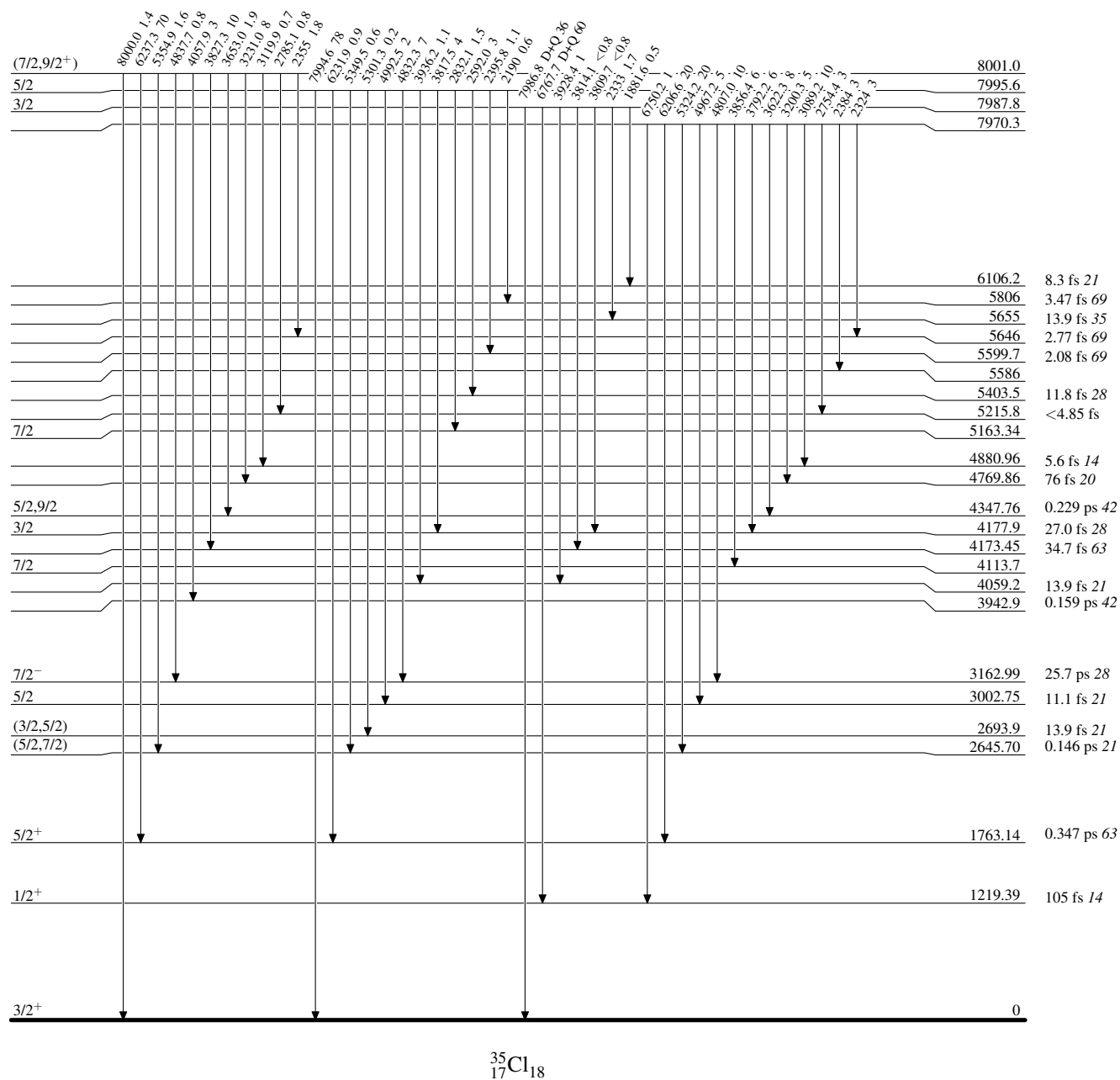
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

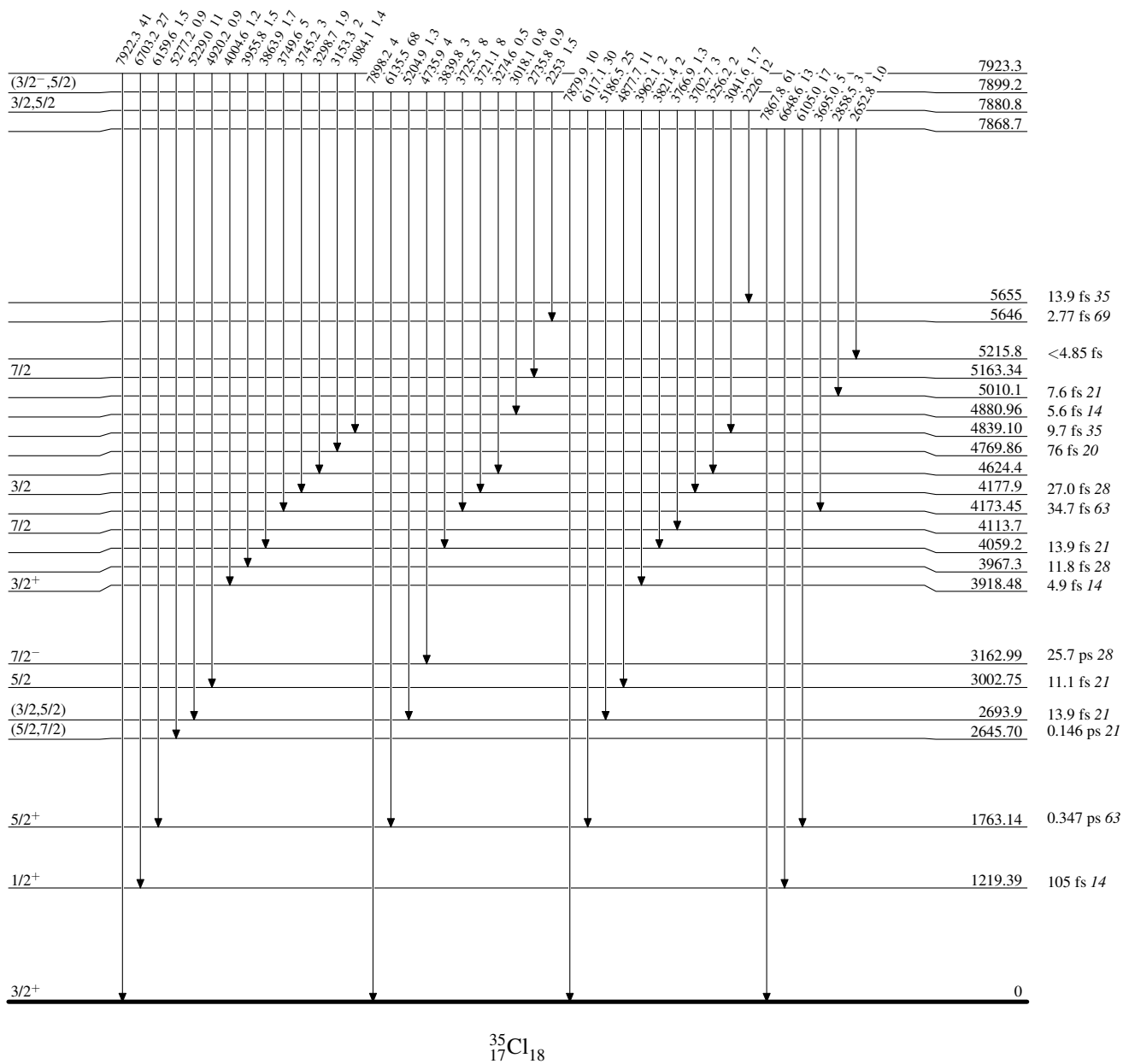
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

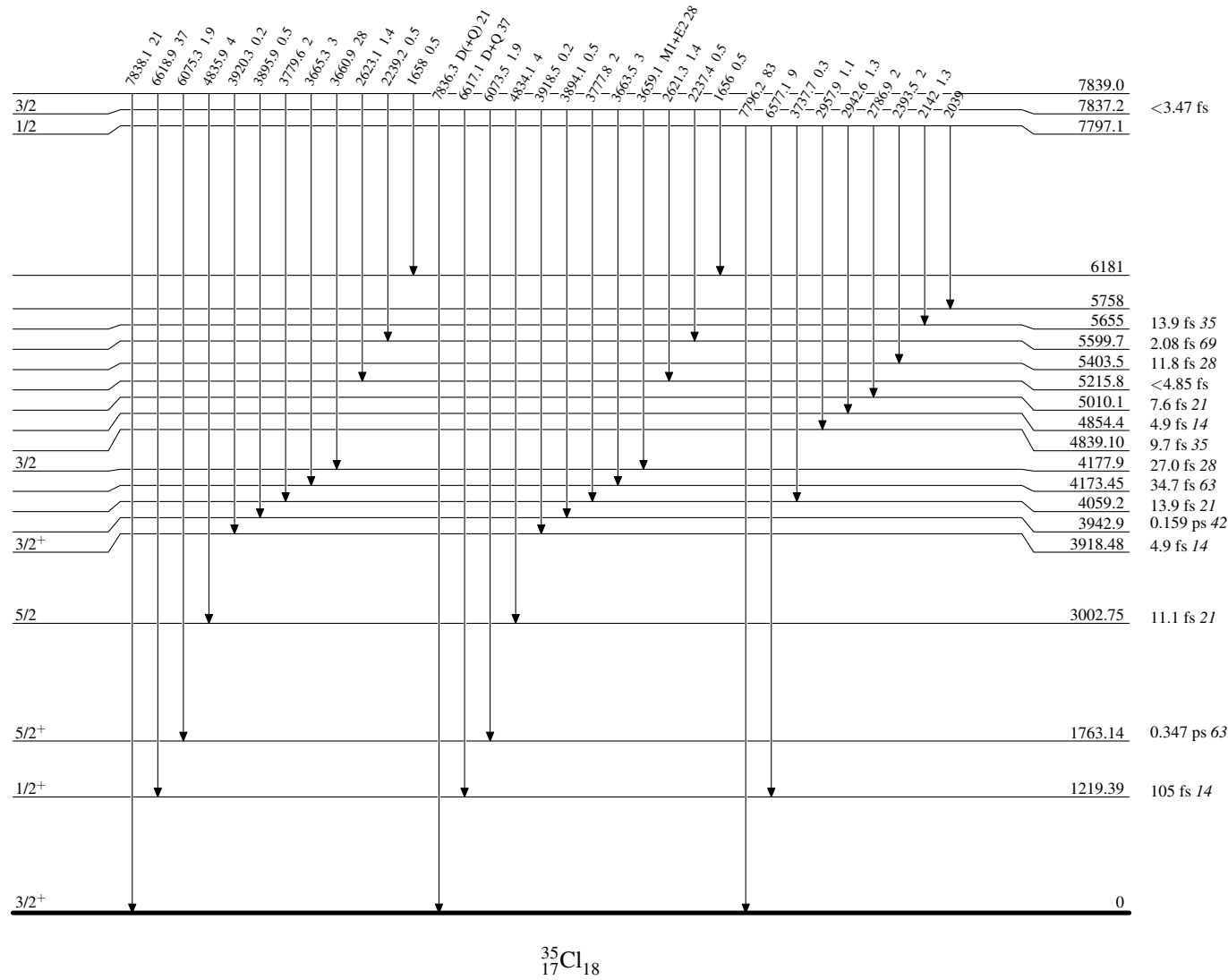
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

³⁴S(p,γ) 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

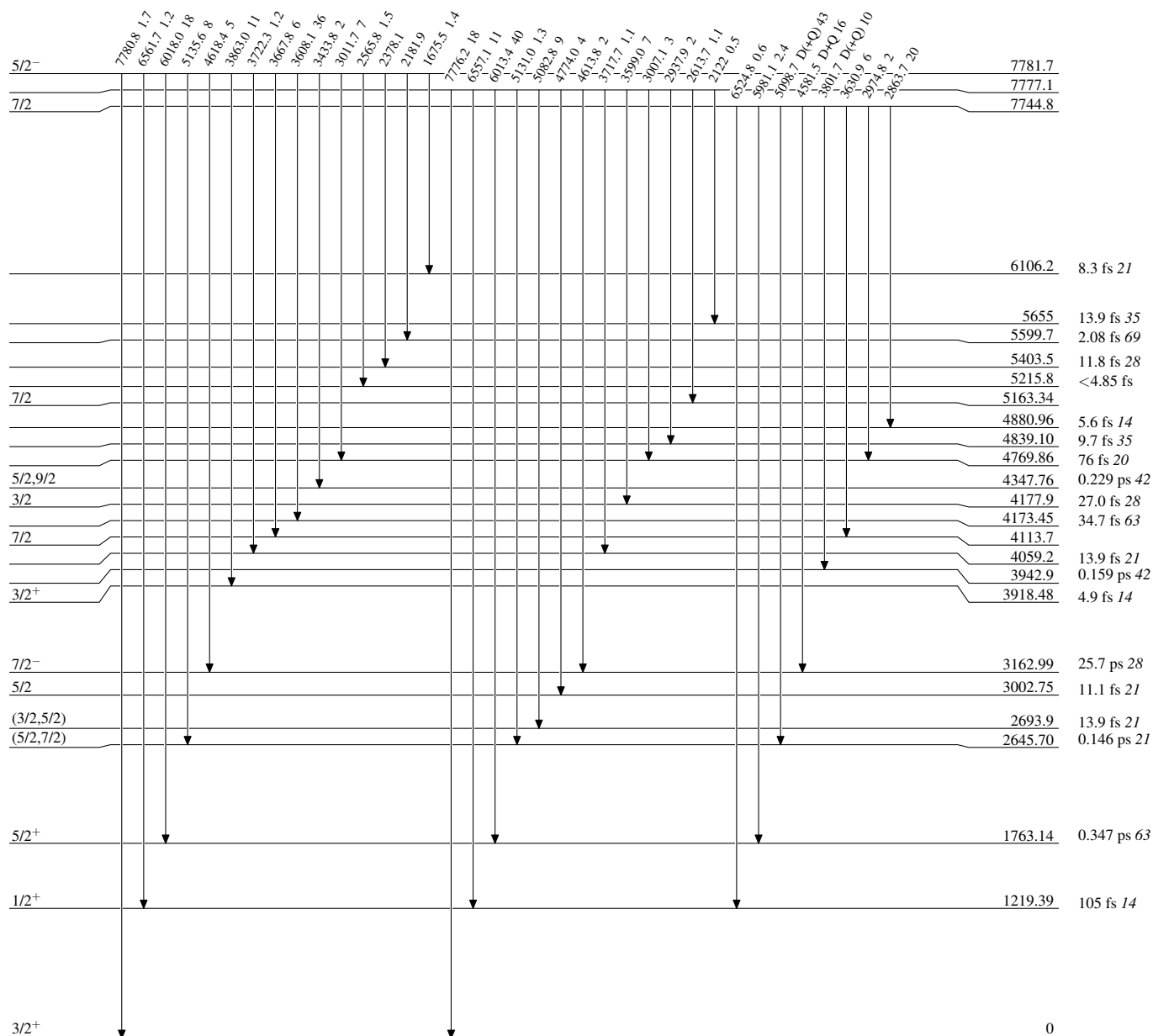


³⁵₁₇Cl₁₈

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

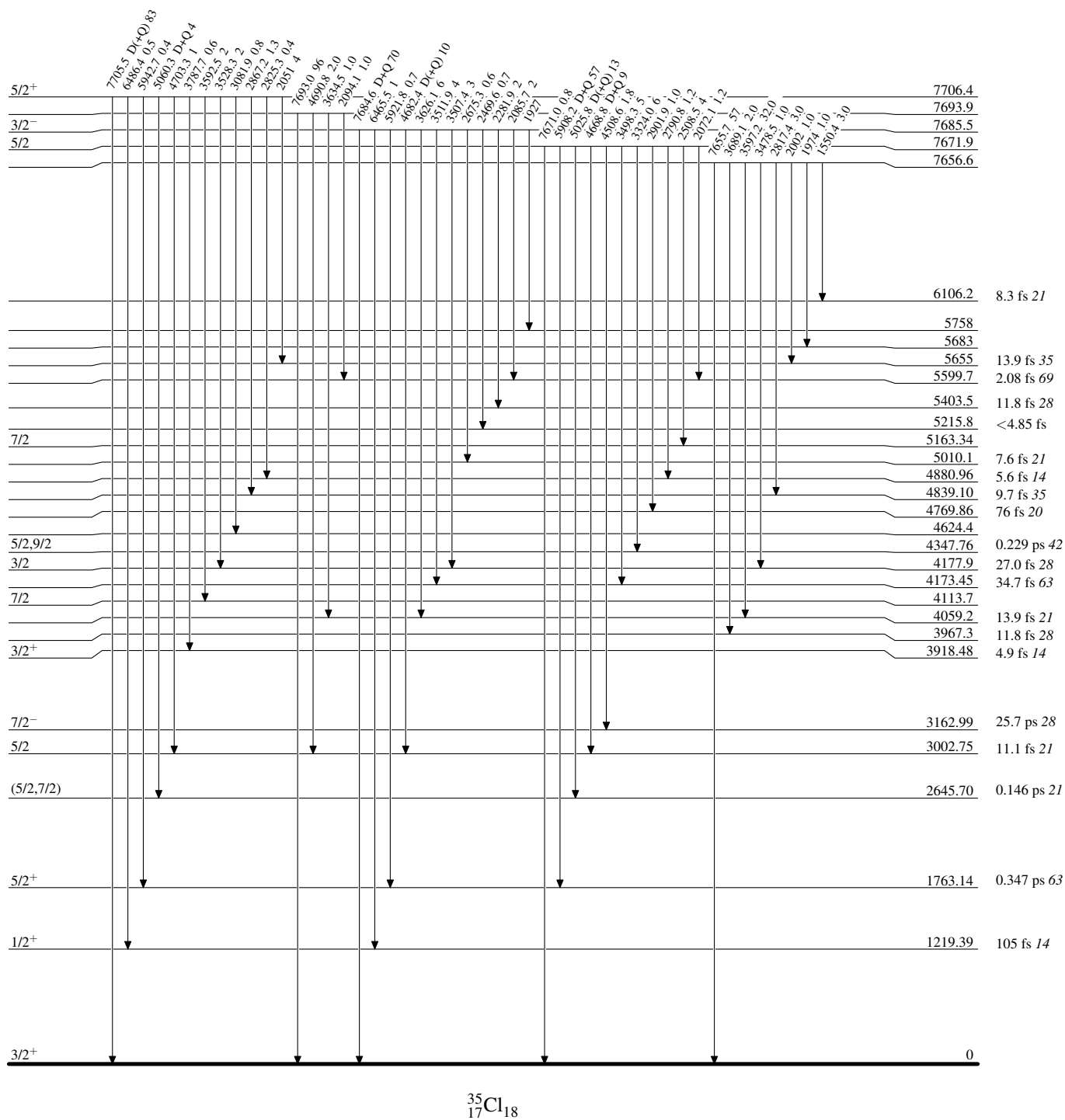
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

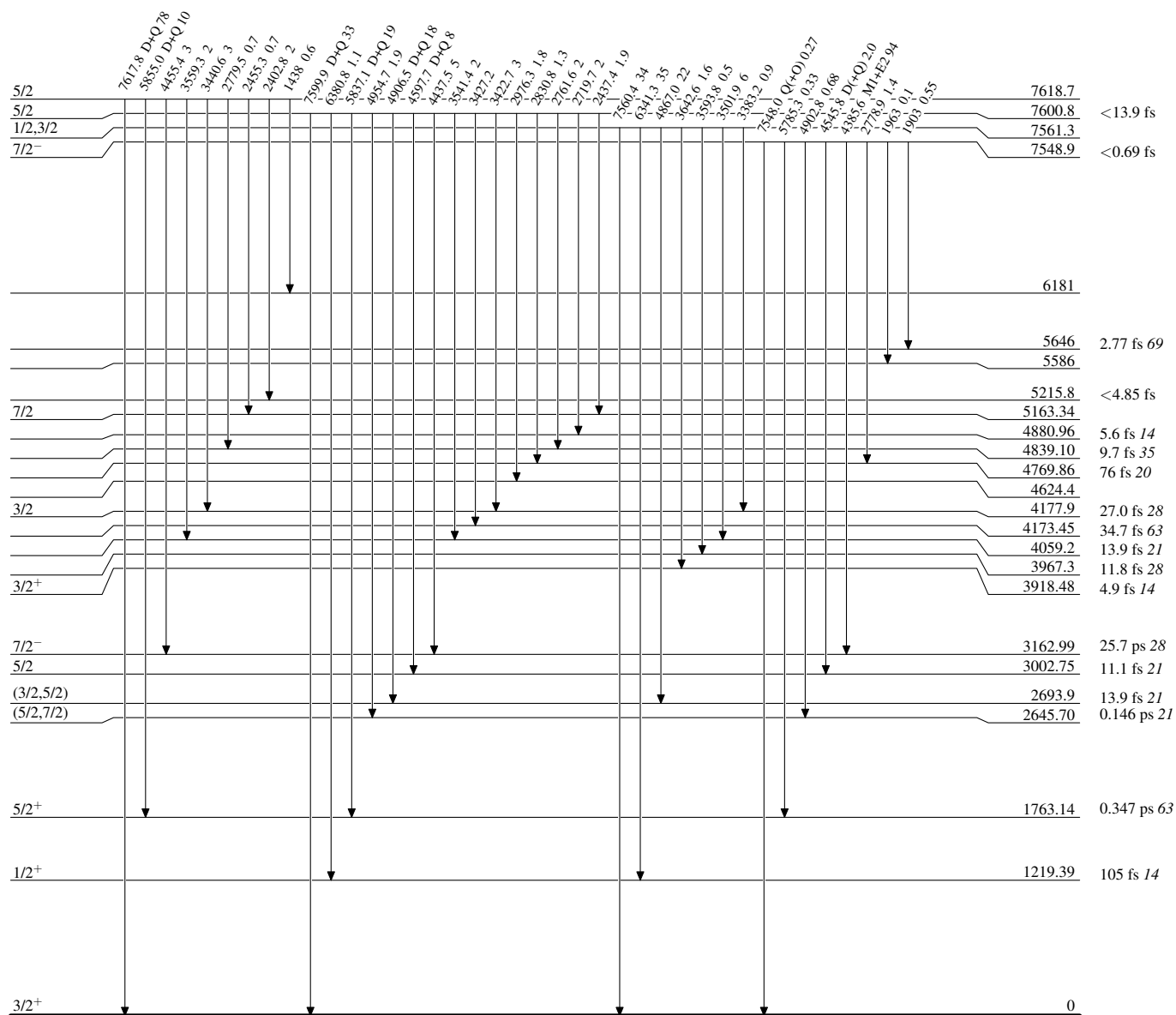
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

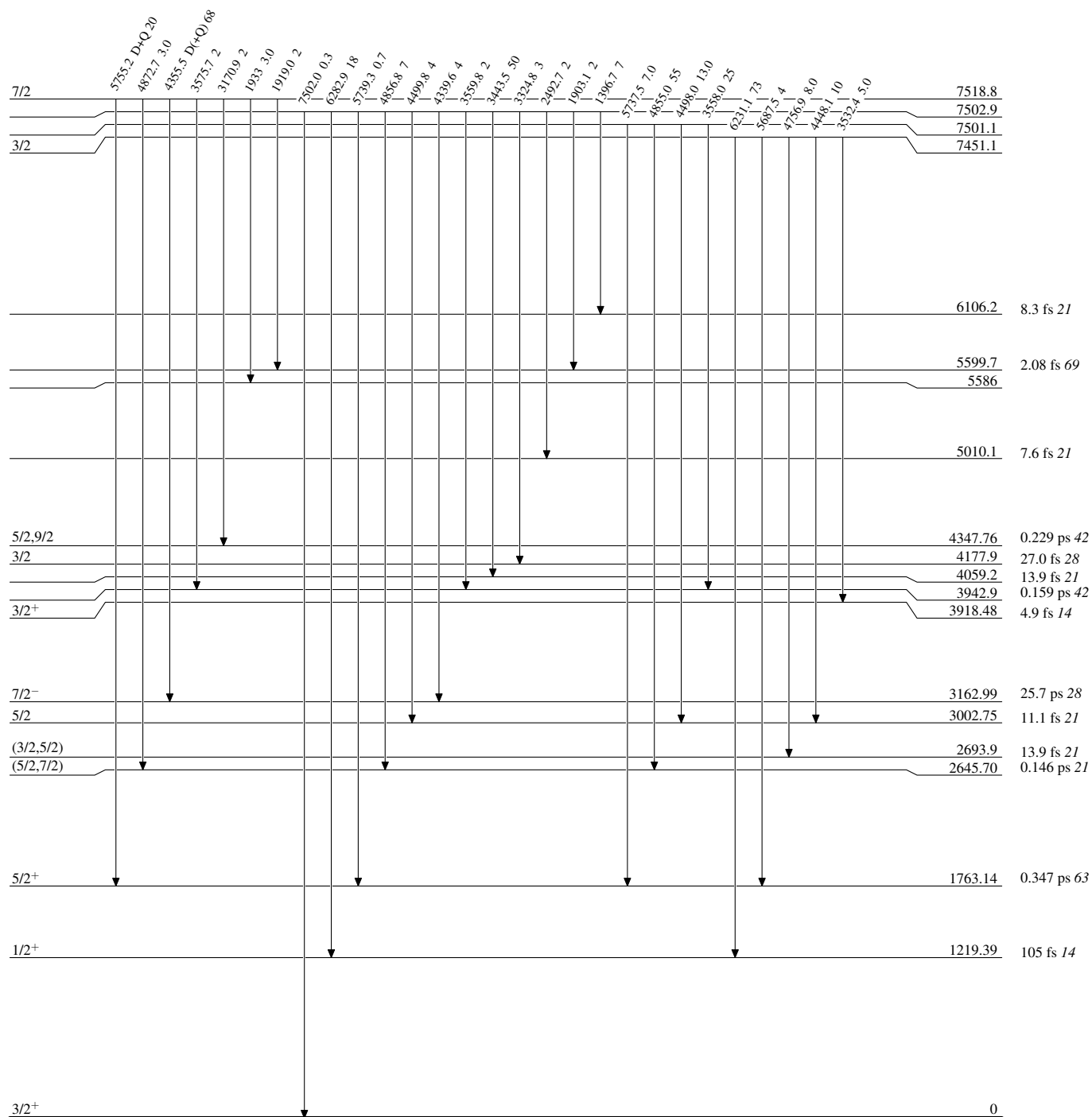
Intensities: % photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

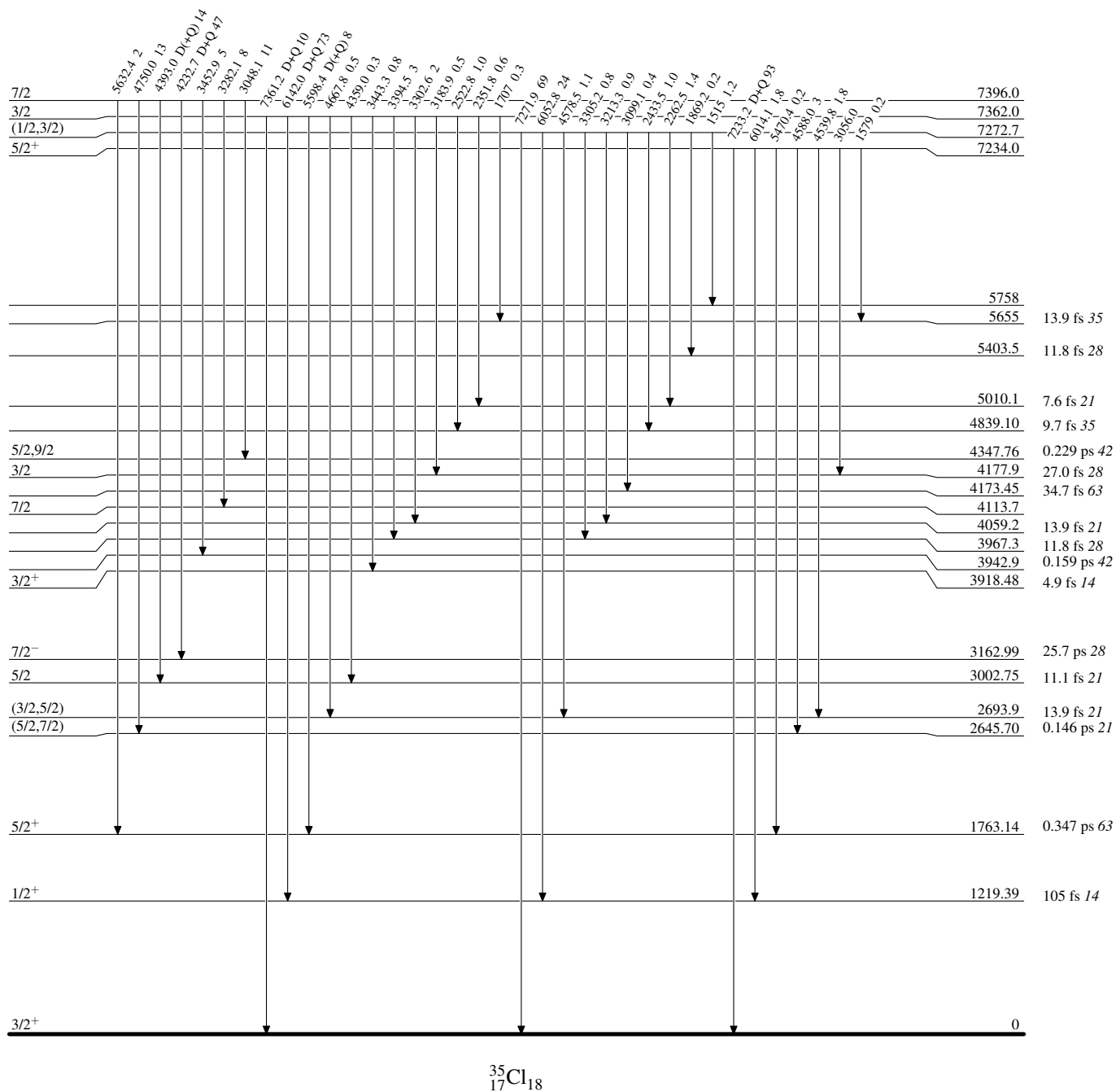
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

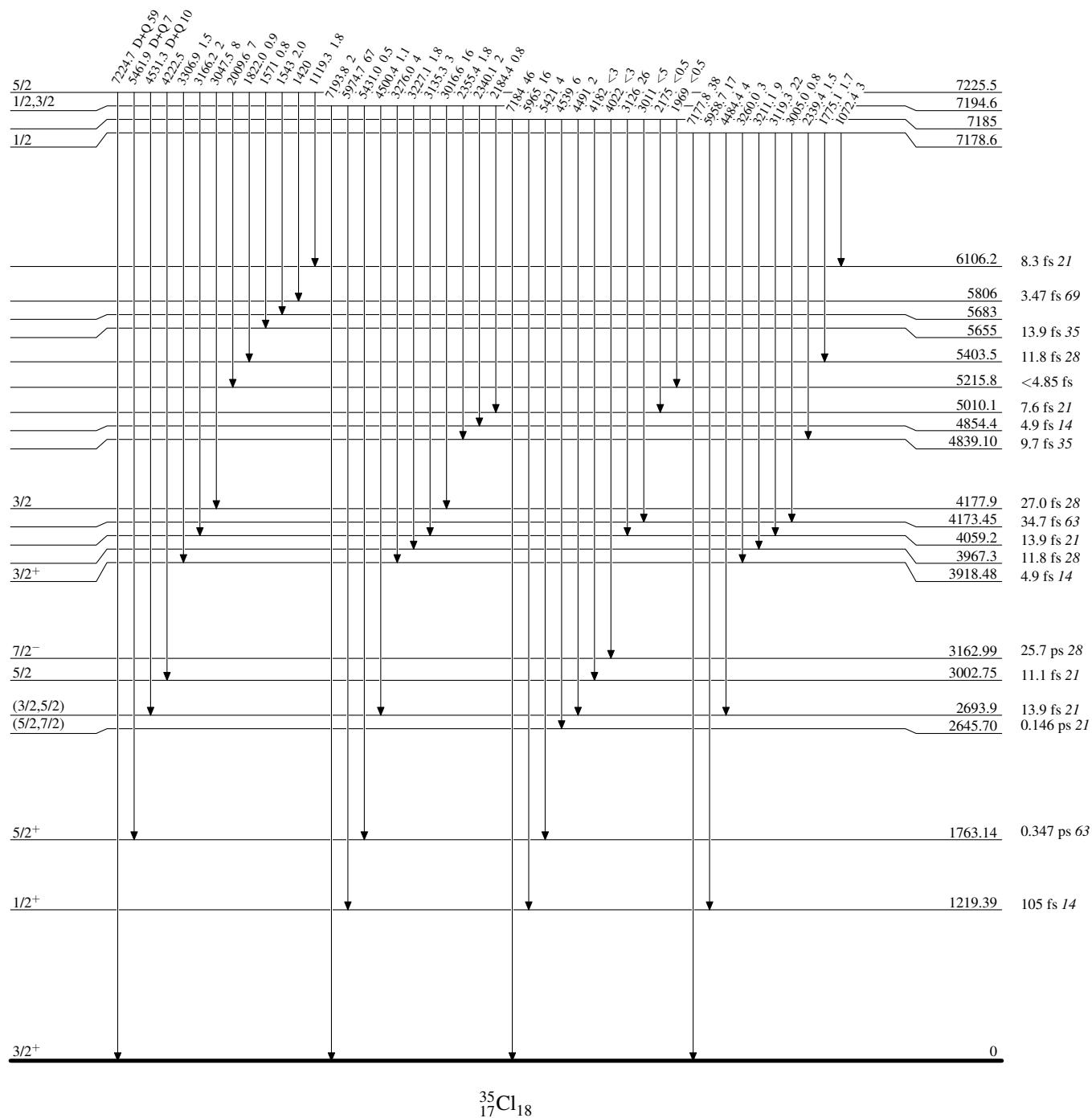
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

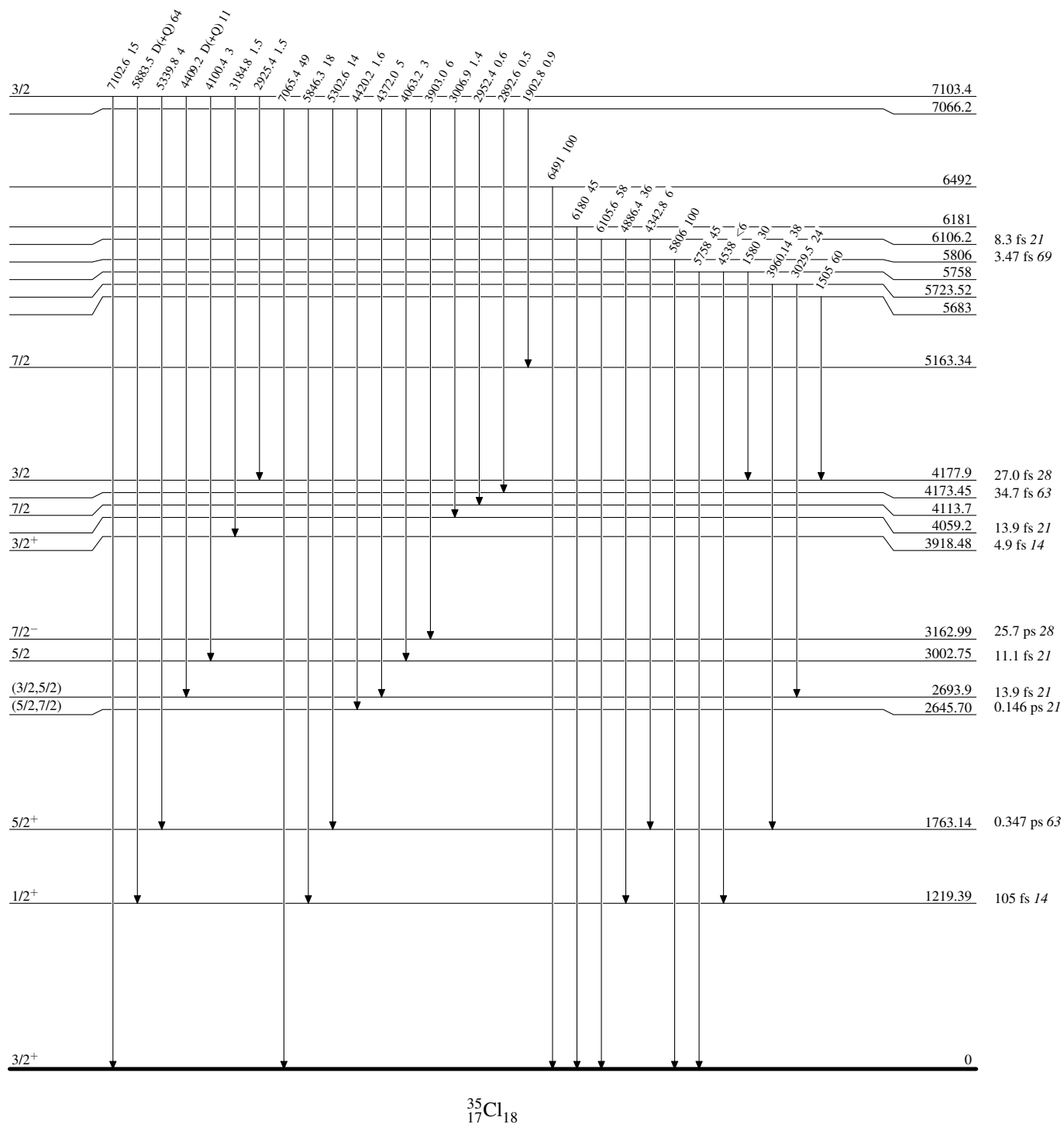
Intensities: % photon branching from each level



$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Level Scheme (continued)

Intensities: % photon branching from each level

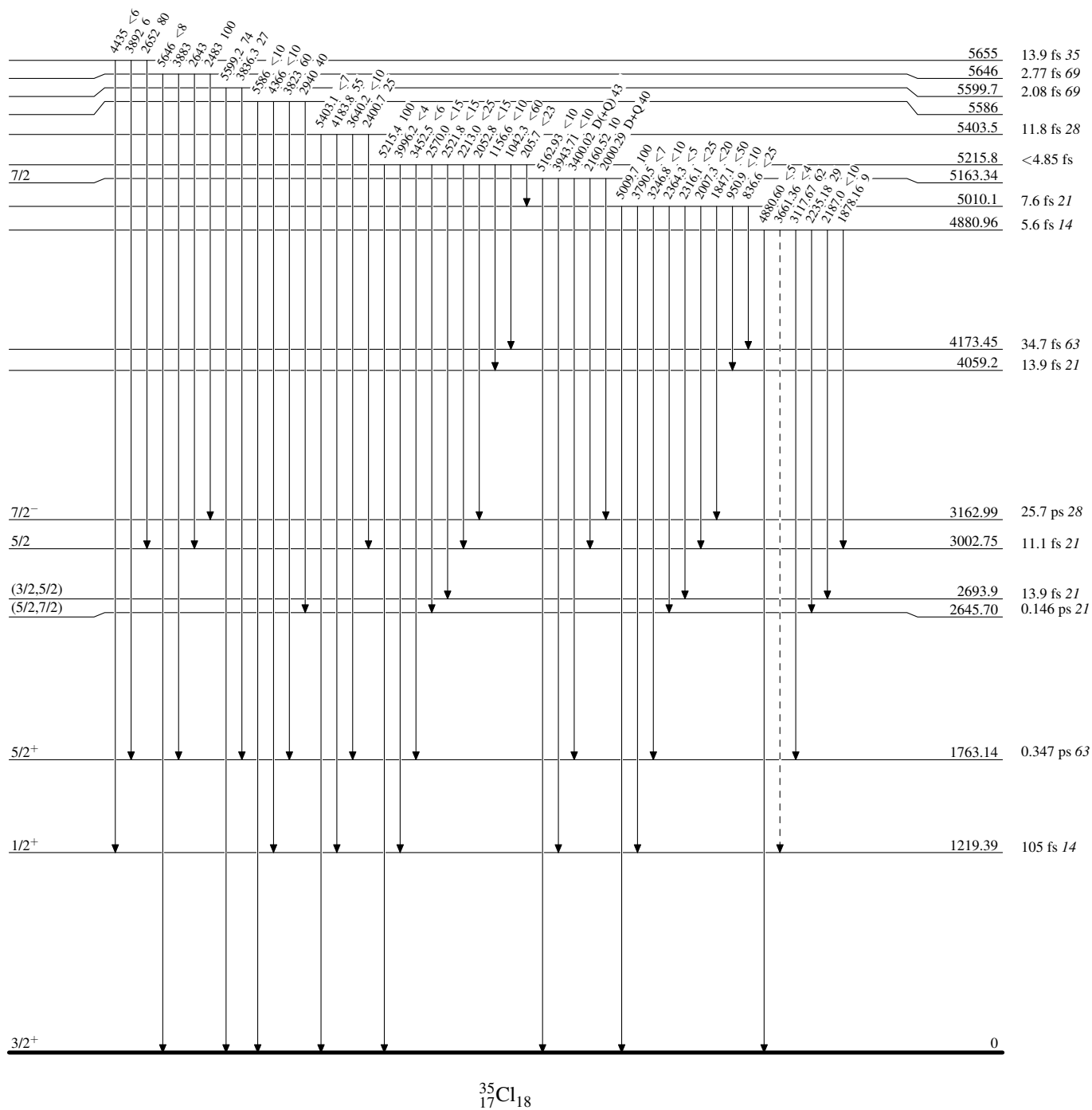


$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

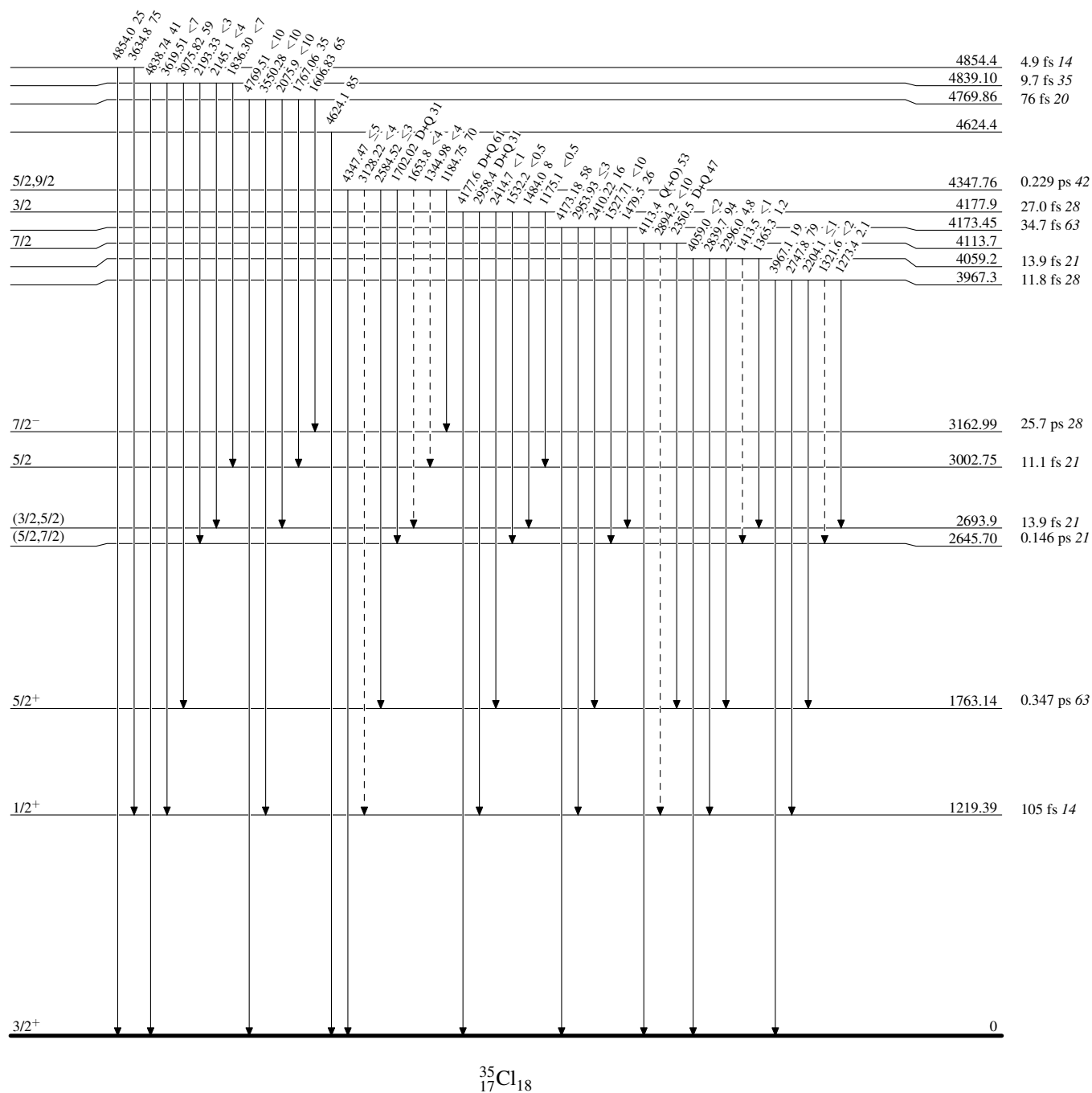
-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

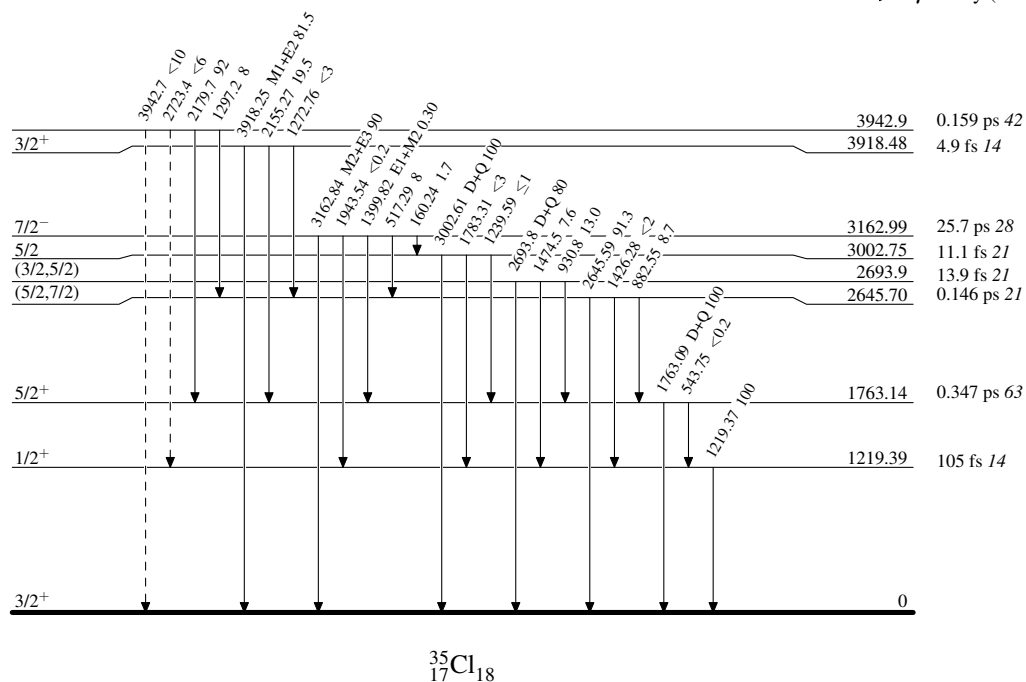
-----► γ Decay (Uncertain)

$^{34}\text{S}(\text{p},\gamma)$ 1976Me12,1976Sp08,1972Hu10

Legend

Level Scheme (continued)

Intensities: % photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

$^{34}\text{S}(\text{p,p}),(\text{p,p}')\text{:resonances}$ **1967Ko19,1977Ou01,1981Bi05** $^{34}\text{S}(\text{p,p}')\gamma$ measurement:

1967Ko19: E=2.78-3.08-MeV proton beams were produced from the 4-MeV electrostatic accelerator of the Physical-Technical Institute of the Ukrainian SSR Academy of Sciences, in energy step of 1 keV. Target was enriched thin ^{34}S . γ rays were detected with a 70 by 60 mm NaI(Tl) crystal for detecting the 2.13 MeV γ -ray of ^{34}S decay, resolution of 11.5% for the 0.622 MeV ^{137}Cs line. Measured $\sigma(E_p, \theta)$ from yields of the 2.13 MeV γ -ray, $\text{py}(\theta)$ of the 2.13 MeV γ -ray. Deduced resonance energies, J, π , relative resonance intensities.

 $^{34}\text{S}(\text{p,p})$ measurements:

1977Ou01: E=1.45-2.82 MeV proton beams were produced from the TUNL 3-MV Van de Graaff accelerator in energy steps of 100 eV for on-resonance and 400 eV for off-resonance, overall FWHM=350-450 eV. Target was CdS, with 90% enriched ^{34}S evaporated onto 5-10 $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected with surface barrier detectors at 90°, 105°, 135° and 160°. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , Γ_p from R-Matrix analysis.

1981Bi05: E=1.4-2.8 MeV proton beams were produced from the Groningen 5 MV Van de Graaff accelerator. Targets were 4 and 1 $\mu\text{g}/\text{cm}^2$ Sb_2S_3 (90% in ^{34}S) evaporated onto 10 $\mu\text{g}/\text{cm}^2$ carbon backings. Scattered protons were detected with three silicon surface barrier detectors at 88°, 124° and 174°. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , Γ_p . Also measured (p, γ).

1971BrXT: Measured $\sigma(E_p)$. Deduced resonance energies, J, π , widths.

1974Bi16: E=1.2-2.1 MeV protons were produced from the Helsinki University 2.5 MV van de Graaff accelerator. Targets were prepared from natural sulphur (4.22% ^{34}S) with ^{34}S ions embedded into carbon foils. Scattered protons were detected with four silicon surface barrier detectors. Measured $\sigma(E_p, \theta)$. Deduced resonance energies, J, π , widths from analysis of the resonances.

Others: **1978Ce02**, **1980Fa07**, **1983Ch50**, **1988Za04**, **2019Ka53**.

 ^{35}Cl Levels

A_2 and A_4 values under comments are from $\text{py}(\theta)$ of the 2.13 MeV γ -ray from ^{34}S (**1967Ko19**).

I(res) under comments are resonance intensities relative to 1 for $E_p(\text{lab})=2799$ resonance (**1967Ko19**).

E(level) [†]	J π [‡]	Γ_p [#]	$E_p(\text{lab})(\text{keV})$ [#]	Comments
7195@	1/2 ⁻ @		848	$\Gamma=27$ eV 8 (1971BrXT).
7274@	1/2 ⁻ @		929	$\Gamma=14$ eV 4 (1971BrXT).
7550	7/2 ⁻		1214	$\Gamma_p=4.6$ eV 10 and $\Gamma=5$ eV +4-2 from a Q^2 minimization analysis of the (p,p) excitation function (1971BrXT). $\Gamma=10.5$ eV 15, $\Gamma_p=5.8$ eV 7, $\Gamma_\gamma=4.7$ eV 17 (1971BrXT); $\Gamma=11.0$ eV 15, $\Gamma_p=7.2$ eV 15, $\Gamma_\gamma=3.8$ eV 15 (1974Bi16), which are deduced using a questionable (p, γ) resonance strength of 21 eV 3 from 1966En04 .
7686&	3/2 ⁻ &		1354	$\Gamma_p=445$ eV 11 for $J^\pi=3/2^-$ (1981Bi05).
7706&	3/2 ⁺ , 5/2 ⁺ &		1375	$\Gamma_p=4$ eV 1 for $J^\pi=5/2^+$, 17 eV 3 for $J^\pi=3/2^+$ (1981Bi05).
7796 3	(1/2) ⁻	0.031 keV 10	1467.3 5	
7837 3	3/2 ⁻	3.7 keV 4	1510.0 5	$\Gamma=3.8$ keV 2, $\Gamma_p=3.8$ keV 2 (1974Bi16).
7880 3	3/2 ⁺ , 5/2 ⁺	0.008 keV 4	1554.0 5	
8004 3	3/2 ⁺ , 5/2 ⁺	0.011 keV 5	1682.1 5	$\Gamma_p=21$ keV 3 for $J^\pi=5/2^+$, 24 eV 3 for $J^\pi=3/2^+$ (1981Bi05).
8034 3	1/2 ⁻ , 3/2 ⁻	0.026 keV 7	1712.9 5	
8038 3	1/2 ⁺	0.30 keV 2	1716.9 5	
8148 3	1/2 ⁻	2.66 keV 27	1830.2 5	
8150 3	3/2 ⁻	0.56 keV 6	1831.5 5	
8209 3	(5/2) ⁺	0.033 keV 10	1893.2 5	$\Gamma=100$ eV 20, $\Gamma_p=120$ eV 20 (1974Bi16).
8211 3	1/2 ⁺	0.094 keV 15	1894.4 5	$\Gamma_p=39$ eV 5 for $J^\pi=5/2^+$, 44 eV 6 for $J^\pi=3/2^+$, 120 eV 11 for $J^\pi=1/2^+$ (1981Bi05).
				$\Gamma=270$ eV 50, $\Gamma_p=270$ eV 50 (1974Bi16).
8216	5/2 ⁺	0.014 keV 3	1900	$\Gamma=14$ eV 4, $\Gamma_\gamma=0.78$ eV (1974Bi16).
8243 3	3/2 ⁻	0.140 keV 15	1928.2 5	
8271 3	3/2 ⁺ , 5/2 ⁺	0.005 keV 3	1956.1 5	
8279 3	3/2 ⁺ , 5/2 ⁺	0.006 keV 3	1964.6 5	
8290 3	(1/2) ⁻	0.04 keV 1	1975.7 5	$\Gamma_p=49$ eV 11 for $J^\pi=3/2^-$, 48 eV 6 for $J^\pi=1/2^-$ (1981Bi05).

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$ **1967Ko19,1977Ou01,1981Bi05 (continued)** ^{35}Cl Levels (continued)

E(level) [†]	$J^{\pi\ddagger}$	$\Gamma_{\text{p}}^{\#}$	$E_{\text{p}}(\text{lab})(\text{keV})^{\#}$	Comments
8300 3	3/2 ⁻	0.073 keV 15	1986.7 5	$\Gamma=30$ eV 7, $\Gamma_{\text{p}}=28$ eV 6, $\Gamma_{\gamma}=2.4$ eV (1974Bi16). J^{π} : 5/2 ⁺ in 1974Bi16.
8383 3	3/2 ⁺ , 5/2 ⁺	0.023 keV 7	2072.2 5	
8404 3	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	2093.9 5	
8406 3	(5/2 ⁻ , 7/2 ⁻)	0.001 keV 1	2095.2 5	
8409 3	1/2 ⁻	0.125 keV 15	2098.7 5	
8418 3	1/2 ⁺	0.026 keV 7	2107.5 5	
8435 3	(3/2) ⁺	0.090 keV 15	2125.9 5	
8486 3	3/2 ⁺ , 5/2 ⁺	0.012 keV 5	2178.2 5	
8488 3	3/2 ⁻	0.150 keV 15	2179.7 5	
8515 3	1/2 ⁻	0.150 keV 15	2208.1 5	
8573 3	(5/2) ⁺	0.08 keV 1	2267.9 5	
8582 3	1/2 ⁺	0.75 keV 8	2276.7 5	
8592 3	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	2287.4 5	
8616 3	5/2 ⁺	0.175 keV 20	2311.6 5	
8620 3	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	2316.2 5	
8633 3	(3/2, 5/2 ⁺)	0.001 keV 1	2328.8 5	
8643 3	3/2 ⁺ , 5/2 ⁺	0.003 keV 2	2339.9 5	
8687 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2385.0 5	
8689 3	1/2 ⁺	0.20 keV 2	2387.2 5	
8691 3	1/2 ⁻	6.4 keV 7	2388.6 5	
8698 3	3/2 ⁻	0.8 keV 1	2396.0 5	
8750 3	3/2 ⁻	0.30 keV 3	2449.5 5	
8775 3	1/2 ⁻	0.57 keV 6	2475.2 5	
8781 3	3/2 ⁻	0.214 keV 25	2481.7 5	
8788 3	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	2489.1 5	
8799 3	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	2500.3 5	
8825 3	1/2 ⁺	1.70 keV 17	2526.8 5	
8829 3	1/2 ⁻	12.2 keV 12	2530.8 5	
8830 3	1/2 ⁺	0.080 keV 15	2532.0 5	
8839 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2541.7 5	$\Gamma_{\text{p}}=4$ eV 1 for $J^{\pi}=7/2^{-}$, 2 eV 1 for $J^{\pi}=5/2^{-}$ (1981Bi05).
8858 3	3/2 ⁺ , 5/2 ⁺	0.010 keV 5	2560.8 5	
8870 3	3/2 ⁺ , 5/2 ⁺	0.027 keV 10	2573.2 5	
8909 3	3/2 ⁺ , 5/2 ⁺	0.002 keV 1	2613.2 5	
8955 3	(3/2) ⁺	0.075 keV 15	2661.3 5	
8959 3	(1/2 ⁻ , 3/2 ⁻)	0.04 keV 1	2665.4 5	
8983 3	5/2 ⁻ , 7/2 ⁻	0.003 keV 2	2690.0 5	
8985 3	3/2 ⁺ , 5/2 ⁺	0.025 keV 10	2692.1 5	
8998 3	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	2705.1 5	
9021 3	3/2 ⁻	3.50 keV 35	2728.4 5	
9031 3	(5/2) ⁺	0.04 keV 1	2739.4 5	
9040 3	1/2 ⁻	0.292 keV 30	2748.2 5	
9048 3	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	2756.7 5	
9050 3	(5/2) ⁺	0.095 keV 15	2758.8 5	
9084 3	(5/2) ⁺	0.057 keV 10	2793.2 5	$\Gamma_{\text{p}}=59$ eV 7 for $J^{\pi}=5/2^{+}$, 67 eV 8 for $J^{\pi}=3/2^{+}$ (1981Bi05). $I(\text{res})=1$. $E_{\text{p}}(\text{lab})(\text{keV})$: from 1967Ko19.
9089 4			2799 4	
9101 3	3/2 ⁻	0.20 keV 2	2810.9 5	
9108 4			2818 4	$I(\text{res})=0.18$.
9114 4			2825 4	$I(\text{res})=0.34$.
9120 4			2831 4	$I(\text{res})=0.31$.
9127 4			2838 4	$I(\text{res})=0.16$.
9136 4			2847 4	$I(\text{res})=0.34$.
9147 4			2859 4	$I(\text{res})=0.27$.
9159 4			2871 4	$I(\text{res})=0.29$.
9168 4	(3/2 ⁻)		2880 4	$I(\text{res})=3.70$.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma):\text{resonances}$ **1967Ko19,1977Ou01,1981Bi05** (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	E _p (lab)(keV) [#]	Comments
9185 4		2898 4	A ₂ =+0.19 133, A ₄ =+0.017 1480 (1967Ko19). I(res)=0.11.
9196 4		2909 4	
9207 4	9/2 ⁺	2920 4	I(res)=1.70. A ₂ =+0.09 18, A ₄ =+0.37 12 (1967Ko19).
9222 4		2936 4	I(res)=0.25.
9227 4	5/2 ⁻	2941 4	I(res)=0.59. A ₂ =+0.59 28, A ₄ =+0.29 32 (1967Ko19).
9241 4	7/2 ⁺	2955 4	I(res)=2.32. A ₂ =+0.24 40, A ₄ =+0.27 53 (1967Ko19).
9255 4		2970 4	I(res)=0.45.
9271 4		2986 4	I(res)=0.34.
9276 4		2991 4	I(res)=0.32.
9284 4	5/2 ⁻	3000 4	I(res)=3.86. A ₂ =+0.72 28, A ₄ =-0.17 29 (1967Ko19).
9294 4		3010 4	I(res)=0.68.
9299 4		3015 4	I(res)=0.45.
9325 4	5/2 ⁺	3042 4	I(res)=1.81. A ₂ =+0.81 28, A ₄ =-0.48 35 (1967Ko19).
9336 4	7/2 ⁺	3053 4	I(res)=1.00. A ₂ =+0.17 29, A ₄ =+0.34 40 (1967Ko19).
9345 4		3062 4	I(res)=0.50.
9349 4		3067 4	I(res)=0.50.
9355 4		3073 4	I(res)=2.23.

[†] From E(level)=E_p(cm)+S(p)(^{35}Cl), where S(p)=6370.81 4 (2021Wa16) and E_p(cm) deduced from E_p(lab) by
E_p(cm)=E_p(lab)*[m(^{34}S)/(m(^{34}S)+m(p))].

[‡] From comparison of experimental differential cross sections with theoretical predictions by R-Matrix calculations (1977Ou01) for levels up to 9101 level and from $\text{py}(\theta)$ (1967Ko19) for levels after that.

[#] From 1977Ou01 up to 9101 level and from 1967Ko19 after that, unless otherwise noted. 1977Ou01 state that although the absolute proton energies are accurate only to about 3 keV, the relative values of E_p(lab) should be reliable to within a few hundred eV, based on which the evaluators have assigned a 0.5 keV uncertainty for E_p(lab) and 3 keV uncertainty for E(level) from 1977Ou01.

@ From 1971BrXT.

& From 1981Bi05.

³⁴S(d,n) **1969Da12**

$J^\pi=0^+$ for ³⁴S ground state.
1969Da12: a 4.95-MeV deuteron beam was produced from the 5.5-MeV Van de Graaff accelerator at University of Alberta and impinged on a CdS target enriched to 37% ³⁴S evaporated onto a 0.05-cm gold backing. Neutrons were detected by NE213 scintillators using the time-of-flight (tof) method with FWHM=1.0 ns. Measured $\sigma(E_n, \theta)$. Deduced levels, J, π , and L-transfers from DWUCK-DWBA analysis of the measured $\sigma(\theta)$.
1993TeZX: ³⁴S(d,n)³⁵Cl at E(d)=25 MeV at CYRJC, Tohoku University.

³⁵Cl Levels

E(level) [†]	L [‡]	Comments
0	2	
1220	0	
1762	2	
2645	(1)	L: weak stripping or possibly two unresolved levels.
2695		
3006	3	
3163	3	L: Discrepancy: L=2 from 0 ⁺ in (³ He,d), (d, ³ He), and (pol d, ³ He).
4060		
4110		
4170		

[†] From **1969Da12**.
[‡] From DWBA analysis of the measured $\sigma(\theta)$ in **1969Da12**.

$^{34}\text{S}(^3\text{He},\text{d})$ 2017Gi09,1969Gr23

$J^\pi=0^+$ for ^{34}S g.s.

- 2017Gi09:** a 20-MeV $^3\text{He}^{2+}$ beam was produced from the MP Van de Graaff tandem accelerator at the Maier-Leibnitz Laboratorium (MLL) and impinged on a $50\text{-}\mu\text{g}/\text{cm}^2$ Ag_2S target 99.999% enriched in ^{34}S with a $8\text{-}\mu\text{g}/\text{cm}^2$ carbon backing. Reaction products were momentum-separated by the Q3D magnetic spectrograph and focused on focal plane detectors consisting of two multiwire proportional counters (MWPCs) and a plastic scintillator. Deduced 22 levels, L-transfers, J, π , spectroscopic factors from FRESKO-DWBA analysis, and resonance strengths. Deduced the thermonuclear $^{34}\text{S}(\text{p},\gamma)^{35}\text{Cl}$ reaction rate through Monte Carlo methods using the STARLIB code. Calculated the $^{34}\text{S}/^{32}\text{S}$ isotopic ratio using the hydrodynamical nova model SHIVA.
- 1969Gr23:** a 11.0-MeV ^3He beam was produced from the MIT-ONR Van de Graaff generator. The target was $9.3\text{-}\mu\text{g}/\text{cm}^2$ ^{34}S made by evaporating PbS (67.6% ^{34}S) onto a formvar backing. Deuterons were detected on $50\text{ }\mu\text{m}$ nuclear emulsions in the MIT multiple-gap spectrograph at 23 different angles from 7.5° to 172.5° . Measured $\sigma(E_d,\theta)$. Deduced 17 levels, J, π , L-transfers, spectroscopic factors from zero-range JULIE-DWBA analysis.
- 1970Mo01:** a 13-MeV ^3He beam was produced from the Argonne tandem Van de Graaff accelerator. The target was $200\text{-}\mu\text{g}/\text{cm}^2$ 86% enriched ^{34}S . Reaction products were detected using two telescopes of silicon surface-barrier detectors with $\text{FWHM}\approx 80\text{ keV}$ and analyzed by a magnetic spectrograph with $\text{FWHM}\approx 20\text{ keV}$ at $\theta<20^\circ$. Measured $\sigma(E_d,\theta)$, $\theta_{\text{c.m.}}\approx 5^\circ$ to 100° . Deduced levels, J, π , L-transfers, spectroscopic factors from local, zero-range JULIE-DWBA analysis.
- 1994Ve04:** a 25-MeV ^3He beam was produced from the Orsay MP Tandem Van de Graaff accelerator. The target was KCl with natural chlorine. Reaction products were momentum-analyzed by an Enge split-pole magnetic spectrograph and detected using a 50-cm long telescope of a position-sensitive proportional counter, proportional ΔE counter, and a plastic scintillator. Measured $\sigma(E_d,\theta)$. Deduced J, π , L-transfer, and spectroscopic factor for the ground state of ^{35}Cl from local, zero-range DWUCK-DWBA analysis.

 ^{35}Cl Levels

For levels below 6 MeV, spectroscopic factor $(2J+1)\text{C}^2\text{S}=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where N is a normalization factor. J is the spin of the populated ^{35}Cl level. $N=4.42$ is used in 1969Gr23 and 1970Mo01 from 1966Ba54. $N=4.43$ is used in 1994Ve04. For levels above 6 MeV, C^2S values are from 2017Gi09 without using the factor N. 2017Gi09 also states that it is not possible to distinguish between $J=L+1/2$ or $J=L-1/2$ coupling given the experimental uncertainties. Values of resonance strengths $\omega\gamma(\text{min})$ and $\omega\gamma(\text{max})$ are calculated by 2017Gi09 using their measured C^2S and calculated Γ_p , assuming the Γ_{tot} is dominated by Γ_γ and thus $\omega\gamma$ depends only on Γ_p .

E(level) [†]	L [‡]	Comments
0	2	S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=3/2^+$; $\text{C}^2\text{S}=1.3$ in 1970Mo01 for $J^\pi=3/2^+$; $\text{C}^2\text{S}=1.07$ in 1994Ve04 for $J^\pi=3/2^+$.
1220 10	0	E(level): other: 1220 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.26$ in 1969Gr23 for $J^\pi=1/2^+$; $\text{C}^2\text{S}=0.28$ in 1970Mo01 for $J^\pi=1/2^+$.
1758 10		
2645 10		
2686 10	2	S: $(2J+1)\text{C}^2\text{S}=0.06$ in 1969Gr23 for $J^\pi=(3/2,5/2)^+$.
3002 10	2	E(level): weighted average of 2997 10 (1969Gr23) and 3010 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.14$ in 1969Gr23 for $J^\pi=(5/2)^+$; $\text{C}^2\text{S}=0.04$ in 1970Mo01 for $J^\pi=5/2^+$.
3158 10	3	E(level): weighted average of 3156 10 (1969Gr23) and 3160 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=2.88$ in 1969Gr23 for $J^\pi=7/2^-$; $\text{C}^2\text{S}=0.66$ in 1970Mo01 for $J^\pi=7/2^-$.
3908 10		
3962 10	0	E(level): weighted average of 3956 10 (1969Gr23) and 3970 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.03$ in 1969Gr23 for $J^\pi=1/2^+$; $\text{C}^2\text{S}=0.06$ in 1970Mo01 for $J^\pi=1/2^+$.
4053 10	1	E(level): weighted average of 4048 10 (1969Gr23) and 4060 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.38$ in 1969Gr23 for $J^\pi=(3/2)^-$; $\text{C}^2\text{S}=0.15$ in 1970Mo01 for $J^\pi=(3/2)^-$.
4167 10	1	E(level): weighted average of 4165 10 (1969Gr23) and 4170 12 (1970Mo01). S: $(2J+1)\text{C}^2\text{S}=0.85$ in 1969Gr23 for $J^\pi=(3/2)^-$; $\text{C}^2\text{S}=0.28, 0.62$ in 1970Mo01 for $J^\pi=(3/2)^-$.
5010 15		
5163 15	(2)	S: $(2J+1)\text{C}^2\text{S}=0.11$ without specifying J (1969Gr23).
5409 12	1	E(level): weighted average of 5407 15 (1969Gr23) and 5410 12 (1970Mo01). L: 1 in 1969Gr23 and L=(1,2) in 1970Mo01.

Continued on next page (footnotes at end of table)

$^{34}\text{S}(^3\text{He},\text{d})$ 2017Gi09,1969Gr23 (continued) ^{35}Cl Levels (continued)

E(level) [†]	L [‡]	Comments
5660 15	2	S: (2J+1)C ² S=0.07 in 1969Gr23 for $J^\pi=(3/2)^-$. E(level): 5670 multiplet reported in 1970Mo01. L: 2 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01.
5684 12	1	S: (2J+1)C ² S=0.47 in 1969Gr23 for $J^\pi=3/2^+$. E(level): 5670 multiplet reported in 1970Mo01. L: 1 in 1969Gr23, possibly a component of the 5670 multiplet with L=(2) in 1970Mo01.
5760 12	(0,1)	S: (2J+1)C ² S=0.11 in 1969Gr23 for $J^\pi=(3/2)^-$. E(level): weighted average of 5760 15 (1969Gr23) and 5760 12 (1970Mo01). L: 1 in 1969Gr23 and L=0 in 1970Mo01.
6284 4	(2)	S: (2J+1)C ² S=0.12 in 1969Gr23 for $J^\pi=(3/2)^-$; C ² S=0.21 in 1970Mo01 for $J^\pi=1/2^+$.
6329 4	(0,1)	S: (2J+1)C ² S=0.086 2.
6377 2	(2,3)	S: (2J+1)C ² S=0.0030 2 for L=0, 0.0044 4 for L=1.
6427 2	(3)	S: (2J+1)C ² S=0.0012 6 for L=2, 0.0024 8 for L=3.
6468 2	(1)	S: (2J+1)C ² S=0.049 2. $\omega\gamma(\text{min})=6.5\times 10^{-24}$ eV 2, $\omega\gamma(\text{max})=2.2\times 10^{-22}$ eV 1.
6491 2	(2)	S: (2J+1)C ² S=0.034 1. $\omega\gamma(\text{min})=4.2\times 10^{-14}$ eV 5, $\omega\gamma(\text{max})=4.7\times 10^{-14}$ eV 5.
6545 2	(0,1)	S: (2J+1)C ² S=0.072 2. $\omega\gamma(\text{min})=3.7\times 10^{-13}$ eV 1, $\omega\gamma(\text{max})=4.1\times 10^{-13}$ eV 1.
6643 2	(1)	S: (2J+1)C ² S=0.004 1 for L=0, 0.0028 4 for L=1. $\omega\gamma(\text{min})=7.4\times 10^{-10}$ eV 13, $\omega\gamma(\text{max})=4.1\times 10^{-9}$ eV 1.
6674 2	(1,2,3)	S: (2J+1)C ² S=0.0144 8. $\omega\gamma(\text{min})=8.5\times 10^{-6}$ eV 10, $\omega\gamma(\text{max})=8.9\times 10^{-6}$ eV 10.
6761 2	(0,1)	S: (2J+1)C ² S=0.0020 4 for L=1, 0.0048 6 for L=2, 0.008 1 for L=3. $\omega\gamma(\text{min})=2.4\times 10^{-8}$ eV 2, $\omega\gamma(\text{max})=4.1\times 10^{-6}$ eV 8.
6778 2	(1)	S: (2J+1)C ² S=0.0056 12 for L=0, 0.0032 4 for L=1. $\omega\gamma(\text{min})=2.4\times 10^{-4}$ eV 3, $\omega\gamma(\text{max})=1.5\times 10^{-3}$ eV 1.
6823 2	(1)	S: (2J+1)C ² S=0.0084 8. $\omega\gamma(\text{min})=9.4\times 10^{-4}$ eV 12, $\omega\gamma(\text{max})=1.0\times 10^{-3}$ eV 1.
6842 2	(2,3)	S: (2J+1)C ² S=0.006 4.
6866 2	(0,2)	S: (2J+1)C ² S=0.00216 6 for L=2, 0.035 2 for L=3. $\omega\gamma(\text{min})=2.7\times 10^{-3}$ eV 3, $\omega\gamma(\text{max})=2.7\times 10^{-3}$ eV 3.
7066 2	(1,2)	S: (2J+1)C ² S=0.0216 6 for L=2, 0.035 2 for L=3. $\omega\gamma(\text{min})=3.3\times 10^{-5}$ eV 1, $\omega\gamma(\text{max})=7.0\times 10^{-4}$ eV 3.
7103 2	(1,3)	S: (2J+1)C ² S=0.0160 2(L=0)+0.0660 6 (L=2). $\omega\gamma(\text{min})=3.0\times 10^{-2}$ eV 3, $\omega\gamma(\text{max})=4.7\times 10^{-2}$ eV 5.
7178 2	(2)	S: (2J+1)C ² S=0.0088 8 for L=1, 0.012 1 for L=2. $\omega\gamma(\text{min})=2.2\times 10^{-2}$ eV 1, $\omega\gamma(\text{max})=2.2\times 10^{-2}$ eV 1.
7194 2	(2)	S: (2J+1)C ² S=0.0184 12 for L=1, 0.025 2 for L=3. $\omega\gamma(\text{min})=6.0\times 10^{-2}$ eV 3, $\omega\gamma(\text{max})=1.30$ eV 7.
7227 2	(0,1)	S: (2J+1)C ² S=0.032 3.
7273 2	(0,1)	S: (2J+1)C ² S=0.065 15 for L=0, 0.044 2 for L=1.
7361 2	(1)	S: (2J+1)C ² S=0.050 12 for L=0, 0.030 1 for L=1. $\omega\gamma(\text{min})=11$ eV 2, $\omega\gamma(\text{max})=34$ eV 3.
7398 2	(2,3)	S: (2J+1)C ² S=0.0040 8. $\omega\gamma(\text{min})=0.30$ eV 4, $\omega\gamma(\text{max})=3.0$ eV 6.
		S: (2J+1)C ² S=0.042 2 for L=2, 0.0656 24 for L=3. $\omega\gamma(\text{min})=0.18$ eV 1, $\omega\gamma(\text{max})=0.19$ eV 1.

[†] From 1969Gr23 for E(level)<6 MeV and from 2017Gi09 for E(level)>6 MeV, unless otherwise noted.

$^{34}\text{S}(^3\text{He,d})$ [2017Gi09,1969Gr23](#) (continued)

^{35}Cl Levels (continued)

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in [1969Gr23](#) for $E(\text{level}) < 6$ MeV and in [2017Gi09](#) for $E(\text{level}) > 6$ MeV, unless otherwise noted. Firm assignments by [2017Gi09](#) are placed in parentheses by the evaluators because they are based on very few data points in a very narrow range of angles in the DWBA fit which may not yield definitive and firm results.

$^{35}\text{Cl}(\gamma, \gamma')$ **1966Ho02, 1962Bo17, 1991Li12**

1966Ho02: a bremsstrahlung photon beam was produced by impinging 1.4-MeV electrons on a thin layer of lead embedded in a water-cooled aluminium disc at the Diamond Research Laboratory in Johannesburg. The target was C_2Cl_6 . Scattered γ rays were detected using a 7.6-cm by 7.6-cm NaI crystal. Deduced the width for the 1.22-MeV first excited state using the self-absorption method of nuclear resonance fluorescence.

1962Bo17: an electron beam was produced from the MIT Van de Graaff generator, passed through a 76- μm aluminum window and a 25- μm platinum foil, and stopped in the cooling water of the electron target. γ rays were detected using a 5.1-cm by 5.1-cm NaI crystal. Lifetimes of ^{35}Cl levels are deduced from the measured intensity of the resonance fluorescence radiation which occurs when a nucleus is bombarded by bremsstrahlung.

1991Li12: a bremsstrahlung photon beam was produced by focusing a 4.1-MeV electron beam onto a radiator target of a gold disc on a water-cooled copper backing at the bremsstrahlung facility at the Stuttgart Dynamitron Accelerator. The target was a stack of discs that contain ^{13}C , ^{23}Na , ^{27}Al , ^{31}P , ^{35}Cl , ^{48}Ti . γ rays were detected using 3 Ge detectors. Deduced σ of photon scattering.

Other: **1972Sh07**.

 ^{35}Cl Levels

$\Gamma_0\Gamma_f/\Gamma$ from **1991Li12**. Γ_0 , Γ_f , and Γ are the decay width to the ground state, the decay width to the final state, and the total width, respectively.

<u>E(level)[†]</u>	<u>T_{1/2}</u>	<u>Comments</u>
0		
1220	90 fs +21-14	T _{1/2} : from lifetime=0.13 ps +3-2 in 1966Ho02 with resonance fluorescence. Other: lifetime=0.30 ps 10 in 1962Bo17 with resonance fluorescence assuming J=1/2.
1760	0.42 ps 15	T _{1/2} : from lifetime=0.60 ps 21 in 1962Bo17 with resonance fluorescence assuming J=5/2.
2693.7		$\Gamma_0\Gamma_f/\Gamma=12$ meV 3 (1991Li12).
3002.7		$\Gamma_0\Gamma_f/\Gamma=26$ meV 11 (1991Li12).
3918.6		$\Gamma_0\Gamma_f/\Gamma=63$ meV 18 (1991Li12).

[†] From E_γ data.

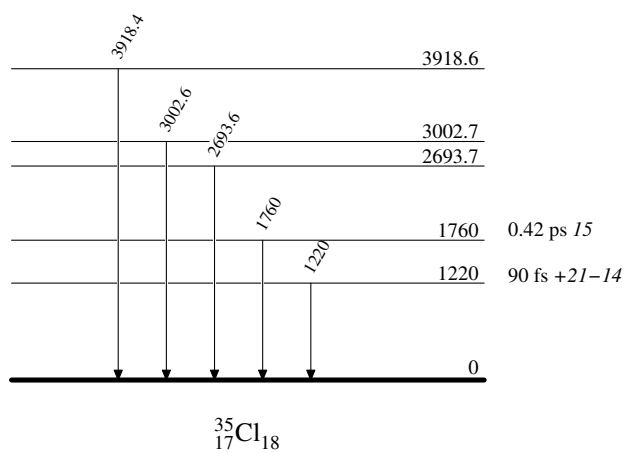
 $\gamma(^{35}\text{Cl})$

<u>E_{γ}</u>	<u>E_i(level)</u>	<u>E_f</u>	<u>Comments</u>
1220	1220	0	E _{γ} : from 1962Bo17 and 1966Ho02 .
1760	1760	0	E _{γ} : from 1962Bo17 .
2693.6 [†]	2693.7	0	
3002.6 [†]	3002.7	0	
3918.4 [†]	3918.6	0	

[†] From **1991Li12**, which quoted from **1978En02** evaluation.

$^{35}\text{Cl}(\gamma, \gamma')$ 1966Ho02, 1962Bo17, 1991Li12

Level Scheme



$^{35}\text{Cl}(\text{n}, \text{n}'\gamma)$ 1989Ge09, 1984El12, 1966Ni03

1984El12, 1989Ge09: fast neutrons of diameter 26 mm were produced from the IRT-2000 reactor at the Institute for Nuclear Research and Nuclear Power of the Bulgarian Academy of Sciences in Sofia. Two ^{35}Cl targets were placed at 45° and 135° with a distance of 70 cm apart. γ rays were detected using a Ge(Li) detector. Measured E_γ . Deduced lifetimes for 4 levels using the Doppler Shift Attenuation Method (DSAM) and transition strengths.

1966Ni03: 2.5-4.2-MeV neutrons were produced via the T(p,n) reactions with protons from the university of Kentucky 6-MeV accelerator. Targets were a cylinder of CCl_4 and a 5.6-cm-diam sphere of C_2Cl_6 . γ rays were detected using a 6.35-cm by 6.35-cm NaI(Tl) detector and 7.5-cc Ge(Li) detector. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J.

1971Fr05: neutrons up to 10 MeV were produced from a 5-Ci Pu-Be neutron source and bombarded table salt targets. γ rays were detected using a Ge(Li) detector. Measured 2 γ rays and deduced 2 levels.

Others: 1955To32, 1977Ko19, 1983Gl07, 1992Be60 (cross sections).

 ^{35}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\#$</u>	<u>$T_{1/2}^\ddagger$</u>
0	$3/2^+$	
1219	$1/2$	139 fs 56
1762	$5/2$	
2646	$7/2$	187 fs 62
2694		43 fs 6
3002		50 fs 9
3163		

† From a least-squares fit to γ -ray energies.

‡ From 1984El12 and 1989Ge09 using DSAM.

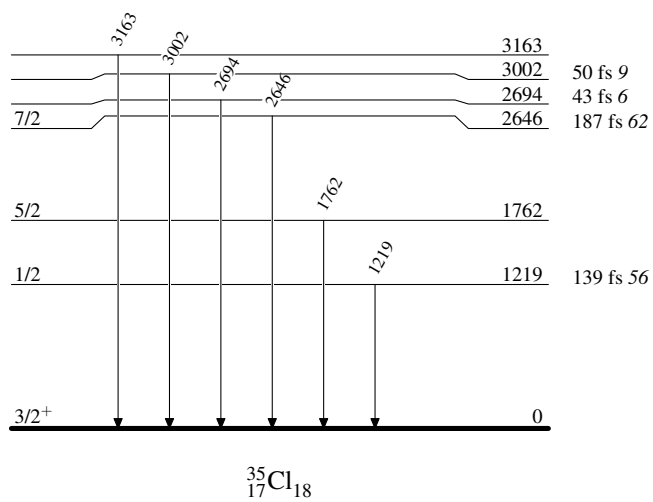
$^\#$ From $\gamma(\theta)$ in 1966Ni03, except for the ground state from the Adopted Levels.

 $\gamma(^{35}\text{Cl})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
1219	1219	$1/2$	0	$3/2^+$	E $_\gamma$: from 1966Ni03. A $_0$ =+0.942 8, A $_2$ =-0.148 23 (1966Ni03).
1762	1762	$5/2$	0	$3/2^+$	
2646	2646	$7/2$	0	$3/2^+$	A $_0$ =+1.078 5, A $_2$ =+0.165 12 (1966Ni03).
2694	2694		0	$3/2^+$	
3002	3002		0	$3/2^+$	E $_\gamma$: from 1966Ni03.
3163	3163		0	$3/2^+$	

$^{35}\text{Cl}(n,n'\gamma)$ 1989Ge09,1984El12,1966Ni03

Level Scheme



$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ 1969Du08,1970Ta02,1972Br45

Also include (p,p') measurements listed below.

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ measurements:

1969Du08: E=4.6-6.2 MeV 0.05 μA proton beam. Thin targets of CdCl_2 (99% ^{35}Cl), 100 $\mu\text{g}/\text{cm}^2$. A 300 μm annular surface-barrier detector on the beam axis and a movable 390 μm surface-barrier detector rotated from 20° to 148° relative to beam direction for detecting protons, FWHM=24 keV; a 7.62 by 7.62-cm-diam NaI(Tl) crystal and 15- and 35-cc Ge(Li) for detecting γ -rays. Measured $\sigma(E_p, \theta)$, E_γ , I_γ , $\text{py}(\theta)$. Deduced levels, J^π , branchings, mixing ratios, half-lives using Doppler Shift Attenuation Method (DSAM).

1970Ta02: E=4.24, 4.82, 5.08 and 5.11 MeV proton beams produced from the Universite Montreal EN tandem accelerator. Targets of 16 $\mu\text{g}/\text{cm}^2$ AgCl evaporated onto 0.013 cm gold foils, 99% enriched in ^{35}Cl . A 12.70-cm-diam by 15.24-cm-long NaI(Tl) and a 30 cm^3 Ge(Li) detector for detecting γ -rays. Measured E_γ , I_γ , $\gamma(\theta)$, $\gamma(\text{lin pol})$. Deduced levels, J^π , branchings, mixing ratios.

1972Br45: E=3.45-6.00 MeV protons produced from the University of Auckland folded tandem accelerator. Natural chlorine targets of AgCl, PbCl_2 and solid Cl (75.77% ^{35}Cl , 24.33% ^{37}Cl). A 25 cm^3 Ge(Li) detector for detecting γ -rays, FWHM=2.3 keV for $E_\gamma=1.332$ MeV. Measured E_γ . Deduced levels, half-lives using DSAM. **1972Br45** reports a few mixing ratios in their TABLE V, but it is unclear where they are taken from. They are definitely not from **1972Br45** since this work is only focused on lifetime measurements.

1972Va06: E=5.47-5.61 MeV protons from the 5.5 MV Van de Graaff accelerator of the Southern Universities Nuclear Institute. Targets of BaCl_2 (99% ^{35}Cl) on gold backings. A 57 cm^3 Ge(Li) detector. Measured E_γ . Deduced half-lives using DSAM.

1961St09: E=2.3-3.25 MeV. NaI(Tl) detector. Targets of AgCl. Measured $\sigma(E_p, \theta)$, $\gamma(\theta)$. Deduced levels, J for the levels of 1.22 and 1.76 MeV.

1966Er06: E=0.733-2.595 MeV protons produced from the Utrecht 3 MV Van de Graaff accelerator. BaCl_2 target (99.5% ^{35}Cl) on a tantalum backing. Cylindrical NaI assemblies, 10-cm by 10-cm. Measured E_γ , $\gamma(\theta)$. Deduced levels, J^π , mixing ratios, mainly for ^{36}Ar and for the ^{35}Cl level at $E_x=1760$.

1970Ta09: E=5.11 MeV proton beam. NaI(Tl) detector. Measured E_γ . Developed an analytical method to interpret the linear polarization measurement of two unresolved γ -rays: 3.00 and 3.16 MeV. Same authors as **1970Ta02**.

1971Ca40: E=4.15-6.15 MeV protons from the Oxford University Tandem accelerator. Targets of 20 $\mu\text{g}/\text{cm}^2$ natural BaCl_2 on gold, lead, and carbon backings. A 32 cm^3 Ge(Li) detector. Measured E_γ . Deduced half-lives for the levels of 2646 and 2695 keV using DSAM.

1974Hu09: E=5.65 MeV protons from the pulsed beam facility of the University of Alberta Van de Graaff accelerator. Targets of natural KCl on tantalum backing and PbCl_2 on gold backing. A 24 cm^3 coaxial Ge(Li) detector. Measured E_γ . Deduced half-life for the level of 3162 keV.

Other (p,p' γ) measurement: **1985Ki07** and **1975De16** (measured 1219 γ yield).

(p,p') measurements:

1969Ta13,1971Ta02: E=3.115 and 3.345 MeV protons produced from the Universite de Montreal Model EN tandem accelerator. AgCl target (99% ^{35}Cl) of 16 $\mu\text{g}/\text{cm}^2$ on gold backing. A 12.70-cm-diam by 15.24-cm-long NaI(Tl) for detecting γ -rays. Measured E_γ , I_γ , $\gamma(\theta)$. Deduced levels, J^π and mixing ratio for the level of 1763 keV. Same authors as **1970Ta02**.

1969An35: E=6.03 MeV proton beams on a CdCl_2 target. Inelastic protons detected by a Si(Li) detector. Measured $\sigma(E_p, \theta)$. Deduced J for a level at 3930.

Other (p,p') measurement: **1993Bo40** (measured σ).

 ^{35}Cl Levels

E(level) [†]	J^π [‡]	$T_{1/2}$	Comments
0	$3/2^+$		
1219.42 10	$1/2^+$	0.150 ps +42–28	J^π : $\text{py}(\theta)$ in 1969Du08 is consistent with spin=1/2. $T_{1/2}$: from $\tau=0.216$ ps +60–40, weighted average of 0.215 ps 75 (1972Va06), 0.21 ps +6–4 (1972Br45), 0.27 ps +19–10 (1969Du08).
1763.26 10	$5/2^+$	0.43 ps 20	J^π : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1966Er06 ; spin=5/2 also from $\text{py}(\theta)$ in 1969Du08 . $T_{1/2}$: from $\tau=0.62$ ps 28, unweighted average of 0.205 ps 75 (1972Va06), 0.49 ps +9–6 (1972Br45), 1.15 ps +45–63 (1969Du08).
2645.74 17	$7/2^+$	0.165 ps 35	J^π : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02 ; spin $\geq 5/2$ from $\text{py}(\theta)$ in 1969Du08 . $T_{1/2}$: from $\tau=0.238$ ps 50, weighted average of 0.185 ps 50 (1972Va06) and 0.26 ps 6 (1972Br45), and 0.35 ps +10–7 (1969Du08 , using 2646 γ). Other: $\tau=0.29$ ps +37–13

Continued on next page (footnotes at end of table)

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45 (continued)** ^{35}Cl Levels (continued)

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}$	Comments
2694.50 28	$3/2^+$	43 fs 11	using 882 γ also reported in 1969Du08, but is considered less reliable by authors; 0.083 ps 21 from 1971Ca40 is considered an outlier. J^π : $3/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02. $T_{1/2}$: from $\tau=63$ fs 16, weighted average of 65 fs 20 (1972Va06) and 62 fs 16 (1969Du08, using 2695 γ). Other: $\tau=115$ fs +95–59 using 931 γ also reported in 1969Du08, but is considered less reliable by authors; <120 fs (1972Br45); 11 fs 7 from 1971Ca40 is considered an outlier.
3002.6 5	$5/2^+$	19 fs 9	J^π : $5/2^+$ from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02. $T_{1/2}$: from $\tau=27$ fs 13, weighted average of 18 fs +26–15 (1972Va06) and 31 fs 13 (1969Du08). Other: <80 fs (1972Br45).
3162.75 10	$7/2^-$	32 ps 4	J^π : spin= $7/2$ from $\gamma(\theta)$ and $\pi=-$ from $\gamma(\text{lin pol})$ in 1970Ta02. $T_{1/2}$: from $\tau=46$ ps 6 in 1974Hu09. Other: $\tau>17$ ps (1972Va06), >10 ps (1972Br45).
3930 30	$3/2$		$E(\text{level}), J^\pi$: from 1969An35. Spin is determined based on J-dependence of measured proton inelastic scattering cross sections and known assignments of 3163 and 3000 levels.

† From a least-squares fit to γ -ray energies with uncertainties, unless otherwise noted.

‡ From Adopted Levels. Supporting arguments from this dataset are given under comments if available.

 $\gamma(^{35}\text{Cl})$

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	Comments
1219.42	$1/2^+$	1219.4 1	100	0	$3/2^+$			E_γ : weighted average of 1219.4 1 (1972Br45) and 1219.3 3 (1969Du08). $\Gamma_\gamma=27.43\times 10^{-4}$ eV (1969Du08). $A_2=-0.025$ 29 for $E_p=3.115$ MeV; +0.007 8 for $E_p=3.345$ MeV (1969Ta13). $A_2=+0.031$ 16, $A_4=+0.014$ 20; $A_2=-0.012$ 22, $A_4=-0.003$ 30; $A_2=-0.009$ 14, $A_4=+0.014$ 6, for different beam energies (1961St09). $A_2=+0.02$ 6, $A_4=+0.05$ 9 (1969Du08). E_γ : weighted average of 1763.2 1 (1972Br45), 1763.3 3 (1969Du08). δ : weighted average of -3.0 5 from $\gamma(\theta, \text{pol})$ in 1966Er06 and -2.64 12 from $\gamma(\theta, \text{pol})$ in 1969Ta13 and 1971Ta02. Others: $\delta<-0.4$ or $\delta>+0.4$ from $\text{py}(\theta)$ in 1969Du08. $A_2=-0.369$ 9, $A_4=+0.213$ 13; $A_2=-0.355$ 14, $A_4=+0.200$ 20, for different beam energies (1969Ta13). $A_2=-0.02$ 2, $A_4=+0.07$ 2 (1969Du08). $A_2=-0.141$ 13, $A_4=+0.107$ 16; $A_2=-0.225$ 29, $A_4=+0.083$ 35; $A_2=-0.049$ 12, $A_4=-0.036$ 14, for different beam energies (1961St09).
1763.26	$5/2^+$	1763.2 1	100	0	$3/2^+$	M1+E2	-2.66 12	I_γ : unweighted average of 8.7 22 from 1970Ta02 and 17.6 35 from 1969Du08. %Branching=8 2 (1970Ta02), 15 3 (1969Du08). $A_2=-0.372$ 34, $A_4=+0.034$ 38; $A_2=-0.533$ 31, $A_4=+0.041$ 34, for different beam energies (1970Ta02). %Branching<1 (1970Ta02). E_γ : weighted average of 2645.7 2 (1972Br45) and 2645.6 5 (1969Du08).
2645.74	$7/2^+$	882.3 [@] 3	13 5	1763.26	$5/2^+$	M1+E2	+0.17 5	
		1426 ^{&a} 2645.7 2	<1 ^{&} 100 2	1219.42	$1/2^+$ $3/2^+$	E2		

Continued on next page (footnotes at end of table)

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45 (continued)** $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
								δ : $\delta(\text{O}/\text{Q})=0$ from $\gamma(\theta, \text{pol})$ in 1970Ta02 and $\text{p}\gamma(\theta)$ in 1969Du08. I_γ : from 1970Ta02. Other: 100 4 from 1969Du08. %Branching=92 2 (1970Ta02), 85 3 (1969Du08). $A_2=+0.22$ 5, $A_4=-0.13$ 6 (1969Du08). $A_2=+0.316$ 22, $A_4=+0.024$ 28; $A_2=+0.233$ 11, $A_4=+0.044$ 10; $A_2=+0.312$ 32, $A_4=-0.117$ 41; $A_2=+0.205$ 18, $A_4=-0.044$ 25, for different beam energies (1970Ta02).
2694.50	3/2 ⁺	931.3@ 3	18.6 25	1763.26	5/2 ⁺	M1+E2	+0.09 3	I_γ : weighted average of 19.0 25 (1970Ta02) and 17.7 38 (1969Du08). %Branching=15 2 (1970Ta02), 14 3 (1969Du08). $A_2=+0.018$ 46 (1970Ta02).
		1475@ 2	8.3 25	1219.42	1/2 ⁺	M1+E2	-0.63 21	I_γ : weighted average of 8.0 25 (1970Ta02) and 8.9 38 (1969Du08). %Branching=6 2 (1970Ta02), 7 3 (1969Du08). $A_2=-0.111$ 45 (1970Ta02).
		2694.1 6	100 3	0	3/2 ⁺	M1+E2	+0.26 2	E_γ : unweighted average of 2693.5 1 (1972Br45), 2694.6 5 (1969Du08). I_γ : from 1970Ta02. Other: 100 4 from 1969Du08. %Branching=79 2 (1970Ta02), 79 3 (1969Du08).
3002.6	5/2 ⁺	1239& 1	2.2& 11	1763.26	5/2 ⁺			I_γ : from 1970Ta02. %Branching=2 1 (1970Ta02).
		1783	3.3 11	1219.42	1/2 ⁺			E_γ : from level-energy difference. I_γ : weighted average of 5.4 32 (1970Ta02) and 3.1 11 (1969Du08). %Branching=5 3 (1970Ta02), 3 1 (1969Du08).
		3002.5 5	100 1	0	3/2 ⁺	M1+E2	+0.07 2	E_γ : weighted average of 3002.5 5 (1972Br45) and 3002.4 10 (1969Du08). I_γ : from 1969Du08. Other: 100 2 from 1970Ta02. δ : others: $-20 < \delta < -0.3$ or $+0.2 < \delta < +5$ (1969Du08, $\text{p}\gamma(\theta)$); $-0.02 < \delta < 0.09$ (1970Ta09, $\gamma(\text{lin pol})$). %Branching=93 2 (1970Ta02), 97 1 (1969Du08). $A_2=-0.30$ 4 (1969Du08). $A_2=-0.166$ 7, $A_4=+0.004$ 10; $A_2=-0.104$ 23, $A_4=+0.050$ 40; $A_2=-0.329$ 13, $A_4=-0.001$ 28; $A_2=+0.342$ 24, $A_4=+0.022$ 43, for different beam energies (1970Ta02).
3162.75	7/2 ⁻	517.0@ 8	19 8	2645.74	7/2 ⁺			I_γ : unweighted average of 11.1 22 (1970Ta02) and 26.6 63 (1969Du08). %Branching=10 2 (1970Ta02), 21 5 (1969Du08).
		1399&a	<1&	1763.26	5/2 ⁺			%Branching<1 (1970Ta02).
		1943&a	<1&	1219.42	1/2 ⁺			%Branching<1 (1970Ta02).

Continued on next page (footnotes at end of table)

$^{35}\text{Cl}(\text{p},\text{p}'\gamma)$ **1969Du08,1970Ta02,1972Br45** (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\&$	Comments
3162.75	3162.6 1	100 2	0	3/2 ⁺	M2+E3	-0.22 5	E_γ: weighed average of 3162.6 1 (1972Br45) and 3162.5 10 (1969Du08). I_γ: from 1970Ta02. Other: 100 6 from 1969Du08. δ: weighted average of -0.14 +8-6 (1969Du08, $\text{p}\gamma(\theta)$) and -0.25 4 (1970Ta02, $\gamma(\theta,\text{pol})$). Other: -0.31<$\delta$<-0.21 from $\gamma(\text{lin pol})$ in 1970Ta09. %Branching=90 2 (1970Ta02), 79 5 (1969Du08). $A_2=+0.48$ 21, $A_4=-0.13$ 73 (1969Du08). $A_2=+0.281$ 68, $A_4=+0.241$ 82; $A_2=+0.453$ 44, $A_4=+0.136$ 63; $A_2=+0.475$ 24, $A_4=+0.096$ 36, for different beam energies (1970Ta02).

[†] E_γ values with uncertainties for the ground-state transitions are not explicitly listed in 1972Br45 and 1969Du08 and the evaluators have taken their listed $E(\text{level})$ values and uncertainties (which are deduced from E_γ values by authors) as E_γ values of the corresponding ground-state transitions, considering ground-state transitions are dominant in each level.

[‡] Relative photon branching ratios deduced by the evaluators based on percentage branching ratios reported in 1970Ta02 and 1969Du08, which are given under comments where available.

[#] From $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in 1970Ta02, unless otherwise noted.

[@] From 1969Du08.

[&] γ reported in 1970Ta02. Energy is from level-energy difference and intensity is from 1970Ta02. The ones with I_γ reported as upper limit are considered questionable by the evaluators.

^a Placement of transition in the level scheme is uncertain.

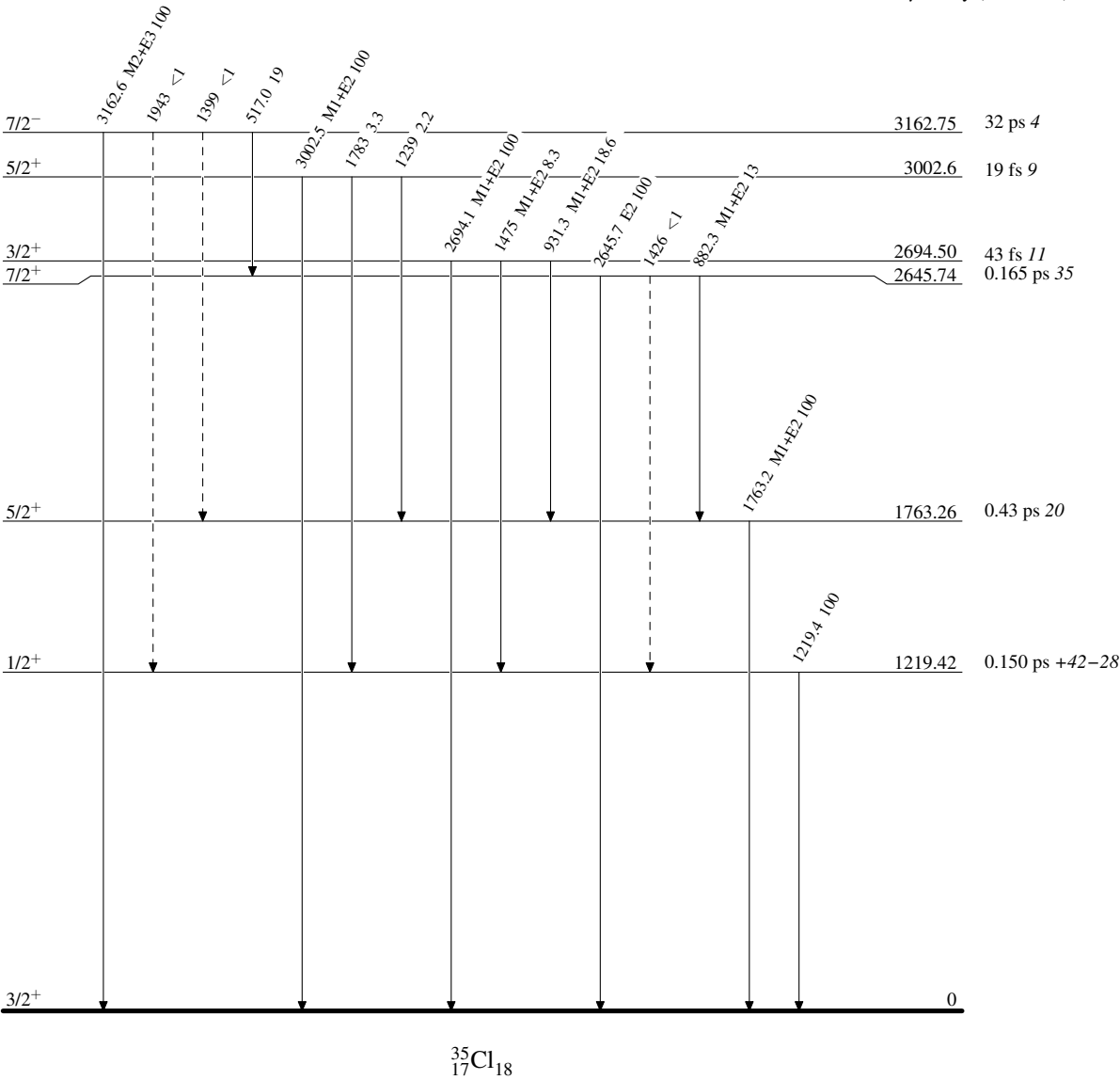
³⁵Cl(p,p'γ) 1969Du08,1970Ta02,1972Br45

Legend

Level Scheme

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)



$^{37}\text{Cl}(\text{p,t})$ 1971Vi02,2017Ch22

$^{37}\text{Cl} \rightarrow ^{35}\text{Cl} + 2\text{n}$ from $J^\pi = 3/2^+$ ^{37}Cl ground state.

1971Vi02: A 40-MeV proton beam was produced from the Grenoble variable energy cyclotron at intensities of 20-200 nA depending on the scattering angle with a 90-keV energy resolution. Targets were natural purified chlorine gas 100 mm in diameter and 25 mm in height and solid AgCl of isotopic ^{37}Cl . The reaction particles were detected using two separate counter telescopes consisting of a 200- μm phosphorous-drifted silicon ΔE detector, a 3-mm lithium-drifted silicon E detector, and a 3-mm lithium-drifted silicon E-reject detector with FWHM=130 keV for tritons. Measured $\sigma(E_t, \theta)$. Deduced L-transfers for 7 levels from zero-range Nelson spin-dependent-DWBA analysis of the measured $\sigma(\theta)$.

2017Ch22: a 30-MeV proton beam was produced from the Holifield Radioactive Ion Beam Facility at Oak Ridge National Laboratory. The target was $\sim 600 \mu\text{g}$ PbCl_2 ($\sim 95\%$ enriched and $\sim 160 \mu\text{g}/\text{cm}^2$ in ^{37}Cl) on a $20 \mu\text{g}/\text{cm}^2$ parylene (C_8H_8) backing. Tritons were detected using the annular silicon detector array (SIDAR) consisting of 100- μm - ΔE and 1000- μm -E telescopes with FWHM=60-100 keV. Measured $\sigma(E_t, \theta)$. Deduced levels, J, π , and L-transfers from TWOFNR-DWBA analysis of the measured $\sigma(\theta)$. Provided the first spin-parity constraint of a 306-keV $^{34}\text{S}(\text{p}, \gamma)$ resonance potentially important for nova nucleosynthesis.

Other: **1985Mi06**.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$L^\ddagger$	Comments
0	0	L: from 2017Ch22 . Other: 0+2 in 1971Vi02 . Relative peak intensity=100 (1971Vi02).
1220 40		Relative peak intensity=0 (1971Vi02). 
1750 40	2	Relative peak intensity=41 (1971Vi02).
2650 40	2	Relative peak intensity=41 (1971Vi02). Unresolved 2645 and 2694 doublet.
3000 40		Relative peak intensity=5 (1971Vi02).
5650 50	0+2	Relative peak intensity=22 (1971Vi02).
6677 15	(2)	E(level): from 2017Ch22 . 2017Ch22 adopts L=2 and states that combinations of L=0+2 or L=1+3 are also possible. Because L=2 and L=0 are better fits than L=1 and L=3, a positive parity assignment for the 6677 level is strong.
7250 50		Relative peak intensity=9 (1971Vi02).

† From **1971Vi02**, unless otherwise noted.

‡ From DWBA analysis of the measured $\sigma(\theta)$ in **1971Vi02**, unless otherwise noted.

$^{36}\text{Ar}(\text{n},\text{d})$ 1977Pa29

$J^\pi=0^+$ for ^{36}Ar ground state.

1977Pa29: 14.1-MeV neutrons were produced via the $^3\text{H}(\text{d},\text{n})^4\text{He}$ reaction at an average rate of 10^9 neutrons per second into 4π sr from the Brown University neutron generator and impinged on a 2.85-mg/cm^2 ^{36}Ar target in the form of a gas cell filled with 99.8% enriched ^{36}Ar . Deuterons were detected using a time-of-flight counter telescope consisting of two proportional counters and a NaI(Tl) crystal. Measured $\sigma(E_{\text{d}},\theta)$. Deduced levels, L-transfers, spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}Cl Levels

<u>E(level)[†]</u>	<u>L[‡]</u>	<u>Comments</u>
0	2	S: 3.83 48 and 3.90 47 using two sets of optical parameters.
1219 [#]	0 [#]	S: 1.94 23 and 3.30 40 using two sets of optical parameters. Other spectroscopic factors calculated using two sets of optical parameters did not show large differences.
1763 [#]	(2) [#]	

[†] From 1977Pa29.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1977Pa29.

[#] 1977Pa29 state that the first and second excited states are recorded together and not resolved, but their angular distributions could be separated due to different L-transfers, however statistical errors prevent an accurate determination of the smaller L=2 component from the second excited state.

$^{36}\text{Ar}(\text{pol d}, ^3\text{He})$ 1993Ma50

$J^\pi=0^+$ for ^{36}Ar ground state.

1993Ma50: a 52-MeV polarized deuteron beam at 15 nA was produced from the Karlsruhe isochronous cyclotron at the Max-Planck-Institut für Kernphysik. The target was 350-mbar 99.86% enriched ^{36}Ar gas. Reaction products were detected using a Si detector telescope consisting of a 250- μm ΔE -strip detector and a 1.5-mm surface-barrier E detector with FWHM=130 keV. Measured angular distributions of cross sections $\sigma(E(^3\text{He}), \theta)$ and angular distributions of analyzing powers $iT_{11}(E(^3\text{He}), \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from DWBA analysis of $\sigma(E(^3\text{He}), \theta)$ and $iT_{11}(E(^3\text{He}), \theta)$.

1993Ma50 states that spectroscopic factors were obtained by fitting the forward-angle maxima. The agreement with the analyzing powers is only qualitative. The determination of spins and parities of final states is mainly based on a comparison with measured angular distributions leading to final states with known spins. The comparison with DWBA predictions serves as additional check. The data were taken only in a relatively small angle interval. This is sufficient for the spin determination but not favorable for the measurement of spectroscopic factors.

 ^{35}Cl Levels

Spectroscopic factor $C^2S = \sigma(\theta)_{\text{exp}} / \sigma(\theta)_{\text{DWBA}} / N$, where $N=2.95$ is a normalization factor used by 1993Ma50 from 1966Ba54.

E(level) [†]	J^π [‡]	L [#]	C^2S [#]	Comments
0	$3/2^+$ @	2	1.72 @	
1213 4	$1/2^+$	0	0.78	
1785 50	$(5/2^+)$ &		0.04 ^a	
2676 7	$(3/2^+)$ &		0.16 ^a	
3002 6	$5/2^+$	2	1.32	
3159 20	$7/2^-$	3	0.19	
3948 40	$(3/2^+, 1/2^+)$ &	2	0.03, 0.08	E(level): possible doublet of 3918 and 3968 (1990En08) suggested by Table 2 of 1993Ma50.
4839 12	$(5/2^+, 1/2^+)$ &		0.08, 0.03 ^a	
5155 11	$(5/2^+)$	2	0.12	J^π : $(7/2^-)$ and $5/2^+$ are given in Table 2 of 1993Ma50. Evaluators adopt $(5/2^+)$, based on DWBA fit of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in Fig. 3 of 1993Ma50.
5583 13	$5/2^+$	2	0.38	
5720 11	$5/2^+$	2	1.10	
6136 8	$5/2^+$	2	0.63	
6745 12	$(1/2^+, 5/2^+)$ &	(0, 2)	0.09, 0.28	
6953 43	$5/2^+$	2	0.49	
7181 60	$5/2^+$	2	0.20	
7565 20	$5/2^+$	2	0.30	
8007 36	$(1/2^-, 5/2^+)$ &	(1, 2)	0.16, 0.20	
8204 34	$5/2^+$	2	0.14	
8580 20	$5/2^+$ &		0.22 ^a	
8921 70	$5/2^+$ &		0.18 ^a	
9.22×10^3 12	$5/2^+$ &		0.12 ^a	
9514 20	$(1/2^-, 5/2^+)$ &	(1, 2)	0.18, 0.30	
9.82×10^3 12	$(1/2^-, 5/2^+)$ &	(1, 2)	0.10, 0.26	
1.25×10^4 5				$1d_{5/2}$ is given for a range of 12.0-13.0 MeV in Fig. 4 of 1993Ma50.

[†] From 1993Ma50.

[‡] L+1/2 transfer from analyzing power measurements with $1d_{5/2}$ proton transfer assumed in DWBA calculations, unless otherwise noted.

[#] From DWBA analysis of the measured $\sigma(\theta)$ and $iT_{11}(\theta)$ in 1993Ma50. C^2S value is extracted for corresponding J^π assignment. Where two C^2S values are given, they correspond to respective J^π values.

Continued on next page (footnotes at end of table)

 $^{36}\text{Ar}(\text{pol d}, ^3\text{He})$ [1993Ma50](#) (continued)

 ^{35}Cl Levels (continued)

@ L-1/2 transfer from analyzing power measurements with $1d_{3/2}$ proton transfer assumed in DWBA calculations.

& Assumed by [1993Ma50](#) for extracting spectroscopic factors.

^a No data on $\sigma(E(^3\text{He}), \theta)$, $iT_{11}(E(^3\text{He}), \theta)$, or L-transfer is given.

$^{36}\text{Ar}(\text{d}, ^3\text{He})$ 1974Do12

$J^\pi=0^+$ for ^{36}Ar ground state.

1974Do12: a 52-MeV deuteron beam was produced from the Karlsruhe isochronous cyclotron at the Max-Planck-Institut für Kernphysik. The target was 200-Torr 99.9% enriched ^{36}Ar gas. ^3He particles were detected using detector telescopes consisting of a 200- μm ΔE and a 2-mm E-surface-barrier counter with FWHM=125 keV. Measured $\sigma(E(^3\text{He}), \theta)$. Deduced levels, J, π , L-transfer, and spectroscopic factors from DWUCK-DWBA analysis.

Other: 1975Wa17 (theory discussions).

 ^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=2.95$ is a normalization factor used by 1974Do12 from 1966Ba54.

<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0	3/2 ⁺	2	2.2	3170 20	7/2 ⁻	3	0.22	6750 20	(5/2) ⁺	2	0.79
1220 20	1/2 ⁺	0	1.34	5150 20	(5/2) ⁺	2	0.09	6930 20	(5/2) ⁺	2	0.46
1763			<0.02	5600 20	(5/2) ⁺	2	0.47 19	8010 20	(5/2) ⁺	2	0.23
2700 20	3/2 ⁺	2	0.39	5720 20	(5/2) ⁺	2	0.73 29	8210 20	(5/2) ⁺	2	0.28
3000 20	5/2 ⁺	2	1.49	6140 20	(5/2) ⁺	2	0.72	8610 20	(5/2) ⁺	2	0.34

[†] From 1974Do12.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1974Do12.

[#] As given in 1974Do12 for levels below 5 MeV. For levels above 5 MeV, 1974Do12 states that they tend to interpret them as 5/2⁺ states because the summed low-lying $C^2S(1d_{3/2})$ already exceeds the simple shell model value for ^{36}Ar and the measured $\sigma(\theta)$ resembles more closely that of the 5/2⁺ state at 3.0 MeV than that of the 3/2⁺ ground state. All C^2S values are given according to the J values.

$^{39}\text{K}(\gamma, \alpha)$ 1987Ar08

1987Ar08: $E_{\text{max}}=32$ -MeV γ -ray beam was produced from the betatron at Moscow University Nuclear Physics Research Institute and impinged on a natural metallic potassium target of mass ≈ 200 g. Photons from (γ, p) , (γ, n) , and (γ, α) from the giant dipole resonance (GDR) in ^{39}K were detected using a 100-cm³ Ge(Li) detector at 140°. Measured E_γ . Deduced partial photonuclear cross sections, levels, J, π .

 ^{35}Cl Levels

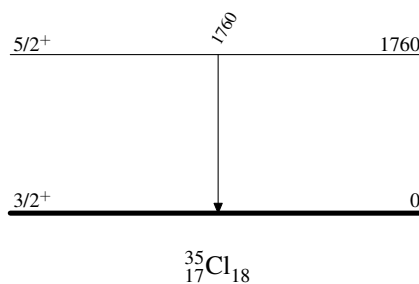
<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	Comments
0	$3/2^+$	
1760	$5/2^+$	T=1/2

[†] From 1987Ar08.

[‡] From Adopted Levels.

 $\gamma(^{35}\text{Cl})$

<u>E_γ</u>	<u>$E_i(\text{level})$</u>	<u>J^π_i</u>	<u>E_f</u>	<u>J^π_f</u>
1760	1760	$5/2^+$	0	$3/2^+$

 $^{39}\text{K}(\gamma, \alpha)$ 1987Ar08Level Scheme

$^{40}\text{Ca}(\mu^-, \nu\alpha n\gamma)$ 2006Me08

2006Me08: Muons were generated from the decay of 90-MeV/c π^- beam in a 6-m, 1.2-T superconducting solenoid and were selected by a bending magnet at the M9B beamline at TRIUMF. The target was 15-g natural calcium. γ rays and x rays were detected using a 186-cm³ HPGe detector. Charged particles caused by electrons from muon decay were tagged by a plastic scintillator in front of the HPGe detector. Measured E_γ , I_γ , $E(\text{x ray})$, $I(\text{x ray})$. Deduced levels, muon capture yields.

Muonic Lyman series for natural Calcium

μ x ray	Energy	Intensity in percent
2p-1s	783.659 25	83.8 10
3p-1s	940.63 10	6.2 2
4p-1s	995.48 10	2.0 1
5p-1s	1020.81 10	2.0 1
6p-1s	1034.62 10	1.8 1
7p-1s	1042.71 20	1.4 1
(8 $^-\infty$)p-1s	1046-1063	2.8 4

Muonic Balmer series for natural Calcium

μ x ray	Energy	Intensity in percent
3d-2p	157.35 13	64.5 9
4d-2p	212.03 10	8.85 20
5d-2p	237.31 10	4.34 20
6d-2p	251.06 10	3.29 20
7d-2p	259.45 10	1.37 20
(8 $^-\infty$)d-2p	261-277	1.4 3

 ^{35}Cl Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>
0	3/2 ⁺
1219.4	1/2 ⁺
1763.2	5/2 ⁺
2645.7	7/2 ⁺
2694.0	3/2 ⁺

[†] From Adopted Levels. Level energies are rounded-off values.

 $\gamma(^{35}\text{Cl})$

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>
1219.4	<0.3	1219.4	1/2 ⁺	0	3/2 ⁺
1763.2	0.5 3	1763.2	5/2 ⁺	0	3/2 ⁺
2645.7	<0.2	2645.7	7/2 ⁺	0	3/2 ⁺
2693.7	<0.2	2694.0	3/2 ⁺	0	3/2 ⁺

[†] Rounded-off values from Adopted Gammas.

[‡] Percent γ -ray yield per muon capture (2006Me08).

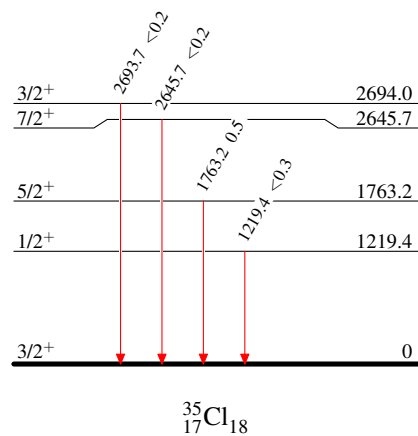
$^{40}\text{Ca}(\mu^-, \nu\alpha n\gamma)$ 2006Me08

Level Scheme

Intensities: percent γ -ray yield per muon capture

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
- $I_\gamma < 10\% \times I_\gamma^{\max}$
- $I_\gamma > 10\% \times I_\gamma^{\max}$



Coulomb excitation **1969Ha17,1977Sc36**

1969Ha17: 51-61-MeV $^{35}\text{Cl}^{5,6+}$ beams of 80 nA were produced from the Chalk River MP tandem accelerator and bombarded 9.6-mg/cm² carbon and 5-mg/cm² ^{24}Mg . Deexcitation γ rays were detected using two 40-cm³ Ge(Li) detectors at 125° and at 0° or 90°. Measured E_γ , I_γ , $\gamma(\theta)$ anisotropy, γ -ray yields. Deduced levels, mixing ratio, transition strengths, $T_{1/2}$ for 2 levels using the Doppler Shift Attenuation Method (DSAM).

1977Sc36: 53-56-MeV $^{35}\text{Cl}^{6+}$ beams of 60 nA were produced from a single MP tandem Van de Graaff accelerator at the three-stage facility of Brookhaven National Laboratory. Targets were ^{23}Na and ^{27}Al . Deexcitation γ rays were detected using a 55-cm³ coaxial Ge(Li) detector at 0°. Measured E_γ , $I_\gamma(\theta)$, γ -ray yields. Deduced levels, mixing ratios, transition strengths, $T_{1/2}$ for the levels of 1220, 1762 and 2650 keV using the Doppler Shift Attenuation Method (DSAM).

 ^{35}Cl Levels

Values of $B(E2)\uparrow$ from **1969Ha17** and **1977Sc36** are from absolute cross-section measurements.

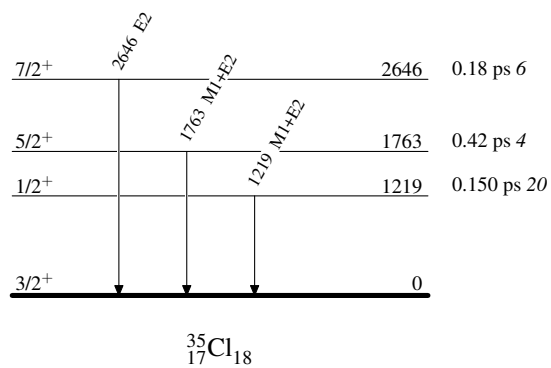
$E(\text{level})^\dagger$	$J^\pi^\dagger$	$T_{1/2}$	Comments
0	$3/2^+$		
1219	$1/2^+$	0.150 ps 20	$B(E2)\uparrow=0.00076\ 7$ $T_{1/2}$: Lifetime $\tau=0.216$ ps 28 from weighted average of $\tau=0.16$ ps 5 (1969Ha17 , DSAM) and $\tau=0.230$ ps 25 (1977Sc36 , DSAM). $B(E2)\uparrow$: weighted average of 0.00081 9 (1969Ha17) and 0.00073 7 (1977Sc36). $B(E2)\uparrow=0.0106\ 7$
1763	$5/2^+$	0.42 ps 4	$T_{1/2}$: Lifetime $\tau=0.61$ ps 5 from weighted average of $\tau=0.57$ ps 7 (1969Ha17 , DSAM) and $\tau=0.63$ ps 5 (1977Sc36 , DSAM). $B(E2)\uparrow$: weighted average of 0.0117 9 (1969Ha17) and 0.0102 5 (1977Sc36). $B(E2)\uparrow=0.0029\ 7$ (1977Sc36)
2646	$7/2^+$	0.18 ps 6	$T_{1/2}$: Lifetime $\tau=0.26$ ps 9 (1977Sc36 , DSAM).

† From the Adopted Levels. Energies are rounded values.

 $\gamma(^{35}\text{Cl})$

$E_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. †	δ	Comments
1219	1219	$1/2^+$	0	$3/2^+$	M1+E2	0.093 10	δ : deduced by the evaluators from adopted $B(E2)\uparrow=0.00076\ 7$ in this dataset and $T_{1/2}=0.119$ ps 14 from Adopted Levels. Other: 0.106 7 deduced by 1977Sc36 based on their measured $B(E2)\uparrow$ and lifetime.
1763	1763	$5/2^+$	0	$3/2^+$	M1+E2	-2.95 45	δ : from $\gamma(\theta)$ anisotropy in 1969Ha17 . Other: $1.8 +10-4$; deduced by the evaluators from adopted $B(E2)\uparrow=0.0106\ 7$ in this dataset and $T_{1/2}=0.36$ ps 3 from Adopted Levels; -2.7 7 from 1977Sc36 deduced from their measured $B(E2)\uparrow$ and lifetime.
2646	2646	$7/2^+$	0	$3/2^+$	E2		

† From Adopted Gammas. Energies are rounded values.

Coulomb excitation 1969Ha17,1977Sc36Level Scheme

Adopted Levels, Gammas

$Q(\beta^-) = -11874.4$ 9; $S(n) = 12740.3$ 7; $S(p) = 5896.2$ 7; $Q(\alpha) = -6429.7$ 7 [2021Wa16](#)
 $Q(\beta^-n) = -29632$ 17, from mass excesses of -1487 17 for ^{34}K measured by [2024Dr01](#); -23047.3 7 for ^{35}Ar from [2021Wa16](#). Value from [2021Wa16](#): $Q(\beta^-n) = -29900$ 200 (syst).
 $S(2n) = 29805.6$ 8, $S(2p) = 11039.4$ 7, $Q(\epsilon) = 5966.2$ 7 ([2021Wa16](#)).
 Isotope discovery ([2012Th10](#)): $^{32}\text{S}(\alpha, n)^{35}\text{Ar}$ at Purdue ([1940Ki12](#), [1941Ki01](#), [1941El04](#)).
 ^{35}Ar production:
[2012Zh06](#): ^9Be , $^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion-ablation, and HIPSE models. Studied target dependence of fragment cross sections.
[2007No13](#): $^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.
 ^{35}Ar decay measurements:
 ^{35}Ar produced via $^{35}\text{Cl}(p, n)^{35}\text{Ar}$ reaction ([1985Da04](#), [1984Ad01](#), [1980Wi13](#), [1979Ha04](#), [1977Az01](#), [1971Ge04](#), [1971De05](#), [1969Wi18](#), [1969Fr09](#), [1960Wa04](#), [1956Ki29](#)). Measured $\epsilon + \beta^+$ -delayed γ rays, decay branching ratios, and parent ^{35}Ar $T_{1/2}$.
 ^{35}Ar radius measurements:
[2002Oz03](#): $\text{C}(^{35}\text{Ar}, X)$ at $E(^{35}\text{Ar}) \approx 950$ MeV/nucleon at RIKEN. Measured interaction cross sections. Deduced effective radii and proton skin features.
[2000Ge20](#): ^{35}Ar produced at ISOLDE. Measured β asymmetry and hyperfine structure using β -NMR spectroscopy. Deduced mean squared charge radii and quadrupole moments.
[1996Ki04](#), [1995KiZZ](#): ^{35}Ar produced by ISOLDE. Measured isotope shifts and hyperfine structure using collinear fast-beam laser spectroscopy. Deduced mean square charge radii and electric quadrupole moments.
 ^{35}Ar mass measurement: [2011Tu09](#).
 Theoretical calculations: [2020Ri06](#), [2020RiZX](#), [2020RiZZ](#).

 ^{35}Ar LevelsCross Reference (XREF) Flags

A	^{35}K $\epsilon + \beta^+$ decay (175 ms)	E	$^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$	I	$^{36}\text{Ar}(p, d)$
B	^{36}Ca ϵp decay (100.9 ms)	F	$^{32}\text{S}(\alpha, n)$	J	$^{36}\text{Ar}(d, t)$
C	$^1\text{H}(^{36}\text{Ar}, d)$	G	$^{33}\text{S}(^3\text{He}, n \gamma)$	K	$^{36}\text{Ar}(^3\text{He}, \alpha)$
D	$^{16}\text{O}(^{24}\text{Mg}, \alpha n \gamma)$	H	$^{35}\text{Cl}(^3\text{He}, t)$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0.0	$3/2^+$	1.7755 s 14	ABCDEFGH IJK	$\% \epsilon + \% \beta^+ = 100$ $\mu = +0.6322$ 2 (2002Ma41 , 2019StZV) $Q = -0.084$ 15 (1996Ki04 , 2021StZZ) μ : β -NMR (2002Ma41). Others: $+0.632$ 2 (1965Ca04), $+0.633$ 7 (1996Ki04) using β -NMR. Q : β -NMR (1996Ki04). J^π : $L=2$ from 0^+ in (p, d) , (d, t) , $(^3\text{He}, \alpha)$, and $^1\text{H}(^{36}\text{Ar}, d)$. Allowed $\epsilon + \beta^+$ feedings to $1/2^+$ levels in ^{35}Cl . $T_{1/2}$: weighted average of 1.83 s 3 (1956Ki29), 1.83 s 2 (1959Al10), 1.79 s 1 (1960Ja12), 1.84 s 10 (1960Wa04), 1.76 s 3 (1963Ne05), 1.770 s 6 (1969Wi18), 1.787 s 12 (1971Ge04), 1.774 s 3 (1977Az01), and 1.7754 s 11 (2006Ia05). Evaluated rms nuclear charge radius $R = 3.3636$ fm 42 (2013An02). XREF: F(890) $E(\text{level})$: 1963Ne05 (α, n) observed the first excited state in ^{35}Ar at 890 50 keV.
1184.08 25	$1/2^+$		ABC FG IJK	J^π : $L=0$ from 0^+ in (p, d) , (d, t) , and $(^3\text{He}, \alpha)$. XREF: F(2030)I(1700)J(1700)
1750.78 22	$(5/2)^+$		A DEFG IJK	

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

E(level) [†]	J ^π	XREF	Comments
			E(level): 1963Ne05 (α, n) observed the second excited state in ^{35}Ar at 2030 80 keV.
			J ^π : L=2 from 0 ⁺ in ($^3\text{He}, \alpha$). Possible mirror level of 5/2 ⁺ at 1763 keV in ^{35}Cl .
2603.22 28	7/2 ⁽⁺⁾	DE G	J ^π : 2603.0γ Q, ΔJ=2 to 3/2 ⁺ .
2638.01 26	(3/2) ⁺	A IJK	XREF: I(2615)K(2649)
			J ^π : L=2 from 0 ⁺ in (p,d) and ($^3\text{He}, \alpha$) and spin=(3/2) from J-dependence in (p,d).
2982.79 12	(5/2) ⁺	A C IJK	XREF: I(2970)
			J ^π : L=2 from 0 ⁺ in (p,d), (d,t), and ($^3\text{He}, \alpha$) and spin=(5/2) from J-dependence in (p,d).
3196.98 [‡] 26	7/2 ⁻	CDE G IJK	J ^π : L=3 from 0 ⁺ in (p,d) and ($^3\text{He}, \alpha$); 3197.0γ Q, ΔJ=2 to 3/2 ⁺ .
3882 5	1/2 ⁺	K	J ^π : L=0 from 0 ⁺ in ($^3\text{He}, \alpha$).
4001 3	1/2 ⁻ , 3/2 ⁻	K	J ^π : L=1 from 0 ⁺ in ($^3\text{He}, \alpha$).
4065.0? 4	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	A	XREF: A(?)
			J ^π : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.6 +4-2.
4113 4		K	
4135 4	1/2 ⁻ , 3/2 ⁻	K	J ^π : L=1 from 0 ⁺ in ($^3\text{He}, \alpha$).
4359.0 5	(9/2 ⁻)	DE K	J ^π : 1162.0γ (D), ΔJ=(1) to 7/2 ⁻ ; mirror level to 4348, 9/2 ⁻ level in ^{35}Cl proposed by 2007De14 in $^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$.
4528.3 4	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	A K	XREF: K(4515)
			J ^π : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.4 +4-2.
4725.9 6	1/2 ⁺	A Hi K	XREF: i(4756)K(4713)
			J ^π : L=0 from 0 ⁺ in ($^3\text{He}, \alpha$). Other: L=0 from 0 ⁺ in (p,d) for a group at 4756 28.
4785.8 11	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A Hi K	XREF: i(4756)K(4774)
			J ^π : allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.2 2. Other: L=0 from 0 ⁺ in (p,d) for a group at 4756 28.
5059 11		i K	
5116 2	3/2 ⁺ , 5/2 ⁺	Hi K	E(level): from ($^3\text{He}, \alpha$). Other: 5102 20 from (p,d).
			J ^π : L=2 from 0 ⁺ in ($^3\text{He}, \alpha$) in 1973Be26 . Discrepancy: L=3 from 0 ⁺ in (p,d) (1968Jo04). 1973Be26 also considered L=3 but found L=2 gives a much better fit of the data than L=3 does.
5207 3		H K	E(level): from ($^3\text{He}, \alpha$).
5384.2 [‡] 4	(11/2 ⁻)	DE HI K	XREF: I(5400)
			J ^π : 2187.1γ Q, ΔJ=2 to 7/2 ⁻ ; band assignment.
5482 2	3/2 ⁺ , 5/2 ⁺	H K	E(level): from ($^3\text{He}, \alpha$).
			J ^π : L=2 from 0 ⁺ in ($^3\text{He}, \alpha$).
5572.67 15	3/2 ⁺	A G	T=3/2
			XREF: G(5537)
			J ^π : isobaric analog state of 3/2 ⁺ ^{35}K g.s. with log ft=3.31 4. L=(0) from 3/2 ⁺ in ($^3\text{He}, n$). $^{35}\text{Cl}(^3\text{He}, t)$ attempted to search this T=3/2 level, but did not find it.
5594 2	3/2 ⁺ , 5/2 ⁺	C HI K	XREF: C(5570)
			E(level): weighted average of 5598 20 from (p,d) and 5594 2 from ($^3\text{He}, \alpha$).
			J ^π : L=2 from 0 ⁺ in (p,d) and ($^3\text{He}, \alpha$).
			Evaluators consider the 5994 level to be different from the 5572.67 T=3/2 level because (p,d) and ($^3\text{He}, \alpha$) from T=0 targets should not populate T=3/2 levels.
5613.6 9	(11/2 ⁻)	E	J ^π : 1254.6γ to (9/2 ⁻); mirror level to 5407, 11/2 ⁻ level in ^{35}Cl proposed by 2007De14 in $^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$.
5765.8 5	(13/2 ⁻)	DE	J ^π : 381.6γ D, ΔJ=1 to (11/2 ⁻), 1406.9γ (Q), ΔJ=(2) to (9/2 ⁻).
5915 3		H JK	E(level): weighted average of 5913 5 from (d,t) and 5916 3 from ($^3\text{He}, \alpha$).

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

E(level) [†]	J ^π	XREF	Comments
5991 3		J	
6037 3	3/2 ⁺ , 5/2 ⁺	HIJK	XREF: I(6024) E(level): weighted average of 6037 3 from (d,t) and 6036 3 from ($^3\text{He},\alpha$). Other: 6024 20 from (p,d). J ^π : L=2 from 0 ⁺ in (p,d) and ($^3\text{He},\alpha$). XREF: J(?)
6055? 3		J	
6076 3		J	
6163 2		JK	E(level): weighted average of 6164 3 from (d,t) and 6162 2 from ($^3\text{He},\alpha$).
6253 3		Jk	XREF: k(6262)
6273 3		Jk	XREF: k(6262)
6302 3		J	
6332 3		J	
6345 3	(1/2, 3/2, 5/2)	A J	E(level): From (d,t). Other: 6348 11 from ^{35}K ε decay. J ^π : ε+β ⁺ feeding from 3/2 ⁺ parent with log ft=7.2 1.
6415 2		J	
6439? 4		J	XREF: J(?)
6460 3		J	
6523 3		J	
6557 3		J	
6585 3		J	
6606 3		iJ	XREF: i(6620)
6616 2	1/2 ⁺	iJK	XREF: i(6620) E(level): weighted average of 6617 2 from (d,t) and 6615 3 from ($^3\text{He},\alpha$). Other: 6620 30 from (p,d). J ^π : L=0 from 0 ⁺ in ($^3\text{He},\alpha$) and (p,d).
6644 3		iJ	XREF: i(6620)
6651 3		iJ	XREF: i(6620)
6673 4	5/2 ⁻ , 7/2 ⁻	IJ	XREF: I(6700) E(level): weighted average of 6700 20 from (p,d) and 6672 3 from (d,t). J ^π : L=3 from 0 ⁺ in (p,d).
6823 2	3/2 ⁺ , 5/2 ⁺	I K	E(level): from ($^3\text{He},\alpha$). Other: 6820 30 from (p,d). J ^π : L=2 from 0 ⁺ in (p,d).
6948 2		K	
7044 4	3/2 ⁺ , 5/2 ⁺	A I K	XREF: I(7030) E(level): weighted average of 7053 11 from ^{35}K ε decay, 7030 20 from (p,d), and 7043 4 from ($^3\text{He},\alpha$). J ^π : L=2 from 0 ⁺ in (p,d).
7117 10		K	
7255 11		A	
7289 10		A K	E(level): weighted average of 7283 11 from ^{35}K ε decay and 7293 10 from ($^3\text{He},\alpha$).
7427 10		A K	E(level): weighted average of 7431 11 from ^{35}K ε decay and 7423 10 from ($^3\text{He},\alpha$).
7509 10	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A K	E(level): weighted average of 7518 11 from ^{35}K ε decay and 7502 10 from ($^3\text{He},\alpha$). J ^π : allowed ε+β ⁺ feeding from 3/2 ⁺ parent with log ft<5.0.
7840 10		K	
8019 10		K	
8109.7 [±] 13	(15/2 ⁻)	E	J ^π : 2725.7γ to (11/2 ⁻); band assignment.
8212.6 8	(15/2 ⁻)	E	J ^π : 2828.3γ Q, ΔJ=2 to (11/2 ⁻) and 2446.6γ to (13/2 ⁻).
8393? 20	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A	XREF: A(?) E(level): From (^{35}K ε decay). J ^π : allowed ε+β ⁺ feeding from 3/2 ⁺ parent with log ft=4.6 +3-2.

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

E(level) [†]	J^π	XREF	Comments
9906.0 [‡] 20	(19/2 ⁻)	E	J^π : 1693.3 γ Q, $\Delta J=2$ to (15/2 ⁻); band assignment.
12277.0 [‡] 32	(23/2 ⁻)	E	J^π : 2370.9 γ Q, $\Delta J=2$ γ -to (19/2 ⁻); band assignment.

[†] E(level) from a least-squares fit to γ -ray energies for levels connected with γ transitions; from particle-transfer reactions or ^{35}K $\varepsilon+\beta^+$ -delayed proton decays for other levels.

[‡] Band(A): Band based on $f_{7/2}$ orbital.

 $\gamma(^{35}\text{Ar})$

$E_i(\text{level})$	J_i^π	E_γ [†]	I_γ [†]	E_f	J_f^π	Mult. [‡]	Comments
1184.08	1/2 ⁺	1184.1 3	100	0.0	3/2 ⁺		E_γ : weighted average of 1184.0 3 from ^{35}K ε decay and 1184.3 4 from ^{36}Ca εp decay.
1750.78	(5/2) ⁺	1750.6 3	100	0.0	3/2 ⁺		E_γ : weighted average of 1750.5 3 from ^{35}K ε decay, 1750.7 4 from ($^{24}\text{Mg}, \alpha n \gamma$), and 1750.8 5 from ($^{16}\text{O}, \alpha n \gamma$).
2603.22	7/2 ⁽⁺⁾	851.9 9	12.3 33	1750.78 (5/2) ⁺			E_γ : weighted average of 852 1 from ($^{24}\text{Mg}, \alpha n \gamma$) and 851.8 9 from ($^{16}\text{O}, \alpha n \gamma$).
		2603.0 5	100 10	0.0	3/2 ⁺	Q	I_γ : weighted average of 10 5 from ($^{24}\text{Mg}, \alpha n \gamma$) and 13.3 33 from ($^{16}\text{O}, \alpha n \gamma$).
							E_γ : weighted average of 2603.0 5 from ($^{24}\text{Mg}, \alpha n \gamma$) and 2602.6 15 from ($^{16}\text{O}, \alpha n \gamma$).
							I_γ : other: 100 22 from ($^{24}\text{Mg}, \alpha n \gamma$).
2638.01	(3/2) ⁺	886.8 [#] 5	16 [#] 6	1750.78 (5/2) ⁺			
		2638.0 [#] 4	100 [#] 13	0.0	3/2 ⁺		
2982.79	(5/2) ⁺	1798.9 [#] 5	3.5 [#] 6	1184.08 1/2 ⁺			
		2982.68 [#] 13	100 [#] 4	0.0	3/2 ⁺		
3196.98	7/2 ⁻	593.7 2	16.4 30	2603.22 7/2 ⁽⁺⁾			E_γ : weighted average of 593 1 from ($^{24}\text{Mg}, \alpha n \gamma$) and 593.7 2 from ($^{16}\text{O}, \alpha n \gamma$).
							I_γ : weighted average of 16 8 from ($^{24}\text{Mg}, \alpha n \gamma$) and 16.4 30 from ($^{16}\text{O}, \alpha n \gamma$).
		1446.2 2	100 8	1750.78 (5/2) ⁺		D	E_γ : weighted average of 1446.2 2 from ($^{24}\text{Mg}, \alpha n \gamma$), 1446.1 6 from ($^{16}\text{O}, \alpha n \gamma$), and 1446.0 6 from ($^3\text{He}, n \gamma$).
							I_γ : other: 100 9 from ($^{24}\text{Mg}, \alpha n \gamma$).
		3197.0 7	21 5	0.0	3/2 ⁺	Q	E_γ : from ($^{24}\text{Mg}, \alpha n \gamma$). Other: 3197 6 from ($^{16}\text{O}, \alpha n \gamma$).
							I_γ : weighted average of 18 5 from ($^{24}\text{Mg}, \alpha n \gamma$) and 24 5 from ($^{16}\text{O}, \alpha n \gamma$).
4065.0?	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	1426.8 [#] 4	100	2638.01 (3/2) ⁺			
4359.0	(9/2 ⁻)	1162.0 8	65 24	3196.98 7/2 ⁻		(D)	E_γ : weighted average of 1162 1 from ($^{24}\text{Mg}, \alpha n \gamma$) and 1162.0 8 from

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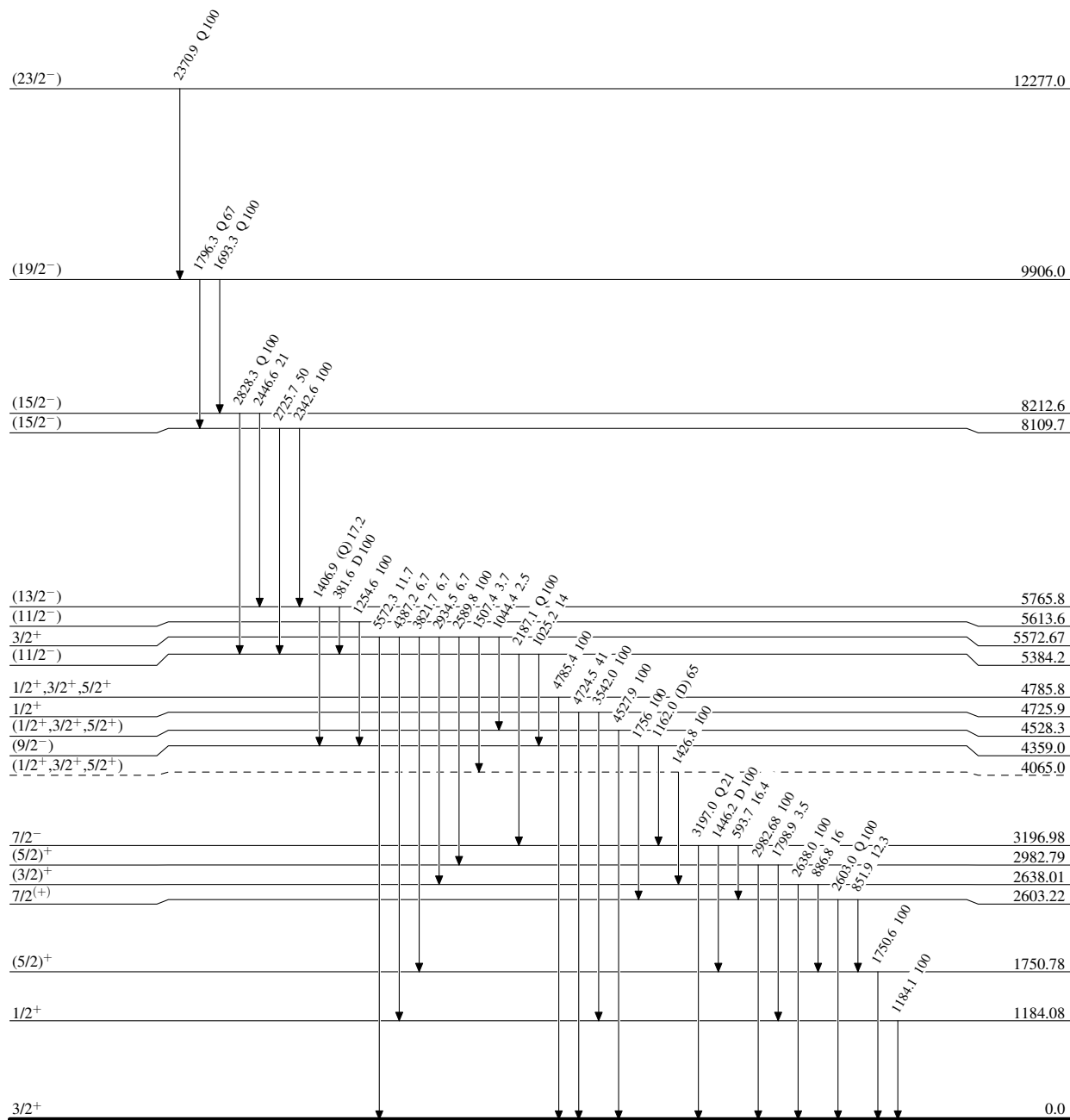
Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Ar})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]
						Comments
4359.0	(9/2 ⁻)	1756 1	100 15	2603.22	7/2 ⁽⁺⁾	
						($^{16}\text{O}, \alpha\gamma$). I _γ : unweighted average of 41 11 from ($^{24}\text{Mg}, \alpha\gamma$) and 88 18 from ($^{16}\text{O}, \alpha\gamma$). E _γ : weighted average of 1756 1 from ($^{24}\text{Mg}, \alpha\gamma$) and 1756.3 14 from ($^{16}\text{O}, \alpha\gamma$). I _γ : from ($^{24}\text{Mg}, \alpha\gamma$). Other: 100 53 from ($^{16}\text{O}, \alpha\gamma$).
4528.3	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	4527.9 [#] 7	100	0.0	3/2 ⁺	
4725.9	1/2 ⁺	3542.0 [#] 6	100 [#] 21	1184.08	1/2 ⁺	
		4724.5 [#] 11	41 [#] 17	0.0	3/2 ⁺	
4785.8	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	4785.4 [#] 11	100	0.0	3/2 ⁺	
5384.2	(11/2 ⁻)	1025.2 4	14 4	4359.0	(9/2 ⁻)	
						E _γ : weighted average of 1025 1 from ($^{24}\text{Mg}, \alpha\gamma$) and 1025.2 4 from ($^{16}\text{O}, \alpha\gamma$). I _γ : weighted average of 21 8 from ($^{24}\text{Mg}, \alpha\gamma$) and 12 4 from ($^{16}\text{O}, \alpha\gamma$). E _γ : weighted average of 2187.4 4 from ($^{24}\text{Mg}, \alpha\gamma$) and 2186.8 4 from ($^{16}\text{O}, \alpha\gamma$). I _γ : other: 100 13 from ($^{24}\text{Mg}, \alpha\gamma$).
		2187.1 4	100 6	3196.98	7/2 ⁻	Q
5572.67	3/2 ⁺	1044.4 [#] 4	2.5 [#] 8	4528.3	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	
		1507.4 [#] 5	3.7 [#] 8	4065.0?	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	
		2589.8 [#] 1	100 [#] 4	2982.79	(5/2 ⁺)	
		2934.5 [#] 5	6.7 [#] 12	2638.01	(3/2 ⁺)	
		3821.7 [#] 7	6.7 [#] 14	1750.78	(5/2 ⁺)	
		4387.2 [#] 9	6.7 [#] 16	1184.08	1/2 ⁺	
		5572.3 [#] 10	11.7 [#] 31	0.0	3/2 ⁺	
5613.6	(11/2 ⁻)	1254.6 8	100	4359.0	(9/2 ⁻)	
5765.8	(13/2 ⁻)	381.6 1	100 10	5384.2	(11/2 ⁻)	D
						E _γ : weighted average of 381.6 1 from ($^{24}\text{Mg}, \alpha\gamma$) and 381.5 3 from ($^{16}\text{O}, \alpha\gamma$).
		1406.9 7	17.2 35	4359.0	(9/2 ⁻)	(Q)
8109.7	(15/2 ⁻)	2342.6 28	100 25	5765.8	(13/2 ⁻)	
		2725.7 14	50 13	5384.2	(11/2 ⁻)	
8212.6	(15/2 ⁻)	2446.6 16	21 7	5765.8	(13/2 ⁻)	
		2828.3 7	100 18	5384.2	(11/2 ⁻)	Q
9906.0	(19/2 ⁻)	1693.3 27	100 20	8212.6	(15/2 ⁻)	Q
		1796.3 25	67 20	8109.7	(15/2 ⁻)	Q
12277.0	(23/2 ⁻)	2370.9 25	100	9906.0	(19/2 ⁻)	Q

[†] From ($^{16}\text{O}, \alpha\gamma$), unless otherwise noted.[‡] From ($^{16}\text{O}, \alpha\gamma$) and/or ($^{24}\text{Mg}, \alpha\gamma$). Stretched quadrupole (Q, ΔJ=2) or stretched dipole (D, ΔJ=1) transitions are assigned by evaluators based on the measured $\gamma\gamma(\theta)$ (ADO) and/or ratios of yields $R(\gamma(\theta))$ compared with the expected values given by the authors.# From ^{35}K ε decay.

Adopted Levels, GammasLevel Scheme

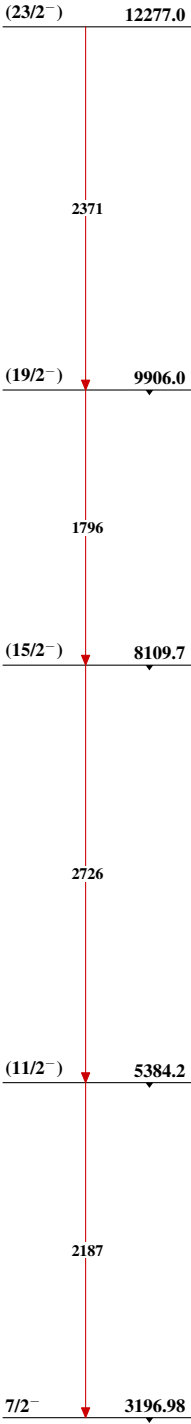
Intensities: Relative photon branching from each level



1.7755 s 14

Adopted Levels, Gammas

Band(A): Band based on f_{7/2}
orbital



³⁵Ar₁₇

^{35}K $\varepsilon+\beta^+$ decay (175 ms) 1980Ew02

Parent: ^{35}K : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=175$ ms 2; $Q(\varepsilon+\beta^+)=11874.4$ 9; $\% \varepsilon+\beta^+$ decay=100

^{35}K - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}K . Adopted $T_{1/2}$ from weighted average of 175 ms 2 (2018Sa54), 178 ms 8 (1998Sc19), and 190 ms 30 (1980Ew02) in this dataset.

^{35}K - $Q(\varepsilon+\beta^+)$: From 2021Wa16.

1980Ew02, 1979Ca15: A 600-MeV proton beam was produced from the synchrocyclotron at CERN-ISOLDE and bombarded a ScC_2 target. The $^{45}\text{Sc}(p,8n3p)$ spallation reaction products diffused out of the target and reached a tungsten surface ionization source where potassium isotopes were selectively ionized. The beam was extracted from the ion source, separated by the ISOLDE analyzing magnet, and collected by a mylar foil for γ -ray measurements and then a carbon foil for proton measurements. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Time for positron activities were determined using a 700- μm thick silicon detector. Protons were detected using a 20- μm -700- μm thick ΔE -E telescope of silicon surface barrier detectors with FWHM=50 keV. Measured $E_\gamma(<5$ MeV), I_γ , $E_p(>0.9$ MeV), I_p . Deduced levels, J , π , decay branching ratios, $\log ft$, parent ^{35}K $T_{1/2}$, and coefficients of the isobaric multiplet mass equation for $A=35$, $T=3/2$ quartets. Comparisons with shell-model calculations and the mirror nucleus ^{35}Cl . Also see abstracts 1978HaYH, 1979HaZY, 1979HaZT, and 1979AnZZ.

2018Sa54: A 36-MeV/nucleon ^{36}Ar primary beam was produced from the K500 cyclotron at Texas A&M University. The secondary ^{35}K beam was produced via the $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2n$ reaction of ^{36}Ar bombarding a LN_2 -cooled hydrogen gas target, separated by MARS, and implanted into a 45- μm DSSD sandwiched between a 140- μm SSSD and a 1-mm Si-pad detector in a pulsed-beam mode. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. γ rays were detected by two HPGe detectors. Measured $E_p(>300$ keV), I_p , E_γ , I_γ , $p\gamma$ -coin, $\gamma\gamma$ -coin. Deduced parent ^{35}K $T_{1/2}$.

2019ChZU: Same beam production as 2018Sa54. ^{35}K was implanted into the AstroBox2 detector filled with 800-Torr P5 gas. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. γ rays were detected by 4 Clover Ge detectors. Measured $E_p(>100$ keV), I_p , E_γ , I_γ , $p\gamma$ -coin, $\gamma\gamma$ -coin.

1998Sc19: A polarized ^{35}K beam was produced via the fragmentation of 500-MeV/nucleon ^{40}Ca impinging on a ^9Be target at GSI, separated using ΔE -tof by FRS, momentum-selected by slits, and implanted into a KBr single crystal placed in the central region of a magnet. Positrons were detected using plastic scintillators. γ rays were detected using a Ge detector. Measured β -decay asymmetry and $\beta\gamma$ -coin. Deduced polarization and g -factor of ^{35}K ground state from β -NMR and ^{35}K $T_{1/2}$ from $\beta\gamma$ -decay time spectra.

2006Me04: A polarized ^{35}K beam was produced via the proton-pickup reaction $^{36}\text{Ar}(^9\text{Be}, ^{10}\text{Li})^{35}\text{K}$, separated by NSCL-A1900, and implanted into a KBr crystal. Positrons were detected using plastic scintillators. Deduced the magnetic dipole moment and g -factor of ^{35}K ground state from β -NMR.

Theoretical studies involving ^{35}K decay: shell model (1985Br29, 2003Sm02).

 ^{35}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$	Comments
0	$3/2^+$	1.7755 s 14	
1184.01 25	$1/2^+$		
1750.72 25	$(5/2)^+$		
2637.99 26	$(3/2)^+$		
2982.79 12	$(5/2)^+$		
4065.0? 4	$(1/2^+, 3/2^+, 5/2^+)$		
4528.2 4	$(1/2^+, 3/2^+, 5/2^+)$		
4725.9 6	$1/2^+$		
4785.8 11	$1/2^+, 3/2^+, 5/2^+$		
5572.66 15	$3/2^+$		$T=3/2$
6348 11	$(1/2, 3/2, 5/2)$		$E(p0)_{\text{c.m.}}=452$ keV 11 (2019ChZU).

Continued on next page (footnotes at end of table)

^{35}K $\varepsilon+\beta^+$ decay (175 ms) **1980Ew02** (continued) ^{35}Ar Levels (continued)

E(level) [†]	J ^π [‡]	Comments
5896+x		E(level): $x < 5978.2$ 5 from $Q(\varepsilon)(^{35}\text{K}) - S(p)(^{35}\text{Ar})$, where $Q(\varepsilon) = 11874.4$ 9 and $S(p) = 5896.2$ 7 from
7053 11	3/2 ⁺ , 5/2 ⁺	E(p0) _{c.m.} = 1157 keV 11 (2019ChZU).
7255 11		E(p3) _{c.m.} = 693 keV 11 (2019ChZU).
7283 11		E(p0) _{c.m.} = 1387 keV 11 (2019ChZU).
7431 11		E(p3) _{c.m.} = 869 keV 11 (2019ChZU).
		E(level): weighted average of E(level) of 7497 20, 7510 20, and 7527 11. The former two E(level) are deduced from E(p0) _{c.m.} = 1601 20 (1980Ew02) and E(p1) _{c.m.} = 1467 20 (1980Ew02), respectively, with the corresponding E(level)(^{34}Cl) (2012Ni10) and $S(p)(^{35}\text{Ar}) = 5896.2$ 7 (2021Wa16). The 7527 11 is from 2019ChZU with E(p3) _{c.m.} = 965 11.
8393? 20	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	E(level): weighted average of E(level) of 8392 20, 8392 20, and 8395 20, deduced from E(p0) _{c.m.} = 2496 20 (1980Ew02), E(p1) _{c.m.} = 2349 20 (1980Ew02), and E(p2) _{c.m.} = 2038 20 (1980Ew02), respectively, with the corresponding E(level)(^{34}Cl) (2012Ni10) and $S(p)(^{35}\text{Ar}) = 5896.2$ 7 (2021Wa16).

[†] From a least-squares fit to γ -ray energies in **1980Ew02** for levels connected with γ transitions.

[‡] From the Adopted Levels.

 ε, β^+ radiations

E(decay)	E(level)	$I\beta^+$ [‡]	$I\varepsilon$ [‡]	Log ft	$I(\varepsilon+\beta^+)$ ^{†‡}	Comments
(3481 20)	8393?	0.062 26	4.3×10^{-4} 18	4.6 +3-2	0.062 26	$I(\varepsilon+\beta^+)$: 0.062 26 $I(p0+p1+p2)$ from Table 3 of 1980Ew02 .
(4356 11)	7518	>0.090	$>3 \times 10^{-4}$	<5.0	>0.09	$I(\varepsilon+\beta^+)$: 0.15 6 $I(p0+p1)$ from Table 3 of 1980Ew02 . Evaluators adopted a lower limit due to unreported $I(p3)$ (2019ChZU).
(5.5×10^3 [#] 55)	5896+x				0.16 16	$I(\varepsilon+\beta^+)$: $I(\varepsilon+\beta^+) = 0.37$ 15 for all E(p)>0.9 MeV is reported in 1980Ew02 . The protons placed from the 7518 and 8393 levels in ^{35}Ar sum to $I(\varepsilon+\beta^+) = 0.21$ 7. Therefore, the remaining $I(\varepsilon+\beta^+) = 0.16$ 16 is assigned by the evaluators to the unplaced protons from the 5896+x levels.
(5526 11)	6348	0.0025 5	2.9×10^{-6} 6	7.2 1	2.5×10^{-3} 5	
(6301.7 14)	5572.66	36.3 24	0.0265 18	3.31 4	36.3 24	
(7088.6 18)	4785.8	1.0 4	5×10^{-4} 2	5.2 2	1.0 4	
(7148.5 15)	4725.9	2.1 4	0.0010 2	4.9 1	2.1 4	
(7346.2 14)	4528.2	0.7 4	3×10^{-4} 2	5.4 +4-2	0.7 4	
(7809.4 14)	4065.0?	0.56 33	2.0×10^{-4} 12	5.6 +4-2	0.56 33	
(8891.6 14)	2982.79	26.0 22	0.0060 5	4.27 4	26.0 22	
(9236.4 14)	2637.99	≤0.4		≥6.2	≤0.4	
(10123.7 14)	1750.72	11.9 9	0.00181 14	4.91 4	11.9 9	
(10690.4 14)	1184.01	2.2 7	2.8×10^{-4} 9	5.8 +2-1	2.2 7	
(11874.4 17)	0	19 4	0.0018 4	5.1 1	19 4	$I(\varepsilon+\beta^+)$: From 1980Ew02 assuming mirror log ft with a small asymmetry correction.

[†] From γ intensity balance at each state, except for proton-emitting states. **1980Ew02** authors state that in complex decay schemes of heavy nuclides this method is known to be suspect since there is significant γ intensity that is unobserved because it lies in a

Continued on next page (footnotes at end of table)

^{35}K $\varepsilon+\beta^+$ decay (175 ms) **1980Ew02** (continued) ε, β^+ radiations (continued)

multitude of very weak γ -ray peaks. In a nucleus as light as ^{35}K the problem is less acute. They have generated a pandemonium test in the same spirit as in [1977Ha51](#) and find that less than one percent of the γ intensity from ^{35}K decay should be missed for that reason.

‡ Absolute intensity per 100 decays.

Estimated for a range of levels.

 $\gamma(^{35}\text{Ar})$

I γ normalization: From $\Sigma\%I\gamma(\gamma \text{ to g.s.})=80.6 \pm 4.0$, deduced from $100-\%(\varepsilon+\beta^+)_{\text{p}}-\%I(\varepsilon+\beta^+)_{\text{g.s.}}$, where $\%(\varepsilon+\beta^+)_{\text{p}}=0.37 \pm 1.5$ ([1980Ew02](#)) and $\%I(\varepsilon+\beta^+)_{\text{g.s.}}=19 \pm 4$ ([1980Ew02](#)), and the latter corresponds to $\log ft=5.07 \pm 0.5$, which was deduced from the ^{35}S (g.s.) to ^{35}Cl (g.s.) mirror $\log ft=5.01 \pm 0.2$ with a small asymmetry correction. $\%(\varepsilon+\beta^+)_{\text{p}}=0.37 \pm 1.5$ for $E(\text{p})>0.9$ MeV ([1980Ew02](#)). Some weak $E(\text{p})<0.9$ MeV branches have also been observed ([2018Sa54,2019ChZU](#)).

Conversion coefficients are considered negligibly small.

$E_{\gamma}^{\dagger}$	$I_{\gamma}^{\ddagger\#}$	$E_i(\text{level})$	J_i^{π}	E_f	J_f^{π}	Comments
886.8 5	0.9 3	2637.99	$(3/2)^+$	1750.72	$(5/2)^+$	$\%I\gamma=0.46 \pm 1.9-1.7$
1044.4 4	1.3 4	5572.66	$3/2^+$	4528.2	$(1/2^+, 3/2^+, 5/2^+)$	$\%I\gamma=0.66 \pm 2.5-2.3$
1184.0 3	14.3 7	1184.01	$1/2^+$	0	$3/2^+$	$\%I\gamma=7.2 \pm 5$
1426.8 4	3.0 5	4065.0?	$(1/2^+, 3/2^+, 5/2^+)$	2637.99	$(3/2)^+$	$\%I\gamma=1.5 \pm 4-3$
1507.4 5	1.9 4	5572.66	$3/2^+$	4065.0?	$(1/2^+, 3/2^+, 5/2^+)$	$\%I\gamma=0.96 \pm 2.7-2.5$
1750.5 3	28 1	1750.72	$(5/2)^+$	0	$3/2^+$	$\%I\gamma=14.1 \pm 9$
1798.9 5	3.5 6	2982.79	$(5/2)^+$	1184.01	$1/2^+$	$\%I\gamma=1.8 \pm 4$
2589.8 1	52 2	5572.66	$3/2^+$	2982.79	$(5/2)^+$	$\%I\gamma=26.3 \pm 1.8$
2638.0 4	5.5 7	2637.99	$(3/2)^+$	0	$3/2^+$	$\%I\gamma=2.8 \pm 5$
^x 2697.7 6						Unplaced γ ray, accounting for no more than 1.2% $\varepsilon+\beta^+$ -feeding (1980Ew02). No ^{35}Ar γ rays at this energy were observed in other reaction studies.
2934.5 5	3.5 6	5572.66	$3/2^+$	2637.99	$(3/2)^+$	$\%I\gamma=1.8 \pm 4$
2982.68 13	100 4	2982.79	$(5/2)^+$	0	$3/2^+$	$\%I\gamma=50.5 \pm 2.7$
3542.0 6	2.9 6	4725.9	$1/2^+$	1184.01	$1/2^+$	$\%I\gamma=1.5 \pm 4$
3821.7 7	3.5 7	5572.66	$3/2^+$	1750.72	$(5/2)^+$	$\%I\gamma=1.8 \pm 5$
4387.2 9	3.5 8	5572.66	$3/2^+$	1184.01	$1/2^+$	$\%I\gamma=1.8 \pm 5$
4527.9 7	2.6 7	4528.2	$(1/2^+, 3/2^+, 5/2^+)$	0	$3/2^+$	$\%I\gamma=1.3 \pm 4$
4724.5 11	1.2 5	4725.9	$1/2^+$	0	$3/2^+$	$\%I\gamma=0.61 \pm 3.0-2.7$
4785.4 11	1.9 7	4785.8	$1/2^+, 3/2^+, 5/2^+$	0	$3/2^+$	$\%I\gamma=1.0 \pm 4$
5572.3 10	6.1 16	5572.66	$3/2^+$	0	$3/2^+$	$\%I\gamma=3.1 \pm 1.0-0.9$
1980Ew02 observed the double escape peak at 4550 keV of this γ ray. 2018Sa54 observed the photopeak at 5572 keV.						

[†] From [1980Ew02](#).

[‡] For absolute intensity per 100 decays, multiply by 0.505 $\pm$ 0.29.

^x γ ray not placed in level scheme.

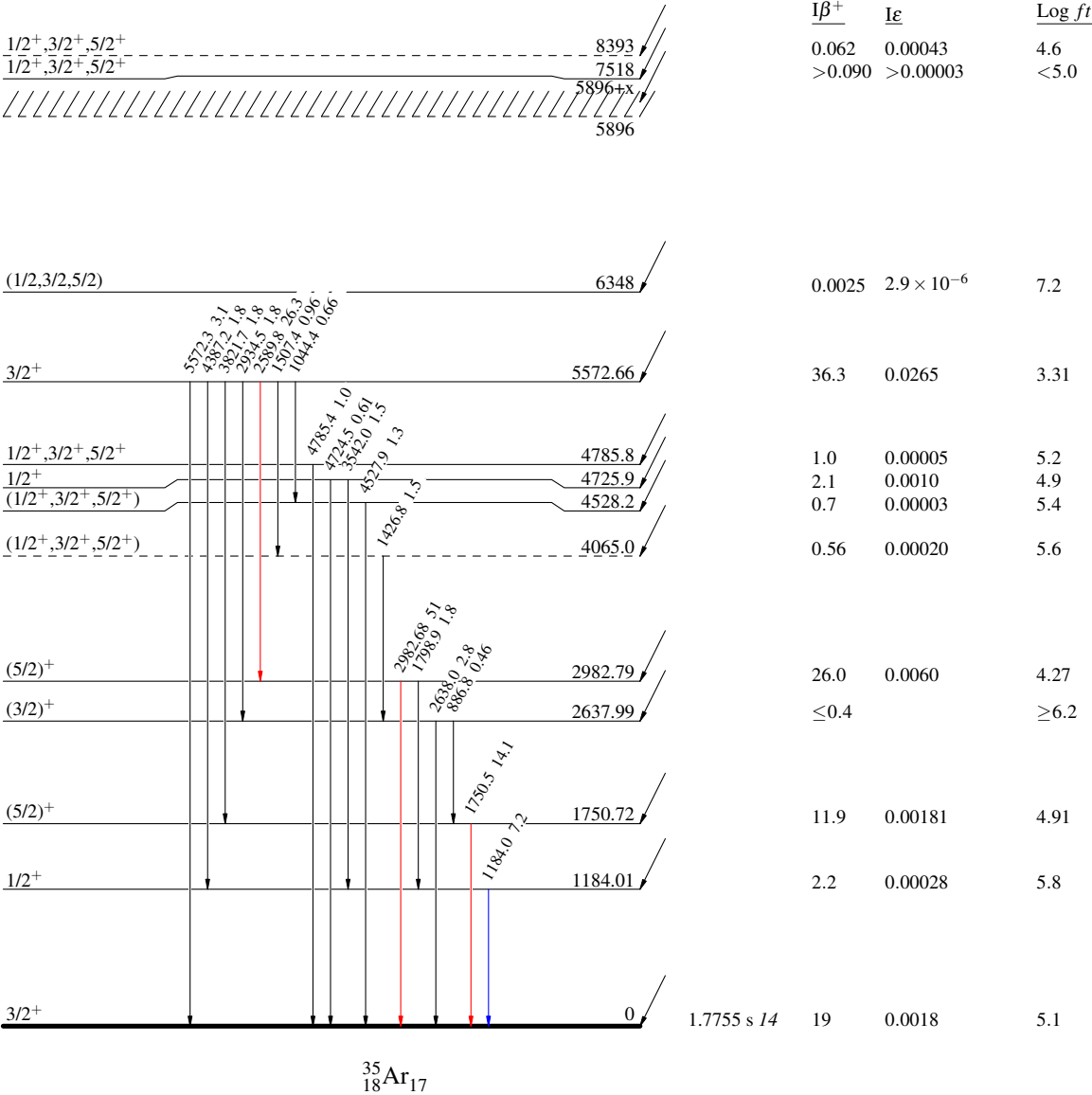
³⁵K ε+β⁺ decay (175 ms) 1980Ew02

Decay Scheme

Legend
→ I_γ < 2% × I_γ^{max}
→ I_γ < 10% × I_γ^{max}
→ I_γ > 10% × I_γ^{max}

Intensities: I_γ per 100 parent decays

$\% \epsilon + \% \beta^+ = 100$
 $\xrightarrow[Q_\epsilon = 11874.4 \text{ 9}]{3/2^+ \quad 0}$ 175 ms 2
³⁵K₁₆



^{36}Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01

Parent: ^{36}Ca : $E=0$; $J^\pi=0^+$; $T_{1/2}=100.9$ ms 20; $Q(\varepsilon\text{p})=9275$ 6; $\% \varepsilon\text{p}$ decay=54.1 12

$^{36}\text{Ca}-J^\pi$: From the Adopted Levels of ^{36}Ca (2012Ni01).

$^{36}\text{Ca}-T_{1/2}$: Weighted average of 102 ms 2 (1995Tr02,1997Tr05), 100.1 ms 23 (2007Do17), and 100.0 ms 24 (2015Su01). Other: 100 ms +90–40 (1981Ay01).

$^{36}\text{Ca}-Q(\varepsilon\text{p})$: Deduced by evaluators from ^{36}Ca mass excess of -6483 6; weighted average of -6483.6 56 (2021Su04), -6480 40 (2021La04), and -6450 40 (2021Wa16,1977Tr03), and ^{35}Ar mass excess of -23047.3 7 (2021Wa16). $Q(\varepsilon\text{p})$ from 2021Wa16: 9310 40.

$^{36}\text{Ca}-\% \varepsilon\text{p}$ decay: Unweighted average of $\% \varepsilon+\beta^+\text{p}=56.8$ 13 (1997Tr05), 54.3 18 (2001Lo11), 51.2 10 (2007Do17), and 53.9 72 (2015Su01).

1997Tr05,1995Tr02: a-300 MeV/nucleon ^{40}Ca primary beam was produced by the GSI heavy-ion synchrotron. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a ^9Be target and was selected using ΔE -tof-B ρ by FRS at GSI, Darmstadt. A total of 2.8×10^4 ^{36}Ca ions were implanted into a 500- μm -thick Si detector. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. β particles were detected by the implantation detector and two 500- μm -thick Si counters. γ rays were detected by two Ge detectors. Measured E_p , I_p , E_γ , I_γ , βp -coin, $\beta\gamma$ -coin, and $\text{p}\gamma$ -coin. Deduced levels, decay branching ratios, $\log ft$, $B(F)$, and $B(GT)$. Deduced parent ^{36}Ca $T_{1/2}$ from the time spectrum of proton events accumulated during the beam-off in the pulsed-beam mode. Comparisons with shell-model calculations.

2001Lo11: A 95-MeV ^{40}Ca primary beam was produced by the SSI facility at GANIL. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a $^{\text{nat}}\text{Ni}$ target and was selected using ΔE -tof by the LISE3 spectrometer and purified by a velocity filter. A total of 102407 ^{36}Ca ions were implanted into a 500- μm -thick Si detector. $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. β particles were detected by two 500- μm -thick Si counters. γ rays were detected by three Ge detectors. Measured E_p , I_p , E_γ , I_γ , βp -coin, and $\beta\gamma$ -coin. Deduced levels, decay branching ratios, $\log ft$, $B(F)$, and $B(GT)$. Comparisons with shell-model calculations.

2015Su01: A 69.42-MeV/nucleon ^{40}Ca primary beam was produced by the Sector Focusing Cyclotron and Separated Sector Cyclotron at the Heavy Ion Research Facility in Lanzhou (HIRFL). The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{40}Ca impinging on a ^9Be target and was selected using ΔE -tof-B ρ by RIBLL. A total of 22890 ^{36}Ca ions were implanted into a 525- μm -thick DSSD. $\varepsilon+\beta^+$ -delayed protons were detected by the DSSD with a threshold of 500 keV. $\varepsilon+\beta^+$ -delayed γ rays were detected by four Clover Ge detectors surrounding the DSSD chamber. Measured E_p , I_p , E_γ , I_γ , βp -coin, $\text{p}\gamma$ -coin, and implant-decay time correlations. Deduced levels, decay branching ratios, and parent ^{36}Ca $T_{1/2}$. Also see 2016Li45.

2007Do17: A 74.5-MeV/nucleon ^{58}Ni primary beam was produced by the SSI facility at GANIL. The secondary ^{36}Ca beam was produced via the projectile fragmentation of ^{58}Ni impinging on a $^{\text{nat}}\text{Ni}$ target and was selected using ΔE -tof-B ρ by the ALPHA-LISE3 separator. A total of 16991 ^{36}Ca ions were implanted into a 500- μm thick DSSD. $\varepsilon+\beta^+$ -delayed protons were detected by the DSSD with a threshold of 60-80 keV. $\varepsilon+\beta^+$ -delayed γ rays were detected by four Ge detectors surrounding the implantation array. A 5-mm thick lithium-drifted Si detector was used as a veto for implantation events and to detect β particles. Measured E_p , I_p , and implant-decay time correlations. Deduced levels, decay branching ratios, and parent ^{36}Ca $T_{1/2}$.

1995Ga16: 60-keV ^{36}Ca was produced by the ISOLDE general-purpose on-line isotope separator at the CERN PS/Booster and implanted into the entrance window of a gas-Si-Si ΔE -E-veto detector telescope. Measured E_p . Deduced coefficients of the isobaric multiplet mass equation for $A=36$, $T=2$ quintets. A by-product of ^{37}Ca decay study (1995Ga03).

1981Ay01,1980AyZZ: ^{36}Ca was produced via the $^{40}\text{Ca}(^3\text{He},\alpha\text{n})$ reaction using a 95-MeV ^3He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. β -delayed protons were detected using a Si surface barrier detector telescope with FWHM=55 keV and a minimum threshold of ≈ 1.5 MeV. Measured E_p . Deduced ^{36}Ca $T_{1/2}$ and coefficients of the isobaric multiplet mass equation for $A=36$, $T=2$ quintets.

Theoretical studies involving ^{36}Ca decay: shell model (1984Mu25,1990Br26), covariant density functional theory (2013Ni09).

 ^{35}Ar Levels

E(level)	$J^\pi^\dagger$	$T_{1/2}^\dagger$	Comments
0	$3/2^+$	1.7755 s 14	E(level): From E_γ data.
1184.3 4	$1/2^+$		

† From the Adopted Levels.

^{36}Ca ε p decay (100.9 ms) [1997Tr05](#), [2001Lo11](#), [2015Su01](#) (continued) $\gamma(^{35}\text{Ar})$

I γ normalization: From 100/% ε + β^+ p.

E_γ	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Comments
1184.3 4	1.2 4	1184.3	1/2 ⁺	0	3/2 ⁺	E_γ : weighted average of 1184.2 4 (1997Tr05) and 1185 1 (2001Lo11). I_γ : From 1997Tr05 .

[†] Absolute intensity per 100 decays.

Delayed Protons (^{35}Ar)

$E(p)^\dagger$	$E(^{35}\text{Ar})$	$I(p)^\#$	$E(^{36}\text{K})^\ddagger$	Comments
1.37×10^3	1184.3	1.2 4	4281.7	$E(p), I(p)$: From 1997Tr05 . $E(^{36}\text{K})$: $J^\pi=0$, T=2 isobaric analog state in ^{36}K .
1648 18	0	8.4 12	3354	$E(p)$: weighted average of 1676 39 (1997Tr05), 1645 21 (2001Lo11), 1660 18 (2007Do17), and 1624 22 (2015Su01). $I(p)$: unweighted average of 11.3 6 (1997Tr05), 9.3 8 (2001Lo11), 7.3 8 (2007Do17), and 5.7 16 (2015Su01).
2549.9 22	0	37.4 10	4281.7	$E(p)$: weighted average of 2519 21 (1981Ay01), 2550.2 22 (1995Ga16), 2548 37 (1997Tr05), 2551 21 (2001Lo11), 2538 18 (2007Do17), and 2594 30 (2015Su01). $I(p)$: weighted average of 37.8 10 (1997Tr05), 32.1 42 (2007Do17), and 34.0 58 (2015Su01). Other: 37 1 (2001Lo11) without separating the weaker proton branch from the same level to ^{35}Ar first excited state.
2713 21	0	2.6 9	4449	$E(^{36}\text{K})$: $J^\pi=0$, T=2 isobaric analog state in ^{36}K . $E(p)$: weighted average of 2713 31 (1997Tr05) and 2713 21 (2001Lo11).
2921 35	0	1.3 2	4663	$I(p)$: unweighted average of 1.7 2 (1997Tr05) and 3.5 5 (2001Lo11). $E(p)$: weighted average of 2937 35 (1997Tr05) and 2895 44 (2001Lo11).
3484 21	0	0.6 2	5242	$I(p)$: weighted average of 1.4 2 (1997Tr05) and 1.0 3 (2001Lo11). $E(p), I(p)$: From 2001Lo11 .
3.98×10^3 7	0	0.9 2	5753	$E(p), I(p)$: From 2001Lo11 .
4.15×10^3 5	0	2.2 5	5927	$E(p)$: weighted average of 4162 45 (1997Tr05) and 4135 44 (2001Lo11). $I(p)$: unweighted average of 2.7 4 (1997Tr05) and 1.7 3 (2001Lo11).
4.99×10^3 7	0	0.4 2	6791	$E(p)$: weighted average of 4989 69 (1997Tr05) and 4983 67 (2001Lo11). $I(p)$: weighted average of 0.7 2 (1997Tr05) and 0.3 1 (2001Lo11).

[†] In lab frame. Evaluators take a weighted average when multiple measurements of a $E(p)$ are available. [2007Do17](#) and [2015Su01](#) reported $E_{c.m.}$. [1997Tr05](#) and [2001Lo11](#) reported the $E(\text{level})$ of ^{36}K proton-emitting levels. Evaluators deduced proton center-of-mass energies $E_{c.m.}=E(\text{level})(^{36}\text{K})-S(p)(^{36}\text{K})-E(\text{level})(^{35}\text{Ar})$ using their original $S(p)=1666$ 8. Evaluators then deduced each $E(p)_{\text{lab}}=E_{c.m.} \times m(^{35}\text{Ar})/[m(p)+m(^{35}\text{Ar})]$.

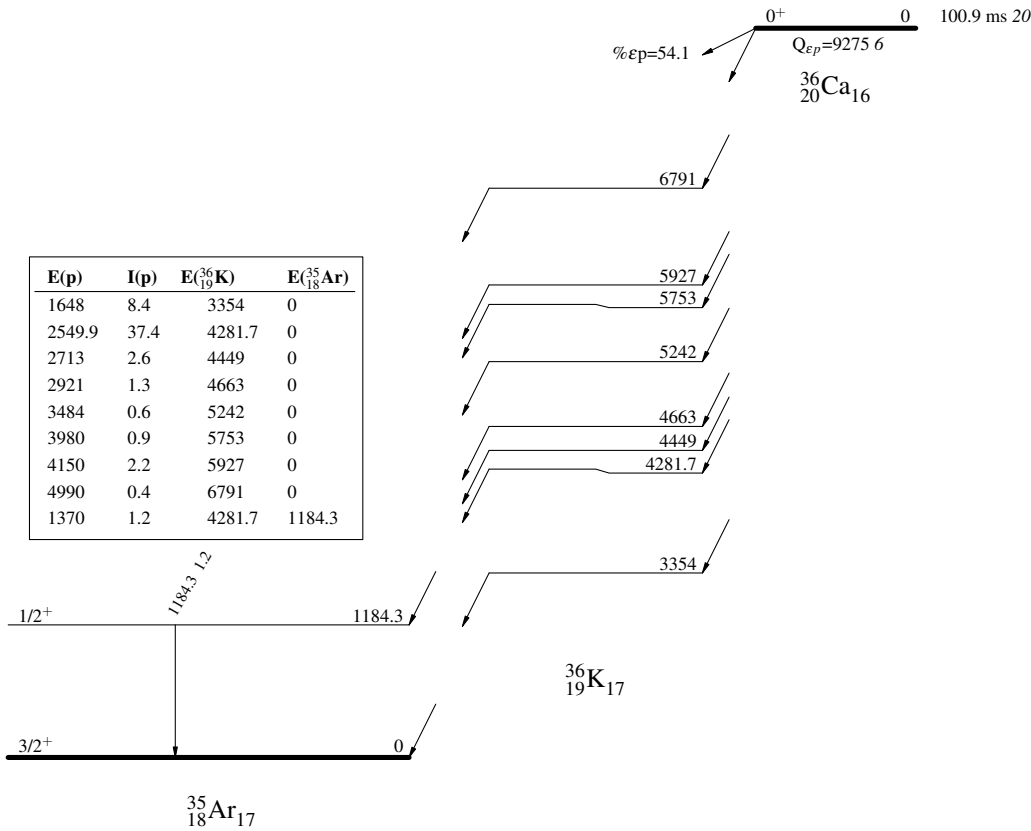
[‡] $E(\text{level})(^{36}\text{K})=E(p)_{\text{lab}} \times [m(p)+m(^{35}\text{Ar})]/m(^{35}\text{Ar})+S(p)(^{36}\text{K})+E(\text{level})(^{35}\text{Ar})$, where $S(p)(^{36}\text{K})=1658.9$ 8 ([2021Wa16](#)).

[#] Absolute intensity per 100 decays.

³⁶Ca εp decay (100.9 ms) 1997Tr05,2001Lo11,2015Su01

Decay Scheme

γ Intensities: I_γ per 100 parent decays
I(p) Intensities: I(p) per 100 parent decays



$^1\text{H}(^{36}\text{Ar},\text{d})$ [2010Le03,2011Le01](#)

$J^\pi=0^+$ for ^{36}Ar ground state.

[2010Le03](#), [2011Le01](#): A ^{36}Ar beam at 33 MeV/u was provided at the National Superconducting Cyclotron Laboratory, MSU.

Targets were polyethylene (CH_2)_n. Deuterons were detected using the High-Resolution Array (HiRA) of Si and CsI(Tl) telescope detectors in coincidence with recoil residues identified in the S800 spectrometer by the focal plane ionization chamber andToF.

Measured $\sigma(E_d, \theta)$ in inverse kinematics. Deduced neutron spectroscopic factors from adiabatic distorted wave approximation (ADWA) analysis of the measured $\sigma(\theta)$ using Chapel-Hill global optical potential parameters (CH89) and JLM optical potentials and geometry for transferred neutron constrained by Hartree–Fock calculations (JLM+HF). Comparisons with shell–model calculated spectroscopic factors.

Theoretical studies involving $^1\text{H}(^{36}\text{Ar},\text{d})^{35}\text{Ar}$: [2011Nu01](#), [2023He15](#).

 ^{35}Ar Levels

<u>E(level)[†]</u>	<u>J^π</u>	<u>$L^\ddagger$</u>	<u>$C^2S^\ddagger$</u>	<u>Comments</u>
0	$3/2^+$	2	2.3 2	C ² S: others: 1.6 <i>I</i> from 2011Le01 using JLM+HF calculations; 2.10 from large basis–shell model calculations (2010Le03); 2.21 <i>49</i> from a reanalysis of the $\sigma(\theta)$ data using finite-range ADWA (2011Nu01), including theoretical uncertainties associated with optical potentials (7%) and the approximate solution of three-body problems (19%); 2.1 +2–4 from a reanalysis of the $\sigma(\theta)$ data using ADWA within a Bayesian framework (2023He15), including theoretical uncertainties associated with optical potentials.
1180		0	1.2 <i>I</i>	
2980 [#]				
3190 [#]				
5570				

[†] From [2010Le03](#).

[‡] From ADWA analysis of measured $\sigma(\theta)$ in [2011Le01](#) using CH89 optical potential parameters.

[#] Doublet in measured spectra.

$^{16}\text{O}(^{24}\text{Mg},\alpha n\gamma)$ 2004Ek01,2005Ek01

2004Ek01,2005Ek01: A 60-MeV ^{24}Mg beam was produced at the Legnaro National Laboratory, Italy. The target was 0.5-mg/cm² enriched ^{40}Ca with a 7-mg/cm² tantalum backing. Oxygen was present in the target, giving rise to the fusion evaporation reactions of $^{16}\text{O}(^{24}\text{Mg},\alpha n\gamma)^{35}\text{Ar}$ and $^{16}\text{O}(^{24}\text{Mg},\alpha p\gamma)^{35}\text{Cl}$. γ rays were detected using the GASP array of Ge detectors and 80 BGO detectors. Charged particles were detected using the ISIS array of 40 Si ΔE -E telescopes. Neutrons were detected using a Neutron Ring replacing the six BGO elements at the most forward angles. The event trigger required one Ge detector, one BGO detector, and one neutron detector, or two Ge detectors and one BGO detector firing. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$, $\alpha n\gamma$ -coin, and $\alpha p\gamma$ -coin. Deduced levels, J, π from the measured ratios of yields $R(\gamma(\theta))$ and comparisons with the mirror nucleus ^{35}Cl .

 ^{35}Ar Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$
0.0	$3/2^+$
1750.8 3	$5/2^+$
2603.2 4	$(7/2^+)$
3197.0 4	$7/2^{(-)}$
4359.2 7	$(9/2^-)$
5384.4 5	$11/2^{(-)}$
5766.0 5	$13/2^{(-)}$

[†] From a least-squares fit to γ -ray energies.

[‡] As given in [2004Ek01](#) based on known assignments of low-lying levels and mirror levels in ^{35}Cl and the measured ratios of yields $R(\gamma(\theta))$. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

 $\gamma(^{35}\text{Ar})$

The ratios of yields $R(\gamma(\theta))$ were measured at 35° and 81° with respect to the beam axis. Expected values are $R(\gamma(\theta))\approx 1.2$ for stretched quadrupole ($\Delta J=2$) and $R(\gamma(\theta))\approx 0.7$ for stretched dipole ($\Delta J=1$) transitions.

$E_\gamma^\dagger$	$I_\gamma^\dagger$	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	Comments
381.6 1	26 3	5766.0	$13/2^{(-)}$	5384.4	$11/2^{(-)}$	D	$R(\gamma(\theta))=0.69$ 18.
593 1	12 6	3197.0	$7/2^{(-)}$	2603.2	$(7/2^+)$		
852 [‡] 1	4 2	2603.2	$(7/2^+)$	1750.8	$5/2^+$		
1025 [‡] 1	5 2	5384.4	$11/2^{(-)}$	4359.2	$(9/2^-)$		
1162 1	11 3	4359.2	$(9/2^-)$	3197.0	$7/2^{(-)}$		
1446.2 2	76 7	3197.0	$7/2^{(-)}$	1750.8	$5/2^+$	D	$R(\gamma(\theta))=0.71$ 9.
1750.7 4	100 7	1750.8	$5/2^+$	0.0	$3/2^+$		$R(\gamma(\theta))=1.41$ 14 indicates $\Delta J=2$, but $\Delta J=1$ from level scheme is inconsistent.
1756 [‡] 1	27 4	4359.2	$(9/2^-)$	2603.2	$(7/2^+)$		
2187.4 4	24 3	5384.4	$11/2^{(-)}$	3197.0	$7/2^{(-)}$	Q	$R(\gamma(\theta))=1.60$ 36.
2603.0 5	41 9	2603.2	$(7/2^+)$	0.0	$3/2^+$		$R(\gamma(\theta))=1.01$ 17.
3197.0 7	14 4	3197.0	$7/2^{(-)}$	0.0	$3/2^+$	Q	$R(\gamma(\theta))=1.45$ 51.

[†] From [2004Ek01](#).

[‡] Tentative placements; later confirmed by $^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)^{35}\text{Ar}$ ([2007De14](#)) and adopted by evaluators.

[#] Not explicitly given in [2004Ek01](#), but deduced by evaluators based on their measured ratios of yields $R(\gamma(\theta))$ compared with expected values as stated in [2004Ek01](#).

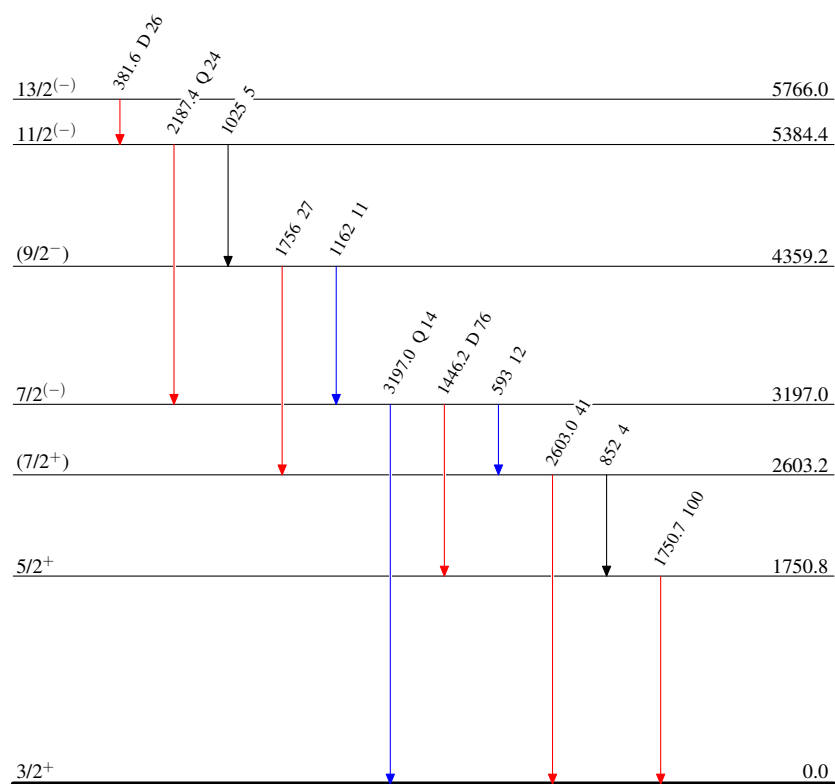
$^{16}\text{O}(^{24}\text{Mg}, \alpha n \gamma)$ 2004Ek01, 2005Ek01

Level Scheme

Intensities: Relative I_γ

Legend

- $I_\gamma < 2\% \times I_\gamma^{\max}$
—→ $I_\gamma < 10\% \times I_\gamma^{\max}$
—→ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{18}\text{Ar}_{17}$

$^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)$ 2007De14,2005DeZZ

2007De14,2005DeZZ: A 70-MeV ^{16}O beam was produced by the XTU-Tandem accelerator at the Legnaro National Laboratory, Italy. The target was a 400 $\mu\text{g}/\text{cm}^2$ self-supporting target of ^{24}Mg . γ ray from fusion evaporation reactions of $^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)^{35}\text{Ar}$ and $^{24}\text{Mg}(^{16}\text{O},\alpha p\gamma)^{35}\text{Cl}$ were detected using the GASP spectrometer, which consists of an array of 40 Compton-suppressed HPGe detectors and a multiplicity filter of 80 BGO scintillators. The GASP spectrometer is operated in conjunction with the 4π charged-particle detector ISIS and a neutron ring of 6 BC501A scintillators. The events were collected when at least two Ge detectors and one BGO scintillator were fired in coincidence. Measured $E\gamma$, $I\gamma$, $\gamma\gamma$, $\alpha n\gamma$ -coin, $\alpha p\gamma$ -coin, $\gamma(\theta)$, $\gamma\gamma(\theta)(\text{ADO})$. Deduced levels, J , π , and transition multipolarities from γ -ray ADO ratios. Comparisons with Shell-model calculations show the multipole Coulomb interaction and the electromagnetic spin-orbit interaction contribute to the observed mirror energy differences. 2007De14 also report data on ^{35}Cl and propose the level scheme of ^{35}Ar based on mirror symmetry.

 ^{35}Ar Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>	<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\ddagger$</u>
0.0	$3/2^+$	3196.7 [#] 7	$7/2^-$	5613.2 11	$(11/2^-)$	8212.1 10	$15/2^-$
1750.8 5	$5/2^+$	4358.6 8	$9/2^-$	5765.3 8	$13/2^-$	9905.5 [#] 21	$19/2^-$
2603.0 7	$7/2^+$	5383.7 [#] 7	$11/2^-$	8109.2 [#] 14	$15/2^-$	12276.5 [#] 33	$23/2^-$

[†] From a least-squares fit to γ -ray energies.

[‡] As given in 2007De14 based on known assignments of low-lying levels and mirror levels in ^{35}Cl , the measured $\gamma\gamma(\theta)(\text{ADO})$ ratios, and the mirror symmetry with levels with similar energies in ^{35}Cl with known assignments. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.

[#] Band(A): Band based on $f_{7/2}$ orbital.

 $\gamma(^{35}\text{Ar})$ 

$R_{\text{ADO}} = [I\gamma(34^\circ) + I\gamma(146^\circ)] / 2I\gamma(90^\circ)$. Expected values are $R_{\text{ADO}} \approx 1.3$ for stretched quadrupole ($\Delta J=2$) and $R_{\text{ADO}} \approx 0.8$ for stretched dipole ($\Delta J=1$) transitions.

<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	<u>Comments</u>
381.5 3	29 3	5765.3	$13/2^-$	5383.7	$11/2^-$	D	$R_{\text{ADO}}=0.81$ 10.
593.7 2	11 2	3196.7	$7/2^-$	2603.0	$7/2^+$		
851.8 9	8 2	2603.0	$7/2^+$	1750.8	$5/2^+$		
1025.2 4	6 2	5383.7	$11/2^-$	4358.6	$9/2^-$		
1162.0 8	15 3	4358.6	$9/2^-$	3196.7	$7/2^-$	(D)	$R_{\text{ADO}}=0.95$ 25.
1254.6 8	15 5	5613.2	$(11/2^-)$	4358.6	$9/2^-$		
1406.9 7	5 1	5765.3	$13/2^-$	4358.6	$9/2^-$	(Q)	$R_{\text{ADO}}=1.3$ 7.
1446.1 6	67 5	3196.7	$7/2^-$	1750.8	$5/2^+$	D	$R_{\text{ADO}}=0.87$ 19.
1693.3 27	15 3	9905.5	$19/2^-$	8212.1	$15/2^-$	Q	$R_{\text{ADO}}=1.41$ 23.
1750.8 5	100 9	1750.8	$5/2^+$	0.0	$3/2^+$		$R_{\text{ADO}}=1.46$ 24 indicates $\Delta J=2$, but inconsistent with $\Delta J=1$ from level scheme.
1756.3 14	17 9	4358.6	$9/2^-$	2603.0	$7/2^+$		
1796.3 25	10 3	9905.5	$19/2^-$	8109.2	$15/2^-$	Q	$R_{\text{ADO}}=1.8$ 3.
2186.8 4	49 3	5383.7	$11/2^-$	3196.7	$7/2^-$	Q	$R_{\text{ADO}}=1.31$ 15.
2342.6 28	8 2	8109.2	$15/2^-$	5765.3	$13/2^-$		
2370.9 25	15 5	12276.5	$23/2^-$	9905.5	$19/2^-$	Q	$R_{\text{ADO}}=1.8$ 4.
2446.6 16	6 2	8212.1	$15/2^-$	5765.3	$13/2^-$		
2602.6 15	60 6	2603.0	$7/2^+$	0.0	$3/2^+$	Q	$R_{\text{ADO}}=1.37$ 20.
2725.7 14	4 1	8109.2	$15/2^-$	5383.7	$11/2^-$		
2828.3 7	28 5	8212.1	$15/2^-$	5383.7	$11/2^-$	Q	$R_{\text{ADO}}=1.7$ 6.
3197 6	16 3	3196.7	$7/2^-$	0.0	$3/2^+$	Q	$R_{\text{ADO}}=1.7$ 8.

Continued on next page (footnotes at end of table)

$^{24}\text{Mg}(^{16}\text{O},\alpha n\gamma)$ [2007De14,2005DeZZ](#) (continued)

$\gamma(^{35}\text{Ar})$ (continued)

[†] From [2007De14](#).

[‡] Not explicitly given in [2007De14](#), but deduced by evaluators based on their measured R_{ADO} ratios compared with expected values as stated in [2007De14](#).

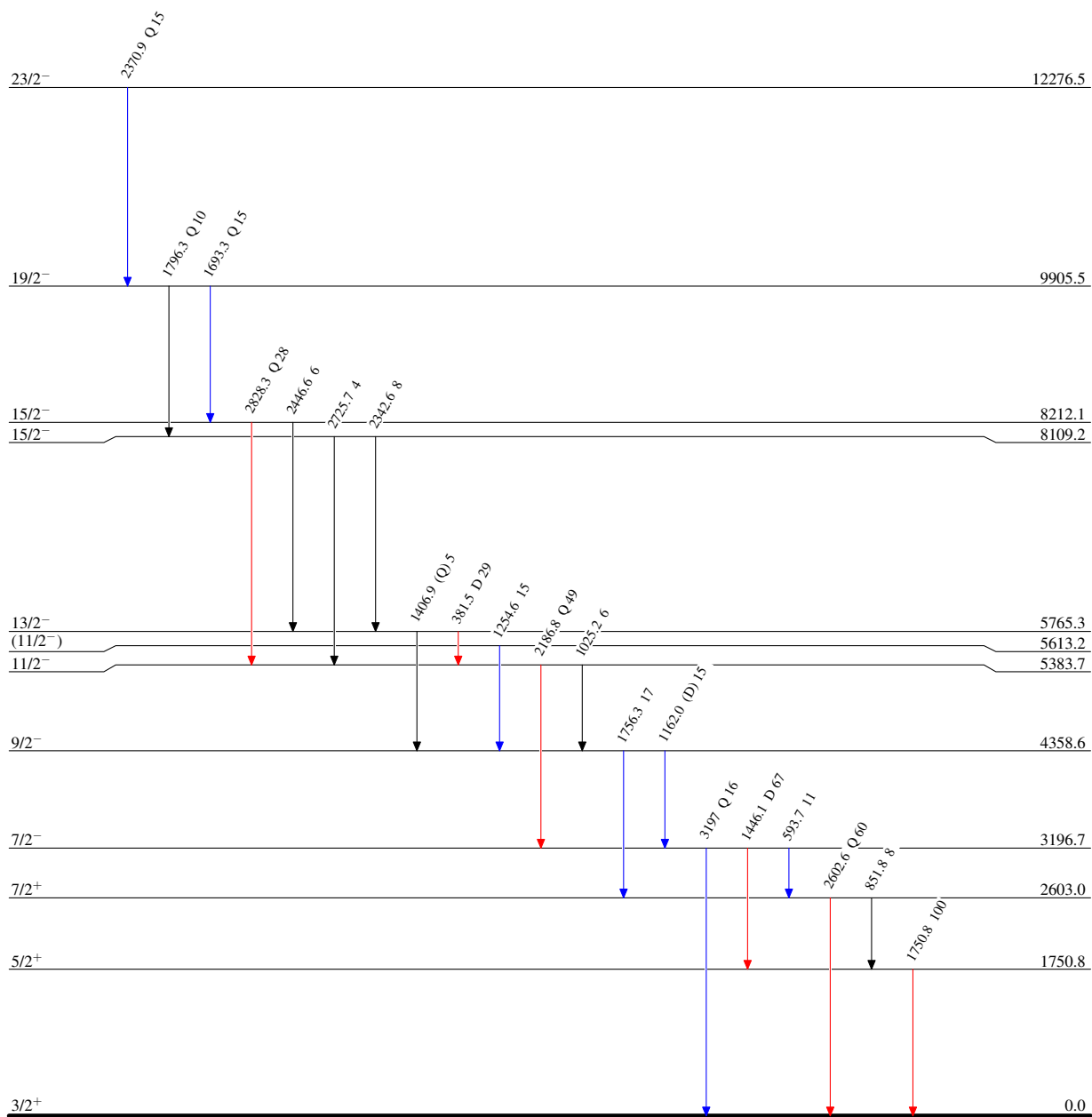
$^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$ 2007De14, 2005DeZZ

Level Scheme

Intensities: Relative I_γ

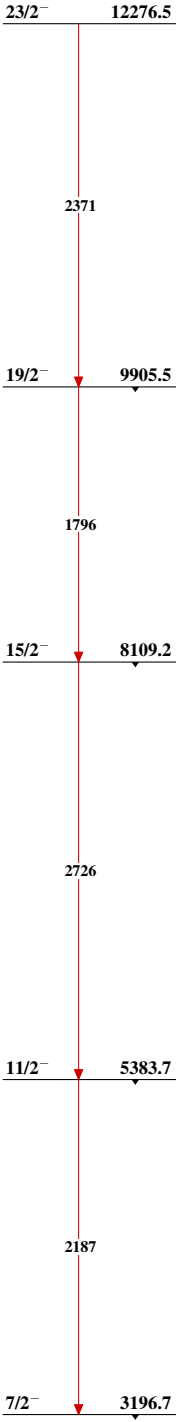
Legend

- $\longrightarrow$ $I_\gamma < 2\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma < 10\% \times I_\gamma^{\max}$
 $\longrightarrow$ $I_\gamma > 10\% \times I_\gamma^{\max}$

 $^{35}_{18}\text{Ar}_{17}$

²⁴Mg(¹⁶O,αnγ) 2007De14,2005DeZZ

Band(A): Band based on f_{7/2}
orbital



³⁵Ar₁₇

$^{32}\text{S}(\alpha, n)$ **1963Ne05**

1963Ne05: A 12.20-MeV α beam was produced by the Tandem Van de Graaff Accelerator at Florida State University. Targets were natural sulfur. Positrons were detected using a Ne102 plastic scintillator. Annihilation γ rays were detected using a NaI crystal. The half-life of $^{35}\text{Ar}(\text{g.s.})$ was measured with a Technical Measurements Corporation 256 channel analyzer and multiscaler logic unit. Counting time per channel was determined by the sweep of a Tektronix Oscilloscope and measured using the standard frequency output of a Hewlett-Packard Electronic Counter Model 524-C. Changes in slope of the excitation curve are interpreted as excited-state thresholds due to excited levels in the product nucleus ^{35}Ar . The threshold of $^{35}\text{Cl}(\text{p}, n)^{35}\text{Ar}(\text{g.s.})$ using a thick AgCl target was also measured to confirm the ^{35}Ar ground state mass.

^{35}Ar isotope discovery: $^{32}\text{S}(\alpha, n)^{35}\text{Ar}$ at Purdue ([1940Ki12](#), [1941Ki01](#), [1941El04](#)).

 ^{35}Ar Levels

<u>E(level)[†]</u>	<u>T_{1/2}</u>	<u>Comments</u>
0	1.76 s 3	T _{1/2} : from decay curve in 1963Ne05 . Threshold E(α) _{lab} =9.846 MeV 20.
890 50		Threshold E(α) _{lab} =10.69 MeV 50.
2030 80		Threshold E(α) _{lab} =11.97 MeV 80.

[†] From [1963Ne05](#).

$^{33}\text{S}(^3\text{He},n\gamma)$ 1975Da14

$^{33}\text{S}+2p \rightarrow ^{35}\text{Ar}$ from $J^\pi=3/2^+$ ^{33}S ground state.

1975Da14: a $^3\text{He}^+$ beam was produced from the University of Alberta Van de Graaff accelerator. Targets were $150 \mu\text{g}/\text{cm}^2$ layers of Ag_2S (59% ^{33}S) on silver backings. At $E(^3\text{He})=6.375 \text{ MeV}$, neutrons were detected using an NE213 liquid scintillator placed at $\theta_{\text{lab}}=0^\circ$, 10° , and 20° . At $E(^3\text{He})=6.660 \text{ MeV}$, neutrons were detected using the NE213 scintillator placed at $\theta_{\text{lab}}=0^\circ$ and 20° . At $E(^3\text{He})=6.390 \text{ MeV}$, neutrons were detected using the NE213 scintillator placed at $\theta_{\text{lab}}=0^\circ$ and neutron-coincident γ rays were detected using a 10% efficient Ge(Li) detector placed at 90° . Measured time-of-flight (TOF) spectra of neutrons, $\sigma(E_n, \theta)$, E_γ , $n\gamma$ -coin. Deduced Q values, levels, L_{2p} , J, π , and isospin.

 ^{35}Ar Levels

$E(\text{level})^\dagger$	J^π	$L^\ddagger$	Comments
0	$3/2^+$	(0)	J^π : mirror $3/2^+$ ^{35}Cl g.s.
1184.2 6			
1749.8 9			
2600.8 15			J^π : 1975Da14 observed $L(^3\text{He},n)=(0)$ for a 2.60-MeV level in ^{35}Ar , implying the existence of a $3/2^+$ level, which possibly corresponds to the $3/2^+$ 2638-keV level in ^{35}Ar .
3195.8 11			E(level): deduced by evaluators from the unplaced 1446.0 γ .
5537 25	$(3/2^+)$	(0)	T=3/2 E(level): from measured neutron energy corresponding to this level. J^π : $(\pi 1d_{3/2})^2(\nu 1d_{3/2})^1$ configuration formed by the $J^\pi=0^+$, T=1 $(\pi 1d_{3/2})^2$ di-proton transfer from ^3He to ^{33}S of $^{32}\text{S}(\nu 1d_{3/2})^1$ configuration. Tentative first T=3/2 state in ^{35}Ar .

† From **1975Da14** based on measured E_γ , but the E_γ values are not provided in **1975Da14**, unless otherwise noted.

‡ the observed maximum at $\theta=0^\circ$ in $\sigma(E_n, \theta)$ implies $L=0$.

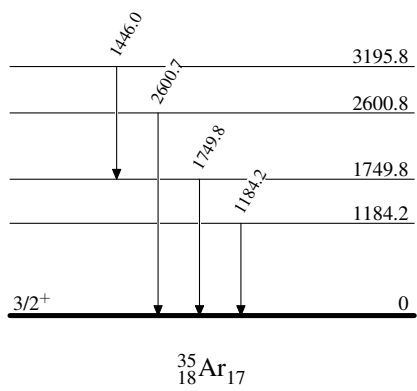
 $\gamma(^{35}\text{Ar})$

$E_\gamma^\dagger$	$E_i(\text{level})$	E_f	J_f^π	Comments
1184.2	1184.2	0	$3/2^+$	
1446.0 6	3195.8	1749.8		Unplaced γ in 1975Da14 . Evaluators placed it based on the Adopted Levels from $^{16}\text{O}(^{24}\text{Mg}, \alpha n\gamma)^{35}\text{Ar}$ (2004Ek01) and $^{24}\text{Mg}(^{16}\text{O}, \alpha n\gamma)^{35}\text{Ar}$ (2007De14).
1749.8	1749.8	0	$3/2^+$	
2600.7	2600.8	0	$3/2^+$	

† E_γ values without uncertainties are deduced by evaluators from the level-energy differences from **1975Da14**.

$^{33}\text{S}(^3\text{He},n\gamma)$ 1975Da14

Level Scheme



$^{35}\text{Cl}(^3\text{He},t)$ **1976Be08**

$J^\pi=3/2^+$ for ^{35}Cl ground state.

1976Be08: $^{35}\text{Cl}(^3\text{He},t)^{35}\text{Ar}$ is studied in an attempt to search for the member of T=3/2 isobaric quartets in ^{35}Ar . A 35-MeV ^3He beam from the Michigan State University cyclotron impinged on a $200\text{ }\mu\text{g}/\text{cm}^2$ Li- ^{35}Cl target. Tritons were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. A T=3/2, $3/2^+$ level at 5537 keV 25 in ^{35}Ar was observed in $^{33}\text{S}(^3\text{He},n)^{35}\text{Ar}$ (**1975Da14**) but causing an IMME breakdown. **1976Be08** measured the ^{35}K g.s. mass using $^{40}\text{Ca}(^3\text{He},^8\text{Li})^{35}\text{K}$ and predicted the T=3/2 member in ^{35}Ar to be 5579 keV 14. **1976Be08** did not find new peaks in $^{35}\text{Cl}(^3\text{He},t)^{35}\text{Ar}$ between 5484 keV 10 and 5591 keV 10 that were already known from $^{36}\text{Ar}(^3\text{He},\alpha)^{35}\text{Ar}$ (**1973Be26**).

 ^{35}Ar Levels

E(level)[†]

4721
4782
5116
5205
5387
5484
5591
5911
6033

[†] Peaks are seen in triton spectrum in **1976Be08** with no energy values reported. The authors mark those peaks with numbers corresponding to those listed in TABLE I of **1973Be26** (not the same authors) in $^{36}\text{Ar}(^3\text{He},\alpha)$. The energy values quoted here are taken by the evaluators from that table.

$^{36}\text{Ar}(\text{p,d})$ 1968Jo04,1968Ko11

$J^\pi=0^+$ for ^{36}Ar ground state.

1968Jo04: a 27.5-MeV I proton beam was produced by the 1.3-m FFAG cyclotron at the Nuclear Physics Laboratory at the University of Colorado. Targets were natural argon gas (99.6% ^{40}Ar) and 99.0% enriched ^{36}Ar gas. Deuterons were detected using a telescope of a 0.228-mm transmission surface barrier detector and a 0.37-cm lithium-drifted stopping counter with FWHM=120 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, L-transfers, and spectroscopic factors from finite-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$.

1968Ko11: a 33.6-MeV proton beam was produced by the 64-in. sector-focusing cyclotron at Michigan State University. The target was >99% isotopically enriched ^{36}Ar gas. Deuterons were detected using a telescope of a 279- μm silicon surface barrier detector and a 3-mm lithium-drifted silicon counter with FWHM=130 keV. Measured $\sigma(E_d, \theta)$. Deduced levels, J , π , L-transfers, and spectroscopic factors from zero-range Macefield-DWBA analysis of the measured $\sigma(\theta)$.

 ^{35}Ar Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=1.65*3/2$ is a normalization factor adopted by **1968Jo04**. The DWBA calculation in **1968Ko11** assumed neutron-well radius parameter $r_{0n}=1.15$ fm for $d_{5/2}$ pickup and 1.35 fm for $d_{3/2}$ pickup, respectively, to reproduce the forward-angle J -dependence observed in the $L=2$ deuteron angular distributions. **1968Ko11** demonstrates that a larger radius results in a larger DWBA-calculated cross section and this gives rise to problems in extracting absolute spectroscopic factors, and therefore, the evaluators put the C^2S from **1968Ko11** in comments. **1968Ko11** states that there is some evidence that the radius parameter corresponding to the best DWBA fit to the data might also result in the most trustworthy value for the spectroscopic factor.

$E(\text{level})^\dagger$	$J^\pi^\#$	$L^\ddagger$	$C^2S^\S$	Comments
0	$3/2^+$	2	2.92	C^2S : others: 3.03 for 1.15 fm and 1.76 for 1.35 fm (better fit) (1968Ko11).
1180 20	$1/2^+$	0	2.50	$E(\text{level})$: also from 1968Ko11 . C^2S : others: 1.29 for 1.15 fm and 1.05 for 1.35 fm (1968Ko11).
1700 30	$(5/2^+)$		0.1	C^2S : from 1968Ko11 for radius parameter=1.15 fm.
2615 20	$(3/2^+)$	2		$E(\text{level})$: weighted average of 2630 20 (1968Jo04) and 2600 20 (1968Ko11). J^π : spin= $(3/2)$ based on the observed J -dependence of the deuteron angular distributions (1968Ko11). L : from 1968Ko11 . Other: 1 from 1968Jo04 is inconsistent. 1973Be26 determined $L(n)=2$ from $^{36}\text{Ar}(^3\text{He}, \alpha)^{35}\text{Ar}$ and states that $L=1$ is probably in error as this level corresponds well with the known $3/2^+$ state at 2695 keV in ^{35}Cl and there are no nearby negative-parity states in ^{35}Cl . C^2S : 0.42 for 1.15 fm and 0.28 for 1.35 fm (better fit) for $L=2$ from 1968Ko11 . Other: 0.12 for $L=1$ from 1968Jo04 .
2970 20	$(5/2^+)$	2	2.47	$E(\text{level})$: weighted average of 2990 20 (1968Jo04) and 2950 20 (1968Ko11). J^π : spin= $(5/2)$ based on the observed J -dependence of the deuteron angular distributions (1968Ko11). C^2S : assuming $5/2^+$ (1968Jo04). Other: 3.10 assuming $3/2^+$ (1968Jo04). 2.31 for 1.15 fm (better fit) and 1.53 for 1.35 fm from 1968Ko11 .
3200 20	$7/2^-$	3	0.63	$E(\text{level})$: weighted average of 3210 20 (1968Jo04) and 3190 20 (1968Ko11). C^2S : others: 0.64 for 1.15 fm and 0.37 for 1.35 fm (better fit) (1968Ko11).
4756 28	$(1/2^+)$	0	0.172	$E(\text{level})$: weighted average of 4770 20 (1968Jo04) and 4700 40 (1968Ko11). C^2S : others: 0.05 for 1.15 fm and 0.04 for 1.35 fm (1968Ko11).
5102 20	$7/2^-$	3	0.46	$E(\text{level})$: weighted average of 5110 20 (1968Jo04) and 5070 40 (1968Ko11). L : From 1968Jo04 . L is not reported in 1968Ko11 . 1973Be26 determined $L=2$ from $^{36}\text{Ar}(^3\text{He}, \alpha)^{35}\text{Ar}$ and states that $L=2$ gives a much better account of the data than $L=3$ does.
5400 50				$E(\text{level})$: From 1968Ko11 only.
5598 20	$3/2^+, 5/2^+$	2	2.93, 2.37	$E(\text{level})$: weighted average of 5610 20 (1968Jo04) and 5570 30 (1968Ko11). C^2S : others: 1.77 for 1.15 fm (better fit) and 1.25 for 1.35 fm, both for $J^\pi=(3/2, 5/2)^+$ (1968Ko11).
6024 20	$3/2, 5/2^+$	2	1.58, 1.31	$E(\text{level})$: weighted average of 6030 20 (1968Jo04) and 6010 30 (1968Ko11).

Continued on next page (footnotes at end of table)

$^{36}\text{Ar}(\text{p,d})$ **1968Jo04,1968Ko11** (continued) ^{35}Ar Levels (continued)

<u>E(level)[†]</u>	<u>J^π[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	Comments
				C ² S: others: 1.18 for 1.15 fm and 0.83 for 1.35 fm (better fit), both for $J^\pi=(3/2,5/2)^+$ (1968Ko11).
6620 30	(1/2 ⁺)	0		E(level),L: From 1968Ko11 only.
6700 20	7/2 ⁻	3	2.60	C ² S: 0.24 for 1.15 fm and 0.19 for 1.35 fm (1968Ko11).
6820 30		2		L,C ² S: From 1968Jo04 only.
				E(level),L: From 1968Ko11 only.
				C ² S: 0.72 for 1.15 fm (better fit) and 0.51 for 1.35 fm, both for $J^\pi=(3/2,5/2)^+$ (1968Ko11).
7030 20	(3/2,5/2) ⁺	2	2.20,1.79	

[†] From 1968Jo04, unless otherwise noted.

[‡] From DWBA analysis of measured $\sigma(\theta)$. Values of L-transfers are from both 1968Jo04 and 1968Ko11 and values of C²S are from 1968Jo04, unless otherwise noted. Where two C²S values are given, they correspond to respective J^π values as given.

[#] Assumed in 1968Jo04 and/or 1968Ko11 for extracting C²S, unless otherwise noted.

$^{36}\text{Ar}(\text{d,t})$ 1970Wh04,2015Fr01

$J^\pi=0^+$ for ^{36}Ar ground state.

1970Wh04: A 21.0-MeV deuteron beam was produced by the Yale MP tandem Van de Graaff accelerator. The target was a ^{36}Ar gas cell. Tritons were detected using a 140- μm -530- μm thick ΔE -E telescope of silicon surface barrier detectors with FWHM=65-70 keV. Measured $\sigma(E_t, \theta)$. Deduced levels, L-transfers, and spectroscopic factors from JULIE-DWBA analysis of the measured $\sigma(\theta)$. Comparisons with shell-model calculations.

2015Fr01: A 22-MeV deuteron beam was produced by the MP tandem Van de Graaff accelerator at the Maier-Leibnitz Laboratorium (MLL) in Garching, Germany. Targets were fabricated by implanting 25-70-keV 3-6 $\mu\text{g}/\text{cm}^2$ of ^{36}Ar ions into 30 $\mu\text{g}/\text{cm}^2$ natural abundance carbon foils. Reaction products were momentum analyzed by a Q3D magnetic spectrograph. Tritons were detected using a multiwire gas-filled proportional counter backed by a scintillator at the focal plane. Measured E_t at $\theta_{\text{lab}}=15^\circ$, 20° , 25° with FWHM $\approx$ 9 keV and at 54° with FWHM $\approx$ 16 keV. Deduced levels, proton resonance energies, level densities. Comparisons with shell-model calculations.

 ^{35}Ar Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=3.33$ is a normalization factor adopted by **1970Wh04** from **1966Ba54**.

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$L^\#$	$C^2S^\#$	Comments
0	$3/2^+$	2	3.4	
1180 10	$1/2^+$	0	1.4	
1700	$(5/2^+)$	(2)	<0.2	
2635 20	$(3/2^+)$	(2)	0.5	C^2S : 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength. 1970Wh04 also gives $S=0.11$ or 0.032 assuming $L=1$.
2986 20	$5/2^+$	2	2.6	
3200 20	$7/2^-$	(3)	0.33,0.11	C^2S : assuming $r_{0n}=1.25$ F and $V_n\approx 60$ MeV, respectively. 1970Wh04 authors state that there is a large uncertainty in the spectroscopic strength.
5913 5				
5991 3				
6037 3				May be a doublet (2015Fr01).
6055? 3				Tentative (2015Fr01).
6076 3				
6164 3				
6253 3				
6273 3				
6302 3				
6332 3				
6345 3				
6415 2				
6439? 4				Tentative (2015Fr01).
6460 3				
6523 3				
6557 3				
6585 3				
6606 3				
6617 2				
6644 3				
6651 3				
6672 3				

† From **1970Wh04** for low-lying states up to 3200 keV and from **2015Fr01** for others.

‡ As given in **1970Wh04**, used for extracting C^2S .

$^\#$ From DWBA analysis of the measured $\sigma(\theta)$ in **1970Wh04**.

³⁶Ar(³He,α) 1973Be26,1998VoZY

$J^\pi=0^+$ for ³⁶Ar ground state.

1973Be26: an 18-MeV ³He beam was produced by the University of Pennsylvania Tandem Van de Graaff accelerator. The target was pure argon gas enriched to 99.8% in ³⁶Ar. α particles were momentum analyzed in a multi-angle spectrograph and detected using Ilford K-1 nuclear emulsions with FWHM=35 keV. Measured $\sigma(E_\alpha, \theta)$. Deduced levels, J, π , L-transfers, and spectroscopic factors from local zero-range DWUCK-DWBA analysis of the measured $\sigma(\theta)$. Comparisons with shell-model calculations and the mirror nucleus ³⁵Cl. Also see **1972MiZO**.

1998VoZY: a 25-MeV ³He beam at 0.5-μA intensity was provided from the FN tandem accelerator at the Nuclear Structure Laboratory of the University of Notre Dame. The targets were made by implanting 75-keV ³⁶Ar into 40 μg/cm² carbon foils from the University of Toledo ion source. The reaction products were momentum analyzed by the Notre Dame broad-range magnetic spectrograph. The particles were subsequently detected at the focal surface of the spectrograph by a position sensitive proportional gas detector backed with a plastic scintillator. Measured E_α . Deduced levels.

³⁵Ar Levels

Spectroscopic factor $C^2S=(2j+1)\times\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where the isospin Clebsch-Gordan coefficient C^2 is 1/2 in this case, j is the total angular momentum of the transferred neutron, and the normalization factor $N=16.8$. **1973Be26** states that the overall normalization for the (³He,α) reaction is not well determined and therefore resort to empirical means to determine N. $N=15.5$ deduced from shell-model calculated total $S=3.52$ for all four 1/2⁺ states and the **1973Be26** measured $NS=54.6$. $N=18.1$ deduced from the ³⁵Cl(³He,d)³⁶Ar(g.s.) $S=4.73$ (**1970Mo10**) and the **1973Be26** measured ³⁶Ar(³He,α)³⁵Ar(g.s.) $NS=85.4$. **1973Be26** adopted the average $N=16.8$.

E(level)	J^π	L [†]	$C^2S^\dagger$	Comments
0	3/2 ⁺	2	2.545	
1175 2	1/2 ⁺	0	1.19	E(level): weighted average of 1179 10 (1973Be26) and 1175 2 (1998VoZY).
1747 2	5/2 ⁺	2	0.025	E(level): weighted average of 1738 10 (1973Be26) and 1747 2 (1998VoZY).
2649 2	3/2 ⁺	2	0.57	E(level): weighted average of 2637 10 (1973Be26) and 2649 2 (1998VoZY).
2983 2	5/2 ⁺	2	1.39	E(level): weighted average of 2982 10 (1973Be26) and 2983 2 (1998VoZY).
3197 2	7/2 ⁻	3	0.39	E(level): weighted average of 3193 10 (1973Be26) and 3197 2 (1998VoZY).
3882 5	1/2 ⁺	0	0.02	E(level): weighted average of 3884 10 (1973Be26) and 3881 5 (1998VoZY).
4001 3	(3/2) ⁻	1	0.065	E(level): weighted average of 4012 10 (1973Be26) and 4000 3 (1998VoZY).
4113 4				E(level): weighted average of 4110 10 (1973Be26) and 4113 4 (1998VoZY).
4135 4	(3/2) ⁻	1	0.025	E(level): weighted average of 4142 10 (1973Be26) and 4134 4 (1998VoZY).
4350 6				E(level): weighted average of 4350 10 (1973Be26) and 4350 6 (1998VoZY).
4515 5				E(level): weighted average of 4530 10 (1973Be26) and 4514 3 (1998VoZY).
4713 6	1/2 ⁺	0	0.05	E(level): weighted average of 4721 10 (1973Be26) and 4710 6 (1998VoZY).
4774 6				E(level): weighted average of 4782 10 (1973Be26) and 4771 6 (1998VoZY).
5059 11				E(level): unweighted average of 5048 10 (1973Be26) and 5069 4 (1998VoZY).
5116 2	(3/2,5/2) ⁺	2	0.25,0.145 [#]	E(level): weighted average of 5116 10 (1973Be26) and 5116 2 (1998VoZY).
5207 3				E(level): weighted average of 5205 10 (1973Be26) and 5207 3 (1998VoZY).
5389 8				E(level): weighted average of 5387 10 (1973Be26) and 5391 8 (1998VoZY).
5482 2	(3/2,5/2) ⁺	2	0.77,0.445 [#]	E(level): weighted average of 5484 10 (1973Be26) and 5482 2 (1998VoZY).
5594 2	(3/2,5/2) ⁺	2	1.98,1.14 [#]	E(level): weighted average of 5591 10 (1973Be26) and 5594 2 (1998VoZY).
5916 3				E(level): weighted average of 5911 10 (1973Be26) and 5916 3 (1998VoZY).
6036 3	(3/2,5/2) ⁺	2	1.3,0.755 [#]	E(level): weighted average of 6033 10 (1973Be26) and 6036 3 (1998VoZY).
6162 2				E(level): weighted average of 6153 10 (1973Be26) and 6162 2 (1998VoZY).
6262 10				E(level): weighted average of 6258 10 (1973Be26) and 6267 12 (1998VoZY).
6615 3	1/2 ⁺	0	0.36	E(level): weighted average of 6631 10 (1973Be26) and 6614 2 (1998VoZY). Probable doublet in 1973Be26 .
6823 2				E(level): weighted average of 6827 10 (1973Be26) and 6823 2 (1998VoZY).
6948 2				E(level): weighted average of 6959 10 (1973Be26) and 6948 2 (1998VoZY).
7043 4				E(level): weighted average of 7055 10 (1973Be26) and 7042 3 (1998VoZY).
7117 [†] 10				
7293 [†] 10				

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$^{36}\text{Ar}(^3\text{He},\alpha)$ **1973Be26,1998VoZY (continued)** ^{35}Ar Levels (continued)E(level)7423[†] 107502[†] 107840[†] 108019[†] 10

[†] From **1973Be26**. L and C²S values are extracted from DWBA analysis of measured $\sigma(\theta)$. Quoted values of C²S are converted from the S values in **1973Be26** with C²=1/2.

[‡] As given in **1973Be26**, used for extracting C²S.

1973Be26 states that the differences for j=3/2 and 5/2 are small in the DWBA-calculated L=2 shapes. It is not possible to differentiate between the two allowed j values for L=2 transitions. Both C²S values are given for each level with two spin values. Assuming that all four levels have spins of 3/2 would lead to a summed L=2 C²S that exceeds the simple shell-model sum rule limit of 8 for combined 1d_{3/2} and 1d_{5/2} pickup, which suggests that all four of these levels probably have 5/2⁺.

Adopted Levels

$Q(\beta^-) = -1.595 \times 10^4$ 11; $S(n) = 17757$ 17; $S(p) = 83.6$ 5; $Q(\alpha) = -6563$ 3 [2021Wa16](#)
 $Q(\beta^-), S(n)$: Deduced by evaluators using mass excesses of 4777 105 for ^{35}Ca measured by [2023La09](#), and -1487 17 for ^{34}K measured by [2024Dr01](#); -11172.9 5 for ^{35}K from [2021Wa16](#). Values from [2021Wa16](#): $Q(\beta^-) = -16360$ 200 (syst), $S(n) = 18020$ 200 (syst). $S(2n) = 34860$ 200 (syst), $S(2p) = 4747.5$ 6, $Q(\varepsilon) = 11874.4$ 9, $Q(\varepsilon p) = 5978.2$ 5 ([2021Wa16](#)).
 Isotope discovery ([2012Th10](#)): $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})^{35}\text{K}$ at Michigan State ([1976Be08](#)).
 ^{35}K decay measurements:
[1980Ew02, 1979Ca15](#): ^{35}K produced via $^{45}\text{Sc}(p, 8n3p)$ spallation at CERN. Measured $T_{1/2}$ and $\varepsilon + \beta^+$ -delayed protons and γ rays.
[2018Sa54, 2019ChZU](#): ^{35}K produced via $^1\text{H}(^{36}\text{Ar}, ^{35}\text{K})2n$ at Texas A&M. Measured $T_{1/2}$ and $\varepsilon + \beta^+$ -delayed protons and γ rays.
[1998Sc19](#): Polarized ^{35}K produced via fragmentation of ^{40}Ca on ^9Be target at GSI. Measured $T_{1/2}$ and g -factor of ^{35}K ground state from β -NMR.
[2006Me04](#): Polarized ^{35}K produced via $^{36}\text{Ar}(^9\text{Be}, ^{10}\text{Li})^{35}\text{K}$ at NSCL, MSU. Measured g -factor of ^{35}K ground state from β -NMR.
 ^{35}K mass measurements: [2023Zh10, 2007Ya08, 1976Be08](#).
 Theoretical calculations (binding energies, moments, radii, levels, J , π , mass, $T_{1/2}$, reaction rates, etc.): [2025Sh19, 2023Bo17, 2023Fo05, 2022Zo01, 2020Ma25, 2016Si02, 2016Me17, 2003Sm02](#).

 ^{35}K LevelsCross Reference (XREF) Flags

- A** ^{35}Ca $\varepsilon + \beta^+$ decay (25.7 ms)
B $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$
C $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})$

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>$T_{1/2}$</u>	<u>XREF</u>	<u>Comments</u>
0.0	$3/2^+$	175 ms 2	A B C	$\% \varepsilon + \% \beta^+ = 100$; $\% \varepsilon p = 0.37$ 15 $\mu = (+)0.392$ 7 (2006Me04, 2019StZV) J^π : $L(^{36}\text{Ca}, ^{35}\text{K}) = 2$ from 0^+ and allowed $\varepsilon + \beta^+$ feeding to 4725.9, $1/2^+$ level in ^{35}Ar . Possible mirror level of $3/2^+$ ^{35}S g.s. $T_{1/2}$: weighted average of 175 ms 2 (2018Sa54), 178 ms 8 (1998Sc19), and 190 ms 30 (1980Ew02). μ : From β -NMR spectroscopy (2006Me04). Other: 0.36 3 (1998Sc19, \beta-NMR spectroscopy). The positive sign is based on the mirror ^{35}S g.s. $\% \varepsilon p$: from 1980Ew02 for $E(p) > 0.9$ MeV. $E(p) < 0.9$ MeV has also been observed (2018Sa54, 2019ChZU).
1553 5	$(1/2)^+$		A C	E(level): other: 1560 40 from $(^3\text{He}, ^8\text{Li})$. J^π : allowed $\varepsilon + \beta^+$ feeding from $3/2^+$ parent; possible mirror level of 1572, $1/2^+$ level in ^{35}S (1976Be08).
2690 50			C	E(level): from $(^3\text{He}, ^8\text{Li})$ only. Possible mirror level of 2717, $5/2^+$ level in ^{35}S (1976Be08).
3781 26	$1/2^+, 3/2^+$		A	
4018 37	$1/2^+, 3/2^+$		A	
4788 49	$1/2^+, 3/2^+$		A	
4982 13	$1/2^+, 3/2^+$		A	
5249 73	$1/2^+, 3/2^+$		A	
5533 49	$1/2^+, 3/2^+$		A	
5710 49	$1/2^+, 3/2^+$		A	
5865 38	$1/2^+, 3/2^+$		A	
6089 62	$1/2^+, 3/2^+$		A	
6335 73	$1/2^+, 3/2^+$		A	
9168 23	$1/2^+$		A	$T = 5/2$ J^π, T : isobaric analog state of $1/2^+$ ^{35}Ca g.s. (1985Ay01, 1999Tr04).

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Adopted Levels (continued)

 ^{35}K Levels (continued)

[†] From ^{35}Ca $\varepsilon+\beta^+$ decay, unless otherwise noted.

[‡] Allowed $\varepsilon+\beta^+$ feeding from $1/2^+$ ^{35}Ca parent, unless otherwise noted.

^{35}Ca $\varepsilon+\beta^+$ decay (25.7 ms) 1999Tr04,1985Ay01

Parent: ^{35}Ca : $E=0$; $J^\pi=1/2^+$; $T_{1/2}=25.7$ ms 2; $Q(\varepsilon+\beta^+)=1.595\times 10^4$ 11; % $\varepsilon+\beta^+$ decay=100

^{35}Ca - J^π , $T_{1/2}$: From the Adopted Levels of ^{35}Ca .

^{35}Ca - $Q(\varepsilon+\beta^+)$: 15950 105 deduced by evaluators from mass excesses of 4777 105 for ^{35}Ca measured by 2023La09 and -11172.9 5 for ^{35}K from 2021Wa16. $Q(\varepsilon)$ from 2021Wa16: 16360 200 (syst).

1999Tr04, 1998Le45: A secondary ^{35}Ca beam at 0.3 ions/s and 98% purity was produced via the projectile fragmentation of a 95-MeV/nucleon $^{40}\text{Ca}^{20+}$ primary beam impinging on a rotating ^{nat}Ni target, selected using ΔE -ToF by the GANIL LISE3 spectrometer, and implanted into a 500- μm thick silicon detector sandwiched between two 500- μm thick silicon detectors for detecting β^+ particles. 3.5×10^4 ^{35}Ca ions were stopped at a depth of 300 μm with FWHM=70 μm (setting 1) and 2.5×10^4 ^{35}Ca ions were stopped at a depth of 450 μm (setting 2). $\varepsilon+\beta^+$ -delayed protons were detected by the implantation detector. γ rays were detected by three Ge detectors and two NaI detectors. Measured E_p , I_p , E_γ , I_γ , E_{2p} , I_{2p} , βp -coin, $p\gamma$ -coin. Built the decay scheme consisting of 1p-emitting states in ^{35}K , a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , 1p daughter states in ^{34}Ar , and a 2p daughter state in ^{33}Cl . Deduced decay branching ratios, $B(\text{GT})$ and $B(\text{F})$, and parent ^{35}Ca $T_{1/2}$ from implant-decay correlations.

1985Ay01: ^{35}Ca isotope discovery. ^{35}Ca was produced by bombarding a natural calcium target using a 135-MeV ^3He beam from the 88-inch Cyclotron at Lawrence Berkeley Laboratory. Recoiling products were slowed down, transported, and collected on a slowly rotating catcher wheel. The $\varepsilon+\beta^+$ -delayed protons were detected using Si detector telescopes. Measured E_p , I_p , pp -coin. Built the decay scheme consisting of a 2p-emitting state ($T=5/2$ IAS) in ^{35}K , sequential 2p intermediate states in ^{34}Ar , and 2p daughter states in ^{33}Cl . Deduced ^{35}Ca $T_{1/2}$ and ^{35}Ca mass using the known members of $A=35$, $T=5/2$ sextuplets IMME.

Theoretical studies involving ^{35}Ca decay: 2003Sm02, 1991De26, 1990Br26.

The decay scheme is considered relatively complete as all observed individual I(1p) and I(2p) add up to almost 100% in 1999Tr04.

 ^{35}K Levels

E(level) [†]	J^π [#]	$T_{1/2}$ [#]	Comments
0	$3/2^+$	175 ms 2	
1553 5	$(1/2)^+$		$E(p0)_{\text{lab}}=1427$ 5, proton line 1 in 1999Tr04.
3781 26	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=3592$ 25, proton line 5 in 1999Tr04.
4018 37	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=3822$ 36, proton line 6 in 1999Tr04.
4520 [‡]			$E(p1)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=4140$ -4900.
4788 49	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=4570$ 48, proton line 9 in 1999Tr04.
4982 13	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=4754$ 38, proton line 10 in 1999Tr04, corresponding to $E(\text{level})=4977$ 39.
			$E(p1)_{\text{lab}}=2727$ 13, proton line 3 in 1999Tr04, corresponding to $E(\text{level})=4982$ 13.
			$E(\text{level})$: weighted average of the two $E(\text{level})$ values of 4977 39 (p0) and 4982 13 (p1).
5249 73	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=5018$ 71, proton line 11 in 1999Tr04.
5493 [‡]			$E(p1)_{\text{lab}}=2947$ -3500, proton group 4 in 1999Tr04, corresponding to $E(\text{level})=5208$ -5778.
5533 49	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=5294$ 48, proton line 12 in 1999Tr04.
5710 49	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=5466$ 48, proton line 13 in 1999Tr04.
5716 [‡]			$E(p2)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=5336$ -6096.
5865 38	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=5616$ 37, proton line 14 in 1999Tr04.
6089 62	$1/2^+, 3/2^+$		$E(p0)_{\text{lab}}=5834$ 60, proton line 15 in 1999Tr04.
6302 [‡]			$E(p3)_{\text{lab}}=1909$ -2647, proton group 2 in 1999Tr04, corresponding to $E(\text{level})=5922$ -6681.
6335 73	$1/2^+, 3/2^+$		$E(p1)_{\text{lab}}=4041$ 71, proton line 7 in 1999Tr04.
6585 [‡]			$E(p0)_{\text{lab}}=5983$ -6649, proton group 16 in 1999Tr04, corresponding to $E(\text{level})=6243$ -6928.
7813 [‡]			$E(p0)_{\text{lab}}=7131$ -7887, proton group 18 in 1999Tr04, corresponding to $E(\text{level})=7424$ -8203.
9168 23	$1/2^+$		$T=5/2$
			$E(\text{level})$: weighted average of the three $E(\text{level})$ values of 9144 92 (p0), 9157 23 (p1), and 9186 27 (2p0).
			J^π, T : isobaric analog state of $1/2^+$ ^{35}Ca g.s. (1985Ay01, 1999Tr04).
			$E(p0)_{\text{lab}}=8802$ 89, proton line 19 in 1999Tr04, corresponding to $E(\text{level})=9144$ 92.
			$E(p1)_{\text{lab}}=6783$ 22, proton line 17 in 1999Tr04, corresponding to $E(\text{level})=9157$ 23.
			$E(2p0)_{\text{lab}}=4305$ 26, proton line 8 in 1999Tr04, corresponding to $E(\text{level})=9186$ 27 with $S(2p)(^{35}\text{K})=4747.5$ 6 (2021Wa16) and adding a +7-keV correction for the difference in the recoil effect between 1p and 2p emissions (1999Tr04).

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^{35}Ca $\varepsilon+\beta^+$ decay (25.7 ms) 1999Tr04,1985Ay01 (continued) ^{35}K Levels (continued)

<u>E(level)[†]</u>	<u>J^π[#]</u>	<u>T_{1/2}[#]</u>	Comments
Others: E(2p0) _{lab} =4089 30, E(2p0) _{c.m.} =4311 40 (1985Ay01), corresponding to E(level)=9059 41 in ^{35}K with S(2p)(^{35}K)=4747.5 6 (2021Wa16). 1985Ay01 observed E(2p1) _{lab} =3287 30 and proposed both 2p0 and 2p1 proceed via a sequential decay mechanism with the first proton E(p) _{lab} =2213 keV, corresponding to an intermediate state in ^{34}Ar at 6807 keV. However, 2p1 has been ruled out in 1999Tr04 due to the nonobservation of expected py coincidences. 1999Tr04 also states that the observed ratio I(2p0)/I(p)=0.98 9 agrees with the calculated branching ratio I(2p)/I(p)=1 for the IAS (1991De26).			

[†] Evaluators deduced E(level)(^{35}K)=E(p)_{lab}×[m(p)+m(^{34}Ar)]/m(^{34}Ar)+S(p)(^{35}K)+E(level)(^{34}Ar), with S(p)(^{35}K)=83.6 5 (2021Wa16), E(level)(^{34}Ar) from 2012Ni10, and E(p)_{lab} from 1999Tr04. For a ^{35}K proton-emitting level with multiple proton branches, evaluators take the weighted average for E(level)(^{35}K) values deduced from each proton branch. 1999Tr04 used S(p)(^{35}K)=78 20 from 1993Au07, which causes a small difference between the original E(level)(^{35}K) in 1999Tr04 and the deduced E(level)(^{35}K) here.

[‡] Unresolved proton-emitting levels corresponding to a group of unresolved protons populating one daughter state in ^{34}Ar , which are not included in the Adopted Levels.

[#] From the Adopted Levels.

 ε, β^+ radiations

<u>E(decay)</u>	<u>E(level)</u>	<u>Iβ^+[#]</u>	<u>Iε[#]</u>	<u>Log ft</u>	<u>I($\varepsilon+\beta^+$)^{†‡}</u>	<u>Comments</u>
(6.78×10 ³ 11)	9168	8.4 4	0.0056 9	3.3 1	8.4 4	I($\varepsilon+\beta^+$): %I(p0)=0.41 6, %I(p1)=3.8 2, %I(2p0)=4.2 3.
(8.14×10 ³ 11)	7813	1.1 2	4.0×10 ⁻⁴ 9	4.6	1.1 [‡] 2	
(9.37×10 ³ 11)	6585	1.08 17	2.5×10 ⁻⁴ 5	4.9	1.08 [‡] 17	
(9.62×10 ³ 13)	6335	2.9 3	6.1×10 ⁻⁴ 10	4.6 1	2.9 3	
(9.65×10 ³ 11)	6302	2.0 7	4×10 ⁻⁴ 2	4.7	2.0 [‡] 7	
(9.86×10 ³ 13)	6089	1.39 19	2.7×10 ⁻⁴ 5	4.9 1	1.39 19	
(1.009×10 ⁴ 12)	5865	1.42 17	2.6×10 ⁻⁴ 5	5.0 1	1.42 17	
(1.023×10 ⁴ 11)	5716	1.0 4	1.7×10 ⁻⁴ 7	5.2	1.0 [‡] 4	
(1.024×10 ⁴ 12)	5710	0.61 15	1.1×10 ⁻⁴ 3	5.4 +2-1	0.61 15	
(1.042×10 ⁴ 12)	5533	0.72 18	1.2×10 ⁻⁴ 3	5.4 +2-1	0.72 18	
(1.046×10 ⁴ 11)	5493	2.2 3	3.6×10 ⁻⁴ 7	4.9	2.2 [‡] 3	
(1.070×10 ⁴ 13)	5249	3.9 3	5.9×10 ⁻⁴ 9	4.7 1	3.9 3	
(1.097×10 ⁴ 11)	4982	10.1 7	0.0014 2	4.32 6	10.1 7	I($\varepsilon+\beta^+$): %I(p0)=4.2 4, %I(p1)=6.0 5.
(1.116×10 ⁴ 12)	4788	2.9 3	3.8×10 ⁻⁴ 6	4.9 1	2.9 3	
(1.143×10 ⁴ 11)	4520	5.4 9	7×10 ⁻⁴ 2	4.7	5.4 [‡] 9	
(1.193×10 ⁴ 12)	4018	3.8 3	4.1×10 ⁻⁴ 6	4.9 1	3.8 3	
(1.217×10 ⁴ 11)	3781	3.0 3	3.0×10 ⁻⁴ 5	5.1 1	3.0 3	
(1.440×10 ⁴ 11)	1553	48.2 13	0.0030 4	4.26 3	48.2 13	

[†] Deduced from the I(p) values in 1999Tr04 multiplied by 0.994. The original % Σ I(p)=96.4 18 and % Σ I(2p)=4.2 3 in 1999Tr04 lead to % Σ I(p)+% Σ I(2p)=100.6. The evaluators perform a renormalization of 100.6 to 100, which yields the factor of 0.994. The original I(p) in 1999Tr04 was determined from the number of observed proton events and the total number of implants, with simulated proton-detection efficiencies.

[‡] Feeding to a group of unresolved levels in ^{35}K . The corresponding Log ft values are displayed without uncertainties.

[#] Absolute intensity per 100 decays.

$^9\text{Be}(^{36}\text{Ca}, ^{35}\text{K})$ 2012Sh21

$J^\pi=0^+$ for ^{36}Ca ground state.

2012Sh21: A ^{36}Ca secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{40}Ca primary beam impinging on a ^9Be target at NSCL, MSU. The ^{36}Ca nuclei were selected by the A1900 separator to a purity of 8%. The ground states of ^{35}K and ^{35}Ca were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Ca beam at a midtarget energy of ≈ 70 MeV/nucleon on a 188-mg/cm 2 ^9Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na) γ -ray spectrometer CAESAR was placed around the Be target position of the S800 to search for γ decay of any excited states of the residuals formed in knockout reactions. Measured the knockout cross sections for producing ^{35}K and ^{35}Ca from ^{36}Ca and the longitudinal momentum distribution of residuals. Deduced J , π , orbital angular momenta of the nucleons removed from ^{36}Ca , and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for proton removal and longitudinal momentum distributions using eikonal models.

 ^{35}K Levels

<u>E(level)</u>	<u>J^π</u>	<u>L</u>	<u>$C^2S^\dagger$</u>	<u>Comments</u>
0.0	$3/2^+$	2	3.19 16	J^π: interpreted as knockout from proton $1d_{3/2}$ orbital. L: 2012Sh21 deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals. C^2S: from eikonal/Hartree-Fock model with $R_s=0.82$ 4, uncertainty only include experimental contributions. Others: 4.37 24 from eikonal/Strong Absorption model with $R_s=1.14$ 6; 3.62 from shell model calculations. $\sigma_{\text{exp}}=51.1$ mb 26. Calculations: $\sigma_{\text{sp}}=16.2$ mb from eikonal/Hartree-Fock model; $\sigma_{\text{sp}}=11.7$ mb from eikonal/Strong Absorption model.

† Spectroscopic factor $C^2S=\sigma_{\text{exp}}/\sigma_{\text{sp}}$.

⁴⁰Ca(³He, ⁸Li) [1976Be08](#)

[1976Be08](#): ³⁵K isotope discovery. ⁴⁰Ca(³He, ⁸Li)³⁵K is studied using 73.7- and 75.8-MeV ³He beams produced by the Michigan State University cyclotron. Targets were 370 and 190 μg/cm² enriched ⁴⁰Ca on a 20 μg/cm² natural carbon backing. ⁸Li particles were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. Measured the cross section for producing ³⁵K g.s. and the mass excess of ³⁵K. Also see [1976BeXJ](#) and [1976BeZJ](#).

³⁵K Levels

<u>E(level)</u>	<u>Comments</u>
0	
1560 40	Assigned by 1976Be08 as the mirror state of 1/2 ⁺ level almost at the same energy in ³⁵ S.
2690 50	Assigned by 1976Be08 as the mirror state of 5/2 ⁺ level almost at the same energy in ³⁵ S.

Adopted Levels

$Q(\beta^-) = -21910$ syst; $S(n) = 17770$ syst; $S(p) = 1.03 \times 10^3$ 11; $Q(\alpha) = -8.94 \times 10^3$ 11 [2021Wa16](#)
 $S(p), Q(\alpha)$: Deduced by evaluators using mass excesses of 4777 105 for ^{35}Ca measured by [2023La09](#), -1487 17 for ^{34}K measured by [2024Dr01](#), and 11290 16 for ^{31}Ar measured by [2024Yu13](#). Values from [2021Wa16](#): $S(p) = 880$ 280 (syst), $Q(\alpha) = -8560$ 280 (syst). $\Delta Q(\beta^-) = 450$, $\Delta S(n) = 360$ (syst, [2021Wa16](#)).
 $S(2p) = 417$ 105, $Q(\varepsilon) = 15950$ 105, $Q(\varepsilon p) = 15866$ 105, from mass excesses of 4777 105 for ^{35}Ca measured by [2023La09](#); -9384.3 4 for ^{33}Ar , -11172.9 5 for ^{35}K , and -18378.29 8 for ^{34}Ar from [2021Wa16](#). Values from [2021Wa16](#): $S(2p) = 00$ 200 (syst), $Q(\varepsilon) = 16360$ 200 (syst), $Q(\varepsilon p) = 16280$ 200 (syst).
 $S(2n) = 41980$ 450 (syst) ([2021Wa16](#)).
Isotope discovery ([2011Am01](#)): $^{40}\text{Ca}(^3\text{He}, \alpha n)^{35}\text{Ca}$ at Berkeley ([1985Ay01](#)).
 ^{35}Ca production: [1986La17](#), [1986AnZV](#): $\text{Ni}(^{40}\text{Ca}, X)$ at GANIL.
 ^{35}Ca decay measurements: [1999Tr04](#), [1998Le45](#): $\text{Ni}(^{40}\text{Ca}, X)$ at GANIL. Measured $T_{1/2}$ and $\varepsilon + \beta^+$ -delayed 1p/2p emissions.
 ^{35}Ca mass measurements: [2023La09](#), [1985Ay01](#).
Theoretical calculations: [2003Sm02](#), [1998Co30](#), [1997Co19](#), [1991De26](#), [1990Br26](#).

 ^{35}Ca LevelsCross Reference (XREF) Flags

A $^1\text{H}(^{37}\text{Ca}, t)$
B $^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$
C $^9\text{Be}(^{37}\text{Ca}, X)$

E(level)	J^π	$T_{1/2}$	XREF	Comments
0.0	$1/2^+$	25.7 ms 2	AB	$\% \varepsilon + \% \beta^+ = 100$; $\% \varepsilon p = 95.8$ 3; $\% \varepsilon 2p = 4.2$ 3 J^π : $L(^{36}\text{Ca}, ^{35}\text{Ca}) = 0$ from 0^+ . $T_{1/2}$: From implant-decay correlation in 1999Tr04 . Other: 50 ms 30 estimated by comparison with the ^{22}Al yield from 1985Ay01 . $\% \varepsilon p$ and $\% \varepsilon 2p$ are derived from the renormalization of $\% \Sigma I(1p) + \% \Sigma I(2p) = 100.6$ in 1999Tr04 to 100. The original decay branching ratios in 1999Tr04 : $\% \Sigma I(1p) = 96.4$ 18 and $\% \Sigma I(2p) = 4.2$ 3.
2.09×10^3 10	$3/2^+$		A C	E(level): weighted average of 2.24×10^3 33 from $^1\text{H}(^{37}\text{Ca}, t)$ and 2.08×10^3 10 from $(^{37}\text{Ca}, X)$. J^π : $L = 0$ from $3/2^+$ in $^1\text{H}(^{37}\text{Ca}, t)$.

$^1\text{H}(^{37}\text{Ca},\text{t})$ 2023La09

$J^\pi=3/2^+$ for ^{37}Ca ground state.

2023La09: A 50-MeV/nucleon ^{37}Ca secondary beam was produced via the fragmentation of a 95-MeV/nucleon $^{40}\text{Ca}^{20+}$ primary beam impinging on a ^9Be target, selected by the LISE3 spectrometer at GANIL, and then impinged on a liquid hydrogen cryogenic target (CRYPTA). Beam ions before target were tracked using two low-pressure multiwire proportional chambers (CATS). Heavy residuals after target were detected by a zero-degree detection system consisting of an ionization chamber, a set of two XY drift chambers, and a plastic scintillator. Tritons from 2n-transfer were detected using a set of six MUST2 DSSD-CsI telescopes in coincidence with the heavy residues of Ca or Ar. Measured $\sigma(E_t, \theta)$ in inverse kinematics. Deduced levels, J, π , L-transfers from FRESKO-DWBA analysis of measured $\sigma(\theta)$.

First measurement of ^{35}Ca mass excess: 4777 keV 105.

First observation of excited states in ^{35}Ca .

Evidence of the magicity of N=16 close to the proton drip line.

 ^{35}Ca Levels

E(level)	J^π	L	Comments
0.0	$(1/2)^+$	2	J^π : mirror level: $1/2^+$ ^{35}P g.s. and shell model calculations. L: removal of one neutron from the $2s_{1/2}$ orbital and the other from the $1d_{3/2}$ orbital, leaving a single neutron in the $2s_{1/2}$ orbital.
2.24×10^3 33	$3/2^+$	0	J^π : mirror level: $3/2^+$ ^{35}P first excited state at 2386.9 11 and shell model calculations. L: removal of two neutrons from the $2s_{1/2}$ orbital, leaving a single neutron in the $1d_{3/2}$ orbital.

$^9\text{Be}(^{36}\text{Ca}, ^{35}\text{Ca})$ 2012Sh21

$J^\pi=0^+$ for ^{36}Ca ground state.

2012Sh21: A ^{36}Ca secondary beam was produced via the projectile fragmentation of a 140-MeV/nucleon ^{40}Ca primary beam impinging on a ^9Be target at NSCL, MSU. The ^{36}Ca nuclei were selected by the A1900 separator to a purity of 8%. The ground states of ^{35}K and ^{35}Ca were populated by the one-proton/neutron knockout reactions, respectively, from the ^{36}Ca beam at a midtarget energy of ≈ 70 MeV/nucleon on a 188-mg/cm 2 ^9Be secondary target. Knockout residues were identified from their energy loss measured by an ionization chamber at the focal plane of the S800 spectrometer and from their ToF measured between two scintillators at the object position and at the focal plane of the S800 spectrometer. The CsI(Na) γ -ray spectrometer CAESAR was placed around the Be target position of the S800 to search for γ decay of any excited states of the residuals formed in knockout reactions. Measured knockout cross sections for producing ^{35}K and ^{35}Ca from ^{36}Ca and the longitudinal momentum distribution of residuals. Deduced J , π , orbital angular momenta of the nucleons removed from ^{36}Ca , and spectroscopic factors. Calculated single-particle cross sections (σ_{sp}) for neutron removal and longitudinal momentum distributions using eikonal models.

 ^{35}Ca Levels

<u>E(level)</u>	<u>J^π</u>	<u>L</u>	<u>$C^2S^\dagger$</u>	<u>Comments</u>
0.0	$1/2^+$	0	0.45 8	J^π: from $L(n)=0$ from 0^+, interpreted as knockout from neutron $2s_{1/2}$ orbital. L: 2012Sh21 deduced L by comparing the measured and eikonal-calculated longitudinal momentum distributions of residuals. C^2S: from eikonal/Hartree-Fock model with $R_s=0.24$ 2, uncertainty only include experimental contributions. Others: 0.49 8 from eikonal/Strong Absorption model with $R_s=0.26$ 2 and also from transfer-to-continuum model; $R_s=0.26$ 2; 1.80 from shell model. $\sigma_{\text{exp}}=5.03$ mb 46. Calculated values: $\sigma_{\text{sp}}=11.1$ mb from eikonal/Hartree-Fock model; $\sigma_{\text{sp}}=10.2$ mb from eikonal/Strong Absorption model; $\sigma_{\text{sp}}=10.3$ mb from transfer-to-continuum model.

† Spectroscopic factor $C^2S=\sigma_{\text{exp}}/\sigma_{\text{sp}}$.

⁹Be(³⁷Ca,X) [2024Dr01](#)

$J^\pi=3/2^+$ for ³⁷Ca ground state.
[2024Dr01](#): a 72-MeV/nucleon ³⁷Ca secondary beam was produced via the fragmentation of a 140-MeV/nucleon ⁴⁰Ca²⁰⁺ primary beam impinging on a 0.5-mm-thick Be target. Experimental setup includes the CAESium-iodide scintillator ARray (CAESAR) for detecting γ rays, a DSSD-CsI(Tl) ΔE -E Ring Telescope for detecting protons, a Scintillating-Fiber Array (SFA) and the S800 spectrograph for detecting heavy residuals. Measured E_p , E_γ , and $E(\text{residual})$. Deduced the first excited state in ³⁵Ca via the $2p+^{33}\text{Ar}$ exit channel from total decay-energy spectra using the invariant-mass method. Comparisons with shell-model calculations with the USDC Hamiltonian.

³⁵Ca Levels

<u>E(level)</u>	<u>J^{π}</u>	<u>Comments</u>
2.08×10 ³ 10	(3/2 ⁺)	E(level): From total decay energy of 2p+ ³³ Ar E _T =1667 keV 20. J ^{π} : shell-model calculations.

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